

Handbook of Clausal Embedding

Edited by

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Empirically Oriented Theoretical
Morphology and Syntax



Empirically Oriented Theoretical Morphology and Syntax

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Acknowledgments

Chapter 1

Adverbial clauses: Some speculations on the division of labour between sentence and discourse

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This chapter focuses on finite adverbial clauses which are typically introduced by so-called subordinating conjunctions including *when*, *while*, *before*, *after*, *as soon as*, *if*, *because*, *as*, *so that*, *in order to/that*. Adopting a broadly generative perspective, we focus on the nature of the syntactic or semantic relation of the adverbial clause to the clause it associates with (cf. Rutherford 1970 for early discussion), the “host clause”. The core issue is to what extent and how an adverbial clause functions as a constituent of the host clause. Our discussion clearly brings to the fore the interaction between empirical data and theory. It also illustrates how both the classification itself and the syntactic analysis of the observed empirical phenomena are determined by the specific criteria used, as well as by developments in the theoretical framework adopted. Haegeman (2003) distinguishes two types of adverbial clauses, for which she proposes the labels “central” and “peripheral”. After reviewing the empirical evidence on which she bases this classification, we show that peripheral adverbial clauses form a heterogeneous set into which more fine-grained semantic and syntactic distinctions must be introduced. Ultimately, we speculate on a classification that merges the insights of Haegeman’s binary classification with those of Frey’s (2018, 2020a, 2020b) ternary classification.



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1 Adverbial clauses: Introduction

In the syntactic literature, the term adverbial clause denotes an embedded or subordinate clause, finite or non-finite, which functions as an adverbial modifier of a superordinate clause. Adverbial modifiers can also be non-clausal. So, for instance, a temporal adjunct can be realised as an adverb (1a), as a prepositional phrase (1b), as a nominal constituent (1c), and as a finite (1d) or non-finite (1e) adverbial clause.

- (1) a. I will be meeting Mary afterwards/later.
- b. I will be meeting Mary after/before lunch.
- c. I will be meeting Mary this evening/tonight.
- d. I will be meeting Mary when I have finished.
- e. I will be meeting Mary while staying in London.

This chapter focuses on finite adverbial clauses. In addition to temporal modification, as in (1d) and (1e), a range of other meanings can be encoded by adverbial clauses, as shown in (2).

- (2) a. Conditional clauses
 I will meet Mary *if I finish on time*.
- b. Concessive clauses
 I will meet Mary, *even if I don't have anything to tell her*.
- c. Causal clauses
 I will meet Mary *because I need to tell her something important*.
- d. Purpose clauses
 I will meet Mary *so that I can inform her about the decisions*.
- e. Result clauses
 I will meet Mary, *so that she will be aware of the decisions*.
- f. Manner clauses
 I did it *as I was told to do it*. (Huddleston & Pullum 2002: 1146, in their (27ii))

Typically, adverbial clauses are introduced by so-called subordinating conjunctions which include *when*, *while*, *before*, *after*, *as soon as*, *if*, *because*, *as*, *so that*, *in order to/that*. The class of subordinating conjunctions (referred to henceforth as *subordinators*) is not homogeneous. For instance, *since*, *before*, and *after* are also

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prepositions, and *if* can introduce an indirect polar question. Indeed, the definition of adverbial clauses itself is problematic; for an overview we refer the reader to Diessel (2013).

Many aspects of adverbial clauses could be discussed, but we will not attempt to present a comprehensive overview here. Adopting a broadly generative perspective, we will instead focus on one area that has received much attention, namely the nature of the syntactic or semantic relation of the adverbial clause to the clause it associates with (cf. Rutherford 1970 for early generative discussion), and which we will refer to as the “host clause”. This label is non-committal in terms of the syntactic analysis ultimately adopted, in contrast with, for instance, the label “root clause” which would seem to commit one to assuming a degree of syntactic integration between the adverbial clause and the associated clause.¹ The core issue, then, is to what extent and how an adverbial clause functions as a constituent of the host clause.

Our discussion clearly brings to the fore the interaction between empirical data and theory. It also illustrates how both the classification itself and the syntactic analysis of the observed empirical phenomena are determined by the specific criteria used, as well as by developments in the theoretical framework adopted. We will refrain from selecting one specific implementation, opting instead to discuss some possible approaches and their implications.

The longstanding research line on the relation of adverbial clauses to the clause they modify was set off, among other things, by the observation that in terms of the relation of the adverbial clause to the host clause, many subordinators seem polyfunctional and come with a range of different interpretations. English *while* is a case in point, as illustrated by the following attestation, in which subscripting is used to identify two distinct uses of *while*.

- (3) [While₂ this ongoing lawsuit probably won’t stop the use of lethal injection], it will certainly delay its use [while₁ the Supreme Court decides what to do]. (adapted from *Guardian*, G2, 12.12.2003, page 4, col. 4)

In (3), the meaning of *while*₁ can be informally paraphrased as ‘during the time that’: the bracketed adverbial clause encodes a temporal modification of the state of affairs expressed in the host clause. The *while*₂-clause, on the other hand, is contrastive, or adversative, or concessive (the term to be used here), and *while*₂ can be replaced by *whereas* or *although*. In this use, *while*₂ introduces

¹A very interesting aspect of adverbial clauses not addressed at all here is the nature of what is often called the subordinating conjunction. For references and recent discussion, see a.o. Blümel & Pitsch (2019).

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a proposition promoted as the privileged context for the interpretation of the proposition expressed by the host clause. The polyfunctionality of *while*-clauses – and other such adverbial clauses – is not English-specific: the readings are replicated with French *tandis que*, Italian *mentre* (Badan & Haegeman 2022), and Dutch *terwijl*.

Similarly, the subordinator *since* allows two distinct readings: in (4a), the *since*₁-clause has a temporal reading; in (4b), the *since*₂-clause develops the argumentation: it forefronts a background justification for the associated proposition (see a.o. Elhadad & McKeown 1990, Charnavel 2020). The difference between the two interpretations of *since* comes out clearly in the following contrast: temporal *since*₁ can be modified by *ever*, while rationale *since*₂ cannot. (The examples in (4) are from Haegeman 2010: 160, her (25).)

- (4) a. The Germans have been trying to get the Rubens back (ever) *since*₁ a consortium of businessmen offered it for sale last year. (adapted from *Guardian* 21.12.2004, page 10, col. 2)
- b. His second is also placed in a recognisable world, but (*ever) *since*₂ this world is perceived through the eyes of Sunless, the reader is entirely at the mercy of his delusions. (adapted from *Observer Review* 09.01.2005, page 11, col. 1)

Similar contrasts arise with *as* (Huddleston & Pullum 2002: 1146), *because* (Lakoff 1970, Hooper & Thompson 1973), and *if* (Declerck & Reed 2001, Kearns 2006). See also Sweetser (1990), Miyagawa (2012). For some discussion in terms of formal syntax see Haegeman (2012) and the references cited there.

Haegeman (2003) drew a basic distinction between two types of adverbial clauses, for which she proposed the labels “central” and “peripheral”. Central adverbial clauses (such as temporal clauses, conditional clauses or causal clauses) directly modify the proposition encoded in the host clause, while peripheral adverbial clauses (such as concessive clauses or rationale clauses) serve to relate the proposition encoded in the host clause to the discourse by providing privileged background assumptions. As the terminology suggests, central adverbial clauses are considered more integrated than peripheral ones.

This chapter first summarises some of the empirical evidence on which Haegeman’s classification into central and peripheral adverbial clauses was based. We then show that peripheral adverbial clauses form a heterogeneous set into which more fine-grained semantic and syntactic distinctions must be introduced. The final part of the chapter reflects on Haegeman’s binary classification and on Frey’s (2018, 2020a, 2020b) ternary classification, and speculates on a classification which merges the two.

2 Central adverbial clauses vs. peripheral adverbial clauses

We start off with the (simplified) hypothesis that two types of adverbial clauses can be distinguished: “central” adverbial clauses and “peripheral” adverbial clauses. Temporal *while*₁-clauses and temporal *since*₁-clauses illustrate the former, and concessive *while*₂-clauses and rationale *since*₂-clauses illustrate the latter. The difference between the two types of adverbial clauses is revealed by a number of diagnostics, all of which confirm the intuition that central adverbial clauses are more integrated into the host clause than peripheral ones.

2.1 Prosody, punctuation, and structural representation

The contrast between the two types of adverbial clauses is reflected in their prosody: central adverbial clauses can be prosodically integrated into the host clause; peripheral adverbial clauses cannot. In writing, the lesser degree of integration of peripheral adverbial clauses is usually reflected by a comma and occasionally by a full stop, as shown in the attested (5):

- (5) a. And yet some popular things are so brilliant, like *The Simpsons* and *the Angel of the North*. While₂ other brilliant things hardly anyone buys – I’d put my friend’s first novel and sherry in this category. (*Observer* 06.12.2009, page 34, col. 2)
- b. She found this extremely restful. Whereas₂ Patrick, with his enormous respect for women and his irreproachable concern for her, she found rather exhausting. (Anita Brookner, *A Missalliance*; Jonathan Brookner (1986, 1987: 94))

Coordination is also suggestive of a difference in integration. Though associated with the same host clause, the two kinds of *while*-clauses in (6) cannot themselves be coordinated.

- (6) a. *[While₂ this ongoing lawsuit probably won’t stop the use of lethal injection] and [while₁ the Supreme Court decides what to do], it will certainly delay its use.
- b. *This ongoing lawsuit will certainly delay the use of lethal injection [while₁ the Supreme Court decides what to do] and [while₂ it probably won’t stop its use].

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Haegeman (2012) attributes the unavailability of coordination to the degree of integration of the *while*-clauses, correlating this with their syntactic positions. Her hypothesis is that coordination is subject to a likeness condition (cf. the Law of the Coordination of Likes of Williams (1981); see also Williams (1978)), where “likeness” is interpreted in structural terms as meaning that constituents merged in distinct structural positions cannot coordinate (Huddleston & Pullum 2006: 201, their (3)).

To represent the different degrees of structural integration of the two types of adverbial clauses, we adopt the (much simplified) representations in Figure 1. The central adverbial clause in Figure 1a is adjoined to either the vP layer or the TP layer of the host clause, depending on which constituent of the proposition in the host clause it modifies. On the other hand, the peripheral adverbial clause in Figure 1b is adjoined to the CP domain, i.e. the layer of the clause which may encode, among other things, illocutionary force.² for a cognitive-functional perspective on such effects). Though preposing of arguments and predicates is mainly attested in postposed adverbial clauses, (18b) illustrates fronting in a preposed clause. In the attestation in (i), negative inversion occurs in an initial *while*-clause:

- (i) While₂ at no point was the safety of the passengers compromised, this falls well short of the high standards EasyJet expects of its pilots. (*Guardian* 24.03.2018, page 9, col. 5).

These representations touch the core of the general question of clausal embedding addressed in this volume, which has been extensively discussed in the literature. Various positions have been proposed, which we will not review here. We refer to Heycock (2006, 2017) for extensive discussion. To give just one example, (Huddleston 1984: 379–380) distinguishes between “embedding” and “subordination”, where embedding implies a higher degree of clausal integration. While all embedded clauses are subordinate, not all subordinate clauses are embedded:

²Note that “CP” is a simplification in that it may (but need not) be associated with illocutionary force for some clause types. As pointed out by an anonymous reviewer for this volume, for instance, factive complements, while endowed with a CP layer, do not encode illocutionary force. The relation between structural layers and illocutionary force can be better formalised in terms of an articulated CP, as initiated in work by Rizzi (1997). We will not discuss the difference between initial and final position of the adverbial clause. For central adverbial clauses, the assumption is that in addition to being right-adjoined (as in Figure 1a), they can also be left-adjoined to TP. They can also be fronted to the left periphery, where they will have a specific function in terms of information structure, such as contrastive focusing or presenting the content of the adverbial clause as given (see also Diessel 2001, 2013)

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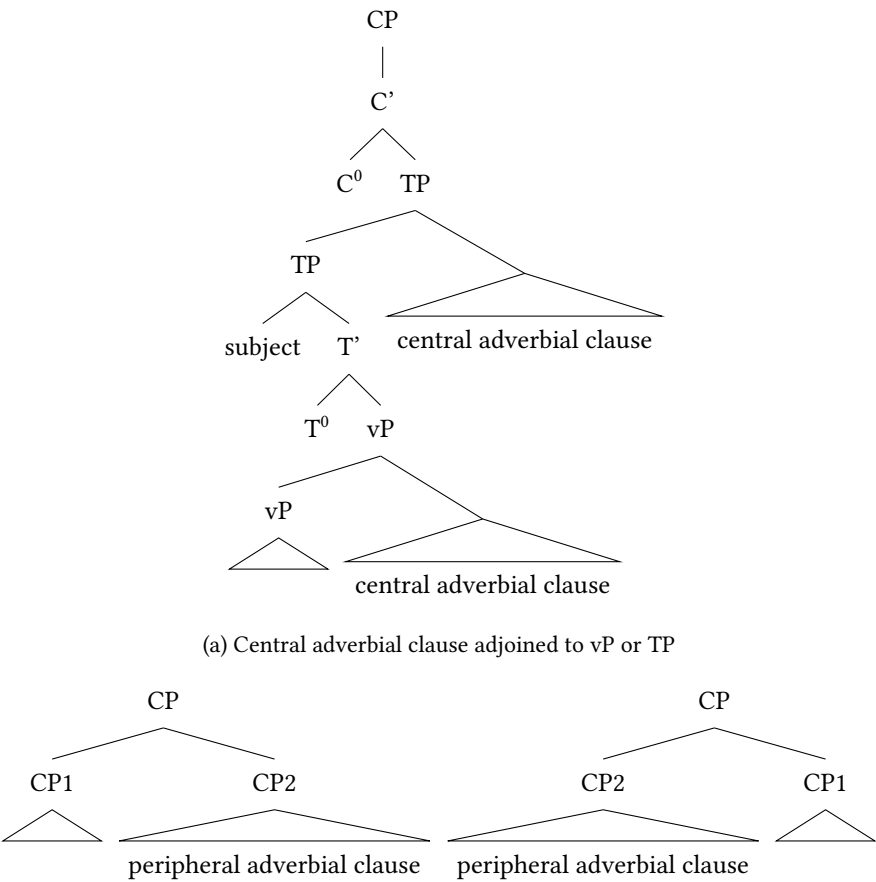


Figure 1: Different degrees of structural integration

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Where a subordinate clause is a constituent of the superordinate clause, we say that it is *embedded*[...] However, contrary to what is assumed in many modern grammars, there is good reason to doubt whether all cases of clause subordination should be handled in terms of embedding. Thus, in [(7a)] we shall want to say that *where as Max thought it appalling* is subordinate [...], but the superordinate clause here is *Ed liked it*: this – unlike [...] *he knew* in [(7b)] – is a clause. The suggestion is that in [(7a)] the subordinate clause is an immediate constituent of the sentence, and in that case it is not embedded, just as *and Ed bought the wine* is not embedded in [(7c)].

- (7) a. Ed liked it, whereas Max thought it appalling.
 b. He knew he was dying.
 c. Liz prepared the food and Ed bought the wine.
 (passage and examples from Huddleston 1984: 379–380)

Huddleston’s *whereas*-clause in (7a) is an example of a peripheral adverbial clause – note that *whereas* can be replaced by *while*₂ – and in terms of our Figure 1b it would be adjoined to a full-fledged CP. On the other hand, the central adverbial clauses in Figure 1a are fully contained within the superordinate CP, and in this sense, they are “embedded”.

Ultimately, whether one considers adverbial clauses as “embedded” or not will be determined by theory-internal arguments. For instance, Diessel (2013) states that “[s]ince (sentential) adjuncts do not fill an obligatory slot in the semantically associated clause, there is no cogent evidence that adverbial clauses are syntactically embedded” Diessel (2013: 342). Diessel adopts a narrower definition of “embedding” than that used here, which would actually only cover argumental adverbials as illustrated in Section 4.3.2 below.

2.2 Structural relations and scope

2.2.1 Temporal subordination

Central adverbial clauses are temporally subordinated to the temporal expressions in the host clause; peripheral adverbial clauses are not temporally subordinated (Hornstein 1993, Haegeman 2019, 2020). With respect to English, the temporal subordination of central adverbial clauses is particularly clear in the encoding of futurity. Consider (3), repeated in (8). In the central *while*₁-clause, the present tense verb *decides* can be assigned a future time reading, which it inherits from the future time expression *will delay* in the host clause. No such temporal

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subordination is available in the concessive *while*₂-clause, in which futurity must be independently realised, here by the future auxiliary *won't*. For temporal subordination in conditional clauses see a.o. Haegeman & Robinson (1979), Haegeman (1984a), Quirk et al. (1985), Schönenberger & Haegeman (2023).

- (8) [While₂ this ongoing lawsuit probably won't stop the use of lethal injection], it will certainly delay its use [while₁ the Supreme Court decides what to do]. (*Guardian*, G2, 12.12.2003, page 4, col. 4)

Similarly, in British English, a temporal *since*₁-clause has been reported to require a perfective tense form in the host clause (Quirk et al. 1985), while no such restriction is operative with rationale *since*₂.

- (9) a. I have been/*am/*was relaxing since the children went away on vacation.
b. He took his coat, since it was raining.
(examples adapted from Quirk et al. 1985: 1070, 15.20)

The different status of central adverbial clauses vs. peripheral adverbial clauses has been recognised in the literature. Hornstein (1993) suggests that in the absence of temporal subordination the relevant subordinators are similar to coordinators:

There is a secondary conjunctive interpretation that all these connectives [*as*, *while*, *when*] shade into. They get an interpretation similar to *and* in these contexts. *And* is not a temporal connective, and these conjunctive interpretations do not tell against the theory [of temporal subordination and complex tense structures] (Hornstein 1993: 206, note 19).

Though a similarity with the coordination relation is also suggested by representation Figure 1b, the relation between a peripheral adverbial clause and its host clause cannot be equated with coordination. Haegeman (2012: 168) signals the following differences: (i) the subject of the second conjunct can be deleted under identity with the first conjunct in (10a), but second conjunct subject ellipsis is not available with a peripheral *while*₂-clause, as in (10b); (ii) gapping is readily available in second conjuncts (11a), but not in a peripheral adverbial clause (11b).

- (10) a. John is doing a Ph.D. in Oxford but did his first degree in Cambridge.
b. *John is doing a Ph.D. in Oxford [while₂ did his first degree in Cambridge].

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- (11) a. John reads the Guardian and Mary the Times.
b. ??John reads the Guardian [while₂ Mary the Times].

In terms of the hierarchical relations represented in Figure 1, the central adverbial clause can be viewed as a constituent of the TP of the host clause while the peripheral clause is “outside” the host TP. Scope-bearing elements in the host clause may scope over central adverbial clauses but they do not scope over peripheral adverbial clauses, as illustrated in (12) with *while*-clauses. In (12a) the *while*₁-clause is in the scope of the quantifying adverb *frequently*: John’s listening to the radio coincides frequently with his wife’s babysitting. The concessive *while*₂-clause in (12b) is not in the scope of *frequently*: there is no claim that John’s listening to the radio frequently coincides with his wife’s playing vinyl records.

- (12) a. John frequently listens to the radio while₁ his wife is in charge of the baby.
b. John frequently listens to the radio, while₂ his wife(sometimes)plays old vinyl records.

This contrast has long been discussed in the literature (see Lakoff 1970: 169 on *because*-clauses).³

2.2.2 Focus

Focal operators in the host clause may scope over central adverbial clauses but not over peripheral adverbial clauses. For discussion of the pattern in English see a.o. Quirk et al. (1985: 1070–1071). The central adverbial clause in (13a) can be clefted, but the peripheral adverbial clause in (13b) cannot be clefted (Haegeman 2006a,b, 2012). Central adverbial clauses can be associated with focusing devices such as *only*, *just*, *simply*, or *mainly*, as in (13c); peripheral adverbial clauses cannot be focused by these elements, as in (13d). See also a.o. Quirk et al. (1972) and Elhadad & McKeown (1990: 99) on *since* and *because*.

- (13) a. It’s [while₁ Bill Clinton was governor of Arkansas] that he developed this programme.
b. *It is [while₂ Bill Clinton was governor of Arkansas] that Hillary Clinton was senator of New York.

³Thanks to an anonymous reviewer for this volume for bringing this reference to our attention.

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- c. Only while₁ Bill Clinton was governor of Arkansas did he develop this programme.
- d. *Only while₂ Bill Clinton was governor of Arkansas was Hillary Clinton senator of New York.

Similarly, a central adverbial clause can function as the answer to a question, but a peripheral adverbial clause cannot. For *since* vs. *because*, Elhadad & McKeown (1990: 99) (14).

- (14) a. Why did Peter leave?
- b. Because₁/*Since₂ he had to catch a train.⁴

2.2.3 Interaction with negation

Central adverbial clauses can be in the scope of the marker of sentential negation in the host clause, while peripheral adverbial clauses cannot. In (15a), the central *while*₁-clause is in the scope of sentential negation; the peripheral *while*₂-clause in (15b) is not.

- (15) a. Bill Clinton did not develop this programme while₁ he was governor of Arkansas.
- b. Hillary Clinton will not become president of the United States, while₂ her husband was president from 1993 till 2001.

See also a.o. Lakoff 1970: 169 and Quirk et al. 1985: 1071.

2.2.4 VP anaphora and VP ellipsis

VP anaphora and VP ellipsis may affect central adverbial clauses (16a), (17a); they do not affect peripheral adverbial clauses (16b), (17b). For instance, in (16a) the second conjunct *so will Bill* can be read as meaning ‘Bill will also leave the room

⁴Specialised conjunctions may bring out the distinct readings. For instance, Dutch *vermits*, French *puisque*, and Italian *poiché* lack the causal reading and only have the rationale reading. Italian *perché*, on the other hand, only has the causal reading. The Italian analogue of (14b) would be as in (i):

- (i) Perché/*Poiché doveva prendere un treno.
 because/*since had to take a train
 ‘Because he had to catch a train.’

Thanks to an anonymous reviewer for this volume for bringing this to our attention.

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while his thesis is being discussed’, in which the possessive pronoun *his* is bound by *Bill*. In (16b), this particular reading is not available, with the second conjunct read as ‘Bill will also be invited, while John’s thesis will not be discussed.’

- (16) a. [While₁ his thesis is being discussed], John will leave the room and so will Bill.
- b. [While₂ his thesis will not be discussed], John will be invited for the interview and so will Bill.
- (17) a. [While₁ his thesis was being discussed], John left the room but Bill didn’t.
- b. [While₂ his thesis will not be discussed], John will be invited for the interview but Bill won’t be.

2.3 Conclusion: External syntax

In this chapter, and based on the data discussed so far, we endorse the hypothesis that the two types of adverbial clauses display a close syntax-semantics fit, and that a difference in external syntax between the two types contributes to the interpretive differences discussed above. Postulating a different level of attachment for central and peripheral adverbial clauses offers a natural way to capture a range of scope phenomena in structural terms. We return to the nature of the syntactic relations in Sections 4 and 5.

For the sake of completeness, we must point out that there is no consensus in the literature that a syntactic characterisation of the relations between different adverbial clause types and their hosts is needed. For instance, Declerck & Reed (2001) deny the relevance of syntax to capture the various interpretive differences discussed in Sections ??–??:

[A] subordinate clause is a syntactically dependent clause. Such questions as the scope of negation, focusing, modality, etc. in the head clause are immaterial to this, as they pertain, not to syntactic, but to semantic integration (Declerck & Reed 2001: 37).

3 The internal syntax of adverbial clauses: Are central adverbial clauses “impoverished”?

Section 2 presented an overview of some diagnostics used to distinguish two basic types of adverbial clauses, referred to as ‘central’ adverbial clauses and

‘peripheral’ adverbial clauses by Haegeman (2003). Sections 4 and 5 will reconsider this binary opposition to reveal that a more fine-grained typology may ultimately be called for. In the present section, however, we continue to adopt the binary distinction as a working hypothesis, and turn to the internal composition of adverbial clauses. We will show that the two types of adverbial clauses differ with respect to clause-internal fronting operations and the availability of certain modal expressions. Informally, the difference could be captured as follows: peripheral adverbial clauses seem to have a richer internal structure than central adverbial clauses.

3.1 Evidence for the structural impoverishment of central adverbial clauses

3.1.1 Argument and predicate fronting

Examples (5a) and (5b) contain peripheral adverbial clauses in which an argument has been fronted to a left-peripheral position. An additional example of argument fronting, drawn from a descriptive grammar, is given in (18a); (18b) is an attestation of predicate fronting.

- (18) a. His face not many admired, while₂ his character still fewer felt they could praise. (Quirk et al. 1985: 1378)
- b. Instead the patriotic duty was dismissing ‘random acts of criminality’. While₂ criminal the rioting indubitably was, random it was not. (*Guardian* 06.09.2012, page 36, col. 2)

In contrast to peripheral adverbial clauses, central adverbial clauses are incompatible with argument or predicate fronting:

- (19) a. *We discovered something else while₁ this paper we were writing. (Quirk et al. 1985: 1378)
- b. *We discovered something else while₁ Bachelor students we were.

Fronting operations like those illustrated in the adverbial clauses in (5), (18) and (19) are referred to as Root Transformations (RTs) (Emonds 1970, 1976, 2004) or Main Clause Phenomena (MCP) (Hooper & Thompson 1973) and have been the subject of much discussion (Haegeman 2012, Heycock 2006, 2017). Other fronting operations with a similar distribution are negative inversion, locative inversion, and inversion around *be*. As the labels suggest, these phenomena are broadly restricted to main clause domains but they are also available in a subset of

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embedded domains. Hooper & Thompson (1973) identify assertion as the key licensing property for these phenomena:

As a *positive* environment we can say that [root] transformations operate only on S[entence]s that are *asserted*. [...] some transformations are sensitive to more than just syntactic configurations. *It does not seem possible to define the domain of an RT in terms of syntactic structures in any general way*. However [...] even if it were possible to define in syntactic terms the conditions under which RTs can apply, [...] the question of *why*[original italics]these transformations can apply in certain syntactic environments and not others would still be unanswered. ((Hooper & Thompson 1973: 495), our italics except where indicated)

In the citation above, Hooper & Thompson seem to say that the environment in which MCP are licensed cannot be defined in syntactic terms. Yet the extract below, taken from an earlier part of the same paper, does seem to imply that the environment in which MCP cannot be licensed can to some extent be defined in syntactic terms:

Though RTs may apply in some complements that are full sentences introduced by the complementizer *that*, *they may never apply in any complements that are reduced clauses*. By reduced clauses we mean *infinitives, gerunds, and subjunctive clauses*, i.e. those complement types which have *uninflected* verbs. (Hooper & Thompson 1973: 484–485, our italics)

In the citation above, MCP are said to be excluded from “reduced clauses”, which Hooper & Thompson correlate with non-finite clauses, but given the observation that MCP are not available in finite central adverbial clauses (19), a subset of finite clauses could then also be argued to be ‘reduced’. See also Maki et al. (1999). Along cartographic lines (Rizzi 1997, Cinque 1999, Cinque & Rizzi 2010), MCP could be argued to be unavailable in domains lacking one or more of the higher structural layers of the clause which are typically used in the encoding of information structure. Such an account was proposed in Haegeman (2006a,b) and also developed by a.o. Bianchi & Frascarelli (2010), Frascarelli & Hinterhölzl (2007), Miyagawa (2012).

3.1.2 The high modal restriction

Central adverbial clauses have been reported to resist speaker-related modal expressions (a.o. Verstraete 2002, Nilsen 2004, Ernst 2001, 2003, 2007, 2009). The

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class of modal expressions singled out roughly corresponds to Ernst's speaker-oriented adverbs (Ernst 2009) and to Cinque's 1999 'high' modals. In Cinque's cartographic terms, the four topmost expressions of modality (Cinque 1999, 2004) are speech-act markers (*frankly*), evaluative markers (*unfortunately*), evidential markers (*allegedly*), and epistemic markers (*probably*, *must*); these modal markers are 'high' because they are structurally located above the temporal layers.⁵ As shown in (20), high modal expressions are not (easily) compatible with central adverbial clauses, while they are compatible with peripheral adverbial clauses. See also Declerck & Depraetere (1995), Verstraete (2002).

- (20) a. *Mary accepted the invitation without hesitation after₁ John probably had accepted it. (based on Verstraete 2002: 149)
 b. The ferry will be fairly cheap, while₂ the plane will probably/might be too expensive.

The restrictions on the availability of the high modals is illustrated in (21) for conditional clauses. Example (21a) from Ernst (2007) illustrates the incompatibility of an evaluative modal with a central *if*₁-clause, and (21b) from Nilsen (2004) illustrates how an epistemic modal is compatible with a peripheral *if*₂-clause, i.e. a conditional assertion. The attested (21c) displays the epistemic modal *may* in a peripheral *if*₂-clause.

- (21) a. *If₁ they luckily arrived on time, we will be saved. (Ernst 2007: 1027, his ex69b))
 b. If₂ Le Pen will probably win, Jospin must be disappointed. (Nilsen 2004: 811, note 5)
 c. But if₂ Sir Richard has been tainted by the affair, and if Mr Sixsmith's role may not have been as entirely well-intentioned as he claims, the individual most damaged by the row remains Stephen Byers. (*Guardian*, 25.02.2002, page 4, col. 3)

In the literature, there is a consensus that the encoding of epistemic modality is related to the speaker, as reflected, for instance, in the following quotation from Zagana (2007) (see also Speas & Tenny 2003, Speas 2004, Tenny 2006).

⁵The label TP as used in Cinque's (1999) model is not to be taken as identical to the label TP in the more parsimonious Minimalist representations. In the latter, at least some modal marking is TP-internal. We do not dwell on this issue, which is theory-internal and beyond the scope of this chapter. Suffice it to say that in an articulated view of clause structure, as espoused by cartography, the domain that is characterised as TP is dominated by the domain hosting speaker-related modal markers, which in turn is dominated by the articulated CP layer.

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[E]pistemic modals should appear only in finite clauses that make statements relative to the speaker’s world of known propositions at the moment of speaking. (Zagona 2007: 230)

The speaker-relatedness of epistemic modality is reflected in its relation to the speech time. For instance, Verstraete notes that “epistemic modals can be morphologically associated with a past tense, ... *this morphological marking does not express the speaker’s past judgement*. Either it is used for tentativeness, [...] or it occurs in a context of indirect or free indirect speech” (Verstraete 2002: 152, our italics). The temporal anchoring of epistemic modals also applies to the other modal expressions located higher than TP in Cinque’s framework. All these modal markers are outside the scope of the event-time operator of the containing clause, provisionally associated with SpecTP, as in (Demirdache & Uribe-Etxebarria 2004, 2012).

3.2 Are central adverbial clauses incompatible with the articulation of information structure?

The observed incompatibility with a range of speaker-related phenomena could lead one to conclude that central adverbial clauses, being non-assertive, cannot encode information structure at all, a conclusion that could be inferred, for instance, from the following quotation:

A particular property of ad[verbial]-subordination is that ad[verbial]-subordinate clauses are always non-asserted or presupposed [...] [T]he claim is that in adverbial clauses, more specifically in situative and relative adverbial clauses in Makhuwa, the proposition of those clauses does not contain new information. This confirms Güldemann’s (1996) findings on the ‘background’ status of asyndetic hypotactic clauses and relative clauses in other Bantu languages. He notes that this status as background “necessitates a compact, largely unstructured presentation of the state of affairs. In order to meet this requirement, the internal (information) structuring of the dependent clause – by focusing the predicate, a term or any other part of the sentence – would be obstructive.” (van der Wal 2014: 61, citing Güldemann 1996: 178)

However, excluding all encoding of information structure from central adverbial clauses is problematic: central adverbial clauses do encode some types of information structure, which we illustrate with examples from English (Section ??) and from French (Section ??).

3.2.1 English

While focus fronting of arguments or predicates is illicit in central adverbial clauses, other focusing devices are available: *in situ* focus is illustrated in (22a), and (22b) shows focal particles such as *only* or *even*. Clefting is also available in central adverbial clauses (see Lahousse & Borremans 2014).

- (22) a. He was always there ready with advice but when₁ I needed money he was nowhere to be found.
 b. When₁ they started to invite only/even Joan, we knew there was something wrong.
- (23) He was always there ready with advice but when₁ it was money that I needed he was not to be found.

Postposing of nominals for focusing purposes is also available in central adverbial clauses, as shown in (24):

- (24) a. I had another shock more recently when₁ I read in the papers that my weakness for the odd square of dark chocolate was not shortening my life but prolonging it. And yet another upset when₁ I read somewhere else [the good news about crosswords]. (*Observer* 26.09.2010, page 4, col. 3)
 b. I have two types of mosquito lotion... But I found that if₁ you put in your pockets [dryer sheets], ...it keeps them away. (Wallenberg 2009: 218, his (14))

3.2.2 French stylistic inversion

French temporal adverbial clauses are compatible with stylistic inversion, in which the subject nominal appears in a non-canonical postverbal position. The reordering has the effect of focusing the subject: in (25a) the subject *les enfants* ‘the children’ occupies a sentence-final position; the example contrasts with (25b), in which it occupies its canonical clause-initial position (Lahousse 2003a,b).

- (25) a. Je voulais partir quand₁ sont arrivés les
 I want.PAST.1 SG leave when be.PRES.3 PLarrive.PART.PL
 enfants.
 the children
 ‘I wanted to leave when the children arrived.’ (Lahousse 2003b: 280, her (1))

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- b. Je voulais partir quand₁ les enfants sont arrivés.
 I want.PAST.1SG leave when the children be.PRES.PL arrive.PART.PL
 ‘I wanted to leave when the children arrived.’ (Lahousse 2003b: 280, her (1))

3.3 How to account for the impoverished structure of central adverbial clauses

Considering the examples in Sections ?? and ??, one might be tempted to conclude, essentially inspired by Hooper & Thompson (1973) proposal cited in Section ??, that the apparent deficiency in central adverbial clauses can be captured in structural terms. Concretely, one might propose that the topmost layers of the clause, dominating the layers encoding temporality (Cinque’s (1999) TP) and used to encode information structure, are missing. Such a proposal is, for instance, developed in Haegeman (2003, 2006a,b), and Frey (2018, 2020a,b). As a result, the articulated CP layer(s) hosting fronted arguments or predicates, would be absent and central adverbial clauses would simply be TPs dominated by a CP shell (as in Frey 2020a: 18). However, this proposal conflicts with the following data drawn from French and Italian.

- (26) a. French
 Quand₁ cette chanson je l’ ai entendue, j’ ai pensé
 when that song I it have.1SG heard.FEM I have.1SG think.PART
 à mon premier amour.
 to my first love
 ‘When I heard that song, I thought of my first love.’ (from Haegeman 2012: 157, her (21c))
- b. Italian
 Se la stessa proposta la fa anche l’ altro candidato, ...
 if the same proposal it make.3SG also the other candidate
 ‘If the other candidate also makes the same proposal ...’ (Cardinaletti 2009: 12, her (22), our translation)

Data such as (26) constitute evidence that central adverbial clauses cannot be analysed as mere TPs immediately dominated by a CP shell, as in Frey (2020a), at least on the cartographic assumption that the nominal constituent to the left of the subject occupies a designated position in the left periphery (see Rizzi 1997 and the literature based on that).

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Note also that the availability of the arguments in the left peripheral space, as in (26), cannot be ascribed to a split between Romance languages and English, according to which the former would be more ‘liberal’ in terms of the availability of the left periphery in central adverbial clauses. As revealed by French (27), fronting operations in central adverbial clauses are also restricted. In a main clause, an infinitival constituent can be fronted without obligatorily having to be resumed by a clitic, as shown in (27a). In central adverbial clauses, on the other hand, fronting of an infinitival constituent with a resumptive clitic (27b) is acceptable, but fronting of an infinitival constituent without a resumptive clitic (27c) is not (see Authier & Haegeman 2015 for extensive discussion).

- (27) a. Fumer dans les bureaux, on pourra (le) faire en toute
to.smoke in the offices we will.be.able.to (it) do with all
impunité.
impunity
‘Smoking in the offices, we’ll be able to do(it)with impunity’
- b. Quand fumer dans les bureaux on pourra le faire en
when to.smoke in the offices one will.be.able.to it do with
toute impunité, ...
all impunity
‘When smoking in the offices we’ll be able to do (it)with impunity ...’
- c. *Quand fumer dans les bureaux on pourra, ...
when to.smoke in the offices one will.be.able.to ...

Garzonio (2008) and Authier & Haegeman (2015) show that the same interaction holds between PP preposing and the need for clitic doubling in Italian and in French. We refer the reader to the relevant papers.

3.3.1 (Selective) truncation

The Romance data introduced in Section ?? might be invoked to propose an account in terms of structural truncation in the sense that one or more structural layers in the clausal domain are missing. Specifically, if the unavailability of speaker-related modal expressions is tied to the absence of the higher modal layer which dominates TP and which is itself dominated by the articulated CP layer, then the availability of the Romance clitic left dislocation patterns (cf. (28)) implies that the layer in the articulated CP which hosts clitic-left-dislocated constituents, Rizzi’s 1997 TopP, is present.

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(28) French

Ce livre-là, il ne l'a probablement pas encore lu.
 this book-there he *ne* it.have.3SG probably not yet read
 'Probably he has not read this book yet.'

Concretely, this requires selective truncation of the lower layers hosting the modals while at the same time preserving some – though not all – of the structural layers that are associated with the CP domain. Haegeman (2003, 2006a,b) develops such an account, which is further developed in Bianchi & Frascarelli (2010), a.o. The drawback of such accounts, however, is that they merely seem to stipulate deficient structure based on the observed data.

3.3.2 Deriving structural deficiency

Haegeman (2010, 2012, 2014) has argued that structural truncation effects can (and hence should) be captured in terms of locality restrictions and intervention effects on movement. We cannot go into her account here, but in essence, her proposal is that adverbial clauses which manifest apparent structural deficiency are derived by the movement of a TP-internal operator to the left periphery and that this movement would be blocked by specific constituents. Haegeman's movement derivation of adverbial clauses revives a long tradition including work by Geis (1970), Reuland (1979), Larson (1985, 1987, 1990), Johnson (1988), Declerck (1997), Lycan (2001), Bhatt (2003), Demirdache & Uribe-Etxebarria (2004, 2012), Arsenijević (2006, 2009a,b), Bhatt & Pancheva (2006, 2017), Stephens (2007), Lip-ták (2009), Tomaszewicz (2009, 2012), Zentz (2011), Williamson (2019)

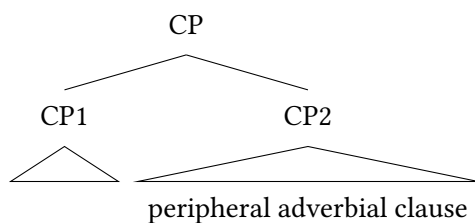
4 The external syntax of adverbial clauses: Proposals and questions

The consensus is that central adverbial clauses are integrated into the host domain. Depending on the degree of functional articulation assumed for the clause, fine-grained levels of attachment of the central adverbial clause in relation to the host clause can be envisaged according to which the adverbial clause is merged with the semantically matching layer in the host clause. For instance, one might envisage that central adverbial clauses which modify the manner of the action encoded in the predicate of the host clause are merged with the latter at a low VP-internal level, that temporal clauses which modify the temporal domain are merged at the TP level, while conditional clauses which modify the modality of

the host clause are associated with the higher modal level (for a detailed implementation in cartographic terms, see Endo (2007, 2011a,b, 2012), Endo & Haegeman (2019). Ultimately, this kind of analysis ‘syntacticises’ the semantic classification of adverbial clauses illustrated in (1d), (1e), (2) in terms of the level of merge with the host clause.

For peripheral adverbial clauses, a number of syntactic analyses have been formulated, which touch upon the general question to what extent constituents encoding discourse relations can or should be integrated at the level of narrow syntax. According to Figure 1b, repeated as (29a), peripheral adverbial clauses are attached at a high level, but the attachment still takes place within the narrow “sentential” syntax. The underlying rationale for the high attachment is that peripheral adverbial clauses manifest root-like phenomena and thus are merged at a level matching the root clause (see Frey 2012, 2018, 2020a,b for the idea of ‘matching’, and also Endo & Haegeman 2019). An alternative proposal is that peripheral adverbial clauses are so-called orphan constituents (Haegeman 1991/2009) which are combined with the host at a discourse level, schematised in (29b).

- (29) a. Orphan analysis: combination with host clause at the discourse level



- b. [_{CP1} host clause][_{CP2} peripheral adverbial clause]

The underlying hypothesis for the orphan approach is that “semantic relations between [peripheral adverbial clauses] and their host sentences are not established at the sentence level at all, but rather at a discourse level” (Haegeman et al. 2011: 348). Similar lines of argumentation were pursued by a.o. Asher (2000) for parentheticals; by Equipe Delic & Véronis (2004) for German conditionals; by Cinque (1982, 2008, 2020), Safir (1986), Fabb (1990) for non-restrictive relatives; and by Cinque (1982), Shaer & Frey (2004), Shaer (2009), Raposo (1996) for hanging topic dislocation.⁶ In a slightly different mode, Espinal (1991) takes such ‘disjuncts’ to be attached to their host clauses in a different dimension of syntactic representation. In this section we discuss some problems for a generalised orphan analysis of peripheral adverbial clauses.

⁶Raposo’s (1996) claim is reported in (Raposo & Uriagereka 2005: 689).

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4.1 Embedding and constituency

Recall that in Section ?? the absence of temporal subordination was invoked as an argument for distinguishing the level of embedding of central adverbial clauses from that of peripheral ones. Haegeman et al. (2011) point out the sequence of tense pattern in (30), their (6), in which the past tense of *had known* in the peripheral adverbial clause is temporally subordinated to the past tense of *claimed* in the superordinate clause.

- (30) As the home secretary, David Blunkett, announced record police numbers, the daughters of Marian Bates *claimed* that while₂ in the past they *had* known their local bobby well, nowadays they *did* not set eyes on a police officer from one week to the next. (*Guardian*, 03.10.2003, page 6, col. 1)

Assuming a syntactic analysis of sequence of tense relations (Hornstein 1993), data such as (30) constitute an argument against the orphan approach, but this is not the conclusion drawn by Haegeman et al. (2011: 352). These authors point out that temporal subordination of the type illustrated in (30) also occurs in non-embedded environments, and at least in these environments it should be viewed as being driven by discourse conditions. Example (30) is reformulated in (31) without the embedding configuration.

- (31) The daughters of Marian Bates no longer knew their local police offers. While₂ in the past they had known their local bobby well, nowadays they did not set eyes on a police officer from one week to the next. (Haegeman et al. 2011: 352, their (7))

However convincing this observation may be, (30) remains problematic: the peripheral adverbial *while*₂-clause is embedded together with its host clause and, at least at first sight, this would suggest that the two form a syntactic constituent. The formalisation of the embedding configuration in an orphan approach is not addressed by Haegeman et al. (2011).

Consider the examples in (32): in each, the bracketed combination of the host clause and a peripheral *while*₂-clause is embedded under a verb of communication – which is also true of (30) – and the *while*₂-clause itself contains a speaker-related modal. In these examples, the sentient mind to whom the modal stance encoded by the modal in the peripheral adverbial clause is being attributed is the subject of the matrix clause, i.e. the speaker introduced by the verb of communication.

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- (32) a. Setting down the new guidelines for privacy cases, Lord Wolf said that [while₂ judges must respect an individual's right to a private life, due weight should also be given to the importance of freedom of expression]. (*Times*, 12.03.2002, page 8, col. 8)
- b. Some said that [while₂ they may not feel British, neither did they feel Pakistani]. (*Guardian*, G2, 28.11.2001, page 3, col. 4)

The effects of temporal and modal subordination, illustrated in (30) and (32) respectively, suggest that while there may well be no c-command relation between constituents in the host clause and those in the peripheral adverbial clause, scopal dependencies obtain between constituents in the higher containing matrix clause and constituents in the embedded peripheral adverbial clause. The proponents of the orphan analysis do not discuss how this relation is established. An approach along the lines of (29a), in which the peripheral adverbial clause is attached at the CP level, would *prima facie* stand a better chance of capturing the embeddability of the combination of the peripheral adverbial clause and its host, given that in this representation the two form a constituent.

Importantly, in the high attachment approach, the embedding patterns in (30) and (32) can only be derived if one adopts an articulated left periphery along the lines of Rizzi (1997) and Haegeman (2012), in which various heads are available above TP. Because the conjunction *that* in (30) and (32) linearly precedes the *while*₂-clause, the peripheral *while*₂-clause cannot be adjoined to a unique CP. In (32b) the conjunction *that* could occupy Force and the inverted auxiliary *did* could occupy the head of the projection whose specifier hosts the negative adverb *neither*, a projection often taken to be a left-peripheral FocP (see Haegeman 1995, 2000, Collins & Postal 2014). This proposal would entail that the peripheral adverbial clause is attached at least as high up as FocP. For completeness' sake, observe also that if, like Frey 2016: 416–417, one were to propose that the peripheral adverbial clause is actually in the specifier of ForceP, the topmost level in the Rizzian articulated CP, then it is unclear how it can ever surface to the right of the complementiser.

4.2 Verb Second (V2)

As shown a.o. by Reis (1997), and recently by Frey (2016, 2018, 2020a,b), a generalised orphan analysis for peripheral adverbial clauses is also challenged by the fact that both central adverbial clauses and peripheral adverbial clauses can be the first constituent of a V2 clause, as shown for Dutch central *sedert* 'since'-clauses in (33a), (33b) and for peripheral *vermits* 'since'-clauses in (33c), (33d).

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Neither clause can standardly function as the extra-sentential constituent in a V3 pattern – also referred to as V2 transgression (Catasso 2015) – in which they would combine with a subject-initial V2 clause, as in (33b) and (33d) (but for dialectal variation see Haegeman & Greco 2018, Greco & Haegeman 2020).

- (33) a. Sedert ik met pensioen ben werk ik thuis.
 since I with pension am work I home
 ‘Since I retired I have been working from home.’
 b. *Sedert ik met pensioen ben, ik werk thuis.⁷
 since I with pension am I work home
 c. Vermits ik met pensioen ben, werk ik thuis.
 since I with pension am work I home
 d. *Vermits ik met pensioen ben, ik werk thuis.
 since I with pension am I work home
 ‘Since I retired I have been working from home.’

In a traditional analysis of V2 according to which the initial constituent occupies Spec-CP, both the central and the peripheral adverbial clause in (33a) and (33c) respectively must be merged in the narrow syntax. In addition, the non-availability of a V3 order, as in (33b) and (33d), shows that these adverbial clauses cannot function as an extra-sentential, i.e. non-integrated, constituent along the lines of Broekhuis & Corver (2016).

4.3 Evaluating the diagnostics

The diagnostic tests discussed in Section 2, which Haegeman (1984a,b, 1991/2009, 2003) used as evidence for a classification of adverbial clauses into central and peripheral adverbial clauses, do not provide conclusive evidence for the analyses proposed by Haegeman in terms of either high CP-attachment or for an orphan analysis. In fact, the diagnostics for differentiating central adverbial clauses (e.g. temporal *while*₁-clauses) from peripheral adverbial clauses (e.g. concessive *while*₂-clauses) also single out a subset of adverbial modifiers standardly taken to be part of the clausal domain, such as the speaker-related epistemic modal adverb

⁷Note that (33b) is a simplification. In West Flemish, patterns such as (33b) are OK:

- (i) West Flemish
 Sedert ik met pensioen ben, ik werk thuis.
 since I with pension am, I work.home

Again, see Haegeman & Greco (2018), Greco & Haegeman (2020)

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probably. As shown in (34), *probably* patterns with concessive *while*₂-clauses for all of the following criteria:

- It is outside the scope of the temporal operator: in (34a) the judgement of probability is located at the ‘present’ speech time and it is not in the scope of the past tense encoded in *took*(cf. the discussion in 3.1.2).
- It is outside the scope of negation: in (34b) negation bears on the event ‘he took early retirement’ and not on the probability of the event.
- It cannot be focused (34c) or clefted (34d).
- It cannot be *wh*-questioned
- It is not affected by VP anaphora. In (34f) the epistemic modality encoded in *probably* is not retrieved in the VP anaphora in the second conjunct (Andrew Radford, p.c.), i.e. the second conjunct reads ‘and his wife has gone home too’ and not ‘and his wife has probably gone home too’.

- (34) a. He probably took early retirement.
 b. He probably did not take early retirement.
 c. He has (*even) probably taken early retirement.
 d. *It is probably that he has taken early retirement, not definitely.
 (with cleft reading)
 e. A: With what likelihood did he leave?
 B: *Probably.
 f. He has probably gone home, and his wife has too.

Importantly, the unacceptability of the patterns in (34) is not due to the categorial status of the *-ly* adverb. Similar patterns would, for instance, be acceptable with the temporal adverb *recently*(cf. Li et al. 2012: 232).

Even if speaker-related modal adverbs such as *probably* are plausibly interpreted higher than their surface position, they are standardly taken to be part of the narrow syntax (cf. Cinque 1999, Ernst 2001, 2003, 2007, 2009). One concrete piece of evidence from English in support of the syntactic integration of these modal expressions is that they can follow the subject (in its canonical position), as in (34a), (34b), and the finite auxiliary, as in (34c), (34f).

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The contrast between speaker-related modal adverbs and temporal adverbs can be related to the level of merger: speaker-related modal adverbs (e.g. epistemic *probably*) differ from lower temporal adverbs (e.g. *recently*) in that the latter are and the former are not within the scope of T(ense)A(spect)M(odal) operators or affective operators. This difference carries over to the syntactic analysis of the contrast between central and peripheral adverbial clauses. Thus, besides the problems posed by the empirical data discussed in Sections ?? and ??, there is a more general problem in the argumentation in that neither the CP-adjunction analysis nor the orphan analysis are actually motivated.

Our findings on the diagnostics that set apart peripheral adverbial clauses from central adverbial clauses reinforce Frey’s 2018, 2020a, 2020b hypothesis, based essentially on German data, that peripheral adverbial clauses pattern syntactically with speaker-related high modals. The observed parallelism suggests that peripheral adverbial clauses are syntactically integrated in the same way that speaker-related modals are. Below we discuss English and Dutch data in support of this hypothesis.

4.3.1 Interactions with V2

Although both central and peripheral adverbial clauses can function as the first constituent in a V2 clause, as discussed in Section ??, they do not pattern in the same way with regard to V2 transgressions (Catasso 2015). Dutch central adverbial clauses, such as temporal *sedert*-clauses can give rise to a V3 pattern in which they combine with a full-fledged V2 root clause, as in (33a), and are picked up by a matching resumptive element left-adjacent to the finite verb, here the PP *sedert dan* ‘since then’. As shown in (33b), this resumption pattern is not available for peripheral *vermits*-clauses.

To account for the acceptability of (33a) we adopt a hypothesis in Haegeman (2010) that central adverbial clauses have referential properties, and we propose that referentiality allows them to be targeted as topics in a resumption pattern. If – by hypothesis – peripheral adverbial clauses lack referential properties, their unsuitability as topics follows.

- (35) a. Sedert ik met pensioen ben, sedert dan werk ik thuis.
 since I with pension am since then work I home
 ‘Since I retired I have been working from home.’
 b. *Vermits ik met pensioen ben, daarom werk ik thuis.
 since I with pension am therefore work I home

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The contrast above is replicated with ‘high’ and ‘low’ adverbs. Relatively low temporal adverbs such as *vroeger* ‘formerly’ or *later* ‘later’ can function as the first constituent in a V2 pattern and may also optionally be doubled by a resumptive element, as in (36), giving rise to a V3 pattern. The choice of the resumptive is determined by the nature of the initial adjunct: *toen* combines with the past adverb *vroeger* and *dan* combines with the future adverb *later*

- (36) a. *Vroeger (toen) werkte ik thuis.*
 formerly (then) worked I home
 ‘Formerly I worked from home.’
 b. *Later (dan) zal ik thuis werken.*
 later (then) will I home work
 ‘Later I will work from home.’

In contrast, higher speaker-related epistemic adverbials such as *waarschijnlijk* ‘probably’ can function as the first constituent in a V2 pattern, as in (37), but they are incompatible with the resumption pattern, as noted by Broekhuis & Corver (2016: 1707)

- (37) *Waarschijnlijk (*dan) is hij weeral ziek.*
 probably(*then)is he again sick
 ‘Formerly I worked from home.’

4.3.2 Adverbial adjuncts and argumenthood

Another reflex of referentiality is observed in the English data in (38) and (39). Central adverbial clauses may function as arguments. Atemporal clause (bracketed in (38a)) or a conditional clause (bracketed in (38b)) can become the complement of a preposition. Similarly, the bracketed *because*-clause in (38c) functions as the subject of the clause. As a subject it can also be dislocated and resumed by a demonstrative *that*. Peripheral adverbial clauses cannot take up such argumental functions, as shown in (39b).

- (38) a. Before you go on vacation, leave yourself something exciting, but not too daunting, for [after you get back]
 (<https://www.smartpassiveincome.com/post-vacation-blues/>)
 b. exactly what it says it is perfect for [if you need some cash in your blizzard wallet].
 (based on <https://www.amazon.com/gp/customer-reviews/R1W5EO8HSSX2CW?ASIN=B012JMS4W2>)

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- c. [Because tennis doesn't have an overt drug problem] (that) doesn't necessarily mean there isn't a problem.
(based on *Observer*, 03.06.2001, page 7, col. 7)

- (39) a. While₁ you are invigilating is not a good time to check proofs.
b. *While₂ you will be invigilating is incompatible with what I expected.

This asymmetry between the two types of adverbial clauses is replicated with non-clausal adverbial modifiers: temporal *-ly* adverbs can function as arguments (40a); on the other hand, the speaker-related epistemic modal adverbs *probably* and *obviously* do not function as an argument in the clause or as a complement in a PP (40b).

- (40) a. They did not book until recently.
b. *I will not book until probably/obviously.⁸

Once again, central adverbial clauses pattern with lower adverbs in the functional hierarchy, while peripheral adverbial clauses pattern with higher speaker-related modal adverbs.

4.4 Summary

Table 1 summarises the similarities and differences between the two types of adverbial clauses. Central adverbial clauses align with lower (temporal) adverbs; peripheral adverbial clauses such as concessive *while*₂-clauses or rationale *since*₂-clauses align with higher (modal) adverbs. Given that higher modal adverbs are syntactically integrated, it follows that the peripheral adverbial clauses that we have discussed so far must also be syntactically integrated.

4.5 Combining high attachment and extra-sentential constituency

So far, we have been distinguishing between central adverbial clauses and peripheral adverbial clauses. Closer inspection reveals that peripheral adverbial clauses are not homogeneous, necessitating a refinement of the classification of adverbial clauses.

⁸(40b) should be read as 'I will not book until it is probable' or 'I will not book until it is obvious'.

Table 1: Two types of adverbial clauses and ‘lower’ and ‘higher’ adverbs

	Central adverbial clause, e.g., <i>while</i> ₁	Lower (temporal) adverbs, e.g., <i>recently</i>	Peripheral adverbial clause e.g., concessive <i>while</i> ₂	Higher modal ad- verbs e.g., <i>probably</i>
Embedded in that-clause	+	+	+	+
In scope of TAM	+	+	-	-
In scope of af- fective opera- tors	+	+	-	-
VPE/VP anaphora	+	+	-	-
First con- stituent in V2	+	+	+	+
Resumptive V3	+	+	-	-
Possible argumental status	+	+	-	-

4.5.1 Re-examining peripheral adverbial clauses

4.5.1.1 Two interpretations

Haegeman (1991/2009), Haegeman et al. (2011) analysed both bracketed strings in (41a) and (41b) as peripheral adverbial clauses.

- (41) a. [If₂ Le Pen will probably win], Jospin must be disappointed.(from Nilsen 2004: 811, npte 5)
- b. [If you don’t mind my saying so], your claims are not very original.

According to the diagnostics laid out in Section 2, the adverbial clauses in (41a) and (41b) both qualify as peripheral rather than central. However, it is also obvious that they function rather differently. The *if*₂-clause in (41a) is a conditional

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assertion, echoing a proposition in the background against which the host proposition is to be processed, whereas the *if*-clause in (41b) expresses what might be called a condition on the speech act: by means of this clause the speaker apologises for the content conveyed in the host clause (see also Sweetser 1990: 76–112 on adverbial clauses as pragmatic markers of politeness, and Declerck & Reed 2001 on conditionals in general).

The bracketed adverbial clauses in (42) are like the adverbial clause in (41b), in that they express a modification of the speech event. The *before*-clause in (42a) functions as a temporal modifier—making it interpretively like a central adverbial clause—of the speech act; the bracketed *since*-clause in (42b), on the other hand, has the rationale reading of peripheral adverbial clauses (see (4b)).

- (42) a. [Before we start], five cabinet ministers will now be voting with the opposition tomorrow.
 b. I possibly have rose-tinged memories because I'd just attained my first girlfriend and earned and spent my first own money (on a small bottle of Brut for Men by Fabergé, [since you ask], and, yes, it still astonishes me how the acquisition of the second did not more violently militate against the acquisition of the first.) (*Observer* 06.11.2005, page 18, col. 1)

Speech-event-modifying adverbial clauses are not within the scope of operators in the host clause, which is why they were classified as peripheral by Haegeman (1991/2009), Haegeman et al. (2011). For instance, the *before*-clause in (42a) refers to the immediate future while the host clause introduces an event that is expected to happen 'after' the speech event. Thus, while the speech-event-modifying temporal adverbial clause in (42a) is arguably 'central' in terms of interpretation, it does not pattern with central adverbial clauses but rather with peripheral adverbial clauses.

However, in contrast to the other peripheral adverbial clauses, speech-event-modifying adverbial clauses with a 'central' interpretation remain incompatible with MCP, just like regular central adverbial clauses:

- (43) a. *Before the meeting we start, five cabinet ministers resigned last night.
 b. *If this you don't mind my saying, your proposals are not very original.

These data suggest that the internal syntax of adverbial clauses is to some extent independent of their degree of integration: even though the temporal interpretation of the *before*-clause in (42a), for instance, is independent of the time

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frame of the matrix clause and thus in those terms the adverbial clause could be said to qualify as ‘peripheral’, its internal syntax remains that of a central adverbial clause. Thus, grouping these speech-event-modifying adverbial clauses with peripheral adverbial clauses like concessive *while*₂-clauses or conditional assertions, as in Haegeman (1991/2009), Haegeman et al. (2011), is problematic. For additional arguments based on conditionals, see also Takami (1988), Ros (2005), Schönenberger & Haegeman (2023).

4.5.1.2 Embedding

Speech-event-modifying adverbial clauses cannot be embedded. In (44a), the (temporal) *while*-clause modifies either the time of the event encoded in the main clause (our *while*₁) or it is recycled as a temporal modifier of the speech event (for which we will use the index ‘?’, i.e. *while*_?). In the former reading, present tense *are* in the *while*-clause is temporally subordinated to that in the matrix clause, i.e. it has a future interpretation. In other words, both the ‘talking about Theresa May’ event and the ‘voting with the opposition’ event happen at some time in the future. In the second reading, present tense *are* refers to speech time, so the ‘talking about Theresa May’ event is taking place now. This ambiguity disappears in an embedded context. The embedded *while*-clause in (44b) must be interpreted as modifying the time of the event encoded in the (bracketed) clause it modifies (*while*₁); a ‘recycled’ interpretation as a temporal modifier of the speech event (*while*_?) is unavailable.

- (44) a. While_{1/?} we are talking about Theresa May, five cabinet ministers will be voting with the opposition.
 b. The Secretary of State will announce [that [while_{1/*?} they are talking about Theresa May], five cabinet ministers will be voting with the opposition].

4.5.1.3 V2 patterns

Speech-event-modifying adverbial clauses like Dutch *terwijl* ‘while’-clauses cannot be the first constituent of a V2 clause, as shown in (45). The speech-event-modifying adverbial clause is an extra-sentential constituent that combines with a full-fledged V2 root clause, leading to a licit V3 order.

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- (45) Terwijl₂ we het over Bart De Wever hebben, ik hoor/*hoor ik dat
 while we it about Bart De Wever have I hear/hear I that
 Antwerpen berucht is voor de invoer van drugs
 Antwerp famous is for the import of drugs
 ‘While we are talking about Bart De Wever, I hear that Antwerp is
 famous for importing drugs.’

As an aside, we note that just like adverbial clauses, some non-clausal adjuncts may be used as speech-event modifiers. By using the speech-event modifier *tussen haakjes* ‘in parenthesis’ in (46) the speaker signals that the statement he/she is about to make is a digression. *Tussen haakjes* combines with a full-fledged V2 clause and it cannot occur as the initial constituent in a V2 clause. In contrast to the speech-event modifier *tussen haakjes*, epistemic *waarschijnlijk* in (47) can occupy the initial position of a V2 clause and cannot combine with a full-fledged V2 clause.

- (46) Tussen haakjes, hij komt/*komt hij morgen ook niet.
 between parentheses he comes/comes he tomorrow also not
 ‘By the way, he is also not going to come tomorrow.’
- (47) Waarschijnlijk, *hij komt/komt hij morgen ook niet.
 probably he comes/comes he tomorrow also not
 ‘Probably he is also not going to come tomorrow.’

4.5.2 A ternary classification

What is common to all of Haegeman’s peripheral adverbial clauses is that they do not contribute to the propositional content of the host clause. One subset of Haegeman’s peripheral adverbial clauses introduces a discourse-accessible proposition which provides the relevant background for the propositional content of the host, while the other subset introduces the relevant background for the speech event, i.e. the communicative act itself. In other words, these two types of adverbial clauses modify quite distinct components of the utterance: the former contribute to the processing or contextualisation of the proposition, while the latter relate to the speech event at the performance level. These two levels of interpretation should be kept apart, as reflected in the literature. Krifka (2017, 2023), for instance, argues that “there must be a distinct semantic layer in-between the layer of the core proposition and the layer of the speech act” (Frey 2020a: 11).

One option to account for the data is to follow Haegeman et al. (2011) and propose that both the high attachment analysis in the narrow syntax and the

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orphan analysis are available in the grammar. In her original analysis, Haegeman (1991/2009) did not fully elaborate on how a discourse-based connection between the orphan constituent and the host clause would be established; neither did Haegeman et al. (2011) offer any concrete formalisation proposals. In the meantime, various proposals have been formulated for establishing the connection between an extra-sentential constituent – an ‘orphan’ – and the host clause. Extra-sentential constituents not only include peripheral adverbial clauses, but also e.g. a subset of non-restrictive relatives and appositives, left-dislocated hanging topics, etc. Several such proposals postulate a merge operation outside the narrow syntax, at the level of discourse grammar, such as Koster’s (1978) *satellites*, his 2000 *ColonP*; Cinque’s (2008) *HP* (also adopted in Giorgi 2014, Frascarelli 2017); de Vries’s (2009) and Griffiths & de Vries’s (2013) *ParP*; Broekhuis & Corver’s (2016) sentence-external constituents; or Haegeman & Greco’s (2018) and Greco & Haegeman’s (2020) *FrameP*.

If one now explores both high attachment in the narrow syntax and extra-sentential merge at the level of discourse, then some of what were called ‘peripheral adverbial clauses’ are attached in the narrow syntax, while speech-event-modifying adverbial clauses – formerly included in the category of peripheral clauses – are outside narrow syntax. Only modifiers merged in the narrow syntax form a constituent with the host clause. Speech-event-modifying adverbial clauses, which, by hypothesis, are merged at the level of discourse syntax, will not form a constituent with the host clause. Moreover, the correlation between richness of internal structure (i.e. MCP, speaker-related meanings) and degree of integration would seem to be relevant only for adverbial clauses merged in the narrow syntax and would not extend to those merged in discourse syntax, an idea to which we return in Section ??.

Based on German data which provide the same type of evidence as that discussed here, Frey (2016, 2018, 2020a,b) develops a ternary classification of adverbial clauses, reproduced in (48).

- (48) The ternary typology of adverbial clauses (Frey 2016, 2018, 2020a,b):
- i. Central adverbial clauses **CACs** modify the **matrixevent**: lower syntactic attachment
 - ii. Peripheral adverbial clauses **PACs** modify the **matrixproposition**: higher syntactic attachment
 - iii. Non-integrated adverbial clauses **NonICs** modify the **speech event**: outside narrow syntax

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Haegeman’s original category of peripheral adverbial clauses is split over PAC (ii) and NonIC (iii). Henceforth, we adopt the label “CAC” for central adverbial clauses, and we reserve the label “PAC” for those peripheral adverbial clauses that modify the main proposition. Departing slightly from Frey’s labelling we use the abbreviation NiC (rather than his NonIC) for those adverbial clauses that modify the speech event. In the next section, we reflect in more general terms on the ternary typology and propose a further revision.

5 Reassessing the classifications

The present section is more speculative and focuses mainly on two issues. First, to what extent is it desirable to posit a clean binary division between CACs and PACs, as in (48)? Second, if speech-event-modifying adverbials (NiCs), by hypothesis, are not syntactically integrated, then how do they become associated with their host clause?

5.1 PACs

5.1.1 An embedding asymmetry

Recall from Section 3 that CACs systematically differ from PACs with regard to their internal syntax in that PACs have a richer left periphery than CACs (a.o. Haegeman 2003, 2006a,b, 2012 for arguments and a survey of the literature; see also Frey 2018, 2020a,b). While Section ?? showed that speaker-related epistemic modal adverbs like *probably* cannot occur in CACs (21a), (49) shows that temporal adverbs like *recently* are compatible with CACs (cf. Li et al. 2012: 232).

(49) When₁ he recently retired, he suddenly discovered new possibilities

In the same way that temporal adverbs such as *recently* are compatible with the internal syntax of CACs, temporal adverbial clauses (e.g. *while*₁-clauses) should also be compatible with CACs, which is to say that they should be embeddable in CACs. Likewise, whatever excludes the availability of speaker-related modals in CACs (structural reduction, or the intervention account discussed in Section ??) should also exclude PACs from embedding in CACs. Both of these predictions are correct.

As discussed in Section ??, both CACs and PACs are embeddable, as also shown for German by Frey (2018, 2020a,b). However, there is an asymmetry in the embeddability of CACs and PACs. In (50a) the temporal *while*₁-CAC is embedded in

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the bracketed (central) event conditional. As expected, the *while*₁-CAC is temporally subordinated, i.e. its present tense *is* inherits the futurity reading from the main clause ('will be very disappointed'). In (50b), the *while*-clause must also be interpreted as temporally subordinated, i.e. it will only receive a temporal CAC reading, leading to the bizarre assumption that 'at some future point in time his sister will no longer have a Cambridge degree'. A concessive PAC reading of the *while*-clause, roughly paraphrased in (50c), is not available.

- (50) a. His mum will be very disappointed [if₁ he quits university while₁ he is still an undergraduate].
 b. His mum will be so disappointed [if₁ he quits university while_{1/*2} his sister has a Cambridge degree].
 c. His mum will be so disappointed [if₁ he quits university despite his sister's Cambridge education].

This embeddability asymmetry is also illustrated in (51), in which *since*-clauses are embedded in central adverbial clauses. Such *since*-clauses are obligatorily interpreted as temporal clauses and cannot receive a rationale reading. A rationale reading of the *since*-clause in (51b) would only become available if the clause becomes a parenthetical, set off by special prosody (comma intonation). In contrast, the *because*-clause in (51b) receives a causal reading, and no special prosody is required since causal *because*-clauses are CACs (see Schönenberger & Haegeman 2023).

- (51) a. [If he hadn't talked to her [since_{1/*2} she came back from China]], he could not have known her phone number.
 b. [If we have to change our holiday plans [because₁/*since₂ there is a deal]], we will have to act promptly.

Observe that the embedding asymmetry replicates the asymmetry in the distribution of adverbials discussed in Section ??.

5.1.2 Cartographic approaches and insertion hierarchies

The parallelisms between PACs and speaker-related epistemic modal adverbs such as *probably* introduced in Section 4, and further elaborated in Section ??, corroborate Frey (2018, 2020a,b)'s hypothesis that PACs, just like these adverbs, must be merged in the narrow syntax and are associated with a modal domain.

To concretise his analysis, Frey adopts Krifka's 2017, 2023 functional hierarchy in (52a). The main properties of the relevant projections are listed in (52b), (52c),

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(52d). Note that in the hierarchy, subjective epistemic modifiers and evidential modifiers are adjoined to JP. We will return to ActP in Section ??.

- (52)
- a. ActP > CmP[CommitmentP] > JP > TP
 - b. “The TP represents a proposition.” (Frey 2020b: 17, his (46i))
 - c. “The judgement phrase (JP) encodes the private assessment of a proposition by a judge.” (Frey 2020b: 17, his (46ii))
 - d. “ActP encodes a speech act performed by a speaker”, i.e. the projection ActP encodes the speaker and the occurrence of a specific speech act (Frey 2020b: 17, his (46iv)); cf. also Pittner 2011 and Freywald 2018 analysis of non-integrated concessives)

In Frey’s classification, the subjective epistemic modal *probably* is characterised as follows:

- “A subjective modal operating on a proposition φ expresses the degree of the judge’s confidence in the truth of φ .” (Frey 2020a: 12, his (32i))
- “Subjective modals, being not-at-issue, are external to the proposition which is communicated by the clause in which they occur.” (Frey 2020a: 10)

Frey’s analysis of subjective modals – our “speaker-related adverbs” – makes the following correct predictions:

- The incompatibility of speaker-related adverbs with CACs follows from the hypothesis that CACs are deficient (whether in terms of structural truncation or as a by-product of locality restrictions), as shown by e.g. Haegeman (2003, 2010, 2012) (see Section 3).
- A speaker-related adverb which is the initial constituent in a V2 configuration, as in (47) above, has moved to a left-peripheral position. The trigger for this movement and the assumed landing site in terms of an articulated clause structure both depend on the theoretical analysis adopted for V2, which we will not go into here (see Haegeman & Greco (2018), Greco & Haegeman (2020) for some discussion and proposals).

Given the parallelism between subjective modals and PACs, the latter too must be JP-adjoined. If we adopt Cinque’s 1999 cartography, reproduced in (53), Krifka’s 2017 JP would be reinterpreted as a “telescoped” variant of two modal projections: MoodP_{evidential} and ModP_{epistemic}

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- (53) MoodP_{speech act} > MoodP_{evaluative} > MoodP_{evidential} > ModP_{epistemic} > TP (Past) > TP (Future) > MoodP_{irrealis} > ModP_{alethic} > AspP_{habitual} > AspP_{repetitive} > AspP_{frequentative} > ModP_{volitional} > AspP_{celerative} > TP (Anterior) > AspP_{terminative} > AspP_{continuative} > AspP_{retrospective} > AspP_{proximative} > AspP_{durative} > AspP_{generic/progressive} > AspP_{prospective} > ModP_{obligation} > ModP_{permission/ability} > AspP_{completive} > VoiceP > AspP_{celerative} > AspP_{repetitive} > AspP_{frequentative} (Cinque 2004: 133, his (3))

If Krifka's functional hierarchy is replaced by a more articulated Cinque-style hierarchy, the classification of PACs and CACs can be rethought in more fine-grained terms. The implications of adopting such an articulated structure for the external syntax of adverbial clauses are discussed in Endo (2012), Endo & Haegeman (2019). On the assumption that adverbial clauses are systematically merged with the domain they modify, CACs would in fact be spread across a range of distinct projections. For instance, temporal clauses would adjoin to TP(Past) or TP(Future) (see Demirdache & Uribe-Etxebarria 2004, 2012), conditional clauses would adjoin to MoodP_{irrealis} (see Haegeman 2010, 2012), aspectual adverbial clauses would adjoin to one of the AspPs, and manner adverbial clauses would adjoin to a still lower projection. Likewise, then, PACs would be associated with two layers of speaker-related adverbials, namely MoodP_{evidential}⁹ and ModP_{epistemic}. In other words, this leads to a more fine-grained analysis of both CACs and PACs.

Adopting Cinque's cartographic implementation has important ramifications in that the binary opposition between CACs and PACs itself dissolves into a fine-grained hierarchy of adverbial clauses. Depending on their exact semantic contribution, PACs would be associated with one of the higher modal projections, while CACs would be associated with lower projections.

5.2 Non-integrated adverbial

Following Frey's argumentation (2016, 2018, 2020a, 2020b), the data on embedding and on V2 discussed in Section ?? led us to the preliminary conclusion that peripheral adverbial clauses come in two shades: one type provides contextual material for the propositional content of the host clause and is syntactically integrated (corresponding to Frey's PAC), and the other type modifies the speech event and is non-integrated, i.e. it is an extra-sentential constituent in the sense of Broekhuis & Corver (2016). By the diagnostics adopted in this chapter, PACs

⁹Charnavel (2020) proposes explicitly that French *puisque* 'since' relates to evidentiality,

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are syntactically integrated while at the same time remaining outside the scope of TP operators.

In Frey’s own analysis, non-integrated dependent adverbial clauses, his Non-ICs, on which we focus in this section, adjoin to the topmost layer, his “ActP”. Just like ActP, NonICs cannot be embedded.¹⁰ Observe that the specifier of ActP hosts the initial constituent (XP in (54)) in an independently occurring V2 configuration. Thus, the presence of a non-integrated clause adjoined to ActP will yield a linear V3 order (i.e. a V2 transgression).

- (54) [_{ActP} NonIC [_{ActP} XP [_{CmP} [_{JP} [_{CP2} peripheral adverbial clause] [_{TP} subject...]]

5.2.1 NonICs: External and internal syntax

In what follows we examine the properties of NonICs described in Frey’s recent publications and compare them with adverbial clauses recycled as speech-event modifiers. Again, we will use the label “NiC” rather than “NonIC” because, though we believe that the diagnostics used are faithful to the spirit of Frey’s proposal, the clause types we identify as non-integrated do not have all the properties of Frey’s (2018, 2020a, 2020b) NonICs.

Frey’s NonICs have the following distributional and interpretive properties:

- (i) While both CACs and PACs can occupy the prefield of a German V2 clause – a position of full integration – the prefield position is not available for NonICs (Frey 2018, 2020a,b: 13, 30, 10–11, respectively). This property carries over to our speech-event-modifying adverbial clauses.
- (ii) While both CACs and PACs can embed together with the host clause, NonICs cannot be embedded (Frey 2018: 14, Frey 2020a: 30, Frey 2020b: 11). Our speech-event-modifying adverbial clauses also fail to embed with the host clause.
- (iii) Regarding interpretation, according to Frey a NonIC encodes an independent speech act ((Frey 2018, 2020a,b: 16, 30, 4, respectively).

Property (iii) has implications for the internal and external syntax of NonICs. As for internal syntax, according to Frey a NonIC is projected to the level of

¹⁰ It is not clear what Frey considers to be the defining characteristic of non-integration: it would appear that he does consider the adjunction of the NonIC as an operation in the narrow syntax, which is contrary to what we will propose in Section ??.

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ActP (Frey 2018, 2020a,b: 27, 30, 21, respectively); a NonIC constitutes a speech act subsidiary or ancillary to the speech act encoded in the host ActP Frey 2020a: 30. Concerning the external syntax of NonICs, Frey writes: “presumably in syntax this ancillary function of NonICs is mirrored by their being adjoined to their hosts” Frey 2020a: 30. Put differently, Frey assumes a matching condition: because of its ancillary function, i.e. because it constitutes a speech act which is subsidiary to another speech act, a NonIC has to be hosted by a speech act, which is to say a constituent projected to the level of ActP. A NonIC will be licensed by adjunction to the root ActP Frey 2020a: 30, his (79); Frey 2020a: 26, his (71). Given that a NonIC is an ActP hosted by the matrix ActP, properties (i) and (ii) above follow: a NonIC cannot be the first constituent in a V2 configuration, which is the position identified as Spec-ActP, and a NonIC cannot be embedded.

PACs can be coordinated; NonICs cannot be coordinated (Frey 2018, 2020a,b: 17, 30, 26, respectively).

- PACs can be coordinated; NonICs cannot be coordinated (Frey 2018, 2020a,b: 17, 30, 26, respectively).
- Strong Root Phenomena (RP) such as tags, interjections and hanging topics (Frey 2018: 10) are licensed by adjunction to ActP (Frey 2018: 52, Frey 2020a: 31, his 39). NonICs, which are projected up to ActP, may therefore host strong RP; since PACs only project up to JP, however, they cannot contain Strong RP (Frey 2018, 2020a,b).
- Among Weak RP, Frey 2018 cites epistemic or evaluative sentence adverbials like German *wahrscheinlich* ‘probably’ and *glücklicherweise* ‘luckily’. Weak RP have a slightly wider distribution than Strong RP and are also compatible with PACs (Frey 2018, 2020b: 27, 27 respectively).

While the properties in (i–vi) characterise non-integrated dependent clauses of various types in German, as shown in Frey’s (2018, 2020a, 2020b) work, to which the reader is referred, they do not extend uniformly to the non-integrated adverbial clauses identified in Section ?? (though other types of clauses in English may well qualify). While adverbial clauses with properties (i) and (ii) can be classified as non-integrated, properties (iv–vi) are not characteristic of English non-integrated speech-event-modifying adverbial clauses.

As for property (iv), in contrast to Frey’s German NonICs, English non-integrated adverbial clauses can be coordinated (thanks to Andrew Radford, p.c., for acceptability judgments). The temporal *before*-clause in (55a) and the temporal *while*-clause in (55b) modify the speech event; as shown in (55c), these two adverbial clauses can coordinate.

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- (55) a. Before we start, next week's class will be cancelled because Monday is a public holiday.
 b. While you are all setting up your laptops, next week's class will be cancelled because Monday is a public holiday.
 c. Before we start, and while you are all setting up your laptops, next week's class will be cancelled because Monday is a public holiday.

Similar evidence is provided by the examples in (56), in which the *while*-clause has a concessive reading. This clause does not provide a background proposition for the propositional content of the host clause; rather it acts as a modifier of the speech event as a whole. As shown in (56c), such recycled *while*-clauses can coordinate:

- (56) a. While I may be prejudiced in this area, students seem nowadays to spend more time on Facebook than on reading.
 b. While I actually do not have systematic evidence to support this, students nowadays seem to spend more time on Facebook than on reading.
 c. While I may be prejudiced in this area and while I actually do not have systematic evidence to support this, students nowadays seem to spend more time on Facebook than on reading.

Turning now to properties (v) and (vi), English argument fronting is a root phenomenon and has a restricted distribution Haegeman (2010, 2012) it is disallowed in CACs but allowed in PACs. Like regular CACs, temporal and conditional clauses which are recycled to modify the speech event continue to disallow argument fronting, as shown in (43), repeated here as (57).

- (57) a. *Before the meeting we start, five cabinet ministers resigned last night.
 b. *If this you don't mind my saying, your proposals are not very original.

Similarly, speaker-related adverbials (see (20a) and (21a)) are incompatible with integrated CACs which modify an event encoded in the host clause. They remain incompatible with temporal and conditional clauses recycled as speech-event modifiers even though these are syntactically non-integrated. This is shown in (58) for a temporal *before*-clause which modifies the speech-event time, and in (59) for a conditional *if*-clause.

- (58) a. Before we start, next week's meeting has been cancelled.

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- b. *Before we probably start, next week's meeting has been cancelled.
- (59) a. If I may change the subject, we need to go over the budget figures once more.
- b. *If I may probably/unfortunately change the subject, we need to go over the budget figures once more.

The data above lead to the conclusion that a CAC recycled as a speech-event-modifying adverbial clause basically retains the internal syntax of the original CAC. But if speech-event-modifying adverbial clauses are non-integrated, as shown by their giving rise to a V2 transgression (in Dutch) and by their non-embeddability (Frey's properties (i) and (ii)), then they ought to be classified with Frey's NonICs. Being non-integrated, one would then expect them to project to ActP and therefore they should allow root phenomena like argument fronting and the presence of speaker-related adverbs. This prediction is not borne out by the data.¹¹

We conclude that, in contrast to Frey's NonICs, CACs recycled as non-integrated speech-event modifiers are not ActPs, i.e. they do not encode speech acts.

In contrast with CACs, English PACs were shown to have a richer structure and to be compatible with fronting operations (19) and with speaker-related modal expressions (21b,c, 22b,c). Again, PACs recycled as speech-event modifiers retain their compatibility with fronting operations, as shown in (60a), and with speaker-related modal expressions, as illustrated in (60b) and (60c).

- (60) a. While prejudiced I may be, the appalling situation is not one that Biden has created.
- b. While I probably would not have systematic evidence to support this, students nowadays seem to spend more time on Facebook than on reading.
- c. Since Jane probably has already told you anyway: Yes, I am going to apply for the new job.

¹¹Frey does not discuss the kinds of CACs or PACs recycled as speech-event modifiers which we focus on here, except possibly in a short note, reproduced here, in which, however, he does not discuss the internal syntax of these clauses: "Different adverbial clauses which usually occur as CACs or PACs may appear as NonICs in front of or following a V2-clause. This is also prosodically marked" (2018: 16, his (13) given below).

- (i) a. /Weil es schon sehr spät ist/, /wir gehen jetzt nach Hause/
because it already very late is we go now to home
- b. /Wir gehen jetzt nach Hause/, /da es schon sehr spät ist/.

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To be clear, the examples in (60) are not, strictly speaking, counter examples to Frey’s system. However, when taken together with the data on CACs presented above, they reveal an inconsistency: when functioning as speech-event modifiers, adverbial clauses should, in Frey’s system, pattern together as a uniform class, i.e. NonICs. However, CACs and PACs do not become syntactically indistinguishable just because they appear as non-integrated speech-event modifiers; rather, the distinction between CACs and PACs “survives” non-integration.

5.2.2 An external discourse syntax for non-integrated adverbial clauses

Haegeman & Greco (2018), Greco & Haegeman (2020) elaborate an analysis for adjunct-initial V3 patterns in West Flemish (see footnote 7 above, example (iii)). In their analysis the initial adjunct is extra-sentential and combines with a full-fledged V2 clause at the discourse level, i.e. beyond the narrow syntax, giving rise to a discourse unit which they label “FrameP”, resembling earlier proposals in the literature. Importantly, being outside narrow syntax, their FrameP, schematised in Figure 2, should not be confused with a further extension of Rizzi’s left periphery (unlike, for instance, proposals by Benincà & Poletto 2004, Wolfe 2020, or Holmberg 2020). Rather, their FrameP, which is merged at the discourse level, is reminiscent of Haegeman’s (1984a, 1984b, 1991/2009) “orphan” approach.

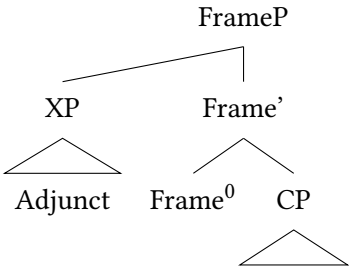


Figure 2: Extra-sentential FrameP merged with V2 clause

As described in Greco & Haegeman (2020), the discourse unit FrameP consists of two speech acts: (i) the illocutionary speech act of assertion, questioning, etc. encoded by the host V2 clause (i.e. “CP” in Figure 2), and (ii) a secondary speech act of framing, encoded by FrameP. The specifier of FrameP introduces an entity (or a set of entities) in the discourse with respect to which the proposition expressed by the host V2 clause is interpreted as relevant (see Jacobs 1984, Endriss

2009, Ebert et al. 2014). For instance, in (61) the speech-event modifier – located in Spec-FrameP – introduces a condition under which the statement made in the host clause will be of relevance to the addressee.

(61) If someone needs me, I am working from home today.

Finally, the speech act associated with FrameP is “subsidiary” or “ancillary” in Frey’s terms (see above), because it remains parasitic on the speech act associated with the V2 clause. Crucially, in this approach, the frame-setting operation itself, encoded by FrameP, is the ancillary speech act – the non-integrated adverbial clause need not itself encode a speech act.

5.3 Towards a revised classification

In a series of papers, Haegeman 1984a,b, 1991/2009, 2003, 2006a,b proposed a binary classification of adverbial clauses, differentiating between central adverbial clauses and peripheral adverbial clauses. Two proposals were developed for the analysis: one analysis was based on the degree of integration/height of adjunction of the adverbial clause (i.e. TP-internal for central adverbial clauses vs. TP-external for peripheral adverbial clauses); the second analysis assumed a more fundamental split between integrated adverbial clauses, characterising central adverbial clauses, and non-integrated “orphan” adverbial clauses, characterising peripheral adverbial clauses (cf. Safir 1986, Fabb 1990). See Table 2.

Endorsing and exploring Frey’s arguments for non-integration for a subset of Haegeman’s original “peripheral adverbial clauses”, one way to proceed is to merge Haegeman’s binary classification with Frey’s ternary classification along the lines of Table 3. The basic opposition is now between integrated and non-integrated adverbial clauses, i.e. merger in the syntax vs. merger in the discourse. Integrated clauses comprise both CACs and PACs (in Frey’s restricted sense).

However, depending on the richness of the clausal articulation assumed, a continuum of levels of merger can be proposed (see Laenzlinger 1996, Cinque 1999 for the integrated adverbial clauses (i.e. CACs and PACs). Assuming a matching condition according to which an adverbial modifier – clausal or non-clausal – adjoins at the level which it modifies, this potentially will lead to a more fine-grained sequencing of integrated adverbial clauses (and adverbials in general) in the narrow syntax. Non-integrated clauses (NiCs) merge as orphans at the discourse level (in Spec-FrameP). Based on the literature, NiCs include an array of subtypes.

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Table 2: The binary classification (top half) and analyses (bottom half) (Haegeman 1984a,b, 1991/2009)

	Central clauses	adverbial	Peripheral clauses	adverbial
Examples	e.g. temporal clause (now: CAC)	while1-	e.g. concessive clause (now: PAC) Including what is now referred to as e.g. speech-event- modifying temporal while-clause, speech- event-modifying concessive while-clause (now: NiC)	while2-
Degree of integration or	TP-internal		TP-external	
Integration vs. orphan analysis	Integrated		Non-integrated	

6 Concluding remarks and further issues

Based on various diagnostic tests relating to scope, such as sentential negation, focus, TAM subordination, Haegeman (1984a,b, 1991/2009) identified two types of adverbial clauses: central adverbial clauses and peripheral adverbial clauses. The former were analysed as adjoined to the host clause in the syntax; for the latter, two analyses were proposed: high adjunction (in the syntax) or merger outside syntax (in the discourse). An examination of peripheral adverbial clauses revealed that a subtype of these – speech-event-modifying adverbial clauses – display specific properties, such as not being embeddable.

Based mainly on German data, Frey (2016, 2018, 2020a,b) argued for a new ternary classification of adverbial clauses, distinguishing central adverbial clauses, peripheral adverbial clauses, and non-integrated clauses. Following Frey (2020a,b), central adverbial clauses and a subset of peripheral adverbial clauses are now taken to be merged in the syntax, with different degrees of embedding. Applying Frey’s diagnostics for non-integration reveals that speech-event-modifying adverbial clauses should be classified as non-integrated. Speech-event-

Table 3: A revised binary classification and analysis

Integrated	Non-integrated
Integrated clauses (CACs and PACs), differentiated by the height of attachment; merged in narrow syntax.	Non-integrated clauses (NiCs), merged in discourse syntax in Spec-FrameP (cf. Haegeman & Greco 2018, Greco & Haegeman 2020). The following is a non-exhaustive list of such sentence-external constituents: • Propositional adverbial clauses: CACs and PACs “recycled” as speech-event modifiers • Frey’s NonICs • Speech-event-modifying non-clausal adverbials • Left-dislocated hanging topics • A subset of non-restrictive relative clauses (Cinque 2008, 2020) • Adverbial clauses modifying the root proposition (with stringent locality conditions) discussed in a.o. Haegeman & Greco (2018), Greco & Haegeman (2020). The reader is referred to the cited papers for examples and discussion.

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modifying adverbial clauses are merged at the discourse level, outside narrow syntax. However, they do not share all the properties of Frey's non-integrated clauses.

This new classification of adverbial clauses (cf. Table 3) leads to many further research questions, of which some specifically relating to non-integration can be mentioned here:

- Which adverbial modifiers and clauses, if any, typically or only function as non-integrated clauses (see Meinunger 2004)? Are there formal constraints?
- In case of co-occurring non-integrated constituents, would these be accommodated by a recursive FrameP?
- Should the observed subtypes of non-integrated clauses be differentiated in terms of their discourse syntax?
- What (if any) are the ordering restrictions among non-integrated constituents, and how are these regulated? Are these subject to discourse-building principles like those in the narrow syntax, or are ordering restrictions governed by a “natural order” in the sense of Wiese et al. (2020)?
- Which property or properties enable an adverbial modifier to function as an extra-sentential constituent?
- How does FrameP as developed in Greco & Haegeman's work (and analogous projections in the literature) fit in with proposals in the literature for the representation of speech act coordination as in Ross (1967), Speas & Tenny (2003), Speas (2004), Tenny (2006), Hill (2007a,b, 2011), Miyagawa (2012), Haegeman & Hill (2013), Wiltschko (2017a,b) (and further references cited there)?

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Chapter 2

Coordination

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Coordination is often referred to as symmetrical linkage of like constituents. However, both phrase and sentence coordination also exhibit quite a number of asymmetries, some of which are relevant to the syntax of coordinate structures and some of which are not. This article gives an overview of the relevant phenomena and discusses their implications for the analysis of coordination. Particular attention is paid to coordinate structures that block coordination ellipses and (only) allow for asymmetric extraction, as both are properties more associated with subordinate structures and are thus of particular importance in delineating coordination from subordination, the central subject of the present handbook.

1 Delimiting coordination from subordination

1.1 Functional (in)dependency

In traditional linguistics, it is generally assumed that there are two modes in building complex sentences: *coordination* and *subordination*. A sentence S2 is called subordinate to a sentence S1, if (and only if) S2 *depends on* S1 in one way or another. Typically, such a dependency is established in one of the following two ways: Either the subordinate sentence functions as an argument to an expression in the matrix clause, see (1a), or it modifies an expression in the matrix clause. Modification, in turn, also comes in two modes. If, on the one hand, the complementizer of the subordinate sentence establishes a semantic relation such as causality between the two clauses, this is called *adverbial modification*, see (1b). If, on the other hand, the subordinate sentence picks up on an expression of the matrix clause, which it restricts or comments on, this is called *relative modification* and the subordinate sentence is a relative clause, see (1c).



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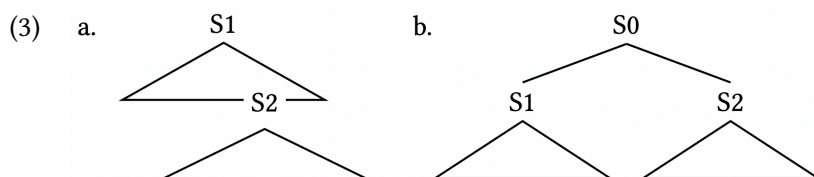
- (1) a. You told me [*you don't like him*].
- b. I like him [*because you don't like him*].
- c. I don't like [the people [*that you like*]].

In the case of coordination, there is (typically) no such dependency. Both conjuncts are, in this respect, independent from each other. This is illustrated with (2a–2c): None of the sentences in (2a–2c) function as an argument to an expression of the other sentence, nor do they modify any expression of the other sentence. This is not to say, of course, that there are no dependencies at all. In (2b) and (2c), for example, the subject of the second clause is pronominalized, since it is coreferent with the subject of the first clause. This kind of dependency, however, is orthogonal to the coordination/subordination distinction.

- (2) a. [John likes hiking], and [Bill plays football].
- b. Either [John is hiking], or [he is at his office].
- c. [Bill plays football], but [he is not a natural].

1.2 Syntactic (in)dependency and syntactic (des)integration

What is crucial to the coordination/subordination distinction, however, is that the dependency criterion has a structural correlate: If a sentence S2 depends on a sentence S1 in the sense sketched above, then S2 is in some structural sense part of S1. If there is no such dependency, there is (typically) also no such part/whole relation: Neither is S2 in any relevant sense part of S1 nor is S1 part of S2. In a first step, this difference can be sketched as follows:



There are several ways to specify this part/whole relationship. One way to do so is the following (see Reich & Reis 2013): Suppose S is your favorite label for sentences in formal syntax. Then, a sentence S2 is a *part of* / *syntactically dependent on* a sentence S1, if there is at least one node labeled S1 that dominates the topmost node labelled S2. Moreover, a sentence S2 is called a *proper part of* / *syntactically integrated into* a sentence S1, if the lowest node labelled S1 dominates

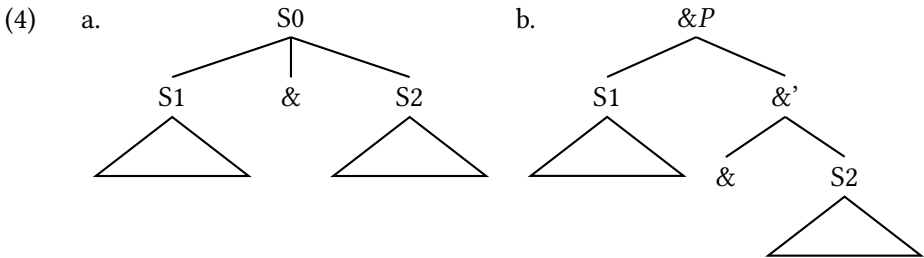
the topmost node labelled S2. In other words, if S2 adjoins to S1, then S2 is still syntactically dependent on S1, but it is not syntactically integrated.

Within the realm of syntactic integration, there are several levels to distinguish (see Reis 1997, and articles 2, 3, 4 and 5 in this volume). However, this is not central to the purpose of the present paper. What is relevant to our discussion is that the proposed definition of syntactic dependency narrows down the possible structures of coordination. The structure of coordination and, more generally, its basic properties is the topic of Section 2. This includes a short discussion of constituent coordination and the question whether constituent coordination can be taken to be an elliptical variant of sentential coordination. After a closer look at the external syntax of coordination (in particular the *External Homogeneity Condition*), it becomes more and more clear that also with respect to coordinate structures there are several types to be distinguished, starting from typical instances of symmetric coordination (Section 3) up to various forms of asymmetric coordination (Section 4) that come rather close to subordinate structures in their syntactic behavior. In this way, this article approaches the topic of this handbook – subordination – successively from its counterpart: coordination.

2 Basic properties of coordinate structures

2.1 Phrase structure

Given a standard approach to phrase structure, coordination can be analyzed as an exocentric (tripartite) structure as sketched in (4a). Alternatively, the conjunction & can be taken to project to a conjunction phrase &P with the second conjunct located in the complement position and the first conjunct in the specifier position, see (4b) (Johannessen 1998). The way I defined dependency rules out structures like the one proposed in Munn (1993), in which the second conjunct plus the coordinating conjunction is adjoined to the first conjunct.



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Empirical support for binary branching comes from the fact that in parenthetical constructions the coordinating conjunction forms a constituent with the second conjunct, see (5a). Moreover, coordinating conjunctions can be used in utterance-initial position, see (5b). Though this might be a somewhat different kind of use (see e.g. Blakemore 2006), it is still suggestive evidence.

- (5) a. Peter runs, and I am really serious here, 100m in 10.2 seconds.
- b. And how do we proceed from here?

The right-branching structure in (4b) with the complement S2 following the head & is typical for English, German and other Germanic languages. Typologically, however, I also observe mirrored, left-branching structures with the complement S1 preceding the conjunction. A case in point is the Korean converb *ko* in (6) cited after Haspelmath (1995: 2):

- (6) Achim mek-ko hakkyo ey kassey yo.
 breakfast eat-CVB school to went PT
 ‘I ate breakfast and went to school.’

When shifting the focus to constituent coordination, one further observes asymmetries in the binding of pronouns. In (7a) the pronoun *his* is naturally interpreted as being co-referent with *John*. This reading seems barely accessible in (7b). Building on the assumption that the binding of pronouns is primarily a matter of hierarchical relations, the most straightforward explanation is that the structure of coordination is in fact asymmetric, the first conjunct *John* being in a higher position from which it c-commands the second conjunct.

- (7) a. [[John_i] [and [his_i colleagues]]] went to the movies last night.
- b. ?? [[His_i colleagues] [and [John_i]]] went to the movies last night.

Again, mirrored structures with the specifier of &*P* following the head and its complement can be observed in the languages of the world, see the following example from Classical Tibetan (cited after Haspelmath 2007: 8), with *dan* (“and”) coordinating *bgegs* (“demon”) and *ndre* (“spirit”):

- (8) Blama-s [[bgegs-dan] ndre] btul.
 Lama-ERG [[demon-and] spirit] tamed
 ‘The lama tamed demons and spirits.’

The above considerations mainly concern simple conjunctions like *and*, *or* and *but*, which link sentences as well as constituents. Besides these frequently discussed conjunctions, there are also complex conjunctions that consist of more than one word (like *as well as*) or that are split in two parts (like *neither ... nor*, *both ... and*). Moreover, in the case of sentences, there is also the possibility of asyndetic coordination, which is the paratactic serialization of sentences without conjunction. These constructions raise important questions concerning the syntax of coordination, ranging from its delineation from discourse up to the question whether all languages of the world do in fact express coordination in one way or another (see Gil 1991, Haspelmath 2007 for discussion). For reasons of space, however, I cannot go into more detail here, but will focus on aspects that are more to the core of the coordination/subordination distinction.

2.2 The law of the coordination of likes

Even though the syntax of coordination in (4b) is structurally asymmetric, coordinate structures are typically symmetric in the sense that both conjuncts are of the same syntactic category. This is true both on sentence level and on the level of constituents. On the level of constituents, there is a strong preference to coordinate, for example, VPs with VPs, PPs with PPs, APs with APs and NPs with NPs, see (9) and (7) above. Similarly, on the sentence level, there is a strong preference to coordinate declaratives with declaratives (10a), interrogatives with interrogatives (10b), and imperatives with imperatives (10c). Moreover, in a language like German, which exhibits two verb positions (fronted, final) and three different verb orders (verb first (V1), verb second (V2), verb final (VL)), this preference for symmetry also extends to verb order, see (11).

- (9) a. They decided [[_{VP} to stay at home] and [_{VP} to watch a movie]].
- b. Sue looked for her keys [[_{PP} in the apartment] and [_{PP} at her office]].
- c. John is a [[_{AP} competent] and [_{AP} popular]] colleague.
- (10) a. John likes hiking, and Bill plays football.
- b. Who likes hiking and who likes playing football?
- c. Come by and bring some food!
- (11) a. *Geh doch bitte runter und hol die Post.*
 Go PT please down and get the mail
 ‘Please go down and get the mail.’

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- b. Ich *putze* die Wohnung und du *machst* das Abendessen.
I clean the apartment and you make the dinner
'I clean the apartment and you make dinner.'
- c. Ich frage mich, ob er zu Hause *bleibt* und einen Film *anschaut*.
I wonder me if he at home stays and a movie watches
'I wonder if he stays home and watches a movie.'

This observation is usually referred to as the “law of coordination of likes,” and it is known in formal syntax since its very beginnings, see Chomsky (1957). It is also known since Chomsky (1957), that this rule is not without exceptions, see the cross-categorial coordinations in (12) from Sag et al. (1985: 117/8).

- (12)
- a. Pat is [NP a Republican] and [AP proud of it].
 - b. Pat is either [AP stupid] or [NP a liar].
 - c. That was [NP a rude remark] and [PP in very bad taste].

In order to account for the full range of coordinate structures, Chomsky (1957: 35) entertains the hypothesis (CR_→), which suggests that the coordination of two constituents can be traced back to the coordination of two sentences plus some kind of “Conjunction Reduction” (CR), see e.g. Ross (1967), Hudson (1973) and more recently Wilder (2018) for discussion.

(CR_→) If we have two sentences $Z + X + W$ and $Z + Y + W$, and if X and Y are actually constituents of these sentences, we can generally form a new sentence $Z - X + Y - W$.

Since in Chomsky’s (1957) original formulation, only sufficient conditions on deletion are given (from the coordination of full sentences we derive the coordination of constituents), this rule is indexed with the symbol $\rightarrow$ for implication. As we will see, however, the more controversial question is whether each (apparent) coordination of constituents can be traced back to the coordination of two sentences. This necessary condition on CR will be indexed with the inverted implication symbol $\leftarrow$ below. If it turned out that both (CR_→) and (CR_←) are viable, we could strengthen Chomsky’s rule to a biconditional (CR_↔).

While (CR_→) deals with the categorially symmetric as well as the cross-categorial cases we saw above, Chomsky also notes that this rule cannot account for the ungrammaticality of **the scene* [[*of the movie*] and [*that I wrote*]] *was in Chicago*. On the other hand, restricting (CR_→) to “constituents of the same type” (Chomsky 1957, 36) excludes the well-formed cases in (12).

2 Coordination

This puts us in a dilemma. On the one hand, the rule (CR $\rightarrow$) that generates coordinate structures seems to be too liberal. Adding a syntactic likeness constraint, on the other hand, is too restrictive. To resolve this dilemma, Sag et al. (1985) start from the observation that all the conjuncts in the cross-categorical coordinate structures in (12) are used predicatively in a copula construction. This suggests, according to Sag et al. (1985), that the copula *be* actually selects for a higher-level category – a predication phrase PredP – rather than specifically for NPs, APs or PPs. What we observe in (12) then is strictly speaking not a cross-categorical coordination of NPs, APs and PPs, but a categorially symmetric coordination of PredPs (or more precisely of different feature extensions of PredPs). This saves the syntactic likeness constraint.

At the same time, this approach rules out cross-categorical coordinate structures like the one in (13a), essentially because the conjuncts have different functions: The adverb *beautifully* modifies the verb *sang*, while the NP *a carol* constitutes its direct object. As a consequence, there is no single syntactic phrase structure rule that generates both the sentence *John sang beautifully* and the sentence *John sang a carol*. The categorially symmetric, but functionally asymmetric structure in (13b) is ruled out by the same reasoning.

- (13) a. *? John sang [_{AdvP} beautifully] and [_{NP} a carol].
 b. *? Hans sang [_{NP} das Lied von Coldplay] und [_{NP} den ganzen Tag].
 Hans sang [_{NP} the song of Coldplay] and [_{NP} the whole day]
 ‘Hans sang the song of Coldplay and the whole day long.’

Rather than taking a syntactic approach, the obvious alternative is to look for an appropriate semantic constraint. Instead of stipulating an underspecified syntactic category like PredP, one might demand that the conjuncts in a coordinate structure are of the same logical type, which is $\langle e, t \rangle$ in the case of copula constructions. In fact, such a constraint is already in place, if I model *and* in terms of generalized conjunction, see Partee & Rooth (1983). Generalizing the sentential connective $\wedge$ to the coordination of two predicates in a copula construction, *and* is essentially taken to map two properties P and Q of type $\langle e, t \rangle$ onto their intersection $P \cap Q$ (i.e. *and* translates as $\lambda Q \lambda P \lambda x (P(x) \wedge Q(x))$). This accounts for the cross-categorical coordinations in (12). At the same time, the coordinate structures in (13) are ruled out, since the nominal objects *a carol* / *das Lied von Coldplay*. (type e), and the adverbial modifiers *beautifully* / *den ganzen Tag* (type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$) are of different logical types. Thus, in a sense Chomsky (1957: 36) was right in stipulating that the conjuncts in constituent coordination need to

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be “constituents of the same type,” though the relevant notion turns out to be semantic rather than syntactic.

However, restricting coordinations to conjuncts of the same logical type still seems to be insufficient. Consider the contrasts in (14) and (15): The conjuncts *slowly*, *with great care*, *yesterday* and *on Tuesday* are adverbial modifiers across the board and thus of the same logical type. Still, crossing manner adverbials with time adverbials (14) seems to be worse than conjoining manner with manner and time with time adverbials (15). Similarly, the subjects *John* and *the water hose* are of the same logical type (type *e*), but (16) is still degraded. This is arguably due to the polysemy of the verb *to lie* that selects either for agentive subjects like *John* or non-agentive ones like *the water hose* (Manfred Krifka, p.c.).

- (14) a. ?? John walked with great care and on Tuesday.
b. ?? John ate beans yesterday and slowly.
- (15) a. John walked slowly and with great care.
b. John ate beans yesterday and on Tuesday.
- (16) ? [_{NP} John] and [_{NP} the water hose] are lying on the lawn.

Even though the contrasts in (14) to (16) might not be as strong as in the type clash cases above, they are there. This suggests that there is a more fine-grained notion of semantic types, which includes types like *manner*, *time*, and *place* as proposed in Munn (1993). Alternatively, one might follow Lang (1984, 1991) in assuming that the concepts expressed by the conjuncts need to relate to a superordinate concept called “common integrator” (CI) in a more or less straightforward way. In the case of *grandpa and grandma* in (17a) this is the concept *grandparents*, in (17b) it is the question (under discussion) “*Who left us what?*”.

- (17) a. Grandpa and grandma leave us the house and the Mustang.
b. Grandma leaves us the house and grandpa leaves us the Mustang.
- (18) a. ? John and his hat took a walk along the Boston river.
b. ? John likes ice cream and the earth revolves around the sun.

In (18) the conjuncts are of the same semantic type (individuals and propositions, respectively), but compared to (17) it is less straightforward to relate the conjuncts to a common integrator. In (18a), one might imagine that the hat is a typical accessory of John, and in (18b) one might interpret the first conjunct in

view of the second conjunct as a matter of course. But this requires some extra reasoning on the side of the hearer, and it is that extra reasoning on the side of the hearer that can be made responsible for the degraded acceptability.

2.3 The limits of “conjunction reduction”

One might thus argue that coordinations, both on sentence and constituent level, are subject to the following syntactic, semantic, and pragmatic constraints:

(CR_→⁺) If $\Phi(X)$ and $\Phi(Y)$ are well-formed sentences, and both X and Y are constituents in these sentences, then the coordination $\Phi(X \& Y)$ is well-formed, too, provided that

- a. X and Y are of the same logical/semantic type. (semantics)
- b. X and Y are easily related to a common integrator. (pragmatics)

The constraint (CR_→⁺) only implements sufficient conditions on the well-formedness of coordinate structures. One might wonder whether it is possible to strengthen the conditional “if” to a biconditional “if and only if” as follows:

(CR_←⁺) Suppose $\Phi(X \& Y)$ is a well-formed sentence and $X \& Y$ is a constituent in this sentence with X and Y being of the same logical/semantic type and easily relating to a common integrator, then both $\Phi(X)$ and $\Phi(Y)$ are well-formed sentences, too.

This property of coordinate structures is also called “*substitution salva grammaticalitate*” (see Höhle 1990) or just “substitutability” (Wilder 2018). Another way to put (CR_←⁺) is this: Is it possible to reduce all instances of constituent coordination to sentential coordination?

This question is certainly the more interesting and has a rather long tradition that is traced back in Krifka (1991) even to Beattie (1788), who gives a negative answer. The crucial argument concerns, like in contemporary linguistics, the existence of collective readings, and thus the need for sum formation or grouping alongside Boolean conjunction. But let’s start with a less problematic case: The coordination of two DP subjects triggers plural morphology on the finite verb; however, the corresponding sentences are in the singular:

- (19) Sue and Bill are originally from Arizona.
 - a. Sue is originally from Arizona.
 - b. Bill is originally from Arizona.

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One way to deal with this problem is to attribute the plural feature in (19) to the conjunction that heads the coordination. Since there is no conjunction in (19a) and (19b), there is also no need to take the plural feature into account.

The assumption that it is the coordinating conjunction & that triggers plural morphology on the verb predicts, however, that the verb is always in the plural. This is (mostly) correct as long as the subject precedes the finite verb. If the subject follows the verb, however, there is also the option of agreement with the first conjunct only, see (20). This phenomenon is called “first-conjunct agreement” (FCA), see e.g. Aoun et al. (1994, 1999) for the discussion of Arabic dialects and Johannessen (1998) for Czech and German.

- (20) a. There was a man and a woman sitting on a bench.
b. There was a man and a woman reading the same book.

Aoun et al. (1994, 1999) propose that what looks like agreement with the first conjunct only, in fact derives from clausal coordination followed by ellipsis (*there was a man sitting on a bench and there was a woman sitting on a bench*). This view is challenged in Munn (1999). Munn (1999) argues that expressions like *same* require a semantic plural, even though they are consistent with a syntactic singular. As a consequence, the clausal coordinate structure **there was a man reading the same book and there was a woman reading the same book* is degraded, and the FCA in (20b) lacks a grammatical clausal source.

This argument relates to a more general phenomenon, namely the distinction between collective and distributive predicates. Consider, for example, the verb *to meet*. Apparently, *to meet* is a collective predicate that makes a statement over a group of people, and thus requires a semantically plural (and non-distributive) subject, see (21). This is why (21a) and (21b) make no sense in terms of semantic interpretation. The verb *to go*, on the other hand, allows for a distributive reading, meaning it licenses the entailments from (22) to (22a) and (22b). It is thus straightforward to relate coordinated subjects with distributive predicates to clausal coordination, but it is less so with collective predicates.

- (21) Sue and Bill met at the museum.
a. * Sue met at the museum.
b. * Bill met at the museum.
- (22) Sue and Bill went to the museum.
a. Sue went to the museum.
b. Bill went to the museum.

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The existence of collective predicates is, therefore, one of the most central arguments against the assumption that constituent coordination of the form $\Phi(X \ \& \ Y)$ can be traced back systematically by way of conjunction reduction to sentential coordination of the form $\Phi(X) \ \& \ \Phi(Y)$. Reversing the reasoning, Munn (1993) proposes a group building semantics for DP-coordination as in (21) and generalizes this semantics to other cases of constituent coordination. Following Link (1983) and others, the basic idea is that if Sue refers to the individual s and Bill to the individual b , the coordination *Sue and Bill* refers to a complex individual $s \oplus b$ that is represented as the ‘sum’ of the two individuals s and b . It is this complex individual $s \oplus b$ that is the logical subject of the predication $\text{MET}(s \oplus b)$ in collective readings. Distributive readings as in (22) are then derived (i) either by attributing the distributivity to the semantics of the predicate and introducing an operator DIST that distributes over atomic parts of complex individuals, or (ii) by avoiding complex individuals and interpreting the coordinate structure by way of generalized conjunction: $\lambda y \lambda x \lambda Q(Q(x) \wedge Q(y))$. While the latter strategy is consistent with the assumption of constituent coordination, it basically implements kind of a semantic equivalent to syntactic deletion. Furthermore, the discussion of iterated coordinations like *Napoleon and Wellington and Blücher fought against each other* raises the question whether besides complex individuals also a notion of groups of (complex) individuals is needed (note that Blücher and Wellington fought on the same side against Napoleon), see e.g. Landmann (1989). According to Krifka (1991), the grouping of (complex) individuals can also be accounted for on the level of discourse by introducing suitable antecedents for anaphoric interpretation along the dynamic interpretation of the syntactic structure of coordination (guided by prosody).

As suggestive as the evidence from collective predicates is, the matter has still not been finally settled. Schein (2017: 3f), for example, reminds us of the fact that in Right Node Raising constructions, like (23a), the bracketed part is likewise a collective predicate (for it contains the reciprocal phrase *each other’s spouses*). As a consequence, the fully spelled out counterpart *Marvin made a grand entrance at the gala with each other’s spouses on their arms and Bernice swept in at the gala with each other’s spouses on their arms* is degraded. The strings that are linked here in Right Node Raising, however, are not necessarily constituents (see Section 3.2), and therefore an analysis in terms of constituent coordination is not readily available. Similarly, the subjects in (23b) cannot refer to a plurality since they participate in two different events.

- (23) a. Marvin made a grand entrance Δ_i and Bernice swept in [at the gala with each other’s spouses on their arms] $_j$.

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- b. Marvin this afternoon from Great Neck and Bernice this evening from Syosset are arriving at Leonard's with each other's spouses in rented Mercedes.

This suggests that the structures in (23) need to be accounted for in terms of sentential coordination. And, so the argument goes, whatever a viable solution for (23) is, it is also a viable solution for all the other cases involving collective predicates. The strategy proposed in Schein (2017) is to shift the problem from individuals to events. The crucial idea here is that (23) is in fact not a predication about a group of people, but a statement about an event e in which each of the subjects participates. According to Schein (2017), an example like (21), repeated here as (24a), is assigned an interpretation along the lines of (24b):

- (24) a. Sue and Bill met at the museum.
- b. $\exists e[\text{PARTICIPATED}(e, \text{Sue}) \ \& \ \text{PARTICIPATED}(e, \text{Bill}) \ \& \ \text{MEET-AT-THE-MUSEUM}(e)]$

This way, we can have our cake and eat it, too: Since *participate* is a distributive predicate, the two predications that Sue participated in e and that Bill participated in e can be linked by sentential conjunction. Since it is the very same event, the collective reading is preserved.

2.4 Homogeneity and (a)symmetric coordination

Whatever the correct analysis of collective readings is, the empirical fact remains that none of the conjuncts in (24a) can substitute for the coordinate structure *salva grammaticalitate*. Still, the conjuncts seem to conform to a constraint called “External Homogeneity Condition (EHC)” in Höhle (1990), which requires that the external syntax of all conjuncts is the same:

- (EHC) *External Homogeneity Condition*: In a coordinate structure $\Phi(X \ \& \ Y)$ the combinatorial properties of the conjuncts X and Y are satisfied by Φ in the same way.

Building on the EHC, Höhle (1990) splits coordinations in *symmetric* and *asymmetric* ones, depending on whether they comply with this constraint, or not. In this sense, conjoined subjects under collective predicates are symmetric, for *neither* of the two conjuncts substitutes for the coordination *salva grammaticalitate*. Typically, cross-categorial coordination is also symmetric, but for a different reason. In cross-categorial coordination, *each* of the conjuncts substitutes for the

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coordination (with the exception of the infamous *you can depend on my assistant and that he will be on time*, see Sag et al. 1985). However, first conjunct agreement is in general asymmetric, since by definition it is only the first conjunct that stands in an agreement relation to the finite predicate.

What Höhle (1990: 222) was aiming at, however, was not phrasal coordination, but the coordination of (dependent) sentences (in German). Consider (25):

- (25) a. Wenn [[jemand nach Hause kommt] und [da der
When [[someone to home comes] and [there the
Gerichtsvollzieher vor der Tür **steht**]], ...
bailiff at the door stands]], ...
'When someone comes home and there is the bailiff at the door.'
- b. Wenn [[jemand nach Hause kommt] und [da **steht** der
When [[someone to home comes] and [there stands the
Gerichtsvollzieher vor der Tür]], ...
bailiff at the door]], ...
'When someone comes home and there is the bailiff at the door.'

The coordination in (25a) is symmetric in the sense that both conjuncts can substitute for the whole coordinate structure *salva grammaticalitate*. This is different in (25b). Within the second conjunct in (25b), the finite predicate *steht* ("stands") is fronted, resulting in a V2 structure. But V2 is inconsistent with the subordinating conjunction *wenn* ("when"), see **Wenn da steht der Gerichtsvollzieher vor der Tür [...]* ("when there stands the bailiff at the door").

A similar point can be made with respect to (26b) from Höhle (1990: 222). Here too, the finite verb in the second conjunct is fronted. This time, however, this results in a V1 structure, since the subject *jemand* ("someone") is external to the coordinate structure. This structure is called SLF-coordination in Höhle (2019[1983]), where SLF is short for "subject lacking in fronted structure."

- (26) a. Wenn jemand [[nach Hause kommt] und [den Gerichtsvollzieher
When someone [[to home comes and [the bailiff
sieht]], ...
sees]], ...
'When someone comes home and sees the bailiff ...'
- b. Wenn jemand [[nach Hause kommt] und [**sieht** den
When someone [[to home comes] and [sees the

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Gerichtsvollzieher]], ...

bailiff]], ...

‘When someone comes home and sees the bailiff ...’

The following English examples, first discussed in Ross (1986: 103), also look unsuspecting at first glance; they show parallel word order as well as categorial symmetry. However, when (27a) is turned into a relative clause, see (28a), and (27b) into a *wh*-clause, see (28b), an asymmetry becomes transparent. The fronted *wh*-expressions are moved out of the second conjunct only.

- (27) a. I [[went to the store] and [bought some whisky]].
 b. She [[has gone] and [ruined her dress now]].
- (28) a. Here’s the *whisky* *which_i* I [[went to the store] and [bought *t_i*]].
 b. *Which dress_i* [[has she gone] and [ruined *t_i* now]]?

This obviously violates the EHC, or to be more precise, a constraint on coordinate structures that is known as “Across the Board movement rule” (ATB) since Ross (1967). This constraint basically states that if a *wh*-expression binds a trace *t_i* in one conjunct, it binds a trace *t_i* in all other conjuncts as well. ATB-movement, therefore, is a corollary of the EHC. This constraint is empirically motivated by the observation that in general *wh*-extraction is not asymmetric, but ATB, see e.g. the following contrast discussed in Sag et al. (1985: 148):

- (29) a. Which books did Robin [[read *t_i*] and [hate *t_i*]]?
 b. * Which books did Robin [[talk to Chris] and [[read *t_i*]]?
 c. * Which books did Robin [[read *t_i*] and [[talk to Chris]]?

With respect to cases like (27a), Ross (1986: 104) observes that the *and*-phrase alternates with an infinitival, see (30), and entertains the hypothesis that what looks like coordination in (30a) is in fact (somehow) derived from the subordinated structure in (30b).

- (30) a. I went to the store [and bought some whisky].
 b. I went to the store [to buy some whisky].

In German, a similar case is the implicative *and*-construction, see Reis (1993): The *and*-phrase in (31a) alternates with the infinitival in (31b), which depends on the adverb *so* (*nett*).

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- (31) a. Er war so nett und fuhr mich nach Hause.
 He was so kind and drove me to home
 ‘He was so kind and drove me home.’
 b. Er war so nett, mich nach Hause zu fahren.
 He was so kind me to home PT drive
 ‘He was kind enough to drive me home.’

It is exactly those constructions that show properties of both coordinated and subordinated structures, and which are crucial, if one wants to understand where and how to draw the line between coordination and subordination (if at all). In the rest of the paper, I will therefore focus on sentential coordinate structures, first elaborating on the properties of symmetric coordination, followed by a more detailed discussion of the above (and more) asymmetric cases.

3 (Some) properties of symmetric coordination

3.1 ATB-movement

The way I introduced the terms symmetric and asymmetric in the last section, the EHC and, in particular, the ATB-criterion are defining characteristics of symmetric coordination. It should be noted that this notion of symmetry is orthogonal to the more logical notion, which requires that in a coordination “A & B” the conjuncts can be swapped without a relevant change in meaning. In this logical sense, the structures in (32) are asymmetric, see (33).

- (32) a. Zuerst hat er Hans eingeladen und **dann** hat sie ihn wieder
 First has he Hans invited and then has she him again
 ausgeladen.
 uninvited
 ‘First he invited Hans, and then she uninvited him.’
 b. Sie hat Hans ausgeladen und **deswegen** hat er ihn wieder
 She has Hans uninvited and therefore has he him again
 eingeladen.
 invited
 ‘She has uninvited Hans, and that is why he has invited him again.’

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- (33) a. * **Dann** hat sie Hans wieder ausladen und zuerst hat er ihn
 Then has she Hans again uninvited and first has he him
 eingeladen.
 invited
 ‘Then she uninvited Hans and first he invited him.’
 b. * **Deswegen** hat er Hans wieder eingeladen und sie hat Hans
 Therefore has he Hans again invited and she has Hans
 ausladen.
 uninvited
 ‘That is why he has invited Hans again and she has uninvited him.’

The reason why the coordinations in (32) are fixed in order is obvious: The adverbs *dann* (“then”) and *deswegen* (“therefore”) express asymmetric relations like temporal order and causality. The same is true of the coordinate structures in (34), which at the same time show ATB-movement of the direct object:

- (34) a. Wen_i hat er zuerst t_i eingeladen und (hat) sie dann wieder t_i
 Who_i has he first t_i invited and (has) she then again t_i
 ausladen?
 uninvited?
 ‘Who did he invite first and she uninvite again?’
 b. Wen_i hat sie t_i ausladen und (hat) er deswegen wieder t_i
 Who_i has she t_i uninvited and (has) he therefore again t_i
 eingeladen?
 invited?
 ‘Who did she uninvite and he invite again because of that?’

This is worth mentioning as it suggests that ATB-movement is consistent with all three kinds of coherence relations (i.e. semantic or pragmatic relations that connect two sentences into a coherent piece of text) distinguished in Kehler (2002): resemblance, cause-effect and contiguity relations. One might suspect, that contrasting the subjects *sie* (“she”) and *er* (“he”) and the predicates *einladen* (“invite”) and *ausladen* (“uninvite”) creates a (possibly additional) layer of semantic parallelism, which might then be argued to constitute a necessary condition on ATB-movement (see e.g. Culicover & Jackendoff 1997). And in fact, most examples that are used to illustrate ATB-movement in the literature show some kind of contrastive relation. The examples in (35), however, suggest that this is not necessarily the case. Apart from the fact that both predicates are transitive and

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share the same direct object, there is no reason to assume any kind of semantic parallelism here. Still, ATB-movement is possible. However, if there is no corresponding object in the second conjunct, asymmetric extraction is also an option, see (36). (One might argue that in (36) the trace t_i is external to the first conjunct. But this is not really an option since, in this case, the first conjunct would be of type $\langle e, \langle e, t \rangle \rangle$ and the second conjunct of type $\langle e, t \rangle$. I take it, following Höhle 1990, that even in the case of asymmetric coordination the two conjuncts need to be of the same logical type / degree of saturation.)

- (35) a. Wen_i hat sie [[t_i eingeladen] und [dann den ganzen Abend über t_i Who_i has she [[t_i invited] and [then the whole evening PREP t_i ignoriert]]]?
ignored]]
'Who did she invite and then ignore all evening?'
- b. Wen_i hat sie [[t_i ausgeladen und deswegen von der Liste t_i nehmen Who_i has she [[t_i uninvited and therefore off the list t_i taken müssen]]]?
must]]
'Who did she uninvite and have to take off the list because of that?'
- (36) a. Wen_i hat sie [[t_i eingeladen] und [sich dann die ganze Zeit Who_i has she [[t_i invited] and [REFL then the whole time darüber gefreut]]]?
it-about rejoiced]]
'Who did she invite and then was happy about it all the time?'
- b. Wen_i hat sie [[t_i ausgeladen] und [sich deswegen schwere Who_i has she [[t_i uninvited] and [REFL therefore heavy Vorwürfe gemacht]]]?
reproaches made]]
'Who did she uninvite and blame herself heavily because of that?'

This suggests that semantic parallelism is not a necessary condition for ATB-movement. But is it a sufficient condition? Kehler (1996) argues that it is: If the relevant coherence relation is a resemblance relation, then movement needs to be across the board, see (37).

- (37) a. Which book_i did Robin [[read t_i] and [hate t_i]]?
b. * Which book_i did Robin [[read t_i] and [[hates *Aspects*]]?

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The contrasts in (34–37) show that there are in fact two kinds of coordinate structures that allow for ATB-movement: Those that allow in addition for asymmetric extraction (type II) and those that do not (type I). Furthermore, if I take asymmetric structures into account that do not allow for ATB-movement at all, see Section 4, this results in the typology in Table 1.

Table 1: Extraction types (in German)

	COORDINATION STRUCTURE	ATB-MOVEMENT	ASYMMETRIC EXTRACTION
1.	symmetric (type I)	yes	no
2.	symmetric (type II)	yes	yes
3.	asymmetric (type III)	no	yes

Examples illustrating type I structures are found all over the place. The minimal contrast in (38) illustrates once more the less known type II structures (see Höhle 2019[1983], Heycock & Kroch 1994):

- (38) a. Diesen Vorschlag_i will die Kommission [[*t_i* prüfen] und
This suggestion_i wants the committee [[*t_i* check] and
[möglichst bald dem Bundestag *t_i* vorlegen.]]
[as-possible soon the Bundestag *t_i* submit]]
‘The committee wants to check this suggestion and submit it as soon
as possible to the Bundestag.’
- b. Diesen Vorschlag_i will die Kommission [[*t_i* prüfen] und
This suggestion_i wants the committee [[*t_i* check] and
[möglichst bald dem Bundestag einen Entwurf vorlegen.]]
[as-possible soon the Bundestag a draft submit]]
‘The committee wants to check this suggestion and submit as soon as
possible a draft to the Bundestag.’

The typology in Table 1 suggests that extraction out of coordinate structures is in fact subject to two constraints: one licensing ATB-movement and the other one licensing asymmetric extraction (see e.g. Reich 2007a). The constraint that licenses asymmetric extraction (both in type II and type III structures) is arguably semantic. Asymmetric extraction is only licensed if the coherence relation in question is either a cause-effect or a contiguity relation. Or, to put it differently, it is only licensed, if the coordinate structure is understood as a description of a complex event (see e.g. Schmerling 1975, Lakoff 1986). ATB-movement, however,

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is syntactically constrained. Meaning it is only licensed if the relevant syntactic structure is a Boolean phrase $\&P$ (rather than e.g. an adjunction structure). Even though (38b) does not conform to the EHC, there is good evidence from verb order (all conjuncts are verb final) that both (38a) and (38b) are to be analyzed as canonical coordinate structures in terms of a Boolean phrase $\&P$. This licenses ATB-movement (if the relevant arguments are coreferential). At the same time, (38a) and (38b) are (narrative) descriptions of a complex event. This in turn licenses asymmetric extraction (in case the relevant arguments are not coreferential). If this is basically on the right track, the question is not whether the conditions on ATB-movement are syntactic or semantic. In order to get the full picture, both levels need to be taken into account.

This leaves us with two questions: Firstly, why is it that canonical coordinate structures (Boolean phrases $\&P$) allow for ATB-movement? Secondly, why is it that in structures like (38) also asymmetric extraction is an option (given that the preconditions are met)?

Let's first have a look at the first question, since there is already a myriad of proposals on the market (see de Vries 2017 for a recent overview). Building on Ross (1967), one might stipulate that in canonical coordinate structures there is an "ATB movement" rule that relates an external phrase to identical traces (copies of that phrase) in each conjunct, see (39).

(39) Which book_{*i*} did Robin [[read *t_i*] and [hate *t_i*]]?



But it is mysterious as to why there is such a rule in the first place and why it is restricted to coordination. Starting with Williams (1978), this motivated multi-dimensional analyses that construe coordination as a kind of secondary layer operating on basic phrase structure rules (see e.g. Goodall 1987, Muadz 1991, de Vries 1992, Moltmann 1992, Grootveld 1994). The basic idea is that the conjuncts in a coordinate structure are located in parallel on different dimensions sharing identical structure like the trace *t_i* of the *wh*-phrase *which book* in (40).

(40) Which book_{*i*} did Robin

read	<i>t_i</i>	and
hate		

In a way, multi-dimensional approaches shift the peculiarity of ATB-movement from extraction to the grammar of coordination: Because of structure sharing moving a *wh*-phrase *across the board* is not a one-to-many relation anymore,

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but a one-to-one relation like any other kind of (standard) movement. The grammar of coordination, however, is considerably complicated by stipulating multiple dimensions. Basically, the same is true of multi-dominance approaches (see Citko 2005, Gračanin-Yuksek 2007, Kasai 2007, Bachrach & Katzir 2009, de Vries 2013) that stipulate that in coordinate structures one and the same node (e.g. the trace t_i) can be dominated by two or more different nodes (e.g. the verbs *read* and *hate*). The sideward movement approach proposed in Nunes (2004) and Hornstein & Nunes (2002) implements basically the same idea, though in a somewhat different way (see e.g. de Vries 2009 for discussion).

Since in both multi-dimensional and multi-dominance approaches it is one and the same phrase that is moved, two crucial predictions are made. Firstly, the trace in each conjunct refers to one and the same object. Secondly, the extracted *wh*-phrase has the same syntactic function in each conjunct. Both predictions are discussed and questioned in de Vries (2017). According to de Vries (2017: 8) the sentences in (41) are fine even though they require (41a), or at least suggest (41b) a non-coreferential reading. (42a) illustrates ATB-movement of a *wh*-phrase that is a direct object in the first and a prepositional object in the second conjunct. And in (42b), cited from Chaves (2012: 482), *some parts* relates to a subject and an object position at the same time.

- (41) a. I wonder *which song* Peter composed _ today and Susan sang
_ yesterday.
b. *How many* matches did Peter play _ and Susan win _?
- (42) a. *Which tree* did you say [Peter hugged _] and [Susan talked to _] ?
b. There were *some parts* that [[I enjoyed _] and [_ were very suspenseful]].

Even though there might be ways to account for such examples also within multi-dominance approaches, see e.g. de Vries (2013), the data situation is not entirely clear. Rating studies on ATB-extraction in German, for example, show a significant decrease in acceptability if (covert) case requirements differ across conjuncts, see Hartmann et al. (2016).

An alternative approach to ATB-movement is traced back to Chomsky & Lasnik (1977) in de Vries (2017). Technical details aside, the basic idea is that what looks like ATB-movement is in fact an artifact of *wh*-extraction from the first conjunct only, in tandem with (i) covert operator movement (see Munn 1993, Franks 1993), (ii) bound covert pronouns (Zhang 2009), or (iii) internal *wh*-movement of ‘identical’ copies in non-initial conjuncts followed by ellipsis (e.g. Wilder

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1994, Reich 2007a, Salzmann 2012). The illustration in (43) follows the proposal in Reich (2007a): First, within each conjunct, a copy of *what* is moved to the conjunct's left periphery. The copy of *what* that resides in the leftmost conjunct then attaches to $\&P$ (by way of traceless movement or, alternatively, it remerges with $\&P$) from where it proceeds to the sentence's left periphery, see (43).

- (43) *What* 3 did Robin [$\&P$ t_3 [$\&P$ [*what* 1 [read t_1]] and [*what* 2 [hate t_2]]]]?
-
- The diagram shows four upward-pointing arrows. The first arrow starts from 'what 1' in the first conjunct and points to the leftmost $\&P$. The second arrow starts from 'what 2' in the second conjunct and points to the $\&P$ immediately before it. The third arrow starts from t_3 and points to the $\&P$ immediately before that. The fourth arrow starts from 'What 3' and points to the very left of the sentence.

While approaches of this sort typically avoid the stipulation of additional grammatical layers or processes, they still have to link the different movement and deletion processes in one way or another, for example by (independently) requiring the conjuncts in a coordinate structure to be of the same semantic type (which is $\langle e, \langle e, t \rangle \rangle$ in the case above).

Although there is not yet a broad consensus on how to model ATB-movement, the above discussion has made it clear that there are several ways to do so. What is not so clear, however, is how to derive the asymmetric extraction in (38b). As we saw above, a necessary condition on asymmetric extraction seems to be that the coordinate structure is understood as a description of a complex event. This suggests that in type I coordinations, like (39), each conjunct is a ‘closed sense unit’ (Chomsky 2000, 2001) in the sense that the event variables e introduced by the finite predicates are (existentially) bound within each conjunct, see (44), where the ATB-movement of the subject *Robin* is ignored for expository reasons. In other words, each conjunct is a phase and, in order to escape from this phase, conjunct-internal movement is required.

- (44) What 3 did Robin₇
 $[\&P\ t_3\ [\text{what 1}\ \exists e_4\ [t_7\ \text{read}(e_4)\ t_1]]\ \text{and}\ [\text{what 2}\ \exists e_5\ [t_7\ \text{hate}(e_5)\ t_2]]]$

- (45) This suggestion 1 $\exists e = e_4 \oplus e_5$ wants the committee₇
 $[\&P\ [t_7\ \text{check}(e_4)\ t_1]]\ \text{and}\ [t_7\ \text{submit}(e_5)\ \text{as soon as possible a draft}]]$

Type II coordinations like (38), however, are descriptions of complex events. This is captured by existentially binding the event variables e externally to the coordinate structure, see the schematic illustration in (45). As a consequence, the conjuncts are not closed sense units and, therefore, are not phases in the relevant sense. This, in turn, means that conjunct-internal movement is not compulsory, and the object *diesen Vorschlag* (“this suggestion”) can be moved in one go to the sentence's left periphery. The fact that extraction is restricted to the first conjunct might be due to locality constraints or, what is more likely, relates to the asymmetry between the two events e_4 and e_5 . In (45), the event e_5 depends on the

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event e_4 , and e_5 is in this sense semantically subordinated to e_4 , see Reich (2007a) and Weisser (2015) for discussion. Two more notes are in order here. Firstly, in (44) as well as in (45), identity in logical type is respected. Secondly, the possibility of asymmetric extraction does not necessarily exclude ATB-movement. (In fact, the subject *the committee* in (45) needs to be moved *across the board*.) This is basically because movement is followed by deletion of identical copies in the c-command domain. Suppose that *diesen Vorschlag* (“this suggestion”) is the direct object in both conjuncts, but asymmetrically extracts from the first conjunct only. The following deletion process then also targets the copy of *diesen Vorschlag* (“this suggestion”) in the second conjunct, which results (without further assumptions) in an uninterpretable structure. It thus follows that if there are identical copies of a phrase in different conjuncts, those copies need to be moved conjunct-internally.

The above discussion should give a general idea of how to model ATB-movement and asymmetric movement in canonical coordinate structures. This leaves us with structures of type III, which allow for asymmetric extraction only. Before I go into details here, however, I have to deal with another crucial property of symmetric coordination: ellipsis.

3.2 Ellipsis in coordinate structures

In Section 2, we already saw that Chomsky (1957: 35) suggested the generalization in (CR $\rightarrow$), repeated here for convenience, to account for the coordination of subsentential constituents.

(CR $\rightarrow$) If we have two sentences $Z + X + W$ and $Z + Y + W$, and if both X and Y are actually constituents of these sentences, we can generally form a new sentence $Z - X + Y - W$.

Now, suppose that $W = \emptyset$, i.e., we have two sentences of the form $Z + X$ and $Z + Y$. The generalization in (CR $\rightarrow$) then (correctly) predicts that the following structures should be perfectly fine:

- (46) *Forward conjunction reduction / Rightward ellipsis / Phrasal coordination*
- a. *He likes to play soccer, but ~~he~~ hates to go hiking.*
 - b. *He likes to play soccer, and ~~he~~ likes to go hiking.*
 - c. *He likes to play soccer, and ~~he~~ likes to play chess.*

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If we, on the other hand, suppose that $Z = \emptyset$, i.e., if we have two sentences of the form $X + W$ and $Y + W$, then the generalization in (CR $\rightarrow$) correctly predicts that the structure in (47a) is fully acceptable, see also the discussion in Section 2. (CR $\rightarrow$) does not license the structures in (47b,c), however, since the strings *John enjoyed* and *Tom picked* are not (canonical) constituents. (Note that e.g. *enjoyed* forms a constituent together with its object *the play*, which excludes the possibility that *enjoyed* also forms a constituent with its subject *John*.) In a footnote, Chomsky (1957: 35f) in fact takes issue with examples like (47b) and points out that they show properties like long pauses (between *liked* and *the*) and contrastive stress that “normally mark the reading of non-grammatical strings.” Without ultimately committing himself, Chomsky entertains the possibility to consider examples like (47b) and (47c) as semi-grammatical.

- (47) *Backward conjunction reduction / Leftward ellipsis / Right node raising*
- a. Tom ~~went to the movies~~ and Sally *went to the movies*.
 - b. John enjoyed ~~the play~~ and my friend liked *the play*.
 - c. Tom picked ~~these grapes~~ and I washed *these grapes*.

To derive structures like (47b) and (47c), Ross (1986: 108) proposed a rule of “conjunction reduction” that ATB-moves the constituents *the play* and *these grapes* to the right periphery of the complex sentence and adjoins it to the coordinate structure. This way of deriving structures like (47b) and (47c) is better known as “Right Node Raising” (RNR) since Postal (1974), see for example Sabbagh (2014) and Wilder (2018) for recent overviews.

RNR is probably one of the most controversially discussed cases of reduced structures since it shows seemingly contradictory properties. The facts that RNR shows no island effects, see (48) from Sabbagh (2014: 29), that RNRred indefinites like German *etwas* (“something”) do not have to be coreferential, see (49), and that the target of RNR does not have to be a constituent (e.g. Wesche 1995, Hartmann 2002), see (50), question a movement analysis.

- (48) Max publicly denounced the senator who wrote __, and Pauline outwardly criticized the magazine editor who published *the speech that encouraged the riot*.
- (49) Peter hat mir __ und ich habe Peter *etwas* *geschenkt*.
 Peter has me __ and I have Peter something given
 ‘Peter gave me, and I gave Peter something.’

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- (50) Peter hat auf ____ und Hans hat neben *dem Sofa* gelegen.
 Peter has on ____ and Hans has next-to the sofa lying
 ‘Peter was lying on and Hans was lying next to the sofa.’

The latter two facts are also potentially problematic for multi-dominance approaches like the ones proposed by Wilder (1999), Kluck & de Vries (2013) and many others, even though there are more or less obvious solutions here, see Wilder (2018) for relevant discussion. Yet, there are (by now familiar) arguments from collective predicates (51) and agreement (52), which are hard to account for in an *in situ* deletion approach; but see Section 2 for discussion.

- (51) John hummed ____ and Mary whistled *the same tune*.
 (52) The pilot claimed that the first nurse ____, and the sailor proved that the second nurse, *were* spies. (cited from Wilder 2018: 708)

Thus, neither a rightward ATB-movement approach nor an *in situ* ellipsis approach account for the relevant data without further assumptions. And since RNR structures apparently “form a class of utterances distinct from ‘John enjoyed the play and liked the book,’ etc., where constituent structure is preserved perfectly” (Chomsky 1957, 36), also an analysis in terms of constituent coordination is not available – at least it seems so. In fact, there are proposals on the market, which argue that the relevant strings are constituents within categorial grammar (see Steedman 2000, 2007), or at some point during the incremental processing of the relevant structure (Phillips 1996, 2003). The appeal of these analyses is that they derive without further ado the characteristic property of RNR that deletion in non-final conjuncts is right-peripheral or left-adjacent to the coordinating conjunction (see also Wilder 1999 for an early proposal to derive this property in multi-dominance approaches based on linearization and see Hartmann 2002, Féry & Hartmann 2005 for a proposal to link this property to the prosodic properties of the RNR target within *in situ* ellipsis approaches). This is even more convincing, if we assume that the structures in (53) are also cases of phrasal coordination rather than sentential coordination followed by ellipsis. The coordination of ‘constituents’ in (47) then results in what is called “right-peripheral deletion” (“leftward ellipsis”); and the coordination of constituents in (53) results in “left-peripheral deletion” (“rightward ellipsis”).

- (53) a. He [[likes to play soccer], but [hates to go hiking]].
 b. He likes [[to play soccer], and [to go hiking]].
 c. He likes to play [[soccer], and [chess]].

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One of the crucial arguments for an analysis of (53a-c) in terms of phrasal coordination is the fact that the overt reconstruction of quantificational subjects, like *nobody*, results in a sentential coordination that has a completely different reading. Compare, for example, (54a) to (54b): While (54b) denies amongst others the existence of people that like to play soccer, (54a) only denies the existence of people that like to play soccer and, at the same time, hate to go hiking.

- (54) a. Nobody likes to play soccer and hates to go hiking.
 b. Nobody likes to play soccer and nobody hates to go hiking.

Thus, to derive the coreferential reading of (54a) one either has to stipulate that rightward ellipsis transforms the quantifier *nobody* into some kind of covert pronominal *e*, that gets bound during the deletion process by the first occurrence of *nobody* in some way or another, or that structures like *nobody likes to play soccer and s/he hates to go hiking* are the source of the ellipsis process from the outset, see e.g. Wilder (1997, 2018) for relevant discussion.

Another class of examples not covered by Chomsky's original proposal concerns what is now called gapping (55a), see Ross (1970), and stripping (55b):

- (55) a. John likes Mary, and Mary John.
 b. John gives a book to his mom, and flowers, too.

Typically, gapping is defined as the elision of the finite verb in non-initial conjuncts (see Repp 2009 for an in-depth discussion of the relevant notion of finiteness), possibly together with additional material, leaving behind at least two major constituents. Stripping is very similar to gapping, except that it strands only one major constituent (which is typically modified by negation or focus particles though), see Johnson (2018) for an overview.

The fact that both gapping and stripping are sensitive to islands – see Neijt (1979) and the examples in (56), cited from Johnson (2018: 568, 573) – suggests that in both cases some kind of movement is involved.

- (56) a. * Some asked who ate seafood once, and others bread.
 b. * Smith asked who ate seafood once, and bread, too.

The observation that the remnants in gapping and stripping are contrastively stressed (see e.g. Hartmann 2002, López & Winkler 2000, Konietzko & Winkler 2010, Konietzko 2016, Winkler 2018) suggests that the kind of movement in question is focus movement. The basic idea then is that contrastively focused constituents are first moved to the left periphery of the sentence (or the coordinate structure) followed by deletion of the sister, see (57).

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- (57) a. John likes Mary, and [Mary]_{1,F} [John]_{2,F} [~~*t*₁ likes *t*₂~~].
 b. John [[gives a book to Ann], and [flowers]_{1,F} [~~*gives t₁ to Ann*~~]], too.

This *move-and-delete* approach has been proposed in Merchant (2001) for sluicing and subsequently extended to other kinds of ellipses like stripping (Merchant 2003) and fragments (Merchant 2004), see also Sag (1976), Depiante (2000), Kolokonte (2008), Konietzko (2016) for similar proposals and Johnson (2018) for the discussion of alternatives that rely on different kinds of movement like heavy NP shift, object shift or ATB-movement.

Within the *move-and-delete* approach, ellipsis is constrained by a condition that requires the antecedent to be semantically equivalent to the target modulo the focused constituents (the e-Givenness constraint proposed in Merchant 2001). This semantic condition nicely captures the fact that gapping and stripping allow for morphosyntactic variation, see (58).

- (58) a. Wir lieben Fußball und du *liebst* Wandern.
 We love football and you love hiking
 ‘We love football and you hiking.’
 b. Wir lieben Fußball und du *liebst* auch Fußball.
 We love football and you love too football
 ‘We love football and you, too.’

The ellipsis approach also captures the fact that indefinites like German *etwas* (‘something’) do not have to be coreferential in gapping and stripping: In both scenarios in (59) it is likely and highly plausible that the books given away are not one and the same.

- (59) a. Hans schenkte Maria ein Buch und Maria ~~schenkte~~ Hans ~~ein~~ Buch.
 Hans gave Maria a book and Maria gave Hans a book
 ‘Hans gave Maria a book, and Maria Hans.’
 b. Hans schenkte Maria ein Buch und ~~Hans schenkte~~ Anna auch ~~ein~~
 Hans gave Maria a book and ~~Hans gave~~ Anna too a
 Buch.
 book
 ‘Hans gave a book to Maria, and to Anna, too.’

What is hard to accounted for in a *move-and-delete* approach is the fact that modal particles in German, which are regarded to be bound to a sentence-internal position, may intervene between two remnants in gapping, see (60a) and the

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discussion on Gapping in Reich (2011: 1861) and on sluicing and short answers in Ott & Struckmeier (2018).

- (60) a. Ich kann Hans mitnehmen und du doch Anna.
 I can Hans take-along and you MP Anna
 ‘I can take along Hans, and you Anna.’
 b. Ich kann Hans mitnehmen und [du]_F ~~kannst~~ doch [Anna]_F
 I can Hans take-along and [you]_F ~~can~~ PT [Anna]_F
 mitnehmen
 take-along

This data is straightforwardly accounted for, however, in an *in situ* deletion approach along the lines of Steedman (1990), Reich (2007b), and Ott & Struckmeier (2018): (60a) is the result of deleting all and only the backgrounded material given semantic identity modulo focus. Deletion thus leaves focused constituents untouched, as well as modal particles and evaluative adverbs like *vielleicht* (“probably”), which communicate the speaker’s attitude (see Ott & Struckmeier 2018). This approach is couched into a question under discussion (QuD) framework (see e.g. Roberts 2012, Onea 2016, Beaver et al. 2017), essentially treating the target of gapping and stripping as a short answer to an implicit *wh*-question that is made salient by the first conjunct in a coordinate structure. In the case of (60b) this is the *wh*-question *Wer kann wen mitnehmen?* (“Who can take along who?”). In this approach, the island sensitivity of gapping and stripping is taken to be an artifact of the island-sensitivity of *wh*-movement within the (accommodated) QuD, and possible remnants are predicted to be in principle identical in form to corresponding short answers (see Reich 2007b for discussion).

In ditransitive constructions like (61), taken from Carlson et al. (2005: 210), gapping creates a global structural ambiguity: *Samuel* can be interpreted either as the indirect object to *insulted* (object reading), see (61a), or as its subject (subject reading), see (61b).

- (61) Somehow, Robert insulted the guests during dinner and Samuel during the dance.
 a. ... and Robert ~~insulted~~ Samuel during the dance. (object reading)
 b. ... and Samuel ~~insulted the guests~~ during the dance. (subject reading)

In experimental work on such structures, Carlson (2002) found evidence for a strong bias towards the object reading in adults. She attributes this to the fact that – alongside a sentential coordination analysis followed by ellipsis – (61a)

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could also be analyzed in terms of phrasal coordination (VP coordination) followed by ATB-movement, which she takes to be preferred for reasons of structural economy. If this is on the right track, this bias is not easily accounted for in QuD approaches to gapping, which presuppose sentential coordination. Bryant's (2006) observation that younger children actually show a subject bias rather than an object bias does not contradict Carlson's approach. On the contrary, as Bryant (2006) argues, the data suggests that younger children follow a semantic parsing strategy (and prefer the semantically simpler sentential conjunction over a conjunction of properties), while adults go for a syntactic parsing strategy, which generally prefers the coordination of smaller conjuncts.

4 (Some) properties of asymmetric coordination

The possibility of RNR, gapping and ATB-movement is, where applicable, certainly one of the crucial diagnostics for symmetric coordination. It can be seen, however, that even within the bounds of the EHC, we may observe certain asymmetries, e.g., in syntactic category or, more crucially, in the ordering of the sentential conjuncts caused by conjunctive adverbs like *dann* ("then") and *deswegen* ("therefore") which indicate asymmetric coherence relations like temporal succession or cause-effect relations. The presence of such asymmetries is consistent with ATB-movement though and thus not a sufficient criterion for asymmetric coordination in a strict sense. If it can be argued that a given construction allows *only* for asymmetric extraction (which typically goes along with degraded acceptability of coordinate ellipses like gapping or RNR), or that the conjuncts do not meet the criterion of *substitutability*, this constitutes a violation of the EHC, and I conclude this construction is asymmetric in the relevant sense.

4.1 Asymmetric coordination and scene-setting

Interestingly, these core cases of asymmetric coordination also show an asymmetry with respect to the events referred to by the conjuncts. One of the events is typically understood as being backgrounded, or as setting the scene for the other event (like in Ross' example or in SLF-coordination). This kind of "scene setting" (see Weisser 2015 for a recent overview) seems to constitute a necessary condition for asymmetric extraction and is frequently associated with either sentence-internal coordination (at some level of VP or vP, see e.g. Höhle 1990), or with adjunction structures that basically parallel adverbial modification (see e.g. Büring & Hartmann 1998). An analysis in terms of adverbial modification seems particularly straightforward if extraction is only asymmetric and restricted to

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the presumed adjunction site (since adverbials are known to be islands for *wh*-movement). If ATB-movement is an option, besides asymmetric extraction (as with type II coordination), the assumption of sentence-internal coordination (&*P*) seems more appropriate, see Table 2 and the discussion in Section 3.1.

Table 2: Parameters of (a)symmetric coordination

	& <i>P</i>	ADJUNCTION
no scene setting	type I	–
scene setting	type II	type III

When I introduced the EHC constraint on symmetric coordination in Section 2.4, I already mentioned some kinds of asymmetric coordination in English and German. The first examples are due to Ross (1986: 103) himself and are repeated in (62–63) for convenience.

- (62) a. I [[went to the store] and [bought some whisky]].
 b. She [[has gone] and [ruined her dress now]].
- (63) a. Here's the whisky which_{*i*} I [[went to the store] and [bought *t_i*]].
 b. Which dress_{*i*} [[has she gone] and [ruined *t_i* now]]?

In all these examples, the first conjunct is in fact backgrounded and sets the scene for the second conjunct, which communicates the primary information, see (62). Extraction seems to be restricted to the second conjunct as e.g. Weisser (2015: 132) argues on the basis of (64a):

- (64) a. * Which phone did he pick up and call John?
 b. Welches Telefon hat er [[sich *t_i* gegriffen] und [(damit) Hans
 Which phone has he [[REFL *t_i* taken] and [(it-with) Hans
 angerufen]]?
 called]]?

While structures corresponding to (63a) are ungrammatical in German, those corresponding to (64a) are not, see (64b). This is probably why I, as a non-native speaker of English, do not reject examples like (64a) or the corresponding variant of Ross' example: *To which store did he go and buy some whiskey?* What is common to both languages is that the relevant constructions – with the notable exception of (62b) – alternate with infinitival constructions, see (65):

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- (65) a. I went to the store *to buy some whiskey*.
 b. Ich habe mir das Telefon gegriffen, *um damit Hans anzurufen*.
 I have REFL the phone taken, to it-with Hans call.
 ‘I took the phone to call Hans with it.’

Ross (1986: 103/4) furthermore observes that the second conjunct is restricted to non-stative verbs, that asymmetric extraction from the second conjunct systematically excludes a subject in this very conjunct, and that tense needs to be identical across conjuncts, see (66):

- (66) a. * The tall friend who Tony has a Fiat and yearns for is cruel to him.
 b. * Here’s the whiskey which I went to the store and Mike bought.
 c. * The excellent whiskey which I went to the store and have bought
 was very costly.

These properties parallel the behavior of the infinitival constructions, which is why Ross (1967) entertains the hypothesis that the coordinate structures in (63) are not derived from coordinate nodes in deep structure (and thus not subject to the CSC). An analysis in this spirit is presented in Weisser (2015) who argues that the backgrounded part is first attached to the (future) second conjunct, thus creating an adverbial structure, which blocks movement out of the (future) first conjunct. This structure then forms the complement of the coordinating conjunction &. In a final step, the previously attached sentence moves to the specifier of &P thus creating a coordinate structure, which allows for ATB-movement of the coreferential subject.

Ross’ whiskey-example and so-called implicative *and*-constructions in German are similar in that both constructions alternate with an infinitival construction. Also, the first conjunct is backgrounded and sets the scene for the second conjunct, see (67), modeled after an example from Reis (1993).

- (67) a. Er war so nett und fuhr mich nach Hause.
 He was so kind and drove me to home
 ‘He was so kind and drove me home.’
 b. Er war so nett, mich nach Hause zu fahren.
 He was so kind me to home PT drive
 ‘He was so kind to drive me home.’

However, in contrast to Ross’ whiskey-example, tense is not required to be the same across the conjuncts, see e.g. *er war so nett und hat mich nach Hause*

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gefahren, with simple past in the first and present perfect in the second conjunct. Moreover, the implicative interpretation is due to the implicative predicate *so nett* (“so kind”), which further selects the infinitive in the subordinate construction, and arguably the second conjunct (in one way or another) in the implicative *and*-construction (since *er war so nett* is ungrammatical in isolation). Reis (1993: 217) further argues that the implicative *and*-construction allows for gapping and RNR, see (68), even though ATB-movement seems to be excluded, see (69a) (my judgement). Extraction is asymmetric and might proceed according to Reis (1993) from the first conjunct, see (69b), or (in particular with *so* + adjective predicates) from the second conjunct, see (69c) (original judgements).

- (68) a. *Würdest du so nett sein und __ hier mal aufräumen?*
 Would you so kind be and __ here PT clean-up
 ‘Would you be so kind and clean up a little bit here?’
 b. *Wenn Peter doch mal so nett sein __ und aufräumen würde.*
 If Peter PT PT so kind be __ and clean-up would
 ‘If Peter was so kind to clean up.’
- (69) a. * *Lass hören, wem_i Peter [[_{t_i} den Gefallen tat] und [_{t_i} öffentlich
 Let hear whom_i Peter [[_{t_i} the favor did] and [_{t_i} publicly
 schmeichelte]].*
 flattered]]
 ‘Tell me, who did Peter a favor and flattered him publicly?’
 b. ? *Lass hören, wem_i Peter [[_{t_i} den Gefallen tat] und [_{ihm_i} öffentlich
 Let hear who_i Peter [[_{t_i} the favor did] and [_{him_i} publicly
 schmeichelte]].*
 flattered]]
 ‘Tell me, who did Peter a favor and flattered him publicly?’
 c. *Für wen_i war Hans so nett und hat _{t_i} die Blumen gegossen?*
 For who was Hans so kind and has _{t_i} the flowers watered
 ‘Who was Hans so kind to water his flowers?’

Since it is uncommon that ATB-movement is strictly excluded, but coordinate ellipsis is not, I would rather suggest an analysis of the data in (68) in terms of constituent coordination along the lines of *[[so nett sein] und [(hier mal) aufräumen]]* (“[[so kind be] and [here PT clean-up]]”). In German, the position right before the finite verb in declarative sentences (the so-called “prefield”) typically hosts one and only one constituent. Therefore, I predict the sentence *[so nett sein*

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und (hier mal) aufräumen] wird er wohl nie (“[so kind be and (here PT) clean-up] will he PT never”) to be perfectly acceptable. And this is exactly what I observe. I take this as evidence that, in (68), there is in fact only one *würde* (“would”), which is external to the coordinate structure and c-commands it. While I share the intuitions with respect to ATB-movement in (69a) and asymmetric extraction from the first conjunct in (69b), I judge the asymmetric extraction from the second conjunct in (69c) as strongly degraded (in the indicated reading).

Reis (1993: 216) further observes that implicative *and*-constructions also occur in the form of what is called SLF-coordination in the literature, see e.g. (70a). According to Höhle (2019[1983]), SLF-coordination is characterized by basically three properties: (i) the second conjunct starts with a finite predicate (it has overt V1-structure), (ii) there is a subject lacking in the second conjunct (indicated by SL in (70)), and (iii) the subject *keiner* (“nobody”) in the first sentence is located in the middle field, i.e. it follows the fronted position of the finite verb, see (70). The first two properties are reflected in the term “Subject Lacking in Fronted structure” (SLF), but it is the third property that distinguishes SLF-coordination from the coordinations in (71).

- (70) a. Hoffentlich ist keiner so blöd und fällt SL auf ihn rein.
 Hopefully is nobody so stupid and falls SL for him PT
 ‘Hopefully no one is so stupid and falls for him.’
 b. Hoffentlich sieht keiner unseren Freund und zeigt SL ihn an.
 Hopefully sees nobody our friend and reports SL him PT
 ‘Hopefully, no one will see our friend and report him to the police.’
 (71) a. Keiner_i [[ist *t_i* so blöd *t_{verb}*] und [fällt *t_i* auf ihn rein-*t_{verb}*]].
 Nobody [[is *t_i* so stupid *t_{verb}*] and [falls *t_i* for him PT-*t_{verb}*]]
 ‘No one is so stupid and falls for him.’
 b. Keiner_i [[sieht *t_i* unseren Freund *t_{verb}*] und [zeigt *t_i* ihn an-*t_{verb}*]].
 Nobody [[sees *t_i* our friend *t_{verb}*] and [reports *t_i* him PT-*t_{verb}*]]
 ‘No one sees our friend and reports him to the police.’

In (71), the subject *keiner* (“nobody”) is in the prefield of the first sentence, and therefore a symmetric analysis in terms of two fronted V1-structures and ATB-movement of the subject *keiner* (“nobody”) to the prefield is readily available. This is different in (70). In (70), the subject *keiner* (“nobody”) is positioned in the middle field of the first sentence, and for reasons of semantic interpretation, the overt subject needs to bind the subject gap SL in the second conjunct in one way

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or another. Unlike in (72), it is necessarily the same person in SLF-coordination that both sees our friend and reports him to the police.

- (72) Hoffentlich [[sieht keiner unsern Freund t_v] und [zeigt ihn keiner
 Hopefully [[sees nobody our friend t_v] and [reports him nobody
 an- t_v]].
 PT- t_v]]
 ‘Hopefully, no one sees our friend and no one reports him to the police.’

Thus, as long as semantic binding is taken to require c-command on the level of syntax, we are forced to assume that the subject *keiner* (“nobody”) is external to the coordinate structure. And since the subject is external to the coordinate structure, I can conclude that the first conjunct is a verbal projection, say VP. The second conjunct, however, is a V1 structure, and thus includes a functional head like I' or C° , depending on your favorite syntax. It follows then that SLF-coordination links a VP and an I'/C' -projection, see Höhle (1990).

One might suggest that the subject *keiner* is generated conjunct-internally and has moved ATB to an external position as sketched in (73a). Höhle (2019[1983]), however, showed that SLF-coordination is inconsistent with ATB-movement, see (74a), and only allows for asymmetric extraction from the first conjunct. If there is a co-referent DP in the second conjunct, it is pronominally resumed, see (74b). This suggests that (if at all) the subject *keiner* is asymmetrically extracted from the first conjunct and binds a covert pronoun SL in the second conjunct (in the prefield or in the middle field), see (73b). Büring & Hartmann (1998) propose this process is mediated by a covert operator OP in the prefield of the second conjunct, which binds the subject gap in the middle field, see (73c).

- (73) a. * Hoffentlich sieht keiner_i [[t_i unsern Freund t_v] und [(t_i) zeigt (t_i)
 Hopefully sees nobody_i [[t_i our friend t_v] and [(t_i) reports (t_i)
 ihn an- t_v]].
 him PT- t_v]]
 b. Hoffentlich sieht keiner_i [[t_i unsern Freund t_v] und [(SL) zeigt
 Hopefully sees nobody_i [[t_i our friend t_v] and [(SL) reports
 (SL) ihn an- t_v]].
 (SL) him PT- t_v]]
 c. Hoffentlich sieht keiner_i [[t_i unsern Freund t_v] und [OP_i zeigt SL_i
 Hopefully sees nobody_i [[t_i our friend t_v] and [OP_i reports SL_i
 ihn an- t_v]].
 him PT- t_v]]

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- (74) a. * Die Briefmarken_i zeigt Karl dem Onkel und bietet SL _{t_i} ihm zum
The stamps_i shows Karl the uncle and offers SL _{t_i} him for
Verkauf an.
sale PT
'Karl shows the stamps to his uncle and offers them for sale to him'.
b. Die Briefmarken_i zeigt Karl dem Onkel und bietet SL sie_i ihm
The stamps_i shows Karl the uncle and offers SL them_i him
zum Verkauf an.
for sale PT
'Karl shows the stamps to his uncle and offers them for sale to him'.

The crucial ingredient of Buring & Hartmann's (1998) account is the assumption that the second conjunct is adjoined to the first sentence at some level within the c-command domain of the subject. This nicely captures the mentioned extraction properties of SLF-coordination since adverbial adjuncts are known to be islands for extraction. From the observation that also objects can bind a pronoun in the second conjunct of an SLF-coordination, see (75), they conclude that the adjunction site is in fact flexible and might even be below the object.

- (75) Im Zirkus Krone serviert der Dompteur jedem Löwen eine Antilope;
In-the Circus Krone serves the tamer each lion an antelope;
und würzt sie_i mit Löwensenf.
and seasons it_i with lion-mustard
'At the Krone Circus, the tamer serves each lion an antelope and seasons it with Löwensenf ("lion mustard").'

However, this assumption is inconsistent with the assumption that both the adjunction site and the adjunct are of the same semantic type (show the same degree of saturation). But if we take it that the relevant level for binding is LF, the data can also be accounted for by asymmetric quantifier raising of the object *eine Antilope* ("an antelope") to a position above VP.

More recently, Mayr & Schmitt (2017: 10) argued that the adjunction site needs to be C' rather than a projection like VP that excludes the subject. The argument is as follows: If the second conjunct is linked to the first sentence at the level of VP (by way of coordination or adjunction), another coordination at a level higher than VP should be fine. The following example instantiating such a structure is, however, judged to be ungrammatical:

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- (76) * Gestern musste [[der Hans morgens mit der Anna
yesterday must.PRET [[the Hans in.morning with the Anna
frühstücken und sollte __ abends mit der Maria ausgehen]
have.breakfast and should.FIN __ in.evening with the Maria go.out]
und [der Peter die Susi treffen]].
and [the Peter the Susi meet]]

However, as far as I can see, there is an alternative explanation. First, note that according to Mayr & Schmitt (2017: 10), the intended reading of (76) is: “Yesterday, Hans had to have breakfast with Anna in the morning and was supposed to go out with Maria in the evening and Peter had to meet Susi.” This suggests that the modal *musste* (“must” / “had to”) is either moved across the board from the first and the last conjunct or gapped in the last conjunct. But ATB-movement is known to typically come with parallelism of the conjuncts (see e.g. Williams 1978), and the same holds for gapping (see e.g. Hartmann 2002, Konietzko & Winkler 2010 and many others). It seems thus straightforward to assume that the second conjunct of the SLF-coordination disturbs this parallelism and blocks ATB-movement / gapping in the final conjunct. If there is no need for either ATB-movement or gapping adding another conjunct is fine, see (77).

- (77) Entweder sieht ihn jemand und zeigt ihn an oder er hat nochmal
Either sees him someone und reports him PT or he has again
Glück gehabt.
luck had
‘Either someone sees him and reports him to the police or he gets lucky again.’

Not least to explain the scene-setting interpretation of SLF-coordination (see Bonitz 2013 for experimental evidence), mixed approaches have been put forward recently. Weisser (2019), for example, extends his above-mentioned analysis of Ross’ whiskey examples (adjunction of the secondary conjunct followed by movement to the specifier of &P) to SLF-coordination. And Barnickel (2017), building on the ideas developed in Weisser (2015), presents an analysis in terms of small clauses, different workspaces, and sideward movement that results in a CP-coordination. The basic idea is that the future second conjunct starts out as a small clause in the specifier of *vP* of the future first conjunct. This complex asymmetric predicate is made responsible for the scene-setting interpretation of SLF-coordination. The second idea is that the subject of the small clause takes over, at the same time, the role of the subject of the future first conjunct. Technically, this is done by moving the small clause (in a last resort operation) to a

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different workspace where it promotes independently of its former host to a full CP, with the subject in the specifier of CP (the prefield). By sideward movement, this subject is remerged as the subject of the ν P of the former host, which also promotes to a full CP. Finally, the two CPs are linked again in a categorially symmetric CP coordination. In order to bind the base position in the second conjunct, the small clause needs to be reconstructed at LF.

Even though this kind of approach to SLF-coordination is rather complex, and requires some probably not fully uncontroversial assumptions, it seems quite promising. What worries me though is the compositional semantics, in particular the interpretation of the trace left by sideward movement in the prefield of the second conjunct, which is neither bound at spell-out nor at LF (as long as the subject is not asymmetrically extracted at LF). With respect to embedded instances of SLF-coordination like the bailiff-sentence in (78a), Barnickel (2017) argues for a categorially asymmetric coordination of a TP and a CP, which is the complement and thus in the scope of the subordinating conjunction *wenn* (“when”). The fact that *wenn* (“when”) triggers verb final order only in the first conjunct is traced back to the special status of the first conjunct as evidenced by the phenomenon of first conjunct agreement (see Section 2.3).

- (78) a. Wenn jemand nach Hause kommt und sieht den Gerichtsvollzieher,
 When someone to home comes and sees the bailiff,
 ...
 ...
 b. Wenn jemand nach Hause kommt und da steht der
 When someone to home comes and there stands the
 Gerichtsvollzieher vor der Tür, ...
 bailiff at the door, ...

The discussion of the bailiff-sentences is in fact less prominent in the literature, and I think this is for a good reason. On the one hand, if we suppose that *wenn* (“when”) takes the coordination as a complement, we are forced to assume, like Barnickel (2017) does, categorially asymmetric coordination (which is probably less problematic), and we need a convincing account for the fact that the verb order in the second conjunct is not affected by *wenn*. On the other hand, if we suppose that *wenn* only heads the first conjunct, the verb order properties follow without further ado, but the compositional semantics gets more complicated (see the discussion in Reich 2009). What is far more challenging, however, is the fact that the bailiff-sentence in (78b) shows the same kind of scene-setting interpretation as the subordinate SLF-coordination in (78a) does, even though the second

conjunct in (78b) has its own subject. And, as far as I can see, the derivation of the scene-setting interpretation relies in the mixed approaches on the presence of a subject gap in tandem with either some kind of small conjunct analysis (predicate coordination), or the proximity of SLF-coordination to participle constructions like e.g. *Wie immer steht Karl im Flur, mit den Kollegen schwatzend* (“As always, Karl is standing in the corridor, chatting with the colleagues”), an example taken from Barnickel (2017: 73). In (78b), however, the scene-setting interpretation can only be due to the V2 verb order in the second conjunct, which apparently pushes this conjunct to the foreground, since V2 is the canonical word order in German main clauses and has (quasi) assertional force, see Gärtner (2002) and Truckenbrodt (2006). In the end, it might thus not be the SL in SLF-coordination that is ultimately responsible for the specific interpretation, but the F.

4.2 Asymmetric coordination and conditional interpretation

All the cases of asymmetric coordination discussed so far show an interpretation that can be characterized as scene-setting, backgrounding or fusioning. But there is yet another class of coordinate structures that show a quite unexpected kind of asymmetric interpretation, namely an interpretation as a conditional. This class of examples has been drawn attention to by Culicover & Jackendoff (1997: 198) and might be called *Big Louie*-sentences after a prominent example in their paper. Consider (79). At first sight, (79) seems to be a run-of-the-mill case of sentential coordination, but it is interpreted as a conditional:

- (79) Big Louie sees you with the loot and he puts out a contract with you.
 (= If Big Louie sees you with the loot, he puts out a contract with you.)

Culicover & Jackendoff (1997: 198f) argue that the conditional reading is lost if both conjuncts contain a complementizer or two VPs are conjoined. They also show that gapping and RNR are not compatible with the conditional interpretation of what they call left-subordinating *and*. Left-subordinating *and* follows the pattern of conditionals when it comes to the binding of pronouns, see (80) and (81): A quantifier in the second conjunct can bind a pronoun in the first one (80), and an indefinite in the first one can bind a pronoun in the second one (81).

- (80) a. You give him_i enough opportunity and every senator_i, no matter how honest, will succumb to corruption.
 b. If you give him_i enough opportunity, every senator_i, no matter how honest, will succumb to corruption.

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- (81) a. You give anyone_i too much money, and he_i will go crazy.
 b. If you give anyone_i too much money, he_i will go crazy.

Since Culicover & Jackendoff (1997) take it that the binding of pronouns takes place at the level of conceptual structure, they argue that the *Big Louie*-sentences are semantically subordinating even though they are coordinating at the syntactic level. Culicover & Jackendoff's (1997: 206) observation that the *Big Louie*-sentences do not like ATB-movement (82a), but allow for asymmetric extraction from the first (82b) as well as from the second conjunct (82c) is accounted for by construing the CSC as a semantic constraint. Since the *Big Louie*-sentences are subordinate at the semantic level, the CSC does not kick in, and asymmetric extraction becomes an option.

- (82) a. ?? This is the thief that_i you just point out *t_i* and we arrest *t_i* on the spot.
 b. ? This is the loot that_i you just identify *t_i* and we arrest the thief on the spot.
 c. ? This is the thief that_i you just identify the loot and we arrest *t_i* on the spot.

The crucial reason why Culicover & Jackendoff (1997) stick with a coordinate structure at the level of syntax is that conditionals do not allow for extraction out of the *if*-clause (see Culicover & Jackendoff 1997: 207):

- (83) a. ? Who did John say Mary goes out with and her father disinherits her?
 b. * Who did John say(,) if Mary goes out with(,) her father disinherits her?

With respect to semantic interpretation, Culicover & Jackendoff (1997) propose that at the level of conceptual structure, the first conjunct of a *Big Louie*-sentence ends up in the restrictor of a generic operator GE and the modalized coordination is interpreted along the lines of (84), see also Keshet (2013) and Weisser (2015) for similar proposals. This way, *and* itself is interpreted as usual as the connector &, but the overall structure is essentially equivalent to $(p \rightarrow (p \& q))$, which in turn is equivalent to the conditional $(p \rightarrow q)$ in propositional logic.

- (84) [GE: *p*] (*p* & *q*)

There is one more thing though. Culicover & Jackendoff (1997) also show that the conditional interpretation is lost if the predicates are in the past tense. This puts them close to a class of categorially asymmetric coordinate structures called pseudo-imperatives, see the examples in (85), taken from Franke (2008). In (85a) an imperative and a declarative are conjoined, and the resulting structure is interpreted as a conditional. (85b) shows essentially the same pattern with the exception that the conjuncts are linked by the conjunction *or*. This structure is now interpreted in an “if you *do not* ... then ...” fashion:

- (85) a. Close the window and I will kill you.
 (= If you close the window, I will kill you.)
 b. Close the window or I will kill you.
 (= If you *do not* close the window, I will kill you.)

If we follow Kaufmann (2012) in assuming that imperatives denote modalized propositions, the link between *Big Louie*-sentences and pseudo-imperatives becomes apparent. Suppose that the modal introduced by the imperative gets scope over the conjunction, then the simplified LF of (85a) can be stated as $\text{MOD}(p \ \& \ q)$, which is pretty close to (84). This is in fact essentially what is proposed in Keshet & Medeiros (2019). They moreover assume that the speech-act related part and the modal part of the directive can be dissociated. If this is correct, then the semantics of the *Big Louie*-sentences is essentially identical to the semantics of pseudo-imperatives like (85a) minus the speech-act related conditions.

5 Concluding remarks

The starting point of this article was a dichotomy that has a long tradition both in traditional and modern linguistics: the distinction between coordination and subordination. In short, a sentence is subordinated to another sentence if it depends on that sentence in a relevant way. If there is no such dependency, we are faced with coordination. Although there are, of course, clear cases of both coordination and subordination, the main purpose of this overview article was to illustrate that between these poles, there is a bouquet of constructions that exhibit, to varying degrees, properties typically attributed to coordination, but also properties typically attributed to subordination. This is not to say that the distinction between coordination and subordination is a continuum. But it shows that the world between these two poles is a complex and exciting one. And that it is crucial to understand why each of these constructions exhibits the properties it does, if one is to gain a deeper understanding of coordination and subordination, or more generally, of the way complex sentences are formed.

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Chapter 3

Typology of argument clauses

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This chapter provides an overview of argument clauses, with focus on (declarative) complement clauses. We compare and explore findings from functional typological as well as formal generative approaches, emphasizing that, despite extensive variation (both in analyses and language-specific properties), there are similar restrictions within and across languages: complement clause formation is determined by a combination of i) the semantics of the matrix predicate, ii) the semantics of the complement clause, and iii) the complement's morphosyntactic properties. The conclusion emerging from various typological approaches is that i)-iii) stand in a reciprocal and interactive relationship with each other, determining the possibilities for complement clause formation in natural language. We focus on the observation that, even though complementation strategies are not uniform cross-linguistically, they follow systematic patterns and are arranged along an implicational hierarchy, determined by an interplay of semantics and morphosyntactic complexity. This universal hierarchy becomes evident only from a comparative perspective. The chapter thus contributes not only to a better understanding of form-meaning interactions as well as the systematic nature of argument clauses, but also highlights the importance of typological studies and approaches bridging theoretical differences.

1 Introduction: Universal questions about argument clauses

Following Dixon (2010), argument clauses can be defined along several dimensions: they have the internal structure of a clause (e.g., a configuration involving



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a predicate–argument configuration; see below for different degrees of clause-hood); they function as a core argument (such as subject, object) of another predicate (the matrix or main predicate); they are restricted to a limited set of matrix predicates; and they are, somewhat simplified, instantiations of semantic concepts such as propositions, facts, activities, states, events or situations (they cannot simply refer to a place, time, entity, object or property). In this chapter we will follow this characterization and summarize major findings of clauses that “fill an argument slot in the structure of another clause” (Dixon 2010: 370). This definition excludes configurations with two verbal predicates in a single clause (a prototypical case would be an auxiliary–lexical verb context), but since the classification of what constitutes a clause (or the distinction between auxiliaries and lexical verbs) is not always clear, we will touch on some of the gray area constructions as well.¹

The overview focuses on the most typical type of argument clause, namely complement clauses, which we define here as clauses that combine directly with a predicate (verb, adjective, others) as an object.² Argument clauses can also occur as subjects (both deep and surface), however the distribution of subject clauses displays a range of language-specific syntactic differences, which make it difficult to state typological claims beyond simple ordering options (such as whether the clause occurs before or after the main predicate). For instance, as suggested in Webelhuth (1992), in the Germanic languages, we find a distinction in finite contexts between complement clauses on the one hand, and subject clauses and topic clauses on the other hand. The former can appear with or without the complementizer *that* (at least in certain contexts), whereas the latter must include the complementizer (when finite). Webelhuth suggests the *External Argument Universal*, which states that “External arguments are nonverbal, i.e., [–V].” (p. 90). A clause with a complementizer is classified as nominal, hence it can occur in subject position, whereas a clause without a complementizer is a verbal embedded root clause, and hence excluded from subject positions. While this covers the Germanic languages well (see adaptations of this approach in, among others, Pesetsky & Torrego (2001)), it is not clear whether it can be extended crosslinguistically. In Zulu, for example, finite clauses with a complementizer are clearly nominal (they show agreement marking), and they can occur in topic position,

¹One set of configurations we will set aside, however, are serial verb constructions. Although some authors (see, e.g., Noonan 2007 below) have suggested complementation configurations for (certain) serial verb constructions, analyses of adjunction and coordination have also been proposed (see Veenstra & Muysken 2017 for an overview), and we are not in a position to settle this debate.

²See Vincent (1999) for other definitions of *complement* clause.

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however not in subject position (see Halpert 2016, 2019). Infinitives, on the other hand, which do not include a complementizer, are possible in subject position. Halpert suggests language-specific properties regarding the subject position to derive such differences. Since the distribution of subject clauses is dependent on factors such as complementizers, null complementizers, nominal structure, the properties of subject positions, whether a clause counts as nominal or verbal, and other properties, many of which do not have clear theory-neutral definitions, typological surveys have so far been limited to questions about the linearization options of subject clauses and the question of how languages syntacticize argument relations (see Dixon 1995, 2006, Schmidtke-Bode 2014).³ However, although these works provide extensive usage-based observations about the distribution of argument clauses, they do not include classifications of subject clause patterns, which is the goal of this overview chapter. We therefore have to set non-complement clauses aside in this overview.

When looking at complement clauses typologically, one striking characteristic is that, despite extensive variation, there are similar restrictions within and across languages, which are determined by a combination of the semantics of the matrix predicate, the semantics of the complement clause, and the complement's morphosyntactic properties. No language follows an "anything goes" strategy, and the combination of matrix predicates and different types and meanings of complement clauses is usually not free. However, these restrictions are not completely uniform. Crosslinguistically, the meaning of a complementation configuration is mapped to different morphosyntactic types of complement clauses and the question arises whether the combinatorial possibilities are systematic. Is the complement clause dependent on the matrix predicate's semantics? Does the semantics of the embedded clause restrict the meaning of the matrix predicate? Does each matrix predicate have a unique morphosyntactic type of complement clause it can combine with? Depending on the theoretical focus of an approach, different properties are foregrounded, and these questions may receive (partly) different answers. However, a conclusion that has emerged – in one form or another – from typological works on complementation (such as the ones summarized in this chapter) as well as corpus studies (e.g., Stiebels 2018) is that the three factors (semantics of the matrix predicate, semantics of the complement, and morphosyntax of the complement clause) stand in a reciprocal and interactive relationship. An important question, the answer to which the different accounts may disagree on, is how to define this relationship, how to implement the typological generalizations or tendencies, and how to build a systematic account of complementation.

³Some of these issues are discussed in [Chapters 2 and 9 of this volume](#)

In this chapter, we summarize a selection of approaches to complementation which all present broadly applicable, typologically relevant classifications of complement clause patterns. Depending on the scope, focus and terminology of an approach, different classification systems arise, which sometimes appear to even contradict each other at first sight. However, despite these differences, we show that putting aside fine-grained distinctions, a unifying observation is that the combination of different types of complement clauses and different types of matrix verbs is restricted and to some extent predictable. These restrictions are either based on the meaning of the matrix predicate and/or the complement clause or on the morphosyntactic coding of the complement clause. Importantly, all approaches take semantics and morphosyntax into account and propose frameworks to characterize the mapping between the two. As we will see, accounts vary in the number of complement clause classes, number of matrix verb classes and number of combinatorial possibilities they allow. However, a common property found in all approaches, in one form or another, is that complementation configurations are ranked along some kind of hierarchy. While surface properties (specific morphological coding such as finiteness or word order) vary across languages, it has nevertheless been observed (in the works we will present here, and beyond) that implicational regularities hold. To give a simple example, Hawkins (1999) notes the hierarchy *clauses in a complex NP* >> *finite clauses* >> *infinitives*, and he suggests the implicational universal that if a language allows a gap (e.g., extraction) in a higher category of this hierarchy, it also allows it in all lower categories. Following the works we summarize, we suggest that both the semantic classification of complement clauses and options for morphosyntactic realizations within complement clauses are ranked according to separate hierarchies. We hypothesize that the semantic complementation hierarchy is universal, whereas the categories of the morphosyntactic hierarchy, but not their order with respect to each other, vary across languages. Similarly, the mapping between the semantic and morphosyntactic hierarchies is not fixed – restrictions in the interaction of the two hierarchies only arise via the implicational relations predicted by the two hierarchies.

Since the form, meaning, and use of argument clauses raises important questions in a variety of linguistic subfields, the topic has attracted significant attention in all frameworks (see also Vincent 1999). Major general studies of complementation include Rosenbaum (1967) (Transformational Grammar), Bresnan (1982) (LFG), Davies & Rosen (1988) (RG), Pollard & Sag (1994) (HPSG), van Valin (1984, 2005), van Valin (2007) (Role and Reference Grammar), Hengeveld (1989), Dik & Hengeveld (1991) (Functional Grammar), Schmidtke-Bode (2014) (Usage-based linguistics), or Kehayov & Boye (2016) (Functional–typological semantics).

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Providing an overview of all works and frameworks is impossible, and we concentrate on a subset of the rich literature. In particular, we have singled out typological works which provide classifications of complementation patterns and whose main aim is to characterize form–meaning interactions in complementation.

With this goal in mind, the chapter concentrates mostly on declarative complement clauses for which the typological research is most developed, notwithstanding that embedded questions are also relevant for the classification of complement clauses. One of the central questions addressed in the literature is why certain verbs (such as *know*) can combine with declarative and interrogative clauses, others can only occur with one clause type (e.g., *believe* does not allow an interrogative, whereas *wonder* does not allow a declarative complement), and yet others can even occur with a surface nominal, yielding a concealed question (as in *to ask the time*). Furthermore, a very rich syntactic research field is the dependency found in questions – what is the nature of the dependency (movement, binding, other mechanism), where do question elements occur (in-situ, the left periphery, a focus position), what is the domain and locality of the dependency, etc. In terms of form–meaning interactions, however, embedded interrogatives show less variation than declaratives. In many languages, embedded interrogatives require the presence of a left peripheral operator domain that can host question expressions, leading to fewer mapping distinctions among different semantic types of interrogative clauses. Indeed a common view since Grimshaw (1979) is that there is a *canonical structure realization* rule which maps interrogative semantics to an operator domain (e.g., a CP). As a result, the form–meaning interactions are more rigid in questions, which is also reflected in the sparser availability of typological surveys of embedded questions. The compositional semantics of embedding, which also covers questions, is discussed in detail in [Chapter 15 of this volume](#), and we refer the reader to this work for further details.

Lastly, although it has traditionally been assumed that imperative complement clauses are impossible, this conclusion has been challenged in the last two decades by observing more and more languages that allow embedded imperatives (see Aikhenvald 2010, Kaufmann 2012, Kaufmann & Poschmann 2013). Among the languages are German (Kaufmann 2012, Kaufmann & Poschmann 2013), Slovenian (Dvořák 2005), Korean (Portner 2005), English (Crnič & Trinh 2011), and several others. While the list of languages is clearly growing, it is still relatively small and typological research is yet developing.

In what follows, we lay out the main observations and conclusions of Givón (1980), Cristofaro (2005), Noonan (2007), Dixon (2010), cartographic approaches, and our own approach in Wurmbrand & Lohninger (2019). In the last section, we

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provide a short discussion of the observed differences and variation, as well as the emerging similarities which point to an abstract notion of universality found in the area of complementation.

2 Approaches to complementation patterns

Although approaches to complementation patterns in natural languages are set in very diverse frameworks (see above), once certain framework-specific terminology and goals are set aside, surprising similarities can be found and formulated. The summaries of the selected approaches presented in this section are aimed to highlight this. We start with the richest literature on the topic which, naturally, is provided by functional–typological approaches. Structural–grammatical works are less rich from a language variation perspective, but some systems have been developed with the aim to bridge frameworks and implement typological findings within formal theories. The accounts summarized here differ in focus, scope, and various terminological choices, but a common goal is to characterize the regularities observed in the mapping between semantics and morphosyntax. Other similarities involve the relevance of different degrees of (in)dependence as a characteristic of different types of complement clauses, and the observation that these types of complement clauses are ranked in relation to each other on specific hierarchies. Although the emerging classifications overlap, they also differ in various ways, and it remains to be seen whether a full unification of these approaches is possible.

Before beginning our summaries, we would like to add a word of caution regarding inconsistencies in the terminology, specifically the term *proposition*, which has been applied to a variety of different semantic and/or syntactic contexts and is used differently by different authors. A common use is the distinction between *proposition* and *property/predicate*, which highlights whether a configuration is saturated or includes an open argument slot, respectively. This distinction plays an important role in many works on non-finite clauses, where different types of infinitival complements have been distinguished along those lines (see, among others, Chierchia 1984, Landau 2015 where this issue is discussed centrally). Most of the works we summarize in this chapter use the term *proposition* as a semantic classification for configurations involving attitude contexts (most typically with verbs like *believe*, *think*), distinguishing *propositions* (or *propositional attitudes*) from other semantic classes such as *desideratives*, *implicatives*, and others. However, within these approaches, we also find variation regarding which subcategories are included under the broad term *proposition*,

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whether *propositions* include speech act information, and which structural part of a clause is associated with this notion. Although this multitude of uses of the same term is unfortunate, the descriptions are usually clear and we do not wish to impose a unified terminology here. Instead we copy the terms as given in the sources we summarize and will only point to certain differences regarding the use of *proposition* in cases where the inconsistencies may lead to confusion or give the impression that certain accounts are not compatible (when in fact they are).

2.1 Givón (1980)

Givón (1980) proposes a functional-typological approach to complement clauses. The main observation is that there is a universal semantic hierarchy regulating the distribution of matrix predicates and corresponding complement clauses. Complement clause taking predicates and their corresponding complement clauses are ordered on a hierarchical scale called the *Binding Hierarchy* which consists of two parts aligned with each other. Matrix predicates are grouped and ordered based on their semantics, which is given as the top branching arrow in the diagram below – the *Binding Scale*. The arrow on the bottom is the *Syntactic Coding Scale* which describes the (in)dependence of a complement clause and thereby its degree of “clausehood.” Both scales also express different degrees of (semantic or morphosyntactic) integration of the complement clause into the matrix clause. The two scales are aligned – the highest point of the scale is to the right which involves complements with the highest integration into the matrix predicate. The lowest point is to the left, involving complements resembling independent clauses – the lower (further to the left) a complement clause is on the *Binding Hierarchy*, the more independent and similar to a main clause it becomes.

The semantic dimension of the *Binding Scale* orders matrix predicates along a semantic scale, defined over the functional concepts *binding*, *independence*, and *success* (see (1); Givón 1980: 335, slightly modified). For instance, *to force someone to do something* exerts a stronger influence on the forcee (the embedded agent is less independent) than *to ask someone to do something* does on the person asked. As a result, the predicate *force* is higher (further to the right) on the *Binding Scale* than *ask*, and the action that the complement describes is more likely to succeed.

- (1) a. *Binding*: The stronger the influence exerted over the agent of the complement clause by the agent of the main-clause verb, by whatever means, the higher the main-clause verb is on the binding scale.

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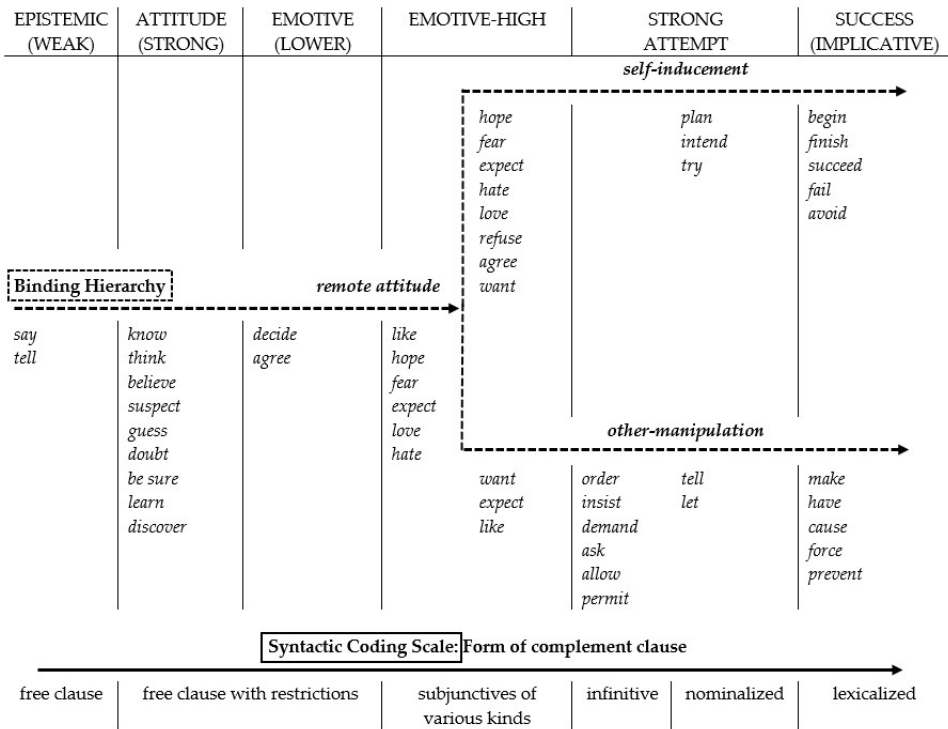


Figure 1: The *Binding Hierarchy* (Givón 1980: 369)

- b. *Independence*: The higher a verb is on the binding scale, the less the agent of its complement clause is capable of acting independently.
- c. *Success*: The less independence possessed by the embedded-clause agent, and the higher the main-clause verb on the binding scale, the more the intended manipulation is likely to succeed.

The scale thus ranges from very strong binding predicates on the right to very weak binding ones on the left: success (implicative) verbs (*begin, finish, make, force,...*) >> strong attempt verbs (*plan, intend, order, insist,...*) >> high emotive verbs (*like, hope, fear,...*) >> low emotive verbs (*decide, agree,...*) >> attitude verbs (*know, think, believe,...*) >> epistemic verbs (*say, tell*).

The syntactic dimension of the *Binding Hierarchy* is defined as the *Syntactic Coding Scale* (see the diagram above) and applies as in (2).

- (2) a. *Syntactic Coding Scale*: The higher a verb is on the binding scale, the less its complement would tend to be syntactically coded as an independent/main clause. (Givón 1980: 337, (11))

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The properties used to describe independent clauses are: i) the degree to which the complement clause agent resembles a main clause agent (how much it reflects the agent-marking of a main clause); ii) the degree to which the tense–aspect–modality (T/A/M) system of the complement clause is reduced; and iii) the presence of predicate raising of the complement verb onto the main verb. Predicate raising describes a process in which the embedded verb is lexicalized together with the matrix verb. A complement clause is the most independent if its agent-marking and T/A/M-marking allow the same configurations they do in the main clause and do not exhibit predicate raising. A complement clause is the least independent if it does not allow an independent agent, has no or only minimal T/A/M marking and its predicate can be (depending on the language) raised. Since the *Binding Hierarchy* is scalar and not binary, these two options describe the ends of the scale with various possibilities in between (from most independent to most dependent): free clause >> free clause with restrictions of different kinds >> subjunctive >> infinitive >> nominalization >> lexicalized clause.

Aligning the *Syntactic Coding Scale* with the semantic *Binding Scale*, the *Binding Hierarchy* emerges. If the complement of a predicate at a particular point on the semantic scale is obligatorily coded by a certain syntactic device (e.g., subjunctive, infinitive), then a complement of a predicate higher on the hierarchy (further to the right) cannot be coded by a syntactically lower device. We illustrate this with examples from Bemba (Bantu). Bemba allows the high emotive predicate *hope* to combine with a subjunctive complement as in (3a), but the lower attitude complement *know/believe* only with a more independent *that*-clause (allowing future tense marking) and not with a subjunctive (see (3b) vs. (3c)). Lastly, the lowest verb on the scale, *say*, is only permitted with a complex *that*-complement clause, exhibiting independent (past) tense as in (3d).

(3) Bemba (Givón 1980: 354–366)

- | | | |
|----|---|----------------------------|
| a. | n-dee-subila umukashi a-inga-isa.
I-PROG-hope woman she-SUBJUNCT-come
'I hope that the woman will/may come.' | <i>High on hierarchy</i> |
| b. | n-dee-sumina uku-ti a-ka-isa.
I-PROG-think/believe that she-FUT-come
'I think/believe that she'll come.' | <i>Middle of hierarchy</i> |
| c. | *n-dee-sumina (nga) a-inga-isa.
I-PROG-believe (SUB) she-SUBJUNCT-come.
Int.: 'I think/believe that she'll come.' | <i>Middle of hierarchy</i> |

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- d. a-a-ebele uku-ti umunaa-ndi a-a-ishile.
 he-PAST-say that friend-my he-PAST-come
 ‘He said that my friend had arrived.’ *Low on hierarchy*

Thus, there is an implicational correspondence between the semantics of the matrix predicate and the syntax of the complement clause, based on the point of the scale where the predicate and its surrounding elements on the scale are. The points on the scale are not entirely discrete but form a gradual increase of binding (i.e., independence) from left to right. Lastly, it is important to note that the matrix predicates are ordered based on their semantic properties, not just their form. The system predicts that predicates that can be interpreted in more than one way, occur in different categories on the hierarchy, depending on the intended interpretation. This is, for example, the case for the Bemba predicate *force/order* which has an implicative reading when combined with an infinitival (dependent) complement as in (4a), but a non-implicative reading when its complement clause is finite (independent) as in (4b).

(4) Bemba (Givón 1980: 347)

- a. n-a-koonkomeshya Robert uku-boomba.
 I-PAST-force/order Robert INF-work
 ‘I forced Robert to work.’ *Implicative*
- b. n-a-koonkomeshya Robert (uku-ti) a-boomb-e.
 I-PAST-force/order Robert (that) he-work-SUBJUNCT
 ‘I ordered Robert to work.’ *Non-implicative*

The interaction between the semantic binding and the morphosyntactic coding scales can thus be described as bidirectional – only if the restrictions of both hierarchies are met and match, a well-formed output arises.

In sum, although the semantic binding and the morphosyntactic coding scales are defined by different notions (in Givón’s system even from different components of grammar – function vs. structure), they nevertheless interact and form one general complementation system, the universal *Binding Hierarchy*. A comprehensive picture of complementation can only be depicted when both components are considered.

2.2 Noonan (2007)

The functional complementation classification proposed in Noonan (2007) ties together complement clause size and meaning with the semantics of the ma-

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trix predicates and their combinatorial possibilities. According to Noonan's approach, languages vary in what type and number of complement clauses they have at their disposal. English, for example, has four options for clausal complementation: *that*-clauses, infinitive clauses, gerundial or verbal noun clauses, and participial clauses. Other languages, like Irish, only have two options of complementation, nominalizations and *that*-clauses:

(5) Irish (Noonan 2007: 54, 116)

- a. Dúirt sé go dtiocfadh sé.
said.3.SG he COMP come.COND he
'He said that he would come.' (that/go-clause)
- b. Is maith liom iad a fheiceáil.
COP good with.me them COMP see.NMLZ
'I like to see them.' (Verbal noun)
- c. D'iarr mé air gan imeacht
asked I on him NEG leave.NMLZ
'I asked him not to leave.' (Nominalization)

According to Noonan, languages may exhibit up to five types of complement clauses. These types differ from each other by the morphosyntax of the complement clause, the syntactic relation between the complement predicate and its arguments, and the semantic relation of the matrix predicate with the complement clause.

The morphosyntax of the complement clause can either be sentence-like (s-like) or reduced/non-sentence-like (non-s-like). S-like complement clauses morphosyntactically resemble a main clause and their lexical predicate has the same relation to its arguments as a main predicate would have in a given language. S-like complement clauses involve either indicative or subjunctive predicates, subjunctive differing from indicative usually in that the inflectional categories are slightly reduced. Subjunctive is at the border between s-like and non-s-like and often comes, similar to indicatives, with a complementizer. Only languages with tense and aspect morphology tend to make the indicative-subjunctive distinction, others usually do not exhibit a subjunctive form. Another s-like complementation strategy is parataxis or verb serialization. These constructions typically consist of a subject NP, followed by a series of fully inflected verb phrases. There is no marker of coordination or subordination (such as a complementizer)

between them and the predicates do not come in a special form other than indicative, making them s-like.⁴

Non-s-like complement clauses, in contrast to s-like ones, do not behave like main clauses. The difference between them usually lies in the morphology of the complement predicate – e.g., infinitive instead of indicative or subjunctive. Infinitives, as opposed to indicatives or subjunctives, lack a subject and typically cannot stand on their own. However, although infinitives are not s-like, they still establish the same predicate–object relation as main-clause predicates and are thus still verb-like. Other non-s-like complementation strategies are nominalizations or participials, which are noun-like instead of verb-like. In noun-like configurations, the embedded subject is often marked like a possessor and when nominalized, predicates typically realize case and number like “regular” nouns. When the embedded predicate is a participial, also called converb, it appears in adjectival or adverbial forms and modifies a noun.⁵

Certain syntactic processes in complementation can render the complement clause morphology to become less s-like. Such operations include equi-deletion (aka control), argument raising (subject raising, raising to object, exceptional case marking), incorporation of reduced complements into the matrix clause (clause union, restructuring), and restrictions on the sequence of tense/mood. Noonan (2007) suggests that if one of these operations applies, the resulting complement clause becomes less s-like, thus the operation itself influences the form of the complement clause.

Since the last property is essential for Noonan’s classification, we discuss it in more detail here. Complement clauses differ regarding tense marking and time-reference. Tense can either be restricted or non-restricted. Noonan (2007) refers to complement clauses with unrestricted tense as having “independent time reference” (ITR). ITR is the ability to refer to a point in time independent of the utterance time and main clause tense, and to display other tense values than the main clause. In contrast to ITR, “dependent/determined time reference” (DTR) contexts show a restriction on the temporal options in the complement clause – the matrix predicate influences the time of the embedded event. In DTR contexts,

⁴Note that it is controversial whether verb serialization is an instance of complementation (see Veenstra & Muysken 2017 for an overview). While Noonan (2007) includes it in his typology of complementation, Dixon (2010), for instance, does not. Similarly Haspelmath (2016) specifically defines serialization as not involving a complementation relation. Furthermore, as pointed out by a reviewer, some of the cases treated as paratactic complementation by Noonan resemble embedded root clauses which may involve less integration into the matrix clause than typical complement clauses (but see Krifka 2023 and Section 2.5 for a short discussion).

⁵As pointed out by a reviewer, such cases may then not qualify as complementation.

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tense can either be the same as the matrix tense or be determined by the matrix predicate (e.g., require a particular tense value, such as future morphology), which depends on the semantics of the matrix verb: certain predicates require DTR whereas others do not.

Not only time-reference but all of the above syntactic operations are enabled by certain semantic values of the matrix predicate. Noonan (2007) characterizes this connection as “the stronger the semantic bond [Givón’s *binding*] between the events described by the matrix and complement predicates, the greater the degree of syntactic integration there will be between the two clauses.” (Noonan 2007: 101). The concept of syntactic integration is related to the degree of reduction of a complement clause and its allowance of operations like equi-deletion or predicate-raising. S-like complements have the lowest degree of syntactic integration, non-s-like complement clauses the highest.

The semantics of complementation is a combination of the semantics of the matrix predicate and the semantic properties of the complement clause. The semantic potential of a complement clause is a compound of its mood distinctions, its degree of reduction, and the type of matrix predicate. In Noonan’s classification, complement clauses can either be subordinated or paratactic. If subordinated, they form one assertion together with the matrix clause, which is usually the case in causative and immediate perception environments. In paratactic complementation, two assertions arise – one for the matrix clause and one for the dependent clause. The semantics of assertion is reflected in different syntactic properties: the predicate inside a paratactic complement can be inflected for subject agreement whereas a subordinated, infinitive predicate cannot, and paratactic complements may occur with a complementizer whereas subordinated complements often are not able to do so.

The two basic mood distinctions of complement clauses are indicative and subjunctive. Indicative complement clauses resemble a main clause in the independence of their time-reference (free), truth-value (realis) and discourse dependency. Subjunctive complement clauses have determined time-reference (DTR, simultaneous or future relative to the matrix time) and are typically irrealis. However, subjunctive and irrealis do not always go together. Some languages, like Russian, mark the realis–irrealis distinction with mood (indicative and subjunctive), other languages code it differently. Noonan (2007) states that if a complement clause has neither indicative nor subjunctive form, it is reduced. One syntactic manifestation of reduction are infinitives, which can only be used when no loss of information arises. Whether an infinitive is licensed depends on the matrix predicate – some contain enough information (*believe, remember, promise,...*) to allow a reduced complement whereas others do not (*regret, know, imagine,...*).

Note that this discussion on the distribution of infinitives and their connection to (loss of) information is of functional nature (see Bianchi 2003, Adger 2007, Bisang 2007, Cristofaro 2007 for other approaches to finiteness).

In addition to the semantics of the complement clause, the semantics of the matrix predicate has a strong influence on the shape of a complementation configuration. Noonan (2007) divides matrix predicates into 13 semantic groups (see Table 1) which differ, on the one hand, in what they mean, and, on the other hand, in what time-reference and proposition their complement clause may have and how syntactically reduced it will be.

Although the classification of predicates in Noonan (2007) is very detailed, the 13 classes can be subsumed into three coarser types, based on whether predicates take s-like or reduced (non-s-like) complements and whether the complement clause involves ITR or DTR. The resulting classes are given in Table 2.

According to Noonan (2007), all languages have the same general set of matrix predicate options combining with the same set of clausal denotations. Additionally, the matrix predicates also behave uniformly in whether they take s-like or reduced complement clauses. However, the number of different sub-distinctions varies across languages. The majority of languages has at least one s-like and one reduced (in whatever form) complement clause type. When a language only distinguishes between two types of complements (the minimal inventory), the difference is usually manifested in time-reference: one type of complement will have ITR, whereas the other type (the reduced one) will have DTR. Languages with three types of complement clauses typically exhibit an s-like indicative form with ITR, an s-like subjunctive form with DTR or ITR, and a reduced form with DTR. Languages with four types of complement clauses usually involve an s-like indicative (ITR), an s-like subjunctive (ITR or DTR) and two reduced types of complements with DTR. Languages with more than four types of complement clauses are rather uncommon.

Noonan's typological observations point towards an implicational universal which is noted in different forms in other works on complementation as well. Complements fall into different classes which form a scale of (in)dependence. On the ends of the scale are, in Noonan's terminology, s-like and fully reduced complements, and languages need to have at least two types of complement clauses to distinguish (in)dependence. One of the main distinguishing properties of s-like vs. reduced complement clauses is determined or independent time-reference. However, (in)dependence is not a binary property, but in-between the two extremes, there is space for other forms, with different degrees of semantic and syntactic integration and dependence.

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Table 1: Semantic types in Noonan (2007)

Class	Predicates	Specifics
Utterance	<i>say, tell, report</i>	simple transfer of information
Propositional attitude	<i>believe, think, assume, deny</i>	attitude towards the complement proposition
Pretence	<i>imagine, pretend, trick into</i>	world described in the complement proposition does not correspond to the real world
Commentative	<i>regret, be sorry, be sad</i>	comments on the complement clause
Semifactives	<i>know, discover, realize, forget; indirect perception verbs</i>	proposition in the form of an emotional reaction, evaluation or judgement.
Predicates of fearing	<i>be afraid, worry, be anxious</i>	complement proposition is presupposed to be true
Desiderative	<i>wish, want, desire, hope</i>	express an attitude towards the complement proposition
Manipulative	<i>cause, force, make, tell, order, command</i>	desire that the complement proposition is realized
Modal perception	<i>epistemic & deontic can, ought, should</i>	express the matrix agent's attempt to manipulate the affectee to do something
Implicatives	<i>manage, dare, remember, try, forget</i>	typically render clause union
Phasal	<i>begin, start, continue, stop, finish</i>	refer to the realization or lack of achievement in the complement proposition
Immediate perception	<i>see, hear, watch, imagine</i>	describe the 'phase' (for example the beginning) of an eventuality
Conjunctive	<i>'and (then)'</i>	matrix agent directly perceives the event of the complement clause
		realized as a verb in some languages (e.g., Lango)

Table 2: Complement clause types in Noonan (2007)

Complement type	Time reference	Matrix predicate
S-like ^a	ITR	utterance, propositional attitude, pretence, commentative
Reduced ^b	DTR	manipulative, modal, achievement, phasal, immediate perception
Both s-like and reduced	ITR/DTR	knowledge, fearing, desiderative

^aindicative/subjunctive

^binfinitive, converb, nominalization

2.3 Cristofaro (2005)

Cristofaro (2005) proposes a functional classification tool for complement clauses, based on the claim that the semantics of the main predicate determines how the embedded predicate is semantically and syntactically shaped. The typology is couched in a *Functional Grammar* approach (van Valin 1984, 2005, van Valin 2007, Dik 1989, 1997, Hengeveld 1989, 1990, Siewierska 1991). The classification of matrix predicates in Cristofaro’s framework is a compressed version of Noonan’s classes (modal, phasal, desiderative, manipulative, perception, knowledge and utterance predicates). By focusing on semantic features of complement relations, an approach of hierarchically built complement clause types is developed. This approach rests on a level-based clause structure, predetermination of embedded semantic features and semantic integration.

As for the first concept, the levels of clause structure (following van Valin (1984, 2005), van Valin (2007) *Role & Reference Grammar*), a clause consists of four layers which are built on each other and thus entail each other. Each of these four layers has different functional properties and is evaluated by different benchmarks. The basic layer involves *predicates* or *terms* (predicates designate properties/relations, terms designate entities). The denotation of terms can be placed in space and time and are evaluated in terms of their reality. The second layer involves predication over the terms of layer one and characterizes an event (or *state of affairs*)⁶, which can be evaluated in terms of its existence. The third

⁶Functional-typological works often refer to events, states or situations as *state of affairs* (see also Siewierska 1991, Dik 1989, 1997). Since the difference does not appear to be essential, we use the term *event* here to not introduce another layer of terminological differences.

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layer involves propositions which are evaluated in terms of their truth value. The fourth and last layer is the entire clause which incorporates the speech situation as a whole by referring to the speech act. It is evaluated in terms of felicity. How these four layers interact can be seen in (6).

- (6) In case you haven't heard, Marilyn allegedly gave the letter to Rob surreptitiously during the staff meeting. (Siewierska 1991: 40)

The lowest level of (6) is the predicate *give* and the terms it requires (*Marilyn, letter, Rob*). The second layer is the predication over *give* and its terms – it is located in time by past tense marking and in space by the satellite (or adjunct) *during the staff meeting*. The third layer is expressed by the quotative addition *allegedly*, making the predication a proposition, indicating that the speaker came to the propositional content indirectly. The fourth layer describes the clause as a whole and is expressed by *in case you haven't heard* and relates the felicity of the speech act to the state of the hearer's knowledge (Cristofaro 2005: 110).

Complement-taking predicates are related to different layers of clause structure. Manipulative and perception predicates act at the predication level; knowledge, propositional attitude and utterance predicates use the proposition level; and modal, phasal and desiderative predicates pertain to the predicate/term level or to the predication level.

Table 3: Functional levels pertained by matrix predicates

Knowledge, propositional attitude, utterance predicates	Proposition-level
Perception, manipulative predicates	Predication-level
Modal, phasal, desiderative predicates	Predicate/ Term-level/ Predication-level

The semantic influence exerted by the matrix predicate over the complement clause manifests itself in two ways: *predetermination* of semantic features and *semantic integration*. Predetermination by the matrix predicate is observable in complement clauses regarding the semantic features time-reference, other T/A/M-specifications, embedded participants (note that these are statistical tendencies rather than universals):

Time-reference: Complement clauses of proposition-level predicates have no pre-determined time-reference. Predication-level verbs require predetermination to some extent: with perception verbs, the embedded time must be

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simultaneous, with manipulative verbs, the embedded time must be posterior to the matrix time. For predicate/term verbs, modals and desideratives, the complement time-reference is predetermined.⁷

Aspect: Proposition-level predicates allow any aspect values in their embedded complement clauses, whereas predication-level predicates and phasal predicates require a specific aspect value (e.g., the requirement of the complement clause to exhibit an on-going event). For complement clauses to predicate/term-level predicates, the embedded aspect value is irrelevant for the complementation relation, however, it is usually determined to some extent.

Mood: Knowledge, perception, manipulative and phasal predicates require factual complement clauses, whereas propositional attitude and utterance predicates demand non-factual complements. For modal and desiderative verbs, the mood value is irrelevant as it is also irrelevant whether the event expressed by the embedded clause even took place (see Meinunger 2017's notion of *contra-factivity*).

Participants: The embedded participants are undetermined with proposition-level predicates but determined with all others. This means that with proposition-level predicates, the embedded clause can have an independent subject, whereas with all other predicates, it tends to have a subject identical to a participant of the main clause.

The second criterion for semantic influence is *semantic integration* of the complement clause into the matrix clause. In complementation, the main and embedded clauses are interconnected (semantically integrated) to a certain degree which varies depending on the type of complement. Semantic integration is higher if the main and embedded clauses share their referents and spatio-temporal configuration. However, semantic integration is not dependent on spatio-temporal configurations; it only describes that the more integrated the content of a complement clause is into the main clause, the more the two events are part of a single event-frame. If the main and embedded clause describe a single event, they tend to overlap spatio-temporally. Utterance predicates, for example, show almost no semantic integration with their embedded clause, perception predicates show integration to some degree, and manipulative predicates

⁷Note that the tense determination presented in Cristofaro (2005) is not completely identical to the one in Noonan (2007). This might be due to a different selection of languages or a different classification of certain observations. We have to leave this open here.

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show a very high degree of integration. The most basic diagnostic of semantic integration is the degree to which the boundary between the main and complement clause is eroded. Take for example a phasal predicate like *start*: in a complementation configuration, there are two events but their boundaries are eroded since starting an action is part of the action itself. Utterance predicates behave differently: a matrix predicate like *say* is not connected to the action in its complement clause and therefore, there are two entirely separate events – the saying event and the embedded event. Modal predicates are in between: the two events are distinct and have low semantic integration, however, they share the same participant and tense values, and thus, semantic integration is given to a certain degree. Predetermination (of referents and time/place) is therefore a reflex of semantic integration (and not the other way around). Cristofaro (2005) proposes that semantic integration is not a binary feature but a scale – it forms the *Semantic Integration Hierarchy* in (7) (note that, since knowledge, propositional attitude and utterance predicates show no semantic integration at all, they are not part of the hierarchy).

(7) Semantic Integration Hierarchy:

Phasals >> Modals >> Manipulatives >> Perception/Desideratives

Combining predetermination and semantic integration, the typological hierarchy of complement clause deranking arises, as shown in (8) (adapted from Cristofaro 2005: 122). “Deranking” (see also Stassen 1985) describes, on the one hand, the degree to which the complement clause has predetermined values, and, on the other hand, the degree of erosion of boundaries between the matrix event and embedded clause event, labeled semantic integration. The elements to the left show the most predetermination and highest semantic integration, whereas the elements on the right exhibit the opposite features. Importantly, the hierarchy describes the combination of the matrix predicate’s semantics and the form of the complement clause.

(8) Hierarchy of Complement Clause Deranking:

Phasals/Modals >> Manipulatives/Desideratives/Perception >>
Knowledge/Propositional Attitude/Utterance

The *Hierarchy of Complement Clause Deranking* is hierarchical in the sense that if deranking of the complement clause is used at any point of the hierarchy, then it is used at all points to its left as well. It holds for all factors of deranking: T/A/M-marking, person agreement and sharing of participants. The hierarchy is scalar and not binary in that the crosslinguistic distribution of complement clauses does

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not involve two parametrical options, but rather a scale which codes decreasing semantic integration and increasing independence of the complement clause.

2.4 Dixon (2010)

Dixon (2010), similar to the three accounts above, provides a classification of complementation which takes into account both the grammatical properties (i.e., the syntax) of complement clauses as well as their semantics. Examining the crosslinguistic variation, patterns of mutual influence between the semantics of the matrix predicate and the form the complement clause are observed, which, according to Dixon, lead to a universal tripartition of complement clause types – *fact*-type, *activity*-type, and *potential*-type (see below).

The main distinctions used to classify the coding of complement clauses are T/A/M marking, the option of peripheral constituents (expressing time or place), the dependence and marking of arguments, and the possibility of constituent movement. Complement clause types vary in the T/A/M specifications on the embedded predicate. Some complement clause types allow equally rich T/A/M specifications as main clauses, whilst others only permit a reduced set. “In English, for instance, a *that* clause has the full set of T/A/M choices, while *-ing* and *to* complement clauses only allow auxiliaries *have* (*-en*) and *be* (*-ing*), not permitting tense inflection or modal verbs.” Dixon 2010: 383).

- (9) a. I imagined [that Mary wrote/has written/had written a book].
- b. I imagined [Mary('s) having written a book].

Another distinction is whether complement clauses can include peripheral constituents like time or place references. This option does not only vary among different types of complementation within one language, it also shows crosslinguistic variation. For instance, there are languages such as Jarawara, an Arawan language, spoken in Brazil, which do not allow peripheral specifications at all in their complement clauses.

Complement clauses also differ regarding whether they require a subject identical to an argument of the matrix clause or allow/require an independent one. Relatedly, marking of the core arguments of a complement clause may be identical to that in a main clause or involve special markers for the embedded subject. For instance, in a number of languages, a complement clause describing an action takes a possessive marked subject, as in English *-ing* clauses (9b).

The above (and other) properties align with certain semantic properties of the complement clause and together form three types of complement clauses. First,

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fact-type complement clauses, which typically indicate that some action took place, involve a structure similar to that of main clauses: they display the full range of T/A/M marking and negation; the embedded subject may or may not be identical to the matrix subject; the time reference of a *fact*-type complement clause is independent (it can refer to a time different from the matrix time), and tense and aspect values may differ from the ones in the matrix clause. Typically, but not necessarily, these complement clauses are marked with a complementizer. However, complementizers may also occur with other clause types and thus are not a reliable indication for *fact*-type clauses. Subtypes of the *fact*-class include clauses that indicate a certain fact, clauses that indicate only a possible fact, and interrogative complement clauses.

Second, *activity*-type complement clauses, which refer to an ongoing activity, involve predicates that have structural similarities to a noun phrase but remain clausal. That is, the verb is not fully nominalized but only receives a reduced, noun-like form. However, it keeps its verbal properties and the complement therefore remains a clause. In English, this is reflected as *-ing* marking. The embedded subject may be different from the matrix subject or be coreferent with it, and it can be marked with a possessive marker. Activity-type complements allow less T/A/M-marking than main clauses, but may have different time-reference. However, tense can only be expressed by a lexical marker (like an adverb) due to the reduced form of the verb. A subclass of *activity* complements are participial clauses.

Lastly, *potential*-type complement clauses refer to the potentiality of the embedded subject getting involved in an activity. They have less structural similarity to a main clause than *fact*-type clauses and less similarity to a noun phrase than *activity*-type clauses. They lack the T/A/M choices available in main clauses, their verb exhibits a special form, usually the infinitive, and they must either have the same time reference as the main clause or refer to a later time (i.e. a future time reference). Some languages require *potential* clauses to have the same subject as the main clause, however this restriction is not universal.

Regarding language variation, Dixon suggests that in single languages, the three types (*fact*, *activity*, *potential*) are instantiated by two to five different configurations, differing in their syntactic behaviour and size. According to Dixon, the potential to express all three meanings is universal, but languages do not always use three distinct complement clause configurations to express them. Jarawara, for example, has only one configuration (the one typical for *activity* complements in other languages), which also instantiates the other clause types in that it can express all three meanings. Akkadian (Semitic) has one *fact*-type construction and one for both *activity* and *potential* complements. Tariana

(Arawakan) has a *fact*-type and a *potential*-type configuration, whereas *activity* meanings are realized as pure nominalizations. Since each type of complement clauses has subtypes, languages can also have more than three complement clause constructions: White Hmong, for instance, exhibits two *fact* configurations, one *activity*, and two *potential* configurations. Fijian (Austronesian), on the other hand, has three configurations for *fact*-types, one for *potential*-types, and one for *activity*-types. However, there are also languages lacking complement clauses altogether (e.g., Dyirbal, Australian Pama-Nyungan), replacing them with *Complementation Strategies* such as serial constructions, relative clauses or nominalizations.⁸

A further point we want to highlight is that Dixon (2010) also observes that the type and meaning of a complementation configuration is not only determined complement clause internally, but also depends on the semantics of the matrix predicate: not every matrix verb can be combined with every complement clause. Dixon suggests five classes of matrix predicates, listed in Table 4. Some predicates require a unique type of complement, others may occur with different types of complement clauses, resulting in different meanings. This combination-based meaning-shift can be observed in almost every language (see also Givón 1980).

Finally, as we pointed out in the introduction, the line between complement clauses and single clauses with more than one verbal element is not always very sharp, and Dixon (2010) describes this via an additional category of secondary concepts. In natural languages, some concepts are expressed as lexemes (like the verbs above), whereas others are closed-class grammatical elements (like pronouns or syntactic function markers). However, according to Dixon (2010), in addition, there are secondary concepts which are in between those two, which are coded as grammatical forms in one language but as lexemes in another. Verbal elements belonging to this class are, for instance, *can*, *must*, *begin*, *try*, *want*, *hope*, *plan*, *make*, *help*, which crosslinguistically can be realized as an affix on the verb, as a secondary affix as part of the T/A/M system, as an independent element modifying the embedded verb/the whole complement clause, or as a lexical verb. The difference between full lexical verbs and secondary elements is that the latter modify a full predicate whereas the former are full predicates themselves. In examples like (10), the full predicate *write* is modified by the secondary class concept *begin* (thus under this approach, cases like (10) would, in fact, not involve a complement clause). We will see in the next section, that a similar concept is suggested in a cartographic approach.

⁸Note that Noonan (2007) classified serial verb constructions as complementation, whereas Dixon 2010 excludes them from complementation and treats them as a complementation replacement strategy. See also fn. (4).

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Table 4: Matrix predicate classes in Dixon (2010)

Predicate type	Examples	Complement clause type
Attention verbs	<i>see, hear, notice</i>	Fact, Activity
Verbs of thinking	<i>think, consider, imagine, dream</i>	Fact, Activity
	<i>assume, suppose</i>	Fact
	<i>remember, forget</i>	Fact, Activity, Potential
	<i>understand, know, believe, suspect</i>	Fact, Potential
Decision verbs	<i>decide, resolve, plan, choose</i>	Fact, Potential
Verbs of liking	<i>like, love, prefer, regret, favor</i>	Fact, Activity
Verbs of speaking	<i>say, inform, tell</i>	Fact
	<i>report</i>	Fact, Activity
	<i>describe, refer to</i>	Activity
	<i>promise, threaten, order, command, persuade</i>	Potential

(10) John began [to write a detective story]. (Dixon 2010: 400)

Dixon’s (2010) tripartite classification of complement clauses is summarized in Table 5.

2.5 Cartography

In contrast to functional grammar approaches, for which structural–grammatical reasoning often plays a more subsidiary role, generative approaches are built on structural–grammatical tools and concepts, and to investigate crosslinguistic variation typically pursue what has been described as *formal generative typology* (Baker 2009, Baker & McCloskey 2007). Formal generative typology allows combining tools from both generative grammar and typology. To make informed claims about language variation and universals, insights from typology are indispensable. On the other hand, to investigate non-surface observable dependencies (e.g., between semantic and morphosyntactic properties), a significant degree of theoretical abstraction is necessary, for which generative grammar offers useful tools. Common to many *Government and Binding* and *Minimalism* approaches to complementation is the idea that syntax regulates, in one way or another, the distribution of complement clauses. Differences between control and raising/ECM

Table 5: Complement clause types in Dixon (2010)

Complement type	Time reference	Matrix predicate
<i>Fact-type</i> ^a	independent	Lexical verbs: <i>think, imagine, assume, remember, forget, know, understand, believe, discover, say, inform, report</i> Secondary concepts: <i>can, wish</i>
<i>Activity-type</i>	time reference often expressed by a lexical element	Lexical verbs: <i>see, hear, like, fear, enjoy, describe</i> Secondary concepts: <i>begin, continue</i>
<i>Potential-type</i>	dependent	Lexical verbs: <i>promise, threaten, order, persuade</i> Secondary concepts: <i>should, try, want, make</i>

^aA reviewer pointed out that the labelling as *fact-type* here is unfortunate since not all of the presented matrix verbs combine with (semantically) factive complement clauses. Dixon (2010) aims to resolve this via describing the *fact* class as also involving *possible facts*, however, these then are difficult to distinguish from *potential* meanings.

have been attributed to concepts of Case and/or clause structure (see Pesetsky 1992 for a very rich study on complementation in English, or Landau 2000 on control crosslinguistically). In the last two subsections, we summarize two formal generative approaches to clause structure and complementation which aim to derive complementation typologies from structural concepts. A shared concept of these approaches is that the structural makeup of clauses is the link between the semantic and morphosyntactic properties of complementation. However, the accounts differ regarding how direct the relation between syntax and semantics is.

We begin with a brief outline of the concept of cartography and more details on its applicability to complementation in the next section.⁹ Cartographic

⁹Cartography was originally not applied to complementation, but rather developed as a model of clause structure (and in its initial form mostly of the left periphery). It then extended to a subclass of complementation, namely restructuring. One aim of this and the next section is to show that cartography and complementation are nevertheless closely connected and that the tools of cartography offer a way to implement the typological patterns. Cartography also offers, as one of only few frameworks in a generative setting, a way to implement a semantic hierarchy structurally (*Nanosyntax* would be another model, but we are not aware of detailed works on complementation in that framework). Finally, it is noteworthy that the cartographic

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approaches aim to capture universal ordering regularities by representing semantic categories via syntactic structure. Sentences are assumed to involve a fine-grained hierarchical structure of functional projections, which are defined syntactically or semantically, and the resulting *functional hierarchy* is hypothesized to be universal.¹⁰ In Rizzi (1997), the left periphery is decomposed into information structural properties such as topic and focus, and syntactic constituents mapped to these functions must occur in these designated positions (either overtly or covertly). The cartography proposed in Cinque (1999, 2004), Cinque & Rizzi (2010) involves a semantic hierarchy of clausal functions such as mood, tense, and aspect (see (11), from Cinque (2004) for the detailed hierarchy), which correspond to functional projections in syntax (we are only concerned here with verbal categories, such as different types of aspect, mood, modality etc., but similar hierarchies have been proposed for the nominal domain). For each particular meaning, the functional head can realize an affix corresponding to that meaning, and the specifier a phrase expressing that meaning. Syntax thus reflects semantics in that (certain) semantic concepts receive their own syntactic projections (cf. the description of the cartographic enterprise in Cinque & Rizzi (2010: 63) as “an attempt to “syntacticize” as much as possible the interpretive domains”). Although single languages do not realize all possible functional heads, the hierarchy is still assumed to be covertly present. Thus, for cartographers, the size of innate clause structure is rather large, providing more structure than needed in individual languages.

- (11) speech act >> evaluative >> evidential >> epistemic >> past >> future >> irrealis >> alethic >> habitual >> repetitive >> frequentative >> volitional >> celerative >> anterior >> terminative >> continuative >> retrospective >> proximative >> durative >> generic/progressive >> prospective >> obligation >> permission/ability >> completive >> Voice >> celerative >> repetitive >> frequentative

The categories can be subsumed into the broader classes Mood >> Modal (epistemic) >> Tense >> Modal (root) >> Aspect, which resemble the hierarchies and semantic classifications in the typological approaches discussed so far. One difference is that cartographic approaches yield a rigid one-to-one syntax–semantics

hierarchy and Givón’s binding hierarchy are very similar (despite terminology), and we take such convergence as exciting support for these ideas.

¹⁰Note that in this section “functional” refers to what is sometimes called the *extended projection* of a head – the part of clause structure that is above the lexical projection of a head (e.g., the functional projection IP, as opposed to the lexical VP).

mapping. Each semantic category corresponds to its own functional head, resulting in a highly detailed syntactic structure. Cartography thus describes the order and the maximal size of functional clause structure possible in language. Changing the order (unless for scope or information structure purposes) or doubling the occurrence of an element of a particular category of the hierarchy is only possible if a new clause is added (e.g., adding the same adverb twice entails two occurrences of the functional hierarchy, i.e., a bi-clausal structure).¹¹ The main motivation for the functional hierarchy comes from the surface ordering of the elements associated with the functional heads, which is highly uniform across languages, for some categories even universal. This is best known for adverbials: Cinque (1999) observes that adverb orders show significant similarities crosslinguistically, and suggests that adverbs occupy specifier-positions along the functional hierarchy. The position of different adverbs is thus determined by the meaning, which must match with the categories of the semantic classes along the functional spine.

In Cinque (2004), an extension of the functional hierarchy to some aspects of complementation is provided. Based on the observation that affixes, auxiliaries, functional particles, and certain verbs show consistency in their ordering, it is suggested that not only affixes, but also auxiliaries and certain verbs may occupy positions along the functional spine of a clause. This is visible, for example, in Italian (12), where the sequence *volere* ‘want’ (volitional) >> *smettere* ‘stop’ (terminative) follows the hierarchy. If the order corresponds to the hierarchy, a monoclausal structure (i.e., a single lexical verb plus its functional domain) is possible, as confirmed by the possibility of the clause-bound phenomenon of clitic climbing in (12a). If the order does not correspond to the hierarchy, as in (12b), only a bi-clausal structure would be possible, which then does not allow clitic climbing.

(12) Italian (Cinque 2006: 19)

- a. Non vi **vuole** **smettere** di importunare.
not 2PL.ACC want.3SG stop.INF to bother
‘He doesn’t want to stop bothering you.’
- b. Non vi **smette** di **voler** importunare.
not 2PL.ACC stop.3SG to want.INF bother
‘He doesn’t stop wanting to bother you.’

¹¹Note, however, that certain categories (e.g., *repetitive*) are listed twice in the Cinque hierarchy, which diverts from the strict one-to-one syntax–semantics mapping and may allow some doubling.

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The functional hierarchy in Cinque (1999, 2004) thus covers only a small aspect of complementation, namely cases that relate to clause union or restructuring (Rizzi 1976, 1978; see Wurmbrand 2001 for an overview and references), by effectively treating the lower predicate *not* as a verbal complement clause, but rather as the main (and only) lexical predicate of a single clause, and any other higher verbal elements as functional elements of the same clause. Although there is no valid definition of what counts as lexical vs. functional, and the assumption of verbs like *try*, *begin* as functional heads is controversial (see Wurmbrand 2001, 2004), it is noteworthy that the Cinque hierarchy in (11) shows surprising similarities to the *Binding Hierarchy* in Givón (1980), which suggests a deeper connection and possible extension to complementation more broadly. Both hierarchies are defined by semantic categories, and although with a different level of detail, the hierarchical order of the categories is the same in both approaches. For instance, both locate speech act predicates (*say*, *tell*,...) very high and ascribe them a high degree of independence. These are then followed by a set of epistemic, attitude, emotive, irrealis predicates (*know*, *decide*, *hope*,....). Lastly, both types of hierarchies locate predicates like (*plan*, *begin*, *try*) low, involving the highest likelihood to lead to dependent complement clauses. Setting aside the specific technical implementation proposed in Cinque (1999, 2004) that (certain) higher verbs are functional elements, the cartographic hierarchy nevertheless appears to be doing some important work, and what is needed is a mechanism that links the height of an element on the hierarchy to different degrees of transparency and dependency between the matrix and embedded predicates. Before turning to such a model, certain other issues of cartography need to be addressed.

While certain aspects of cartography are widely accepted in structural approaches (e.g., that CP-level categories occur above TP-level categories), the universality, applicability, evolvability, and innate nature of cartographic hierarchies have also been questioned. Ramchand & Svenonius (2014), for instance, address a tension between cartographic approaches which, as laid out above, employ rigid clausal templates with a rich inventory of ordered projections, and minimalist approaches, which often utilize the concept of free Merge and consider predetermined templatic structures as impossible. The concerns are how such a complex system could have evolved, be acquired, and be implemented genetically. Ramchand and Svenonius conclude that there is an irreducible part of clause structure, which is stable across languages, however instead of an array of functional projections it consists of only three broad clausal domains which are in a containment configuration. This universal hierarchy of broad functional domains is semantically necessitated (to yield certain interpretations) and reflects a syntactic residue of cartographic structure. However, since it can be motivated

by the conceptual–intentional interface, it is compatible with minimalist tenets. Ramchand and Svenonius’s approach to clause structure hence unifies the advantages of both cartographic and minimalist approaches, since the clausal hierarchy as defined is minimal and (largely) predictable, but clausal computation is also not entirely free. Since the Ramchand and Svenonius clause structure model allows for a direct application to complementation, we lay out their model here first, before returning to the structures of clausal embedding.

The Ramchand and Svenonius hierarchy consists of three broad clausal domains which are defined semantically as *e(vent)*, *s(ituation)* and *p(roposition)*. These semantic ‘sorts’ are seen as conceptual primitives, which map uniformly to syntactic structure: the semantic *e*-sort corresponds to the syntactic *v/V*-domain, the semantic *s*-sort to the syntactic *T*-domain, and the semantic *p*-sort to the syntactic *C*-domain. The three broad domains are hypothesized to be universal, however, in contrast to (full) cartographic approaches, within these domains, there could be language-specific, individual ordering.¹² This is observable, for example, in the flexible ordering of perfect and root modals in many languages, such as the Norwegian examples in (13), in contrast to English, where these categories are rigidly ordered.

- (13) Norwegian (Ramchand & Svenonius 2014: 160)
- a. Kari kan ha gått på ski.
Kari can have gone on ski
‘Kari might have gone skiing.’ (Mod >> Perf)
 - b. Kari har kunnet gå på ski til jobb hver dag.
Kari has could.PTCP go on ski to work every day
‘Kari has been able to ski to work every day.’ (Perf >> Mod)

Hierarchy effects do arise, however, in Norwegian (as in all other languages we are aware of) between epistemic modals and root modals – epistemic modals are always higher than root modals. In a domain approach, this is captured by the hypothesis, that epistemic modals, on the one hand, and root modals and perfect, on the other hand, occur in different domains: the former in the propositional domain, the latter in the situation domain. As such, epistemic and root modals are rigidly ordered, whereas root modals and perfect can alternate (at least in principle – different languages may impose additional restrictions).

¹²The resulting clausal organization is very similar to the three “prolific” domains developed in Grohmann (2003) – theta-domain, phi-domain, and operator-domain, as well as the categoric–structural accounts in Platzack (2000), Wiltschko (2014).

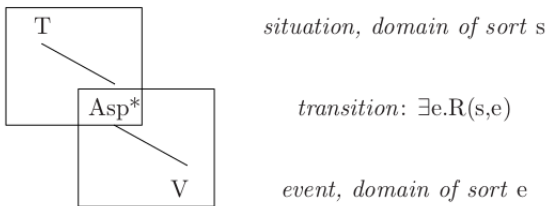
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The three domains are connected to each other via containment such that higher domains contain and are elaborations of lower domains: *p* is built from *s*, and *s* is built from *e*. *Events* correspond to the syntactic argument structure domain (e.g. VoiceP, *v*P, VP). Within the *event*-denoting constituent, thematic roles are assigned and the single components, namely participants and subevents, are bound together. In particular, causation and resultativity unite subevents into a single, complex event. *Events* can be stative or dynamic, they can be quantified over, and they have no temporal dimension (see (14)).

$$(14) \quad \llbracket [\text{VP verb}] \rrbracket = \lambda e.\text{VERB}(e) \quad (\text{Ramchand \& Svenonius 2014: 162})$$

Situations correspond to the syntactic IP/TP-domain (or equivalent label). They are created by combining an *event* with time- and world-parameters, allowing them to refer to specific times and worlds. A *situation* is an elaboration of an *event*, thus presupposes the existence of an *event*, and is necessarily more complex than an *event*. In order to build a *situation*, an *event* has to be existentially closed (or bound by another operator) and transferred to the next domain. The transfer point is a functional head, e.g., *Asp**, which combines with an *event*, existentially closes it, and renders a *situation* with an aspectual parameter. *Asp**P may then be further combined with a modal or temporal head, which adds further T/A/M parameters and fully converts an *event* into a *situation*. This is shown schematically in (15) (for the detailed computation the reader is referred to the original paper).

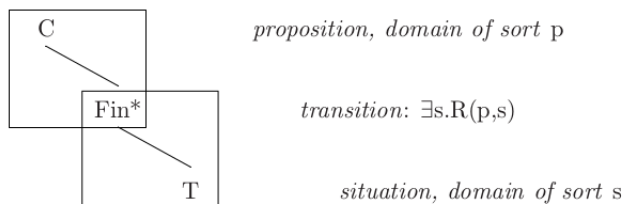
$$(15) \quad \text{Ramchand \& Svenonius (2014): 164}$$



Lastly, *propositions* correspond to the syntactic CP-domain. They are created by combining an existentially closed *situation* with speaker-oriented parameters, which anchors the *situation* to an utterance-context. A *proposition* is thus an elaboration of a *situation*, presupposing its existence. As show in (16), the transfer point between a *situation* and a *proposition* is given as *Fin** (but nothing hinges on this particular label), which adds temporal and world parameters to a *situation*, including speaker-oriented variables or pronouns, thereby anchoring the *situation* in the discourse.

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(16) Ramchand & Svenonius (2014): 164



Thus, the three sortal domains, *event*, *situation*, and *proposition* are built in a way that higher sorts are expansions of lower sorts, with certain syntactic functional heads acting as transition points.

A significant benefit of this model is that the broad containment relations are not stipulated but follow from the semantic composition of *events*, *situations*, and *propositions* (see also Ernst 2001 for similar building blocks deriving adverbial hierarchies). The most important insight lies in the way syntax and semantics interact. Although there is a tight connection between the broad syntactic domains and the resulting semantic sorts, there is no general one-to-one mapping between syntax and semantics. The broad domains are uniform (e.g., to create a configuration that is interpreted as a *situation*, some layer of the T/A/M-domain must be present). However, the internal composition of these domains, as well as the specific heads that make up the domains in any given language (or sentence) may vary, as long as some basic heads responsible for transition and elaboration into higher sorts are present. The resulting hierarchy of clausal domains is basic enough to be learnable, and flexible enough to allow for language variation. It imposes fewer universal restrictions than Cinque’s fine-grained hierarchy, which is warranted by the crosslinguistic variation found.

At this point it is useful to come back to the notion of *proposition*. In the Ramchand and Svenonius model, the term *proposition* refers to a combination of syntactic and semantic properties – a full clause that is anchored to the context. However, they do not discuss details of the domain that turns a *situation* into a *proposition*. Traditionally, the CP-domain included projections for clause typing and force (interrogative, declarative), information structure (topic and focus), and in cartography also finiteness. Another trend has been to enrich the propositional domain to also include representations of the speaker and the addressee, information about speech acts and illocutionary force (see, among others, Speas & Tenny 2003, Wiltschko 2014, 2017, Krifka 2023, Miyagawa 2022, and [Chapters 2 and 16 of this volume](#)).

In these contexts, the notion of *proposition* becomes even more complex and it is important to separate terminological differences from content differences.

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Whereas *proposition* refers to the entire clausal domain (i.e., the CP-domain) in the Ramchand and Svenonius model (with TPs mapping to *situations*), *proposition* refers to the TP-domain, excluding the illocutionary domain above, in Krifka (2023). In Krifka (2023), following a tradition starting with Stenius (1967), the CP-domain of assertive clauses is decomposed into the layers *speech act* [Act] >> *commitment* [Com] >> *judgement* [J], where *judgements* correspond to epistemic and evidential attitudes, *commitments* to social commitments related to assertions, and *speech acts* to updates or changes of the common ground. Note, however, that the two views are fully compatible – the Ramchand and Svenonius model (as well as our model in the next section) can either be extended by including another domain above the CP (a fourth clausal domain to encode speech act and illocutionary force), or by a more structured organization of the CP domain itself including the categories suggested in Krifka’s model. We leave the choice of technical implementation open here, but note that a combination appears very promising as both approaches naturally lead to different complementation patterns (see next section). Furthermore, Krifka’s structure is also defined via containment. By adding the syntactic head *Judge* to a *proposition* (or in Ramchand and Svenonius’s terms *situation*) a propositional function is created specifying that a judge considers the proposition expressed in the clause to be true; by further adding the syntactic head *Commitment*, the JP function is turned into another propositional function specifying that the speaker is committed to the judged proposition. Lastly the syntactic head *Act* includes an operator that determines whether the ComP is a question or assertion. Given these containment relations, the categories suggested by Krifka thus form a syntactic and semantic hierarchy with a predictable order ActP >> ComP >> JP >> TP.¹³

Krifka (2023) also provides a new approach to the phenomenon of embedded root clauses, such as embedded *that*-less clauses in English or embedded verb second clauses in German (see Heycock 2017 for an overview). Embedded root clauses have raised interesting questions about the nature of complementation in these cases. On the one hand, embedded root clauses show similar properties to

¹³As the reader can verify, these categories are very similar to the highest categories in the Cinque hierarchy in (11) and thus the two approaches are compatible. A main advancement of Krifka’s approach is that the structure is defined via containment, thus necessarily built in the way it is, whereas (11) is merely given as a list without motivation. As shown in Krifka (2023), the semantic hierarchy is also supported by syntactic differences, both within simple clauses and different types of *proposition* complements. Furthermore, Wurmbrand (2024) suggests that the hierarchy is reflected in an implicational ECM hierarchy. In this paper we cannot go into details of finer-grained structures of the broad clausal domains, but refer the reader to these works.

regular embedded clauses (extractability¹⁴, functions as a semantic argument of the matrix predicate), but on the other hand, they appear to be less integrated into the matrix clause, thus less complement-like (see Reis 1997). This discrepancy is resolved in Krifka's model by representing embedded root clauses as ActPs, since they show properties of illocutionary acts and have the potential to update the common ground of the conversation. True embedded clauses, on the other hand, lack the ActP-layer and thus do not constitute independent speech acts.¹⁵ The existence of embedded root phenomena thus makes it very plausible that embedded clauses include a structural aspect of speech act information, and further typological research will show whether their distribution allows us to gain further insights into the question of whether a syntactic separation of what Ramchand & Svenonius refer to as *proposition* and a distinct speech act domain is supported or whether the layers could be seen as finer-grained structure of the proposition domain. Although the proposal in Ramchand & Svenonius 2014 is not designed as a model for complementation but has as its goal a theory of basic clause structure, we show in the next section that it can be extended directly to complex clauses and that the hierarchy observed for clausal domains in simple clauses is replicated in the complementation hierarchies that have been observed in the typological works in one way or another.

Although the proposal in Ramchand & Svenonius (2014) is not designed as a model for complementation but has as its goal a theory of basic clause structure, we show in the next section that it can be extended directly to complex clauses and that the hierarchy observed for clausal domains in simple clauses is replicated in the complementation hierarchies that have been observed in the typological works in one way or another.

2.6 Wurmbrand & Lohninger (2019)

Wurmbrand & Lohninger (2019) propose a formal generative typology approach to complementation which can be seen as the structural–grammatical counterpart to Givón (1980)'s *Binding Hierarchy*. Both works follow the hypothesis that there is a possibly universal implicational complementation hierarchy which is defined semantically and detectable through a diverse set of grammatical properties. While the distribution of morphosyntactic properties varies significantly

¹⁴Note that there is a controversy with respect to extractability out of embedded verb second complements (see Reis 1997, Holmberg 1986).

¹⁵Similarly, embedded direct speech contexts exist (see [Chapters 15 and 17 of this volume](#)), but it is not clear yet how wide-spread (with respect to different languages) this phenomenon is. Another phenomenon that is relevant in this context is reported speech, in particular the question of which verbs allow it (see Spronck & Nikitina 2019 for a typology).

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across languages, the semantic grouping of complement types shows a (more) stable distribution. The core observation in Wurmbrand & Lohninger (2019) is that the categories of the implicational complementation hierarchy [ICH] can be defined as *propositions*, *situations*, and *events*, adopting the terminology and definitions in Ramchand & Svenonius (2014) (similar classifications have been proposed in Rochette 1988, 1990, Pesetsky 1992 under different labels). These three semantic types are supersets of the categories suggested in Givón (1980) and align with the hierarchy given there. Since the three-way distinction has been robustly attested across languages, whereas the division into sub-classes shows variation, the ICH is defined only for these broad classes.

Proposition complements comprise speech and epistemic contexts (possibly also factive complements). The truth of the complement clause can be asserted via predications such as *which is true/false* (e.g., *Nova claimed that she bought salad, which is true* can mean that it is true that Nova bought salad) or be presupposed as in the case of factive complements. Furthermore, *proposition* complements are temporally independent in that the embedded event can occur simultaneously to, before, or after the matrix event. The freedom of the embedded time-reference comes from the fact that the complement clause has its own reference time (typically, the attitude holder's "Now") which anchors the embedded tense. Lastly, *proposition* complements may involve speaker-oriented and/or discourse-linking parameters, an independent embedded subject, and, if infinitival, partial control (see Landau 2000). Matrix predicates typically combining with *proposition* complements are *admit*, *affirm*, *announce*, *assume*, *believe*, *claim*, *consider*, *discover*, *figure*, *find*, *forget* (factive), *imagine*, *know* (factive), *observe*, *say*, *suppose*, *tell* (speech), *wager*.

Situation complements, which involve emotive and irrealis contexts, refer to temporally grounded events that are not evaluated for truth (no assertion or predication of their truth is possible), but for other properties (such as, likelihood, value to the speaker; for instance, *Nova promised to buy salad* can be followed up (when referring to *to buy salad*) with *...which will never happen/which is a good idea/not easy to do on Sundays*). *Situation* complements elaborate *Events* without speaker- or utterance-oriented properties. However, they show time- and world-parameters and refer to a specific, possibly predetermined, time. The most common type of *situation* complements are unrealized irrealis events ('to decide to do something'; the decision is to bring about a not yet realized situation) in which the time of the complement clause is set in the future with respect to the matrix time. Lastly, infinitival *situation* complements often allow partial control. Matrix verbs combining with *situation* complements are *agree*, *ask*, *choose*, *decide*,

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demand, desire, know (modal), *need, plan, promise, refuse, tell* (imperative), *want, wish*.

Lastly, *event* complements involve implicative and strong attempt contexts. They lack speaker- and utterance-properties, as well as world- and time-properties. They are therefore tenseless – the embedded *event* occurs simultaneously with the matrix event and in contrast to *situation* complements, cannot refer to a time in the future relative to the matrix time (e.g., **She forgot/failed/managed to leave tomorrow*). Furthermore, *event* complements may have reduced argument structure and/or event properties, and typically trigger obligatory exhaustive control. Lexical matrix verbs which take *event* complements are *avoid* (implicative), *begin, continue, fail, finish, forget* (implicative), *manage, start, stop, succeed, try*.

The temporal and subject properties are summarized in Table 6. While languages may display finer-grained distinctions, the three broad classes in Table 6 are reflected, in some form or another, in all typologies summarized in this article. Importantly the classes are defined semantically and not necessarily by specific matrix verbs. Certain verbs can occur in more than one context, such as factive vs. implicative *forget*, factive vs. modal *know* or speech vs. order *tell*. Depending on the meaning, these verbs would thus be in different classes, and their properties *under a particular interpretation* follow (only) the properties of the class associated with that interpretation. Thus, the tripartite classification in Table 6 is about the semantics of complementation configurations, not about specific verbs, nor about syntactic properties of the complement clause.

Table 6: Complement clause types (Wurmbrand & Lohninger 2019: 194)

Properties	<i>Proposition</i>	<i>Situation</i>	<i>Event</i>
Temporal interpretation	embedded reference time; no pre-specified tense value; anchored in an utterance or embedding context; may involve speaker-oriented parameters	no speaker/utterance properties; no embedded reference time; pre-specified tense value (most common future)	no speaker/utterance properties; no time and world parameters; tenseless, simultaneous
Infinitives	partial control possible	partial control possible	exhaustive control

A main property of the three semantic types of complements is that they stand in a hierarchical relation with each other as illustrated in Table 7. Independence

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refers to properties such as the presence and/or interpretation of an independent subject or tense in the embedded clause. Transparency describes the permeability of the embedded clause for certain operations or dependencies. Integration is the degree to which the embedded predicate is an integral part of (e.g., incorporated into) the matrix predicate. The implicational nature is reflected in the scale arising from the hierarchy: *proposition* complements are more independent, less transparent, and less integrated than *situation* complements, which in turn are more independent, less transparent, and less integrated than *event* complements.

Table 7: *Implicational Complementation Hierarchy* (ICH)
(Wurmbrand & Lohninger 2019: 188)

Most independent		Least independent
Least transparent	<i>Proposition > Situation > Event</i>	Most transparent
Least integrated		Most integrated

The three-way distinction is motivated by a range of what Wurmbrand & Lohninger (2019) refer to as *ICH-signature* effects, which can be observed within and across languages. As indicated in the last row of Table 8, for any property [P] that distinguishes between different types of complements, *proposition* and *event* complements show the opposite values, and *situation* complements either allow both values or ‘side’ with one or the other (depending on the property). ICH-signature effects have been observed for a range of morphosyntactic and semantic properties. An illustration is given in Table 8 for the distribution of clause introducers in Greek and Bulgarian. The two languages lack infinitives and all non-factive declarative complement clauses are realized as finite, introduced either by *oti/na* (Greek) or *če/da* (Bulgarian). The distribution of the introducers is not random, however, but reflects the ICH-signature: *proposition* complements obligatorily require *če/oti*, whereas *event* complements only allow *da/na*. *Situation* complements permit both options.

The mapping between the hierarchy and morphosyntactic properties of complement clauses is not absolute but relative. How independence, transparency and integration are manifested in different languages varies and there are no universal morphosyntactic properties that can be associated with any of the different classes of complements. However, there are implicational relations – for any independence property (e.g., complementizers, finiteness), a type of complement can never be more independent than complements of the classes to its left. While in certain *Government and Binding* approaches, (non-)finiteness is seen as a defining characteristic for particular clause types and structures (e.g., ECM and

Table 8: Clause introducers in Greek and Bulgarian, ICH-signature (Wurmbrand & Lohninger 2019: 196)

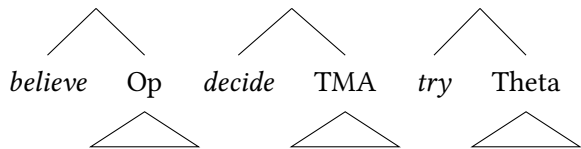
	<i>Proposition</i>	<i>Situation</i>	<i>Event</i>
Bulgarian	<i>če</i> , * <i>da</i>	<i>če</i> (+FUT), <i>da</i>	* <i>če</i> , <i>da</i>
Greek	<i>oti</i> , * <i>na</i>	<i>oti</i> (+FUT), <i>na</i>	* <i>oti</i> , <i>na</i>
ICH-signature (general)	+P	+/-P, +P, -P	-P

subject raising clauses must be non-finite in English), the ICH model does not prescribe a strict mapping between finiteness (or other morphosyntactic properties) and specific types of complements. While there is a crosslinguistic tendency for *proposition* complements to be finite and *event* complements to be non-finite, this is not universally the case, and non-finite *proposition* complements as well as finite *event* complements are possible in certain languages (see also Boye 2010 making a similar observation with respect to tense). The distribution is not entirely arbitrary, however, but follows the ICH indirectly. As shown in Wurmbrand et al. (2020), a type of complement can never be more finite than the complement(s) to its left on the ICH. This implicational universal is derived by finiteness being the realization (not the cause or trigger) of different syntactic structures associated with different ICH categories.

While there is no direct connection between the ICH and morphosyntactic properties, Wurmbrand & Lohninger (2019) propose that there are mapping restrictions between them. Classifying complement clauses in terms of the conceptual primitives and sorts defined in Ramchand & Svenonius (2014) (see the previous section), an (in)dependence scale automatically arises: *propositions* are most independent, as they are anchored in an utterance/embedding context and contain time and world parameters; *events* are most dependent as they lack all T/A/M and context parameters; and *situations* are in-between since they contain T/A/M but no context parameters. Furthermore, adopting the syntax–semantics mapping modeled in Ramchand & Svenonius (2014), a syntactic complexity scale can be defined, which yields different *minimal* structures for the three types of complements, illustrated in (17).

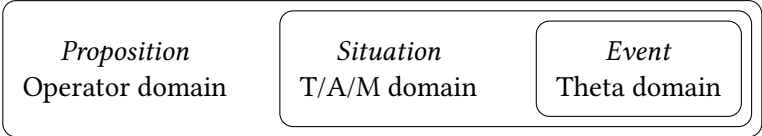
(17) (Wurmbrand & Lohninger 2019: 214)

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Finally, the system predicts the implicational relations among different clause types by the same containment configurations found in simple clause structure: *situations* are elaborations of *events*, and *propositions* are elaborations of *situations*, as illustrated in (18).

(18) (Wurmbrand & Lohninger 2019: 214)



Proposition, *situation*, and *event* complements may thus differ in structural size (in addition to different semantic complexities).¹⁶ However in contrast to cartographic approaches, this mapping is not absolute and, as long as the relevant semantic relations are maintained, the syntactic structure can also lead a partially independent life. This autonomy of syntax is an important concept, since there is variation across languages in the structure of complement clauses with the same meaning. In particular, the system allows variation in the composition of the clausal domains as well as complement clauses to grow larger than the minimally required structures in (17). This is for example the case in Dutch *try* complement clauses which can occur with or without the complementizer *om*, but the alternation does not induce a meaning-change of the configuration (see Besten & Rutten 1989, Rutten 1991). Thus, in these cases, the *event* clause can be structurally larger than the minimally required Theta/Voice-domain. The ICH model only demands that structures cannot be too small or lacking crucial components to yield the desired output.¹⁷ For instance, to create a configuration that should be mapped to a semantic *situation*, the complement needs to contain some property of the T/A/M domain. However, it is not prescribed how this parameter

¹⁶Wurmbrand & Lohninger (2019) adopt the terminology in Ramchand & Svenonius (2014), however, as pointed out above, the *proposition* domain may be enriched by speech act categories.

¹⁷A reviewer pointed out that smaller than required structures could arise if “coercion” does not only affect the meaning of matrix predicates but also complements and that “concealed questions” (Heim 1979)) might be a case in point. We leave this open to further research.

is expressed – it can be added by various syntactic operators like tense, modality or aspect. Similarly, in such a case it would also be possible to add a syntactic operator domain, as long as the resulting configuration can still be interpreted as a *situation*. Hence, there is no one-to-one correspondence between meaning and syntactic structure, but different structures can, in principle, map to the same semantic concept.

The (partial) autonomy approach of syntax is paired with what Wurmbrand & Lohninger (2019) refer to as a *synthesis* condition on complementation. In contrast to cartographic approaches that assume a syntactic template, the ICH-model is compatible with a *free merge* system, where the compatibility of verb–complement configurations is determined at the output (when syntax feeds into semantics). The specific hypothesis is that complementation configurations are computed freely in syntax and that the semantic output is determined jointly by the specifications imposed by the matrix predicate and the complement. A semantic counterpart of this model can be found in semantic decomposition approaches (see Kratzer 2006, Moulton 2009a,b), where it is suggested that the meaning of attitude contexts is ‘spread’ over the matrix and embedded clause and aspects of the attitude meaning are attributed to the embedded complementizer rather than the matrix verb. A crucial aspect of the synthesis approach is that in complementation the influence is mutual – a matrix verb can impose properties on the embedded clause, but properties of an embedded clause can also affect the matrix predicate. An area where complement influences on the matrix predicate have been observed involves alternating verbs such as *tell*, *forget*, or *know*, which in many languages occur in two frames. To illustrate, *tell* has a speech meaning (*I told him that she left*) or an irrealis command meaning (*I told him to leave*). Importantly, different meanings often correlate with different morphosyntactic coding: in English (and many other languages), the speech meaning occurs with a finite complement whereas the infinitive has only the command meaning; in Greek, Bulgarian, or Macedonian, different clause introducers are used for the two meanings. The synthesis model opens the door for an approach which avoids duplicating lexical entries for these verbs. Instead, these verbs have an underspecified semantics (see White 2014 for a related idea) and freely combine with different types of complements. Depending on which type of complement is chosen, which is often reflected in morphosyntactic coding differences, different meanings are computed (i.e., the type of complement dynamically contributes to the meaning of the matrix predicate). An open question is how exactly the meaning of these verbs can be characterized to allow the flexibility to yield different combinatorial meanings when combining with different types of complement clauses.

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The effect of synthesis can also be observed in class switches where verbs are shifted into different interpretations based on the composition of the embedded clause. An issue researchers have encountered especially in field research on complementation is that speakers sometimes allow verbs to be reinterpreted and coerced into a different meaning, in particular when forced by morphosyntactic coding that points to a different complementation class. For instance, the verb *decide* typically requires a future irrealis complement, even when it occurs with a finite complement: *I decided to leave/that I would leave* are possible, but *I decided to have left/that I/she left* are very odd. However, in some contexts, *decide* can be coerced into an attitude meaning, such as the performative use in cases like *I decided that he is a nice person* where the individual denoted by the matrix subject evaluates or assigns truth to the embedded proposition (other attitude meanings are possible for some speakers as well). Whenever class switches are possible, the resulting configurations obey the ICH and the properties exclusively follow the properties of the switched into class. For instance, a (regular) *decide* configuration with an irrealis interpretation shows the properties of *situation* complements, whereas configurations with a (coerced) attitude interpretation follow the properties of *proposition* complements. The extent of such class and meaning switches is not known yet, and further research is necessary to determine the crosslinguistic distribution of (im)possible class switches and their semantic and morphosyntactic properties.

3 Variation and universality

As the above summaries have shown, examining complementation from different vantage points renders a heterogeneous picture. Approaches to complementation differ depending on the focus, theoretical setting, methodologies employed, and goals of an account. The resulting classifications vary from very rich and detailed sets of verb classes and complement clause types to more reduced and abstract inventories. These differences, among others, reflect the tension between universality and variation: coarse classifications with broader classes and hence fewer categories as in Cristofaro (2005), Ramchand & Svenonius (2014) or Wurmbrand & Lohninger (2019) apply to more languages and are easier to observe crosslinguistically; rich and fine-grained classifications as in Givón (1980), Noonan (2007), Dixon (2010) or Cinque (1999, 2004) cover more language-specific properties, but are less suitable to describe crosslinguistic patterns. The first part of this section presents points of variation found across languages as well as differences in the theoretical conclusions. In the second part, we show that there

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are nevertheless also many encouraging similarities which can be extracted from the different approaches.

3.1 Variation and differences

The works on complementation have brought to light a wealth of interesting facts, observations, and generalizations. Unfortunately, not all of the patterns observed are uniform across languages and many only express crosslinguistic tendencies. Natural languages often behave heterogeneously when it comes to clausal complementation, and especially morphosyntactic properties may show language-specific idiosyncrasies and crosslinguistic variation. Thus, building a framework covering complementation universally is a difficult task and, as seen in this chapter, can have different outcomes.

Two core concepts found in most works on complementation are T/A/M-marking and (in)dependence of the embedded subject, however, accounts differ regarding the significance of these concepts in shaping a theory of complementation. To illustrate, Givón (1980) for example provides a classification based on subject dependence, whereas Noonan (2007) builds an account based on T/A/M-marking and time-reference. The two approaches lead to complement clause systems which not only differ in their number of complement clause types but also in their distribution. The third main concept of most accounts is that complement clauses differ in the degree of clausehood – complements may be full clauses or be reduced in some way. However, as we will see, there is no agreement on how notions such as ‘clausehood’ or ‘reduced clauses’ are defined. Since complementation is an interplay of clausehood, semantics (of the matrix verb and/or the complement clause) and morphosyntactic coding, a multitude of analytical options arises, depending on the definitions of these concepts and their interactions. In what follows, we summarize differences and issues arising for T/A/M-marking, subject (in)dependence, the definition of clausehood, and the classifications of matrix verbs. We conclude that T/A/M-marking restrictions and (in)dependence properties of the embedded subject are not sufficient to define complement clause types, and neither does there seem to exist a uniform definition of (reduced) clausehood that applies across frameworks. Morphosyntactic marking and subject dependence show certain tendencies in the distribution of complement clauses, but are not rigidly aligned with different complement types or the semantics of complementation configurations.

T/A/M-marking plays a core role in many typologies and approaches to complementation. The concept of *reduced clausehood* overlaps, and is sometimes even seen as equivalent, with the notion of reduction of T/A/M-marking. Givón

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(1980) claims that the higher a complement clause is on the *Binding Hierarchy*, the more reduced its T/A/M-possibilities are. Dixon (2010) uses T/A/M-marking as a defining criterion for complement clause types: one class has full, one reduced and one no T/A/M-possibilities. Similarly, Cristofaro (2005) proposes that complement clauses either have independent, predetermined or absent T/A/M-specifications, rendering three complement clause options. Lastly, Noonan (2007) employs finiteness (although he does not use that specific term) as a defining criterion: fully specified indicatives or partly specified subjunctive form s-like complements, whereas infinitives are reduced complement clauses.

While the resulting three-way distinctions suggested in the different accounts (clauses with full T/A/M-marking, some T/A/M-marking and no or very little T/A/M-marking) are suggestive and seem to present a clear crosslinguistic tendency, reduction in T/A/M-specifications is not a universal property of complement clauses. It is not always the case that languages have different T/A/M-possibilities for different types of complements. For example, in languages like Greek, Bulgarian, and Macedonian (and many others), all complement clauses are finite and none of them has reduced T/A/M-marking, as illustrated for Bulgarian in (19) where even *event* clauses exhibit full T/A/M-marking. While T/A/M-marking does not necessarily distinguish between different clause types, the existence of ICH-categories, however, can nevertheless be observed by the distribution of clause introducers (see Wurmbrand & Lohninger 2019, Wurmbrand et al. 2020).

- (19) Bulgarian (Wurmbrand et al. 2020: 122)
 Lea se opitvaše da čete kniga.
 Lea REFL try.PRF.3SG DA read.PRS.3SG book
 ‘Lea tried to read a book.’

A related controversy concerns the concept of ‘finiteness’ itself. Due to the extensive crosslinguistic variation in the distribution of finiteness in complement clauses, many works have concluded that there is no single morphosyntactic definition of finiteness, nor a single semantic function associated with it (see e.g., Cristofaro 2007, Bisang 2007, Nikolaeva 2007b). Among the categories that have been suggested to reflect finiteness in different languages are tense, aspect, mood, illocutionary force, person marking, politeness, special forms which cannot be used in independent clauses, and/or nominal morphology on the verb (see the works in Nikolaeva 2007a). Although finiteness often reflects different degrees of clausehood, there is no specific feature or property of finiteness that applies crosslinguistically and that could be used to classify different types of complements.

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Thus, using only morphosyntactic criteria to divide complement clauses into different classes does not yield a uniform picture typologically as there is a significant amount of variation in the distribution of morphosyntactic coding of complement clauses. Languages may or may not use morphosyntactic strategies to mark different types of clauses, and even for languages that do display differences, the strategies are diverse and not uniform. Although reduced T/A/M-possibilities are a valid, efficient way to distinguish between complement clause types, they are largely language-specific and most likely only the surface reflection of a deeper, more abstract division of complement clause types. What can be concluded, however, is that if a language makes use of T/A/M-reduced complement clauses, their degree of reduction aligns with complementation hierarchies as developed in the different accounts.

The second main characterizing property of different types of complement clauses is the *independence of the embedded subject*. Noonan (2007) suggests that raising and equi-deletion are both instantiations of dependent subjects, which yield non-s-like complement clauses. Whereas infinitives lack a subject, finite (s-like) complement clauses always have one. Having no subject is thus equivalent to clauses being reduced. Givón (1980)'s *Binding Hierarchy* is based on a (functional) dependence between the embedded and matrix subjects and integration of the embedded clause depends on the independence of its subject. Dixon (2010), too, uses subject dependence as defining criterion for complement clause classes, and Cristofaro (2005) argues that the determination of the embedded subject is fully dependent on the matrix predicate.

Most approaches relate subject dependence to clause reduction – if the subject is dependent on an argument of the matrix predicate, the embedded clause is reduced in some form and tends to show less specific T/A/M-marking. However, similar to T/A/M-reduction, subject-dependence is not a universal indication of clause reduction, nor a reliable tool to distinguish between complement clause types crosslinguistically. While it is common that certain types of complements require an interpretation in which the embedded subject is identical to the matrix subject (or lack a subject even semantically), the morphosyntactic instantiations associated with subject (in)dependence (case marking, agreement, control or equi-deletion, ECM, raising to subject or object) show wide variation. Phenomena such as raising to subject or object, or ECM, may indicate clause reduction in some languages since they are only possible with infinitival complements (Noonan 2007, Givón 1980), but this is not the case universally. Raising to subject or object from a finite (i.e., non-reduced) complement clause, also known as *Hyperraising* is found in several languages (see Wurmbrand & Lohninger 2019 and

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Lohninger et al. 2022 for an extensive overview), and examples like the Brazilian Portuguese (20) contradict the claim that subject raising always renders a non-s-like reduced clause.

- (20) Brazilian Portuguese (Martins & Nunes 2010: 145)
Os meninos parecem [que viajaram ontem]
 the boys seem.3.PL [that traveled.3.PL yesterday]
 ‘The boys seem to have traveled yesterday.’

Lastly, even control/equi-deletion has been shown not to be restricted to infinitives but be possible in finite non-reduced complements as well (see Landau 2004 for an overview). Thus, subject dependence is often linked to clause reduction, but it is only a tendency. Similar to T/A/M-marking, it is a way for individual languages to distinguish between their complement clause types, but it is not a universal tool.

Although most approaches to complementation employ the concept of clausehood, even the mere *definition of clause(hood)* is not uniform. Basic questions – such as how to define a clause, what a complement clause is and what does not count as a clause anymore, or where the threshold between monoclausal and biclausal structures is – often remain open or a gray area. Givón (1980) and Noonan (2007), for example, include nominalizations into the set of complement clauses, whereas Dixon (2010) does not. Noonan (2007) includes serial verb constructions into his complementation patterns, but Dixon (2010) considers them as different complementation strategies, set aside from complement clauses. Givón (1980) treats clause union and predicate raising as involving a complement clause construction, whereas Noonan (2007), Dixon (2010), and Cinque (2004) analyze the two verbal predicates as constituting a single clause.

The diversity of the definitions of “clause” carry over to notions built on that concept, such as *monoclausal* and *biclausal*. Cinque (2004), for example, distinguishes between lexical and functional verbs, the former occurring in biclausal complementation configurations, the latter in monoclausal configurations. Functional verbs (such as modal, aspectual and motion verbs) are assumed to be located in the functional spine of a clause, and the “complement” is the main lexical predicate of the clause (i.e., there is only one lexical VP). Such functional restructuring thus creates a monoclausal and monopredicate configuration which, for all intents and purposes, behaves like a simple clause with a single predicate. This view, and a binary clausehood distinction in general, is challenged in Wurmbrand (2001, 2004), Wurmbrand & Lohninger (2019), where it is argued that monoclausality comes in several forms and degrees: functional restructuring, lexical

restructuring lacking TP and CP domains, and lexical restructuring lacking only the CP domain.

Furthermore, the works on complementation have brought out that the distinction between “functional” and “lexical” verbal elements is not sufficient, and that between the two there is a class which language-specifically patterns with one or the other side. Dixon (2010) observes that some verbal elements referred to as secondary semantic types (among them *can*, *must*, *begin*, *try*, *want*, *hope*, *plan*, *make*, *help*) behave like functional elements in some languages, but as lexical elements with certain functional behavior in other languages. Noonan (2007) describes the same phenomenon as *clause union* where the matrix predicate (usually a manipulative verb) is unified with the embedded predicate, creating a monoclausal structure where the matrix verb becomes a kind of functional appendix to the embedded lexical verb. Finally, Wurmbrand (2001) and Wurmbrand & Lohninger (2019) observe that, in addition to fully lexical and fully functional verbal categories, there are also semi-lexical/semi-functional verbs (e.g., causatives and perception verbs), which occur as part of the functional clausal spine but nevertheless show thematic properties usually reserved to lexical verbs.

Given the diversity of the theoretical frameworks investigating complementation, the tools, strategies, and grammatical primitives of the different approaches diverge significantly, and as a result, the definitions of **matrix verb classes** that trigger complementation also show some degree of variation. All accounts observe that there are different semantic classes of matrix predicates taking complement classes, however, the number of classes singled out varies. Some accounts give coarser-grained classifications, others more detailed ones, depending on the empirical scope, as well as theoretical focus of the different studies. Givón (1980) bases the classification on the emotional commitment of the matrix agent towards the complement proposition and distinguishes six types (*implicative*, *strong attempt*, *high emotive*, *low emotive*, *attitude*, *epistemic*). Noonan (2007) proposes a finer-grained division into 13 classes based on the meaning of the matrix predicates (*utterance*, *propositional attitude*, *pretence*, *commentative*, *knowledge*, *fearing*, *desiderative*, *manipulative*, *modal*, *achievement*, *phasal*, *immediate perception*, *conjunctive*). Cristofaro (2005) adopts these 13 types but further groups them into seven classes (*modal*, *phasal*, *desiderative*, *manipulative*, *perception*, *knowledge*, *utterance*). Dixon (2010) distinguishes five types of semantic classes of lexical verbs (*attention*, *thinking*, *deciding*, *liking*, *speaking*) and three types of secondary semantic classes involving semi-lexical/functional verbs. Finally, Wurmbrand & Lohninger (2019) suggest that there are only three general classes of matrix predicates, defined by the predicate’s and complement clause’s

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semantics (*proposition, situation, event*), with possible further language-specific subdivisions.

At a more general level, the framework diversity is reflected in the mechanisms and tightness of the *linking between semantics and morphosyntax*. Most accounts propose that the semantics of the matrix predicate and the morphosyntactic form of the complement clause are more or less tightly linked. Givón (1980), Cristofaro (2005) or cartographic approaches suggest that certain matrix predicates only combine with complement clauses of a specific morphosyntactic form (or clausal layer), i.e., that the linkage between the two is rather rigid. A looser connection is assumed in Wurmbrand & Lohninger (2019) where it is proposed that there are minimal matching requirements for the (morpho)syntactic computation of different types of complement clauses, but mismatches are possible as well, making the semantics–morphosyntax mapping relative and not absolute. Crosslinguistically, morphosyntactic properties align along a complementation hierarchy, but not in a rigid one-to-one fashion (see also Cristofaro 2005).

3.2 Universalities

Given the fundamental differences in frameworks and variation across languages, the question of what the common properties of complementation are and what theoretical observations and conclusions are shared among accounts can only be answered at an abstract level by setting aside many details. Broadly speaking, the properties that play essential roles in different accounts are morphosyntactic (usually T/A/M) marking distinctions of the embedded predicate, (in)dependence of the embedded subject, the distribution and restrictions of embedded time-reference, and the semantics of the matrix predicate. All accounts also assume some, possibly relative, linking relation between the morphosyntactic properties of the embedded clause and the meaning of the matrix predicate. As we have seen in the previous sections, however, the weight and relevance of these properties varies in different approaches.

In this section, we summarize some of the similarities in clausal complementation. Abstractly, the generalizations that hold across languages and frameworks are: i) the combination of different semantic classes of matrix predicates with different types of complement clauses typically yield three *complementation configurations*; ii) the semantics of the matrix predicate interacts with the semantics and (to some degree) the morphosyntax of the complement clause in a predictable manner; iii) the different complement types are in some sort of hierarchical relationship.

All accounts observe that matrix predicates form (more or less fine-grained) groups based on their meaning, which is related to the semantics of the embedded clause. By combining different classes of matrix verbs and different semantic and morphosyntactic types of complement clauses, generally three types of complementation configurations arise. Dixon (2010) classifies complement clauses into three classes in the first place and then divides matrix predicates depending on what complement clause class they accept. In Wurmbrand & Lohninger (2019), a three-way classification is suggested based on the meaning of the entire complementation configuration (the combination of the complement clause with the matrix predicate). Remarkably, the three classes can also be observed in accounts with more detailed classification systems of verbs and/or morphosyntactic coding options. Combining, for example, Noonan (2007)'s 13 matrix predicate classes with the two complement clause types proposed there (s-like and non-s-like), three classes emerge: matrix predicates which take s-like complements, those which take non-s-like ones and those which take both. Similarly, in Cristofaro (2005), matrix verbs are divided into seven classes, and complement clauses into four levels. However, if combined, only three classes emerge: those verbs taking proposition-level complement clauses, those taking predication-level complement clauses and those taking predicate- and predication-level clauses. Thus, viewed from the crosslinguistic perspective, focusing on the combination of matrix verbs and embedded clauses, which Wurmbrand & Lohninger (2019)'s synthesis approach decidedly does, provides the crucial key to understanding a deep similarity across languages, despite the extensive variation found in the details.

An interesting phenomenon in this context, which is mentioned in most accounts, is the possibility of double membership of certain verbs. Since matrix verb classifications are based on the semantics of a verb, verbs with more than one meaning are assigned to different classes, depending on the interpretation used. A generalization observed in Wurmbrand & Lohninger (2019) and summarized under our synthesis approach (see above) is that when a verb has alternating meanings, the morphosyntactic coding of the complement clause tends to correspond to different meanings (as for example with English *tell* which shifts between a speech verb (indicative complement) and a command (infinitival complement)).¹⁸

¹⁸ A strict form–meaning mapping is, however, not always the case. There are also predicates like (*to be glad*) which take different types of complement clauses (finite or non-finite), but do not shift their meaning. This challenges the view that verbs that can occur in two different morphosyntactic frames always come as two lexical entries.

- (i) a. I'm glad to be teaching at the university.

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Wurmbrand & Lohninger (2019) suggest that such patterns indicate that there is mutual influence between the matrix predicate and the embedded clause. It is not solely the matrix predicate that determines the properties of the complement clause, but the complement clause can also affect the meaning of the matrix predicate.

A second observation that applies crosslinguistically is that the meaning of the matrix verb and the meaning (and to some extent also the form) of the complement clause are not independent of each other but work together to form a well-formed complementation configuration. Matrix predicates and complement clauses do not combine in an “anything goes” fashion, but there are restrictions, typically definable only implicationally. These restrictions consist of the meaning (and possibly other selectional restrictions) of the matrix predicate, the meaning of the complement clause, and, usually connected to the latter, the morphosyntactic coding of the complement clause.

Independently of the number of classes distinguished for matrix predicates and complement clauses, one property that is found in various instantiations in most works on complementation is the time-dependence between matrix predicates and embedded clauses. Complement clause types differ in how dependent their time-reference is in any given complementation configuration – some complements can be independent in that they can be interpreted as before, after or simultaneously with the matrix predicate; others receive an independent time reference, however, one with a predetermined tense value; yet others are fully dependent on the matrix time and have no time reference on their own. Noonan (2007), for instance, divides complement clauses into independent time-reference (ITR) and dependent time-reference (DTR), which lead to three combinatorial possibilities: verbs that combine only with DTR complements, only with ITR complements, or with either DTR or ITR. Cristofaro (2005) and Dixon (2010) propose that there are three options for embedded time-reference: no determination, future predetermination, and same time as matrix. According to those authors, predetermination of time-reference solely depends on the meaning of the matrix verb. Lastly, Wurmbrand & Lohninger (2019) also propose a similar three-way split (independent time-reference, embedded future or irrealis orientation, and no time-reference), but do not assume that the determination comes just from the matrix predicate. For a successful complementation configuration, the time restrictions and specifications of both the matrix and embedded clauses must match. If there is a mismatch, the combination is either impossible, or, in certain cases the meaning may be shifted to accommodate the mismatch.

b. I'm glad that I'm teaching at the university.

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The last uniformity emerging across accounts of complementation is the observation that different types of complementation stand in a (usually implicational) hierarchical relation to each other. Givón (1980) develops the *Binding Hierarchy*, Cristofaro (2005) the *Hierarchy of Complement Clause Deranking*, Cinque (1999, 2004) the *Functional Hierarchy*, and Wurmbrand & Lohninger (2019) the *Implicational Complementation Hierarchy*. Despite differences in terminology and details, the common conclusion is that there is an implicational, scalar hierarchy which orders complementation types along a semantically determined scale. The hierarchies are typically also ordered along notions of clausehood such as (in)dependence and integration of the complement, and play a direct or indirect role in the mapping of semantic categories to morphosyntactic types of complements.

When looking at complementation configurations, two relevant restrictions emerge across all accounts: i) the matrix predicate and the embedded clause must be semantically compatible, ii) a complement clause needs to be seen in combination with its matrix predicate, and iii) the semantics of the complement clause is linked to its morphosyntax, however not fully necessarily in a one-to-one fashion. As mentioned above, how this linkage is derived differs between functional–typological accounts (where extragrammatical functions motivate structure) and structural–grammatical accounts (where selectional restrictions or synthesis does), however, the core insight, namely that there is a linkage, is similar across frameworks. Abstracting away from specific definitions, but focusing on the similarities, the common insight about complementation configurations crosslinguistically can be presented as in Table 9.

Table 9: Hierarchy of Complementation

Most independent	<i>Type 1 > Type 2 > Type 3</i>	Least independent
Least integrated		Most integrated

Properties associated with (in)dependence and integration may differ across languages and frameworks, but the implicational nature can be seen as a true universal. Once a dependency/integration property is defined, the systems predict that Type 3 configurations can never be more independent and less integrated than Type 2 configurations, which in turn can never be more independent and less integrated than Type 1 configurations. Morphosyntactic properties associated with (in)dependence and integration (such as T/A/M-marking, complementizers, incorporation) align along the hierarchy but are not rigidly mapped to

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specific categories (although there are crosslinguistic tendencies). This (abstract) hierarchy is, in our view, encouraging – despite the vastly different frameworks and methodologies used to understand complementation, a shared concept has crystallized which can be employed and developed in future approaches.

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Chapter 4

The typology of relative and adverbial clauses

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The present contribution surveys prominent patterns in the typology of relative and adverbial clauses, with particular reference to the notion of embedding. To this end, we unfold the notion of embedding into a three-dimensional space consisting of a functional, a syntagmatic and a formal axis along which a clause may be argued to be embedded in a (constituent of a) matrix clause. We show how these dimensions of embedding interact with each other to yield cross-linguistically recurrent subtypes of relative and adverbial clauses. In doing so, we also discuss the processing and diachronic dynamics of clausal embedding in these domains as well as correlations between the degree of embedding and other grammatical properties of the subordinate clause, such as accessibility to relativization or the reduction of overt markers of the embedding relationship.

1 Adjunct clauses and the notion of embedding

1.1 Goal and structure of the paper

The present chapter is concerned with clauses that function as adjuncts within a matrix clause of a complex sentence. As such, they typically provide further information on a certain element of the matrix clause, but they are not an argument of the main-clause predicator. The traditional classification of adjunct



clauses rests on whether they relate to a nominal of the matrix clause (= prototypical relative clauses) or the predicate or the entire proposition expressed by the main clause (= prototypical adverbial clauses). Accordingly, our goal in the present chapter is to provide an overview of prominent cross-linguistic patterns of relative-clause and adverbial-clause formation.¹

As it proves impossible for a single article to do justice to the enormous amount of typological research on these clause types that has accumulated over the last decades, we will pursue a narrower, more focussed agenda here. Specifically, we will attempt to systematize the typology of adjunct clauses from the central perspective of the present handbook, that of embedding. To this end, we begin the paper (in Section 1.2) by unfolding the notion of embedding into a three-dimensional space consisting of a functional, a syntagmatic and a formal axis along which a clause may be argued to be embedded into a (constituent of a) matrix clause. In doing so, we also intend to lay the terminological groundwork for all subsequent sections. The ensuing Sections 2 and 3 will then be concerned with the typology of relative and adverbial clauses, respectively, before Section 4 concludes the paper by situating commonalities between the two domains in a dynamic view of clausal embedding and grammatical structure more generally.

1.2 Dimensions and reflexes of embedding

In Lehmann's (1988) typology of clause linkage, embedding is the endpoint of a continuum of "hierarchical downgrading", a situation where a clause comes to behave as a "well-defined constituent" (ibid.: 184) of another unit, i.e. of a clause or even a phrase. In the present paper, we understand this to mean that a clause is *functionally*, *syntagmatically* and *formally* integrated with that other unit. In the following subsections, we introduce each of these three dimensions and begin to carve out their relevance for the typology of relative and adverbial clauses.

1.2.1 The functional dimension

A clause is fully integrated from a *functional* point of view if it has a specific semantic and syntactic function inside another unit. This is arguably the case when

¹For conceptual clarity, we want to point out that we understand a MATRIX CLAUSE to be a superordinate clause in a complex sentence in which a subordinate clause has a dedicated syntactic function (more on this below). The term MAIN CLAUSE, by contrast, is here used to refer to the unit that remains of the matrix clause when the subordinate clause is removed. Using this terminological distinction, we would say, for example, that a given instance of an adverbial clause is part of a matrix clause and that it either precedes or follows its associated main clause.

4 The typology of relative and adverbial clauses

a clause “elaborates a site” (Langacker 1987: 304) or fills a gap (Lehmann 1983, Diessel 2019b: Ch. 9) projected by a relational element of the main clause, i.e. by a verbal, nominal or adjectival predicator in the main clause. Accordingly, such subordinate clauses are known as ARGUMENT CLAUSES. NON-ARGUMENT CLAUSES, by contrast, are not projected in the same way. This holds for all of the following clauses, taken from typologically diverse languages (a list of glosses is provided at the end of the paper):

- (1) relative clause from Persian (Indo-Iranian: Iran; Mahootian 1997: 33)
 [*Film-i-(ro)* [*ke hæme dust+dašt-ænd*]] *næ-did-im*.
 film-DEM-(OBJ) REL all friend+have-3PL NEG-saw-1PL
 ‘We didn’t see the film that everyone liked.’
- (2) adverbial clause from Noon (Atlantic-Congo, North-Central Atlantic, Cangin: Senegal; Soukka 2000: 278)
 [*Balaa mi le’ kaad-aa*], *mi híd-oh-ha na húnísmun*.
 before 1SG arrive house-IRR 1SG meet-RECP-NARR with friend
 ‘Before I arrived home, I met with a friend.’
- (3) clause chaining from Mauwake (Nuclear Trans New Guinea, Madang: Papua New Guinea; Berghäll 2015: 335)
 [*Nainiw ekap-ep*] *maa me sesenar-e-mik*.
 again come-SS.SEQ food not sell-PST-1/3PL
 ‘They did not come back and sell food.’

In the literature, it is thus said that the presence of the dependent clauses in (1–3) is not specifically licensed by a predicator. Since it has been pointed out, however, that the difference between arguments and non-arguments is gradient rather than categorical (see, e.g., Deutscher 2000 on Akkadian), we prefer the term *projection* to the more categorical *licensing* here. We will see later on that some relative and adverbial clauses are more closely associated with, or more strongly projected by, their head element than others, with interesting typological consequences (see Sections 3.2–3.3).

Despite not being projected by a main-clause element, non-argument clauses can still fulfil a semantic and syntactic role in another unit and thus be functionally embedded in that unit. This is what happens most clearly in (1) and (2) above. In (1), the dependent clause functions as an attributive modifier of a nominal, and hence has a dedicated function inside an NP; in (2), the dependent clause spells out part of the setting of the main-clause situation and thus functions just like

a phrasal constituent with similar semantics. In both cases, then, the dependent clause acts as a modifier of an element in the matrix clause. This contrasts with the situation in (3), where we see a succession of events in which the formally dependent clause – the one lacking tense and person inflection on the verb – cannot be said to function as a modifier of the final finite clause. The grammatical construction in (3) has aptly been called *CLAUSE CHAINING* in the typological literature, best-known from (but not limited to) languages of Papua New Guinea (see, e.g., Roberts 1988 for an overview). The dependent clauses in such chains have been argued not to have a syntactic function in a larger clause and hence not to be embedded within it (see van Valin 1984, 2005: Ch. 6, who uses the term *COSUBORDINATION* for this phenomenon).² Accordingly, while the typological domain of “non-argument clauses” also includes dependent clauses like the one in (3), the latter are not functionally embedded like the dependent clauses in (1–2). Since (1) and (2) fulfil the syntactic function of adjunct (to a nominal and a clause, respectively), we refer to them as *ADJUNCT CLAUSES*, and this is the domain that the present paper seeks to investigate from a typological perspective.

It is widely recognized that adjuncts are extremely heterogeneous themselves (see, e.g., Maienborn & Schäfer 2011 for a survey). A coarse-grained but useful distinction is that between adjuncts that modify the descriptive meaning of an element of the matrix clause (= *MODIFIERS*) and adjuncts that provide additional comment on the element in question (= *SUPPLEMENTS*). Modifiers of a nominal element typically restrict the referential potential of the NP headed by that nominal and hence go by the name of *RELATIVE CLAUSES*; modifiers that restrict a situation to a specific setting (time, place and manner) or specific contingent circumstances (condition, cause, purpose, result, etc.) are traditionally known as *ADVERBIAL CLAUSES*. They are often said to spell out “peripheral” information on the “core” expressed by the main-clause proposition (e.g. in van Valin & LaPolla 1997: 26’s terms).

Adjuncts that provide additional comment are called supplements here (extending a descriptive term used by Huddleston & Pullum 2002: 59 for English). Relative clauses can be used to express supplementary information on both an

²The term “cosubordination” is supposed to capture the fact that, in the languages in question, clause-chaining constructions are grammatically distinct from both coordinate and subordinate clauses. Unlike subordinate (and like coordinate) clauses, they do not fulfil a syntactic function in another clause and rather encode sequential (and often non-backgrounded) events in a narrative; unlike coordinate clauses (and like subordinate clauses in these languages), they are grammatically dependent on another clause, i.e. the final one in the chain, notably for the interpretation of tense, aspect and mood operators. The dependent clauses in such chains are also called ‘medial clauses’.

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NP in the matrix clause (4) and a whole clause (5); they are also known as NON-RESTRICTIVE, SUPPLEMENTARY and – in the case of (5) – SENTENTIAL relative clauses. In many languages, the clausal constructions used as adverbial modifiers can also be employed as adverbial supplements (see also Haegeman et al. in the present volume). In this function, they often provide the speaker's attitude towards the main-clause proposition or relate to the speech act (rather than the propositional content) expressed by the main clause (6–8):

- (4) supplementary relative clause (with a nominal antecedent) from Basque (isolate: Spain, France; Hualde & de Urbina 2003: 804)

Gure Jainkoak, [*beti xuxen dabilana*], *ongi gidatzen baitu*.
our god.ERG always right acts.SUB.DET well guide.IMP SUB.AUX
'Because our God, who always acts rightly, guides well.'

- (5) supplementary relative clause (with a sentential antecedent) from Avatime (Atlantic-Congo, Volta-Congo, Kwa Volta-Congo: Ghana; van Putten 2014: 85)

Áà-gu kpɛ [*gì lí-tá-bubɔ*].
C1:SG.POT-speak put.in REL C3:SG.NEG-INT-easy
'He will be speaking into it, which will not be easy.'

- (6) supplementary (conditional) adverbial clauses from English (Huddleston & Pullum 2002: 774)

- a. *Dick is coming to the party*, [*in case you're interested*].
- b. [*If you must know*], *I wasn't even shortlisted*.

- (7) supplementary (purposive) adverbial clause from Yakkha (Sino-Tibetan, Himalayish, Kiranti: India, Nepal; Schackow 2015: 450)

[*Yeppa cok-ma bhoŋ*] *i-ha-ca im-ma por-a*
true do-INF COND/PURP what-NMLZ.NSG-ADD buy-INF must-NATIV
n-joŋ-me-ŋa-n.
NEG-do-NPST-1SG-NEG

'To be honest/If I'm honest, I do not have to buy anything.'

- (8) supplementary (concessive) adverbial clause from Georgian (Kartvelian: Georgia; Kojima 2018: 436)

Ak sač'mel-i=a, [*tumca es šen i-c-i*].
here food-NOM=be.PRS.3SG although this.NOM 2SG.ERG i-know-THEM
'There is food here, although you know this.'

1.2.2 The syntagmatic dimension

There are several ways in which a clause can be more or less *syntagmatically* integrated into another unit. First, the more internally coherent the unit in question, the more likely it is to be considered a constituent of a higher unit. Coherence is achieved by keeping the constituting elements of the unit in linear adjacency (Behaghel 1932); the opposite pattern is to separate these elements from one another (see e.g. Hale 1983 for classic examples). Cross-linguistically, adverbial clauses seem averse to being split up by their main clause, even in languages that otherwise license “discontinuous constituents” (see, e.g., Reinholtz 1999 on Swampy Cree). If adverbial clauses are interrupted at all, it is usually by further adverbial (or other dependent) clauses, as in (9):

- (9) Koyra Chiini (Songhay: Mali; Heath 1999: 281)
 [*jaa* [*nda baana kar*], *ganji-ije di yo o ñin*]
 since if rain hit forest-child DEF PL IMPF drink
 ‘because, if rain falls, the wild animals will drink’

Discontinuities are more common in relativization, when the relative clause is separated from the head nominal it is meant to modify. This so-called *EXTRAPOSITION* from NP is illustrated in (10):³

- (10) Slave (Athabaskan-Eyak-Tlingit, Northern Athabaskan: Canada; Rice 1989: 1327)
John łi wehk’é [ʔeyi ts’ódani kayihk’a yilé i].
 John dog 3.shot the child 3.bit PST REL/NMLZ
 ‘John shot the dog that bit the child.’

This leads us to the position of adjunct clauses more generally, the second and more important indication of their syntagmatic integration with the main clause: the more tightly a dependent clause can be joined to the main clause, the clearer its status as a constituent of the matrix clause and hence its degree of embedding. Along this continuum, we find dependent clauses that (i) are able to disrupt the

³One could argue that a similar pattern arises when adverbial clauses are represented by a correlate in the main clause, as in German *Ich habe es deshalb gesagt, weil es mir wichtig war* ‘I said it (therefore) because it was important to me.’ We consider these different from extraposed relative clauses precisely because the relative clause in (10) modifies its head nominal and thereby restricts the reference of ‘dog’, while the relationship between an adverbial clause and its correlate is not one of modification but of co-reference, with the correlate being a mere placeholder for the adverbial clause.

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core predicate-argument structure of the main clause (CENTRE-EMBEDDING, as in (11)), (ii) attach to the left or right of the this core (12), (iii) are relegated to more marginal positions of the matrix clause, such as strictly before or after other adverbial modifiers of the core (13), or (iv) are removed from the main clause into a left- or right-detached position (14). The latter pattern typically goes hand in hand with the intonational separation of the two clauses (see also van Valin 2005: §6.2 for discussion of these positional types):

- (11) Urarina (isolate: Peru; Olawsky 2006: 690)

Nii hetau=te [ahe-uru-a=ne hana] turu-a ku-e.
 that HRS=FOC get.drunk-PL-3.DF=SUB when arrive-NTR go-3.EF
 ‘That [man] arrived when they [the people] were getting drunk.’

- (12) English

Anna was angry [after she left the party] because of Mike’s sexist jokes.

- (13) Martuthunira (Pama-Nyungan, Pilbara: Australia; Dench 1995: 252)

Ngayu kartungu parla-marta purra-rninyji pal.ya-a, [
 1SG.NOM 2SG.ACC stone-PROP hit-FUT temple-ACC
pungka-waa-rru].
 fall-PURP.S=P-now
 ‘I’ll hit you in the temple with a stone, so that you fall down.’

- (14) Supyire (Atlantic-Congo, North Volta-Congo, Senufo: Burkina Faso; Carlson 1994: 492)

[Myàhíí u a cèè gé], ci náhá mii fúnjí í.
 song.DEF.3PL she PRF sing REL 3PL be.here my inside in
 ‘The songs which she sang, they are here inside me.’
 (‘I remember the songs which she sang.’)

In (13), we see a typical representative of what has been called an “adjoined clause” (Hale 1975) in Australian languages, a dependent clause that precedes or follows the main clause, may be “bound intonationally” to it (e.g. Nordlinger 1998: 217 on Wambaya), but is not normally found inside it. In (14), finally, the relative clause is found in a left-detached position.⁴ As Carlson (1994: 488) states, “relative clauses in Supyire are unembedded,” and they are “typically followed

⁴For the sake of clarity, it should be pointed out that this relative clause is of the internally-headed type (see Section 2.2.1 below), so that it is the whole unit in brackets that is left-detached from the main clause here.

by a short pause before the main clause.” In contrast to the relative clause in (10) above, constructions like the one in (14) are not due to an option of extraposition; they standardly occur in a left- or right-adjoined position and are not normally inserted between elements of the main clause (for further discussion, see Section 2.2.2 below).

Examples (11) to (14), then, illustrate a continuum from clearly integrated to clearly non-integrated adjunct clauses. As we shall see, this parameter is a crucial typological variable that distinguishes different construction types of relative and adverbial clauses cross-linguistically.

1.2.3 The formal dimension

When clauses are functionally and syntagmatically integrated into another unit, it is common for them to also approximate the form of the most typical syntactic constituent, i.e. a phrase. Haiman & Thompson (1984) and Lehmann (1988) introduce a wide range of dimensions that reflect the “decategorization” (Malchukov 2004) of a clause and its “recategorization” as a phrasal unit, a phenomenon also known in typology as *DESENTENTIALIZATION* (Lehmann 1988) or *DERANKING* (Cristofaro 2003, building on Stassen 1985). In the present paper, we will use the term *deranking* to refer to this dimension of clausal embedding.

Among the hallmarks of *deranking* are the restriction (or the lack) of clausal and sentential operators (illocutionary-force marking (see Verstraete 2007, Bickel 2010 for specific parameters), tense-aspect-mood marking, negation), the use of dependent verb forms, and the reduction and/or special coding of the arguments and modifiers. Dependent verb forms that are specifically associated with relativization have been called *PARTICIPLES* (see Shagal 2019), while dedicated adverbial verb forms have been labelled *CONVERBS* in the typological literature (Haspelmath 1995, Nedjalkov 1998), although it must be stated that there is considerable diachronic interaction and hence synchronic overlap between participles, converbs and nominalizations (Ylikoski 2003; on nominalization, see also Alexiadou and Kornfilt in the present volume). In (15–20) below, we illustrate different degrees and facets of *deranking* in adjunct clauses.

- (15) weakly deranked concessive clause: internal structure as in independent clauses, except for a clause-initial subordinator
 Tz’utujil (Mayan: Guatemala; Dayley 1985: 371)
[Maanaan xtipeeti ja Aa Lu?], majun nuub’an.
 although 3ABS.might.come the youth Pedro nothing 3ABS.3ERG.do
 ‘Even though Pedro might come, he won’t do anything.’

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- (16) weakly deranked relative clause: no subordinator, internal structure as in independent clauses, except for reduced person indexation
Musqueam (Salishan: Canada; Suttles 2004: 76)

kʷəθə swəj̥qəʔ [céw-ət-Samx-Ø ceʔ]

ART man help-TR-me-Ø FUT

‘the man who will help me’

(compare *céw-ət-S(amx)-əs ceʔ* = help-TR-me-3TR FUT = ‘He will help me.’)

- (17) more strongly deranked temporal/causal clause: subordinating conjunction, non-canonical subject coding (zero Accusative instead of overt Nominative case), dependent verb form (phonologically reduced), but retention of tense-aspect inflections

Wappo (Yuki-Wappo: USA; Thompson et al. 2006: 155)

Kʼanihtučʼm-i naleʔ-iš-khiʔ [kʼəšu-ø pulu:mek-ta wen].

chief-NOM angry-INCH-STAT deer-ACC run:away-PST.DEP when

‘The chief got angry when/because the deer ran away.’

- (18) strongly deranked temporal clause: semantically specific subordinating suffix *-ka* attached to a bare verb stem (*aka* = converb), reduction of TAM and person inflection

Awa Pit (Barbacoan: Colombia; Curnow 1997: 272)

[*Santos a-ka*]=na, kula-ta-w.

Santos come-when=TOP hide-PST-LOCUT:SBJ

‘When Santos came, I hid.’

- (19) strongly deranked purpose clause: nominalized verb form, reduction of TAM and person expression, dative case marker (= “purposive phrase”)
Konso (Afro-Asiatic, Cushitic: Ethiopia; Mous & Oda 2009: 349)

Innaannó [golpaytá pidd-a-ʼé] urmala-pá aan-é.

brother:1PL.POSS goat buy-NMLZ-DAT market-DEST go-PRF

‘Our brother went to the market in order to buy a goat.’

- (20) strongly deranked relative clause: nominalized verb form, reduction of tense, mood, person-number inflection and lack of separate negative suffixes

Dolakha Newar (Sino-Tibetan, Himalayan: Nepal; Genetti 2007: 390)

[*Am pipāna ye-u*] mi=pen gun?

that veranda come-NMLZ/REL person=PL who

‘Who are the people who came to the veranda?’

In recent typological studies, the gradient nature of deranking has been captured empirically by developing fine-grained structural variables along which language-specific clause-linking constructions may differ (e.g. Cristofaro 2003, Bickel 2010, Schmidtke-Bode 2014, Hetterle 2015, Shagal 2019). This, in turn, allows a more rigorous comparison of the degree to which different types of adjunct clauses (e.g. different adverbial relations) are structurally downgraded and thus approximate the formal make-up of phrases rather than clauses (as in (19) above).

Care must be taken not to interpret deranking as a unidirectional diachronic process. While it is true that a fully-fledged independent clause may come to be structurally reduced or dependent, the opposite pathway from smaller to larger units is also well-known: as outlined by Heine (2009), a phrase may be expanded into a (more) clausal unit. For example, when the NP governed by a benefactive adposition ('for my wife') is replaced by a deverbal noun with its own arguments and modifiers, the result can be a fully deranked adverbial expression of purpose ('for (the) feeding (of) the children'). Such constructions can, in turn, gradually rid themselves of nominal properties and acquire a more clause-like treatment of their arguments and modifiers (e.g. Disterheft & Viti 2010 for infinitives in Indo-European, Fanego's (2004) detailed study of English gerunds). For further remarks on the diachrony of embedding, see also Viti (this volume).

1.2.4 Interim summary

We have introduced three dimensions of clausal embedding that can be seen as three related scales in a typological space, as illustrated in Figure 1.⁵ The idea behind this illustration is that language-specific dependent-clause constructions vary in terms of ...

- how strongly they are associated with (or "projected" by) a relational element in the matrix clause (yielding a range from typical argument clauses to completely independent states-of-affairs as in typical coordinate structures);
- how closely they can be positionally integrated with their associated main clause (ranging from centre-embedded to detached positions);
- how strongly they are deranked in comparison to independent clauses (ranging from fully clausal to fully phrasal structures).

⁵Figure 1 was created by applying the package `scatterplot3d` (Ligges et al. 2018) in R 3.6.3 (Team 2019) to purely fictitious data points.

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The first two criteria are thus concerned with “external” properties of a dependent clause, while the third one relates to its “internal” structure.

A common result across typological studies is that each language-specific construction represents a unique point in a gradient multidimensional space such as Figure 1, but that certain clusters of points emerge as cross-linguistic types of dependent-clause formation. In this spirit, the following sections will show how the three dimensions of embedding combine with one another to yield typologically prominent patterns of relative-clause and adverbial-clause formation.

2 Relative clauses in typological perspective

Loosely speaking, relative clauses (RCs)⁶ are often characterized as clauses which function as noun modifiers. But in order to discuss the cross-linguistic variation in the forms and functions of RCs insightfully, we offer a more technical definition here. A relative clause is a clausal construction that is semantically oriented towards one of its participants (Lehmann 2014: 4), in order to specify an entity of the matrix clause in such a way that the designatum of that entity occupies a participant position in the situation encoded by the relative clause. The participant towards which the relative clause is oriented is known as the RELATIVIZED POSITION.

The matrix-clause entity that is specified by the relative clause is most commonly a NOMINAL (e.g. *old man*) centred on a NOMINAL HEAD (*man*); in these situations, relative clauses are typically used to restrict the referential potential of the NP headed by the nominal head, as in *the old man I saw at the pub last night*, or in our earlier example (1). These relative clauses thus act as modifiers of a nominal and are known as RESTRICTIVE RELATIVE CLAUSES. By contrast, SUPPLEMENTARY RELATIVE CLAUSES specify an NP, the main-clause predicate or the main clause by providing elaborative information on it without restricting its referential potential, as in (4–5) above. In the following overview, we will concern ourselves primarily with the typology of restrictive relative clauses, as most cross-linguistic work has been done on this type of relativization.⁷

⁶The most comprehensive typological analysis of relative clauses is Lehmann (1984), crucial aspects of which are succinctly summarized in various follow-up publications (Lehmann 1986, 2008, 2014). Non-monographic systematizations of the typology of relativization can be found in work by Bernard Comrie (e.g. Comrie 1981, 1998, 2006), in Downing (1978), Keenan (1985), de Vries (2005, 2018), Andrews (2007), and in a number of volumes on (functional) syntax in cross-linguistic perspective (e.g. Dik 1997: Chs. 3–4, Givón 2001: Ch. 14, Dixon 2010: Ch. 17, Croft 2022: Ch. 19). Finally, Hendery (2012) provides a diachronic typology of relative clauses.

⁷In fact, many typological studies take restrictiveness as a crucial part of their comparative

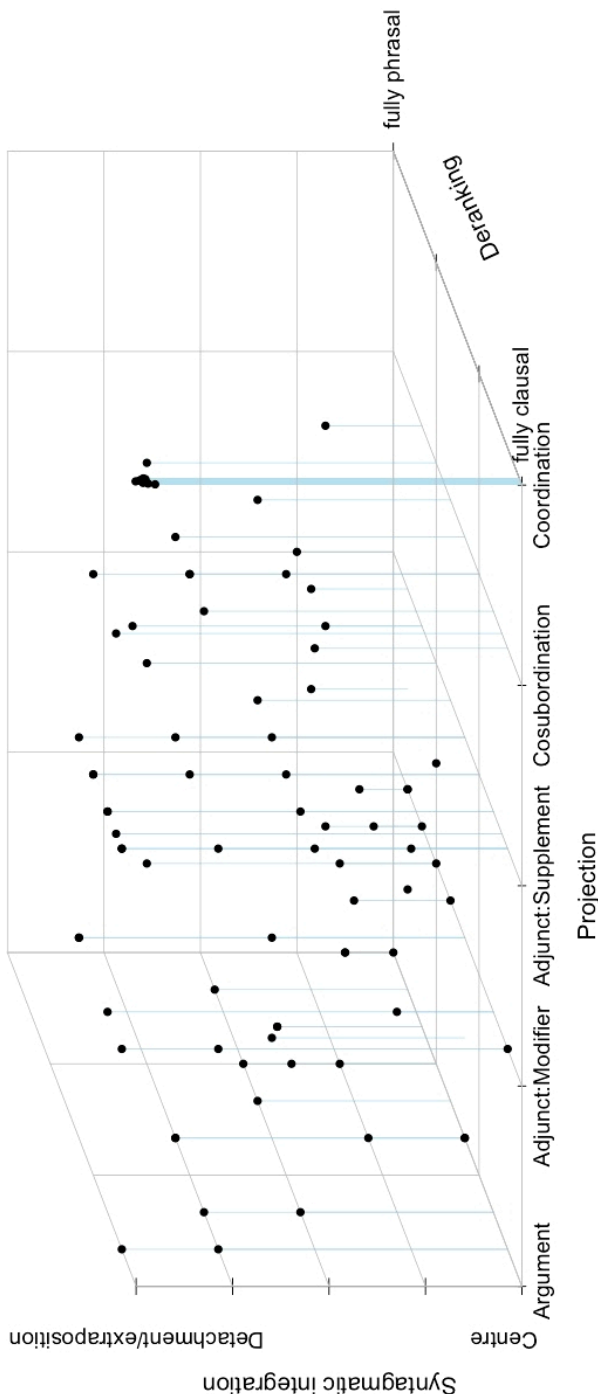


Figure 1: A three-dimensional space of embedding

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2.1 Headedness

The first parameter of cross-linguistic variation for RCs is whether the nominal head to be modified is syntactically part of the RC, appears outside of it, or is left implicit. Accordingly, many typologies distinguish between INTERNALLY-HEADED, EXTERNALLY-HEADED and HEADLESS RCs. The externally-headed type is the familiar one from languages like English; we illustrate it for a typologically different language in (21):

- (21) externally-headed RC from Chiapas Zoque (Mixe-Zoque: Mexico; Faarlund 2012: 162)

NP [*Te'* [*jo'nchi* REL [*y-ku't-u=pü='is* *te' tüm*]]]
 DET bird.ABS 3ERG-eat-COMPL=REL=ERG DET fruit.ABS
Ø-kek-u.
 3ABS-fall-COMPL
 'The bird which ate the fruit fell down.'

In this example, the noun *jo'nchi* is external to the relative clause: it appears before the relative clause and is case-marked for its syntactic function in the main clause, that of absolutive S-argument.⁸ In (22) below, by contrast, the noun to be modified by the relative clause, *úsa*, appears in the case form that is appropriate to its syntactic function *inside* the relative clause, namely its absolutive S-role. It is the RC as a whole that bears an ergative flag for its syntactic function in the matrix clause. This is, therefore, an internally-headed RC:

- (22) internally-headed RC from Epena Pedee (Chocoan: Colombia/Ecuador; Harms 1994: 167)

[*Úsa-Ø t^hot^hóo bí*]-*pa peróra pee-pa-čí.*
 dog-ABS white be-ERG spotted.cavy kill-HAB-PST
 'The dog that is white used to kill spotted cavies.'

Because of its treatment as a RC-internal argument, *úsa* is bracketed in (22) as a proper part of the RC. The fact that it appears at the beginning of the RC does not

concept for relative clauses, such as Comrie & Kuteva (2013) or Andrews (2007). For some typological notes on supplementary relative clauses, readers are invited to consult a more elaborate version of this article here: <https://doi.org/10.5281/zenodo.7971218>.

⁸In the remainder of this paper, we will be using the typological labels S, A and P as comparative concepts for the nominative or absolutive argument of intransitive clauses (S), the nominative or ergative argument of transitive clauses (A) and the absolutive or accusative argument of transitive clauses (P); see Haspelmath (2011) for further clarification of these concepts.

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make it external: it simply reflects the normal position of S-arguments in Epena Pedee clauses; if the relativized position were that of absolutive P argument (e.g. *the dog I saw*), *úsa* would appear in its canonical P position in the relative clause (e.g. I-ERG dog-ABS saw).

A headless RC is illustrated in (23):

- (23) Ute (Uto-Aztecan, Numic: USA; Givón 2011: 288)

[*kani-vaa-tʰ* *wúʰka-qha-tʰ*]
house-LOC-DIR work-ANT-NMLZ
‘the one who worked at the house’

In this example, the proposition of the subordinate clause is oriented towards one of its participants by virtue of a participant nominalizer, the suffix *-tʰ*, yielding the English translation ‘the one who ...’ (rather than the unoriented proposition ‘someone worked at the house’). Recent work on different areas of the world has brought out the importance of (participant) nominalizations for the typology of relative clauses. In particular, it has been pointed out that when such nominalizations are placed in apposition to an NP in the main clause, the resulting structure and initial interpretation is that of a supplementary relative clause – a looser, often phonetically unintegrated juxtaposition of two NPs with the same referent:

- (24) Ute (Uto-Aztecan, Numic: USA; Givón 2011: 291)

Mamachi, [(’ú) *kani-vaa-tʰ* *wúʰka-tʰ*], *tʰgʰvʰ-n* ’*ura-’ay*.
woman.SBJ that.SBJ house-LOC-DIR work-NMLZ friend-1SG be-IMM
‘The woman, the one who works at home, is my friend.’

Given the right circumstances, however, such appositions can come to be reanalyzed as restrictive, by the processes summarized succinctly in Lehmann (2008: 219ff., 2014: 3–4). This may or may not be accompanied by the structural integration of the participant nominalization with the nominal head, yielding a difference in the degree to which the two nominals form a single NP. In Ute, for example, there is also a prosodically integrated version of (24) above, even though it is less common in discourse than the appositive type in (24):

- (25) Ute (Uto-Aztecan, Numic: USA; Givón 2011: 288)

mamachi ’*u* [*kani-vaa-tʰ* *wúʰka-qha-tʰ*]
woman.SBJ 3SG.SBJ house-LOC-DIR work-ANT-NMLZ
‘the woman who worked at home’

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2.2 Integration with the main clause

In the context of the present volume, perhaps the most crucial parameter of cross-linguistic variation is that RCs of all headedness types above can be either EMBEDDED in the matrix clause, where they form a syntactic constituent with the nominal head (if there is one), or be ADJOINED to a main clause. We will discuss this difference separately for internally- and externally-headed RCs.

2.2.1 Internally-headed RCs (IHRCs)

2.2.1.1 The circumnominal type

IHRCs are embedded in the matrix clause if they directly fulfil a syntactic function in it, as is the case in (22) above.⁹ Lehmann (1984, 1986) reserves the term CIRCUMNOMINAL RC for this embedded variant of an IHRC. In much of the typological literature, the label IHRC is often used more narrowly to designate only this type of RC (see also Cole 1987, Culy 1990 and Basilico 1996 for relevant work). In Dryer's (2013b) sample of more than 800 languages, 24 exhibit the circumnominal type as their primary RC construction; these languages cluster conspicuously in North America and, moreover, they are overwhelmingly OV languages (all 24 except for Kutenai and Oneida).¹⁰ A further 39 languages have the circumnominal type as one of several RC constructions. Since the hallmark of circumnominal RCs is that they operate like an NP in the matrix clause, they typically bear some nominal morphology at their right boundary, such as a case marker in (21) or a case-marked demonstrative in (26):

(26) Choctaw (Muskogean: USA; Broadwell 2006: 50)

[*Hattak-mat* *ofi'* *chōpa-tok-mã*] *pīsa-li-tok*.

man-DEM.NOM dog buy-PST-DEM.ACC see:NGR-1SG.I-PST

'I saw the dog that the man bought.' (or: 'I saw the man that bought the dog.')

2.2.1.2 The correlative type

Circumnominal RCs contrast with those internally-headed constructions whose nominal head is also represented in the main clause, in different degrees of ex-

⁹Strictly speaking, therefore, such RC constructions are actually argument clauses in terms of their syntactic status, but they are semantically oriented towards one of their participant positions and hence to be interpreted as a modifier of a nominal.

¹⁰VO languages with circumnominal RCs beyond Dryer's (2013c) sample that have been mentioned in the literature include Mooré (Culy 1990: 214) and Dagbani (Lehmann 1984: 118)

plicitness. In all cases, this leads to the RC and the main clause containing a CORRELATIVE element, and typologists thus often use the term CORRELATIVE RC. The most explicit but arguably least economical way of creating this structure is by having a full noun in each clause, either a direct copy of the head nominal or a corresponding light noun like ‘thing’ or ‘person’. Only a single language in Dryer’s (2013b) database, Kombai, exhibits such **DOUBLY-HEADED RCS** as its primary RC type, and only four others (Jamsay, Mina (both Africa), Kobon and Yagarian (both Nuclear Trans-New Guinea)) as a secondary type. Two examples beyond Dryer’s database come from Gooniyandi (27), where doubling occurs occasionally” (McGregor 1990: 438), and from Santali (28), where doubling is part of a distinct RC construction called a CORRELATIVE DIPTYCH (Haudry 1973):

- (27) Gooniyandi (Bunaban, Australia; McGregor 1990: 438)

[*Thangarndi* garndiwangooddoo gooddooimba-ya yoodjidi]
 word many paper-LOC we.put.it
thangarndi binaddigmiloona.
 word I.taught.them
 ‘I taught them (some of) the many words we had put on paper.’

- (28) Santali (Austroasiatic, Mundaic: Bangladesh; Neukom 2001: 199)

[*Oka disom-re onko gadel hɔr-ko*
 which country-LOC those.ANIM.PL crowd person-3PL.SBJ
jarwa-akan-tahēkan], *ona disom-ren raj-dɔ*
 gather-PRF.MID-COP.PST that.INAN country-GEN.ANIM king-TOP
tis-re cɔ-e gɔc’-akan-tahēkan.
 when-LOC ever-3SG.SBJ die-PRF.MID-COP.PST
 ‘The king of the country where these crowds of people had come together had died some time previously.’

(28) represents a correlative construction that is familiar from Latin or Hittite and which is a well-known areal feature of RC formation in South Asia as well as in Mande languages and their neighbours (Dryer 2013b). In this ‘question-and-answer’ construction, the IHRC contains a relative/interrogative determiner accompanying the nominal head, while the main clause either exhibits a demonstrative determiner with a copy of the head nominal (as in (28)) or else simply a (demonstrative or personal) pronoun. In fact, the latter pattern, with only a pronominal correlate in the main clause, is also found in Santali and even preferred to double heads in Gooniyandi. We encountered this type of correlative clause earlier, in (14) from Supyire. It was mentioned there that the IHRC in

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Supyire is relegated to a left-detached (and less commonly, a right-attached) position, usually with a slight intonation break, and obligatorily correlated with a personal pronoun in the main clause. Another representative of this type is Bambara:

- (29) Bambara (Mande, Manding-Vai: Guinea; Givón 2001: 183)

[*Cε min ye muru san*], *n ye o ye*.
 man REL PST knife buy I PST him see
 ‘The man who bought the knife, I saw him.’

The pattern that unites all correlative constructions is their (preferred) occurrence at the left sentence boundary (“preposed”, “left-dislocated”), in a topical or expository discourse function (Lehmann 1984, 2008); this also holds for the RC illustrated in (27) from Gooniyandi (McGregor 1990: 438). What distinguishes the constructions in (27) and (28–29) is thus chiefly their relative degree of deranking: the RC in Gooniyandi shows all vestiges of an independent clause, while the ones in (28–29) bear a relative marker that would not be present if these were independent clauses. Some authors have thus described (27) as a PARATACTIC relative clause (e.g. Comrie & Kuteva 2013). Overall, however, correlative RCs show a low degree of deranking, which reflects their diachronic origin as a discourse strategy in which a (typically indefinite-specific) referent is “introduced as something to be resumed” in the main clause (Lehmann 2008: 217).

In the present context, it is interesting to point out how these adjoined clauses may develop into embedded structures. In fact, depending on the specific type of correlative RC, only small adjustments might be needed to turn it into a circumnominal one, notably the combination of the two clauses without a prosodic break and the omission of the anaphoric demonstrative in the second clause. If the demonstrative carried case markers, these may come to attach to the RC as a whole (which would account, for example, for a pattern like (22) above). Alternatively, the anaphoric demonstrative may be reanalyzed as the final element of the first clause, also yielding a clausal nominalization (which would account for structures such as (26) above).

Also, as observed by Lehmann (1984: 388), some languages allow their correlative diptychs to change the order of relative determiner and noun, making the head noun topical (e.g. [*book which you gave me*], *I lost it*). This structure may then be reanalyzed as postnominal, and if the correlative element in the main clause is dropped, the result is an embedded subordinate clause.

Finally, it is possible for preposed correlative clauses, as a whole, to be attracted to other positions in the matrix clause. Firstly, a correlative clause may

be postposed (i.e. right-adjoined) to the main clause, as an “inverted diptych” that codes presupposed information, from which it can develop into a postnominal RC (Lehmann 2008: 218; see also Section 2.2.2 below). Secondly, the normally preposed RC may be inserted directly into the main clause right after the nominal to be modified; this latter option is illustrated by Comrie (2006: 139) for Hittite, and he argues that these constellations gave rise to postnominal RCs with relative pronouns (Section 2.2.2), which subsequently spread across Europe through language contact. Thirdly, correlative constructions may show up, perhaps rather unexpectedly, in centre-embedded positions, as in Wappo (Thompson et al. 2006: 115) or in Cabecar:

- (30) Cabecar (Chibchan: Costa Rica; Gonzáles Campos & Lehmann 2019: 10)

Bá te [jǐjǐ tk-á=ju yíkí] jé s-á?
 2SG ERG earthquake cross-PRFV=AM yesterday DEM feel-PRFV
 ‘Did you feel the earthquake that happened yesterday?’

With regard to their form, RCs in Cabecar are of the paratactic type, i.e. there is no trace of deranking, and they are often (but not obligatorily) resumed by the medial demonstrative *jé* in the main clause. Gonzáles Campos and Lehmann argue that this construction evolved from a sequence of independent main clauses, functionally equivalent to the preposed correlative clauses of the present section. By prosodic integration and omission of the demonstrative, one can easily obtain a ‘standard’ circumnominal RC which directly occupies a syntactic slot in the main clause. But as (30) shows, it is also possible to centre-embed the RC while retaining the co-referential demonstrative. As a result, the RC is no longer left-adjoined, unlike a typical correlative clause. This yields an interesting surface combination of a syntagmatically embedded clause that seems apposed to an anaphoric demonstrative. However, Gonzáles Campos & Lehmann (2019: 25) argue that “the sole function of the demonstrative” in these cases “is to mark the final boundary of the relative clause,” which effectively makes (30) exactly parallel to the circumnominal RC in (26) and thus marks the “last step” (ibid.) in the development from a paratactic to an embedded structure.

2.2.2 Externally-headed RCs (EHRCs)

EHRCs can, in principle, occur before or after the nominal they modify. The latter option was exemplified in (1), (4) and (10) above, while the former option is illustrated in (31):

- (31) Lezgian (Nakh-Daghestanian, Lezgi: Azerbaijan; Haspelmath 1993: 340)

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[*Mu'minat-a ktab ga-ji*] *ruš-a q^hfe-na*.
 Muminat-ERG book give-AOR.PTCP girl-ERG go.away-AOR
 'The girl to whom Mu'minat gave the book went away.'

Cutting across this basic ordering distinction is, once more, the parameter of embedding: specifically, the question here is whether the EHRC occurs *ADNOMINALLY*, i.e. adjacent to the nominal it modifies (as in (31)) or *EXTRAPOSED* from it (as in (10)), and whether it is syntactically (and not just semantically) an attributive modifier of the head.

2.2.2.1 The adnominal type: Postnominal and prenominal RCs

We begin by inspecting the *ADNOMINAL* type of EHRCs, which is the typologically dominant relative construction by a clear margin. However, there is a strong asymmetry in the distribution of the *PRENOMINAL* and the *POSTNOMINAL* subtype: whereas postnominal RCs are extremely widespread (in geographical, genealogical and typological respects), prenominal RCs are much more confined. First, barring a few exceptions in mainland China and Taiwan (e.g. Bai (Tibeto-Burman), Chinese languages, Amis (East Formosan, see Comrie 2008a), their occurrence is restricted to OV languages (Dryer 2013c), while postnominal RCs are common in both VO and OV ordering types. Second, among the OV languages, prenominal RCs are dominant mainly in Asia, while OV languages in the other macro areas appear to prefer different kinds of RC constructions (*ibid.*).¹¹ Finally, while languages with prenominal RCs sometimes have a postnominal alternative construction (29/170 languages = 17% in Dryer 2013b), only very few languages with postnominal RCs have a prenominal alternative (31/610 = 5% in Dryer 2013b).¹²

Several factors are responsible for this asymmetry. To begin with, it appears that attributive modification has a certain postnominal bias, as attributive adjectives – the structurally smaller equivalents of RCs – are cross-linguistically also preferred after the noun they modify (Dryer 1992). Furthermore, in contrast to postnominal RCs (and the preposed internally-headed types we surveyed above), prenominal RCs are oriented towards a certain participant before that entity is

¹¹Local pockets of OV and prenominal RCs are found in "(i) New Guinea; (ii) Ethiopia and Eritrea; and (iii) southern Colombia and the adjacent area of Brazil" (Dryer 2013c). However, all of these regions are parts of larger macro areas used for typological studies, and the OV languages of these macro areas do not generally favour prenominal RCs.

¹²These figures are not controlled for geographical or genealogical dependencies, and neither do they imply that the languages in question use the *same* RC construction in pre- and postnominal order. All that matters is that there is *some* prenominal counterpart to the dominant postnominal construction or vice versa.

actually fully mentioned. This order of “gap before filler” (Hawkins 2004: 175) goes against the grain of the usual management of discourse referents, which is primarily anaphoric, i.e. backward-looking. Prenominal RCs have hence sometimes been argued to be less efficient for online processing than their postnominal counterparts (Antinucci et al. 1979, Hawkins 1994, 2004, 2014). All other types of RC avoid this problem by introducing the nominal first and resuming it, implicitly or explicitly, in the ensuing clause. The most explicit anaphoric devices are ...

- (i) RELATIVE PRONOUNS, which typically occur at the beginning of the post-nominal RC, combining the function of a subordinator and an anaphoric pronoun. As argued by Comrie (2006) and documented in Comrie & Kuteva (2013), relative pronouns are rather common in Eurasia but rarely used elsewhere. (32) below provides an example from Hungarian, where the clause-initial relative pronoun inflects for the case role of the relativized position inside the RC:

- (32) Hungarian (Uralic, Hungarian: Hungary; Kenesei et al. 1998: 38)

A könyv, [amely-et Anna olvas-ott], érdekes volt.
 the book **which-ACC** Anna read-PST interesting was
 ‘The book that Anna was reading was interesting.’

- (ii) RESUMPTIVE PRONOUNS, i.e. in-situ pronominal representations of the antecedent inside the RC, as in (33):

- (33) Krongo (Kadugli-Krongo: Sudan; Reh 1985: 253)

Òdínkò bílyáatà [ɲ-òkúró-n-tú iʔiŋ].
 fall.M.PRFV child CONN.M-push.PRFV-TR2SG him
 ‘The boy you pushed has fallen down.’

Indexing-prominent languages, i.e. those with verbal agreement markers for more roles than the subject, may show an index on the RC predicate instead, as in (34):

- (iii) RESUMPTIVE VERBAL INDEXES

- (34) Modern Standard Arabic (Afro-Asiatic, Semitic; Holes 2004: 283)

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ʔal-qişşatu [*llati: qaraʔa-ha:*]
 ART-story.F REL read.PST.3SG.M.SBJ-3SG.F.OBJ
 ‘the story that he read (it)’¹³

Even where the most economical and hence most widespread technique of co-reference, namely a so-called GAP in the relativized position, is chosen, postnominal RCs are still often marked by a clause-initial subordinating morpheme that has its diachronic roots in a topical pronoun, notably a demonstrative, such as *inde* ‘this’ in Mayogo:

- (35) Mayogo (Atlantic-Congo, Volta-Congo, Ubangi: The Democratic Republic of the Congo; Sawka 2001: 170)

Ma dje engú [inde mɛ nga-pa-e ma _____] de.
 1SG PST.hear affair REL 2SG PROG-say-REF 1SG NEG
 ‘I don’t understand (the) problem that you are talking to me about.’

In Hendery’s (2012: 267ff.) sample, demonstrative and interrogative pronouns constitute the most common sources of subordinators of postnominal RCs (see also Kuteva et al. 2019 and Diessel 1999). Therefore, while invariant markers like Mayogo *inde* are not synchronically pronominal and merely serve to mark the RC as such rather than a particular referent inside the RC, they tie in diachronically with the more explicit anaphoric elements we find in other postnominal RCs.

The situation in prenominal RCs is rather different, as has been pointed out by numerous authors. Wu’s (2011) recent survey of the syntax of prenominal RCs confirms the absence of relative pronouns, the prevalence of other types of subordinators (which we will turn to below), and a less systematic use of resumptive pronouns. As for the latter, it is telling, firstly, that in Comrie & Kuteva’s (2013) sample, all languages with resumptive pronouns have postnominal relative clauses. Wu (2011) makes out only three clusters of prenominal-RC languages in which resumption is used more regularly for certain relativized positions, namely Semitic, Chinese and “some Causasian languages” (Wu 2011: 594). Secondly, as noted by Lehmann (1984: 230), the resumptive elements of prenominal RCs can be markedly different from the anaphoric pronouns in postnominal RCs; in particular, they can take the form of (long-distance) reflexive pronouns (e.g. Turkish (Kornfilt 1997: 61), Korean (Sohn 1994: 66) or Tanti Dargwa (Sumbatova & Lander

¹³As discussed in Comrie (1981: 220), resumptive pronouns and indexes are person forms whose presence is not required in the corresponding simple sentence. This is the case in both (33) and (34).

2014: 192)), whose antecedent is unlikely to be found in the preceding sentence and thus expected in the upcoming main clause.

What these observations and the general filler-before-gap preference suggest is that prenominal RCs are the most difficult type of RC to grammaticalize from a sequence of referentially coherent (independent) clauses. In keeping with this, we find that the internal structure of prenominal RCs resembles that of independent clauses much less commonly than that of postnominal RCs. On the contrary, they characteristically show dependent verb forms with reduced verbal and clausal operators, as in the nominalized RC from Dolakha Newar in (20) above and the participial RC from Lezgian in (31). In fact, of all RC types, the prenominal one is associated with the highest degree of deranking and thus formal embedding into the matrix clause. In this respect, prenominal RCs also differ crucially from IHRCs: circumnominal RCs are typically ‘externally’ nominalized, i.e. they are sentential structures flagged by appropriate case markers and demonstratives at their right boundary; and correlative RCs are not normally nominalized at all, i.e. they, too, are basically sentential structures (Keenan 1985: 161).

Therefore, it has been proposed that prenominal RCs chiefly develop by processes of expansion of and substitution within nominal constructions rather than the integration of successive sentences in discourse. The expansion scenario accounts for RCs which carry morphological traits of other noun modifiers, such as attributive adjectives and genitives. Lehmann (1984: 376f.) mentions the Turkish *-An*, the Dravidian *-a* and the Dyirbal *-ŋu* participles as representative examples (see also Aristar 1991 for diachronic relationships between adjectives, genitives and relative clauses). Along the same lines, Wu (2011: 581) points out that what looks like a general “complementizer” in some relatively sentential prenominal RCs is actually identical to a genitive marker (e.g. Mandarin *de* or Alamlak *-ho*; see also Trask (1997: 247) on a possible genitive origin of the Basque relativizer *-en*). The general thrust of these examples is thus that a phrasal attributive construction becomes expanded into a clausal one, either by adding arguments and adjuncts to a deverbal adjective or by inserting a clause into the slot of a genitival or other modifying construction.

The second prominent diachronic source of prenominal RCs is that of participant nominalizations, such as agent-, patient- or various oblique-role nominalizations. A widely cited example comes from Lhasa Tibetan, where the suffixes used in relativization go back to lexical nouns in compounds:

(36) Lhasa Tibetan (Sino-Tibetan, Bodic: Bhutan, Nepal; DeLancey 1986)

- a. *mkhan*
‘teacher, expert’

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- b. *s'i'n-mkhan*
lit. wood-expert = 'carpenter'
- c. [*stag gsod-mkhan*] *mi*
tiger kill-NMLZ man
'the man who killed the tiger'

While *-mkhan* is, quite predictably, a relativizer for the subject position, locative and instrument relativizers harken back to the nouns for 'place' and 'tool', respectively. Similar uses of participant nominalizers for relative-clause formation have been reported for many languages, particularly in Asia (Yap et al. 2011) and the Americas (Comrie & Zarina Estrada-Fernández 2012). In fact, many of the "oriented participles" in Shagal's (2019) world-wide study of relativization are participant nominalizations.

Importantly, while the diachronic pathway from (36a–36c) involves successively larger structures, and in this sense an expansion, it still often proceeds from looser to tighter syntactic configurations at the final stages: as we saw in (23–25) from Ute, the RC interpretation arises when an expanded participant nominalization is placed in apposition to another nominal element, i.e. 'man' in (36c) and 'woman' in (24). At least at the beginning, no dependency is involved, as the nominalization constitutes a separate NP rather than a modifier embedded into an NP (see also Comrie & Thompson 2007 [1985]: 378). This is nicely illustrated by Awa Pit, where the nominalization can be placed in either order with regard to the noun it modifies semantically (and can be completely separated from it); it can also be marked independently as a clausal topic and for its syntactic function in the matrix clause:

- (37) Awa Pit (Barbacoan: Colombia; Curnow 1997: 287–288)
- a. [*A-mtu=mika*]=**na** *ashaŋpa=na wan pyan-ti-zi*.
come-PTCP=NMLZ.SG=TOP woman=TOP all hit-PST-NLOCUT
'The woman who was coming hit everyone.'
- b. [*Santos=ta pyaŋta-ta=mika*]=**ta** *pyan-ta-w, ashaŋpa=ta*.
Santos=ACC kill-PTCP=NMLZ.SG=ACC hit-PST-LOCUT.SBJ woman=ACC
'I hit the woman who killed Santos.'
(lit. 'I hit the one who killed Santos, the woman.')

In (37b), the RC and the 'head noun' are non-adjacent and hence they do not form a syntagmatic unit. In (37a), they are adjacent and hence the RC could be said to be 'adnominal', but it is still not embedded within a single NP by the

criteria above. Thus while Awa Pit has retained the syntactic independence of the semantic modifier, Ute has kept this pattern (24) alongside a more integrated one where the nominalization is embedded as a RC (25).

Example (37) is also revealing for another reason: while the stages in (36) suggest a diachronic expansion, it has recently been argued that nominalizers in prenominal RCs can also come from nominal classifiers or semantically light and hence frequent head nouns in earlier RC constructions, such as ‘thing’, ‘place’, etc. These head nouns may come to be bound to the RC to yield exactly the participial nominalizers that are characteristic of prenominal RCs. This would account for =*mika* in Awa Pit, which synchronically cliticizes to a participial suffix. Curnow (1997: 290) speculates that this may be a remnant of an earlier RC construction which is no longer grammatical by itself, and Epps (2009) argues that these developments may generalize to various languages in South America (see also Genetti et al. 2009 for similar developments in Tibeto-Burman languages).

Whatever the precise direction by which these nominalizations emerge, our claim in the present section is *not* that this pathway, just like the expansion of deverbal adjectives and genitive modifiers, is restricted to prenominal RCs. As Ute demonstrates, it can also underlie the genesis of postnominal RCs (see also Lehmann 1984 on German, Deutscher 2009a on Akkadian and Cristofaro 2014 on Ewe). But the current typological picture suggests that these developments of expansion, “insertion” (Kibrik 1992) and apposition are *relatively* more important, and perhaps even the only ones, for the emergence of prenominal RCs.

2.2.2.2 The postposed (right-adjoined) type

We round off our discussion of EHRCs with a type of construction that is externally headed but attaches to the right of the main clause rather than the nominal head; in other words, this type of RC is not adnominal but POSTPOSED or RIGHT-ADJOINED. It can, however, appear in this position for two rather different reasons.

On the one hand, a postposed RC can result from a performance option of extraposition, as introduced in (10) above from Slave. Very similarly, “it is common” for the IHRC from Choctaw in (26) above “to be extraposed to the end of the [sentence], leaving the head of the relative clause in situ” (Broadwell 2006: 301):

(38) Choctaw (Muskogean: USA; Broadwell 2006: 301)

Ofi-it balii-it kaniiya-h [ipiita-li-k-aash-m-at].
 dog-NOM run-PTCP go.away-TNS feed-1SG.I-TNS-PREV-NOM
 ‘The dog that I fed ran away (lit. went away running).’

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Where extraposition affects nominalized RCs (as in Slave and Choctaw), the result looks like the mirror image of the NP apposition pattern we saw in (37b) above, i.e. ‘The dog ran away, the one that I fed.’

Sometimes the pattern of extraposition is such that the whole relative construction, i.e. the RC along with its head noun, is shifted to the right sentence boundary. In Evenki, for example, nominalized (or participial) RCs are typically prenominal constructions, as in (39a), but the order can be changed in such a way that the modified nominal and the RC appear sentence-finally (39b):

- (39) Evenki (Tungusic: China, Mongolia; Nedjalkov 1997: 137)
- a. [*himat agi-tki tuksa-d’ari-va*] *asi-va*.
 quickly forest-ALL run-PTCP-ACC.DEF woman-ACC.DEF
 ‘a/the woman who quickly ran into the forest’
- b. *Bi archa-ø-m asi-va* [*himat agi-tki*
 1SG meet-NFUT-1SG woman-ACC.DEF quickly forest-ALL
tuksa-d’ari-va].
 run-PTCP-ACC.DEF
 ‘I met a/the woman who quickly ran into the forest.’

While each language has its own specific set of motivating factors for RC extraposition (see, e.g., Nefedov 2012 on Ket, Strunk 2014 on German, Rasekh-Mahand et al. 2016 on Persian), a factor that appears to play a role in all of the languages for which corpus data have been scrutinized is the relative weight of clausal as opposed to single-word or phrasal modifiers. By shifting the RC, one can “minimize domains” (Hawkins 2004: 31) for processing the fundamental predicate-argument structure of the main clause. As this applies to languages of all basic word-order types, patterns of RC extraposition are productive in both OV and VO languages.

The second type of postposed RC is not due to a performance option of extraposition, but one that is canonically adjoined to the main clause in this position. In (13) above, we introduced this as a characteristic type of subordinate clause in Australian languages. And indeed, in Dryer’s (2013b) database, there are only two languages outside of Australia in which adjoined RCs are the dominant RC construction, namely Mekens (Tupian) and Kuikúro (Cariban), both spoken in Brazil. Another example of an adjoined RC from Australia is given in (40b):

- (40) Kayardild (Tangkic: Australia; Evans 1995: 512, 517)
- a. *Ngada birrmurdami* [*ngijin-inja kajakaja-ntha bukawa-thurrk*].
 1SG.NOM sad.NOM my-COBL daddy-COBL die-IMM.COBL
 ‘I can feel that my daddy has just died.’

- b. *Jina-a maku, [ngijuwa kurri-jurrk].*
 where-NOM woman.NOM 1SG.SBJ.COBL see-IMM.COBL
 ‘Where’s this woman who I’m seeing?’

The construction illustrated in (40) is characterized by special (“complementizing oblique”) case marking on all elements of the dependent clause. And as in other Australian languages, this subordinate clause has a number of different grammatical functions and hence interpretations in Kayardild, as reflected by (40a–40b). But interestingly, when it is supposed to function as a relative clause, the free constituent order of Kayardild is usually adapted in such a way that the modified nominal appears as the last element of the main clause, effectively achieving an adnominal structure without centre-embedding it in the matrix clause (just as in (39b) from Evenki).

According to Hale (1975), this is a signal of the “attraction” of the subordinate clause to a nominal element in the main clause: Hale argues that such adnominal environments at the right sentence boundary enable a reanalysis of adjoined subordinate clauses as postnominal modifiers, and when this pattern is analogically extended to non-final NPs in the main clause, the result is a (centre-)embedded RC. This, then, constitutes another way in which embedding can emerge from syntagmatic adjunction.¹⁴

2.3 Orientation of the RC

A defining property of RCs is that they are semantically oriented towards one of their participant positions. While a typical argument clause designates a situation (‘(that) I read the book’), a relative clause designates an entity participating in a situation (‘which (book) I read’). The relative clause is thus said to be oriented towards that entity (Lehmann 2014: 2). Therefore, a major parameter of cross-linguistic variation is how explicitly this orientation is coded morphosyntactically, so as to enable the unambiguous and efficient processing of the RC and the complex sentence of which it is part. The different types of RC formation surveyed above present different challenges in this regard.

¹⁴In fact, Hale’s “attraction”-and-extension scenario is also involved when postnominal RCs develop from independent clauses with topicalized anaphoric pro-forms, e.g. ‘We met a woman. That one was feeding her child.’ > ‘We met a woman that was feeding her child.’ (attraction) > ‘The woman that was feeding her child looked happy.’ (extension to an embedded position). This pathway is discussed, for example, by Lehmann 1984, 2008 for Homeric Greek, Haader 2002 for Hungarian and McConvell 2006 for Ngumpin-Yapa. Here, too, the overall process is one of combining and successively integrating two clauses

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For IHRCs, the major processing challenge is to discover that a nominal element inside the RC is meant to be modified by the clause as a whole; in other words, one needs to discern the presence of a nominal head. This problem is illustrated by our earlier example (26) from Choctaw: as the translation shows, there are two possible ways of interpreting the relative clause, which results from the fact that there is no explicit means of orientation. It is thus not surprising that languages with IHRCs often develop grammatical strategies of what Lehmann (1984: 207) calls *NUCLEUS FORMATION*. The most common strategy is to move the relativized NP to the beginning of the RC; this is illustrated by our earlier example (14) from Supyire again, and found in a similar fashion in, for instance, Slave and Menya. Alternatively, speakers of Supyire can also leave the relativized NP in situ, but then it needs to be marked by a relative determiner (Carlson 1994: 491ff.), just as we saw for Bambara in (29) and in the typical correlative diptych in (28). Further means of nucleus formation include special agreement markers on the RC verb (Lavukaleve), an accompanying indefinite article (Lakota) or tonal adjustments (Jamsay).

For EHRCs, the processing challenges are, in a way, the opposite of those in IHRCs: the modified noun is given externally, but now the crucial task is to figure out the relativized position inside the RC. This is sometimes called the “*CASE RECOVERABILITY PROBLEM*” (Givón 2001: 180), and it becomes apparent whenever the RC does not contain an explicit representation of the relativized NP, i.e. in what we introduced as the gap strategy above. The syntactic function associated with such gaps can be recovered comfortably in languages with (i) high “referential density” (Bickel 2003), i.e. overt coding of even contextually salient participants, and/or (ii) fixed constituent order and/or (iii) the possibility of stranded case markers. In Abun, for example, constituent order is rather rigidly SVO, and adpositions can be left in situ without an overt complement:

(41) Abun (West Papuan: Indonesia; Berry & Berry 1999: 153–156)

- a. *ii bi buku [gato an gre _____ mo bot]*
 1SG POSS book REL 3SG burn LOC fire
 ‘my book that she burned in the fire’
- b. *ʃi mu kenyak nu [gato Isak sem mo _____ ne].*
 1SG go locate house REL Isak sleep at DET
 ‘I went to find the house that Isak slept at.’

In these examples, it makes sense to speak of a gap *strategy*, as is often done, but where constituent order is more flexible and adposition stranding is ungram-

mathematical, the gap itself cannot be called a strategy for role recovery. In such cases, the relativized position needs to be identified on the basis of other devices.

One such tool can emerge directly as a by-product of grammaticalization, namely when different participant nominalizers (or contrasting participles) come to be associated with different relativized positions. This is what we saw in (36) from Tibetan above, and a similar system is found, for example, in Mapudungun. (42) shows the agent nominalizer *-t* yielding relativization on a subject and the “objective verbal noun” in *-el* yielding relativization on an object, but there are several other nominalizers in addition to these.

- (42) Mapudungun (Araucanian: Argentina, Chile; Smeets 2008: 216, 200)
- a. *Kim-nie-fi-y-m-i* *ti wentrú* [*mi*
know-PROG-OBJ-IND-2-SG the man 2SG.POSS
leli-nie-e-t-ew]?
watch-PROG-OBJ-AVN-SBJ:DAT
‘Do you know that man who is watching you?’
- b. *Fey-chi chanchu* [*eymi mi* *ngilla-el*] *trongli-le-y-Ø*.
that-ADJz pig you 2SG.POSS buy-OVN lean-STAT-IND-3
‘That pig you bought is lean.’

In a similar fashion, many languages distinguish (at least) between a subject-oriented affix (often an agent nominalizer or an ‘active participle’) and a non-subject-oriented affix (e.g. from a patient or an action nominalization or a ‘passive participle’), e.g. Quechua, Wolaytta, Dolakha Newar, Finnish, Aguaruna, Kolyma Yukaghir and Maricopa. Some languages, like Yuracaré, exploit their switch-reference markers to make exactly the same contrast, i.e. here, too, a grammatical system that evolved for the purposes of reference tracking can function as a cue to serve the role recovery problem when such constructions are used for relativization. All of these cases have thus been referred to collectively as demonstrating a “verbal affix” strategy (e.g. Comrie 2003).

Considering the potential processing difficulties and the nominalizing/adjectival origin of prenominal RCs we discussed above, one would expect that such role-distinguishing dependent verb forms are particularly common in prenominal RCs. However, Shagal (2019) finds exactly the opposite: in her study, languages which *only* have “contextually oriented participles” (i.e. verb forms that can be flexibly oriented towards several core roles) strongly tend to have them in prenominal RCs (28:6 languages), while languages which *only* have role-specific verb forms are more likely to have them in postnominal RCs (23:12 languages).

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A similar observation can be made with regard to resumptive pronouns, the second device that can be used when a gap is judged not to be sufficient for role recovery. In their seminal study, Keenan & Comrie (1977) established an implicational hierarchy for the distribution of gaps and resumptive pronouns in RCs. A modified version of this hierarchy, adapted from Lehmann (1984: 211), is provided in (43):

- (43) The accessibility hierarchy of relativization¹⁵
 SBJ/ABS > DO/ERG > IO/'time', 'place' > oblique complements > adjuncts

According to (43), whenever a language uses resumptive pronouns instead of a gap for a certain position on the accessibility hierarchy, it will also use resumption on all lower positions. This is a statistical generalization that applies to all types of RC. However, it is noteworthy that prenominal RCs do not only make less frequent and less systematic use of resumptive pronouns (as we mentioned above), but that their resumptive elements do not extend as highly on the hierarchy than those of postnominal RCs: it is very uncommon to find prenominal RCs with resumptive elements for core syntactic functions (S, A, P), whereas resumptive pronouns for P are often at least licensed, if not required, in postnominal RCs (e.g. East Ambae, Arabic, Kera, Persian, Tongan).

It appears, therefore, that prenominal RCs are overall less prone than postnominal RCs to code their orientation explicitly, given the lack of relative pronouns, less systematic use of resumption, use of unoriented verb forms, infrequent adposition stranding and flexibility of constituent order in the OV languages in which they typically occur.¹⁶

In the absence of orienting devices, the relativized position of EHRCs can be “infer[red] by subtraction” Givón (2001: 184), notably from the semantic and syntactic requirements of the RC predicator and the arguments that are already supplied in the RC. In this connection, it is noteworthy that some languages restrict gaps to those arguments that can be indexed on the dependent verb (e.g. Ungarinjin, Basque) or restrict relativization to the core syntactic functions more

¹⁵Keenan & Comrie's (1977) original hierarchy included possessors and objects of comparisons as the two lowest roles. But as pointed out by Lehmann (1984: 211), one needs to distinguish at least between an *ad-verbal* accessibility hierarchy (where the relativized position is a syntactic function in a *clause*) and an *ad-nominal* accessibility hierarchy (where the relativized position is a syntactic function in a *phrase*, like possessor or standard of comparison). In (43) above, we limit our attention to the *ad-verbal* hierarchy.

¹⁶This also ties in with the observation that prenominal RCs are often more general “noun-modifying constructions” (Matsumoto et al. 2017) whose specific semantic link to the nominal head can be highly diverse and needs to be inferred from the context.

generally (e.g. Jarawara, Mangarrayi). Furthermore, where resumption occurs across argument and adjunct positions (e.g. Hebrew), corpus evidence shows that it is significantly more common for adjuncts (Ariel 1999).

All of this suggests, in accordance with (43) above, that RCs are most frequently oriented towards core syntactic positions, and hence core participants, whose involvement in the RC event is easiest to retrieve and in this sense most “accessible”. Deviations from these most accessible roles correlate typologically with special marking strategies, either more explicit ways of indicating the relativized position (i.e. various forms of resumption) or by deriving relativizable core roles through valency-changing operations. The latter strategy is illustrated in the following example from Yakan, where only S and P are relativized on directly while A and lower roles require appropriate means of promotion to a higher argument position:

- (44) Yakan (Austronesian, Malayo-Polynesian, Sama-Bajaw: Philippines; Brainard & Behrens 2002: 166–167)

- a. {A} becomes {derived-S} by means of antipassivization and then takes the nominalized relative clause appropriate to {S}

Iyan nakanak [*ma-N-polong* *buwa'-buwa'-in*].
that child NMLZ-ANTIPASS-break toy-DEF
'That is the child who broke the toy.'

- b. {OBL} recipient ('to the dog') becomes {derived-P} by means of applicativization and then takes the non-nominalized relative clause appropriate to {P}

Iyan asu [*-in-teppad-an-ne* *tolang-in*].
that dog -TR-throw-APPL-3SG.ERG bone-DEF
'That is the dog that she threw a bone to.'

There has been considerable debate in the literature as to how exactly constraints on the relativized position can be explained. On the one hand, there are discourse-pragmatic preferences that have been argued to lead to different degrees of accessibility even within the core roles (S, P > A, see Fox 1987). On the other hand, there is a wealth of psycholinguistic research attempting to motivate accessibility differences between the core roles (subject > object) and between these and the non-core roles (see Norcliffe et al. 2015 for an overview of recent work). But as summarized succinctly in Hawkins (2014: 22–28), even the processing motivations are multifactorial in nature, involving several different ways in which the lower roles on the accessibility hierarchy create more processing effort and hence profit from more explicit coding strategies.

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While these different facets of processing are beyond the scope of the present overview, we can conclude our discussion of RCs with a more general observation related to language processing. It has been argued, most prominently by Hawkins (2004, 2014) again, that OV languages face conflicting pressures in the grammatical expression of relativization. On the one hand, these languages favour structures in which dependent elements like NP complements of verbs and adpositions as well as subordinate clauses appear before their respective heads; this would promote the grammaticalization of prenominal RCs. On the other hand, the nature of reference management in discourse favours anaphoricity (“filler before gap”) and hence actually militates against prenominal RCs. The result of these conflicting pressures is what we have seen throughout the present section, namely a greater diversity of RC types in OV languages: the internally-headed types deliver a dependent clause early in a sentence while adhering to the filler-gap principle; the right-adjoined RC also ensures that fillers come first while still allowing efficient processing of all main-clause dependencies; and the two major externally-headed RC types resolve one of these pressures each (see also Lehmann 1984 and Dryer 2013c). In VO languages, by contrast, no conflicting pressures arise in the first place, and different pathways of grammaticalization all converge on the postnominal type most straightforwardly.

3 **Adverbial clauses in typological perspective**

For the purpose of cross-linguistic comparison, adverbial clauses (ACs)¹⁷ are commonly defined extensionally, as subordinate clauses which encode (more or less explicitly) one or more of the following semantic relations between situations, henceforth adverbial relations (ARs):

- temporal simultaneity (WHEN/WHILE), anteriority (AFTER), posteriority (BEFORE), end points (UNTIL) and beginnings (SINCE)
- realis, irrealis and counterfactual CONDITIONALITY

¹⁷ Monographic treatments of the typology of adverbial clauses include Kortmann (1997) on European languages and Hetterle (2015) with a world-wide scope. For non-monographic typological surveys of the adverbial domain, the reader is referred to Thompson et al. (2007) and Dixon (2009). In contrast to these latter overviews, we will not survey the different semantic relations and their coding properties individually, but focus instead on the comparison of the major adverbial relations from the perspective of the different facets of embedding introduced in Section 1.2 (see also Diessel 2013 for a like-minded overview).

- CAUSE and REASON¹⁸
- PURPOSE
- RESULT
- CONCESSION
- ADVERSATIVITY ('whereas/while')
- MANNER

For reasons of space, we will deal exclusively with ACs in a modifying function, leaving aside their uses as supplementary adjuncts (as in (6–8) above; see, e.g., the papers in Tsunoda 2018 for relevant work).

3.1 Coding, function and position of ACs

Just as some languages have been reported to make very little use of relative clauses in discourse (e.g. Donohue 1999 on Warembori), some have been said to make little use of grammaticalized adverbial clauses, even if the latter are available in the language system (see Deutscher 2009b on Akkadian and similar assessments for Lango (Noonan 1992), Warao (Romero-Figeroa 1997) and again Warembori). In Akkadian, for example, ARs are commonly inferred from an underspecified and hence highly versatile coordinate clause combination (*-ma* 'and'). And even where ARs are coded more explicitly, some of them may behave formally like coordinate constructions (see also Verstraete 2007 and Cristofaro 2014 on this issue) or simply be rendered by coordinate alternatives (Mithun 1984). This is particularly common with result clauses and concessive clauses: for example, the logical relation expressed by concessive clause linkage can equally well be expressed by adversative (BUT) coordination (see Dixon (2009: 28–30)). It is not surprising, therefore, that distinct ACs of result and concession are less frequently grammaticalized than those encoding the other ARs from above.

ARs are most commonly encoded by a specific marker at the boundary of the AC, such as an adverbial conjunction (as in (2), (8), (9) and (11) above) or a bound marker that attaches to the AC verb stem (18), to a nominalized or other dependent form of the verb (19) or to the whole clause:

¹⁸The terms CAUSE and REASON are not always distinguished systematically, but where they are, the term CAUSE tends to be reserved for external causality (*She couldn't go skiing because she had broken her leg*), while the term REASON captures internal causality, i.e. motivation (*She left the party because it was boring*). We will have to gloss over this distinction here, but Verstraete (2008) shows that it has empirical typological consequences.

Table 1: Positioning patterns of ACs in corpus data and a typological sample

(a) Barbareño Chumash corpus (Wash 2001: 345)

AR	PREPOSED	POSTPOSED	TOTAL
TEMPORAL	142 (66%)	72 (34%)	214
CONDITION	76 (86%)	12 (14%)	88
CAUSE	3 (6%)	47 (94%)	50
PURPOSE	8 (9%)	83 (91%)	91
CONCESSION	3 (27%)	7 (73%)	10
MANNER	2 (13%)	13 (87%)	15

(b) Diessel’s (2019a: 101) sample of 100 languages (numbers count constructions)

AR	PREPOSED	FLEXIBLE	POSTPOSED	TOTAL
TEMPORAL	119 (60%)	68 (34%)	12 (6%)	199
CONDITION	94 (91%)	9 (9%)	0 (0%)	103
CAUSE	40 (35%)	24 (21%)	49 (43%)	113
PURPOSE	33 (29%)	19 (17%)	63 (54%)	115

itate towards postposed position. The postposing preference of concessive and manner clauses in Chumash, a VO language, may be due to an overall proclivity of dependent clauses towards the postposed position; from a typological perspective, concessive and manner relations are more typically coded by preposed ACs (Hetterle 2015: 124). And result clauses, which are missing in Table 1, show the strongest cross-linguistic bias towards the postposed position (ibid.). In Ojibwe (Algic: USA, Canada), for instance, ACs show very flexible positions, but result clauses strictly follow the main clause (Valentine 2009: 207). Importantly, when the category of time clauses is broken up into more specific temporal relations, the picture usually changes such that UNTIL- and BEFORE-clauses deviate from the general preposed trend (in Chumash, for example, 93% of all UNTIL-clauses are placed after the main clause). This is all the more striking when it happens in verb-final languages that generally keep dependent clauses in preverbal position. Thus in Akkadian, it is only UNTIL- and BEFORE-clauses that can appear in postposed position while all other ACs precede the main clause (Deutscher 2009b: 64).

The primary semantic motivation for these ordering preferences is that of

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iconicity of sequence. As already observed by Greenberg (1963), AC order typically mirrors the underlying cause-effect or antecedent-consequence order of the AR in question, and this would account for the positioning preferences of conditional, SINCE/AFTER/BEFORE/UNTIL as well as purpose and result clauses (see also Diessel 2008). In addition to semantic considerations, however, there are important aspects of discourse organization that affect the placement of ACs (e.g. Chafe 1984, Thompson 1985, Ramsay 1987, Verstraete 2004). As Wash (2001: 551) remarks on her corpus of Chumash, “initial adverbial clause tokens often played a global role in structuring the discourse, while the final adverbial clauses tended to be limited to the more localized role of qualifying the main clause.” This “global” role is typically reflected by preposed ACs having scope over several ensuing clauses (see also Thompson 1985).

The discourse-structuring or grounding function is particularly prominent with WHEN- and conditional clauses (Haiman 1978), and when other ARs assume such expository roles, they are sometimes explicitly marked with topic morphology (e.g. preposed purpose clauses in Awa Pit (Curnow 1997: 280)). Conversely, the fact that causal clauses are often used to provide a more “local” justification for an immediately preceding statement (Diessel & Hetterle 2011) may explain their anti-iconic ordering preference across languages: As cause-effect relationships, we would expect causal clauses to precede their main clause, but as Table 1 shows, this is strongly dispreferred in Chumash and mildly dispreferred from a typological perspective. In contrast to temporal and conditional relations, then, causal relations are less frequently used to provide an interpretative framework for the ensuing main clause (see also Ford 1993, Thompson et al. 2007: 280ff.). Similarly to purpose and result clauses, they rather tend to encode new information. In fact, when languages have several types of causal clauses that differ in their discourse-pragmatic functions, it is usually the case that preposed causal clauses encode presupposed information (‘since’) while postposed causal clauses are reserved for new or focal information (‘because’), as in Iquito (Zaparoan: Peru; Michael 2009: 157f.).

In sum, we have seen that the semantic and discourse-pragmatic functions of the different ARs naturally lead to mixed ordering patterns in the adverbial domain. The only factor working against such heterogeneity is a consistent ordering preference for all ACs, and this appears to happen virtually only in strong verb-final languages (like Korean) that keep all ACs in preverbal position (see Diessel 2019a: 100). This makes ACs crucially different from relative and argument clauses, which have an overall cross-linguistic bias towards postnominal and postverbal position, respectively; and where languages exhibit mixed ordering patterns of relative clauses or argument clauses, they will be OV languages.

In other words, homogeneity of clause order is typical of relative and argument clauses in VO languages, but for adverbial clauses, it can only be found in OV languages (if at all). VO languages always allow at least some types of AC to precede the main clause.

3.2 Deranking of ACs

Another important aspect of the typology of ACs is the systematic correlation between specific ARs and their relative degree of deranking. Cristofaro (2003) and Hetterle (2015) provide ample empirical evidence for a deranking hierarchy of ARs, i.e. a systematic implicational scale which allows us to predict that the degree of deranking of any given AR is at least as pronounced as that of all ARs higher on the hierarchy. Figure 2 is the empirical result of Hetterle’s (2015) study; the different positions on this hierarchy show statistically significant differences in their mean degree of deranking across languages, while the ARs that share a position do not differ in a significant way:

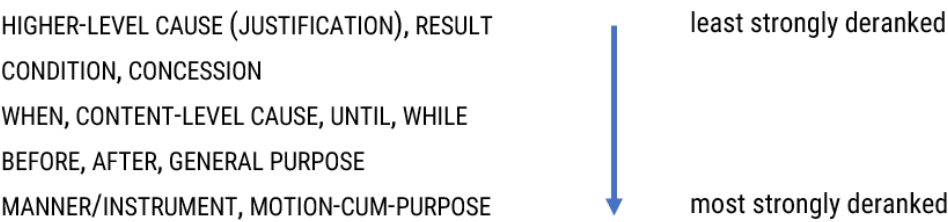


Figure 2: The deranking hierarchy of ACs (Hetterle 2015: 179)

There are (at least) two strands of functional explanation that can account for these patterns of grammatical coding, and they are convergent rather than competing motivations.

Firstly, building on a proposal by O’Dowd (1992), both Cristofaro (2003) and Hetterle (2015) invoke the idea that some ARs are open to reification, i.e. a construal as a THING in Langacker’s (1987) sense. In keeping with this hypothesis, Hetterle (2015: 84) finds that the different ARs are unevenly associated with nominalized verb forms: AFTER, BEFORE, purpose, cause and WHEN relations show a significantly higher degree of nominalization than, for example, condition, result and WHILE relations. Cristofaro uses this event-as-object metaphor to explain why conditional relations are the most sentential type of AC in her study; indeed, there are languages like Mapudungun in which conditional clauses are less deranked than any of the other ACs.

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Conversely, it is easy to see how some of the strongly deranked ARs can emerge as expansions of NPs and PPs (see also Genetti 1991). In (46), we see a nominalized causal clause with ablative case morphology that treats the event of the AC as the metaphorical source for the event in the main clause:

- (46) Kolyma Yukaghir (Yukaghir: Russian Federation; Maslova 2003: 432)
 [*Tīne tet ibil'-ōl-**get***] *ediŋ nodo-pe āj kel-ŋi*.
 recently you cry-NMLZ-ABL this bird-PL CONN come-3PL:INTR
 'Because you cried before, these birds came again.'

In a similar fashion, purposes can be seen as metaphorical goals (Lakoff & Johnson 1980), and events can also easily mark temporal reference points (AFTER, BEFORE, SINCE, UNTIL). This is again reflected in their diachronic origins: purpose clauses commonly bear allative, dative or benefactive marking (Thompson et al. 2007, Rice & Kabata 2007, Schmidtke-Bode 2009), and AFTER-, BEFORE-, SINCE- and UNTIL-clauses are transparently based on the corresponding adpositions or case affixes (e.g. Harris & Campbell 1995: 291f.). This line of explanation is also consistent with the fact that converbs (i.e. special adverbial affixes rather than case-marked nominalizations) are preferably found with ARs that are not as easily conceptualized in nominal terms, notably manner and temporal WHILE-clauses (Hetterle 2015: 92f.).

A second line of explanation for Figure 2 relates to the fact that the different ARs exhibit different degrees of what we called functional embedding in Section 1.2 above. Whilst ACs are generally not seen as argument clauses (because they are not selected by a main-clause predicator), some ACs do bear a closer semantic relationship with that predicator than others because the ARs they encode elaborate more crucial aspects of the main-clause situation. Manner clauses, to begin with, directly specify in which way the verbal action of the main clause is carried out and are hence particularly relevant, in Bybee's (1985) sense, to the main-clause verb.¹⁹ Time clauses are sometimes explicitly referred to as "temporal arguments" (see, e.g., van Valin 1984, who uses this criterion to say that temporal ACs are embedded in the main clause while chaining constructions are not). Incidentally, this is reflected in the ordering pattern in our earlier example (12)

¹⁹Bybee (1985) observed that, across the world's languages, aspectual morphology tends to appear closer to the verb stem than tense morphology. She argued that this is because aspectual meaning has a stronger impact on the construal of the verb meaning than its mere location in time by tense marking. In this sense, Bybee deems aspect to be more "relevant" to verb meaning than tense.

above, where the temporal clause attaches to the core of the main-clause predication while the causal clause appears only afterwards (see van Valin's (2005: Ch. 6) difference between 'ad-core' and 'ad-clausal' relations). Finally, a particular subtype of purpose clause, the so-called "motion-cum-purpose" construction (Aissen 1984), could likewise be seen as an argument of a main-clause verb of directed motion: wherever we go deliberately, it is for the sake of carrying out an intended action, and hence it is not surprising that many languages have a distinct motion-cum-purpose construction which formally behaves like an argument clause (Schmidtke-Bode 2009: 94ff.). In line with Figure 2, we thus find that motion-cum-purpose clauses can be more reduced than general purpose clauses, as in the following example:

(47) Huallaga Quechua (Quechuan: Peru; Weber 1989: 117)

- a. [*Miku-na-nchi:-paq*] *yanu-ku-shka-:*.
eat-NMLZ-1INCL.POSS-PURP cook-REFL-PRF-1
'I cooked it so that we could eat it.'
- b. [*Rika-q*] *aywa-shka-:*.
see-NMLZ go-PRF-1
'I went to see (him/her).'

As can be seen, the construction in (47b) employs a different (and shorter) nominalizer than the one in (47a) and the purposive marker as such is optional and often missing; we will return to this observation in Section 3.3.

Note, however, that even beyond motion-cum-purpose clauses, purposive relations are often more strongly deranked than all other types of AR, as some important semantic features of purpose clauses are easily predictable from the main clause and do not need overt coding in the purpose clause. Specifically, purpose relations involve an intentional component, have fixed time reference relative to the main-clause event and a high probability of subject sharing. As a result, purpose clauses are prone to the omission of grammatical formatives (TAM, person marking) and to syntactic reduction (e.g. control constructions) even in languages without explicit deranking morphology (Schmidtke-Bode 2009). Therefore, as "the semantic difference between purpose relations and the other ARs is greater than the difference between any of the other ARs" (Cristofaro 2003: 179), we repeatedly find systems of ACs in which purpose clauses stand out as the only structurally deranked, or a significantly more deranked, construction type (e.g. in Lango, Mekens, Ungarinjin, to name but three typologically and geographically diverse languages).²⁰

²⁰Note that Cristofaro's (2003) study does not include manner clauses, and these are, of course,

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The two parameters of nominal construal and elaboration of the main-clause predicator are logically independent, but they converge on the same result, namely ACs that behave like typical constituents of the matrix clause. As such, they also stand a higher chance to be syntagmatically embedded in the matrix clause, as illustrated by the centre-embedded purpose clause in (19) above. What we thus observe is that the more deranked types of AC can occur in centre-embedded position while the less deranked ACs in the same language need to occur at the periphery of the matrix clause (see, e.g., Warao and Wambaya). Once again, there is a correlation between the formal and the syntagmatic dimension of embedding, both of which may themselves be determined by the degree of functional embedding of an AR into the matrix clause.

The opposite patterns are found with the ARs at the top of Figure 2. As we saw above, conditional (and often also concessive) clauses provide an interpretative frame for the proposition in the main clause and are hence more peripheral – both functionally and syntagmatically – to the matrix clause than temporal, manner or purpose specifications of the main-clause predicator. Causal clauses, especially in their discourse function of providing post-hoc justification of the main-clause proposition, are equally peripheral and, like result clauses, resemble an independent (i.e. coordinated) discourse move (Diessel & Hetterle 2011). And since, additionally, none of these relations predetermines any particular TAM or person constellation, they tend to be fully-fledged clauses, or at least less deranked, than the ARs discussed above.

Interestingly, Hetterle's (2015) study also shows that when causal and concessive clauses appear in preposed position (and hence with topical/presupposed rather than focal function), they are significantly more deranked than in postposed position (see also Hengeveld 1998 on this correlation). For purpose, manner and some temporal relations, by contrast, deranking is equally prominent in pre- and postposed position. This suggests that causal and concessive clauses, in keeping with their different pre- and post-main-clause functions, may also be associated with different diachronic origins in the two positions: a scenario of expanding phrases into clauses in preposed position, and a scenario of integrating successive clauses in postposed position. This is exactly parallel to the developmental trajectories of pre- and postnominal RCs from above.

It is clear, then, that there is a systematic triangular relationship between the formal, functional and syntagmatic degree of embedding of ACs across the world's languages, just like in relative and argument clauses.

not unlike purpose clauses in many of the above respects (fixed TAM constellation, same-subject constraint) and thus show a predictably high degree of deranking, as we see in Figure 2.

3.3 Properties of adverbial subordinators

The final correlation we wish to point out is that between the position of the AC and the primary marker of the AR in question. Based on what is known from other types of complex sentence, the typologically expected position of adverbial subordinators would be at the clause boundary that links the AC to the main clause, such that postposed ACs are introduced and preposed ACs are terminated by an adverbial marker. This expectation is nicely borne out, for example, in Abun, as illustrated in (48):

(48) Abun (West Papuan: Indonesia; Berry & Berry 1999: 195, 209)

- a. [*ʃi ma sa*] *an yo ma nde tó*.
1SG come **when.RL** 3SG NEG come NEG INCOMPL
'When I came, he had not yet arrived.'
- b. *ʃi yo ma nde* [*we ji bi nggon i*].
1SG NEG come NEG **because** 1SG POSS woman sick
'I did not come because my wife was sick.'

It turns out, however, that preposed ACs are actually often found with initial markers. In particular, as shown by Diessel (2001, 2019a), preposed conditional and temporal clauses (but not causal and purpose clauses) are often marked by free initial subordinators. This is in stark contrast again with relative and argument clauses, which only rarely have clause-initial markers in prenominal or preverbal position, respectively (Lehmann 1984, Hawkins 1994, Dryer 2009).

These patterns can arise when an AR is grammaticalized in postposed position but then commonly shifted to sentence-initial position for discourse-pragmatic reasons. However, a further explanation lies in the fact that a sizeable number of preposed ACs have their diachronic roots in postnominal RCs on "adverbial" head nouns like 'time'. This is illustrated by (49) from Fongbe:

(49) Fongbe (Atlantic-Congo, Volta-Congo, Kwa Volta-Congo, Gbe: Benin, Togo; Lefebvre & Brousseau 2002: 171)

- [*Hwènú dè-è à xá àtín jí ó*], *ùn mò wè*.
time OP-REL 2SG climb tree on DEF 1SG see 2SG
'When you climbed up the tree, I saw you.'

In (43) above, we saw that 'time' ranks relatively high on the accessibility hierarchy of relativization, thus making it an easily available source for grammaticalization into a temporal AC construction. Crucially, it is this source construction

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itself that is commonly placed sentence-initially because of its expository function, even in VO languages, and the temporal AC resulting from it simply retains this positional preference. In this scenario, the combination of sentence-initial ACs and clause-initial subordinators is thus a “persistence effect” (Hopper 1991) from a source construction that was subject to similar discourse-pragmatic pressures. In keeping with this hypothesis, Hetterle (2015) finds that the adverbial linkers in her study that had a clearly identifiable origin in nouns (= 10% of all linkers) are distributed unevenly over the different ARs: they are common for *WHILE*, *BEFORE* and *WHEN* relations, but rather uncommon, for example, for *PURPOSE* and *RESULT*.

For conditional clauses, too, such persistence effects can be found. On the one hand, many sources have pointed out that conditional markers are often indistinguishable (or develop) from temporal *WHEN*-markers, thus occupying the same position in the clause. On the other hand, conditional connectors also derive from copulas (‘is (the case) that ...’, ‘it being that ...’; see Hopper & Traugott (2003: 186) on Swahili, Japanese and Chickasaw), modal markers of various kinds (‘perhaps’, ‘suppose’) and markers of polar interrogatives (ibid., Kuteva et al. 2019: 479). Where these sources are clause-initial structures, they can give rise to preposed conditional clauses with initial markers. And as also shown by Hopper & Traugott (2003: 187f.), temporal and conditional clauses are themselves sources for other ARs, thereby bequeathing their (initial) markers to causal (‘since’) and concessive (‘(even) if’) clauses.

Overall, then, it appears that synchronic patterns in the position of adverbial connectives and the resulting AC as a whole are often directly explicable in terms of their diachronic origins. As argued by Diessel (2019a), this motivates the cases in which we find a neat correlation between the position of the AC and the placement of the connector, as in (48) above, as well as (at least some of) the non-correlating cases of preposed ACs with initial markers and postposed ACs with final markers.

A final aspect of subordinators worth pointing out concerns the reduction of their substance depending on contextual factors. We saw above that ACs may contain multiple markers that cue the specific AR in question; consequently, the primary adverbial connector may be dropped if the remaining markers are judged sufficiently informative (see Hetterle 2015: 108). In the following example from Turkish, the ablative marking on the nominalized verb licenses the optionality of the causal marker at the end of the AC:

- (50) Turkish (Turkic: Turkey; Kornfilt 1997: 69)

Karsten Schmidtke-Bode & Holger Diessel

Hasan [*kitab-ı san-a ver-diğ-im-den (dolayı)*] çok kız-dı.
 Hasan book-ACC you-DAT give-NMLZ-1SG-ABL (because) very angry-PST
 ‘Hasan got very angry because I gave the book to you.’

In the recent typological and psycholinguistic literature, such patterns have attracted increasing attention under the label of redundancy management in grammar (e.g. Hawkins 2014, Jaeger & Buz 2018). As such, they join many other grammatical devices whose optional or differential occurrence is said to be motivated by efficiency: whenever a particular grammatical meaning is predictable (from the context or as a long-term frequency effect), it tends to receive shorter expression (relative to the less predictable contexts, see Haspelmath 2021 for an overview).

Following this line of argumentation, it has recently been shown that a strong predictor of subordinator omission is the relative likelihood that an argument clause or relative clause will appear given a certain main-clause verb or noun phrase, respectively (see Jaeger 2010, Wasow et al. 2011 for relevant corpus-linguistic work, and Norcliffe & Jaeger 2016 for experimental evidence). In this light, it is certainly no accident that ARs which are more intimately connected to the main-clause predicator show the same trend towards shortening and omission. First and foremost, this applies to the motion-cum-purpose clauses we encountered in (47b) above; as we saw there, the purposive connector as such is optional in purpose clauses with a main-clause verb of directed motion, and the nominalizer used in these constructions is also shorter than the one in general purpose clauses. Schmidtke-Bode (2009) shows that this is a more general typological pattern. In Korean, for example, the purposive connector commonly used with motion verbs *-(u)le* is shorter than the general-purpose connector *-ki wiha-ye* (Hwang 1997), and in Amele and Hausa (to name but two relevant languages), the purposive marker can be omitted particularly with a main-clause verb of directed motion.

Moreover, the fact that purpose clauses (in general) have a strong usage bias towards subject co-reference across main clause and subordinate clause leads, in some languages, to exactly the same reduction effects: where same-subject and different-subject purpose clauses are distinct constructions, the different-subject marker will usually be at least as long as the same-subject marker (e.g. Awa Pit); and a purpose connector may become optional only in same-subject constellations (e.g. Lango). These coding patterns are thus not only based on immediate contextual predictability but also on long-term effects of frequency-sensitive linguistic memories (see also Haspelmath 2013 for the same effects in desiderative argument clauses).

4 Conclusion: The dynamics of clausal embedding

In this chapter, we have laid out some prominent typological patterns in the structure, function and use of relative and adverbial clauses. Apart from presenting the major cross-linguistic types of RCs and ACs as such, we have attempted to situate them in the more general context of clausal embedding. Specifically, we have argued that, for both RCs and ACs, there are systematic correlations between *functional*, *formal* and *syntagmatic* dimensions of embedding: prototypically, the more a clause can be conceptualized as being functionally embedded into another unit (i.e. into a phrase or a clause), the more it will assume the formal and distributional properties of typical constituents of that unit. It needs to be emphasized, however, that there can also be mismatches between form and function (e.g. syntactically appositive nominalizations used with restrictive meaning, as in (37)), and that our three dimensions of embedding are not to be understood as discrete entities but as continua: as we have seen throughout the chapter, clauses can be more or less embedded along all of the three dimensions, and the manifold combinations that can result from this reflect the enormous diversity of clause-combining constructions across the world's languages (Figure 1; see also Lehmann 1988 on this point).

Therefore, the view of embedding espoused in this paper is a synchronically gradient and diachronically dynamic one, in keeping with the more general understanding of grammar as a usage-based dynamic system (Beckner et al. 2009, Diessel 2019b). On this view, clausal embedding is an emergent phenomenon: for both RCs and ACs, we observed that looser syntactic configurations – such as juxtaposed clauses with anaphoric links, afterthoughts or appositions of NPs and clauses – can come to be integrated into formally more asymmetrical structures over time. On this account, many RCs follow similar diachronic trajectories as other constituents of NPs; determiners, for example, often emerge from appositive demonstrative pronouns (Himmelmann 1997). Alternatively, clausal embedding may come about by successively expanding the elements of phrasal constituents: RCs can thus be derived from the expansion of other noun modifiers, such as genitival or adjectival slots, and ACs can be expansions of flagged NPs and adverbial PPs. And once grammaticalized, embedded clauses continue to be dynamic entities, being reanalyzed into semantically and discourse-pragmatically related construction types (e.g. relative > temporal clauses, temporal > conditional clauses, purpose > result clauses, etc.). They may also be analogically extended from one domain to the next (e.g. from purposive to argument environments (Haspelmath 1989), from reported speech to various adverbial relations (Heine et al. 1991, Saxena 1995) or from relative to both adverbial and argument

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relations (e.g. Givón 1991)). These developments yield dynamic networks of poly-functionality (see van Lier 2009 and Hetterle 2015 for detailed studies), and as we have argued for ACs in particular, they can often directly account for larger typological regularities (such as word-order correlations between adpositional phrases, dependent clauses and subordinators; see Diessel 2019a).

In surveying different types of RCs and ACs, we have also seen that both domains of clause linkage are subject to very similar communicative and processing-related pressures. Concerning the former, we demonstrated how considerations of information structure, such as reference management (“filler before gap”) and thematic grounding, influence the grammaticalization of different types of RC and AC. As far as processing is concerned, we saw that the more predictable the dependent clause is from its associated main-clause element (i.e. the head nominal or the main predicator) or from other aspects of the construction, the less overtly it needs to be marked itself. This principle extends from the subordinator to other aspects of economical coding, such as the gaps in relativization: the more recoverable the relativized position is from other cues in the RC construction (or the context), the more easily it can be gapped. Communicative pressures motivate which semantic roles are most commonly chosen as relativized positions, and these, in turn, are again the ones with the least overt coding.

Overall, then, both RCs and ACs cannot only be embedded in a structural sense, but they are always embedded (more metaphorically) into a language-specific ecosystem that is shaped by the cognitive, communicative and social demands of the respective language users.

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Abbreviations

The paper follows the Leipzig Glossing Rules (Comrie et al. 2008). Additional abbreviations are as follows:

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I, II, III	agreement classes (e.g. Choctaw)	NARR	narrative (tense)
ADD	additive focus (Yakkha)	NATIV	nativizer (verbal affix in Yakkha attached to certain loan words from Nepali)
ADJZ	adjectivalizer		
AM	autonomous motion (Cabecar)	NGR	n-grade (aspectual marker for duration and incompleteness in Choctaw)
ANIM	animate		
ANT	anterior (aspect)		
AOR	aorist (perfective past tense)	NLOCUT	non-locutor (3 rd -person index in Awa Pit declarative clauses)
AVN	action verbal noun (Mapudungun)	NTR	neutral form (specific inflectional form of Urarina)
C _{NUMBER}	noun class		
COBL	complementizing oblique case (Kayardild)	OP	operator
CONN	connector	OVN	object verbal noun (Mapudungun)
DEP	dependent form	POT	potential
DEST	destination	PREV	previous mention (Choctaw)
DIR	direction	PROP	propriative
DF	D-form (specific inflectional form of Urarina)	REF	reference marker (Mayogo)
EF	E-form (specific inflectional form of Urarina)	RL	realis
HRS	hearsay (evidential)	SEQ	sequential
INAN	inanimate	SS	same subject
INT	intentional	STAT	stative
LOCUT	locutor (1 st -person index in Awa Pit declarative clauses)	SUB	subordinator
		THEM	thematic suffix (Georgian)
		TNS	tense marker
		TR	transitivizer

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Chapter 5

The diachrony of embedding

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Embedded finite clauses are usually presented as the most typical form of subordination, since they are dominant in English and in most modern Indo-European languages of Europe. However, the situation is quite different if we consider not only other language families and areas, but also Indo-European itself in its early stages, as attested by Hittite, Vedic, Ancient Greek, Latin, etc. In these languages, embedded finite constructions compete with (and are often superseded by) participial and nominalizing strategies on the one hand, and non-embedded forms of clause linkage, such as parataxis and correlative structures on the other hand, so that an increase of embedded finite subordination is often observable in the history of the IE languages. We analyze this morphosyntactic tendency, as well as alternative patterns of change, with consideration of different relative, adverbial, and completive functions. We will show that factors of semantic complexity, as well as more or less formal sociolinguistic registers, also play a role in the selection of embedded or non-embedded constructions.

1 Introduction

Embedded clauses, that is, subordinate clauses (SCs) functioning as a constituent of the matrix clause, present a wide variety of structures, both across different languages and in the subordinating system of a language, which also reflect a variety of diachronic developments. In most cases, embedded clauses develop from non-embedded clauses (although the opposite change is also attested, as we will see). Their diachrony can be therefore better analysed in the framework of subordination and of clause linkage in general. In the following, we will discuss the diachronic change of embedded relative clauses (Section 2), adverbial clauses (Section 3), and complement clauses (Section 4), before drawing some



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conclusions in the final paragraph Section 4. Our materials are especially drawn from the Indo-European (IE) languages, not only because these languages represent my main field of expertise, but also because they offer a longer attested diachronic record with respect to other language families.

2 The development of embedded relative clauses

2.1 Main strategies of relativization

An embedded relative clause (RC) modifies a noun phrase of the matrix clause (MC). It is often marked by a relativizer, as the relative pronoun *who* in example (1a). The RC and its nominal head noun form together a relative construction (RCON), as indicated in (1b). The nominal antecedent can also be missing, as in (1c); in this case, RC and RCON coincide.

- (1) a. *The girl*_{RC}[*who is reading a book over there*] *is Peter's sister.*
- b. *RCON*[*The girl who is reading a book over there*] *is Peter's sister.*
- c. *RC/RCON*[*Who has written that letter*] *is certainly not a native English speaker.*

Embedded RCs can be distinguished according to whether the head noun (when present) is external or internal to the RC itself and, in the former case, whether they precede or follow their head noun. Moreover, embedded RCs also vary according to the type of relativizer (when present), which can be more or less explicit (cf. Lehmann 1984, Keenan 1985, Smits 1989, Andrews 2007, etc.; cf. also Meinunger & Alexiadou, this volume). Typological studies usually consider external postnominal RCs, as in (1a), to be the most common structure of RC (cf. Keenan 1985: 143ff). From a functional point of view, they also privilege restrictive RCs, which specify the extension of the RC domain, with respect to non-restrictive or appositive RCs, which depend instead on definite head nouns such as proper names or nouns modified by a possessor, e.g. *Peter's sister, who arrived yesterday, likes reading a lot*. In the case of non-restrictive RCs, we often have a pause in intonation between the head noun and the RC, which is graphically represented by a comma in many languages. Some languages also use different relative markers for restrictive and non-restrictive RCs, as in English, where the relative pronoun *that* can only be used with a restrictive function, or in Italian, where the relative pronoun *il quale* can be only non-restrictive instead. This syntactic variety of RCs is, however, mitigated by some typological correlations. For

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example, it is well-known that the RC is marked by more or less explicit strategies, such as full noun phrases, personal pronouns, relative pronouns or gapping, according to the position of the relativized syntactic function on the Accessibility Hierarchy (cf. Keenan & Comrie 1977). Higher positions, such as subject and direct object, are usually relativized by means of more implicit markers, while relativized functions of direct object, object of preposition/postposition, and possessor usually require heavier anaphoric resumption.

On the basis of the data of the ancient IE languages, I could observe that what is often considered to be the norm in typological studies of the RC represents a secondary development of alternative subordinating constructions, which were originally not embedded or not even structured as RCs. In particular, embedded RCs can replace, or be derived from, two kinds of relativization strategies, such as participles and correlative constructions, in the history of IE.

2.2 Relativizing participles

The participle is a non-finite verbal form so called by Graeco-Roman grammarians (*metokhé* in Ancient Greek, *participium* in Latin) because it “participates” in both verbal morphology and nominal morphology, in having tenses/aspects like verbs and grammatical cases like nouns. Its case inflection allowed to modify noun phrases with different syntactic functions in the MC. Its verbal inflection also permitted to express different relations of anteriority, posteriority or simultaneity with respect to the MC situation as well as to the speech situation. For Proto-Indo-European (PIE), we can reconstruct at least a present participle, an aorist participle, and a perfect participle, in addition to various verbal adjectives. As a strategy of RC, the participle is considered by Keenan (1985: 144) to be marginal, owing to the fact that it may relativize a limited number of syntactic functions in the languages of his database. In German, for example, the present active participle can only paraphrase a RC that relativizes the function of the subject, e.g. *ich sehe den in seinem Büro arbeitenden Mann* = *ich sehe den Mann, der in seinem Büro arbeitet* “I see the man who works in his office”.¹ Moreover, this use of the participle is quite marked in German, mainly limited to the written language. When, however, a language has morphological devices available to convert non-agentive semantic roles into the subject function, e.g. a morphologically marked voice system, the productivity of participles may increase. In PIE, while the present participle in *-nt-* (later extended to the aorist) and the perfect participle in *-wos-* are active, the participle in *-meno-* is middle-passive and therefore

¹Instead, German perfect participles are passive and can therefore promote the original object to subject position, e.g. *ich sehe das von Klee gemalte Bild* “I see the picture painted by Klee”.

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compatible with various non-agentive roles (e.g. Latin *alumnus* “nursling, pupil” from **alo-menos*, the passive participle of *alō*, -ere “to nurture”). Moreover, some verbal adjectives, such as the formations in -no- and in -to-, which originally were not linked to any specific tense/aspect, later acquired functions similar to proper participles. The verbal adjective in -to-, in particular, was associated to the passive voice in various IE languages (e.g. Latin *laudātus* “praised” vs. *laudāns* “who praises”). This also increased the possibilities of participial relativization.

These possibilities were not equally extended in all IE languages. Hittite, for example, only disposes of a participle in -ant-, which is typically active with intransitive predicates and passive with transitive predicates (e.g. *pānt-* “gone” vs. *kunant-* “killed”), and is in general the ancient IE language where participles are minimally productive. This may be related to the fact that, beyond the participle, the whole verbal system is also simple in Hittite, where the finite verb has two tenses (present and preterit) and two moods (indicative and imperative). Ancient Greek, instead, has a very elaborate verbal inflection, and may be considered to be the IE language which is also richest in participles – already in the ancient grammatical tradition, Ancient Greek was celebrated for its *philo-metokhē* “love for participles”. Especially in Homeric Greek, we can observe how participles represent the favourite strategy to express a RC, in both restrictive and non-restrictive functions, although they may also have numerous other subordinating functions beyond relativization (they may also express adverbial relations of time, condition, cause, concession, purpose, and consecution, in addition to various complement relations, cf. Section 4). An example of this is illustrated in (2), where the participle *eóntas* “being” is translated with a RC in modern European languages such as English. A proper RC, instead, introduced by a relative pronoun, is preferred in Homeric Greek with syntactically complex constructions, as in (3), where the head noun *ámpeloi* “vines” is coordinated in a series of subjects in a right-dislocation.²

- (2) énth' huías Priámoio dúō lábe Dardanídaο *ein heni*
 then son(M):ACC.PL Priam:GEN two took Dardanus:GEN in one:DAT.M
díphrō *eóntas...*
 chariot(M).DAT.SG be:PART.PRES.ACC.M.PL
 “Then he took two sons of Priam, Dardanus’ son, who were in one
 chariot” (lit. “being in one chariot”). (Homer, *Iliad* 5.159–160)
- (3) allà tá g' ásparta kaì anérota
 but this:NOM.N.PL PTC unsown:NOM.N.PL and unploughed:NOM.N.PL

²In these and in the following examples, the / indicates the end of a verse in a poetic text.

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pánta phúontai, puroi kai krithai
 all:NOM.N.PL grow:PRES.IND3PL wheat(M):NOM.PL and barley(F):NOM.PL
 ēd' ámpeloi, haí te phérousin /
 PTC vine(F):NOM.PL which:NOM.F.PL PTC bring:PRES.IND3PL
 oînon eristáphulon
 wine(M):ACC.SG made.of.fine.grapes:ACC.M.SG
 “But all these things grow there without sowing and ploughing – wheat,
 barley, and vines which bring a wine made of fine grapes.” (Homer,
Odyssey 9.1091-111)

As nominal morphology has decayed to a larger extent than verbal morphology in most IE languages, the adnominal possibilities of the participle have been reduced with time. It may be hypothesized that the change has started in relative constructions characterized by complex head nouns or by complex RCs. Eventually, even head nouns with syntactically short modifiers may have been rendered by relative constructions with a relative pronoun. This is an aspect of the more general change from synthesis to analysis which can be observed in the history of IE as well as of other language families, and which may be related to case loss or case reduction. Relativizing participles may resist longer in languages where participles represent a synchronically productive category and where nominal inflection is complex. The Baltic languages are an example of this. Moreover, contact factors may interact in the (non-)implementation of the syntactic replacement of participial RCs with embedded RCs marked by a relative pronoun. I consider the flexible use of the participle in the Baltic languages, in particular, to represent an old inheritance from PIE and, at the same time, to have been retained thanks to a Finno-Ugric substrate, as well as to a Finnish adstrate, along the lines of Heine & Kuteva (2003) “contact-induced grammaticalization”. Finnish and Uralic in general have a very flexible use of the participle, which is often rendered by RCs in English (cf. Viitso 1998: 113).

2.3 Correlative relative clauses

Correlative constructions have been often considered to be a marginal relativization strategy in typological studies in that they do not represent a noun phrase and are not embedded (cf. Keenan 1985: 163–164, Dryer 2013, etc.) – again, embedding is more or less consciously presupposed as the standard in the literature, as this is the syntactic structure typically used in English. In a correlative construction, the RC is rather adjoined to the MC. It is placed at the periphery (usually the

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left periphery) of the MC, and it does not take determiners, case markers or adpositions, which are typical for a noun phrase instead. Although the presence of a relative marker, usually a relative pronoun, signals subordination, the adjoined RC of a correlative diptych is not morpho-syntactically deranked. Its verb has a finite form, with no concord in tense or mood with the verb of the MC, and all arguments present the same form as expected in a non-subordinate clause. The RC of a correlative construction often has an internal head, which is also considered to be aberrant with respect to external RCs. The syntactic linkage between RC and MC is loose, and the MC usually present a correlative element, typically a demonstrative pronoun with anaphoric function, to resume the relativized element. The structure is like that: “which story you have told me last time, this (story) is very interesting” (instead of the corresponding embedded construction “the story you have told me last time is very interesting”). De Vries (2002: 20) analyses the structure of correlative relative clauses as the following:

correlatives [S-matrix [RC (...) N ...] [S-matrix ... (Dem) ...]]

Correlative clauses are favoured in languages having an unmarked SOV major constituents order, especially loose verb final languages, which allow right-dislocation of syntactically heavy constituents, as in Hindi. They are also common in languages with a syntactically free word order. Instead, rigid SVO or VSO languages do not have a productive use of correlative RCs. Keenan (1985: 165) mentions Sanskrit and Medieval Russian as examples of IE languages with correlative constructions. Lipták (2009a,b) presents further case studies from Dutch, Polish, Basque, Hungarian, Tibetan, etc.³, have rightly shown the importance of correlative RCs, typological studies continue to present them as a minor relativization strategy as compared to externally headed embedded RCs (cf. Dryer 2013). This is probably due to the fact that typological studies especially rely on synchronic data from modern languages, which in IE as in other language families have often developed embedding from earlier patterns of correlation.

Unlike in most modern languages of Europe, correlative RCs represent the norm in early IE. We especially find them in Vedic, as in (4) and in Hittite (5), the two earliest attested IE languages. It is not by chance that the very term “correlative diptych” (*diptyque correlative*) has been introduced by an Indologist, Armand Minard (1936), who identified various forms of it in the prose of the Śatapathabrāhmaṇa. One of his students, Jean Haudry (1973), later recognized the same construction in more or less residual expressions of other early IE languages.

³It must be remarked that, although new publications, such as de de Vries (2002) and Lipták (2009a)

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- (4) *yám* *bhadréṇa* *śávasā*
 whom:ACC.M.SG auspicious:INSTR.N.SG force(N):INSTR.SG
codáyāsi *prajāvatā* *rādhasā*
 rouse:CAUS.SUBJ2SG rich.in.offspring:INSTR.N.SG generosity(N):INSTR.SG
té *syāma*
 these:NOM.M.PL be:OPT1PL
 “Whom you will rouse with auspicious force and with generosity rich in
 offspring – may we be these ones.” (*Rig-Veda* 1,94,15cd)
- (5) *kuiš* *ammēl āppan LUGAL-uš* *kīšar[i]* *nu* ^{URU}*Hattušan*
 who:NOM me:GEN after king:NOM.SG become:PRES3SG CON Hattusha:ACC
āppa ašāš[i] *n=an* *nepišaš* ^D*IŠKUR-aš*
 again settle:PRES3SG CON-him:ACC heaven:GEN stormgod:NOM
ḥazziē[(ttu)]
 strike:IMP3SG
 “Who becomes king after me and again settles Hattusha, let the stormgod
 of heaven strike him!” (CTH 1.A, edited by Neu 1974 in StBoT 18)⁴

The Vedic sentence in (4), drawn from a hymn to Agni, the god of fire, presents a binary structure of unequal extension. The first member, a headless RC introduced by the relative pronoun *yá-*, identifies a discourse topic (the one whom the god favours), while the subsequent MC presents the resumptive *te* “these one” and expresses a comment about it (i.e. the speaker wishes to be one of these favourites of the gods). The grammatical inconsistency here observable between the singular relative pronoun and the plural anaphoric pronoun – a construction *ad sensum*, as the RC has a generic referent – underlines the loose connection between RC and MC. The Hittite sentence in (5) is taken from a historical text, the famous celebration of king Anitta (18th or 17th century BC), who destroyed the city of Hattusha and laid a curse on it. The preposed and headless RC, marked by the relative pronoun *kuiš* “who”, is resumed by the demonstrative clitic pronoun *-an* in the subsequent MC. The latter is introduced by the typical clause connecting particle *nu* (etymologically a temporal adverb meaning “and then, and”, related to Ancient Greek *nûn* “now” and **ne(w)ós* “new”), which marks the syntactic separation between the two clauses.

⁴In Hittite transcription, the sign [...] may indicate material lost in text break. Superscript writing indicates determinatives, that is, signs of the semantic category of the next noun. In (5), the superscript ^{URU} is the determiner of cities, and the superscript ^D is the determiner of gods.

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In Indic, the correlative diptych has remained as the typical relativization strategy from the earliest texts of the Veda up to the modern era. In fact, many modern studies of correlative RCs are based on Hindi, where this construction is unmarked for relativization (cf. Srivastav 1991, Dayal 1996, Bhatt 2003 among others). That is, no change from adjoining to embedding is necessary. However, *if* a change occurs in the domain of relativization, then the adjoined RC of the correlative diptych turns out to change into an embedded RC much more typically than the other way round, on the basis of my data. That is, an anaphoric resumptive of the relative construction is more easily lost than added in the MC. Since anaphoric resumptives are especially common for RCs that rank low on Keenan & Comrie (1977) Accessibility Hierarchy, it may be argued that the change from correlative to embedded RCs, if any, starts with RCs having a subject function, where anaphora can be more easily dispensed with.

Further restrictions may apply in embedded RCs, e.g. the RC tends to be adjacent to the head noun, usually postposed to it. The verb and the arguments of the RC, however, continue to be marked independently from the MC, as it would be expected in a non-subordinated clause. The reason is that on the one hand a RC presents an independent time reference with respect to the MC (e.g. *the cake that you ate / are eating / will eat is done with cricket flour*), that is, the tense/aspect of the MC does not automatically condition the tense/aspect of the RC, as it happens in some kinds of complement clauses instead (cf. Section 4). On the other, the RC only concerns a noun phrase of the MC and leaves other arguments or adjuncts unaffected. This change from correlative RC to embedded RC can be seen in the history of Latin, for example. The earliest documents of this language often offer examples of correlative diptychs, as in (6), besides examples of embedded RCs. With time, embedded RCs generalize (7), and correlative diptychs remain only in fixed expressions, e.g. in proverbs.

- (6) SEI. QVES ESENT. QVEI. SIBEI. DEICERENT.
 if any:NOM.M.PL be:SUBJ.IPF3PL rp:NOM.M.PL REFL say:SUBJ.IPF3PL
 NECESVS. ESE. BACANAL. HABERE. EEIS. VTEI. AD.
 necessary to.be Bacchanal:ACC to.have they:NOM.M.PL CONJ to
 PR. VRBANVM ROMAM. VENIRENT.
 praetor(M):ACC.SG urbanus:ACC.M.SG to.Rome come:SUBJ.IPF3PL
 DEQVE. EEIS. REBVS. VBEL. EORVM.
 about.and these:ABL.F.PL thing(F):ABL.PL where of.them
 VER[B]A. AVDITA. ESENT. VTEI. SENATVS
 words(N):NOM.PL heard:NOM.N.PL be:SUBJ.IPF3PL CONJ senate(M):NOM

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NOSTER. DECERNERET

our:NOM decide:SUBJ.IPF3SG

“If there are any who say that it is necessary to have a bacchanal, they are to come to Rome to the praetor urbanus, and the senate is to decide on those things, when their words have been heard.” (*Senatus Consultum de Bacchanalibus*, CIL I² 581, 3–6)

- (7) Gallia est omnis divisa in partes tres,
 Gaul(F):NOM is all:NOM.F.SG divided:NOM.F.SG in part(F):ACC.PL three
quarum unam incolunt Belgae,
 RP:GEN.F.PL one:ACC.F.SG inhabit:PRES.IND3PL Belgae:NOM.PL
 aliam Aquitani, tertiam *qui*
 another:ACC.F.SG Aquitani:NOM.PL third:ACC.F.SG RP:NOM.M.PL
 ipsorum lingua Celtae, nostra Galli
 their:GEN.PL language(F):ABL.SG Celts:NOM.PL our:ABL.F.SG Gauls:NOM.PL
 appellantur
 call:PRES.IND.PASS3PL

“All Gaul is divided into three parts, one of which is inhabited by the Belgae, another by the Aquitani, the third by those who are called Celts in their own language and Gauls in our language.” (Caesar, *De bello Gallico* 1,1)

The text in (6) is drawn from the *Senatus Consultum de Bacchanalibus*, an Old Latin senatorial decree, engraved on bronze and dating to 186 BC, which prohibited the popular festivals in honour of Bacchus. Here the relative pronoun *quei* (corresponding to Classical Latin *quī* “who”) depend on a previous SC (the conditional “if there are any”) and is resumed in the subsequent MC by the demonstrative pronoun *eeis* (corresponding to Classical Latin *iī* “these, they”).⁵ The text in (7), instead, belongs to Classical Latin. It presents the beginning of Caesar’s commentary on the Gallic war. Here we have two RCs, both embedded. The former, introduced by the genitive relative pronoun *quārum* “of which”, is immediately postposed to its head noun *partēs trēs* “three parts” (lit. “three parts, of which”

⁵The form *eeis*, from a demonstrative stem **eyo-*, represents the hyper-characterization of the regular NOM.M.PL marker *-ī* with the plural ending *-s*, common for many other nominal and pronominal stems, cf. the corresponding Oscan form *insc* “*iī*”. The section of the *Senatus Consultum de Bacchanalibus* in (6) would be rendered in Classical Latin as: *si qui essent, qui sibi dicerent necesse esse Bacchanal habere, iī ut ad pr(aetorem) urbanum Romam venirent, deque iis rebus, ubi eorum verba audita essent, ut senatus noster decerneret*. The correlative elements of the relative pronoun and of the demonstrative pronoun are here in bold.

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etc.). The second RC, marked by the nominative relative pronoun *quī* “who”, is headless (“who are called” etc.). This does not imply that correlative RCs disappear in Caesar or in Classical Latin in general – they do not – or that embedded RCs are not found in Old Latin. As I said, both constructions coexist, partly overlapping and partly preferred in different functional domains. For example, correlative RCs seem to prevail in Latin in restrictive RCs with a generic interpretation, according to my data, while embedded RCs are rather favoured with restrictive RCs which identify a specific referent, as well as with non-restrictive RCs. I rather mean that correlative RCs are more common in Old Latin than in Classical Latin, and that embedded RCs are more common in Classical Latin instead.

The correlative diptych decays in Romance. However, it may be interesting to notice that in spoken communication and informal registers, Romance languages such as Italian may still present examples of adjoined RCs. I have recorded the example in (8) in a natural conversation in Italian, where a mother addresses her son who was used to take a lot of food without eating it up. The Standard Italian correspondent construction would be embedded (*Devi mangiare la pasta che prendi*, with the RC after the MC). Corresponding constructions are also acceptable in the spoken form of other Romance languages. Spoken French, for example, also admits a sentence like (8) : *les pâtes que tu prends, tu dois les manger* (besides the Standard French construction *tu dois manger les pâtes que tu prends*). This suggests that the choice of an embedded or adjoined RC (the latter being characterized by an anaphoric resumptive) is subject to sociolinguistic variation, in addition to its diachronic and regional underlying factors. Synchronically, sentences like (8) are analysed as left dislocations of an NP with a regular RC plus resumption of the relative construction by a clitic (cf. Cinque 1990). It is important to notice, however, that they represent the same structure as the ancient correlative diptych.

- (8) La pasta che prendi poi la devi mangiare.
 DEF.ART pasta:F.SG which you.take then it:F.SG you.must eat
 “You must eat the pasta you take.” (lit. “The pasta that you take, then you must eat it.”)

2.4 Paratactic constructions with demonstrative pronouns

Correlative RCs are not the only finite constructions that developed into embedded RCs. Otherwise, embedding may arise from the reanalysis of independent clauses, for example when a demonstrative pronoun with anaphoric function is

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reinterpreted as a relative pronoun. Accordingly, a construction consisting of two asyndetically linked independent clauses, such as [*I see a man.*] [*That is sitting over there*], may be reanalysed as one complex sentence, formed by a MC and a postposed RC, such as [*I see a man* [_{RC} *that is sitting over there*]]. This is probably the origin of the RCs with the D-type of relative markers in Germanic, that is, of the RCs marked by a relativizer that is etymologically derived from a demonstrative pronoun, e.g. German *der, die, das*. D-relatives represent in fact the most ancient type of relative markers in Germanic – they are dominant not only in the whole history of German, but also in Gothic and in Old English.⁶: 420ff, and more recently Bacskai-Atkari 2020, among others). For the development from correlative adjunction to embedding in Old English and Middle High German, cf. Bianchi (1999), who uses this diachronic scenario to justify Kayne’s antisymmetric view of syntax. A paratactic structure especially characterizes what Gärtner (2001) calls “integrated verb second” clauses, that is, a type of RC which presents the verb in the second position, like independent clauses and unlike regular RCs (the latter share the same verb final word order as other subordinates). Gärtner shows how a regular RC (e.g. *das Blatt hat eine Seite, die ganz schwarz ist*) may be replaced by a verb second integrated clause (e.g. *das Blatt hat eine Seite, die ist ganz schwarz*), and how further syntactic properties nonetheless may distinguish these clauses. For example, verb second integrated clauses must be verb final and immediately follow the finite verb of the MC, rather than their head noun. Despite the presence of a *rp*, verb second integrated clauses are therefore not really RCs in the syntax – according to Gärtner, they may be better analysed in terms of parataxis or as hybrid constructions between parataxis and hypotaxis. Cf. also Zifonun (2001).

Other IE branches also attest cases of the change from a sequence of independent clauses, with an anaphoric pronoun in the second clause, into a complex sentence having an exocentric and postnominal RC. Homeric Greek, for example, presents RCs marked by a demonstrative pronoun *hō, hē, tō* “this” (M, F, N), the source of the definite article in Classical Greek, besides RCs introduced by the proper relative pronoun *hós, hē, hó* “who, which” (M, F, N), cf. Schwyzler (1950: 19ff; 642). We probably deal with a polycentric syntactic change, independently occurring in different times and in different areas of IE (cf. Heine & Kuteva 2002: 113–115, Schmidtke-Bode & Diessel 2025 [this volume]). We therefore do not nec-

⁶In Germanic, the ancient correlative construction has only left few traces, and other kinds of relative markers, such as W-relatives (derived from an interrogative-indefinite pronoun, e.g. English *who, which*) and 0-relatives or gapping, emerge later and are also synchronically more constrained (cf. Helgander 1971, Harbert 2007)

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essarily have to presuppose the reconstruction of RCs marked by a demonstrative relative marker for PIE.

3 The development of embedded adverbial clauses

3.1 Main strategies of adverbial clauses

An adverbial clause (AC) is a subordinate clause that modifies the MC like an adverb modifies the verb and that has therefore the syntactic function of a circumstantial element in the complex sentence. ACs are typically embedded, that is, they represent a constituent of their complex sentence, as in (9), where different ACs are written in brackets for different adverbial relations. Like adverbs, ACs are semantically heterogeneous and may code relations of time, as in (9a), condition (9b), cause (9c), concession (9d), purpose (9e), consecution (9f), etc.

- (9) a. Temporal AC: [*After arriving home*], *Peter takes a shower*.
- b. Conditional AC: [*If he arrives late*], *Peter skips his dinner*.
- c. Causal AC: [*Since it is raining*], *Peter stays at home*.
- d. Concessive AC: [*Although it is raining*], *Peter goes out*.
- e. Purposive AC: *Peter goes out* [*to buy some food*].
- f. Consecutive AC: *It rained a lot*, [*so that the street is completely wet*].

ACs are quite diverse from a formal point of view, as they may be marked by a particular subordinating conjunction, a form of the verb which is different from a verb in an independent sentence, or a special word order (cf. Thompson 1985, revised 2007 with Hwang). In (9), for example, we have both finite and non-finite subordinates, in the latter case marked with a participle, as in (9a) or with an infinitive, as in (9e). However, while adverbial relations of time, condition, cause, etc. are universal, ACs are not. In some languages, these relations may be left implicit and recovered from the context. In some others, adverbial relations may be explicitly expressed by non-subordinating strategies, such as coordination or clause-chaining. Embedded and non-embedded constructions may compete in the same language for the expression of an adverbial relation. The relation of cause, for example, can be rendered in English not only by embedded ACs marked by subordinators such as *since*, *because*, *as*, etc. but also by the conjunction *for*, which rather belongs to coordination. This can be clearly seen in German, where SCs and matrix/independent clauses are associated to different word orders, and where the particle *denn*, the functional equivalent of English *for*, selects the verb-second word order typical of independent

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clauses, unlike causal subordinators such as *weil* or *da*. Of course, subordinating and coordinating adverbial relations are not synonymous, and their usage is conditioned by various semantic and pragmatic features. For example, coordinating conjunctions of cause, such as English *for*, German *denn*, French *car*, etc. introduce focused information, while their subordinating counterparts commonly mark clauses representing a presupposed situation. That is, clauses introduced by subordinating connectives are usually backgrounded in their entirety, while clauses with non-subordinating connectives are not. Moreover, subordinating and non-subordinating strategies of adverbial relations may go back to different diachronic stages. As in the case of RCs, for ACs as well we can observe that non-embedded constructions are generally used earlier than embedding, and are modified into (or replaced by) embedded structures only at a later stage of a language. In my IE data, the earliest constructions used to express adverbial strategies are RCs (Section 3.2) and parataxis (Section 3.3), the latter presenting asyndeton or coordinating conjunctions.

3.2 Correlative adverbial clauses

The adjoined structure of a RC in the form of the correlative diptych, illustrated above in Section 2, can also be used to convey various adverbial relations. In this case, an adverbial subordinator of time, cause, purpose, etc. in the ACs is correlated with a semantically corresponding demonstrative adverb in the MC, e.g. “When he arrives home, then he watches the news” or “The sooner you’ll come, the quicker we will be finished”. This old correlative pattern is attested in most early IE languages. Axel-Tober (2023), for example, suggests that, in Middle High German, correlative *da*-clauses were reanalysed as embedded adjunct clauses because left-peripheral correlatives are surface-structurally identical to left-dislocated embedded clauses. Correlative adverbial clauses are still quite productive in Lithuanian, a language known for being conservative in several aspects of its grammar. Lithuanian presents correlations not only between *tàs* “that” and *kàs* “who” in the domain of relativization, but also between various adverbial elements such as *taip* “so” and *kàd* “that”, *tiék* “so much” and *kíék* “how much”, *tadà* / *tuomèt* “then” and *kaì* / *kadà* “when”, *tòl* “until” and *kòl* “until”, *teñ* “there” and *kuR* “where”, etc. According to Ambrazas (1997: 723), these correlations “serve to stress and reinforce the link between the clauses. In a way, they represent the subordinate clause in the structure of the main clause”. Among the many examples, he reports the following (p. 738):

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- (10) kalbėsiu tiesą ir tada, kai pavojinga
 speak:FUT1SG truth:ACC even then when dangerous
 “I will speak the truth even when it is dangerous.” (lit. “even then when”
 etc.)

My interpretation is different from that of Ambrazas (1997). I do not think that correlative structures have the function of reinforcing the linkage between SC and MC. Otherwise we cannot explain the difference from other ACs favouring embedding, which represents a more integrated clause linkage with respect to correlation. I rather maintain that they serve to express separation and balance between the situations represented in the two clauses.⁷

If we consider the functional load of correlative constructions, we can observe that they are not equally common for all adverbial relations. Diachronically, they turn out to be the structural source especially for relations of space (introduced in English by the subordinator *where*), time (cf. English subordinators such as *when*, *while*, *as*, *before*, *after*, *since*, *until*, *once*, *as soon as*, etc.), and manner (cf. English subordinators such as *like* and *as*), in both ancient and modern IE languages. In Ancient Greek, for example, a language where embedded ACs largely prevail over adjoined ACs in general, correlative structures are especially found for 1) expressions of location (*hoû / hópou / hóthi* “where” ... *éntha / entháde / entaûtha / ekeî / autoû* “there”), direction (*hoî / hópoi* “where” ... *ekeîse* “there”, etc.), origin (*hóthen / hopóthen* “from where” ... *énthen / enthénde / enteûthen / ekeîthen* “from there”), perlocative movement (*hêi / hópēi* “through where” ... *têide / taútēi* “through there”) – all expressing space; 2) expressions of time (*hóte / hopóte* “when” ... *tóte* “then” or *tēníka / autíka* “immediately at that point”) and age (*hēníka / hopēníka* “at which age” ... *tēníka / tēnikáde / tēnikaûta* “at that age”); 3) expressions of manner (*hōs / hópōs* “how” ... *hōs / hódē / hoútōs* “so”). Now, ACs of space, time, and manner are considered to be simple or primary in typological studies with respect to other ACs, since they can be paraphrased by a RC modifying a head noun of space, time or manner (cf. Thompson 1985: 177ff). For example, the temporal AC *When he arrives home* can be reformulated as the relative construction *The time (at which) he arrives home*. Similarly, the space AC *Where you can buy croissants* can be reformulated as the relative construction *The place at which you can buy croissants*. The manner AC *How you do it* can be reformulated as the relative construction *The way (in which) you*

⁷It may be remarked that the sentence in (10) would be much better with the main verb *sakysiu* (the FUT.1SG OF *satyti* “say”, a cognate of English *say* and German *sagen*). The omission of the correlative element *tada* is possible but considered to be less natural than the correlative variant above (Daniel Petit, p.c.).

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do it, etc. As correlative structures represent a major source of embedded RCs, it is understandable that variants of the correlative diptych also serve to form those ACs which are semantically associated with RCs, as in the case of space, time, and manner. Alternatively, among other types of ACs, correlative constructions especially emerge for consecutive clauses, which express the consequence of the situation represented in the MC. The consecutive relation structured as an embedded AC in (9f) may have a correlative variant: *It rained so much that the street is completely wet*. The latter may be even more natural than the embedded construction in (9f).

I find it interesting that, although in most IE languages adverbial correlative structures have been superseded by embedded ACs introduced by an adverbial subordinating conjunction, they especially persist precisely in the two domains of space, time, and manner ACs on the one hand, and of consecution on the other. This can be seen in Italian, for example. In Italian, as in other Romance languages and in Latin already, embedded ACs are much more frequent than correlative ACs. However, correlative ACs are common in the specific domain of consecutive clauses (*così ... che* “so ... that”, *tanto ... che* “so much ... that”) and of comparative clauses (*tanto ... quanto* “so much ... how much”, *tale ... quale* “as ... such”, *più ... che* “more ... than”, *meno ... che* “less ... than”, etc.), where the form *che* is to be analysed an adverbial subordinator rather than as a complementizer. Comparative clauses are a kind of subordination of manner, whereby the situation of the MC is presented as equal, major or minor, better or worse than the situation of the subordinate with respect to a certain parameter, as in (11).

- (11) a. Il paesaggio era così bello come lo
 the landscape be:IPF.IND3SG so beautiful how it
 ricordavo.
 remember:IPF.IND1SG
 “The landscape was as beautiful as I remembered.”
- b. Alla fine è andata meglio di quanto (non) mi
 in.the end be:PRES.IND3SG gone better than how.much NEG me
 aspettassi.
 expect:IPF.SUBJ1SG
 “In the end, it went better than I had expected.”

Comparative clauses intrinsically have a binary structure, which is properly underlined by the correlative pattern. The presence of a correlative element in the MC usually indicates that the sentence is not concluded and that another clause will follow (in (11b) as well, the MC *alla fine è andata meglio* implies the

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mention of a situation that has the function of the standard of comparison). The position of the two correlatives may vary – they can be adjacent, as in (11b) *meglio di quanto* “better than that”, or non-adjacent, as in (11a) *così ... come* “so ... how” – but syntactically one belongs to the MC and one to the AC. Complex sentences with consecutive clauses also have a semantically balanced structure, describing in this case a relation of cause-effect, as in (12), again from Italian.

- (12) Ha passato **così tante** ore al computer **che** gli fanno
 has spent so many hours at.the computer that to.him hurt:PRES.IND
 male gli occhi.
 3PL the eyes
 “He has spent so many hours at the computer that his eyes are hurting.”

Other ACs instead, are less prone to be conceptualized in a binary, balanced structure. They usually express a situation that is presented on the background of the MC situation. Purposive clauses, for example, are semantically similar to consecutive clauses, in that both refer to a situation that follows the MC situation. However, while consecutive clauses are factual, purposive clauses are not. They refer to an aim or a wish of the MC subject. The two clauses are clearly on different levels. This may be the reason why other ACs may lose the ancient correlative structure and are embedded earlier than consecutive clauses. According to my data, conditional clauses are the ACs that are least associated with correlative constructions in the ancient IE languages. That is, unlike many other ACs, clauses introduced by a conditional marker such as “if” usually do not trigger a resumptive such as “in that case”, “then”, or “so”, in the MC (while correlative elements such as “then” are quite common in MCs connected to temporal subordinates, for example). We have seen in Section 2 that correlative constructions are especially common in Hittite relativization, and that the typical RC in Hittite is adjoined and not embedded. The same holds for most Hittite ACs – but not for conditional clauses. This can be clearly seen in the laws, in which a conditional protasis and its following apodosis represent an offense and its punishment, respectively. E.g. “If anyone blinds a male slave in a quarrel, he shall pay 20 shekels of silver” (cf. KBo 6.4). In these cases, the conditional marker “if” (*takku* in Old Hittite, *mān* especially in New Hittite) is usually not accompanied by a correlative element (and in Old Hittite neither by an introductory conjunctive particle) in the MC, cf. Friedrich (1959: 88); Hoffner (1997: 12).⁸ Among the modern IE

⁸In New Hittite, instead, the introductory conjunctive particle *nu* is used with conditional clauses as well, by analogy with the dominating correlative pattern used in subordination in this language.

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languages, even in a language rich in correlative clauses such as Lithuanian, the conditional conjunction *jėi / jėigu* “if” does not trigger a correlative. Condition is a logical operator (although the use of conditional clauses cannot be reduced to logical meanings). A conditional clause typically expresses an implication or a prediction which affects the realization of the MC situation. One situation depends on another. They are not represented on the same conceptual level. This may explain why conditional clauses are much more commonly embedded than structured in the adjoined structure of a correlative diptych.

3.3 Paratactic adverbial clauses

As anticipated above, in addition to correlation, ACs can also develop from originally paratactic constructions. The juxtaposition or coordination of two clauses referring to a series of chronologically ordered events can be easily reinterpreted as a temporal AC, in particular as a *when*-clause or as an *after*-clause (*Peter goes to the restaurant. He orders lasagne* > *When/after Peter goes to the restaurant, he orders lasagne*). When the situation of one of the two clauses is meant as depending on the occurrence of the other, we may also have the interpretation of a conditional clause (*If Peter goes to the restaurant, etc.*). The logical fallacy “post hoc ergo propter hoc” brings about the reinterpretation of a temporal clause into a causal clause (*Since/as Peter goes to the restaurant, etc.*). The etymology of adverbial subordinators may bear traces of these reinterpretations. In English, the subordinator *since* originally expressed a temporal relation (cf. it derives from Old English *sīþþan* “afterwards, since, ago” < *sīþ þæm* “after that (time)”), and only later it subsumed a causal relation. In German, the subordinator *wenn* (cognate of German *wann* “when”) marks habitual temporal clauses and conditional clauses, and so on. An adverbial relation that implies another adverbial relation is semantically more complex. A causal relation is more complex than a temporal relation because a causal relation implies a temporal relation of anteriority, while a temporal relation does not necessarily entail a causal component. According to my data, the use of embedded rather than paratactic ACs is established according to a cognitive path of increasing semantic complexity. Embedding emerges earlier for temporal ACs (both *after*-clauses and semantically less specific *when*-clauses), for example, than for causal clauses. We have seen already that causal relations are still represented in most modern IE languages by coordinating conjunctions, such as English *for*, German *denn*, French *car*, Italian *infatti*, etc., besides the corresponding adverbial subordinators. Nowadays, coordinate causal clauses are rather marginal with respect to subordinate causal clauses. In English, *for*-clauses are rarer than *as*-, *since*- or *because*- clauses. In the ancient

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languages, instead, we have the opposite situation. In Vedic texts, for example, causal clauses introduced by the particle *hí*, the functional correspondent of English *for* or French *car*, are much more common than causal clauses marked by the subordinator *yád* (cf. Viti 2007a: 134ff).

3.4 Factors of semantic complexity

Different degrees of semantic complexity can also be identified within the domain of temporal relations. *When*-clauses and *after*-clauses refer to states of affairs that are anterior to, or simultaneous of, the MC situation. They are therefore semantically less complex than *before*-clauses, which locate the MC situation with respect to an event that has *not* yet happened. The sentence *Before I go to work I take a rich breakfast* implies that, when I take my breakfast, I have not yet gone to work. Occurred events are much more natural than non-occurred events as temporal frames. This may explain why the explicit coding of *before*-clauses by embedded strategies establish at a relatively late in languages. Vedic, for example, possesses several subordinating structures for *when*- and *after*-clauses, introduced by (non-synonymous) subordinating conjunctions such as *yád*, *yadā*, *yádi*, *yád*, etc., which are also used for relations other than temporal. Despite their polysemy, however, Vedic temporal subordinators do not express *before*-clauses. Although we must be careful in evaluating the absence of a structure from a dead language – it may occur that a structure is not attested by chance – we have to consider that Vedic is attested by a huge corpus. When we therefore observe that *when*-subordinate clauses (more commonly meant as *after*-clauses, in the sense of temporal anteriority, than in the sense of temporal simultaneity in Vedic) are largely attested while *before*-subordinate clauses are not attested at least as a productive strategy, we can well infer a different weight of these subordinate clauses in this language. This does not mean that relations of temporal posteriority lack expression in Vedic. They were rather expressed by non-subordinating strategies. An embedded sentence such as *The sacrificer eats before other men have eaten* would be rendered in this language by means of paratactic constructions such as *The sacrificer eats; other men have not eaten*, or by a single clause where this temporal relation is coded on an adjective or adverb meaning “first, anterior, before”. Similarly, Satapathabrāhmaṇa 1,1,1,8, defines as unbecoming the situation of somebody “who takes food before men (i.e. his guests) have eaten”: *yó manuṣyēṣv ánaśnatsu pūrvo ’śnīyād*, lit. “who takes food earlier than men that do not eat”.⁹ The compositional and often transparent form of subor-

⁹Here the adjective *pūrva*- “in front, previous to”, inflected in the nominative in agreement with the clause subject, governs the locative complement *manuṣyēṣv ánaśnatsu* “men that do not

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dinators of *before*-clauses confirm their relatively late emergence. In Classical Sanskrit, for example, *before*-clauses receive an explicit marking by means of the polyrhematic conjunction *yāvan na*, where the negation *na* precisely expresses that the denoted situation has not yet happened. This complex subordinator has usually a correlative element *tāvat* in the matrix clause. Since the correlation *yāvat ... tāvat* typically means “as long as ... so long”, in Vedic already, the Classical Sanskrit complex marker of *before*-clauses literally means “as long as not ... so long”. That is, even at this relatively late stage, *before*-clauses are not embedded. They are adjoined in the form of the correlative diptych, as in (13).

- (13) tad yāvan na kaścīd vetti tāvac chīghraṃ
 then as.long.as NEG anybody:NOM know:PRES.IND3SG so.long quickly
 gamyātam iti
 go:imp.pass3sg QUOT
 “Then before anybody knows (of it), go away quickly!” (*Pañcatantra* 1.276,
 edited by Ramchandra Jha, 6th edition, Varanasi, Vidyabhavan (1991))

Similarly, Hittite possesses many more resources for *when*-clauses (introduced by the subordinators *mān* in Old Hittite, and *maḥḥan* and *kuwapi* in New Hittite), as well as for *while*- and *until*-clauses (marked by the subordinator *kuitman* in both stages of Hittite) than for *before*-clauses. The latter may be expressed by some uses of the relatively recent composite subordinator *kuitman ... nāwi*, lit. “while ... not yet”, also in the opposite order *nāwi ... kuitman*. E.g. *kuitman ... nāwi eṣḥat* “before I sat” (KBo III 4 I 3f), which some scholars still prefer to render “as long as I had not yet sat” (cf. Friedrich 1960: 138). The use of expletive negation with *before*-clauses can find several linguistic parallels. Krifka (2010) analyses this phenomenon in German sentences such as *Bevor wir nicht gegessen haben, gehen wir nicht nach Hause* “Before we have eaten, we do not go home”. He relates this construction to the specific type of negation: expletive negation is possible with a negation working as an adverbial modifier, as German *nicht*, but is not possible in English, where negation is expressed by a separate head such as the *do*-support. The latter, Krifka observes, is rather tied to assertion.

A semantic component of negation is also implied in concessive clauses, which express a contrast with respect to a certain presupposed situation. The concessive sentence *Although it is raining, Peter goes out*, reported in (9d), presupposes

eat”. Alternatively, if the latter phrase is interpreted as a locative absolute, the sentence above may be literally translated as “who eats first, while men do not eat”. Absolute constructions are usually motivated by clausal syntax in Vedic, in this case by the locative meaning implied by *pūrva*- (with a metaphor from space to time). In any case, the relation of temporal posteriority is expressed by an adjective of time rather than by subordination in this passage.

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that under normal conditions people do *not* go out when it rains. In the domain of coordination, a similar relation is conveyed by the adversative conjunction *but*. It is therefore understandable that concessive relations, as other semantically complex adverbial relations, are coded by embedded SCs at a relatively late stage of a language. This can also be seen in the transparent form of their subordinators, which are often polymorphemic, or even polyrhematic, containing expressions of negation, contrast or focus. In the modern languages of Europe, consider concessive subordinators such as English *although*, *even though*, German *obwohl*, *wenngleich*, French *bien que*, *malgré que*, Italian *nonostante*, Spanish *aunque*, etc. (cf. König 2006). A similar situation can be observed in the ancient IE languages, where concessive subordinators are also synchronically transparent. Moreover, in the ancient languages, embedded SCs explicitly marked by a concessive subordinator are much rarer than implicit subordination, especially represented by the participle; they are also rarer than coordination or juxtaposition, optionally reinforced by discourse particles, for the concessive function. Originally, a complex sentence such as *Although they were few, they won the war* would have been better reformulated into a participial structure such as *Being few, they won the war* or into paratactic constructions such as *They won the war. And yet they were few*, or *They were few, but they won the war*. Correlation was also possible, in the form *Although they were few, nevertheless they won the war* – recall that correlative structures are not embedded.

Even Ancient Greek, which we have seen to privilege embedding over correlation in general, attests concessive subordinators formed on the basis of the coordinator *kaí* “and”, cf. *kai ei* and *kai eán*, meaning “although, even though”, as well as participial constructions marked by *kaíper*, *kai mala*, *kai taúta*, etc. In these cases, Ancient Greek *kaí* can also have the function of an additive focus adverb, in addition to its basic coordinating function. In Hittite, the polysemic subordinator *mān*, meaning “when, if, although” typically presents a correlative structure in concessive clauses, such as *mān ... (y)a* “although ... nevertheless”. Similarly, in Vedic, concessive subordinates introduced by the composite adverbial conjunction *yác cid hí* “although” (formed by the relativizer *yád*, the indefinite particle *cid*, and the causal coordinator *hí*) are largely minoritarian with respect to concessive participles, optionally marked by particles. In Classical Sanskrit, the concessive marker *yady api* “even if, although” (based on the subordinator *yadi* “when, if” with the focus particle *api* “and, also, even, very”) is usually accompanied in the MC by the correlative *tathāpi* “nevertheless” (formed correspondently by affixing the particle *api* on the demonstrative adverb of manner *tathā* “in that manner, so, thus”). Again, Old Indic prefers the adjoined structure of the correlative diptych to embedding to code an adverbial relation, as can be seen in

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(14), illustrating this time a concessive conditional. All this indicates a syntactic change from parataxis or correlation to embedding for concessive relations.

- (14) **yady api** tvadiya-vacanam na karoti **tathāpi**
 although your-word(N):ACC.SG NEG make:PRES.IND3SG nevertheless
 svāmī sva-doṣa-nāśāya vācyaḥ
 master(M):NOM.SG own-fault-destruction:DAT to.be.blamed:GV.NOM.M.SG
 “Even if he does not listen to your words, you must blame your master, in
 order that he may amend his faults.” (*Pañcatantra* 1.171)

3.5 Adverbial infinitives

This example can also illustrate the early non-subordinating coding of another type of adverbial relation, that is, purpose. In (14), the purposive sentence of the English translation “in order that he may amend his faults” is expressed in Classical Sanskrit by the dative of an action noun (in this case a compound literally meaning “for the destruction of his own fault”). The verbal noun or infinitive was the primary expression of purposive relations in Old Indic, as well as in other early IE languages, and in PIE. Embedded subordinates with a purposive function are also attested, but are clearly secondary with respect to the infinitive in IE. In Vedic, embedded purposive subordinates marked by conjunctions such as *yád* or *yáthā* are much rarer than purposive infinitives. In IE in general, purposive SCs are formed with language-specific strategies, which cannot be reconstructed for PIE.

It must be stressed that the infinitive was originally more a noun than a verb in IE, according to a scalar view of parts of speech, as it was marked for case, like nouns, but not for tense, mood or voice, like verbs. In Old Indic, the infinitive usually had the case ending of the dative, which is called *sampradāna-* “bestowal” in the Sanskrit grammatical tradition (cf. Pāṇini I.4.32), and which expresses the recipient, the beneficiary, and the aim of an action. The subject of the purposive infinitive is usually the same as the subject of the MC situation, and is therefore left implicit.¹⁰ For the direct object of the infinitive, if any, we have more vari-

¹⁰In (14), the different subjects of the MC (“you”) and of the purposive clause (“in order that he” etc.) are just in the translation, as the passive construction would be unnatural in English in this case. In the Sanskrit original, however, the name of the master (svāmī) is the subject of a nominal passive clause, expressed by the gerundive vācyaḥ “that is to be blamed” (“the master is to be blamed, in order that he may destroy” etc.). That is, the MC and the action noun share the same subject. In case they differ, we can have a double dative (a sentence such as *The Hotṛ sings a poem in order that Indra may drink the soma* would be rather formulated as *The Hotṛ sings a poem for Indra, for drinking the soma* (or *for drinking of the soma* with a genitive complement).)

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ation. In (14), the direct object is coded as the first member of a compound. In Vedic, where compounding is much more restricted than in Classical Sanskrit, we would rather have an infinitive with a genitive object or an accusative object (*sva-doṣasya nāśāya* or *sva-doṣam nāśāya*). The genitive is the form expected in a possessive structure, and represents the originally nominal syntax of the verbal noun, while the accusative indicates the acquisition of a verbal syntax for this category. Irish, where verbal nouns are extremely productive from the earliest texts up to now, only maintains the objective genitive in this use.

Later, on the one hand, the IE infinitive attained an increasing number of verbal features in syntax and in morphology, e.g. tense and voice marking, as in Greek and in Italic. Once integrated in the verbal system, the infinitive could express many more subordinating relations (especially in the domain of complementation, as we will see below) which were originally impossible for this category. This may be due to the fact that a verb is more typically used than a noun to represent events. It is therefore understandable that verb-like infinitives may be more prone to express clausal meanings. On the other hand, purposive embedded clauses introduced by subordinating conjunctions also become more common with time, especially in case of different subject between MC and SC. We can therefore observe the development from a circumstantial element or a clause adjunct expressing aim, syntactically included into the MC in origin, to an embedded subordinate. It may be argued that this relatively late use of embedding for purposive relations can be also explained by their semantic complexity, as for other ACs analysed above. Similarly to *before*-clauses, purposive clauses refer to a situation that has not yet occurred at the time the MC situation takes place. The sentence *Peter goes out to buy some food*, illustrated in (9e), implies that the action of buying some food has not yet happened when Peter goes out. Also in this case, it is more natural to frame a situation with respect to an already occurred event than with respect to an event which has not yet happened – and which we do not know whether it will happen, as purposive clauses are not factual. Also in this case, as for *before*-clauses, it is understandable that embedded adverbial clauses are neither the earliest nor the most common strategy attested for purposive clauses.

4 The development of embedded complement clauses

4.1 Main strategies of complementation

Similarly to AC, complement clauses (CCs) are also extremely diverse in meaning and, consequently, in structure – not only across different languages, but

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also within the same language. They may depend on different complement taking predicates, such as predicates of utterance (cf. English *say, tell, ask* (in order to know), *report*, etc.), propositional attitude (*think, believe, suppose, assume, doubt, deny*, etc.), knowledge (*know, discover, realize, find out, forget*, etc.), immediate perception (*see, hear, feel*, etc.), pretence (*pretend, imagine, trick (into thinking)*, etc.), fearing (*fear, be afraid, worry, be anxious*, etc.), achievement (*manage, chance, dare, happen to, try, fail, avoid*, etc.), as well as on commentative predicates (*regret, be odd, be important*, etc.), desiderative predicates (*want, desire, wish, hope*, etc.), manipulative predicates (*make* someone do something, *let* something occur, *force, persuade, ask* (to do), etc.), modal predicates (*can, be able, may, must, should*, etc.), phasal predicates (*start, begin, continue, stop, finish*, etc.), etc. Cf. Noonan (1985), Cristofaro (2003: 95ff), Dixon (2006). Different complement taking predicates may select different types of CCs, represented by embedded subordinate clauses, paratactic clauses, nominalizations, participles, as well as by morphological strategies. Sometimes, even the same complement taking predicate may appear in more than one construction, as in the following examples, where the utterance predicate *say* is associated to a paratactic structure (15a), to an embedded subordinate clause characterized by a complementizer and a finite verb (15b), and to an embedded infinitive (15c), the latter with a passive verb in the MC. Competing structures are not synonymous, and may imply formal adjustments in the expression of the subordinate situation. The passive construction in (15c), for example, is used when one wants to demote the agent of the complement taking predicate, as it is typically the case for this grammatical voice. The two variants of parataxis (15a) and of embedding with a finite subordinate (15b) differ in the absence or presence of anaphoric shifters, which are typical of the indirect speech. Moreover, direct speech tends to be more common in the spoken or informal language, while indirect speech and subordination in general have a tighter connection with the writing and with the formal register (cf. Givón 1979: 207ff, Coulmas 1986, Chafe 1994: 215ff, etc.).

- (15) a. *Peter said to Sandra: "You are a good friend."*
 b. *Peter said to Sandra that she was a good friend.*
 c. *Sandra is said to be a good friend.*

Parataxis, finite subordination, nominalizations, participles, etc. are all attested for complement relations since the earliest stages of IE (cf. Viti 2015: 315ff). As we have seen for RCs and ACs, CCs as well often attest a development towards embedding in the history of a language. This development, however, proceeds with

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different pace for different taking predicates. According to my IE data, the earliest embedded constructions attested for complementation are implicit CCs represented by participles depending on predicates of immediate perception (Section 4.2).

4.2 Non-finite, participial complement clauses

Despite their amazing variety in the expression of complement relations, most early IE languages agree in their constructions of predicates of immediate perception such as “see” or “hear”, as in the following examples from Vedic (16a), Homeric Greek (16b), and Old Latin (16c), where the participial clause is a syntactic constituent embedded in the MC.

- (16) a. yátredānīm páśyasi jātavedas
 where.now see:PRES.IND2SG Jātavedas:VOC
tíṣṭhantam agna utá vā
 standing:PART.PRES.ACC.M.SG Agni:VOC and or
cárantam / yád vāntárikṣe
 moving:PART.PRES.ACC.M.SG when or.atmosphere(N):LOC.SG
 pathíbhiḥ **pátantam** tám
 path(M):INSTR.PL flying:PART.PRES.ACC.M.SG him:ACC.M.SG
 ástā vidhya śárvā
 archer(M):NOM.SG pierce:IMP2SG dart:INSTR.SG
 śísānaḥ
 sharpen:PART.PRES.NOM.M.SG
 “Where you see him now standing or moving, o Jātavedas, or flying
 along the paths in the atmosphere, pierce him, as archer, with your
 dart, sharpening (it).” (Rig-Veda 10,87,6)
- b. ò pópoi ê phílon
 alas say:IPF3SG dear:ACC.M.SG man(M):ACC.SG
 ándra **diōkómenon** peri
 pursued:PART.PRES.MED.ACC.SG around wall(N):ACC.SG
 teíkhos / ophthalmóisin horômai
 eye(M):DAT.PL see:PRES.IND.MED1SG
 “Alas,” he said, “I see a man who is dear to me being pursued around
 the walls.” (Homer, *Iliad* 22.168–169)
- c. Sed quis hic est,
 but who(INT):NOM.M.SG this:NOM.M.SG be:PRES.IND3SG

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quem	astantem	video
who(RP):ACC.M.SG	standing:PART.PRES.ACC.M.SG	see:PRES.IND1SG
ante ostium?		
before door(N):ACC.SG		

“But who is it this one that I see standing before the door?” (Plautus, *Bacchides* 451)

Complement taking predicates of immediate perception typically share an argument with their subordinate clause, as the MC object is the subordinate subject (e.g. *I see you running* implies that I see you and that you are running; this is also clear when the object is a nominalization, as in *I see your running*). Moreover, the subordinate situation has a determinate time reference which is simultaneous with the situation represented in the MC. As it is typically the case in complementation, sharing of information determines syntactic deranking of the CC, that is, the arguments or the verb of the CC are formally different from their correspondent categories in an independent clause. In this case, the verb is implicit, or non-finite, lacking information for person, and the subject of the CC is in the oblique case – in the ancient IE languages, typically in the accusative, as the predicate “see” is regularly transitive.¹¹

Formally, the participial clauses in (16) cannot be analysed as relative constructions, which in IE studies are limited to subordinate clauses marked by a relative pronoun. From a functional point of view, however, these clauses are similar. In fact, I consider the use of the participle depending on immediate perception predicates in CCs to be derived from the participle used as a strategy of relativization. We have seen in Section 2 that participles, similarly to adjectives, originally agree in gender, number and case with their head noun, and may be functionally equivalent to RCs in ancient IE. This construction is maintained when the head noun is the object of a complement taking predicate. In this case, the modifying participle and its head noun form together a noun phrase, which is the direct object of an immediate perception predicate. Everything remains in the domain of the simple sentence. A sentence such as *I see a man running* in (17a) had originally the same constituent structure of a sentence such as *I see a blond man* in (17b). It may be hypothesized that the change occurs when the verbal features

¹¹The predicate “hear”, instead, displays more variation. It can govern an accusative but also another oblique case in IE. In Ancient Greek, for example, the verb *akoúō* “I hear” properly selects the accusative of the thing heard and the genitive of the person from whom it is heard, but one also finds the genitive of the thing, as well as a prepositional phrase for the person. The lower transitivity of “hear” with respect to “see” may be explained by the fact that hearing implies a less direct source of information.

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of the participle, such as tense/aspect, mood or voice, prevail on its nominal features of gender, number and case. As a verbal constituent, the participle is now to be connected with a new subject, that is, with the object of the immediate perception predicate, which is its syntactically closest noun phrase. Together, this noun and the participle are reinterpreted as a subordinate clause, depending on the immediate perception predicate, as in (17c). At this point, the sentence does not mean that I see a man who is running or a running man (English may use a different word order to express the attributive or predicative function of the participle). It means that I see the event of a man that is running. An originally clause constituent, consisting in the participle and its head noun, is externalized and acquires clause status.

- (17) a. [S [NP I] [VP [V see] [NP a man running]]]
 b. [S [NP I] [VP [V see] [NP a blond man]]]
 c. [S1_MC [NP I] [V see]] [S2_SC [NP a man] [V running]]

Later on, participles extend from their original relativizing domain to other complement relations, as well as to other subordinating relations outside complementation. Their old use with complement taking predicates of immediate perception is maintained in several IE languages, such as English, Lithuanian, and Russian, up to the present. In Germanic languages such as German and Icelandic, instead, we have the infinitive in this function. Cf. German *ich sehe einen Mann rennen*, Icelandic *ég sé mann hlaupa* (I see man run:IF), with an infinitival complement clause, meaning ‘I see a man running’, vs. German *ich sehe einen rennenden Mann*, Icelandic *ég sé hlaupandi mann* (I see running man), with an attributive, relativizing participle, meaning ‘I see a man who runs’. Some Romance languages also use the infinitive with complements of immediate perception predicates, cf. Italian *vedo un uomo correre*, French *je vois un homme courir*, as Latin used both the infinitive and the participle with these predicates (*video hominem currentem*, *video hominem currere*). The construction of the infinitive, however, represents an innovation with immediate perception predicates – the original use of the infinitive, as we have seen in Section 3, was circumstantial, to mark purposive relations, in IE.

4.3 Paratactic and correlative complement clauses

Complement taking predicates of immediate perception imply an intermediate level of syntactic integration with respect to predicates of utterance, propositional attitude and knowledge, on the one hand, and to modal, phasal and desiderative predicates, on the other (cf. Lohninger & Wurmbrand, this volume). The

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former group of predicates is originally characterized by loose types of clause linkage, such as parataxis or correlation. This is because they express situations that are scarcely conditioned by the situation represented in the MC. The two clauses do not necessarily share arguments (*I say that I / you / he go(es) to the city*). Moreover, the time reference of the subordinate is also not conditioned by the time reference of the MC (*I say that I go / went / will go to the city*). Utterance predicates selecting paratactic structures can be observed virtually everywhere in IE, but in some languages, as in ancient Indo-Iranian and in Hittite, they represent the most common strategy for reporting someone's words. That is, these languages favour direct speech over indirect speech. In Hittite, in particular, direct speech is marked by the quotative particle *-wa(r)-*, which commonly occupies the second position (after the conjunctions *-a / -ia* "and" and *-ma-* "but", and before the enclitic pronouns, if any) in the clause-initial chain of clitics that is so typical of the Anatolian languages. The following text is drawn from the annals of the king Muršiliš II (14th-13th century BC), who informs a local chieftain about his intention to conduct a war. As can be seen, the particle *-wa(r)-*, here written in bold, is repeated in each reported clause, to indicate that the direct speech continues.

- (18) ANA ^MU^hha-LŪ=ma ^{LŪ}TEMU uīyanun nu=ši
 to Uhhaziti-but messenger send:PRET1SG CON-him:DAT
 hatranun ARAD^{MEŠ}=YA=**wa**=ta=kan kuēš anda
 write:PRET1SG servant:PL-MY-QUOT-you:DAT-PTC who:NOM.C.PL PREV
 uer nu=**war**=ašta EGIR-pa kuit wewakkinun
 come:PRET3PL CON-QUOT-PTC back because demand:PRET1SG
 nu=**war**=aš=mu EGIR-pa UL pāišta
 CON-QUOT-THEM:ACC-me:DAT back NEG give:PRET2SG
 nu=**wa**=mu=za DUMU-lan halzeššešta nu=**wa**=mu=za
 CON-QUOT-me:ACC-REFL child:ACC call:PRET2SG CON-QUOT-me:ACC-REFL
 tepnuškit kinun=a=**wa** ehu nu=**wa** zahhiyawaštati
 belittle:PRET2SG now-and-QUOT EXCL CON-QUOT fight:PRES.MED1PL
 nu=**wa**=naš ^DU BELÍ=YA DINAM hannāu
 CON-QUOT-us:ACC Stormgod lord-my case decide:IMP3SG
 "But to Uhhaziti I sent a messenger, I wrote him: 'My servants who came to you, because I now demanded (them) back, you have not given them back to me. And you have been calling me a child, and you have been belittling me! And now we will fight each other. And the Stormgod, my lord, decide our case.'" (Kbo.III.4)

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A syntactic change from parataxis to embedded hypotactic structures is traditionally identified in Germanic, where utterance predicates select a complementizer, such as English *that*, German *dass*, Icelandic *að*, which goes back to an originally demonstrative pronoun (cf. Hopper & Traugott 1993: 185ff). Similarly to what we have seen in Section 2 for relative pronouns derived from demonstrative pronouns in Germanic, in complementation as well we may have a reanalysis. Originally, two paratactic clauses connected by asyndeton are reconstructed, e.g. [*The man said that.*] [*We have a cold weather.*]. The demonstrative pronoun, direct object of the utterance predicate, refers cataphorically to the content of the subsequent clause. Later, the two clauses may be reinterpreted as a unitary complex sentence, with a shift in the clause boundary, i.e. [*The man said [that we have a cold weather].*]. Here we have a complementizer, with no argument function in the MC. Actually, this traditional reconstruction has been recently challenged by Axel-Tober (2017), among others, who considered Germanic complementizers to be derived from relative pronouns, rather than from demonstrative pronouns. Originally, we would have a correlative construction, e.g. *The man said that, that we have a cold weather*, where the subordinate clause is functionally complete. Later, the resumptive element in the MC would have been omitted. The relationship with the demonstrative source of the subordinator would be indirect in this case.¹² A similar analysis has been offered for the Hungarian complementizer *hogy* “that” (É. Kiss 2023). However, as correlative constructions are adjoined and not embedded, this hypothesis does not substantially contradict the change

¹²The use of CC derived from correlative clauses is not limited to the case of subordinators (more or less directly) derived from the stem of demonstrative pronouns, as in Germanic. Ancient Indo-Iranian, for example, attests “explicative clauses” (*Explicativsätze*, cf. Delbrück 1888: 573ff; 1900: III, 324ff) that go back to correlative constructions and still have a relative pronoun (**yó-*) that synchronically differs from demonstrative pronouns (**tó-* etc.). In this case, a predicate of utterance, propositional attitude or knowledge in the MC presents a noun phrase which is further elaborated in the subsequent subordinate. E.g. *We know this deed of Indra, that he killed the dragon* or *This deed of you is renown, o Indra, that you killed the dragon*. Like a relative pronoun, the subordinator “that” agrees here in gender, number, and case with a nominal constituent of the MC. Unlike a relative pronoun, however, and like a complementizer, this subordinator has no syntactic function in the subordinate clause. It is enough to omit the head noun in the MC to have embedding with predicates of utterance, propositional attitude and knowledge – this is an easy step, since the head noun usually denotes a discourse topic that is recoverable from the context. E.g. *We know this deed of Indra, that he killed the dragon* > *We know this deed of Indra, that he killed the dragon. This deed of you is renown, o Indra, that you killed the dragon* > *This deed of you It is renown, o Indra, that you killed the dragon*. For the development of complement clauses in Indo-Iranian, cf. Viti 2007a: 211ff and Lühr (2008, 2014). The latter hypothesizes that the structural change from relative pronoun to complementizer in Old Indic subordination is facilitated by the presence of null objects in the sentence.

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from a loose clause linkage to embedding, or from less subordinate to more subordinate structures – subordination, as any other syntactic category, is often a matter of degree, rather than a dichotomy (cf. Haiman & Thompson 1984).

4.4 Deranking strategies of complement clauses

Unlike parataxis, embedding typically implies the use of shifters from deictic into anaphoric elements in indirect speech (first- or second-person pronouns > third person pronouns; *here* > *there*; *now* > *then*; *yesterday* > *the day before*, cf. Jakobson 1971), as we have seen in (15a)-(15b). Also in this case, as the difference between coordination and subordination is not discrete, we may have various hybrid structures. Classical Sanskrit, for example, attests utterance predicates selecting a complementizer (*yad*, from the stem of the *rp*) without shifters. A sentence such as *The man said to the woman_i that you_i are a good friend* is perfectly possible in Classical Sanskrit, albeit less common than the strategy with direct speech. The latter is marked by the particle *īti*, lit. “thus”, used once at the end of the reported speech, cf. example (13). In Viti (2007b), I explained the synchronically aberrant cases of complementation without shifters in Old Indic as the result of a syntactic calque from Dravidian, where complementizers (also beyond the domain of utterance complement taking predicates) commonly derive from a verb of saying. Tamil *en-ru* “that”, Malayalam *en-nə* “id.”, Telegu *an-i* “id.”, etc. are originally perfect participles of the Proto-Dravidian verb **aHn-* “say” (cf. Krishnamurti 2003: 451). This development from verbs of saying to complementizers, which is uncommon in IE, is common in many other language families and areas of Asia, Africa, and Oceania (cf. Sauerland et al. 2020, for example, on the form *wa* “tell; that” in Teiwa, a Papuan language spoken on the Pantar Island in eastern Indonesian). This represents another instance of the change from direct speech to indirect speech and more generally from parataxis to embedding.

In embedding, various morpho-syntactic strategies may be used to represent the tighter linkage between MC and subordinate clause. Agreement between the tenses or moods of the two clauses is one of the most common strategies that are cross-linguistically used for this purpose. Accordingly, the tense or mood of the subordinate is conditioned by the tense or mood of the MC. This is not limited to complementation, but seems to be much more typical of CC than of other types of subordinates, according to my data. In Ancient Greek, for example, a historical tense in the MC (i.e. an imperfect, an aorist or a pluperfect) typically triggers an optative mood in the subordinate, in the so-called use of the “oblique optative”. In Latin, agreement in tenses and moods (*consecution temporum*) becomes even

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more elaborated, according to the distinction into two aspects, imperfective (*infectum*) and perfective (*perfectum*), and three tenses (present, past and future), which characterizes the whole Latin verbal system. If, for example, the MC has a verb in the indicative, the subordinate will have the subjunctive present to express simultaneity between the two clauses, a subjunctive perfect to express anteriority with respect to the MC, and a periphrastic structure (with *sim*, *sis*, etc.) for posteriority, cf. Hofmann & Szantyr (1965: 550) for details. In Latin, the subjunctive is associated to subordination, and its productivity has been inherited by the Romance languages, which attest the subjunctive in complementation to a much larger extent than most Germanic languages.¹³ Usually, when complementation admits variation between the indicative and the subjunctive in a language, predicates expressing a report are associated to the indicative. The subjunctive, instead, is preferred for the expression of more subjective thoughts, doubts or scarce commitment. Cf. Italian *Dico che viene* ‘I say that he comes’, with an indicative verb in the subordinate, vs. *Penso / dubito / non credo che venga* ‘I think / I doubt / I don’t believe that he comes’, with the subjunctive. Italian preserves the subjunctive more than other Romance languages. Agreement in tense or mood between MC and complement clause represents an innovation in IE. Most IE languages have no such agreement, and each clause presents the tense and mood that is more appropriate for its situation independently of the surrounding clauses. We can observe this in ancient Indo-Iranian, for example. Even nowadays, many IE languages, such as Lithuanian, do not have this kind of agreement. Thus, no agreement in tense or mood can be reconstructed for PIE. This may be due to the fact that the common paratactic or correlative loose clause linkage of the earliest IE languages is not so compatible with any deranking strategy that tends to present the subordinate clause as different from a correspondent independent clause.¹⁴

Embedding is especially evident when the structure of the subordinate clause is deranked, in the form of both a non-finite verb and oblique arguments, with

¹³ An exception to this may be seen in Icelandic, where the subjunctive is quite productive e.g. in expressions of belief (cf. Sigurðsson 2010, Gärtner & Eyþórsson 2020).

¹⁴ In some languages, the higher or lower commitment on the information conveyed by the subordinate clause is expressed by different subordinating conjunctions, rather than by different moods. This is a rarer strategy with respect to mood alternance, but is already attested in some ancient IE languages, such as Ancient Greek, again by independent innovation. In Ancient Greek, predicates of utterance, propositional attitude or knowledge tend to select the subordinator *hóti* ‘that’ (from the stem of the *rp*) when the MC subject or the speaker is confident that the situation of the complement clause has occurred or will occur. In case of doubt, instead, the cognate subordinator *hōs* is usually privileged.

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respect to an independent sentence. This can be seen in English, where complement taking predicates of propositional attitude, for example, may be structured in infinitival patterns, e.g. *I believe him to be sincere*. This is also an innovation, independently occurred in different IE branches, which cannot be reconstructed for PIE. It appears already, however, in some ancient languages such as Latin, where this use, called *accusativus cum infinitivo*, is found with many more types of complement taking predicates than in English. The following passage, from Old Latin comedy, contains two examples of this construction with a predicate of utterance (*dicere* “to say”) and a predicate of propositional attitude (*negāre* “to deny”) at short distance. In particular, the MC subject says or denies in the present that something happened in the past. The anteriority of the subordinate situations is rendered by the infinitive perfect in Latin, which extends its typical agreement in tense to the domain of non-finite complement clauses.

(19) AMPHITRUO:

Tun **me** heri **advenisse** dicis?
 you me:ACC yesterday come:IF.PF say:PRES.IND2SG

ALCMENA:

Tun **te** **abiisse** hodie hinc **negas**?
 you:NOM.INT you:ACC go.away:IF.PF today from.here deny:PRES.IND2SG

“AMPHITRUO: Do you say that I came yesterday?

ALCMENA: Do you deny that you went away from here today?” (Plautus, *Amphitruo* 758)

The accusative with infinitive is used in several ancient and modern languages, such as Latin, Ancient Greek, English, etc. also with complement taking predicates other than utterance, propositional attitude and knowledge. It is even more common with desiderative predicates, manipulative predicates, phasal predicates, and modal predicates. In example (20), again from Old Latin, the desiderative predicate *volō* “I want” in the MC selects an infinitive complement. The MC subject and the subordinate subject differ, and the latter is coded in the accusative (*adfinitem* “alliance”). The Latin structure corresponds quite well to the English translation *I want this alliance to continue* etc. (but not to its Romance continuations, e.g. Italian *voglio che questa alleanza continui*, with a finite subordinate marked by a complementizer, see below).

- (20) **manere** **adfinitem** **hanc** inter nos
 continue:IF.PRES alliance(F):ACC.SG this:ACC.F.SG between us:ACC

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volo, si ullo modo est ut possit,
 want:PRES.IND1SG if any way:ABL be:PRES.IND3SG that may:PRES.SUBJ3SG
 quod spero fore
 which hope:PRES.IND1SG be:IF.FUT
 “I want this alliance between us to continue, if by any way it possibly
 may, which I hope will be the case.” (Terence, *Hecyra* 723–724)

Manipulative predicates meaning “make” or “let” also usually select an infinitive complement, e.g. Latin *facio te canere* “I make you sing”, *sino te canere* “I let you sing”, where the primary argument of the subordinate situation, if explicit, is coded by an oblique case, mainly the accusative. This syntactic construction remains in Romance, e.g. Italian *ti faccio cantare*, Spanish *te hago cantar*, French *je te fais chanter* (apart from the gradual change from second position clitics to verb-adjacent clitics). We can find plenty of parallels of infinitival causative predicates in other IE branches.¹⁵ Similarly, an infinitive is usually required by modal predicates, expressing possibility or necessity, with both deontic and epistemic functions. Auxiliaries of possibilities are originally verbs meaning “be strong, have power”, such as Latin *possum* “I can” from **potis sum* “I am the master”. Auxiliaries of necessity are expressions of lacking or taking away, such as Latin *debeō* “I must” from **dē-habeō* “I hold away from someone, I withhold”. In this case as well, we have a substantial syntactic continuity from Latin *facere possum* “I can do”, *facere debeo* “I must do” and the Romance reflexes, such as French *je peux faire*, *je dois faire* (apart the change in word order). Phasal predicates, as anticipated, imply the inception, continuation or completion of a process. The infinitive is the most common strategy used for this, e.g. French *je commence à / continue à chanter* “I start / continue to sing”, *j’arrête de / termine de chanter* “I stop / finish singing”. Other IE languages, such as English, use the participle in this case, another non-finite and deranked construction.

Deranking was even more pronounced for these predicates at a previous stage of IE. Originally, in PIE, desiderative predicates having the same subject as the subordinate situation (*I want to do something*) could be expressed by means of morphological strategies, such as a series of suffixes in sibilant. We can recon-

¹⁵This does not imply that infinitive complement clauses are the only expression of causative predicates. We can also have syntactically heavier constructions with subordinator and a finite verb, such as Latin *efficio ut*, *curo ut* + subjunctive. These constructions often have a connotation of less direct causation with respect to their more integrated infinitival correspondents. A similar situation holds in Romance. In Italian, for example, the sentence *faccio in modo che tu canti* (which can also be translated as ‘I let you sing’, or ‘I make sure that you sing’) iconically implies a less direct causation than the more deranked sentence *ti faccio cantare*.

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struct a desiderative formation with the suffix *-se-* added to an accented reduplication in *i* and to a root in the zero grade. E.g., from the PIE root *tek^w* - “run”, we have PIE **tí-tk^w-so-nti* “they want to run” > Young Avestan *tixšənti* “they want to run”. Similarly, suffixation was originally used in PIE to express causative predicates (*I make you do something*). In this case, a root in the full grade *o* could be added with an accented suffix *-éye-*. E.g., from the PIE root **wert-* “turn”, we have PIE **wort-éye-ti* “it makes turn” > Gothic *fra-wardeiþ* “it corrupts”.

Morphological strategies of prefixation, suffixation or inflection could also serve to represent modal predicates and phasal predicates. Modal functions were originally coded by inflection in different grammatical moods. Besides the indicative, expressing realis, and the imperative, used for commands, PIE also had an optative and a subjunctive, for example. The forms of these two moods still co-exist in ancient Indo-Iranian and in ancient Greek, despite some functional overlaps.¹⁶ The optative (formed with the PIE suffix **-yéh₁/ih₁-* and with historical endings) especially expressed the speaker’s wish, as well as the possibility that the denoted situation takes place. E.g., from the PIE root **h₁es-* “be”, we have PIE **h₁s-yéh₁-t* “it may be” (Vedic *syāt*). The subjunctive, a thematic formation instead, was mainly associated with expectation and with situations that shall or should happen in the future (this is the reason why several IE languages form their future tense, inexistent in PIE, on the basis of the subjunctive). E.g. PIE **h₁és-h₁e-t* “he shall be, let him be, he will be” (Vedic *ásat*). In the domain of phasal predicates, durative and completed events were originally coded by the imperfective and the perfective verbal stem, respectively. Alternatively, continuation could be expressed by various iterative formations, such as an iterative suffix **-éye-* associated to the *o* grade of the root. E.g. from the PIE root **g^{wh}ed^h-* “beg”, we have the PIE injunctive **g^{wh}od^h-éye-t* “he continued begging” > Homeric Greek *póthei* “he missed, longed for”. Inception could be coded by an inchoative suffix **-sk-*, e.g. Latin *florēscō* “to begin to blossom”. Moreover, originally adverbial particles with a locative or directional meaning may also be prefixed and acquire various actional or aspectual values. We can see this especially in Slavic, e.g. Russian *bezhát’* “to run” vs. *pobezhát’* “to start running”. For reasons of space, I cannot describe in detail all these phasal formations, which develop in different ways in different IE languages, and which reveal phenomena of interaction between aspect and actionality.

The integrated constructions used for desiderative, causative, phasal and modal predicates can be explained by their semantics, which imply shared ar-

¹⁶From the functional point of view, instead, the term “optative” has been less properly extended to some uses of the subjunctive, as in Latin (cf. van der Auwera & Schalley 2004 for such terminological problems).

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guments and determinate time reference for the subordinate situation. Modal predicates and phasal predicates imply the same subject for the MC and the subordinate. In the sentences *He can sing*, *He must sing*, the person who can or must is also the person who sings. In the sentences *He starts / continues / stops / finishes singing*, the person who starts, continues, stops or finishes is also the person who sings. The time reference of the subordinate situation, in this case the action of singing, is simultaneous with the time reference of the MC. In complement clauses headed by causative predicates, the object of the MC is the subject of the subordinate. In the sentence *I let him sing*, the form *him* indicates the shared argument. Moreover, the time reference is the same, at least in case of immediate causation, which is in fact the unmarked situation for causative predicates. Desiderative predicates also have a determinate time reference, in this case following the MC situation. Their arguments are also shared. In sentences such as *I want to sing*, the MC and the subordinate share the same subject. In sentences such as *I want him to sing*, the subordinate subject is the MC object. The former case is more common in usage, as people prefer to express their own desires than an aim of somebody else.

Morphological strategies of complementation are quite productive in ancient Indo-Iranian. In Old Indic, for example, morphological desiderative and causative predicates are regular (e.g. *pípāsāmi* “I want to drink”, *dhārāyāmi* “I make you hold”). In most other languages, instead, they decline and have just left residues that are not transparent anymore at a synchronic level. This is due, on the one hand, to the decay of verbal inflection, e.g. to the loss of some tenses or moods in the specific IE languages, and on the other hand to the increasing opacity of derivational affixes. Morphological formations of desiderative, causative, modal, and phasal predicates are replaced by periphrastic constructions, according to a common drift from synthesis to analysis. In this case as well, however, the competition between more and less integrated syntactic constructions is sensitive to semantic factors of shared information. In case of same subject between MC and subordinate, we often have the infinitive, as we have seen. In case of different subject, finite clauses marked by a subordinator prevail. In French, for example, constructions with desiderative predicates such as *je voudrais savoir* vs. *je voudrais que tu saches* present the same pattern as structures of propositional attitude such as *je pense d’avoir eu de la chance* vs. *je pense que tu as eu de la chance*. In case of different subject, the accusative with infinitive tends to be lost.¹⁷

¹⁷Note that the loss of the accusative with infinitive in Romance is not due to the downfall of the ancient case system. Cases remain in Romance pronouns (cf. French *je* “I” vs. *moi*, *me* “me”), and could be in principle employed with the infinitive similarly to what happens in English.

4.5 Increasingly specific complement taking predicates

The category of complement taking predicates widely differs in extension across different languages. While English allows embedding with a large variety of complement taking predicates expressing utterance, propositional attitude, knowledge, pretence, fearing, etc., we have seen that in ancient IE languages such as Old Indic or Hittite complementation is much more restricted. Remarkable differences may emerge even within the same class of complement taking predicates. Utterance complement taking predicates, for example, represent in English a large class including predicates that express specific manners of verbal communication, such as *shout* or *whisper*. This, however, is not possible in many other languages. Manfred Krifka (p.c.) observes that in Daakie (an Oceanic language spoken in Ambrym Island, in the archipelago of Vanuatu) one cannot say a sentence such as “He shouted that there is a whale out there”. One would rather say “He shouted, he said that there is a whale out there”. A study of the diachronic development of complement taking predicates in languages would require a precise corpus analysis, which to my knowledge is still a desideratum, and which is beyond the scope of this paper. It may be argued that semantically unspecific utterance predicates, such as *say*, which do not express the manner of verbal transfer, are associated first to embedding, and that other predicates may follow, according to whether subordination is more or less productive in the language at issue. We have a gradual extension in this. Similarly, among desiderative predicates, generic predicates such as *want* better establish morphological devices of verbal suffixation with respect to predicates that specify the manner of desire, such as *hope*, *wish*, *crave*, etc. A grammaticalization process starts with forms characterized by a generic semantics.

5 Conclusions

In this paper, we have discussed some common developments of embedded SCs in the history of IE. We have seen that embedding, which is so typical of subordination in English and in most modern languages of Europe, is often secondary from a diachronic point of view. On the one hand, embedded subordinates tend to develop from originally paratactic constructions (usually by reanalysis of demonstrative pronouns or adverbs into subordinators), from originally adjoined correlative structures (by deletion of the anaphoric resumptive in the MC), or more generally from looser to tighter clause linkage. On the other, implicit embedded subordinating structures, represented by verbal adjectives or participles and by

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verbal nouns or infinitives, tend to be replaced by explicit embedding with a finite verb and a subordinator. We have also seen that the attainment of explicit strategies of subordination often follows a path of relative cognitive complexity in languages, as semantically more complex subordinating relations, such as concession, receive a specific subordinator later than simpler relations, such as those expressed by temporal clauses or conditional clauses. Of course, components of semantic complexity must be independently identified and motivated by specific cognitive factors – there is a rich linguistic literature on that, based on both synchronic and diachronic evidence, as well as on first and second language acquisition and pidgin studies, cf. Dahl (2004); Miestamo et al. (2008), etc. A subordinate clause which implies a semantic component of negation, for example, is more complex than a subordinate clause without negation. Moreover, within embedding, we may have more or less deranked constructions, according to whether the MC and the SC share pieces of information, such as arguments or time reference.

We can also observe that RCs are diachronically primary with respect to ACs and to CCs, that is, RCs have been expressed by specialized constructions earlier than ACs and CCs, as far as we can reconstruct by comparing the subordinating system of the different IE languages. ACs and CCs are often marked by subordinators based on the stem of the relative pronoun, for example. While RCs longer retain the ancient adjoined pattern of the correlative diptych, embedding turns out to develop for ACs first. Most ACs do not present any anaphoric resumptive element in the matrix clause. This may be explained by the fact that ACs refer to an event or state, and are therefore intrinsically less referential than relative constructions, which refer instead to an object or person and hence more easily trigger the use of an anaphoric pronoun. We have seen that the old correlative pattern is retained only in few ACs, such as clauses expressing space, time, and manner (which can be functionally equivalent to RCs), as well as consecutive clauses. The latter present a cause-effect sequence of events as a balanced structure. CCs turn out to be the last acquisition in the subordinating system of the IE languages. CC often go back to paratactic or adjoined constructions, which may leave traces in the morphology of the complementizer or in the clause structure. Consider the following construction, drawn from the Old Persian story of Darius killing the usurper Smerdis, where a complement taking predicate of fearing is rendered by parataxis: *kāra-šim hacā daršam atarsa kāram vasiy avājaniyā hya paranam Bardiyam adānā* “The people feared greatly that he could slay at will the people who previously had known Smerdis”, lit. “The people feared him greatly. (They thought that) he would slay”, etc. (Darius inscription at Behistun, 1 column, 50–51). Within complementation, we have seen that the earliest embedded

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pattern is represented by the participle with predicates of immediate perception. It may be interesting to investigate to which extent these diachronic tendencies, based on the IE languages, may be applied to other language families.

All this does not imply that a change from implicit to explicit subordination is necessary. In Mandarin Chinese, for example, subordinating relations are often left implicit and must be inferred from the context. Participles have been always the main strategy to express RCs in Dravidian, and nominalizations have been always the main pattern of RCs in Turkic. Neither we imply that paratactic patterns necessarily develop into subordination and embedding. In Arabic, subordination has been always less common than coordination in clause linkage. One of the first things one learns in doing translations from Arabic into English is that the Arabic coordinator *wa* is much more frequent than the English coordinator *and*, and can be often rendered in English by means of subordinating conjunctions as *when*, *even though*, etc. We rather imply that, *if* a change occurs in the subordinating system of a language, implicit strategies of subordination are replaced by explicit strategies much more commonly than the other way round, and that the path from parataxis to embedding, from correlation to embedding, and more generally from a looser to a tighter clause linkage is much more common than the other way round.

Some potential criticism must be briefly addressed. Hendery (2012), for example, talks of an alleged “ideological investment” to an “earlier ‘received view’ of adjunction and parataxis as primitive and embedding and hypotaxis as sophisticated, modern devices” (p. 165). She contests the idea of a general change from juxtaposition to syntactic bondedness – an idea which has originated in the past (e.g. by the Neogrammarians) and is often still supported nowadays (cf. Hopper & Traugott 1993, Givón 2001, etc.). Now, we do not have to confound ideology with philology. The Neogrammarians may be corrected in some uses of the terminology – their statements must be understood in the framework of the positivistic culture of their time, and currently nobody is anymore opposing primitive to mature languages in IE studies or in historical linguistics. The Neogrammarians, however, had a profound mastery of the ancient IE languages – especially Delbrück (1900) has identified syntactic phenomena and mechanisms that have been proven to be valid also in other language families and to stand to the scrutiny of modern linguistic theory (cf. Cotticelli-Kurras & de Decker 2023). The data of the IE languages substantially *confirm* the idea that – in case a syntactic change occurs – clause linkage usually proceeds from a looser to a tighter connection, where syntactic bondedness can be measured on the basis of precise syntactic criteria, such as the use of more or less deranked forms for the predicate and for the arguments (cf. Lehmann 1988). Whoever has some familiarity with Vedic or

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Hittite finds this absolutely clear. But also in other IE languages – even languages with dominant embedding, such as Latin and Greek – traces of ancient paratactic or adjoined structures appear, especially if we consider the earliest texts.

The fact that adverbial subordinators and complementizers may go back to relative pronouns does not impinge upon this diachronic tendency. Firstly, this makes the process just less direct, as relative pronouns usually go back to demonstrative or interrogative-indefinite pronouns which had a syntactic function in independent, paratactic or adjoined sentences. Secondly, one thing is the morphology of a subordinator and another thing is the clause structure. Again, correlative structures are *not* embedded – this is worth stressing since correlation is often wrongly presented as a piece of evidence against the idea of a relatively late established hypotactic system. And again, the difference between parataxis and hypotaxis must be seen as a matter of degree. The gradual development of embedding can be explained by syntactic theories of grammaticalization, as we have seen, as well as by reanalysis and extension (cf. Harris & Campbell 1995 on the latter two mechanisms).

Neither is this diachronic tendency contravened by the change from one kind of embedding to another, rather than directly from paratactic constructions. As not all relations, in any linguistic domain, are grammatically coded at the same time, it is also expected that some subordinating relations are associated to embedding earlier than others. We have seen that some embedded constructions, especially in the domain of complementation, emerge relatively late as compared to relative clauses and to adverbial clauses. While the change from relative clauses to complement clauses prevails in the history of IE, cases of complement clauses derived from the syntactic pattern of adverbial clauses are also quite common. In the Romance languages, for example, complementizers such as Italian *come* ‘how’, Spanish *como* ‘id.’, French *comment* ‘id.’, Romanian *cum*, etc. (all derived from Latin *quō-modo* ‘in what manner’) may introduce complement clauses of indirect questions (e.g. French *je voudrais savoir comment tu as pu faire cela*), and clearly derive from manner clauses, which belong to adverbial subordination. Beyond IE, a similar change is widely attested in Semitic, where the conjunction *k-* expressing manner and time (cf. Ugaritic *k-*, *ky*, Hebrew *kī*, Phoenician *k-*, Aramaic *ky*, Arabic *ka-*, *kay-*, etc.) passed to adverbial relations of cause (Old Akkadian and Assyro-Babylonian *kī*, *kīma*, Phoenician *k-šm ‘ql(’)*, etc.) and later to complement relations, especially after verbs of command and perception (cf. Hebrew *kī*, Middle Babylonian *kī*, Ge’ez *kama*, Tigre *kə*, etc.), cf. Brockelmann (1913: 606; 6013ff); von Soden (1995: 275ff); Lipiński (2000: 473-474; 538-539; 546), etc. Deutscher (2000) has studied in details an instance of this change, that is, the use of the Akkadian comparative and temporal conjunction *kīma* ‘as, when’

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first as a causal subordinator meaning “because” and later as a complementizer meaning “that”. In his view, this change may seem “unusual” and “very problematic” (p. 37). In fact, there is no problem at all in this change. There is no reason to believe that finite sentential complements develop *straightforwardly* from paratactic constructions, and no historical linguist, to my knowledge, has ever claimed the regularity of such a direct step. Again, we speak of a tendency from a looser to a tighter type of clause linkage, to be measured according to precise criteria of sharing of argument, time reference, control, etc. All these changes considered so far do *not* represent an opposite change from tighter to looser connection, or from subordination to parataxis. Only this opposite change would be really problematic. Incidentally, consider that Akkadian *kīma*, as all the above-mentioned subordinators, ultimately go back to a deictic and asseverative particle *k-* (Lipiński 2000: 473), that is, not to a subordinating conjunction.

The claim that the syntactic change from loose clause linkage to tight clause linkage may be equally common than the opposite one, or that embedding may be equally common in ancient as in modern languages, relies on a wrong interpretation of uniformitarianism. Nobody is contesting the fact that what is observed in the present can also be proposed in the past. This, however, does not imply that the textual expressions of ancient languages, associated to a fundamental oral mode of communication, have to follow the same pragmatic principles as the texts of modern languages, spoken in societies where the written mode is dominant, especially in urban cultures. In a functional view of language change, as adopted here, discourse pragmatics may also affect syntax, as different contexts and different formal or informal registers may be related to different constructions. In historical linguistics, by definition, syntactic constructions must be contextualized in their history, society and culture. On the one hand, although the ancient languages are documented by written texts, these texts were usually composed for an oral performance, and often were recorded much later than their original oral form. On the other, it is well-known that the use of subordination is much more common in the written language; Givón (2001), for example, has provided important evidence on this. To argue that no particular directionality may be identified in the diachrony of embedding and subordination in general assumes a comparison among syntactic systems that are completely extrapolated from their socio-cultural context.

Genuine exceptions to the change from loose to tight linkage occur, of course. It is well-known that clauses introduced by the conditional coordinator “if”, for example, can be used as independent clauses, as in English *If you’ll just come next door* or in French *Oh, si j’étais riche!* (cf. Patard 2014). These are instances of what Evans & Watanabe (2016) calls “insubordination”, that is, the formation

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of independent clauses from subordinating structures, which seems to be a well attested phenomenon cross-linguistically (cf. also Evans & Watanabe 2016, Beijering et al. 2019). Moreover, we may also have constructions that diachronically acquire more paratactic functional features. Converbs, for example, traditionally called “gerunds” in IE studies, are typically subordinating and adverbial in Old Indic (especially with the function of an *after*-clause), but are mainly coordinating in New Indic. These cases are rarer, however, as compared to the change from parataxis or correlation to embedding. The very fact that insubordination has been only recently identified is revealing. These exceptional cases deserve to be explained, but they do not impinge upon the prevalent directionality towards embedding. For Indic converbs, contact factors may have played a role, as coordinating converbs are common in Dravidian, as well as in numerous other Asian languages.

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Abbreviations

ABL	ablative	IF	infinitive
AC	adverbial clause	IMP	imperative
ACC	accusative	IND	indicative
ART	article	INDEF	indefinite
C	(genus) commune	INT	interrogative
CAUS	causative	IPF	imperfect
CON	connective	INSTR	instrumental
CONJ	conjunction	LOC	locative
DAT	dative	M	masculine
DEF	definite	MC	matrix clause
EXCL	exclamative	MED	middle voice
F	feminine	N	neuter
TUT	future	NEG	negation
GEN	genitive	NOM	nominative
GV	gerundive	OPT	optative
IE	Indo-European	PART	participle

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PASS	passive	REFL	reflexive
PF	perfect	RP	relative pronoun
PIE	Proto-Indo-European	SC	subordinate clause
PRES	present	SG	singular
PRET	preterit	PL	plural
PREV	preverb	QUOT	quotative
PTC	particle	SUBJ	subjunctive
RC	relative clause	VOC	vocative
RCON	relative construction		

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Chapter 6

Prosody and syntax of argument and adverbial clauses

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In this chapter, we discuss the relation between clausal embedding and phonological structure. Since Downing's (1970) seminal work on English intonational phrases (IP), it is commonly assumed that only *root* clauses are visible to the phonological component and necessarily introduce IP breaks. Nevertheless, embedded clauses can sometimes clearly form their own IP and give rise to prosodic realizations that closely mirror the syntactic embedding. The main question we address is whether there is a systematic correlation between the nature of the embedded clause (i.e., argument or adverbial clause) and/or its syntactic position (e.g., extraposed, high or low-attached, verb adjacent) and its prosodic status.

1 Introduction

When it comes to the prosody of complex sentences, it has long been observed that English tends to display an asymmetry between so-called *root* and *non-root* sentences or clauses, where only the former constitute their own higher level prosodic unit, while the latter only optionally do so.¹ Syntactically speaking, Emonds (1969) offers the definition of root sentences given in (1).

¹We use both the terms *sentence* and *clause* to reflect that in early works such as Emonds (1969) and Downing (1970) the term *root sentence* was used for any constituent that is directly dominated by the root node, which is labelled by the term *S* for *Sentence*. In more recent work, such constituents are more commonly called *root clauses*.



(1) Root sentence (Emonds 1969: 6)

A root sentence will mean either the highest S in a tree, an S immediately dominated by the highest S or the reported S in direct discourse. (Hooper & Thompson 1973: 465)

Downing (1970), whose work is based on Emonds' insight that root clauses are obligatorily set off by *commas* (i.e., pauses) and whose main goal it is to predict them, offers a slightly revised definition of root sentences, given in (2).

(2) Root sentence (Downing 1970: 30)

A root sentence is any sentence which is not dominated by a predicative sentence (where "A predicative sentence is any sentence in which the S node immediately dominates a VP").

Downing also alternatively offers the definition in (3), using the notion of *command* (Langacker 1969 and Ross 1967: 338).²

(3) Root sentence (Downing 1970: 31)

A root sentence is any sentence that is not commanded [dominated] by a VP node.

In addition to simple sentences, root-clauses are understood to include parenthetical expressions (4), non-restrictive relative clauses (5), tag questions (6), vocatives (7), as well as some left/right dislocated phrases (8) and (9) (Nespor & Vogel 1986: 188).

(4) Lions [as you know] are dangerous.

(5) My brother [who absolutely loves animals] just bought himself an exotic tropical bird.

(6) That's Theodore's cat [isn't it?]

(7) [Clarence] I'd like you to meet Mr. Smith.

(8) [Good heavens] there's a bear in the back yard.

(9) They are so cute [those Australian koalas].

Theses bracketed fragments, which do not all seem to constitute clauses/sentences of their own in a strict syntactic sense, constitute domains onto which "an intonational contour is spread" (Selkirk 1978: 130, Nespor & Vogel 1986: 187).

^{2b}Node A of a phrase marker commands node B if neither node dominates the other, and if node B is dominated by the first node S above A" (Downing 1970: 197).

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In this respect, they are typically considered to contrast with restrictive relative clauses (10), complement clauses (11) and at least some adverbial clauses (12), which are intonationally integrated to their context (Nespor & Vogel 1986: 196–198).

- (10) [That kind old lady always buys fresh meat for the stray cats that live in the park].
- (11) [I thought that you already knew that Gertrude was moving to southern Italy].
- (12) [Paul called Paula before Carla called Carl].

How to best capture the above relationship between clauses and major prosodic chunks is still a matter of debate. A number of studies have argued that the speech flow is organized into a finite set of hierarchically organized phonological domains to which phonological rules are sensitive (Selkirk 1978, Nespor & Vogel 1982, 1986, among others). These domains more or less reflect syntactic constituency although other factors such as speech rate and prosodic weight have been shown to play a role too (e.g., Gee & Grosjean 1983, 1987: on prosodic weight). In fact, different traditions place a different amount of emphasis on the contribution of syntax. In those works that regard the role of syntactic constituency as central in determining postlexical prosodic domains, a number of different proposals have been put forward. At earlier stages, prosodic categories often reflected particular properties of the language that was studied. For example Minor and Major Phrase were used for Japanese, also sometimes called Accentual and Intermediate Phrase (e.g., Haraguchi 1977, Beckman & Pierrehumbert 1986, Kubozono 1988). Nowadays, a consensus has been reached in Prosodic Phonology to distinguish only two different prosodic categories above the word level: the phonological phrase and the intonational phrase (Ito & Mester 2012, Selkirk 2009, 2011).

As a rule of thumb, the phonological phrase (PP or ϕ) corresponds to lexical XPs (Truckenbrodt 1999, Selkirk 2011) and the intonational phrase (IP or ι) to syntactic clauses (Truckenbrodt 2005, Selkirk 2005, 2009, 2011, Hamlaoui & Szendrői 2015). From this perspective, the above discussed root and non-root clauses differ on whether they map onto an intonational phrase of their own. What exactly constitutes a *clause* and a fortiori a *root clause* has been regularly debated. Although there is considerable overlap between the theories, a consensus has not yet been reached. We will come back to this issue in Section 3.

As far as the relationships between the prosodic units are concerned, it was originally assumed that they constitute exocentric categories (i.e., they are not organized around a head) arranged in a hierarchical fashion: that every unit would

only contain units of the immediately lower level. This is known as the Strict Layer Hypothesis (SLH) (Selkirk 1984: 26, Nespor & Vogel 1986). But already in the 1980s, certain phenomena were identified that called into question a strict formulation of the SLH (Ladd 1986, Inkelas 1989, Ito & Mester 1992). In particular, Ladd (1986) noted that a more elegant analysis can be given for structures involving certain appositives and parentheticals in English if one allows for recursivity, i.e., the idea that any prosodic category could include a prosodic category of the same type. Under this view, a weaker version of the SLH that prohibits higher level categories to be included inside lower level categories still remains (Selkirk 1996). This move, which is widely accepted by now (Truckenbrodt 1995, 1999, 2005, Féry & Truckenbrodt 2005, Wagner 2005, 2010, Ito & Mester 2007, 2009, Selkirk 2009, 2011, Elfner 2012), brings prosodic structure closer to syntactic structure in the sense that it introduces an intrinsically hierarchical organization in what has been previously perceived as a flat structure. Nevertheless, crucial differences remain. First, prosodic structure remains exocentric. Second, prosodic phrasing can be and often is influenced by non-syntactic considerations such as prosodic well-formedness constraints (e.g., size constraints), information-structural constraints (e.g., ALIGNTOPIC, STRESSFOCUS) or processing considerations (e.g., saliency of domain edges).

In the present chapter, we concentrate on two types of embedded clauses, i.e., arguments and adverbials, and consider whether there is a systematic correlation between the nature of the clause (subject/complement/adjunct) and/or its syntactic position (e.g., extraposed, high or low-attached, verb adjacent) and its prosodic status. The chapter is structured as follows. In Section 2, we discuss the prosodic realization of these types of embedded clauses in English and the mapping proposals that have recently been made and their respective predictions. Section 3 concentrates on cross-linguistic variation in the realization of these embedded sentences and the challenges it brings for the various mapping algorithms/constraints that relate clauses to intonational phrases. Section 4 discusses more complex cases of intonational phrasing, involving information structural considerations. Section 5 concludes the paper.

2 Syntax-phonology mapping of argument and adverbial clauses

2.1 Some empirical facts from English

In English, intonational phrase boundaries are often identified by means of various tonal and durational phenomena, most often associated with their *terminal*

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portion or *right edge*. Based on a number of previous studies (Lieberman 1967, Gleason 1961, Trager & Smith 1957), Downing (1970: 7–8) identifies intonational phrases as having their own intonational contour and terminal juncture and as realizing only one primary stress (also called *nuclear* or *sentence stress*). In more recent works, such as Selkirk (2005: 12), a final rising contour, noted L-H% in Pierrehumbert's (1980) theory of English intonation, or its alternative deep final fall (L-L%) are also central in diagnosing intonational phrases. In this theory, boundary tones (noted with the % symbol) only appear at intonational phrase edges. Additionally, words preceding a major prosodic break tend to show an increased duration, and more specifically a syllable-final lengthening (Selkirk 1984, Ladd 1986, Beckman & Edwards 1990, Price et al. 1991, Rob van den Berg & Rietveld 1992, Taglicht 1998).

As mentioned in the Introduction, there is a widespread tendency to associate intonational phrases with the presence of pauses. Studies like Price et al. (1991: 2968) find that major prosodic boundaries are indeed often associated with a pause (in 23% (out of 212 utterances) of level 4 and 67% (out of 25 utterances) of level 5 break indices), whereas minor prosodic breaks are not. According to Selkirk (2005: 12), citing work by Beckman & Edwards (1990) and Beckman & Ayers-Elam (1997), the temporal juncture is greater at an intonational phrase edge than at the edge of phrases lower down in the prosodic hierarchy. Note however that, as made clear in Downing's dissertation, a perceived juncture does not necessarily imply an actual pause in the sense of a *cessation of phonation*.

According to Ladd (1986, 1988), the intonational phrase is also the domain of declination, i.e., "the gradual F₀ decline often observed over the course of phrases or utterances" (Ladd 1988: 530) or, in an Autosegmental approach to intonation, the "setting of register for the realization of tone" (Selkirk 1995: 556). Concomitant to this, an upward pitch reset is indicative of the start or the left-edge of an intonational phrase, with non-initial intonational phrases showing only a partial reset (Ladd 1988). Depending on the language, (partial) resets can also be found at the left-edge of other prosodic domains (i.e., phonological phrases), but the ones at the beginning of intonational phrases generally reach higher tonal targets (Rob van den Berg & Rietveld 1992).

2.1.1 Subject clauses

Although a lot of work has been done on the prosody of English, there remains a number of gaps as to the obligatory and optional prosody and phrasing of some of the clauses that are of interest to us in this chapter. Subject clauses, in particular, do not seem to have been the object of as much systematic attention

as other types of embedded clauses. From a syntactic perspective, and as extensively discussed, for instance, in Lohndal (2014: and references therein), there is no agreement as to whether sentential subjects occupy the canonical subject position – and should thus be expected to prosodically behave as other non-root clauses – or whether they are more akin to topics and occupy a higher position within the clausal spine, one that would potentially make them a root clause. We will come back to the realization of this type of argument clause in Section 3, as experimental data has been discussed in other (Germanic) languages.

2.1.2 Complement clauses

Complement clauses, when in their base position, are a typical example of non-root clauses, i.e., they normally do not introduce their own intonational phrase boundaries. This is illustrated in the example in (13) where, despite its significant length, the clausal complement does not form a separate intonational phrase.

- (13) $IP(PP(Bi^{H^*}lly\ thought\ his\ fa^{!H^*}\ ther\ was\ a\ me^{!H^*}rchant^{L-})PP\ PP(\wedge\ and\ his\ mo^{H^*}\ ther\ was\ a\ secret\ a^{!H^*}gent^{L-L\%})PP)\ IP.$ (adapted from Selkirk 2005:11)

Metrically strong syllables carry a high pitch accent (H^*). Every H^* is downstepped (symbolized with $!$) with respect to the preceding one within the same phonological phrase (PP) and an (minor) upward reset (noted with the $\wedge$ symbol) takes place at the start of the second PP. The phrasing of the example in (13) is to be contrasted with the one of the example in (14), where the conjoined clause is a root clause and introduces its own intonational phrase.

- (14) $IP(PP(Bi^{H^*}lly\ thought\ his\ fa^{!H^*}\ ther\ was\ a\ me^{!H^*}rchant^{L-H\%})PP)\ IP\ ||\ IP(PP(\wedge\wedge!\ and\ his\ fa^{H^*}\ ther\ was\ a\ secret\ a^{!H^*}gent^{L-L\%})PP)\ IP.$ (adapted from Selkirk 2005:11)

According to Selkirk, example (14) differs from (13) in that a boundary tone is found on *merchant*, indicating the right edge of an intonational phrase. The reset at the start of the conjoined clause is more significant than in (13), without however going back to the register of the first intonational phrase and thus being downstepped with respect to it, as indicated by the $!$ symbol following the $\wedge$ symbols. A pause (noted $||$) is also perceived between the two conjuncts.

Interestingly, and as already noted by Downing (1970: 90–91), direct quote complements of the type in (15) do insert their own intonational phrase boundaries.

- (15) $[[Ann\ said]\ [\"I'll\ make\ you\ some\ sandwiches\"]].$

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Although they are not generally included in the lists of root clauses, direct quote complements behave like ones and seem to constitute a challenge for the definitions of root sentences given in (1) to (3). Rather than altering the definition of root clauses to fit these complement clauses in, Downing proposes a “Quote Detachment” operation by which these complements are syntactically extraposed and (Chomsky-)adjoined to the highest S. He acknowledges, though, that this is problematic in examples like (16) and (17), in which the quote is not sentence final.

(16) John reported that Ann said “I feel better” rather weakly yesterday.

(17) His saying “you are another” was uncalled for.

In (16) and (17), the quotes do not however, according to him, form separate intonational phrases. More investigations seem needed regarding both the syntax and the prosody of these sentences. If direct quote complement clauses however happen to occupy a similar structural position as their non-quotative alternative, i.e., in the scope of the quotative verb, and systematically form an intonational phrase of their own, this would suggest that the phrasing of some embedded clauses is not due to their syntactic location, but rather to their semantic/discursive status. We come back to this point in Section 2.2, when we discuss the role of illocutionary force and speech acts.

2.1.3 Adverbial clauses

Adverbial clauses represent a much larger and diverse set than argument clauses. When it comes to their prosodic realization, the type of relation they express (e.g., adversative, causative, consecutive, causal, manner etc.) does not seem to play a central role. The examples in (18) and (19), from Selkirk (2005), suggest that their structural position and in particular their attachment height is however crucial.

(18) $IP(PP(Ci^{H*}ndy\ isn't\ pla^{!H*}nting\ a\ ga^{!H*}rden^{L-})_{PP}(becau^{H*}se\ she\ lo^{!H*}ves\ toma^{!H*}toes^{L-L\%})_{PP})_{IP}.$

(19) $IP(PP(Ci^{H*}ndy\ isn't\ pla^{!H*}nting\ a\ ga^{!H*}rden^{L-H\%})_{PP})_{IP}$
 $\parallel IP(PP(\wedge\wedge\ becau^{H*}se\ she\ lo^{!H*}ves\ toma^{!H*}toes^{L-L\%})_{PP})_{IP}.$

Example (18), where the embedded clause is in the scope of the negation, excludes the content of the *because*-clause as the reason for planting a garden. With this interpretation, the adverbial clause is usually treated as a VP modifier, i.e., an instance of low adjunction (Rutherford 1970, Sæbø 1991, Charnavel 2017). In

that case, it does not form a separate intonational phrase, which is consistent with Emonds and Downing's prediction as, in that syntactic configuration it is not a root clause. In contrast, when the *because*-clause provides the reason for not planting a garden, as in (19), it is a case of high attachment (to the root node) and the embedded clause comes with its own intonational phrase breaks.

Rutherford (1970: 97), who focuses on the structural analysis of the contrast illustrated in (18) and (19), provides numerous examples in which a comma intonation allows to distinguish between a *restrictive* interpretation of adverbial clauses, in (20a) to (27a), and a *non-restrictive* one (in his terminology), in (20b) to (27b).

- (20) a. He's not coming to class because he's sick.
b. He's not coming to class, because he just called from San Diego.
- (21) a. She loves her husband (even) though he beats her.
b. She loves her husband, (al)though (I know) he beats her.
- (22) a. Mary won't marry John if I have anything to say about it.
b. Mary won't marry John, if I have anything to say about it.
- (23) a. Mary will marry John unless the fortune teller is too pessimistic.
b. Mary will marry John, unless the fortune teller is too pessimistic.
- (24) a. He'll take his umbrella in case it rains.
b. He'll take his umbrella, in case you're wondering.
- (25) a. Mary will marry John whether the fortune teller predicts it or not.
b. Mary will marry John, whether the fortune teller predicts it or not.
- (26) a. He kept looking at me as if I had {something/*anything} to do with his punishment.
b. He kept looking at me, as if I had {?something/anything} to do with his punishment.
- (27) a. Thou shalt not kill as the Bible says.
b. Thou shalt not kill, as the Bible says.

Additionally, in the case of *while*-clauses, Downing (1970: 82) observes that they only phrase separately from the main clause when they express a coordinate adversative clause, as in (28), and not an adverbial clause of duration, as in (29).

- (28) The men worked, / {while/whereas/but} the woman talked.
- (29) The men worked while the sun was shining.

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In Rutherford's analysis, the non-restrictive adverbial clauses are treated as coming from a high sentence, headed by a performative that has been deleted. Their relation to the main clause is thus looser than the restrictive adverbials'.

Left-peripheral *if*-clauses are also described by Selkirk (2005) as phrasing separately from the main clause, as illustrated in example (30). According to her, this is consistent with Emonds' treatment of this type of clauses as root clauses.

(30) $_{IP}(\text{If you had a llama})_{IP}, \text{ }_{IP}(\text{could you ride it})_{IP}?$

Downing (1970: 49), who assumes that the base position of English adverbial clauses is within VP (i.e., to the right of the main verb and its complements) and that the subordinate-matrix order is obtained by leftward extraposition (Ross 1967: 309), also reports a difference in intonational phrasing between (31) and (32) (adapted from Downing).

(31) $_{IP}(\text{We can talk after we eat})_{IP}.$

(32) $_{IP}(\text{After we eat})_{IP} \text{ }_{IP}(\text{we can talk})_{IP}.$

In (31), the embedded clause is attached low and, according to him, phrases together with the main clause, while in (32), it is attached to the root and phrases separately. Again, this seems consistent with the root/non-root clause distinction. Downing however notes that being separated from the main clause by a pause is not a property of leftward adverbial clauses only, but of any leftward adverbial, be it a clause or not. This is illustrated with the examples in (33) to (37), where the break following the adverbial is symbolized with /.

(33) While sleeping / I heard the phone ringing.

(34) When empty / the container weighs 14 ounces.

(35) Empty / the container weighs 14 ounces.

(36) In the afternoon / everyone went swimming.

(37) Tonight / I want to relax home.

Downing also contrast sentences (38) and (39) which, according to him, provide evidence for the fact that a root clause inserts its own intonational phrase breaks and that a break is only found if the adverbial is moved out of it. The perceived break in (38) is thus simply the left edge of the root clause.

(38) Tomorrow/I promised that he would be there.

(39) I promised that tomorrow he would be there.

As this difference in phrasing is also observed with clausal adverbs in (40) and (41), the same conclusion can be reached that the perceived break is the left edge of the main clause rather than associated with the right edge of the adverbial clause. Note that Downing's (1970: 52–53) account of the phrasing of left-peripheral *if*-clauses thus differs from Selkirk's in (30). Downing further notes that an intonational break is only obligatory if the adverbial clause originates from a root clause, i.e., in (40a) and (41a). It is optional in (40b) and (41b).

- (40) a. If you go to that meeting, / you may be arrested.
 b. I wonder if you are aware of the fact that if you go to that meeting (/ you may be arrested.
- (41) a. Because they went to the meeting, / they were arrested.
 b. If because they went to that meeting (/ they were arrested, / the situation is worse than we thought.

Further examples of left-peripheral adverbial clauses from Downing (1970: 53) are given in (42) to (44), which share a similar phrasing. He notes that the equivalent participial phrases also display this prosody.

- (42) When he had finished his task, / he locked up and went home.
- (43) Since you are an old friend of the family / you have a right to know.
- (44) Then John turned to me / and (he) remarked how hot it was.

In sum, in English complex sentences, both argument and adverbial clauses seem to be prosodically integrated to the main clause when they are in situ or attach in a position that is in the scope of the main verb. Whenever clauses are right or left-extraposed or their attachment site is simply higher within the sentence structure, they tend to phrase separately from the main clause. In the latter case, it is not always clear whether they form an intonational phrase of their own (i.e., introduce both their own left and right intonational phrase edges), or whether they are simply embedded in a prosodic domain that encompasses the entire sentence and contains an intonational phrase corresponding to the main clause (i.e., the break that separates them from the rest of the sentence originates from the main clause and not from the subordinate clause itself). Before turning to cross-linguistic variation in intonational phrasing, let us first turn to the theoretical treatments that have been proposed to account for intonational phrasing.

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2.2 Proposed theoretical treatments

Inspired by Emonds's (1969) observation that root clauses are set off by a comma intonation, Downing (1970: 31) formulates the rule given in (45).

- (45) **Obligatory Boundary Insertion (OBI)**
 Phrase boundaries [termed Intonational Phrase (IP) boundaries in later literature] are inserted as leftmost and rightmost immediate constituents of every root S node that appears in any postcyclic derived P-marker.

Translated into the Prosodic Hierarchy Theory (e.g., Selkirk 1984: and subsequent work), this means that a root clause is taken by Downing to insert both a left and a right intonational phrase boundary. Example (46) to (52) schematize the phrasing of the various types of complex sentences considered so far.

- (46) IP(subject clause + main clause)_{IP}
 (47) IP(IP(topicalized subject clause)_{IP} IP(main clause)_{IP})_{IP}
 (48) IP(main clause + complement clause)_{IP}
 (49) IP(IP(main clause)_{IP} IP(extraposed complement clause)_{IP})_{IP}
 (50) IP(main clause + in situ adverbial clause)_{IP}
 (51) IP(IP(main clause)_{IP} IP(coordinate adversative clause)_{IP})_{IP}
 (52) IP(adverbial/adverbial clause IP(main clause)_{IP})_{IP}

Given the definition in (45), this means that the complement clause in (48) or the subject clause in (46) do not form their own intonational phrase, as they are not directly connected to the root. In contrast, the extraposed complement clause in (49) and the topicalised subject clause in (47) do form their own intonational phrase as they are directly attached to the root. In addition, in both cases there is an intonational phrase encompassing the entire complex sentence introduced by the topmost S-node in Downing's approach. This correctly reflects the intuition that two clauses that form two separate sentences are phrased differently compared to two clauses that are part of the same sentence.

As visible in (46) to (52), Downing's approach is compatible with a recursive view of phonological structure. In that sense, it contrasts with a number of subsequent proposals, which assume the Strict Layer Hypothesis (Selkirk 1984, Nespor & Vogel 1986). This is the case for instance of the approach proposed by Nespor & Vogel (1986), which assumes that intonational phrases are formed by the rule in (53).

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(53) Intonational phrase Formation

a. *I domain*

An I domain may consist of

- i. all the ϕ s in a string that is not structurally attached to the sentence tree at the level of S-structure, or
- ii. any remaining sequence of adjacent ϕ s in a root sentence.

b. *I construction*

Join into an n-ary branching I all ϕ s included in a string delimited by the definition of the domain of I.

When several intonational phrases belong to the same larger prosodic domain, this domain is distinct and called the *phonological utterance* (U) (54).

(54) Phonological Utterance Formation

a. *U domain*

The domain of U consists of all the Is corresponding to X^n in the syntactic tree.

b. *U construction*

Join into an n-ary branching U all Is included in a string delimited by the definition of the domain of U.

Although Nespor & Vogel acknowledge previous observations by Downing and Emonds as to the connection between syntactic fragments of a certain type and obligatory intonational breaks, their own approach does not incorporate a privileged relation between intonational phrases and a specific syntactic category. They assume, as illustrated in (55), that any fragment surrounding an intonational phrase (here the parenthetical) can constitute an intonational phrase of its own (adapted from Nespor & Vogel 1986: 189).

(55) $_{IP}(\text{Lions})_{IP} \text{ }_{IP}(\text{as you know})_{IP} \text{ }_{IP}(\text{are dangerous})_{IP}$.

But as Ladd (1986) observed, such utterances actually support the case for recursive, nested intonational phrases once, as noted by Cooper & Sorensen (1981) and Elanah Kutik & Boyce (1983), we take into account the declination observed in such sentences. What they observe is that the declination in the matrix clause is the same with or without the parenthetical, suggesting the recursive prosodic structure in (56).

(56) $_{IP}(\text{The book on the table, }_{IP}(\text{it seems to me,})_{IP} \text{ was a gift from my mother})_{IP}$.

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This declination could also be viewed as evidence for the presence of a higher level category, Utterance Phrase, wrapping the whole utterance, but as Ladd (1986) eloquently argues, this is not a desirable option for several reasons. First, he reviews the phonetic markers of alleged Utterance Phrases compared to Intonational Phrases and remarks that they do not seem to be distinct enough to warrant a categorical difference between the two. Rather, it seems that there is a bunch of phonetic markers, which seem to cluster more, the larger the intonational phrase is. So, he argues for a quantitative, rather than a qualitative difference between the two.

Second, examples can easily be constructed, as in (57), where more than one level of embedding of intonational phrases seems to be warranted by the data.

- (57) $?(U(Lions_{IP}(as\ you\ know)_{IP}\ are\ dangerous)_{U}\ U(and\ the\ book\ on\ the\ table_{IP}(it\ seems\ to\ me)_{IP}\ was\ a\ gift\ from\ my\ mother)_{U})?$

One would then be forced to invent yet another category. Given that recursivity is intrinsically potentially infinite, this will not be practical.³

In Ladd's view, thus, prosodic structure is much more similar to syntactic structure than assumed before. Intonational phrases are not distinguished from Utterance Phrases, just as modern syntax does not distinguish S from CP. Although, the presence of such recursive, nested Intonational Phrases violates the Strict Layer Hypothesis in its strong form in (58), it is nevertheless compatible with a weaker formulation, which simply prohibits lower level categories to dominate higher level ones.

- (58) A category of level i in the hierarchy immediately dominates a (sequence of) categories of level $i - 1$. (Selkirk 1984: 26)

This weaker definition, in the form of violable Optimal-Theoretic constraints given in (59) (Selkirk 1996), and the ensuing availability of nested, recursive Intonational Phrases has since been widely adopted.⁴

- (59) Constraints on Prosodic Domination (where C^n = some prosodic category)

³See for instance Myrberg (2013: 110) for a recent, more detailed discussion of why declination is not an argument for the Utterance category and additional evidence from Swedish.

⁴Although proposals have been developed outside OT for the syntax-phonology mapping of other prosodic domains (e.g., phonological phrases), most of the recent work on the mapping of intonational phrases is set in this framework. For an alternative approach based on recent developments of the minimalist program and, in particular the notion of *termination of derivation* (Chomsky et al. 2019), the interested reader is referred to Dobashi (2018).

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LAYEREDNESS: No C^i dominates a C^j , $j > i$,
e.g., "No σ dominates a Ft."

HEADEDNESS: Any C^i dominates a C^{i-1} (except if $C^i = \sigma$),
e.g., "A PWd must dominate a Ft."

EXHAUSTIVITY:

No C^i immediately dominates a constituent C^j , $j < i-1$,
e.g., "No PWd immediately dominates a σ ".

NONRECURSIVITY:

No C^i dominates C^j , $j = i$,
e.g., "No Ft dominates a Ft".

Having settled this issue, let us now consider how different approaches propose to account for Downing's main findings. There are essentially two main issues that need an explanation. First, Downing showed that embedded clauses in their canonical in situ position typically do not map onto separate Intonational Phrases, despite having a syntactic structure that would correspond to an intonational phrase in a free-standing position. Second, the same embedded clauses nevertheless do map onto separate Intonational Phrases once they occupy a high extraposed position in the structure. Finally, we should also note that certain left/right asymmetries also seem to play a role in determining whether a particular embedded clause corresponds to its own Intonational Phrase.

Assuming the edge-alignment theory developed in Selkirk (1986, 1996) – according to which, in a specific language, only one syntactic edge (i.e., left or right) systematically aligns with a detectable prosodic edge –, as well as the Generalized Alignment in Optimality Theory (McCarthy & Prince 1993, Prince & Smolensky 2004), Truckenbrodt (2005: 287) and Selkirk (2005) respectively formulate the syntax-prosody alignment constraints given in (60) and (61).

(60) ALIGN-CP, RIGHT

The right edge of a CP must coincide with the right edge of an intonational phrase.

(61) Interface Constraint for Intonational Phrase in English

ALIGN R (CommaP, IP)

Align the R edge of a constituent of type Comma Phrase in syntactic (PF) representation with the R edge of a corresponding constituent of type Π_{CommaP} (= Intonational Phrase, IP) in phonological (PR) representation.

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In the former approach, primarily motivated by data from German to be discussed in Section 3, any clause can form an intonational phrase and the notion of *clause* is simply equated with CP. To ensure the distinction between root and non-root clauses, Truckenbrodt additionally offers the constraint in (62), reminiscent of the Wrap-XP constraint of Truckenbrodt (1999). In the case of embedded clauses, if one disregards the possibility of nested intonational phrases, this constraint conflicts with ALIGN-CP,L/R and, if it outranks it, has the effect of blocking the introduction of clause-internal intonational phrase boundaries which would have the effect of splitting a root CP into several Intonational Phrases.

(62) WRAP-CP

Each CP is contained in a single intonational phrase.

If nested intonational phrases are considered, a ranking of NONRECURSIVITY above ALIGN-CP and WRAP-CP would have the effect of favoring the phrasing in (63), with only one large Intonational Phrase, over the one in (64), in which the embedded clause forms an intonational phrase of its own and splits the root CP into two Intonational Phrases.

(63) $\text{IP}(\text{CP main clause } (\text{CP complement clause}))_{\text{IP}}$

(64) $\text{IP}(\text{CP main clause } \text{IP}(\text{CP complement clause})_{\text{IP}})_{\text{IP}}$

As we have seen above, (63) seems to be the correct phrasing in English. In Truckenbrodt's theory, it is to be expected that in other languages the more complex phrasing in (64) is manifested. In such languages, clauses would generally map onto Intonational Phrases, whether they are stand-alone or embedded in a larger complex sentence. Indeed such languages arguably exist. We will investigate different typological possibilities in the next section.

Remaining with English for the moment, we observe that the ranking NONRECURSIVITY >> ALIGN-CP, WRAP-CP also correctly predicts the phrasing of English in situ subject and adverbial clauses, repeated below for convenience.

(??) $\text{IP}(\text{subject clause} + \text{main clause})_{\text{IP}}$

(??) $\text{IP}(\text{main clause} + \text{in situ adverbial clause})_{\text{IP}}$

In the following configurations, however, there seems to be evidence for more complex intonational phrasing, as in the configurations repeated below for convenience. It is not immediately obvious that the ranking NONRECURSIVITY >> ALIGN-CP, WRAP-CP alone can provide an account.

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- (??) $IP(IP(\text{topicalized subject clause})_{IP} IP(\text{main clause})_{IP})_{IP}$
- (??) $IP(IP(\text{main clause})_{IP} IP(\text{extraposed complement clause})_{IP})_{IP}$
- (??) $IP(IP(\text{main clause})_{IP} IP(\text{coordinate adversative clause})_{IP})_{IP}$
- (??) $IP(\text{adverbial/adverbial clause } IP(\text{main clause})_{IP})_{IP}$

In such situations, the spirit of an Optimality Theoretic analysis should lead one to search for an independent higher-ranking constraint that would impose the complex phrasing in these cases, and these cases only. An obvious candidate would be one that refers to the high-extraposed position of the embedded clauses. Indeed, it has been independently proposed that constituents that are topical from an information-structural perspective form their own intonational phrases, as in (65) (Frascarelli 2000, Feldhausen 2010).

- (65) ALIGN-TOPIC, R (Feldhausen 2010)

Align the right edge of a [dislocated] topic constituent with the right edge of a prosodic phrase [*ι*/Intermediate phrase]

Although it is not trivially true, it is arguable that the high-extraposed clauses are topical in nature. If so, an account can be pursued invoking this information structure constraint, ALIGN-TOPIC; the ranking ALIGN-TOPIC >> NONRECURSIVITY would give rise to the desired phrasing. Without going into further details, we can conclude that an Optimality Theoretic account making use of generalised alignment constraints, WRAP-CP, NONRECURSIVITY and some higher-ranked information structural constraints can be constructed to account for the data Downing observed, and that this account would also open up interesting typological possibilities through the possible different rankings of the constraints in question.

Direct quotes need a similar treatment, in terms of an appropriate higher-ranked constraint, as they too, as Downing observed, form their own Intonational Phrases (see (15)). One possibility would be to adopt Downing's proposal and assume that direct quotations are syntactically adjoined to the root and then to assume some kind of more general constraint like ALIGN-TOPIC, which would also encompass non-topical root-adjoined clauses.

A potentially different route is offered by Selkirk's (2005) approach. She proposed that the syntactic constituent that is relevant to the formation of obligatory intonational phrases is Potts, Potts, Potts's (2002, 2003, 2005) [+ comma]-marked phrase or Comma Phrase (CommaP), where both simple sentences and *supplements* (i.e., Downing's root sentences and root-like fragments) belong to

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this category. What [+ comma]-marked constituents have in common, according to Potts and Selkirk, is the fact that they express a speech act of their own.⁵

This unifying feature is an attractive side of the proposal. But we note that it rests on the need to find an independent and objective way, to determine what does or doesn't constitute a speech act, which is not always a simple matter. Nevertheless, as far as the English in situ data are concerned, this approach successfully predicts that in situ embedded clauses (i.e., (46), (48) and (50) above) do not form their own intonational phrase as they do not form separate Speech Acts.

Regarding the examples with high-extraposed clauses, the proposal is partially successful. As Selkirk points out, the constraint in (61) makes an interesting prediction. Whenever material is adjoined to the root sentence, an asymmetry between the intonational phrasing of right and left adjunction is predicted. Whereas (root-level) right adjuncts necessarily follow the intonational phrase break introduced at the right edge of the root clause (see e.g., example (19)), the phrasing of (root-level) left adjuncts depends on their own status as a CommaP. If they are not themselves a CommaP (i.e., if they do not form their own Speech Act), it is predicted that they should not phrase separately as they do not insert an intonational phrase right edge of their own (Selkirk 2005). This however seems insufficient to account for Downing's intuition regarding examples in (33) to (38) according to which the left-adjoined adverbials and adverbial clauses are separated from the main clause by the left edge of an Intonational Phrase, one introduced by the root clause itself. As the main clause minus the adverbial (clauses) do not seem to constitute a separate Speech Act, they are predicted, in Selkirk's approach, to simply phrase in the same intonational phrase as the preceding adverbial (clause). Note, however, that equally, an approach based on NONRECURSIVITY, WRAP-CP, such as Truckenbrodt's, would need to be augmented to account for the phrasing difference between (47) and (52).

As far as direct quotes are concerned, it seems that these could easily be subsumed under the definition of CommaP, as they express a speech act of their own. This is apparent for instance if one observes that a question can be a direct quote inside a declarative main clause. If indeed direct quotes are CommaPs, Selkirk's proposal immediately accounts for their Intonational Phrase-status.

Overall, Selkirk's (2005) approach tackles the issue that embedded clauses do not necessarily have the same prosodic status as free-standing ones by proposing an additional requirement for clauses to map onto Intonational Phrases, a

⁵From this perspective, the intonational phrase is not only formed based on syntactic but also on discourse/pragmatic considerations. We will come back to this point subsequently.

semantic one, i.e., that they form their own Speech Acts. Typological differences, then, in this case, could arise from how important this additional requirement happens to be in a particular language.

A yet different approach was put forwards by Selkirk (2009, 2011) in her recent theory of the syntax-prosody mapping. In this proposal, called Match theory, it is argued that prosodic structure can show as much recursivity as syntactic structure. The most relevant constraint for the phrasing of complex sentences is the one given in (66).

(66) Match Clause

A clause in syntactic constituent structure must be matched by a constituent of a corresponding prosodic type in phonological representation, call it ι [Intonational Phrase].

Prosodic structure is thus, by default, assumed to be as faithful as possible to syntactic structure. Assuming minimalist phase theory (Chomsky 2001) and that CP is a phase of the syntactic derivation, Selkirk (2009: 14) proposes that a clause and thus an intonational phrase correspond to CP's Spell-Out domain, i.e., the complement of C. But then how does this theory propose to account for the facts observed by Downing, i.e., that embedded clauses sometimes fail to form Intonational Phrases on their own? Selkirk (2009) proposes to identify the notion of 'syntactic clause' with one of the functional heads of Rizzi (1997)'s split CP, and more particularly Force⁰, which represents the illocutionary force of the sentence. It is specifically assumed that only the clauses that are a complement of Force⁰, i.e., those that have an illocutionary force of their own, would match with an Intonational Phrase. Going back to Selkirk (2005) and Potts (2005)'s idea of a CommaP, Selkirk (2009: fn.13) also maintains that the constituents that form an intonational phrase of their own constitute a Speech Act (see also Truckenbrodt 2015). In situ embedded clauses would be the complement of a different C head. Thus, Selkirk (2009: 15) offers the two versions of Match clause given in (67).

(67) Match Clause: Spelling out the Complements of complementizer heads as ι

a. Match Force⁰ Clause

$ForceP[Spec\ Force' [Force^0\ CP\ [\dots\dots\dots]]]$
 $\Downarrow$ SPELL-OUT on the ForceP phase
 $\iota (\dots\dots\dots) \iota$

b. Match Comp⁰ Clause

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$$CP[Spec\ C' [Comp^0_{FncP} [\dots]]]$$

↓SPELL-OUT on any Comp-level phase
 $\iota (\dots) \iota$

(where $Comp^0$ designates any functional head of the 'complementizer layer')

So, the fact that embedded clauses are sometimes different from free-standing ones is taken to be a direct reflex of their assumed difference in syntactic structure, one corresponds to ForceP, the other to some other kind of ComplementizerP.⁶

As pointed out by Selkirk, typological differences between languages can be accounted for by different constraint rankings. Just like in Truckenbrodt's proposal involving WRAP-CP and NONRECURSIVITY, here if MATCH-COMP⁰-CLAUSE ranks high enough in a particular grammar, one would expect all clauses (and not only root clauses) to form Intonational Phrases. We will come back to this point in Section 3, as it seems that it is indeed the case that in some languages, e.g., Japanese, some non-root clauses also systematically form their own Intonational Phrase.

Turning now to the case of the high-extraposed clauses, in Match theory, a natural way to account for the fact that they are prosodically set off from the main clause by an intonational phrase left-edge corresponding to the left-edge of the main clause would be to assume that they are attached higher than the complement of ForceP. This would account for their phrasing directly based on their syntactic positioning. Taking a closer look at Rizzi (1997: 297)'s structure of the complementizer system, given in (68) one of the issues facing this extension of Selkirk's approach is that ForceP itself is already the highest assumed category of the complementizer system.

(68) ForceP >> TopP* >> FocP >> TopP* >> FinP

But perhaps one could posit that high-extraposed clauses sit in [Spec, ForceP]. This syntactic configuration would result in the desired phrasing in (69) and (70)

(69) $IP(\text{embedded clause } IP(\text{main clause})_{IP})_{IP}$

(70) $IP(IP(\text{main clause})_{IP} \text{ embedded clause})_{IP}$

⁶A reviewer notes that a different way of looking at Selkirk's proposal is to assume that clause always corresponds to CP, in which case (66) is equivalent to (67b) and Selkirk's proposal boils down to assuming (67a) alongside it.

In addition, in Match Theory, any clause that corresponds to a separate Speech Act, regardless of its position or size, also corresponds to an intonational phrase prosodically. This would give rise to the phrasing in (71) and (72). Direct quotes for instance fall under this category, as they come with their own illocutionary force (i.e., one can quote a question inside a declarative).

(71) IP(IP(embedded clause)_{IP} main clause)_{IP}

(72) IP(IP(main clause)_{IP} IP(embedded clause)_{IP})_{IP}

To sum up, the most innovative feature of Selkirk's Match Theory is that it assumes a more precise correspondence between syntactic and prosodic structure. By making reference to specific syntactic phrases (i.e., Force⁰ and Comp⁰) it introduces the potential for typological differences being the direct result of syntactic differences. Similarly, the specific syntactic position of a clause, i.e., high-extraposed or in situ, would have direct repercussions for its prosodic phrasing in this theory. In addition, the theory incorporates the idea that Speech Acts automatically map onto Intonational Phrases from earlier approaches.

The final approach we would like to discuss is similar in the sense that it also assumes a more direct link between syntactic structure and prosodic structure than earlier approaches. Hamlaoui & Szendrői (2015, 2017), propose that the notion of 'clause' is tightly linked to the position of the verb, and particularly the highest projection occupied by the root verb (see (73)). This projection can vary both within and across languages, depending on the particular type of sentence considered.

(73) a. Syntax-to-prosody mapping

i. ALIGN-L (HVP, *ι*)

Align the left edge of the highest projection whose head is overtly filled by the **root** verb, or verbal material with the left edge of an *ι*.

ii. ALIGN-R (HVP, *ι*)

Align the right edge of the highest projection whose head is overtly filled by the **root** verb, or verbal material with the right edge of an *ι*.

b. Prosody-to-syntax mapping

i. ALIGN-L (*ι*, HVP)

Align the left edge of an *ι* with the left edge of the highest projection whose head is overtly filled by the verb or verbal material.

6 *Prosody and syntax of argument and adverbial clauses*ii. ALIGN-R (*ι*, HVP)

Align the right edge of an *ι* with the right edge of the highest projection whose head is overtly filled by the verb or verbal material.

This approach makes the prediction that any embedded clause that is in the scope of the root verb should be prosodically integrated into the intonational phrase matching with a root clause. This is the case for English complex sentences containing an in situ subject clause, an in situ complement clause or an adverbial clause attached lower than the root verb. In contrast, any clause that attaches higher than the specifier of the projection hosting the root verb should be outside of the intonational phrase formed by the root clause. Given that it is the high attachment position of the extraposed clause that is assumed to be directly responsible for its phrasing, the default phrasing for complex sentences containing a right or left high-extraposed embedded clause is one in which the main clause remnant forms an intonational phrase, the entire sentence forms an intonational phrase, but the embedded clause itself is not an intonational phrase (cf. (69) and (70) above).

In this proposal the asymmetry between free-standing and embedded clauses, which as we noted several times above is potentially also a source of typological variation, is captured by an asymmetry between the syntax-to-phonology and the phonology-to-syntax mapping constraints. While syntax-to-phonology mapping only recognizes root verbs (i.e., main clause verbs in complex clauses, the only verb in a free-standing clause) and obligatorily maps root clauses' edges with Intonational Phrase boundaries (73a), the phonology-to-syntax mapping constraints see both root and non-root (i.e., free-standing or embedded) clauses and simply ensure that Intonational Phrase boundaries, if present, correspond to syntactic clause boundaries (73b). This has the effect that embedded clauses are not required to map onto their own intonational phrase to satisfy the prosody-to-syntax mapping constraints, only root clauses are. But if other constraints (e.g., prosodic or discursive) favor them to do so, this does not violate any of the mapping constraints in (73).

In addition, just as Truckenbrodt's and Selkirk's earlier proposals, Hamlaoui and Szendrői's proposal also needs to be augmented to account for phrasing where the high-extraposed clause does not only phrase separately from the following material, but itself forms a separate intonational phrase, as in (71) and (72) above. As far as direct quotes are concerned, Hamlaoui and Szendrői's proposal needs to be augmented to account for these too (see Section 3). In addition to the

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mapping principles listed in (73) above, they also assume a set of mapping principles adopted from Selkirk (2011) and Truckenbrodt (2015), which ensure that Speech Acts correspond to Intonational Phrases. This ensures that direct quotes form their own Intonational Phrases.

To sum up, Hamlaoui and Szendrői's proposal is similar to Selkirk's Match Theory in that it advocates for a more direct correspondence between syntactic and prosodic structure. In particular, it argues that the surface position of the root verb is directly relevant for determining the syntactic chunk that corresponds to an intonational phrase. Its innovative feature compared to the previous proposals is that it accounts for potential typological differences by assuming a difference between the syntax-to-phonology and the phonology-to-syntax mapping constraints.

To summarize, the various syntax-phonology mapping theories discussed in this section differ in the following respects:

- whether only one edge or both syntactic edges map onto an intonational phrase boundary AND whether prosodic recursion is the universal default or not,
- the exact definition of *clause* (i.e., CP, complement of Force, complement of C, highest projection occupied by the root verb),
- how the root/non-root distinction in terms of Intonation Phase-mapping is to be captured (i.e., a combination of WRAP, NONRECURSIVITY and ALIGN constraints; separate constraints for two types of clauses; syntax-phonology mapping associated with syntax-mediated discourse-phonology mapping constraints)

3 Cross-linguistic variation in the syntax-prosody mapping of argument and adverbial clauses

Data on the syntax-prosody mapping of argument and adverbial clauses in situ position and in extraposed positions is not available systematically for many languages. Rather, there is relevant data available from many languages, which together are still informative in a theoretical sense about the breadth of variation in this area.

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3.1 Subject clauses

Just like for English, subject clauses do not seem to have attracted a lot of attention and little seems to be known about whether and how systematically they are prosodically integrated to the rest of the clause. According to Downing's definition of a root clause, subject clauses might have different status depending on their structural position in different languages. The prosody of complex sentences containing a subject clause has been described by Truckenbrodt (2005), who examines the productions of a speaker of Austrian German. German is an interesting case as it is a V2 language. If the sentential subject is located in Spec,CP and the verb in C, it seems to us that the constraints discussed in Section 2.2 make different predictions as to the default phrasing of this type of complex sentence (based on their syntax only). Truckenbrodt (2005) can account for the phrasing in (74), in which the right edge of the intonational phrase corresponding to the sentential subject (and its corresponding left edge) is optional (so long as ALIGN-CP,RIGHT outranks NONRECURSIVITY). Selkirk (2009, 2011) predicts two different structures, corresponding respectively to (75) and (76), depending on how her two constraints rank.⁷ Another factor that might be relevant for Selkirk's analysis is the assumed syntactic analysis of the subject clause itself. Given that subject clauses always start with an overt complementizer, it is possible that sometimes that complementizer would be of the type that triggers obligatory phrasing of the subject clause as its own Intonational Phrase. Hamlaoui & Szendrői (2017) predict the obligatory presence of a single intonational phrase, as in (77). However, their proposed syntax-to-prosody mapping allows for an additional rightward intonational phrase boundary at the edge of the subject clause, as in (74), if required by some other constraint.

(74) $IP(IP(\text{subject clause})_{IP} \text{ rest of main clause})_{IP}$

(75) $IP(\text{subject clause } IP(\text{rest of main clause})_{IP})_{IP}$ (MATCH-FORCE)

(76) $IP(IP(\text{subject clause})_{IP} IP(\text{rest of main clause})_{IP})_{IP}$ (MATCH-C)

(77) $IP(\text{subject clause} + \text{rest of main clause})_{IP}$

Using declination and (Intonational Phrase-final) upstep as the main correlates for intonational phrasing, Truckenbrodt reports that in this dialect of German,

⁷Note, however, that some additional assumptions need to be applied to make sure that the finite main verb sitting in C will phrase together with its linearly following sister TenseP, given the exact wording of the definition in (67). One could perhaps assume it is a right-leaning clitic.

sentential subjects form their own intonational phrase. Example (78), adapted from Truckenbrodt, illustrates this type of sentence. Note that, in contrast to what is observed in example (14), the upward reset (noted with the $\wedge$ symbol) here takes place before rather than after the relevant boundary.

- (78) [CP [CP Dass die Leh^{L*+H}rerin dem Leh^{!L*+H}rer eine War^{^^L*+H}nung
geben will^{L-H%}] hat die Han^{L*+H}nelore gewun^{H*+L}dert^{L%}]

The prosody of subject clauses is also briefly discussed in Kandybowicz (2017) who focuses on four Tano languages, spoken in Ghana: Krachi, Bono, Wasa and Asante Twi. Using final L% as well as pause duration to diagnose the right edge of intonational phrases, Kandybowicz (2017: 126) argues that subject clauses also form their own intonational phrase in Krachi. An example is given in (79).

- (79) IP(Kε Kofi é-kyá-wũ)IP IP(mē ódum é-fwɪ)IP.
COMP Kofi PST-dance-CL.DET 1ST.SG heart PST-boil
'That kofi danced angered me (i.e., made my heart boil).'

According to Kandybowicz (2017: 129), a similar phrasing is observed in Bono. Example (80) illustrates a complex sentence containing a clausal subject in this language.

- (80) IP(Sé Kofi kùm akoko kē)IP IP(yě Áma nwanwa)IP
COMP Kofi kill.PST chicken the do Ama strange/surprise
'That Kofi slaughtered the chicken surprised Ama.'

Data from more languages and speakers are needed to establish the systematicity of this pattern and how to best account for it, so we leave the issue of the phrasing of subject clauses open.

3.2 Complement clauses

Turning to complement clauses, many languages seem to behave exactly like in English in prosodically integrating them with the main clause. This is the case of Turkish *ki*-headed finite complement clauses (Kan 2009), illustrated in (81), Hungarian (Hamlaoui & Szendrői 2017), in (82), or Basaá (Hamlaoui & Szendrői 2017), in (83).

- (81) IP(Duy-du^{L+H*}-k ki^{H-} Numan-lar^{!H*} Alman^{!H*} ya-ya
hear-PAST-1PL comp Numan-PL Germany-DAT
yerleş-iyor-muş^{L-L%})IP.
settle-FUT-EVID
'We heard that the Numans are settling in Germany.' (Kan 2009: 67)

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- (82) $\text{IP}(\text{Le}^{\text{L}*}\text{jla}^{\text{HL-}} \text{me}^{\text{L}*} \text{gkérdézte}^{\text{HL-}} \text{EL}^*\text{leonórától}^{\text{HL-}} \text{hogy a maláj}$
 Lejla PRT-asked Eleonora-from that the Malay
 $\text{lá}^{\text{H}*}\text{ny}^{\text{L-}} \text{el}^{\text{L}*} \text{menekült-e}^{\text{HL-}} \text{EH}^*\text{míliahoz}^{\text{L\%}}\text{)IP.}$
 girl PRT-escaped-Q Emilia-to
 ‘Lejla asked Eleonora whether the Malay girl escaped to Emilia.’
- (83) $\text{IP}(\text{mè} \text{ń-sòmból} \text{jí} \quad \text{lóngé} \text{'lé} \text{mbómbó} \quad \text{à} \text{-ń-lò})\text{IP}$
 I want to.know well that 1.grandmother 1.AGR-PST1-MH-arrive
 ‘I really want to know that the grandmother came.’

Some languages however seem to differ from the English-type of languages in that complement clauses systematically form their own intonational phrases. This is the case of the Fukuoka dialect of Japanese, discussed by Selkirk (2009). In this language, *wh*-questions are characterized by a H tone plateau that extends from the *wh*-word to the right-edge of the clause (Hayata 1985, Kubo 1989, Selkirk 2009). The words that belong to this so-called *wh-domain* (Selkirk 2009: and reference therein) do not carry their typical H*+L pitch accent. This pattern is observed in both matrix (as in (84)) and embedded *wh*-questions (as in (85)).

- (84) dare-ga kyoo biiru nonda?
 who-NOM today beer drank
 ‘Who drank beer today?’
- (85) dare-ga kyoo biiru nonda ka sitto?
 who-NOM today beer drank Comp know
 ‘Do you know who drank beer today?’

Additionally, the complementizer *ka*, in (84), carries a L tone and the matrix verb *sitto* a H*+L pitch accent. Selkirk (2009) proposes that this prosody is consistent with the phrasing in (86) and (87), which is predicted by a ranking of the Match constraint in (67) that places the constraint MATCH-COMP⁰-CLAUSE constraint higher than any constraint restricting the proliferation of intonational phrases in the structure, (e.g., NON-RECURSIVITY, Selkirk 1996).

- (86) $\text{IP}(\text{dare-ga kyoo biiru nonda})\text{IP}$
- (87) $\text{IP}(\text{IP}(\text{dare-ga kyoo biiru nonda})\text{IP} \text{ka sitto})\text{IP}$

Other languages have been reported to display a systematic prosodic separation of in situ complement clauses. This is the case of Luganda (Bantu, Uganda)

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and Huave (isolate, Mexico), in which according to Pak (2008) in situ complement clauses form their own *tone domain*. It is however not clear whether these tone domains correspond to Intonational Phrases or, rather, Phonological Phrases.

Kandybowicz (2017) argues that in Krachi and Bono, in (88) and (89) respectively, in situ complement clauses phrase separately from the main clause, which distinguishes them from Wasa and Asante Twi, two other Tano languages, in (90) and (91).

- (88) IP(Fe kwáré fi-gyɪ fě́é)IP IP(ɔkyí wũ é-mɔ bwaté wũ)IP
 2ND.SG collect 2ND.SG-eat COMP woman the PST-kill chicken the
 ‘You think that the woman slaughtered the chicken.’

- (89) IP(Wó dwene sě)IP IP(mméma kě be-kùm akoko kě)IP
 2ND.SG think COMP man.PL the 3RD.PL-kill.PST chicken the
 ‘You think that the men slaughtered the chicken.’

- (90) IP(Wó dwéne sé mɛ ré ma no be-kùm
 2ND.SG think COMP man.PL the 3RD.PL-kill.PST chicken the
 akóko no)IP

 ‘You think that the men slaughtered the chicken.’

- (91) IP(Yaw kaa sě Kofi bóɔ̃ Áma)IP
 Yaw say.PST COMP Kofi hit.PST Ama
 ‘Yaw said that Kofi hit Ama.’

Note that in Krachi and Bono, the complementizer phrases together with the matrix rather than with the embedded clause. A question that emerges is whether the difference in phrasing between Krachi and Bono on the one hand and Wasa, Asante Twi and more generally what we have called the English-type languages is at the syntax-phonology interface level (whether the former simply maps more clauses into their own intonational phrases) or whether there are other differences, syntactic or pragmatic in nature, that would explain why these complement clauses form their own intonational phrase. Some of the ideas that should be explored in this regard concern the information structural import of the embedded clause. If it were topical in nature, then perhaps the additional boundaries are due to that, as enforced by ALIGN-TOPIC. Also, observe that the Japanese examples involve *wh*-questions. As we will see in the next section, focal elements seem to sometimes have the effect of ensuring the presence of extra boundaries in

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Japanese. It should be explored whether the extra boundaries in the *wh*-questions are perhaps linked to their focal status.

Except for Japanese, the languages discussed so far happen to display a VO word order. Interestingly, some OV languages obligatorily extrapose complement clauses to a postverbal position. This is the case of German and Bangla. In his data from one Austrian German speaker, Truckenbrodt (2005) finds that the extraposed complement clauses do not form an intonational phrase of their own. In an experiment with more participants and different items, Truckenbrodt & Darcy (2010) however find evidence that German extraposed complement clauses consistently form their own intonational phrase. The authors offer an interesting discussion as to the phrasing preferences that emerge from the two experiments: whenever the main verb is stressed, the embedded complement clause preferably constitutes its own intonational phrase. This is illustrated in the examples (92) to (94), where italics indicate phrasal stress and small capitals nuclear stress (Truckenbrodt & Darcy 2010: 205).

- (92) $\text{IP}(\text{Der } \textit{Werner} \text{ hat auf dem } \textit{Treffen} \text{ gesagt, dass er der } \textit{Lola} \text{ das}$
 DET Werner has at the meeting said that he DET Lola the
 WEBEN zeigen will) IP
 weaving show wants
 ‘Werner has said at the meeting that he wants to show Lola weaving.’
- (93) $\text{IP}(\text{Der } \textit{Werner} \text{ hat auf dem } \textit{Treffen} \text{ GESAGT})\text{IP}, \text{IP}(\text{dass er der } \textit{Lola} \text{ das}$
 DET Werner has at the meeting said that he DET Lola the
 WEBEN zeigen will) IP
 weaving show wants
 ‘Werner has said at the meeting that he wants to show Lola weaving.’
- (94) $\text{IP}(\text{Der } \textit{Werner} \text{ hat dem } \textit{Maler} \text{ gesagt, dass er der } \textit{Lola} \text{ das WEBEN}$
 DET Werner has the painter said that he DET Lola the weaving
 zeigen will) IP
 show wants
 ‘Werner has said to the painter that he wants to show Lola weaving.’

Whereas the main verb is unstressed when preceded by an object (94), it is optionally stressed when preceded by an adjunct, as in (92) and (93). According to Truckenbrodt & Darcy (2010: 206), this difference is the central one in the intonational phrasing of the extraposed complement clause, and not possible differing landing sites across sentences. Based on evidence provided by binding relations

between a quantifier in the subject position of the main clause and a pronoun in the complement clause, they briefly argue that the extraposed clauses must occupy a low adjunct position, somewhere within the matrix CP. As the (low-adjoined) complement clause does not constitute a root clause, the possibility of matching it with its own intonational phrase goes against expectations and indeed suggests that other constraints may be at play that force a sentence like (93) to deviate from default syntax-phonology mapping.

Bangla is similar to German in displaying postverbal complement clauses in a language in which objects otherwise precede the verb. According to Hsu (2015), the position of complement clauses depends on their information-structural status. Postverbal ones are part of a broad focus, immediately preverbal ones are contrastively focused, and sentence-initial ones are topicalized clauses. According to Hsu, postverbal complement clauses form one intonational phrase with the main clause, as in (95).

- (95) $\text{IP}(\text{Jon bol-echi-lo je dadubhai kal rate oSudh khey-eche})_{\text{IP}}$
 John say-PERF-PST that grandfather last night medicine eat-PERF
 ‘John said that grandfather took medicine last night.’

The complex sentence in (95) contrasts with the ones in (96), in which the (non-discourse neutral) preverbal complement clause phrases separately. Unfortunately, the complete prosodic structure of the sentence is not provided.

- (96) Jon $\text{IP}(\text{dadubhai je kal rate oSudh khey-eche})_{\text{IP}}$ bol-echi-lo.
 John grandfather that last night medicine eat-PERF say-PERF-PST
 ‘John said that grandfather took medicine last night.’

Data from both languages suggest, again, that it might be too early to conclude that there is a systematic relation between the syntactic status of a particular type of embedded clause and its prosodic phrasing. More typological data is needed. When such data is collected, it seems important to bear in mind the syntactic structure, the attachment site and the information structural make-up of the complement clause.

Perhaps surprisingly, there are not that many studies that explore the prosody of direct quotes cross-linguistically. Hamlaoui & Szendrői (2017) discuss cases of direct quotes in Hungarian and show that, just like in English, direct quotes are independent intonational phrases.

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3.3 Adverbial clauses

The intonational phrasing of adverbial clauses is also an area that generally remains to be further explored. Among the available descriptions, a number of languages do not seem to differ from what has been observed in English and discussed in Section 2. Using the typical Eastern European (H-) L*H-L% intonational contour of *yes-no* questions, Hamlaoui & Szendrői (2017) observe that in Hungarian, complex sentences containing a *while*-clause form a single intonational phrase with the main clause when they are in situ, as in (97).

- (97) IP ([_{TopP} Péter_j IP([_{VP} el-viszi_i [_{VP} *t_i t_j* a gyerekeket a múzeumba
Peter PRT-takes the children-acc the museum-to
[_{DP} (addig) [_{CP} amíg Mari dolgozik?]]]]))IP
D while Mary works
'Does Peter take the children to the museum, while Mary is working?'

Whenever the *while*-clause is left-extraposed, as in (98), it is phrased outside the intonational phrase formed by the main clause. There is no evidence that it forms an intonational phrase too.

- (98) IP([_{TopP} [_{CP} Amíg Mari dolgozik], ([_{addig_i}) [_{TopP} Péter_j [_{VP} IP(el-viszi
while Mary works (D) Peter PRT-takes
a gyerekeket a múzeumba *t_i t_j*?]]]]))IP
the children-acc the museum-to
'While Mary is working, does Peter take the children to the museum?'

A similar pattern is observed in Basaá: the temporal clause in (99) is prosodically integrated to the main clause when appearing in situ, but is not when left-extraposed, as in (100) (Hamlaoui & Szendrői 2017).

- (99) IP(sóyól à-ŋ-ké 'í 'ŋgɛŋ Lingom à-ŋ-lɔ̃)IP
1.grandfather 1.AGR-PST1-leave at hour Lingom 1.AGR-PST1-arrive
'The grandfather left when Lingom arrived.'
- (100) IP('í 'ŋgɛŋ Lingom à-ŋ-ké IP(sóyól à-ŋ-lɔ̃)IP)IP
at hour Lingom 1.AGR-PST1-leave 1.grandfather 1.AGR-PST1-arrive
'When Lingom left, the grandfather arrived.'

Just like in Hungarian, there is no evidence so far that the left-extraposed adverbial clause forms an intonational phrase of its own. In Hamlaoui & Szendrői's (2017) approach, this phrasing is accounted for through the fact that only

main clauses obligatorily insert their intonational phrase boundaries. The embedded clause, when attached high enough (i.e., above the highest projection containing the root/main verb), simply sits outside the intonational phrase constituted by the main clause. For it to form an intonational phrase of its own, other constraints, for instance the prosodic constraint *STRONGSTART* (Selkirk 2011) in (101), need to prosodically promote it. As long as the extra intonational phrase edges match the edges of a syntactic clause, this more complex phrasing neither constitutes a violation of syntax-phonology nor phonology-syntax mapping constraints.

- (101) *STRONGSTART* (Selkirk 2011: 122)
 A prosodic constituent optimally begins with a leftmost daughter constituent which is not lower in the prosodic hierarchy than the constituent that immediately follows.

A more complex prosodic phrasing however emerges from Stockholm Swedish *if*-clause in V1 position, investigated by Myrberg (2013). These adverbial clauses, illustrated in example (102), occupy the initial position of V2 sentences, and have been analyzed as sitting in Spec,CP (Platzack 1998: 89–92).

- (102) Om 'sebrorna kom 'närmare så skulle 'Ida kunna 'röra vid dem
 if zebras-the came closer so would Ida be.able to.touch at them
 'If the zebras came closer, Ida would be able to touch them.'

In contrast with coordinated clauses, used as a baseline for comparison in her experiment, Myrberg (2013: 14) observes that complex sentences of the type in (102) receive variable phrasing. The main and embedded clause can either form two intonational phrases embedded within a larger one as in (103), be phrased within a single intonational phrase as in (104), or show the phrasing in (105), in which only the *if*-clause forms its own intonational phrase and is embedded in a larger one corresponding to the entire sentence. Each of her three speakers shows a clear preference for one of these strategies, using it for at least 6 out of 9 utterances.

- (103) IP(IP(*if*-clause)IP IP(main clause)IP)IP. (7/27 cases)
 (104) IP(*if*-clause + main clause)IP (7/27 cases)
 (105) IP(IP(*if*-clause)IP main clause)IP (13/27 cases)

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A high ranking of Selkirk's MATCH-COMP⁰ and, alternatively, Truckenbrodt's ALIGN-CP would favor the phrasing in (105). This phrasing is unexpected, as a default phrasing, under Hamlaoui & Szendrői (2017)'s approach. Rather, (104) is the one expected under the strict application of their default syntax-phonology mapping constraints, as the *if*-clause is not a root clause under their definition and should thus not, as a default, map onto an intonational phrase of its own. The phrasing in (103) seems problematic for all accounts in Section 2.2, as the second intonational phrase does not correspond to the main clause but is only a part of it. This phrasing calls for the purely prosodic constraint EQUALSISTERS in (106), proposed by Myrberg, which together with MATCH constraints, allows her to derive all and only the grammatical prosodic structures in (103) to (105).

- (106) EQUALSISTERS (Myrberg 2013: 75)
Sister nodes in prosodic structure are instantiations of the same prosodic category.

To model the intonational variation observed in Swedish, Myrberg casts her analysis in a version of Optimality Theory that allows variable ranking of constraints in (107) to (109), which respectively derive the phrasings in (103) to (105).

- (107) EQUALSISTERS >> MATCH-CP(S-P) >> MATCH-CP(P-S)
(108) EQUALSISTERS >> MATCH-CP(P-S) >> MATCH-CP(S-P)
(109) MATCH-CP(S-P) >> MATCH-CP(P-S) >> EQUALSISTERS

Crucially, any of the three rankings in (107) to (109) correctly predicts the invariable phrasing observed in Swedish sentences containing two coordinated clauses and given in (110).

- (110) IP(IP(clause)IP IP(clause)IP)IP

As pointed out by Myrberg, clausal embedding of the type discussed here is generally expected to present more intonational variation than e.g., complex sentences involving coordinated clauses, as it gives rise to a conflict between the need for a prosodic structure that reflects syntactic embedding on the one hand (i.e., recursive prosodic structure) and prosodic well-formedness constraints that favor a more balanced (i.e., flat) structure on the other.

4 Effects of information structure on the phrasing of argument and adverbial clauses

In this final section, we consider the effect of the discourse context, and more particularly information structure, on the prosodic phrasing of complex sentences. It has been argued that information structural categories such as focus and topic have the ability to insert extra prosodic boundaries and are sometimes responsible for the lack of isomorphy between syntax and phonology. Let us see whether and how this applies in complex sentences and take a glimpse at the various accounts that have been proposed to capture the interaction between the components of grammar involved.

4.1 Focusing

As we have seen in Section 3, Hungarian complement clauses do not generally align with their own intonational phrase edges, but are prosodically integrated with the main clause. They however do whenever the embedded complement clause contains a focused constituent, as in (111).

- (111) Péter AZT mondta/utálta/bánta meg, hogy MARY választottuk
 Peter D-ACC said/hated/regretted PRT that Mary-ACC selected
 be a bizottságba.
 PRT the committee-to
 ‘What Peter said/ hated/ regretted was that we selected MARY to the committee.’

It has been argued that foci generally need to satisfy the constraint in (112) (e.g., Reinhart 1995, Szendrői 2001). In Hungarian simple sentences, they do so by moving to the immediately preverbal position, where they align with the left edge of the intonational phrase and realize its head (i.e., sentence stress).

- (112) Focus rule or Stress-Focus Correspondence Principle
 ‘The focus of a clause is a(ny) constituent containing the main stress of the Intonational Phrase, as determined by the stress-rule.’

As discussed in Hamlaoui & Szendrői (2017), long focus movement to the edge of the matrix clause is possible with some verbs. There however seems to be a preference for embedded foci to remain in their clause. This, according to the authors, motivates the selection of a prosodic structure that, under the pressure of satisfying (112), contains extra intonational phrase edges. In contrast with

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approaches like Kanerva (1990) or Frascarelli (2000), information structure is not taken to directly influence prosodic structure. As the extra edges do align with the highest projection to which the embedded verb moves (here FocP), they simply do not violate the phonology-syntax constraints in (73b), while ensuring that (112) is satisfied.

An effect of focusing on prosodic structure is also found in Schuboe (2020) who examines German complex sentences of the type discussed in Truckenbrodt (2005) and Truckenbrodt & Darcy (2010). As we have seen in Section 3 in connection to German, the prosodic status of the items preceding the embedded clause might have an effect on its (lack of) prosodic integration to the main clause. Relatedly, Schuboe investigates the effect of focus and givenness on the phrasing of German complex sentences containing a complement clause, comparable to the ones in Truckenbrodt & Darcy (2010). He concentrates on three information-structural configurations: broad focus on the entire sentence in (113), narrow focus on the object of the main clause in (114) and narrow focus on the subject of the (extraposed) complement clause in (115). The condition in (114) differs from the other two in that the verb is in postfocal position and should thus be destressed. What is predicted by both Schuboe (2020) and Truckenbrodt & Darcy (2010) is that in this condition, the embedded clause should be prosodically integrated with the main clause.

- (113) [Ja/Nein Cornelius will dem Lehrer melden, dass Manuel eine
yes/no Cornelius wants the teacher report that Manuel a
Brille gestohlen hat]_F.
glasses stolen has
'Yes/No, Cornelius wants to report to the teacher that Manuel stole a
pair of glasses.'
- (114) Ja/Nein [Cornelius will]_G [dem Lehrer]_F [melden, dass Manuel eine
yes/no Cornelius wants the teacher report that Manuel a
Brille gestohlen hat]_G.
glasses stolen has
'Yes/No, Cornelius wants to report to the teacher that Manuel stole a
pair of glasses.'
- (115) Ja/Nein [Cornelius will dem Lehrer melden, dass Manuel]_G [eine
yes/no Cornelius wants the teacher report that Manuel a
Brille]_F [gestohlen hat]_G.
glasses stolen has
'Yes/No, Cornelius wants to report to the teacher that Manuel stole a

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pair of glasses.’

Despite a certain amount of variability, his data show a clear preference for the realization of an internal intonational phrase boundary in the broad focus condition, confirming Truckenbrodt & Darcy’s (2010) findings. In both narrow focus conditions, in contrast, there was a preference for the absence of internal intonational phrase boundary, which was more pronounced for the condition in (114). This latter result however tends to indicate that verb stress does not reliably predict the phrasing of the complement clause. What the two narrow focus conditions have in common, according to Schuboe, is that one of the two clauses contains only given material and there should thus be a dispreference for phrasing it separately. The phrasing in which both clauses are prosodically integrated is selected under the ranking of the information structural constraints *STRESSFOCUS* (similar to (112)) and *DESTRESSGIVEN* (which militates against stressing discourse-given items), as well as the prosodic constraint *RIGHTMOST* (which requires to keep nuclear stress rightmost) above syntax-phonology and phonology-syntax mapping constraints. Whenever nuclear stress shifts to the focus in (114), keeping stress rightmost as well as destressing post-focal material is better achieved by not mapping the complement clause into its own intonational phrase. In (115), in contrast, destressing pre-focal material is responsible for dephrasing and thus prosodic integration. Information structural requirement relating to the expression of focus thus seems to have an (indirect) effect on the phrasing of complex sentences.

4.2 Topicalization

In his study of complex sentences containing a complement clause in Catalan, Feldhausen (2011) observes that a prosodic break often separates the embedded subject from the rest of the complement clause. Just like in Myrberg’s study of Swedish, experimental data show that there is considerable variation in the phrasing of the complex sentences investigated. In 40% of the time, an intonational phrase break separates the matrix and the embedded clause (including the category of Intermediate Phrase, used by Feldhausen, the complement clause phrases separately in 80% of the time). An intonational phrase break is also sometimes found to separate the embedded subject from the embedded verb and object, grouping the embedded subject and the preceding complementizer with the matrix clause. Feldhausen (2008: 175) and Feldhausen (2010: 93) report that embedded left-dislocated phrases fail to phrase with the embedded clause, and also tend to phrase with the matrix clause while being followed by an intonational

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phrase break (over 65% of the time at normal speech rate). To account for this phrasing, schematized in (116), Feldhausen proposes the ALIGN-TOP, R constraint given in (65), which is responsible for inserting the right edge of an intonational phrase after the topic and separating it from the rest of the embedded clause.

(116) (... main V C Topic) YP

More prosodic structure than predicted by default syntax-phonology mapping constraints is thus found when an embedded clause contains a topic. This is also observed in Bàsàá and discussed by Hamlaoui & Szendrői (2017). An embedded topic, here *síngâ*, also fails to phrase together with the embedded clause in the example (117).

- (117) [TP [TP (_i hálà à-jè lóngé)] [CP ⁻¹ lé [TopP *síngâ* [TP (_i sóyól] à-n-ǵé_i [_{vP} *t_j* *t_i* jǵ)]]]]]]
 hálà à-jè lóngé lé síngâ sóyól à-n-ǵé jǵ
 so 1.AGR-be.PRES well that 9.cat 1.grandfather 1.AGR-PST1-eat 9.PRO
 ‘This is good that the cat the grandfather ate it.’
 (= This is good that the cat was eaten by the grandfather)

What is seen in (117) through the failure of Falling Tone Simplification, a phenomenon by which a sequence of HL-H tones becomes H-⁻¹ H when no left intonational phrase edge intervenes, is that the topical phrase sits outside of the intonational phrase constituted by the rest of the embedded clause. As argued by Hamlaoui and Szendrői, there is however no evidence that the topic itself forms an Intonational Phrase. Rather, and as visible in (117) and just like in Catalan, it phrases with the material that precedes it. The position of the left intonational phrase break aligning with the embedded TP rather than CP is, according to Hamlaoui and Szendrői, consistent with their idea that the syntactic projection relevant to the syntax-phonology and phonology-syntax mapping of the intonational phrase is the one to which the verb moves (here the embedded verb) and not generally CP, as proposed for instance by Truckenbrodt. In their approach, the constraint in (118) simply requires for a topic to align with the edge of an intonational phrase but not for it to form one.

- (118) ALIGN-TOPIC (Hamlaoui & Szendrői 2017: 23)
 Align the left or right edge of a topic with the left or right edge of an Intonational Phrase.

The embedded intonational phrase edge required to satisfy (118) is not however free to appear just anywhere. It has to satisfy the prosody-to-syntax constraints in (73b), which it does by aligning with the left edge of TP (as this is the highest projection to which the verb moves in this structure). This approach, according to the authors, better accounts for the cross-linguistically limited distribution of topics, which tend to appear at clausal edges (i.e., where intonational phrase edges appear to satisfy syntax-phonology and phonology-syntax mapping constraints) rather than clause medially.

5 Conclusion

In this paper we explored the prosodic realisation of complex sentences involving argument and adverbial clauses. We started by reviewing a body of evidence about English complex sentences, including complement clauses, subject clauses, adverbial clauses and direct quotations, – the work of Downing (1970). The first important finding was that embedded clauses do not always form independent intonational phrases, despite their syntactic clausal status. The second important observation from this body of evidence was that the attachment site of the embedded clause affects its prosodic phrasing, with high-extraposed clauses typically being separated from the main clause by intonational phrase boundaries.

Next, we reviewed a series of proposals from the literature enumerating their main tenets and exploring the predictions they make with respect to Downing's findings. In particular, we looked at Selkirk's earlier work using syntax-prosody alignment constraints for all clauses and its extension involving the idea of CommaP (Potts 2005), which ensures that clauses that form separate Speech Acts form their own Intonational Phrases. We also discussed Truckenbrodt's WRAP-CP proposal, to account for the fact that in situ embedded clauses typically do not form their own intonational phrases. Next we turned to two proposals that advocate a more direct, more detailed correspondence between syntactic and prosodic structure. Selkirk's (2005, 2009, 2011) Match Theory involves a more fine-grained mapping between different kinds of Complementizers (i.e., Force⁰, Comp⁰); Hamlaoui & Szendrői (2015, 2017) argued for the relevance of the surface position of the main or root verb in the structure, and a difference between syntax-to-prosody and prosody-to-syntax mapping constraints.

In Section 3, we expanded the empirical basis of our discussion to other languages, with an aim to formulate typologically valid generalizations regarding the different types of embedded clauses (i.e., subject clauses, complement clauses, adverbial clauses and direct quotations). This proved difficult, due to the lack of

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systematic data on all of these domains in the literature. In the final section, we explored the effect of information structure on prosodic phrasing involving complex clauses. As has been observed also for simplex sentences, arguably, prosodic phrasing is sometimes affected by information-structural considerations. Focal and topical elements can trigger additional prosodic boundaries (see ALIGN-TOPIC and ALIGN-FOCUS). From the reviewed evidence it seems that information-structural considerations also play a role in determining the prosodic structure of complex sentences.

In sum, we would like to draw the conclusion that the current existing theories fare well when faced with data involving complex sentences with argument and adverbial clauses. It also seems to be the case that systematic data collection in this area from a typologically wide array of languages would be immensely helpful to advance our understanding of the prosody of complex sentences, and consequently our quest for the best theoretical framework.

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Chapter 7

The syntax-prosody mapping of coordinated clauses, relative clauses, parentheticals and left-dislocations

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This chapter reviews findings and suggestions about the syntax-prosody mapping of particularly non-integrated syntactic collocations, i.e., coordinated clauses, relative clauses, parentheticals and left-dislocations. Downing (1970) usefully structured this terrain. His suggestion in terms of root sentences can be seen as a precursor to the more recent proposal of Kan (2009) and Selkirk (2011) that separate illocutionary acts must be separate IPs. The discussion also reveals some amount of disconnection between searches for generalizations in experimental work and what generative analyses there may be to account for them. The chapter attempts to cover some of both and points out some possibilities for further theorizing.

1 Overview

This chapter addresses the prosodic detachment of elements that can also be said to be syntactically and/or semantically detached from each other or from their host constituent to some larger extent: coordinated clauses, non-restrictive



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relative clauses, parentheticals and dislocated constituents. This is discussed by comparison with elements that show less syntactic or semantic detachment, such as restrictive relative clauses.

One of the anchor points of the discussion is a classical suggestion of Bruce Downing (1970), according to which the relevant syntactic detachment leads to prosodic detachment in the form of “obligatory pause”, today often analyzed as obligatory IP boundaries (Nespor & Vogel 1986). In its modern form due to Kan (2009) and Selkirk (2011), this takes the form that illocutionary clauses are separated by obligatory IP boundaries.

We begin with this background in Section 2. In the following sections on coordinated clauses (Section 3) and relative clauses (Section 4), we review related studies from the syntax-phonology interface, phonetics, and parsing. These seem to largely call for an additional analysis of the initial picture without challenging it. The section on parentheticals (Section 5) broadens the theoretical basis we set up with our discussion of Downing (1970) and speech acts. We outline on the basis of Bing (1985, 1979) what a possible generative approach to integrating parentheticals that lack nuclear stress might look like and we present a review of experimental studies showing the variation that exists when it comes to parentheticals. In our discussion of left-dislocations (Section 6), we report on recent literature that argues for the alignment of topics with prosodic boundaries and discuss this in relation to the theoretical claims of the initial sections of this chapter. Section 7 sums up the paper.¹

2 Theoretical background

2.1 Root sentences

The prosodic boundaries of primary interest in the current chapter are Intonation Phrase (IP) boundaries². We structure our discussion with reference to an early piece of work on the IP boundaries, namely, the PhD thesis of Bruce Downing (1970). He referred to the relevant perceived phonological juncture as “pause” (not necessarily implying a phonetic pause, i.e. a stretch of silence). He investigated the distribution of pauses that are obligatory, and analyzed obligatory pauses in terms of the assignment of phonological phrase boundaries (using a term

¹A closely related chapter is that of Hamlaoui & Szendrői (2025 [this volume]). It focuses on argument and adverbial clauses.

²In this chapter, we use for the intonation phrase the abbreviation IP in the running text and iota in the examples. For the syntactic inflectional phrase, we use the abbreviation *InflP* (so that it can be distinguished from the intonational phrase).

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from Chomsky & Halle 1968) relative to the syntactic structure. What Downing referred to as phonological phrase later came to be called intonation phrase (IP), see, e.g., Bing (1979), Nespor & Vogel (1986). The term phonological phrase is used by Nespor & Vogel (1986) and others for a lower level of prosodic constituency. The following pieces of motivation for the central role of Downing (1970) in our discussion are also a first introduction to Downing's work.

1. It is broad, addressing the prosodic separation of conjoined clauses, non-restrictive relative clauses, appositions, vocatives, left- and right-dislocated elements, topics at the clausal periphery, speech-commenting parentheticals and parenthetical adverbs.
2. At the length of more than 200 pages, Downing defends a single principled suggestion across all these collocations: phrase boundaries (today's IP boundaries) are inserted at the edges of syntactic root sentences, for which Downing gives a specific syntactic definition.
3. Downing's syntactic definition of root sentences foreshadows better-known later syntactic analyses such as the syntactic orphan analysis by Rotenberg (1974) and Safir (1986) and a hypothesis of syntactic attachment with crossing syntactic lines by McCawley (1982), recently adopted by Schlenker (2015).
4. Modern prosodic suggestions such as Kan (2009) and Selkirk (2011, 2009) build on Downing's suggestion in a fairly direct manner: the syntactic constituents giving rise to obligatory IP boundaries (Downing's root sentences) are here characterized as illocutionary clauses.

Downing's suggestion is rendered in (1), where (1a) is a syntactic definition and the principle of the syntax-phonology mapping in (1b) builds on it.

- (1) a. *Root sentences*:
A clause is a root sentence if it is not the case that it is embedded in a higher clause with a predicate of its own (cf. Downing 1970:30).
- b. *Obligatory boundary insertion convention (OBI)*:
Phonological phrase boundaries "\$" (today's IP boundaries) are inserted at the left and right edge of every root sentence (cf. Downing 1970:31).

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For example, Downing makes the standard assumption that subject clauses and object clauses are parts of a higher matrix clause with a predicate of its own. They are therefore not root sentences and do not trigger obligatory pauses (p.26).

On the other hand, consider an example with coordinated sentences from Downing (1970: 27), to which we have added some modern CP nodes for the purpose of discussion Figure 1.

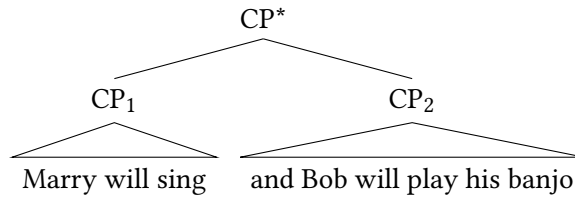


Figure 1: Coordinated root sentences. The example is from Downing (1970), the CP nodes are from us.

The way Downing looked at it is still relevant today: The two sentences (here notated CP₁ and CP₂) are dominated only by a higher sentence-node (here notated CP* for ease of reference) that does not have a predicate of its own. Both CP₁ and CP₂ are therefore root sentences as defined in (1a). Therefore (1b) inserts a phrase boundary between them (today: an IP boundary). The phrase boundary gives rise to the perceptual division that Downing calls “pause” and notates “/”, so that (2) is obtained, which Downing defended as the empirically correct result.

(2) Mary will sing / and Bob will play his banjo.

Importantly, Downing envisaged a similar syntax for expressions where this is not as obvious, at a more abstract level of representation. For example, the non-restrictive relative clauses in (3) are separated by obligatory pauses (p.148, 152), while the restrictive relative clauses in (4) may show a pause at its right edge but need not (p.14). Restrictive relative clauses (RRC), like *you met* in (4), restrict the nominal’s denotation: Not all the boys are members of X, but only the boys the person referred to as “you” did meet. A RRC thus restricts the class of referents of the noun. Non-restrictive relative clauses (NRRC), in turn, provide additional information about the referent without such a restriction (see Stark 2016: 1030, Lehmann 1984: §V.2, Jackendoff 1977 and many others).

- (3) a. Ralph, / *who has always been a good student*, / failed the exam.
- b. The library, / *which is a large stone and glass building*, / is on the east side of the campus.

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- (4) The boys *you met* (/) are all members of the same fraternity.

Downing (1970: 145ff) argued that non-restrictive relatives are represented as in Figure 2 at an abstract level of representation. For him, this was a post-cyclic representation, though one that undergoes further transformations to (5a) and then to (5b) before the surface. We continue to use the modern syntactic label CP for illustrating Downing's ideas.

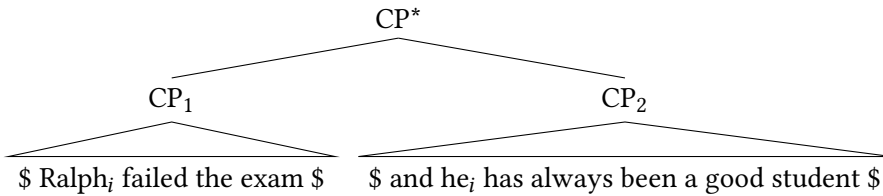


Figure 2: Illustration of the insertion of Downing's (phonological) phrase boundaries "\$" at the edges of a root sentence that is later transformed into a non-restrictive relative clause (see (5)). The example is from Downing (1970), the CP nodes are from us.

- (5) a. $Ralph_i$, \$ and he_i has always been a good student \$, failed the exam.
 b. $Ralph_i$, \$ who_i has always been a good student \$, failed the exam.

Here again CP_1 and CP_2 are root sentences by (1a), since CP^* does not have a predicate of its own. Therefore, CP_1 and CP_2 come with markers for phrase boundaries at their edges by (1b). The phrase boundaries are retained when CP_2 is placed within CP_1 as in (5a) and when it is further transformed to (5b).

Downing (1970: 115) also discusses speech-commenting parentheticals as in (6d) and derives obligatory pause for it at both its edges as shown in (6). The derivation starts with a single root sentence (6a), which is then divided into two root sentences (6b). Based on this, phrase boundaries are established (6c). After that, a transformation to (6d) applies, placing one root sentence within another in parallel to (5).

- (6) a. I believe John has sold his Mustang.
 b. [I believe] [John has sold his Mustang] (two root sentences)
 c. \$[I believe]\$ \$[John has sold his Mustang]\$\br/>
 d. John, \$ I believe \$, has sold his Mustang.

Various later accounts have retained the syntactic detachment of the parentheticals from Downing’s suggestion, which triggers prosodic detachment. McCawley (1982), for example, suggested untypical attachment at the sentence level, a suggestion also recently argued for by Schlenker (2015). We retain our notation, and we represent this suggestion as in Figure 3. The hierarchical relations are parallel to Downing’s structure Figure 2. With crossing lines assumed, the detached element (here CP₂) occurs in its surface position.

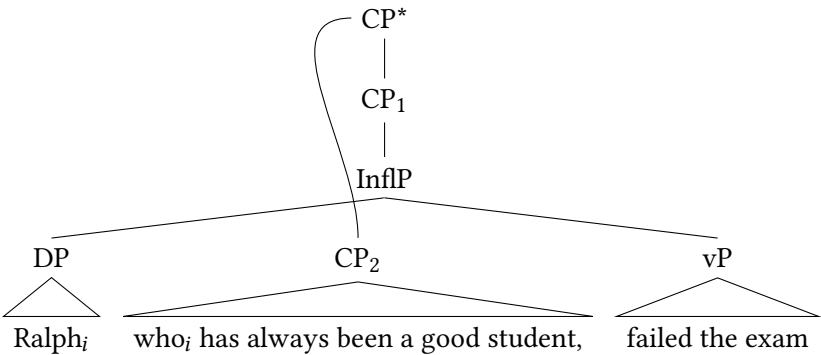


Figure 3: Untypical attachment of CP₂ at the sentence level following McCawley (1982) and Schlenker (2015)

Yet another suggestion is the syntactic orphan analysis of Rotenberg (1974), which was also argued for with appositive relatives by Safir (1986). In this analysis, the elements under discussion do not have a syntactic mother-node at all and are orphans in that sense. In the example used here, CP₂ then does not have a syntactic mother, as in Figure 4.

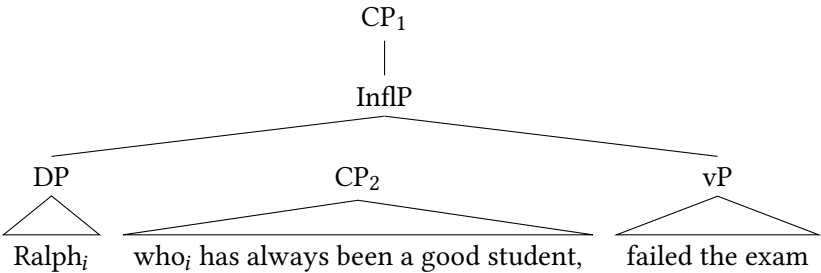


Figure 4: Orphan analysis of CP₂ following Rotenberg (1974) and Safir (1986)

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We mention two further implementations of these special elements in different terms. Potts (2005) suggested that certain constituents in this class, his *supplements*³, have standard syntactic attachment (without orphanage or crossing lines) and are marked by special features; these features trigger separate semantic interpretation at LF and separate prosodic interpretation at PF. de Vries (2006, 2012) developed the idea that there is a special separate merge-operation that merges such elements in standard configurations but with special detachment consequences. The analysis of Downing (1970) also has interesting consequences for further phenomena that do not necessarily include sentential material such as dislocations. While he doubts the grammaticality of right-dislocations in English and rather re-interprets them as “amplificatory parentheses” (*afterthoughts* in modern terms), which are added only in the performance of a sentence (Downing 1970: 141), he provides a detailed analysis of English left-dislocations (which he also labels *theme topicalization*) as in his examples in (7), cf. Downing (1970: 41 ff.).

- (7) a. John, / he never does anything right.
b. Me and Albert, / what would you do without us?

As shown in these examples, Downing (1970) diagnosed an obligatory prosodic boundary (obligatory pause) between the left-dislocated constituent and the main clause. He derived this from a syntactic analysis in which the left-dislocated constituent adjoins to the root sentence, and is outside of the root sentence in the relevant respect. The root sentence is delimited by prosodic phrase boundaries due to the OBI, so that the left-dislocated constituent is separated from the root sentence by an obligatory phrase boundary.

By contrast, a fronted focused constituent as in (8a) is phrased with the following clausal material, according to Downing. Downing (1970: 47 f.) accounted for this by taking sentences such as (8a) to be derived from a corresponding cleft sentence (as in (8b)).

- (8) (What do you like to eat?)
a. Fried zucchinis I like to eat.
b. It is fried zucchinis that I like to eat.

³Potts (2005) subsumes under the notion of *supplements* what we have called parentheses in the narrow sense (= speech-commenting parentheticals) and also appositive relative clauses (=NRRC).

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In the cleft sentence, the clause *that I like to eat* is not a root sentence, and thus not delimited by obligatory prosodic boundaries.

In this analysis, other fronted theme elements (elements not part of the focus) cannot be derived from a cleft sentence and so must be left-adjoined to the root sentence. The OBI predicts that they are also separated by an obligatory prosodic boundary. Downing accepts and endorses that consequence. An example from him with a fronted theme argument (without pronominal resumption) is shown in (9a), an example with a fronted adverbial in (9b), cf. Downing (1970: 44, 50).

- (9) a. That book, / I never managed to finish.
- b. In the afternoon, / everyone went swimming.

In later studies, dislocations (and other elements) are labeled “extra-sentential elements” (e.g. Huddleston & Pullum 2002, Astruc-Aguilera 2005, Astruc-Aguilera & Nolan 2007) and this notion nicely reflects the idea by Downing: Because the elements are realized sentence-externally, i.e. outside the root sentence, an obligatory prosodic boundary exists between them and the root sentence (triggered by the root sentence).

2.2 Illocutionary acts

Potts (2005) initially attributed a speaker-oriented interpretation to his supplements, which was later refined in Harris & Potts (2009). Selkirk (2011, 2009) and Kan (2009) then developed the idea that what is special about root sentences is that they carry a relation to illocutionary force. Kan (2009), in a study on Turkish intonation, analyzes them as a special illocutionary ForceP and suggests that this ForceP is identified with an IP by (10).

- (10) During Spell-out, a Force_{Illoc}P is mapped to an IP (Kan 2009: 145).

For Kan, the syntactic category Force_{Illoc}P is a syntactic reflex of speaker-intended illocutionary force.

Selkirk (2011, 2009) suggested that the syntactic complement of a Force head is an illocutionary clause and is matched to IP boundaries. She casts this in the format of her more general match constraint format for the syntax-prosody mapping (11).

- (11) Match(illocutionary clause, IP):
The left and right edges of an illocutionary clause must correspond to the left and right edges of an IP (Selkirk 2009, 2011).

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We will refer to this constraint as Match-iCP. These suggestions have found some support since then.

Güneş (2015) built on Kan (2009) and Selkirk (2011) and looked at a variety of non-integrated expressions in Turkish, assessing their edges in terms of measurements of final lengthening, pause length and fundamental frequency. Finite clauses infixed into the main assertion (e.g. *please turn to the side*, when unrelated to the main clause) and appositive relative clauses introduced by *ki* are similar to sequences of unembedded clauses in being separated by IP boundaries. Güneş argued that it is the illocutionary act status that is decisive for IP formation, not their status as finite clauses. For example, vocatives show values comparable to these latter categories with IP boundaries. Building on Hill (2013) and others, vocatives were argued to be separate illocutionary acts. This can form the basis of an understanding of their boundary strength. Conversely, the category of comment clauses (infixed *I swear, it is unbelievable*) include finite verbs but pattern more closely with phonological phrases (= the prosodic level below the IP) at their edges. They may be classified as not constituting separate speech acts. Güneş interpreted her results more generally in the light of the possible correlation between illocutionary acts with IPs (where they are not finite clauses, they are augmented to Force_{Illoc}P, in her account), and showed how the results support that correlation.

Truckenbrodt (2015) also supported the correlation between illocutionary acts, or, for him, speech acts, and IPs. He mostly discussed cases at the fringes of Downing's separate root sentences and argued that some of them are neither separate IPs nor separate speech acts. On the other side of this divide, when they are separate speech acts, they also seem to be separate IPs.

Ishihara (2022) concentrated on the role of CPs that are not illocutionary acts. He argued that they do not play a special role in the syntax-phonology mapping. His more general argument that the status of CP or ForceP is not accessible to the syntax-phonology mapping then led him to a revision of Match-iCP (and the OBI, if you will) in terms of speech acts alone, with no reference to CP or ForceP (12).

- (12) A speech act is realized as an IP in the prosodic representation (Ishihara 2022).

Next to the obligatory boundaries inserted by the OBI and its modern versions such as Match-iCP, there are also optional but also somewhat systematic IP boundaries at other syntactic edges. We return to them and to Ishihara's point about regular CPs in connection with restrictive relative clauses below (see Section 4.2).

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3 Coordinated clauses

As was seen in Section 2, coordinated unembedded clauses are separate root sentences and are separated by obligatory pauses in Downing's account. The example (2) is repeated here as (13).

(13) Mary will sing / and Bob will play his banjo.

On the other hand, embedded coordinated clauses in complement position are not root sentences by Downing's definition (1b) and are not separated by obligatory pauses by (1a). That can be seen in the following example from Downing (1970: 27).

(14) I hope that Mary will sing (/) and Bob will play his banjo.

The following sections address coordinated root sentences (Section 3.1), coordinated root sentences with multiple layers (Section 3.2) and embedded coordinated clauses (Section 3.3).

3.1 Coordination of root sentences

Kawahara & Shinya (2008) experimentally investigated coordinated clauses in unembedded position in Japanese. They showed F0-raising initially in the clauses and F0-lowering in final position of the clauses, which they plausibly analyzed as correlates of separate IPs for the separate clauses. This analysis is endorsed in Selkirk (2009). The results by Kawahara & Shinya (2008) can be seen as experimental confirmation that separate coordinated clauses constitute separate IPs.

An aspect of the results of Truckenbrodt (2002, 2004, 2007) can be seen in the same way. A phenomenon of intonational upstep occurred with considerable regularity at the right edge of a clause when a coordinated clause follows, but not internal to the clauses. This upstep is analyzed as correlated with an utterance-medial IP boundary.

In the analysis of Selkirk (2011, 2009) and Kan (2009), coordinated declaratives can be analyzed as separate illocutionary clauses with separate IP boundaries, due to Match-iCP.

For German, Truckenbrodt (2015) showed apparent exceptions to the notion that coordinated declaratives are necessarily separate illocutionary clauses and separate IPs. In German, the standard form of a declarative is a V2-clause with the finite verb in second position. It is possible, however, for a sentence adverb

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in the initial position of a V2-clause to scope over a coordinated following V2-clause, as in (15). The finite verbs in clause-second positions are italicized to help the reader with the syntactic analysis of this example.

- (15) Angeblich *hat* der Mond geschienen und es *ist* kalt gewesen.
 allegedly has the moon shone and it is cold been
 ‘It was claimed that the following both hold: the moon shone and it was cold.’

These structures can also be coerced, using focus, into a single IP (marked by *ι*), as in B’s utterance in (16). The evidence that there is only a single IP here is that there is only a single nuclear stress, namely the one on the conjunction, shown by underlining. (In response to an anonymous reviewer’s question, we note that there is no causal or temporal relationship between the two propositions in connection with this semantic and prosodic integration.)

- (16) A: Did the moon shine? Was it cold?
 B: (Angeblich *hat* der Mond geschienen und es *ist* kalt gewesen)*ι*
 ‘Allegedly, the moon shone and it was cold’

Since this class of cases involves neither a separate illocutionary act nor separation by obligatory IP boundaries, it does not challenge the interface claims that separate illocutionary clauses correlate with obligatory separate IPs. Truckenbrodt (2015) analyzes it in terms of untypical coordination of the second clause with a part of the first clause, as in (17). The two clauses constitute one illocutionary act together, which thus does not strictly require an internal IP boundary. Notice also that there is no right edge of a full CP preceding the conjunction.

- (17) [_{CP} Angeblich [[_C *hat* der Mond geschienen] und [_{CP} es *ist* kalt
 allegedly has the moon shone and it is cold
 gewesen]]].
 been
 ‘Allegedly, the moon shone and it was cold.’

German has a set of other phenomena of “asymmetric coordination” in which a second conjunct in the form of a declarative V2-clause shows behavior that is unexpected for a coordinated declarative (Höhle 1990, Reich 2009).

Güneş (2015: 279ff.) addresses related issues in Turkish with a related suggestion in terms of the syntactic label Force_{SENTP}.

3.2 Multiple layers of coordinated root sentences

Ladd (1988) compared the intonation of sentences like (18a) with sentences like (18b) in English.

- (18) a. Allen is a stronger campaigner, and Ryan has more popular policies, but Warren has a lot more money.
- b. Ryan has a lot more money, but Warren is a stronger campaigner, and Allen has more popular policies.

The structures are [AB]C on the one hand, and A[BC] on the other, where A, B, and C are clauses. The respective sisters in [AB]C and in A[BC] are syntactically coordinated at two levels: At a higher level, the bracketed constituent is coordinated with its sister. At a lower level, the two elements inside of the brackets are sisters. Ladd found systematic distinctions in tonal height between the two conditions. He accounted for them using phonological structures for the two conditions that mirror the syntactic structures, i.e. the phonological structures are likewise [AB]C vs. A[BC]. The tonal height phenomena are captured with these structures if a sister configuration of X and Y leads to lowering of the tonal register of Y relative to that of X. In A[BC], this amounts to lowering from A to B (sisters A and [BC]) and lowering from B to C (sisters B and C). In [AB]C, it amounts to lowering from A to B (sisters A and B) but not from B to C. Instead, C is lowered relative to its sister [AB], which amounts to lowering relative to the initial reference height of A.

Ladd's main results were reproduced for German in Truckenbrodt & Féry (2015). The tonal height distinctions were also shown to interact with German upstep in an expected fashion. Match-iCP can provide the isomorphic mapping from CPs to IPs.

The updated discussion of Ladd's account there also shows how Ladd's account bears on recursion in prosody, an issue that has also received attention in other recent work. For details, we refer the interested reader to Wagner (2005), Selkirk (2011), Ito & Mester (2013), Féry (2017), Bennett & Elfner (2019), Feldhausen et al. (2021) and references cited therein.

3.3 Genuinely embedded coordinated clauses

Downing (1970) strengthened the prosodic distinction between unembedded and embedded coordinated clauses with the example (19).

- (19) Bill believed his father was older than his mother / and his mother was older than his father.

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Downing's point is as follows. With a pause in the position of “/”, it has a reading in which Bill has a wrong belief, but not a contradictory one. In this reading, the clausal structure is as in (20a). Here the final clause Y is a root sentence and requires the pause to its left. Importantly, when the sentence is read with the pause omitted, the only reading left is one in which Bill has contradictory beliefs. Here the structure is as in (20b). The final clause is part of an embedded coordination. It is not a root sentence and therefore does not require an obligatory pause to its left.

- (20) a. [Bill believed X] & Y
 b. Bill believed [X & Y]

Downing's account correctly predicts that the omission of the pause leaves only the reading involving contradictory beliefs, since only this reading has a possible phrasing without the internal pause. The structure and reading in (20a) require an obligatory pause to the left of Y.

4 **Relative clauses**

This section is dedicated to relative clauses. As was seen, Downing (1970) predicts that non-restrictive relative clauses (NRRC) are root sentences and are thus separated by obligatory pause. He does not exclude that pauses exist where his account does not insert them, such as at the edges of restrictive relative clauses (RRCs). However, it is expected that they are not obligatory in the same way. In this area of pauses that are less strict, we already saw the tendency for pauses at the boundaries of embedded clauses. In this section, we turn to studies that further bear on the stricter boundaries of NRRCs and on factors that encourage boundaries at the edges of RRCs. In Section 4.1, we address studies dealing with the prosodic realization of the edges of these two types of relative clauses, and in Section 4.2, we take a closer look at prosodic boundaries at the edges of RRCs. In Section 4.3, we turn to some relevant factors from studies on the interplay between prosodic edges and parsing preferences. Finally, in Section 4.4, we briefly address another prosodic effect in relative clauses: syntactic movement for relative-clause formation interacts with stress-assignment.

4.1 **The prosodic edges of relative clauses: informal and experimental findings**

Traditional descriptions converge with the expectations of the account of Downing (1970) in the following respects: non-restrictive relative clauses (NRRCs) cre-

ate a prosodic unit of their own and their edges are crucially marked by pauses, while restrictive relative clause (RRCs) create a prosodic unit together with the preceding material and are not accompanied by pauses (see, e.g., Smith 1964: :38, Carlson 1977 for English; Lehmann 1984: 263 for German; Bello & Cuervo 1970:128 [1898], Hadlich 1971: 136, Franch & Blecua 1975: 1024, D’Introno 1979: 214 for Spanish). In the prosodic literature, these distinctions are analyzed at the level of the IP (e.g. Nespor & Vogel 1982, 1986/2007: 88, Selkirk 1984, Vigário 2003, Kan 2009, Hualde et al. 2020: 265, Güneş 2015).⁴

Garro & Parker (1982) phonetically investigated restrictive and non-restrictive relative clauses in English, and Garro & Parker (1983) reported results of a related experimental study in Spanish (from the *Río de la Plata* region). They compared RRCs and NRRCs for the length of pauses at the left and right edge of the relative clauses, as well as the length of the vowel preceding these edges, i.e. in a position where final lengthening can be expected if the edge of a relative clause coincides with a prosodic boundary. In both languages, and at both edges, NRRCs showed greater pause length and greater length of the preceding vowel than in RRCs. Their results support the original description of Downing (1970) or the analysis in terms of illocutionary-act related clauses by Selkirk (2005, 2009) and Kan (2009): NRRCs show stronger boundaries at both their left and right edges than restrictive relative clauses. (To the extent that CPs that are not illocutionary acts also trigger prosodic boundaries in RRCs, they do so in this data in a way that does not neutralize this effect.)

We would like to mention that the study on Spanish is based on productions of two speakers, and that in our reading of their paper, one of the cues for triggering the distinction between RRCs vs. NRRCs was the presence of orthographic commas on both sides of NRRCs vs. the absence of commas with RRCs. It is conceivable that the orthographic representation, in addition to the mental syntactic representation, played a role in triggering these prosodic effects (see Potts 2005 and Selkirk 2009: fn.13 for “comma intonation”; see Feldhausen 2016a: 300 for further comments on test items and punctuation marks).

Closely related to the last point, Poschmann & Wagner (2016) investigated differences in phonetic realization and acceptability between RRCs and NRRCs in German. They considered subject relative clauses in unextraposed position and in extraposed sentence-final position, and their stimuli were assessed under subject focus, object focus and sentence focus. Of particular interest here is that non-extraposed NRRCs were preceded by stronger prosodic boundary cues than

⁴For general references and discussion of the syntax of relative clauses, we refer the reader to Alexiadou & Meinunger This Volume.

Restrictive RCs show a mixed picture: While the right edge of an RRC is always separated by an IP boundary accompanied by “a pause (often acoustic silence)”, the left edge of the RRC is never realized with a pause and the boundary tone seems to be optional (Vigário 2003: 267). Due to this mixed picture, Vigário concludes that RRCs are part of a compound IP, illustrated by $\iota^{\max}$ in (22), where the relative clause and its preceding material are grouped together in a higher IP – indicating that they have a stronger connection than the same constituents in NRRCs – and that each constituent is also an IP of its own – allowing for the attested phenomena at the left edge of the RRC (see Ladd 1992, 2008: 297 for compound prosodic domains). According to Vigário (2003), the advantage of this assumptions lies in the following facts: (i) the last word before the RC is obligatorily accented – which is typical for heads of IPs in EP (inversely, units that are not heads are often not accented in EP); (ii) the uncertainty of the boundary tone before the RC can be explained.

As mentioned in Section 2.2, Güneş (2015) includes Turkish appositive relative *ki*-clauses in her study of prosodic boundaries in Turkish. They showed values for final lengthening preceding the left and right edge of the relative clause as well as pause frequency at both edges and pause length at the right edge that was clearly more similar to the comparison case of IPs than to the comparison cases of the smaller phonological phrase.

Van Heuven & Kaland (2017) conduct a perception study on the prosody of RCs in Dutch and German to investigate “which prosodic patterns are (i) preferred, (ii) allowed and (iii) illegal as markers of” NRRCs and RRCs (p. 159). For each language they had 20 native hearers that listened to a prosodically modified set of RCs. The results show some differences between the two languages. For Dutch listeners NRRCs are more acceptable with prosodic breaks at both edges, while RRCs are more acceptable when the left boundary is not marked by a pre-boundary lengthening and a pause – independent of the occurrence of a high boundary tone (p.167). The authors conclude that temporal marking of a boundary is the primary cue for prosodic cohesiveness. In German, the results did not show a clear picture and thus indicate – according to the authors – that there is no systematic difference between NRRCs and RRCs (p.167). In another perception study on German RCs, however, Schubö et al. (2015) found evidence that the two types of RCs can be distinguished by native listeners. Also, Poschmann & Wagner (2016) found clear differences between the two types of RCs in German (see above). Despite the contradicting findings on German, the perception experiments seem to also indicate that NRRCs need to mark both edges, while RRCs need only a marked right edge.

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In summary, recurring patterns in the more detailed studies on relative clauses are: (a) confirmation for the asymmetry between stricter boundaries at the edges of NRRCs than at the edges of RRCs; (b) in RRCs, a greater affinity to such a boundary at the right than the left edge; and (c) variation and open issues regarding the prosody of the left edge of RRCs. The results also support the decision of the literature after Downing (1970) to transpose his “obligatory pause” to “obligatory IP boundary” rather than to “obligatory acoustic pause”. There are regularly cues for IP boundaries at the edges of NRRCs, but an actual acoustic pause is only an optional accompaniment of such boundaries. These boundaries can be derived by the OBI, construed in terms of IP boundaries, or in terms of Match-iCP. In our overview, a variety of language-specific observations were made, which we hope will be pursued and refined in the future.

4.2 The boundaries at the edges of restrictive relative clauses

While Match-iCP inserts the boundaries at the edges of non-restrictive relative clauses, it does not insert any IP boundaries at the edges of restrictive relative clauses. We report two suggestions about what might motivate these additional boundaries.

Truckenbrodt (2005) had included subject clauses, object clauses and restrictive relative clauses in a small study of IP boundaries in German. The study was based on recordings of designed sentences by a single speaker. He found regular IP boundaries at the right edges of all these embedded clauses, even where their right edges do not coincide with the edges of root sentences. His tentative suggestion was that the grammar also contains a constraint that aligns all clauses, not just root sentences, with IP boundaries. Selkirk (2009) similarly argued for such a constraint with an analysis of *wh*-questions in Fukuoka Japanese. Feldhausen (2011) also argued for a constraint aligning all clauses with prosodic constituents. He conducted a study on Catalan SVO structures with both DP objects and sentential objects (the latter being either short (1 prosodic word) or long (3 prosodic words)). His results showed that object clauses are typically separated by an IP boundary from the preceding SV part in contrast to DP objects. A constraint in that spirit, then, would also be able to insert IP boundaries at the edges of restrictive relative clauses in other languages. Selkirk (2011) suggested that the grammar includes two constraints building IP boundaries relative to syntactic constituents. Match-CP for the regular clauses and Match-iCP for the illocutionary clauses.

On the other hand, Ishihara (2022) argues that the grammar does not make reference specifically to CP. Instead of a constraint matching illocutionary clauses, he suggests the constraint in (12), Match (speech act, *ι*), or MatchSA for short,

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which matches speech acts to IPs. He also argues against the grammar containing a constraint like Match-CP. In this context, he offers alternative analyses for some of the cases mentioned above. For Fukuoka Japanese, he argues that the relevant phenomenon is not bounded by clauses but by the scope of the wh-phrases. For German, he points out that the data could also be captured in terms of a constraint that aligns long syntactic constituents with IP boundaries. This could then also be the source of the IP boundaries at the right edge of restrictive relative clauses in other languages. In the examples used there, as in Truckenbrodt (2005) data, the right edge of such an embedded clause is normally the only internal right edges of a longer embedded constituent. However, the alternative analysis in terms of length does not carry over to the Catalan data of Feldhausen (2011), who shows that clausal status affects the prosody regardless of length. While the Catalan data is thus a considerable problem for a length-based approach, Ishihara (2022) discusses a further possibility to account for IP boundaries without referring to Match-CP: *stylistic promotion* (in the sense of Selkirk 2005), which refers to a prosodic promotion of phonological phrases to IPs. However, the weak point of *stylistic promotion* is the fact that this approach is arbitrary. It does not offer a principled explanation, instead, it rather collects cases that cannot be explained otherwise – but the Catalan pattern is not arbitrary.

4.3 Prosodic edges and parsing preferences with relative clauses

While research on boundary differences between RRC and NRRCs seems to be relatively rare, there are many studies dealing with attachment ambiguities in relative clauses (see Alexiadou & Meinunger This Volume, for detailed information). These also relate to the prosody of the left edge of RRCs. The relevant structure of attachment ambiguities is given in (23). The question, informally put, is whether the relative clause modifies the first or the second noun, as shown by coindexing.

- (23) a. [DP_1 the servant₁ of [DP_2 the actress₂ who₂ was on the balcony]]
 b. [DP_1 the servant₁ of [DP_2 the actress₂] who₁ was on the balcony]

Languages differ in their attachment preferences. Non-local or high attachment preferences as in (23b) are attested for Spanish (Cueto & Mitchell 1988, Carreiras & Jr. 1993, 1999, Carreiras et al. 2004, Bergmann et al. 2008, Acuña-Fariña et al. 2009, de la Cruz-Pavía & Elordieta 2015), European Portuguese (Vigário 2003 [NRRC], Soares et al. 2010), Galician (Fraga et al. 2005), French (Zagar et al. 1997, Fodor 2002), and German (Konieczny 2000, Hemforth et al. 2015). Local or low attachment preferences as in (23a) are attested for Italian (Vinzenzi &

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Job 1993), Basque (Gutierrez et al. 2004), English (Cuetos & Mitchell 1988, Fodor 2002), Swedish and Norwegian (Ehrlich et al. 1999, Fodor 2002), and Brazilian Portuguese (Miyamoto 1998, 1999; but see Miyamoto & Finger 2003, who argue that BP shows high attachment). The relevant point of this research for the present discussion is the fact that prosody is used for disambiguating the structure (Fodor 2002, Vigário 2003): The default phrasing pattern of languages with high attachment preferences is the one given in (24), in which a prosodic break (indicated by //) occurs immediately before the RC (Maynell 1999, Lovrić et al. 2000, 2001, Fodor 2002, de la Cruz-Pavía & Elordieta 2015).

(24) $N_1 N_2 // RC$

De la Cruz-Pavía & Elordieta (2015), for example, argue for (24) based on a production study with native speakers of Spanish spoken in the Basque country. Their results show four possible phrasing patterns (1: a single boundary after N_2 (= (24)); 2: a single boundary after N_1 ; 3: two boundaries (one after each noun); and 4: no sentence-internal boundaries), but (24) is the most common one.

They have furthermore shown that the length of the RRC also plays a role for phrasing. The longer the RC the more probable a boundary between NP_2 and RC (p.196). Fodor (2002), in the same vein, discusses the finding that short RRCs show a greater tendency to local attachment concluding that the prosodic breaks are interpreted as a mark of non-local attachment (Fodor 2002: 3).

Thus, the research on attachment ambiguities also adds information about prosodic boundaries in places where they are not generally obligatory by Downing's initial classification, such as at the edges of restrictive relative clauses.

While most studies on ambiguities deal with RRCs, Vigário (2003) considers NRRCs (in EP). She concludes that "although the prosodic break is independently triggered by the syntax-prosody algorithms it seems to be reused for resolving the attachment ambiguity" (p. 271) – which is in line with Fodor (2002). That reuse of the IP-boundary (marked by ι) does not occur, though, when the lower constituent is semantically the only possible antecedent as in (25), taken from Vigário (2003: 272). Here, the non-local preference is blocked, because *carro* 'car' is not a possible antecedent.

(25) (O carro da Joana) ι (que vive em Coimbra) ι (foi essencial para ela poder trabalhar em Portimão) ι

'The car of Joana, who lives in Coimbra, was crucial for her to be able to work in Portimão'

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In summary, the prosodic marking of the left edge of a relative clause interacts with parsing preferences. A boundary preceding a RRC seems to have an affinity to high attachment in case of attachment ambiguities and in case the relative clause is long.

4.4 Interaction of stress-assignment and movement in relative clauses

In this section, we review an interaction of syntactic wh-movement with the prosodic structure that also occurs in relative clauses. While it is unrelated to the OBI and Match-iCP, it is an interesting quirk in connection with the prosody of relative clauses. Further, a more general suggestion we offer in Section 5.2 will be based on these interactions.

Like wh-questions, relative clauses show interaction of stress-assignment with syntactic movement as described by Bresnan (1971). This is illustrated in (26) with examples from Bresnan (1971). (26a) illustrates the stress-pattern of English clauses when narrow focus does not retract the stress: Stress is assigned rightmost in the sentence, i.e. rightmost in the IP. (26b) is one of a class of systematic exceptions: Where movement has taken place from rightmost position, the sentence stress is in certain cases instead on an earlier element – broadly speaking, an earlier element that is syntactically connected to the sentence-final position. In (26b) this is the head noun of the relative clause, with the trace of relativization in final position.

- (26) a. Mary liked the proposal that George leave.
 b. Mary liked the proposal_i that George left t_i.

Bresnan saw these effects as evidence for stress-assignment on a syntactic cycle, where rightmost stress is correctly assigned by the Nuclear Stress Rule (Chomsky & Halle 1968) on the lowest cycle (*that George left the proposal*). Truckenbrodt (2019) argues that they are instead evidence for multi-dominance structures, which can naturally be derived by Chomsky's internal merge (Chomsky 2000, 2001, 2008). In (27), a multiply linked representation of (*the*) *proposal* is indicated by means of stricken-through copies. The structure satisfies the stress-requirements because of the link to the object position in the embedded clause, and the stress surfaces in the position in which the NP *proposal* is spelled out.

- (27) Mary liked the proposal_i [~~the proposal~~_i that George left ~~the proposal~~_i].

See Truckenbrodt (2019) and Buring & Truckenbrodt (2021) for arguments for this analysis. The effect is generally found for movement to Spec,CP and

movement to C in German, though not for English *tough*-movement (Richards 2017) and not for German scrambling (Truckenbrodt 2019). The cases of stress-movement interaction studied by Zubizarreta (1998), where stress is not “taken along”, are in a natural class with the latter phenomena.

In summary, relative clauses may show prosodic effects of relativization over and above the boundary-effects addressed by Downing (1970). These relate to the interaction of stress-assignment with syntactic movement.

5 Parentheticals

In this section, we take a closer look at parentheticals and go beyond our theoretical questions based on Downing (1970) and speech acts. In Section 2.1, we have already shown how Downing deals with speech-commenting parenthesis (6) and non-restrictive relative clauses (5) (which are often considered to be parentheticals). Thus, from a general point of view, Downing (1970) approach is able to account for parentheticals. As we will see here, there is much more to say on that issue. We therefore broaden our considerations and integrate generative issues as well as findings from various empirical studies that go beyond our previous theoretical base.

In Section 5.1, we first clarify the notion of parentheticals (in a wide and in a narrow sense). In Section 5.2, we discuss the class 0 elements of Bing (1979, 1985) and the need for further theory in a generative analysis of parentheticals. In Section 5.3, we review a rich literature with empirical approaches to parentheticals. Section 5.4 concludes the part on parentheticals.

5.1 Introduction: the notion of parentheticals

As we already mentioned, the class of elements triggering obligatory pauses at their edges according to Downing (1970) overlaps with the class of parentheticals, such as the parenthetical *I believe* in the example *John, I believe, has sold his Mustang*.

According to Burton-Roberts (2006), a “parenthetical (P) is an expression of which it can be argued that, while in some sense ‘hosted’ by another expression (H), P makes no contribution to the structure of H.” (p.179). In this wide sense, parentheticals also include non-restrictive relatives, and a parenthetical might be as syntactically disjoint from its host clause as one of two coordinated declaratives is from the other. Burton-Roberts (2006: 180) also states that these parentheticals (in the wider sense) “are marked off from their hosts by some form of punctuation in writing and special intonation contour in speech”. He gives an example

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for a P (p. 180) that is neither syntactically nor discursively integrated in H, see (28). Despite this non-contribution to H, P's position within the host clause is "quite severely constrained" (Burton-Roberts 2006: 180).

- (28) The main point – *why not have a seat?* – is outlined in the middle paragraph.

In line with this definition, parentheticals have also been labelled extra-sentential elements (e.g. Huddleston & Pullum 2002, Astruc-Aguilera 2005, Astruc-Aguilera & Nolan 2007) and their extra-sentential nature is typically taken to correspond to some amount of prosodic separation (see also Pierrehumbert 1980, Nespor & Vogel 1986/2007: 188, Nespor 1993).

In many cases, the extrasentential elements are not quite as dramatically unrelated as in (28). Dehé (2014) investigates six selected types of parentheticals in British English. They are presented in (29) in italics in sentences excerpted from the International Corpus of English.

- (29) a. Full parenthetical clauses, e.g.
 So what we can do in fact *I'll just turn it off* is to use that signal to train people's ability to perceive voicing distinctions in speech and their ability to actually produce them (ICE-GB: s2a-056 #87; Dehé 2014:45)
- b. Non-restrictive relative clauses
 But what has amazed me about a lot of Elgar's music is that since then I don't think uh a great deal of his music's played in Germany uh *which I think is a great pity* (ICE-GB: s1b- 032 #50; Dehé 2014:48)
- c. Nominal appositions
 Captain Richard Sharp *editor of Jane's Fighting Ships* is on the line (ICE-GB: s2b-012 #36; Dehé 2014:56)
- d. Comment clauses
 Well they dropped cards in *I suppose* the doors (ICE-GB: s1a- 020 #177; Dehé 2014:65)
- e. Reporting verbs
 It wasn't a round table *President Gorbachev said later* it was a square table (ICE-GB: s2b-040 #092–093; Dehé 2014:75)
- f. Question tags
 We had a big chat about this before *didn't we* (ICE-GB: s1a- 006 #172; Dehé 2014:80)

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We already discussed the non-restrictive relative clauses (point (b) of Dehe's list) earlier. Before turning to other parentheticals and their phrasing properties, we highlight the special subclass consisting of comment clauses (d) and reporting verbs (e). This subclass has also been called *parentheticals* in a narrow sense. They are short clauses that provide evidence about the source of the speaker's assertion or that attribute the content of their host clause to the speech or to the thoughts of an individual or a group of individuals (see, e.g. Reinhart 1983). Downing (1970) treated them as syntactic root sentences of their own, separated prosodically by obligatory pauses. An example is given in (30). See (6) for the treatment of this case in the account of Downing (1970).

(30) John, *I believe*, has sold his Mustang

5.2 The class 0 elements of Bing (1979, 1985)

The generative analysis in terms of the OBI of Downing (1970) and its modern versions from Kan (2009) and Selkirk (2011) in the form of Match-iCP have given our discussion up to here principled theories to connect to. In a generative account, additional analysis is needed when it comes to the prosody of parentheticals in the narrow sense and various parentheticals in the wider sense. We think that sensible starting points for such additional analysis can be found in Bing (1979, 1985) and we review some relevant elements in this section. We also outline a direction for such analysis to encourage further theorizing in this domain.

In our discussion, we tentatively relate parentheticals in the wide sense to illocutionary clauses as follows: If an element E has a host clause that is an illocutionary clause and if that illocutionary clause does not genuinely syntactically contain E, then E is a parenthetical. That allows that a parenthetical can itself be an illocutionary clause or not.

In standard cases, an English or German IP comes with a nuclear stress or nuclear prominence (the strongest stress or prominence in the IP) and with a nuclear pitch accent (a pitch accent on that nuclear stress) (see Pierrehumbert 1980, Beckman & Pierrehumbert 1986 and later literature). A range of parentheticals, however, seem to lack the nuclear stress and the nuclear pitch accent, as discussed in the following.

Bing (1979, 1985) argued for a distinction of parenthetical elements into class 0 and class 1 expressions. Non-restrictive relative clauses are members of class 1. This is the well-behaved class in which the intonation phrase delimiting a parenthetical normally includes the intuitive presence of a nuclear stress and a pitch-accent on it that can be documented in the F0 contour. An example of our own is shown here:

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(31) John visited Mary, who lives in Austria.

On the other hand, class 0 elements, when occurring in medial or final position, are allowed to lack pitch accents in nuclear (and other) positions, and intuitively also seem not to have a nuclear stress when they do. Bing takes the allowed lack of pitch-accents to be the defining factor of class 0 elements. Many different kinds of parentheticals belong to class 0, as shown with examples from Bing (1979) in (32). We added simple underlining for the nuclear stress in the main clause. There is not a second instance of such intuitive nuclear stress on the class 0 parentheticals later in the utterances. (Examples that show that these class 0 expressions are not inherently stressless will be seen below in (38).)

(32) Class 0 elements (Bing 1979, 1985)

- | | |
|---|------------------------------------|
| a. Pray be <u>seated</u> , <i>Miss Faulkner</i> . | Vocatives |
| b. I'd like a box of <u>chocolates</u> , <i>please</i> . | Polite expressions |
| c. John wouldn't give me his <u>car</u> , <i>the stupid bastard</i> | Expletives |
| d. She already <u>knows</u> , <i>apparently</i> . | Sentence adverbials |
| e. You're <u>coming</u> , <i>aren't you?</i> | Tag questions |
| f. It's not <u>possible</u> , <i>I suppose</i> . | Parentheticals in the narrow sense |

Bing argues that these parentheticals are nevertheless separate IPs, though in a special way since they are allowed to lack pitch accents.⁶ To us, they also seem special in that they seem to lack nuclear stress.

The lack of a nuclear stress was independently observed for German by Reis (1995) for parentheticals in the narrow sense. She cast this issue in terms of the number of focus-background structures in a configuration with two clauses. Dehé (2007) made a related point that will be reviewed in the next section. In her analysis, there is no separate IP for her category of integrated parentheticals, which could be viewed as close to Bing's class 0 elements.

One would like to know what derives the distinction between class 0 and class 1 elements. We are not aware of principled suggestions on this matter. We briefly outline one of our own. We begin by illustrating the analysis with parentheticals in the narrow sense.

A sketch of the slifting analysis for a relevant class of parentheticals by Ross (1973) is shown in (33a). An implementation of this in the copy-theory of movement (Chomsky 1993) is shown in (33b). A variant of this analysis from Truckenbrodt & Sode (2020) is sketched in (33c–33d). Here (33c) gives the idea of the

⁶Bing uses the term *prominence tones* instead of *pitch accents*. The latter is mostly used since Pierrehumbert (1980).

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analysis in terms of a classical operator and (33d) gives the fuller version using copies. The operator is a copy of its antecedent host clause and the trace of the operator is a copy of the operator.

- (33) a. I suppose [that it's not possible]. => [it's not possible]_i I suppose e_i
 b. [it's not possible]_i I suppose [~~that it's not possible~~]_i
 c. [it's not possible] Op_i I suppose e_i
 d. [it's not possible] [~~it's not possible~~]_{Op,i} I suppose [~~it's not possible~~]_i

Syntactic arguments for these structures are the original island arguments for a movement analysis in Ross (1973) as well as the Condition C effects, which can be argued to support the presence of the rightmost copy at LF (see the discussion of subject-oriented parentheticals in Reinhart 1983 and see Pittner 1995 on German, for the descriptive binding facts that we have in mind).

For the following discussion, either one of the structures in (33b–33d) or a variant of them in terms of multidominance can be assumed.⁷

- (34) it's not possible (...) I suppose [~~it's not possible~~]

We now combine this with stress-assignment and with what was discussed in Section 4.4, and we obtain (35).

- (35) It's not possible (...) I suppose [~~it's not possible~~]
 (_i) (_i)
 = It's not possible , I suppose . (this line only shows overt elements)
 (_i) (_i)

Without further ado, we derive a parenthetical that is an IP but lacks nuclear stress. It contains an empty element with nuclear stress (~~possible~~), though this stress surfaces on the copy in the main clause (on possible). We think this is a very reasonable approach to class 0 elements since its parts are independently motivated. We could then leave open, for the moment, whether the intonation phrase of the parenthetical is retained (closer to Bing's analysis) or whether phonological readjustment leads to its reduction (with a result closer to Dehé's analysis).⁸

⁷See Kluck (2011) for an analysis of a different case, amalgams, in terms of multidominance. See Griffiths (2015) for an analysis of constituent answers with parentheticals in terms of ellipsis.

⁸We think that this approach is promising even though it takes a different stand on the interaction of deletion and deaccenting than does Tancredi (1992), whose perspective would not allow him to integrate the account outlined here. See Truckenbrodt (2013: 139ff) for discussion of deletion and deaccenting that supports the perspective taken here.

Bing (1979) characterizes class 0 elements semantically as relating to the interaction of the speaker and the hearer. Following up on this, we might tentatively say that this class more generally embeds a silent copy of the main clause, embedded in a speech-act related meaning component that includes or integrates the class 0 element. For example, in (36a), the class 0 vocative expression is the hearer of this speech-act related meaning component, and in (36b), the class 0 element *please* can be construed as a short form of such a meaning component that embeds a silent copy of the main clause.

- (36) a. S asks Mrs. Faulkner: Pray be seated, S asks Mrs. Faulkner to (pray) ~~be seated~~
 b. S asks H politely: I'd like a box of chocolates S asks H politely: I'd like a box of chocolates
 "please"

According to Bing, class 1 elements do not relate to the speaker or the hearer of the main clause but to its content. We might say that class 1 elements constitute a separate speech-act that does not include a copy of the host clause, see (37).

- (37) S tells H: [John visited Mary] [~~S tells H:~~ who lives in Austria]
 (1) (1)

Since this parenthetical does not include a silent copy of its host clause with silent nuclear stress, it requires overt nuclear stress in the IP of the parenthetical.

Bing (1985, 1979) also observes that while class 0 elements are allowed to lack pitch accents in medial and final position, they will carry pitch accents in initial position. In (38), we indicate this by single underlining, so that there are two nuclear pitch accents.

- (38) a. Mrs. Faulkner, pray be seated.
b. Apparently, she already knows.

If, in (32a), the two speech-acts are constrained to be identical in meaning, then the overt occurrence of *Mrs. Faulkner* may count as *contextually given* relative to the preceding, covert one. This would be a reason for it to reject stress and a pitch accent (Ladd 1983, Féry & Samek-Lodovici 2006, Büring 2016). This reason would not be present in (38), assuming that the first occurrence of *Mrs. Faulkner* is spelled out here. In an analysis along these lines, the deletion analysis could account for the lack of a *need* of a pitch-accent on the overt parenthetical

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elements, but the actual deaccentuation would also be due to a hidden givenness factor that emerges under these circumstances.

We sum up this section. If a generative analysis explores a constant base-line for mapping syntactically unintegrated elements to separate IPs, additional analysis is needed for the class 0 elements of Bing (1979, 1985). They seem not to require a nuclear stress or pitch accents. We outlined a possible direction for such an analysis.

5.3 Acoustic findings on parentheticals

There are a range of experimental studies (not necessarily with a generative orientation) that have detailed acoustic results in the domain of parentheticals. In the following, we review such studies that bear on the amount of separation of the parentheticals. Two questions accompany our investigation:

- (39) a. Are parentheticals prosodically separated from the matrix sentence?
- b. How are parentheticals realized phonetically?

In her acoustic analysis of spontaneous British English data taken from the British component of the International Corpus of English (ICE_GB, Nelson et al. 2012), Dehé (2007) argues that the edges of syntactic parenthetical constructions do not always coincide with prosodic boundaries. Dehé (2007) makes a distinction between syntactic and prosodic parentheticals. While the former “are linearly represented in the syntactic string of utterance but are not part of its hierarchical structure” (Dehé 2007: 264), the latter have to be “prosodically different from its immediate environment in at least one prosodic parameter” (Dehé 2007: 266). Following these criteria Dehé proposes a classification of parentheticals into (i) prototypical parentheticals, (ii) integrated parentheticals and (iii) other types. In the prototypical case, where the parenthetical matches the syntactic and prosodic criteria, it goes straightforwardly along with the characteristic prosodic features such as pauses, faster speech rate or the relevant pitch movement, e.g. in the sentence “C is therefore important *and this is how the argument runs* to do things or to prevent other things, Dehé (2007: 267)” the middle part <this is how the arguments runs> creates the prototypical parenthesis. Furthermore, in the case of integrated parentheticals, the construction is part of another intonation domain. For instance, in the sentence: “The voters *I think* just have an opportunity” both parts exhibit a nucleus and a fall-rise pattern. The parenthetical <I think> is prosodically fully integrated with the first part of the host clause. This category also includes pause fillers. Thus, it is apparent that prototypical

parentheticals resemble class 1 elements and integrated parentheticals class 0 elements proposed by Bing (1979) (see Section 5.2) Finally, there are cases which cannot be classified to either of the categories described above. In such cases it is difficult to find at least one prosodic parameter differentiating the parenthetical from its environment (Bolinger 1989); at the same time such syntactic parenthetical constructions are not prosodically integrated in the sense described above. This category also includes structures which meet prosodic but not syntactic criteria of a parenthetical construction, e.g., in the sentence “...to put the philosophy into practice *because I suppose* uh in in a way what Marx was uh uh having a go at” (Dehé 2007: 278) the parenthetical <*because I suppose*> forms one intonation domain with a nuclear fall on the stressed second syllable of *suppose*. This intonation domain is complemented by some syntactic material from the host clause as the prehead.

In a more recent study Dehé (2014) investigated six selected types of parentheticals in British English (see (29) above). Her prosodic analysis is based on 1160 excerpted items and reveals that full parenthetical clauses, non-restrictive relative and to a lesser degree clause nominal appositions are prosodically separated while comment clauses and reporting verbs are prosodically integrated by default. Variation seems to be, however, an inherent characteristic of parenthetical constructions. But as Dehé (2014) observed, more variation is observed in the comment clauses and reporting verbs than in full parenthetical clauses and nominal appositions. Finally, question tags are more prosodically separate than integrated.

The study by Peters (2006: 1) investigates whether parentheticals obligatorily form an IP and break up the IP of the matrix sentence as claimed by Selkirk (1980) and Nespor & Vogel (1986). By examining a spontaneous speech corpus of Hamburg German (17 speakers), Peters proposed four types of prosodic integration of parenthesis, as shown in (40). The first type (40a) represents parentheticals which break up the host clause into two IPs creating the IP on its own. Thus, such a sentence is realized with three IPs. The second type (40b) shows a full prosodic integration of the parenthesis with the matrix sentence. The third type (40c) illustrates the insertion of one IP into another. Finally, the fourth type (40d) presents the parenthesis and the following part of the matrix sentence as separate IPs. It is a combination of prosodic parataxis and prosodic parenthesis.

- (40) a. Prosodic parataxis {...} {...} {...}
 {Paul} {glaub ich} {ist es gewesen}
 Paul think I is it been
 It was Paul, I think.

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- b. Prosodic incorporation {...}
 {Paul glaub ich ist es gewesen}
- c. Prosodic parenthesis {... {...} ...}
 {Paul {glaub ich} ist es gewesen}
- d. Mixed type {... {...}{...}
 {Paul {it is true} {he is dead}}

To identify prosodic integration Peters (2006) defines the IP as the domain of a single intonation contour considering pauses, discontinuities in speech scaling and varying speech rate as optional cues in defining IP (see also Wichmann 2000; Fagyal 2002a). In order to determine the number of contours in a given sentence the inventory of nuclear contours established for Hamburg German was used (Peters 2004). The study reveals that out of 96 syntactic parentheticals found in the corpus prosodic parenthesis (40c) was the most frequent type (more than 40% cases) followed by prosodic parataxis and mixed type (ca. 25 % each). The prosodic incorporation type (40b) was attested in less than 10 % of the data. Another finding was that shorter parentheticals are more likely to be prosodically incorporated than longer ones. Finally, the study also shows that the prosodic integration of parentheticals is related to their syntactic integration. First, prosodic parataxis is most frequently found in parentheticals inserted between clauses. Second, prosodic parenthesis and prosodic incorporation are attested in parentheticals inserted within a clause. Finally, the mixed type is most frequently found in parentheticals which are followed by syntactic resumption. To sum up, the phonetic evidence does not support the claim that parentheticals always form IPs on their own, as suggested by Downing (1970), Fortmann (2007), Burton-Roberts (2006), Selkirk (1980) and Nespor & Vogel (1986).

In a similar vein, Gachet & Avanzi (2008) and Avanzi (2012) show that the prosody of parenthetical constructions in French is less prototypical than expected: Even though parentheticals can be separated from the preceding and following material, they can also build a prosodic unit with adjacent material. More concretely, in their corpus study on spoken French, Gachet & Avanzi (2008: 63ff.) identify four possible phrasing options for center-embedded parenthetical clauses, see (41).⁹ An example for a parenthetical clause (PC) is given in (42). Here, the PC *je me trompe peut-être* ‘I might be wrong’ is located inside the clause *Est-ce que ça correspond à D.J. ou D.J.* ‘Does it correspond to D.J. or D.J.’ extracted from

⁹Most of the data in Gachet & Avanzi (2008) are based on the corpora *Choix de Textes de Français Parlé* (cote CTFP) (Blanche-Benveniste et al. 2002) and *Corpus de Référence du Français Parlé* (cote CRFP) (DELIC 2004).

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an interview (Gachet & Avanzi 2008: 59). “D.J.” stands for “disc jockey” and the speaker wonders whether it has to be pronounced as [didʒe] or [didʒi].

- (41) a. [matrix clause] [PC] [matrix clause]
 b. [matrix clause PC] [matrix clause]
 c. [matrix clause] [PC matrix clause]
 d. [matrix clause PC matrix clause]
- (42) Est-ce que ça correspond je me trompe peut-être à D.J. ou D.J.
 ‘Does it correspond – I might be wrong – to [didʒe] or [didʒi]’
- (43) J’ai eu la chance d’avoir à Besançon parce que j’étais + à Besançon dans
 l’enseignement + j’ai eu la chance d’avoir..
 ‘I had the good luck in Besançon to have because I was + in Besançon in
 teaching + I had the good luck to have...

In the first type (41a), the parenthetical clause is realized as an IP of its own. The PC is characterized by a lower fundamental frequency than the surrounding material and the left and the right edges can optionally be marked by pauses. In the second type (41b), the PC is prosodically not separated from the preceding material by an IP boundary (there might be a boundary of a lower prosodic level, though). Gachet & Avanzi attest a further realization option for the second type: The IP boundary (indicated by +) might occur within the PC, as shown in (43). This realization typically occurs in case the PC is very long. In both realization options of the second type, the PC displays an IP boundary at its right edge. Both the first and the second type are typical prosodic realizations of parenthetical clauses in the corpus of French investigated by Gachet & Avanzi (2008). The third and the fourth type occur only to a very small amount (if at all) and can thus not be considered as typical realization patterns. The third type is characterized by the absence of an IP boundary at the right edge of the PC. As a consequence, the PC creates one prosodic unit with the following material. In the fourth type neither a left nor a right IP boundary occurs at the edges of the PC.

Schneider (2007) investigated parenthetical constructions focusing on the so-called reduced parenthetical clauses in spontaneous speech of French, Italian and Spanish; see examples in (44) where the parentheticals are underlined (examples and translations from Schneider 2007:1).

- (44) a. Ah ben c’est plus libre maintenant, je crois
 ‘It’s freer now I believe’

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- b. c'était euh je me rappelle/dans les années après-guerre/ [...]
 'that was, I remember, in the years after the war'

According to Schneider (2007: 215) syntactically defined parentheses may or may not interrupt the utterance's intonational contour. If they do, they are marked by the sudden change of the utterance's initial contour and a following low and almost flat F0 slightly rising at the end. Their speech rate is higher and they can be separated by pauses. Such characteristics allude to a separate intonation units. If the parentheses do not interrupt the intonational contour of the utterance which is often but not always the case in reduced parenthetical clauses, their prosodic characteristics are different: they neither induce a sudden change of the intonational contour nor pauses. In other words, they do not create separate intonation units.

However, it should be also stated that there are studies which speak in favour of Downing (1970) claims. For instance, Barros & Frota (2015) demonstrate that in Porto and Évora, varieties of European Portuguese, parentheticals form independent IPs and their intonational contour is similar to the one described for the Standard European Portuguese (Frota 2000). In particular, it was reported that utterances with inner parentheticals (i.e. appearing host-medially) tend to form 3 IPs and show the following contours: L*(+H) H%, L*(+H) H%, or (H+)L* L%. In total 16 sentence-internal parentheticals taken from a corpus of spoken speech (SEP, Frota 2000) underwent the analysis. The authors conclude that on the one hand there is no variation in prosodic phrasing in the sense that parentheticals form the IPs in both varieties but on the other hand there is variation concerning the effect of constituent length on phrasing and boundary marking strategies.

What the study of Stoltenburg (2003: 20) adds about the location of parentheticals is that most parentheticals, at least in German, are found in the middle field (46% out of 531 parentheses found in corpora of German spontaneous speech), followed by locations at sentence boundaries (12%), and after conjunctions (8%). Furthermore, 4 % of the parentheses were found after the right sentence boundary and before the postfield and 4% between pre-prefield and prefield; 12 % between the pre-field and the left sentence boundary and 11% after the left sentence boundary or the following pronoun.

In summary, regarding question (39a) we conclude based on empirical cross-linguistic evidence that not all parenthetical constructions are prosodically separated from the matrix sentences. The phonetic evidence does not entirely support the claim that parentheticals form IPs on their own, as suggested by Downing (1970), Fortmann (2007), Burton-Roberts (2006), Selkirk (1980) and Nespor & Vogel (1986) because other types of prosodic integrations were found as well. If

viewed from a more narrowly generative standpoint, it remains to be seen how well refinements such as the one sketched in connection with Bing (1979) above are adequate for capturing the observations reviewed here.

There are also other possibilities including, e.g., prosodic integration which speakers seem to frequently employ.

Regarding question (39b) parenthetical constructions are found to be usually faster and quieter than the remaining part of the sentence. They are also produced with a lowered F0 and pauses. This is related to the assumption that parentheticals are constructions with inherent conflict: on the one hand they are interpolated into a sentence (often interrupting the linear order of that sentence), but on the other hand they are somehow detached from that sentence. This syntactic independence is often reflected in prosody.

Döring (2010: 44) gives an overview of different cues theoretically assumed or experimentally found in the phonetic realization of parenthetical constructions; see Table 1.

Table 1: Prosodic characteristics of parenthetical constructions (Döring 2010: 44).

Faster tempo	Brandt (1994), Bolinger (1989), Crystal (1969), Jung (1980), Payà (2003), Pittner (1993), Schwyzer (1939)
Varying tempo	Erben (1980), Öhlschläger (1996)
Lowered F0	Bolinger (1989), Crystal (1969), Jung (1980), Payà (2003), Pittner (1993), Reinhart (1983)
Mostly lowered F0	Schwyzzer (1939), Wichmann (2001)
Quieter	Bolinger (1989), Crystal (1969), Jung (1980), Payà (2003)
Pauses	Altmann (1981), Bolinger (1989), Dehé & Kavalova (2006), Erben (1980), Grewendorf (1988), Helbig & Buscha (1996), Hoffmann (1998), Lampert (1992), Payà (2003), Pittner (1995), Reinhart (1983), Schindler (1990), Taglicht (1998), Toepel (2001)

The acoustic cues provided in Table 1 are by no means exhaustive. According to Bolinger (1989: 186), a parenthetical construction is typically characterized by “lower pitch than the matrix sentence, it is set off by pause(s) and it has a rising terminal” (Bolinger 1989: 188). Wichmann (2001: 188) also notices that a

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parenthesis can be marked by higher pitch. However, it is also possible that the presence of these cues is contingent upon the function, length and position of the parenthetical construction (Wichmann 2001; see also reports of other studies given below).

Several experimental studies also report variation among speakers in the production of parentheticals. For instance, in her study on parenthetical constructions in German, Döring (2010) analysed a corpus of political speeches given in the Bundestag. Her analysis based on 67 participants clearly shows that parentheticals are not only produced faster, quieter, with a lowered F0 and pauses (cf. Table 1). On the contrary, her data point to a substantial interspeaker variation: 1/3 of the data is not produced at a higher rate of speech, but rather at a lower or unchanging rate of speech. Furthermore 1/4 of the data are produced either with higher F0 or do not show any F0 differences. Next, 40% of the data (or even 60% depending on the assumptions) are not produced with a lower intensity. Finally, 80% of the parentheticals are produced without pauses. Thus, Döring (2010) shows that the interspeaker variation is an inherent part of the production of the parentheticals as far as speech rate, F0, intensity or pauses are concerned.

Furthermore, in her investigation of sentence medial French parenthetical constructions, Fagyal (2001) studied pitch range defined as the speaker's maximal range of F0 variations (calculated as the highest minus the lowest pitch values). Based on four speakers' production Fagyal reports that all non-final IP pitch targets are lowered but the degree of lowering is "entirely speaker-dependent" (Fagyal 2001: 159). The interspeaker variation found in the study let Fagyal conclude that there is no unified strategy in compressing pitch range and lowering H and L pitch targets in the parenthetical IP. In addition, the resetting of the pitch range also varied among speakers. Finally, she also suggests that the issue of modelling pitch range as gradient or categorical phonetic/phonological features might be not of phonological nature but is rather triggered by the pragmatic-phonetic interface, see also Fagyal (2002b) for similar conclusions.

5.4 Conclusions on the prosodic phrasing of parentheticals

We discussed parentheticals from two different angles. Bing (1979, 1985) analyzed class 0 elements as IPs that do not need to have a pitch accent (or, we think, nuclear stress) and there seems to be a not unreasonable way of deriving such structures in a principled way. There is, on the other hand, a rich experimental literature in which parentheticals have often been analyzed as smaller prosodic categories than IPs. The prosodic reduction of parentheticals has been found in

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almost all languages examined so far, including English, German, French, Turkish and Portuguese. The phonetic studies also convincingly show that there is no single cue (e.g. pauses) which would define the prosody of a parenthetical. Instead, a repertoire of various cues is employed by speakers and their choice is subject to considerable variation.

6 Left-dislocated constituents

In this section we would like to take a closer look at left-dislocations, which belong to the class of the so-called extra-sentential elements (ESE, see Astruc-Aguilera & Nolan 2007, and Section 2.1). Left-dislocations, like other ESE, are not necessarily sentential – but they can be. This is one reason why it is worth looking at them in more detail in this chapter. Another reason has to do with the role of ESEs in phonological research: they have been used to define IPs (cf. Pierrehumbert 1980, Nespor & Vogel 1986: 188, Nespor 1993, Astruc-Aguilera & Nolan 2007: 85) – the level with which we are primarily concerned in this chapter. The third reason is based on their specific role in the sentence: they are not part of the root sentence, but they form the entire clause with it, and thus should be taken into account when considering the intonational pattern of a (complex) sentence at the syntax-prosody interface. Recall that Downing (1970) accounts for the obligatory boundary between the (left-)dislocated element and the following sentence by means of the root status of the latter (see Section 2.1). Our discussion of class 0 and class 1 elements (in the sense of Bing 1979) or extra-sentential elements (in the sense of Astruc-Aguilera 2005, Astruc-Aguilera & Nolan 2007) has shown that the assumption that these elements constitute an independent intonational unit cannot be fully maintained. Modern approaches, in turn, typically go a different way than Downing (1970): they attribute the occurrence of the prosodic boundary to the dislocated element and not to the following root sentence. By doing so, they are able to deal with more complex data, which includes dislocations in embedded sentences and the varying prosodic nature of dislocations. In Section 5.2, we discussed Bing's 1979, 1985 class 0 and 1 elements and we could observe that a central difference between the two classes was the role of the elements as a speech act. The question of whether LDs can count as a speech act in their own right, which then motivates their prosodic separation will be discussed at the end of this section.¹⁰

¹⁰We do not elaborate on right-dislocation here; firstly for reasons of space and secondly because the phenomenon has been less studied so far from a prosodic point of view compared to LD (nevertheless, here are several references on the prosody of right-dislocations for the in-

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Another reason to consider left-dislocations in this chapter is to be found in relatively recent approaches that assume that sentences with dislocations are biclausal, i.e., they consist not of one sentence but of two sentences that are connected to some degree (cf. Ott 2014, 2015). The idea of a complex, biclausal structure for dislocations has received a new boost thanks to Ott's work. However, biclausal approaches for dislocations already existed before the current upswing and can be found, for example, in studies on right-dislocations (cf. Kuno 1978, cited in Whitmann 2000). Recently, Fernández-Sánchez (2017, 2020) proposed a biclausal analysis for right-dislocations in Romance languages.

Let us start with a definition of left-dislocations: According to Alexiadou (2006) left-dislocations are characterized by “the presence of a phrase in the first position of the clause which is connected with that clause through the intermediary of some anaphoric element referred as the resumptive element” as illustrated in (45) for English (resumptive element: *it*) and Spanish (resumptive element: *lo*; see, e.g., Casielles-Suárez 2003: 327 for syntactic differences between English left-dislocation and Spanish/Romance clitic left-dislocation). From an information structural point of view, dislocations are typically characterized as expressions of topicalization (see, e.g., Gundel 1988, Lambrecht 1994, 2001, see López 2009a,b for a different view), (45).

- (45) a. I think that **the book**, I read (it) (English)
 b. Pienso que **el libro**, lo he leído (Spanish)

As Astruc-Aguilera & Nolan (2007) point out, due to the traditional assumption that these elements are syntactically and prosodically independent, they “have been used as evidence of the effect of syntactic constraints upon phrasing” (p. 85). In general, these elements are typically considered as establishing independent units on the level of the IPs (Downing 1970, Nespor & Vogel 1986: 188, López 2016). The idea that dislocations are independent IPs is formalized by Frascarelli's (2000) TOPIC PROSODIC DOMAIN or Féry's (2011) TOPIC constraint, see (46) and (47), which proposed their formulations based on sentence-initial dislocations as shown in (48). The resulting prosodic structure is illustrated in (48) by parenthesis. In contrast to Downing (1970), these formulations concentrate on the information structural component of the dislocated material and not on the status of the following sentence.

terested reader: for Germanic languages see Truckenbrodt 2015, Kalbertodt 2019; for Romance languages see: Fant 1984, Zubizarreta 1998, Astruc-Aguilera 2005, Samek-Lodovici 2006, 2015, Astruc-Aguilera & Nolan 2007, Avanzi 2009, 2012, Feldhausen 2010, Crocco 2013, Feldhausen & Lausacker 2018; for Bantu languages see: Zerbán 2007, Downing 2011, Embanga Aborobongui & Rialland 2014).

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(46) TOPIC PROSODIC DOMAIN (Frascarelli 2000: 63)

A Topic is minimally and exhaustively contained in an Intonational Phrase.

(47) TOPIC (Féry 2011: 1910)

A topic is entirely contained in a separate ι (=IP).

- (48) a. (**The book**) ι (I read) ι (English)
 b. (**El libro**) ι (lo he leído) ι (Spanish)

(48) shows that the constraints (46) and (47) work well in case of unembedded sentences. The crucial test case for constraints that require prosodic separation of a topic, however, are dislocations occurring in embedded clauses: Only when there is material both preceding and following the dislocation as in (45) above, the question about the correct generalization can be answered. Recent experimental research, which was conducted mainly on Romance and Bantu languages, provides several interesting results with respect to the type of prosodic unit, the obligatoriness of the right boundary, and the obligatoriness of the left boundary.

First, the relevant prosodic unit of a *clitic left-dislocation* (CLLD) can be both an intermediate phrase (ip) or an intonation phrase (IP), and the prosodic cues for the prosodic boundary are hardly pauses (in contrast to the standard assumption in the syntactic literature). This is attested for French (Delais-Roussarie et al. 2004, Avanzi 2012, Delais-Roussarie & Feldhausen 2014), Catalan (Feldhausen 2010, Feldhausen & Villalba 2020), Italian (Bocci 2013: 147ff., Feldhausen 2014a), and Spanish (Feldhausen 2016a,b, Feldhausen & Lausecker 2018, Sequeros-Valle 2019). Despite the variable strength of the boundary at the right edge, it is obligatory in these languages. The boundary is typically marked by a rising intonation, only Italian allows for low boundary tones (for an overview see Feldhausen 2016a: 298).

The variability in the strength of the prosodic boundary leads to the question as to why this variation shows up. Delais-Roussarie & Feldhausen (2014) suggest that some sort of prosodic neutralization might occur when the correct interpretation of the sentence does not depend on a clear distinction between different layers of prosodic phrasing (such as ip and IP). The neutralization between ip and IP might show up as long as the boundary is the highest or most important one in the sentence. The question as to why the CLLD boundary is the most essential one in the sentence is captured by Feldhausen's (2016a) *IS-over-Alignment* hypothesis: It says that prosodic boundaries satisfying information structural (IS) demands (like ALIGN-TOPIC,R in (52) below) play a more important role in the

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n-kha-kumana na-βo pa-kw-áamba.
I-TAM-meet with-2 LOC-INF-begin.
‘I remember when I first met these good friends.’

Third, the little research on CLLDs in embedded clauses shows a mixed picture of phrasing possibilities, which is illustrated by (49), (50), and (51). While the left boundary is obligatory in Murcia Spanish (49) (Feldhausen 2016a), it is optional in Cusco Spanish (Feldhausen & Lausecker 2018), and not realized in most of the cases in Catalan (50) (Feldhausen 2010: 162). Downing (2011), expanding the research question of Feldhausen (2010) to Bantu languages, shows that the left boundary is optional in Chichewa and Tumbuka (51). The embedded left-dislocations are marked by bold letters; the relevant location is the one at the [CP₂] edge.

Based on Catalan, Feldhausen (2010) proposes the constraint ALIGN-TOPIC,R (52), which demands the alignment of the right edge of a topic phrase to the right edge of a prosodic phrase (either ip or IP). This constraint guarantees the obligatory right boundary of CLLDs, but does not call for a left boundary (in contrast to TOPIC PROSODIC DOMAIN or TOPIC, see (46) and (47), respectively). This constraint also has an advantage over Downing’s (1970)’s OBI analysis, since he encounters problems accounting for the obligatory prosodic boundary preceding an embedded CLLD in data such as the one from Murcia Spanish or in sentences with several left-dislocated constituents.

(52) ALIGN-TOPIC,R (Feldhausen 2010: 174)

Align the right edge of a topic constituent to the right edge of a prosodic phrase.

ALIGN-TOPIC,R can be reranked in order to account for the attested patterns in several Romance and Bantu languages (see Patin et al. 2017 for an implementation of that idea) and it has been adapted for Hungarian (Surányi et al. 2012) and in a slightly revised version for the Bantu language Bàsàá (Hamlaoui & Szendrői 2017: 23). The constraint ALIGN-CP,L, which calls for the alignment of the left edge of a CP to the left edge of a prosodic phrase (Feldhausen 2010: 113, Feldhausen 2011: 1954), accounts for the cases in which a boundary occurs between the matrix clause and the embedded left-dislocation. A high-ranked ALIGN-CP,L leads to an obligatory boundary preceding the embedded CLLD, whereas a low-ranked ALIGN-CP,L allows for an optional boundary in that position. In order to

account for the frequency-dependent patterns, Feldhausen uses Stochastic Optimality Theory (Boersma & Hayes 2001), a variant of the classical Optimality Theory. The combination of ALIGN-TOPIC,R and ALIGN-CP,L suggests that the boundary directly preceding the embedded left-dislocation is triggered by the sentential nature of the verb’s complement rather than by the mere existence of an (embedded) left-dislocation.

In her approach on LDs in Bantu languages, Downing (2011) also refers to the sentential nature of the verb’s complement by proposing a constraint which is similar to ALIGN-CP,L (p. 781). However, she does not assume a constraint specific for topics nor a stochastic candidate evaluation. Instead, she proposes two different syntactic positions for the topic constituent, see (53). LDs without a right boundary are located in the sentence-internal position *Topic*₂, while LDs with a right boundary are located in the position *Topic*₁. The position *Topic*₁ is the structural position for embedded topics in Tumbuka, since they phrase with preceding but not with following material.

(53) Topic positions in Downing (2011: 780)

(	)	Prosody
[FP Topic ₁ [CP [TP Subject [TopP Topic ₂ [vP...]]]]]		Syntax

This analysis works well for Bantu languages, but it encounters problems when dealing with Romance languages. Not only are sentence-internal topic positions reserved for right-dislocations (see, e.g. Cecchetto 1999, López 2009a,b, Villalba 2009, Feldhausen 2010: ch.4; see Ott & de Vries 2014, Ott & Onea 2015 for a different view based on a biclausal analysis), but the proposal cannot account for the pattern found in Murcia Spanish nor for any other language languages in which the left-dislocation has a right and a left boundary in embedded clauses. While the right boundary is generated by Downing’s version of ALIGN-CP,L, there is no possibility to generate the left boundary, which separates the entire embedded clause (FP in (53)) from the matrix clause.

A recent approach to embedded left-dislocations is proposed in Hamlaoui & Szendrői (2017). They assume a flexible ALIGNLEFT constraint which calls for the alignment of the left edge of an IP with the left edge of the highest projection whose head is overtly filled by the verb or verbal material (p. 5). The left-dislocation is either inside that highest projection (and no boundary occurs after the LD, as in Northern Sotho) or it is outside (and the LD is immediately followed by that boundary, as in Bàsàá). They additionally assume a version of ALIGN-TOPIC (p. 23), that is fulfilled when a topic is adjacent to a boundary inserted by ALIGNLEFT. As in Downing, this approach encounters difficulties when

the embedded LD is preceded by a boundary (as in Murcia Spanish). Hamlaoui & Szendrői's version of ALIGN-TOPIC cannot introduce the necessary boundary, because it is already saturated by the boundary introduced by ALIGNLEFT.

In summary, while Downing (1970) analyzed the prosody of left-dislocations in terms of a prosodic demand by the root sentence following the left-dislocated constituent, more recent literature has analyzed it in terms of prosodic requirements stemming from the topic (e.g. TOPIC PROSODIC DOMAIN, TOPIC or ALIGN-TOPIC). The difference may be interesting for future research when the speech-act perspective is extended into this domain: Is this boundary related to a speech-act nature of the following sentence, or to a speech-act nature of the left-dislocated topic, or both?

Following ideas by Endriss (2009) and Bianchi & Frascarelli (2010), Ishihara (2022) comments on the idea that (certain types of) topics perform a speech act. If topics express a separate speech act, MatchSA would generate an IP boundary at its edges – regardless of whether dislocations are sentential or not. The advantage of this analysis is obvious: it is no longer necessary to refer to the syntactic structure of the dislocation or its environment. Moreover, MatchSA provides a broader generalization than a constraint that refers only to topic (such as TOPIC PROSODIC DOMAIN, TOPIC or ALIGN-TOPIC). However, several points are open. What evidence is there that dislocations/topics really do constitute speech acts in their own right? Can embedded dislocations also express speech acts? MatchSA generates both a left and a right boundary, but there are several languages in which the left boundary is typically not realized or in which boundaries are optional; how is the empirical issue resolved?¹¹ How does MatchSA deal with varying nature of the prosodic boundary strength?

While the previous studies more or less explicitly assume that structures with dislocations are monoclausal, there are approaches that argue that the structures consist of two sentences. Ott (2014, 2015), for example, has argued for this at length, giving prosodic reasons for the analysis in addition to syntactic ones. For reasons of space, we briefly present the core idea of the analysis and then address prosodic aspects.

Ott (2015: 239) assumes that a structure as the Italian one in (54) (taken from Cinque 1983) consists of two separate CPs (55) that are directly adjacent to

¹¹Ishihara & Myrberg (2023) deal extensively with this issue, which they call the Asymmetry Problem, and propose an analysis within the framework of Match Theory. They argue “that prosodic asymmetry arises from the directionality of prosodic heads (i.e., right- or leftheadedness) rather than asymmetric syntax–prosody mapping, and that asymmetry can be explained through the interaction of Match constraints with markedness constraints that determine the distribution of prosodic heads” (p.1).

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each other (juxtaposition). Since these CPs are maximally redundant, material is deleted on PF ((56), as typical for ellipsis), yielding the surface pattern. The two separate sentences are connected by endophoric relations (the resumptive element is connected by anaphoric linkage to the dislocation; the ellipsis in CP₁ is connected by a cataphoric link to its postcedent domain in CP₂. (See Ott 2014, 2015 for the question whether the dislocation moves to the edge of CP₁ prior to deletion or not).

- (54) Su questo lavoro, non riesco a concentrarmi.
 On this work not I.can to concentrate.on-it
 ‘This work, I can’t concentrate on it.’
- (55) [CP₁ non riesco a concentrarmi su questo lavoro][CP₂ non riesco a concentrarmi]
- (56) [CP₁ ~~non riesco a concentrarmi~~ [su questo lavoro]_i][CP₂ non riesco a concentrarmi[ci]_i]
- (57) ([CP₁ ~~non riesco a concentrarmi~~ [su questo lavoro]_i] [CP₂ non riesco a concentrarmi[ci]_i])_i

Ott (2015: 244–246) assumes that LDs are prosodically separated from the host clause and that this typically occurs at the level of the IP (marked by *ι* in the example). With Selkirk (2011, 2005) and Downing (1970), he assumes that IPs are the prosodic correlates of root clauses, and since LD (= CP₁) and the host clause (= CP₂) are distinct root clauses in his account, the syntax-prosody mapping is straightforward in his approach, (57). The fact that the prosodic boundary between the LD and the rest of the sentence is not always an IP and sometimes does not even exist (as also described in the previous paragraphs of this subsection) is attributed to prosodic restructuring by him (see Ott 2015: 244, fn. 27).

Ott (2015: 272–280) also applies his approach to embedded LD and always takes as a standard result the prosodic (not the syntactic!) structure in (49). The fact that the languages of the world also show prosodic patterns as in (50) and (51) is mentioned by him (see fn. 57), but pushed aside with a reference to the occurrence of prosodic regrouping. What is desirable now is a detailed elaboration of the syntax-prosody mapping of his approach for it is unclear, for example, whether one can really speak of prosodic restructuring when languages like Haya and Northern Sotho never show a prosodic boundary at the right edge of a LD.

To sum up, we conclude that (clitic) left-dislocations are not minimally and exhaustively contained in an IP. First, the relevant prosodic level can also be lower,

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namely the intermediate phrase (or the phonological phrase). Second, there are languages that do not show a right boundary at all (like Northern Sotho and Haya). Third, in several languages the embedded CLLD constituent creates a prosodic unit together with the preceding main clause – indicating that the left boundary is not obligatory either (like in Catalan, Peruvian Spanish, Chichewa, and Tumbuka). We further conclude that the combination of ALIGN-TOPIC,R and ALIGN-CP,L (Feldhausen 2010, 2016a, Patin et al. 2017) accounts for the empirical facts in a more appropriate way than the competing approaches. This approach together with a constraint like STRONGSTART or EQUALSISTER might account for the correct picture concerning the prosodic category (ip vs. IP) of the dislocated element (as proposed in Selkirk 2011 and Hamlaoui & Szendrői 2017). Ott’s (2015) highly interesting syntactic approach still requires a detailed elaboration of the integration of the prosodic features of LD attested across languages.

7 Conclusion

The principled OBI convention by Downing (1970) in which root sentences require obligatory prosodic divisions (that we may take to be intonation phrase boundaries today) at their edges, is still relevant today, and deserves renewed attention in the context of the more recent suggestions of Selkirk (2011, 2009), Kan (2009) and Ishihara (2022) that these root sentences, or something close to them, are syntactic constituents corresponding to speech acts. In that perspective, obligatory intonation phrase boundaries derive from a syntactic distinction that also has consequences at the syntax-semantics interface. The prosodic separation of coordinated clauses, of non-restrictive relative clauses, and a range of other constituents can be traced back to this underlying distinction. Clauses that are not root sentences have also been reported to show an affinity to intonation phrase boundaries, which has been captured in terms of additional constraints. Unlike Downing’s boundaries, these can be overridden by prosodic consequences of focus assignment.

Coordinated root sentences thus show obligatory intonation phrase boundaries between them (and apparent exceptions seem to be related to unexpected syntax which exempts them from the relevant constraints). Coordinated genuinely embedded clauses on the other hand, show a weaker and more irregular tendency to showing intonation phrase divisions.

In the domain of relative clauses, non-restrictive relatives show the OBI-demands of obligatory intonation phrase boundaries, while restrictive relatives seem to exhibit only the weaker and optional effects of the affinity of clause-

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boundaries to intonation phrase edges. Parsing preferences show that a boundary preceding a restrictive relative clause supports an interpretation of high attachment in case of attachment ambiguities and in case the relative clause is long.

For a variety of other syntactically and/or semantically detached elements, here discussed under the label of parentheticals, this analysis on its own is clearly not enough. A rich literature that is independent of the OBI and related claims has developed. From a generative perspective, we think that the division between class 0 and class 1 elements of Bing (1979, 1985) provides useful descriptive elements for structuring this terrain: Class 0 elements are special parentheticals that are allowed to lack pitch accents and/or the nuclear stress when not occurring in initial position. While Bing argued that class 0 elements are nevertheless intonation phrases, other authors have taken a different stand, given the prosodic and intonational incompleteness of the assumed intonation phrases. Whichever approach is correct, it seems that a generative analysis that invokes something related to the OBI or the modern version Match-iCP needs additional analysis for deriving the special behavior of class 0 elements. We outlined a possible path for such additional analysis.

Phonetic studies have shown a range of details on the prosody of parentheticals. They have also shown that a range of languages exhibit such parentheticals that are not fully regular intonation phrases. Furthermore, a repertoire of different cues is employed by speakers to signal the presence of such a parenthetical construction.

Finally, left-dislocations, which also often shows an intonation phrase division, has been looked at from two different perspectives: While the OBI analysis related the boundary to the left edge of the sentence following the dislocated constituent, more recent literature has analyzed it in terms of prosodic demands stemming from the left-dislocated topic. This approach allowed for accounting for the complex picture that occurs when considering embedded dislocations. Finally, we presented ideas on the relation between dislocation and speech acts and we thus close the circle between that extra-sentential element and the role of speech acts addressed in the paper.

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Chapter 8

Projective content: Presuppositions and conventional implicatures

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This chapter deals with a group of sub-sentential meaning components that can be divided into two main groups: presuppositions and clausal implicatures. These include, for example, the factive meaning of certain clausal verbs, non-restrictive relative clauses, and parentheticals. This seemingly diverse group of phenomena is linked by common properties. The first is their *projective* behaviour. This means that their contribution to utterance meaning is not affected by embedding under truth-cancelling operators. The second is their status as *not-at-issue* content. This means, pre-theoretically, that they do not contribute to the speaker's main point. This chapter provides an overview of the phenomena and discusses approaches that attempt to sharpen the pre-theoretical notion of *at-issueness* through the notion of subordination to a *question under discussion* (QUD). In particular, it discusses the extent to which the concept thus explicated helps to explain the projective behaviour of the individual phenomena.

1 Introduction

There are some phenomena that are included into virtually any introduction to the semantics-pragmatics interface. These include presuppositions, conventional implicatures, expressive content, information structure, and many more. Each of them come with their own specific research topics and puzzles and each of these



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bundles of phenomena can be and has indeed been studied extensively in linguistic research over the past few decades within various frameworks. Oftentimes, however, the most successful theories are those that manage to bridge over individual phenomena and explain seemingly unrelated ones in a unified way. The main topic of this chapter is one of those ideas that has electrified research at the semantics-pragmatics interface by bringing together precisely the phenomena mentioned above.

Consider two fairly well-known cases: presuppositions illustrated in (1) and conventional implicatures introduced by non-restrictive relative clauses illustrated in (2).

- (1) Skylar knows that Ashanti is very progressive.
 - a. asserts: Skylar rightfully believes that Ashanti is very progressive.
 - b. presupposes: Ashanti is very progressive.
- (2) Josh told Ashanti, who is very progressive, a funny joke.
 - a. asserts: Josh told Ashanti a funny joke.
 - b. conventionally implicates: Ashanti is very progressive.

These two examples have something in common. In particular, if we were to embed them under some operator like negation, a modal operator or the antecedent of a conditional, the relevant asserted proposition is no longer entailed. However, the second entailment, the presupposed/conventionally implicated one, remains entailed. This is shown in (3). Such cases are usually known under the label of projection, i.e. presupposed and conventionally implicated content can *project*.

- (3) a. Skylar does not know that Ashanti is very progressive.
 - i. does not entail: Skylar rightfully believes that Ashanti is very progressive.
 - ii. presupposes: Ashanti is very progressive.
- b. Skylar did not tell Ashanti, who is very progressive, a funny joke.
 - i. does not entail: Josh told Ashanti a funny joke.
 - ii. conventionally implicates: Ashanti is very progressive.

However, the two examples in (1) and (2) are also quite different in important ways. For example, one can assert (1) in a context in which the presupposed content has just been established, but not (2) in a context in which the conventionally implicated content has just been established. This is shown in (4). Further

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differences that we do not elaborate upon here include differences with respect to binding of variables, speaker-orientation and more.

- (4) a. Context: Did you know that Ashanti is very progressive?
- b. Yes, and Skylar knows that Ashanti is very progressive.
- c. Yes, and Ashanti (# who is very progressive) studied comparative literature.

The idea that we will explore in this chapter is the following: the projectiveness of these constructions has a common explanation that is widely independent of the syntactic (and semantic) peculiarities of the respective structures. In particular: all of the inferences that project are so called not-at-issue inferences and they project for that very reason.

The intuition underlying this idea is this: when asserting (1) and (2), the speaker will usually not have the presupposed or conventionally implicated inferences as her main point. At an operational level, this means that this inference will not constitute the answer to the question the speaker may have had in mind when making the respective utterance, the so called *question under discussion* (QUD).¹ Projection is claimed to be a side-effect of not being the answer to the question under discussion. This immediately raises the question of how not being the main point of an utterance actually leads to projection.

This will be the main topic of our investigation in this chapter. We adopt the following presentational strategy. In a first step, in Section 2 we introduce some of the main empirical phenomena that are usually discussed under the label of projection and explain their basic characteristics. Given the topic of this volume, we will try to limit our discussion to cases of clausal embedding modulo some phenomena that can at least be conceptualised as such, even if they are not prototypical cases. However, we will occasionally fail in this attempt for the simple reason that it is in fact one of the main points of the theoretical approach we explore that many phenomena that seem remote and unrelated have in fact a common core. Hence, we will need to occasionally talk about epithets, appositives and other phenomena that really fall outside of clausal embedding. In the next step, in Section ??, we get into more technical details showing how the idea can be put in terms of a predictive theory and discuss some examples. This discussion will incorporate not only the question of what the QUD is and how it

¹The QUD is an important notion in many recent theories within formal pragmatics, developed among others by Klein & von Steutterheim (1987), Roberts (1996), van Kuppevelt (1995) and especially popular since Beaver & Clark (2008). We will return to this notion in section 3 in more detail.

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should be modelled but also how exactly projection is predicted or not depending on the context. One of the important aspects of the discussion in this section is the exploration of possible implementations and limits of the theory that projection is based on not-at-issueness. In Section 4, we sum up the discussion of the features and limits of the theoretical approach discussed and outline some ways in which the main idea can be connected to some of the other approaches from the literature. Thereby, we suggest some possibilities for future research. Finally, the paper ends with some concluding remarks in Section ??.

Before concluding this introductory section, we need to point out that there are several very well-written handbooks and overview articles on presuppositions that we want to mention here. Levinson (1983: Ch. 4) is a standard reference that summarizes the discussion up to the end of the 1970s. Beaver (1997) provides an excellent overview of the literature up to the end of the 1990s. It is particularly useful as a guide to the dynamic theories of Karttunen, Heim, and van der Sandt. There are very accessible accounts by Geurts (1999) and van der Sandt (2012) on the presupposition as anaphora account, that is embedded in *Discourse Representation Theory* (Kamp & Reyle 1993: DRT). The experimental literature has been reviewed by Schwarz (2019). Conventional implicature had been introduced by Potts (2003, 2005). The phenomena are exhaustively described in Gutzmann (2013). Further accessible introductions to phenomena and theory are provided in (Liu 2012, Gutzmann 2015, 2019). In light of this, we do not make it our purpose to add to this body of literature; instead we focus on a detailed analysis of the relation of (not-)at-issueness and projection. Thus, the overview of the literature and theoretical discussions in Section 2 will remain limited and focus mainly on a comprehensive description of the main empirical facts.

2 Projective phenomena

2.1 What is projection?

What all linguistic phenomena that we address in this chapter have in common is that they involve *projective meaning*. The relevant *meaning* is a proposition that is implied by a certain *trigger*. For example, in (1), *Skylar knows that Ashanti is very progressive*, the verb *know* triggers the inference that the embedded proposition *Ashanti is very progressive* is true. Projective meaning is characterised by a peculiar property of these inferences:

- (5) **Definition of projection** (Simons-Tonhauser-Beaver-Roberts).

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An implication projects if and only if it survives as an utterance implication when the expression that triggers the implication occurs under the syntactic scope of an entailment-cancelling operator (Simons et al. 2010: 309)².

As an example let us consider (6). In the philosophical literature *knowledge* is often identified with *justified true belief* (originally going back to Plato and more recently proposed, e.g. by Chisholm 1957 but see also Gettier 1963 and subsequent literature). Hence, one may be tempted to paraphrase *Skylar knows that* with *Skylar correctly/rightly believes that* and expect that (6a) and (6b) are almost synonyms.

- (6) a. Skylar knows that Ashanti is very progressive.
- b. Skylar correctly/rightly believes that Ashanti is very progressive.

For both (6a) and (6b), one would ordinarily infer that the speaker who utters (6a) or (6b) also believes that Ashanti is very progressive. However, when the sentences are negated, as in (7a) and (7b), then we see a difference between the two versions.

- (7) a. Skylar does not know that Ashanti is very progressive.
- b. Skylar does not correctly/rightly believe that Ashanti is very progressive.
- c. Skylar does not believe that Ashanti is very progressive.

It can, pre-theoretically speaking, still be inferred from (7a) that the speaker believes that *Ashanti is very progressive*, whereas it is not possible to infer it from an utterance of (7b). An utterance of (7b) still implies that Skylar believes that Ashanti is very progressive. What is negated is that this belief is correct. If the adverbial is dropped, as in (7c), then there is a strong tendency to infer that Skylar believes that Ashanti is not very progressive, while nothing can be inferred about the beliefs of the speaker. Hence, although *know* and *correctly believe* may seem near synonyms, they turn out to behave differently when syntactically embedded under negation. As the proposition *Ashanti is very progressive* is entailed by both (6a) and (7a), where the trigger of this entailment is syntactically embedded under an entailment-cancelling operator, one says that it *projects*.

From the perspective of clausal embedding, projection is the most interesting property of the linguistic phenomena that are discussed in this chapter.

²This definition goes back to a large body of literature, primarily the *family of sentences* test proposed in Chierchia & McConnell-Ginet (1990), which involve the embedding under entailment-cancelling operators, as shown below, viz. examples (8) and (9).

Embedding under negation is the traditional test for projective meaning. Besides embedding under negation (8a), the most widely used ones are embedding under epistemic operators (8b), embedding in an antecedent of a conditional (8c), and embedding in questions (8d).

- (8) a. **Negation:** Skylar does not know that Ashanti is very progressive.
- b. **Epistemic operators:** It is possible that/Maybe/Probably Skylar knows that Ashanti is very progressive.
- c. **Antecedent of conditionals:** If Skylar knows that Ashanti is very progressive, she will be at her party.
- d. **Questions:** Does Skylar know that Ashanti is very progressive?

As in the case of negation, speakers of English would agree that normally someone who says (8b), (8c), or (8d) must believe that the proposition embedded under *know*, i.e. *that Ashanti is very progressive*, is true. If *know* is replaced by *believe*, then this is not the case any more:

- (9) a. **Negation:** Skylar does not believe that Ashanti is very progressive.
- b. **Epistemic operators:** It is possible that/Maybe/Probably Skylar believes that Ashanti is very progressive.
- c. **Antecedent of conditionals:** If Skylar believes that Ashanti is very progressive, she will be at her party.
- d. **Questions:** Does Skylar believe that Ashanti is very progressive?

If a proposition *p* entails *q*, then *not p*, *possibly p*, *if p*, and *p?* do not entail *q*. The definition in (5) says that an inference $p \Rightarrow q$ is called *projective meaning* if it has the surprising property of surviving these embeddings. The notion goes back to Langendoen & Savin (1971). The tests in (8) and (9) are collectively known as the *family-of-sentences diagnostic* (Chierchia & McConnell-Ginet 1990: 29f). In the following, we consider two types of projective meaning that have been extensively discussed in the literature: *presuppositions* and *conventional implicature*. Examples of presuppositions are the inferences from the factive *know* *p* to *p* or from the definite description *the dog* to the existential *there exists a dog*. An example of a conventional implicature is the inference that the speaker who uses the expressive adjective in *the damn book* has a dislike of the book. In the next section, we address presuppositions. In Section 2.3 we consider the main phenomena related to conventional implicature. These are two broad classes of phenomena. In Section 2.4 we address the issue of whether there are projective phenomena that do not fall under the labels of presupposed or conventionally implicated content. We present a proposal by Tonhauser et al. (2013) that addresses

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this issue. They sub-divide projective phenomena into four classes based on two semantic conditions.

2.2 Presuppositions

2.2.1 Defining presuppositions

The modern notion of *presupposition* can, at least, be traced back to Frege (1892). For Frege, a presupposition (*Voraussetzung*) is a proposition that must be true for a sentence to have a meaning. The presupposition has to be distinguished from what a sentence *asserts*. For example, the sentence in (10) presupposes that there is someone who discovered the elliptic form of the planetary orbits and asserts that this person died in misery.

- (10) Whoever discovered the elliptic form of the planetary orbits died in misery.

As it was Kepler's (1937) who discovered the elliptic form of planetary orbits, sentence (10) is true if, and only if, Kepler died in misery. If the presupposition of (10) is false, i.e. if the elliptic form of the planetary orbits had never been discovered, then it is not clear which truth value to assign to (10), the intuitive reason being that it is unclear what the sentence asserts³. The same intuition provides a reason for why presuppositions survive embedding under entailment-cancelling operators: the embedding operator targets the asserted content; it can only target the asserted content if the content is a well-defined proposition; it can only be a well-defined proposition if its presuppositions are satisfied; hence, embedding under an operator can only be felicitous if the presuppositions are true.

It is far from obvious, however, that all phenomena that are traditionally subsumed under the notion of *presupposition* fall under Frege's concept of a *Voraussetzung* that has to be satisfied for a sentence to have a well-defined truth value. For example, the sentence (6a) *Skylar knows that Ashanti is very progressive* presupposes the proposition that *Ashanti is progressive* and asserts that Skylar believes it. However, the truth of the presupposed part is not a requirement for the asserted proposition to have a truth value. It seems rather that the speaker who says (6a) takes it for granted that the presupposed proposition is true. Stalnaker (1974) distinguished between *semantic* presuppositions, which are a property of

³For simplicity's sake, we follow here Strawson (1950) who argued that sentences like (10) lack a truth-value if the definite description has no referent. There was a famous debate between him, Russell (1905, 1919), and others about this issue.

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sentences and correspond to the Fregean conception, and *pragmatic* presuppositions which are propositions that the speakers assumes to be shared beliefs between her or him and the addressee. Stalnaker shows that semantic presuppositions are also pragmatic presuppositions if one assumes that speakers normally assert sentences only if they have well-defined truth conditions. For example, if someone asserts (10), then she or he must assume that it is shared knowledge between speaker and hearer that the Fregean presupposition that there is someone who discovered the elliptic form of planetary orbits is true. Hence, the semantic presupposition of (10) is also a pragmatic presupposition.

2.2.2 Presupposition triggers

The issue of how to define the notion of *presupposition* is notoriously difficult, and it is safe to say that there is no universally accepted candidate. There is, however, a core of phenomena for which there is a general agreement that they *trigger* presuppositions. This means that they are linguistic constructions where an asserted and a presupposed proposition can be distinguished. The constructions can be divided into semantic and syntactic presupposition triggers. In the following list, we indicate the presupposition by $\gg$.

- (11) a. **Definite Descriptions**
 The King of France is bald. $\gg$ There is a King of France.
The φ $\gg$ There exists an x such that $\varphi(x)$ is true.
- b. **Possessives**
 Toni and Niki come with their children. $\gg$ Toni and Niki have children.
P poss x $\gg$ There exists an x such that P possesses x .
- c. **Quantifiers**
 Every/Some/No Athenian is wise. $\gg$ There are Athenians.
 Similar: *most, many, few, etc.*
- d. **Factive Verbs**
 Skylar *knows* that Ashanti is progressive. $\gg$ Ashanti is progressive.
 Also: Skylar *discovered/noticed/accepts/regrets/etc.* that φ $\gg \varphi$ is true.
- e. **Aspectual Verbs**
 Ashanti *stopped* smoking. $\gg$ Ashanti smoked before.
 Similar: *began, started, ceased, continued.*
- f. **Implicative Verbs**
 Sasha *managed* to open the door. $\gg$ Sasha tried to open the door.

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Similar: Sasha *failed to V* >>Sasha tried to V; Sasha *forgot/remembered to V* >>should have Ved.

g. Iteratives

Niki visited me *again*. >>Niki visited me before.

Similar: *a second time, before*; also: Niki visited me *too*.

h. Clefts and Pseudo-clefts

It was Bilge who wrote the programme. >>Someone wrote the programme.

What Bilge did was write the programme. >>Bilge did something.

In this list, only factive verbs embed a finite clause. Definite descriptions, possessives, quantifiers, and clefts trigger an *existence* presupposition (*there exists ...*). Sometimes, pronouns are also counted among the triggers of existence presuppositions (it is on the shelf >>something is on the shelf).

If the family-of-sentence-test is applied to them, it can be seen that the presuppositions do project. In (12), this is demonstrated for a cleft sentence. In each case, the addressee can infer the presupposition that there is someone who wrote the programme.

- (12)
- a. **Negation:** It wasn't Bilge who wrote the programme.
 - b. **Epistemic operators:** It is possible that it was Bilge who wrote the programme.
 - c. **Antecedent of conditionals:** If it was Bilge who wrote the programme, she will get a reward.
 - d. **Questions:** Was it Bilge who wrote the programme??

There is also an important test that is usually used to target specifically presuppositions. It is the *hey-wait-a-minute-test* (von Fintel 2004). While this test was not originally designed to distinguish presuppositions from other projective content, but rather from assertions, it works best for presupposed content. In von Fintel's example (Section 2.2.2) it is applied to a sentence containing a definite description.

- (13)
- A: The mathematician who proved Goldbach's Conjecture is a woman.
 - B: Hey, wait a minute. I had no idea that someone proved Goldbach's Conjecture.
 - B': #Hey, wait a minute. I had no idea that that was a woman.

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The idea is that an addressee who disagrees about the presupposition of an utterance cannot simply reply with *no*. Negation can only target asserted content. *Hey, wait a minute* is a device that indicates that the reply targets something else. The contrast between B and B' shows that *hey, wait a minute* can introduce an objection to the presupposed content of an utterance, but not the asserted one.

2.2.3 Properties of presuppositions

Presuppositions have properties that would normally classify them as pragmatic in nature. The semantic meaning of a linguistic construction that contains no anaphoric or deictic element is context independent. If presuppositions are a purely semantic phenomenon, then a linguistic trigger should create the presupposition whatever the context is. In the early semantic theory of Langendoen & Savin (1971), presuppositions are locally triggered by lexical or syntactic elements, and automatically projected to the global utterance context. What is locally embedded is only the asserted part; the presupposition becomes a presupposition of the global utterance. There are, however, cases that show that presuppositions are context sensitive, and do not automatically project. For example, none of the following sentences presupposes that Bilge wrote a programme:

- (14) a. If anybody managed to write the programme, then it was Bilge who did it.
 b. Either no-one wrote the programme, or it was Bilge who did it.

In response to these examples, Karttunen (1973) proposed general rules that specify when presuppositions triggered in embedded clauses become presuppositions of the embedding sentence. For example, if a presupposition is triggered in the consequent of a conditional, then it projects (becomes a presupposition of the whole conditional), if the truth of the antecedent does not entail the presupposition. This rule explains why the presupposition triggered by the cleft sentence in the consequent of (14a), that someone wrote the programme, does not become a presupposition of the whole conditional: if the antecedent is true, then the presupposition is entailed by it.

In (14), the features that cancel the presupposition are sentence internal. Presuppositions can, however, also be cancelled by features of the wider discourse context, which points to their pragmatic nature. In the following example the critical implicature trigger is the factive verb *aware* (see also Levinson 1983: 189f).

- (15) A: Well, I simply want to know whether Bilge wrote the programme.

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- B: The only person who would know for sure is Imani; I've talked to him, and he isn't aware that Bilge wrote the programme. So I think that she didn't write it.

In B's response, *aware* triggers the presupposition that *Bilge wrote the programme*. The trigger is embedded under negation in a conjunction, hence, according to standard assumptions, it should project. However, it does not. The context makes clear that the speaker B does not know whether Bilge wrote the programme or not. Examples like these show that presupposition cancellation cannot be explained purely on syntactic and semantic grounds.

Since Karttunen's seminal work, a wide variety of theories has been developed that tried to explain when and why presuppositions of embedded clauses project. All of them consider a mixture of semantic and pragmatic factors that can lead to activation, cancellation, or non-activation of presuppositions (e.g. Karttunen 1974, Gazdar 1979, Karttunen & Peters 1979, Heim 1983, 1992, van der Sandt 1992, Geurts 1999, Beaver 2001; see also Singh 2020 for a recent overview). In the new millennium, a novel kind of explanation emerged that explained presupposition projection on information theoretical grounds. The core ingredient is the *question under discussion* that dominates a sentence or a stretch of discourse (Simons et al. 2010, Tonhauser et al. 2013, Simons et al. 2017). These accounts will be the main topic of section (??).

2.3 Conventional implicatures

The term *conventional implicature* was coined by Grice (1975). It denoted meaning of linguistic expressions that is conventionalised but not truth-conditionally relevant. Examples are connectives like *but* and *therefore*. In the sentence *It is beautiful but expensive*, the truth-conditional contribution of *but* is equivalent to that of *and*. In addition, it signals a contrast between the two connected propositions and makes an argumentative contribution by weighing the second proposition more than the first. As conventional implicatures have no effect on truth conditions, it was widely held that their proper place is in pragmatics and not semantics. This changed with the influential work of Potts (2005). As conventional implicatures are *conventional*, as the name says, he argued that the proper place for studying them is in an extended framework of compositional semantics. Before sketching this framework, we first consider general properties of conventional implicatures and linguistic phenomena that have been identified there.

To build up some intuition about conventional implicatures, we consider the expressive adjective *damn* in (16) (see Potts 2005: 17).

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- (16) We bought a new electric clothes dryer [...] Nowhere did it say that the *damn* thing didn't come with an electric plug!

The word *damn* clearly contributes lexical meaning to the sentence. *The damn N* can be paraphrased as: the speaker is frustrated by whatever *the N* refers to. As lexical meaning, it is by definition a form of conventional meaning, i.e. meaning that is attached to the linguistic material and understood by language speakers just on the basis of this material. Furthermore, conventional implicatures are not cancellable, in contrast to presuppositions and conversational implicatures. In Potts's words, the speaker is committed to their conventional meaning. As the meaning can be paraphrased by a declarative sentence, it can be represented by a proposition that allows for further inferences. The main difference between conventional implicature and asserted truth-conditional content of a sentence is their deletability: if *damn* is deleted from the sentence, as in (17), then the shortened sentence states the same fact as the original one, i.e. both variants are true or false in the same contexts.

- (17) Nowhere did it say that the thing didn't come with an electric plug!

The non-cancellability is demonstrated by the following examples:

- (18) a. #Nowhere did it say that the *damn* thing, if it is indeed the case that the thing is frustrating, didn't come with an electric plug..
b. #Nowhere did it say that the *damn* thing didn't come with an electric plug; but, in fact, the thing is not frustrating.

Potts himself defines *conventional implicatures* (CIs) as follows:

(19) **Properties of conventional implicature**

- a. CIs are part of the conventional meaning of words.
- b. CIs are commitments, and thus give rise to entailments.
- c. These commitments are made by the speaker of the utterance 'by virtue of the meaning of' the words he chooses.
- d. CIs are logically and compositionally independent of what is 'said (in the favored sense)', i.e. independent of the at-issue entailments.

(Potts 2005: 24; see also Potts 2003: 2)

Compositional independence causes a problem for semantics. The expressive adjective *damn* is syntactically part of the determiner phrase *the damn thing*. The

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adjective takes the meaning of *thing* and delivers an expressive proposition. The article *the*, however, ignores the adjective and only combines with the meaning of the noun *thing* to deliver a referent for further compositional meaning computation. As the expressive proposition is not involved in these further computations, it remains unbound and floats above the meaning of the remaining sentence. Christopher Potts, and others after him (McCready 2010, Gutzmann 2015), therefore, developed a *hybrid* semantics with a two-layered representation of meaning, where the lower layer represents truth-conditional meaning, and the higher layer expressive content. Potts also introduced the notion of *at-issue* content for the truth-conditional content of a sentence, which is set in opposition to the *expressive* content. Both types of content can be represented as propositions. In standard model-theoretic semantics, a proposition is a set of *possible worlds*: the set of worlds in which the sentence that expresses the proposition is true, which is usually called the *truth-conditional meaning*. Gutzmann (2013) proposed to represent the expressive or use-conditional content in parallel fashion by a set of *utterance contexts*, and called it the *use-conditional* meaning of a sentence.

The carriers of conventional implicatures are syntactically but not semantically integrated in the sentences. In contrast to Fregean presuppositions, they are not necessary for calculating truth conditions of their surrounding hosts. With respect to clausal embeddings, conventional implicatures pose a problem similar to presuppositions: they are predominantly not represented by clauses but have a meaning that has to be paraphrased by a full sentence. So, even if the carrier of the conventional implicature is not sentence-like, on another layer, there is a full proposition embedded. In the following, we concentrate on clausal carriers of conventional implicatures, and on *appositives* as an example of a non-sentential carrier with clause-like semantics.

Potts (2005) distinguished between two broad classes of phenomena: *supplements* and *expressives*. Supplements comprise non-restrictive relative clauses, appositives, and parentheticals. Potts also subsumed certain adverbs under supplements:

(20) Types of supplementary conventional implicatures

- a. Ames, a successful spy, is now behind bars. (*nominal appositive*)
- b. Ames was, as the press reported, a successful spy. (*as-parenthetical*)
- c. Ames, who was a successful spy, is now behind bars. (*supplementary relative*)
- d. Amazingly, they refused our offer. (*speaker-oriented adverb*)

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- e. Thoughtfully, Ed destroyed the evidence for us. (*topic-oriented adverb*)
- f. Just between you and me, Aldo is a dangerous spy.
(*utterance-modifying adverb*)

(Potts 2005: 90)

Among them, only parentheticals and non-restrictive relative clauses (NRRC, Potts's *supplementary relative*) embed a clause on surface level. The nominal appositive in (20a) is semantically equivalent to the NRRC in (20c), so that there is a semantically embedded proposition. The NRRC and the appositive take a nominal (*Ames*) from their host sentence as argument. The adverbs, and also the parenthetical, however, take as argument the whole proposition that is expressed by their host. Hence, they are embedding a clause, but not being embedded by a clause. From a semantic perspective, the meaning of the adverbs can be represented as properties of utterance contexts. For the given example, this is also plausible for the parenthetical (20b). For the meaning of the nominal appositive and the NRRC this is, however, not plausible. Their meanings are propositions of the same type as the propositions representing asserted content. In general, as parentheticals also do not express properties of the utterance context but truth-conditional meaning, as can be seen in (21).

- (21) Ames, as a successful spy, was a respected employee of the secret service.

Expressives in the narrow sense are, for example, expressive adjectives (*damn*), honorifics (*Sir*), ethnic slurs (*Lessing is a Kraut*; Kraut = derogatory for German), and interjections as, for example, *ouch*.⁴ They can broadly be considered as expressing properties of the utterance situation, for example, as propositions about speaker attitudes. Expressives are not clausal and, although they may take arguments, for example expressive adjectives, they are not clause embedding.

Some interjections and particles arguably take propositional arguments:

- (22) a. Man, that is hot! (*interjectional use of man*)
 b. Riku ist *wohl* in Japan. (*particle*)
 Riku is *perhaps* in Japan.

The interjectional *man* intensifies the gradable adjective *hot* and turns the whole utterance (22a) into an expressive exclamative. With the German modal particle *wohl* in (22b), the speaker communicates that she/he assumes that the asserted proposition is true, without being certain about it.

⁴For a thorough overview, see Gutzmann (2013). For expressive adjectives see (Liu 2012).

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2.4 A classification of projective content

Potts (2005) distinguished between *at-issue* and *not-at-issue* content of utterances. The distinction was mainly an intuitive one. In later research, however, an attempt was made to give it a precise meaning. Simons et al. (2010) proposed to equate at-issue content with that part of a sentence that answers a possibly implicit *question under discussion* (QUD). The not-at-issue content can then be considered as the material that is not answering the QUD. This led to the following proposal about projective content:

- (23) Hypotheses about what projects and why
- a. All and only those implications of (embedded) sentences which are not-at-issue relative to the question under discussion in the context have the potential to project.
 - b. Operators (modals, negation, etc.) target at-issue content.
- (Simons et al. 2010: 315)

The (23b) condition says that not-at-issue content is ignored by embedding truth-conditional operators. This entails that not-at-issue content is ignored by further calculations of compositional meaning. This is a generalisation of Potts's observation that expressive content is logically and compositionally independent of what is asserted. The qualification *have the potential* in the (23b) condition says that being not-at-issue is a necessary condition for projection, but not a sufficient one. For content to actually project, further pragmatic conditions have to be satisfied. These conditions will be discussed in more detail in section (??).

We have seen two large groups of projective phenomena: presuppositions and conventional implicatures. Both the presuppositions of a sentence and its expressive and supplementary content are not-at-issue and, hence, are not answering the (implicit) QUD to which the asserted content answers. Based on this insight, Tonhauser et al. (2013) proposed a taxonomy of projective phenomena on the basis of two conditions: the satisfaction of a *strong contextual felicity constraint* and the existence of *obligatory local effects*. The taxonomy classifies items *t* for which the family-of-sentences diagnostic shows that *t* triggers projective content *m*, where *m* is some proposition. Tonhauser et al. (2013: 75) distinguish between two types of contexts that they called *m*-positive and *m*-neutral contexts. An *m*-positive context is a context in which it is known that *m* is true, and an *m*-neutral context is an utterance context in which it is both possible that *m* is true or that it is false. If an occurrence of trigger *t* is felicitous only if the utterance context makes clear that *m* holds, then *t* satisfies the *strong felicity constraint*:

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- (24) **Strong contextual felicity constraint:** If utterance of trigger t of projective content m is acceptable only in an m -positive context, then t imposes a strong contextual felicity constraint with respect to m .
(Tonhauser et al. 2013: 76)

Examples are pronouns that presuppose the existence of a referent, and the English particle *too* that presupposes an alternative, e.g. *Ashanti came too* presupposes that someone else than Ashanti also came. Clausal or clause embedding triggers, for example factive verbs, non-restrictive relative clauses, the focus particle *only*, and aspectual verbs come without strong contextual felicity constraints.

The next condition is satisfied if projected content m is semantically embedded under belief operators that embed the trigger t :

- (25) **Obligatory local effect:** A projective content m with trigger t has obligatory local effect if and only if, when t is syntactically embedded in the complement of a belief-predicate B , m necessarily is part of the content that is targeted by, and within the scope of, B .
(Tonhauser et al. 2013: 93)

The existence presuppositions of pronouns and the particle *too* are embedded under belief operators and therefore satisfy this condition:

- (26) Dayo believes that *it* is in the drawer *too*.
 $\leadsto$ Dayo believes that something is in the drawer.
 $\leadsto$ Dayo believes that something else but the referent of *it* is also in the drawer.

Among clausal or clause embedding triggers, some have obligatory local effects, and some do not. For example, non-restrictive relative clauses do not have them:

- (27) Dayo believes that Ames, a successful spy, is now behind bars.
 $\leadsto$ Dayo believes that Ames is a successful spy.

Factive verbs, the focus particle *only*, and aspectual verbs, for example *stop*, have obligatory local effects:

- (28) a. Dayo believes that Ashanti knows that Riku is in Japan.
 $\leadsto$ Dayo believes that Riku is in Japan.
 b. Dayo believes that only Riku is in Japan.
 $\leadsto$ Dayo believes that nobody else but Riku is in Japan.

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- c. Dayo believes that Bilge stopped programming.
 $\leadsto$ Dayo believes that Bilge programmed before.

By cross classification, Tonhauser et al. (2013) predict the existence of a class of projective phenomena that satisfy the strong contextual felicity constraint but do not have obligatory local effects. This is a small class of phenomena, none of which are clausal or clause embedding.

3 Interaction between at-issueness and projection

With this background we turn to the question raised in the introductory section: Is there a common explanation for projection across the categories shown in Section 2.4? As suggested in the introduction, we shall explore whether at-issueness as a semantic/pragmatic category can explain projection in a unified way, as suggested by Simons et al. (2010) and subsequent literature.

We start by briefly introducing the core idea that it is the not-at-issue status of an entailed inference that is responsible for projection. The core hypothesis is postulated in (5), repeated as (29).

(29) The Simons-Tonhauser-Beaver-Roberts projection hypothesis

- a. All and only those implications of (embedded) sentences which are not-at-issue relative to the question under discussion in the context have the potential to project.
- b. Operators (modals, negation, etc.) target at-issue content.

(Simons et al. 2010: 315)

This formulation contains two aspects that we will ignore for now. Firstly, the hypothesis does not state that whenever an implication is not-at-issue it will invariably project, but merely that it has the potential to project. Essentially, this only guarantees that at-issue inferences do *not* project. However, usually, scholars interpreted the hypothesis as a biconditional: all and only embedded not-at-issue inferences project. This is partly because it is not obvious what would stop something that has the potential to project from projecting if operators do not. We will also not focus particularly on the second part of the thesis in this paper. This is in part because we are not aware of a complete compositional implementation of the theory within which this claim can be fully evaluated, but see compositional implementations in (AnderBois et al. 2015) and the more recent (Koev 2022) for a compositional implementation for a subset of the cases.

Focusing then on the core of the projection hypothesis, two important questions come to mind that need to be addressed, as well as a third one that emerges from the answers to the first two questions.

The first question is this: when exactly is an inference at-issue? How can we know? This is in fact not a trivial question and there are some misunderstandings that need to be addressed in this paper. We deal with this question in section (??). The second question is an elaboration on the biconditional nature of the generalisation: if it holds that that which projects is not-at-issue and that which is not-at-issue projects, what is really the part that needs to be explained? Are, for example, presuppositions inferences that ‘want’ to project but at-issueness prevents them from doing so, or are presuppositions entirely harmless inferences that are sent on to the way of projection precisely by their not-at-issue status? Whether and how the second part of the theory stated in (29) can help here will be discussed in section (??).

3.1 What is at-issueness?

If one looks at the literature after the dissertation of Potts (2005) and especially Simons et al. (2010), it would seem that at-issueness has become something like the it-girl of the semantics-pragmatics interface. It seems to play a role in quite a range of seemingly unexpected places. We name just a few of the domains in which it is being applied: semantics of it-clefts (Velleman et al. 2012, Destruel 2013, DeVaugh-Geiss et al. 2018), word order in embedded sentences in German (Antomo 2016), or discourse coherence (Hunter & Asher 2016). Also, like a veritable it-girl it is not only everywhere but there are also some reservations with which the success story is being viewed. Part of the problem boils down to a simple question: how do we know what is at-issue and what is not?

Koev (2018, 2022) discusses at length a range of possible notions of at-issueness alongside specific empirical tests for these notions. He distinguishes at least between Q-at-issueness, which is essentially the notion that both we use in this paper and Koev (2022) adopts eventually, P-at-issueness (essentially referring to propositions that have been proposed for a discourse update, e.g. in these of Farkas & Bruce 2010 but whose discourse acceptance or rejection is not yet settled, cf. Murray 2014, AnderBois et al. 2015 and others), and C-at-issueness (essentially reducing to the right frontier in *Segmented Discourse Representation Theory*, SDRT, see Asher & Lascarides 2003, Hunter & Asher 2016). We will ignore the notion of C-at-issueness in this paper since it is not connected to the notion of projection, even though it has important merits in discourse structuring.

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Syrett & Koev (2015) use the anaphoric potential of propositions as a diagnostic for at-issueness, essentially targetting P-at-issueness. For example, in (30) the speaker B's contradiction can only target the proposition asserted by the main clause, but not the one expressed by the non-restrictive relative clause. This is often called the dissent diagnostics, cf. Amaral et al. (2007). Importantly, Syrett & Koev (2015) argue that – among others – based on this test, one could show that in fact sentence final non-restrictive relative clauses can be at-issue, as shown in (31); however this only applies to the P-at-issue status.

- (30) a. A: S
kylar, who is on vacation in South Africa, is a big fan of Joe.
b. B: T
hat is not true, she is not a fan of Joe.
c. B: #
That is not true, she is in Mongolia.
- (31) a. A: S
kylar is a big fan of Joe, who is on vacation in South Africa.
b. B: T
hat is not true, she is not a fan of Joe.
c. B: T
hat is not true, he is in Mongolia.

Syrett & Koev (2015) suggest that double-questions are a reasonable diagnostic for what it is for an inference to be at-issue, this time clearly targeting Q-at-issueness.

- (32) a. Where is Skylar and whom does she vote for?
b. # Skylar, who is on vacation in South Africa, is a big fan of Joe.
c. The greatest fan of Joe is Skylar, who is on vacation in South Africa.

However, Snider (2018) suggests that all the at-issueness diagnostics used by Syrett & Koev (2015) to show that sentence-final non-restrictive relative clauses can be at issue may allow for independent explanations, either involving more elaborate discourse structures or independently motivated anaphoric properties of propositions.

Another diagnostic that has been suggested, for example in Tonhauser et al. (2018), is the rephrasing of a polar question that focuses on the 'main' partition introduced by the question, or put differently, on finding the primary speech act

performed by the speaker. Accordingly, in the example (33), (33a) is expected to be dispreferred as a paraphrase of the speech act performed by Patrick as compared to (33b).

- (33) Patrick asks: Was Martha's new car, a BMW, expensive?
- a. Is Patrick asking whether Martha's new car is a BMW?
 - b. Is Patrick asking whether Martha's new car is expensive?

With all existing diagnostics discussed above, it is not entirely obvious whether they are actually based on the definition of at-issueness, understood as Q-at-issueness (in fact at least for the assert-dissent-test, Koev 2022 actually agrees that it targets a different notion of at-issueness), or whether both the diagnostics and the definition are attempts to capture some deeper intuitive notion. This is surely a problem, two remarks, however, are in order: firstly, the diagnostics do for better or worse correlate with the range of phenomena that have been claimed to be at-issue and not-at-issue respectively. Moreover, especially Tonhauser et al. (2018) convincingly argue that when it comes to experimental settings both at-issueness and projection appear to be gradient notions. Whether or not these are intrinsically gradient or whether the gradient comes from the variability with which participants in experiments interpret utterances in the existing context or imagine a non-given one, remains to be seen in future.

For the purposes of this overview paper, we will focus on the main definition of at-issueness given in Simons et al. (2010), as in (34) and the respective auxiliary definition in (35).

- (34) **The Simons-Tonhauser-Beaver-Roberts definition of at-issueness.** A proposition p is at-issue relative to a question Q iff $?p$ is relevant to Q .
(Simons et al. 2010: 317)

- (35) **Relevance to the QUD**
- a. An assertion is relevant to a QUD iff it contextually entails a partial or complete answer to the QUD.
 - b. A question is relevant to a QUD iff it has an answer which contextually entails a partial or complete answer to the QUD.
- (Simons et al. 2010: 316)

We show the workings of the definition using a conjunction as an example, as in (36).⁵

⁵Thereby, we note in passing that the a. part of the definition in (35) is not necessary, as the relevance of an assertion plays no further role in the theory.

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- (36) a. Q: W
ho danced?
b. A: A
shton smoked and Jaden danced.

Now, the theory itself does not specify the exact theory of questions and answers used, but for explicitness, we assume that questions are modelled as sets of Hamblin alternatives (cf. Hamblin 1973 and Onea & Zimmermann 2019 for a recent overview) and these alternatives are in fact the possible answers to the question. Hence, we model (36a) as $\{\lambda w.danced(x, w) | x \in C\}$ for some contextually given domain C of individuals. We further consider the at-issue status of the following four entailments of .

- (37) a. Ashton smoked = $\lambda w.smoked(A, w)$
b. Jaden danced = $\lambda w.danced(J, w)$
c. Jaden danced and Ashton smoked = $\lambda w.danced(J, w) \wedge smoked(A, w)$
d. Jaden danced or Ashton smoked = $\lambda w.danced(J, w) \vee smoked(A, w)$

By definition, a proposition p is relevant to the QUD Q if $?p$ is relevant to Q . The relevant relations between the two questions are given in (38). Thereby, we use the symbol $\cong$ for relevance of a question to another question.

- (38) a. $\{\lambda w.smoked(A, w), \lambda w.\neg smoked(A, w)\} \not\cong \{\lambda w.danced(x, w) | x \in C\}$
b. $\{\lambda w.danced(J, w), \lambda w.\neg danced(J, w)\} \cong \{\lambda w.danced(x, w) | x \in C\}$
c. $\{\lambda w.danced(J, w) \wedge smoked(A, w), \lambda w.\neg danced(J, w) \vee \neg smoked(A, w)\} \cong \{\lambda w.danced(x, w) | x \in C\}$
d. $\{\lambda w.danced(J, w) \vee smoked(A, w), \lambda w.\neg danced(J, w) \wedge \neg smoked(A, w)\} \stackrel{??}{\cong} \{\lambda w.danced(x, w) | x \in C\}$

Let us consider (38a) first. Here, the polar question has two alternatives. It is relevant to the QUD if at least one of these answers (contextually, but we ignore this part here) entails one of the question alternatives of the QUD. Thus, we have to check whether any of the following holds:

- (39) a. $\lambda w.smoked(A, w) \subseteq X$ for some X in $\{\lambda w.danced(x, w) | x \in C\}$
b. $\lambda w.\neg smoked(A, w) \subseteq X$ for some X in $\{\lambda w.danced(x, w) | x \in C\}$

Since neither is the case – unless very special contextual rules apply – we can conclude that Ashton smoked is indeed not-at-issue. By using similar steps, we

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can easily check that Jaden danced is in fact at-issue, as it is identical to one of the question alternatives. Moreover, we can show that Jaden danced and Ashton smoked as a conjunction is also at-issue, since it entails that Jaden danced. Finally, the disjunction Jaden danced or Ashton smoked is more complicated. Here, we can establish that the positive alternative, as it does not entail any of the QUD-alternatives, is not relevant. When it comes to the negated version, the situation is more complicated. In particular, while there is still no direct question alternative that is entailed by the negated disjunction, as shown in (40), the negated disjunction, written as a conjunction of negations, entails the negation of one of the question alternatives.

$$(40) \quad \lambda w. \neg \text{danced}(J, w) \wedge \neg \text{smoked}(A, w) \models \lambda w. \neg \text{danced}(J, w)$$

If, for example, we assume that the question *Who danced* presupposes that someone danced, then it follows that someone other than Jaden danced. Then, at least a disjunction of a proper subset of the question alternatives is entailed. If, however, we do not take the question to presuppose that someone danced, it is hard to show that (37d) or its negation license any entailment regarding the question alternatives. Since our discussion, is of course, limited by our decisions above regarding question alternatives, and since the intuitions are also not entirely clear, we leave it undecided whether the disjunction is at-issue or not, marked by the question marks in (38d).

Let us then use the example in (41) to check the predictions regarding projection. For this, we embed the conjunction under an entailment-cancelling operator such as *it is possible that*.

(41) A: Who danced?

B: It is possible that Jaden danced and Ashton smoked.

Applying the theory bluntly, the prediction is that it should project that *Ashton smoked*, it should not project that *Jaden danced*, and it should not project that *Ashton smoked and Jaden danced*. It is somewhat unclear whether the disjunction *Ashton smoked or Jaden danced* is also predicted to project as it is unclear as to whether it is at-issue. Importantly, it is not obvious that the inference that *Ashton smoked* (or the disjunction) projects in (41). Hence, we have an example of at least one not-at-issue inference that fails to project. We would then like to know why. Yet, an anonymous reviewer correctly points out that the acceptability of the answer in (41) is questionable altogether which may raise doubts on the conclusiveness of the example. However, the problems do not stop here: let us consider another example, (42), explicitly discussed in Simons et al. (2010).

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(42) Q: Why aren't you inviting any boys from your class to the party?

A: I don't like the boys in my class.

Here, according to Simons et al. (2010), the inference that there are boys in the speaker's class is in fact at-issue since the negation of this information, i.e. there are no boys in the class, would answer the QUD (in fact it would provide a complete answer to the QUD) in (42). Still, arguably, the inference that there are boys in the class projects in this case. Hence, we also have an example in which the at-issue inference projects.

The way Simons et al. (2010) address this question is by revising the definition of at-issueness somewhat. The revised definition is in (43):

- (43) a. A proposition *p* is at-issue iff the speaker intends to address the QUD via ?*p*.
 b. An intention to address the QUD via ?*p* is felicitous only if:
 i. ?*p* is relevant to the QUD, and
 ii. the speaker can reasonably expect the addressee to recognise this intention.

This definition leaves quite a bit of space for interpretation. However, it allows the following solution to the problem in (42). The proposition that there are boys in the class could in principle address the QUD but it is not intended to answer the QUD. There are in fact several reasons for this. One is that there is a more direct and more complete answer which is entailed: the speaker does not like the boys. In fact, if there were no boys in the class, this sentence would not even make sense. Another one is that a definite description is conventionally marked to not address the QUD by its existence presupposition (though examples such as (44) can easily be constructed, even if such examples seem to be somewhat rhetorically loaded because they, at least intuitively, require the hearer to use presupposition accommodation as a question-answering strategy):

(44) Q: Is Kim a bachelor?

A: Her wife is talking with Michael and Jada in the living room.

Returning to (41), it is not really obvious how the revised version of the theory in and of itself helps correct the prediction. After all, there is no reasonable way in which the inference that Ashton smoked could be taken to address the question asked in the previous turn. Hence, by any account, this inference will be not-at-issue and thus should project.

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At this point, a final complication comes into the picture: Simons et al. (2010) explicitly state, in the discussion of examples such as (45), that there are cases in which even inferences that are conventionally marked as being not-at-issue, such as epithets (cf. Potts 2005, Gutzmann 2012), do in fact sometimes provide the answer to the question explicitly asked in the dialogue:

- (45) Q: What do you think of Bailey?
A: I've never met the son-of-a-bitch.

In order to deal with such examples, a very important feature of the QUD needs to be considered. In particular, following Roberts (1996) and subsequent research, scholars have assumed that every single utterance in discourse answers a question. However, if one looks at naturally occurring discourse, quite obviously, questions are far less frequent than assertions. Hence, clearly, the question addressed by a speaker can also remain implicit. So, how can we know what the question under discussion is? The answer is that the way in which an utterance made by a speaker is performed, be it in terms of word order, syntax, prosody or even morphosyntax, can be indicative of the question the speaker had in mind – the so-called congruent question. Usually, scholars do not assume a one-to-one relation between the way in which assertions are articulated and the questions that they address. Still, given that part of the construction of the contextual representation is driven by general domain inference patterns in addition to QUD-marking, the situation cannot be generally deemed hopeless; see e.g. Roberts (2012), Onea (2016) and many others. Thus, we assume that in (45), B's answer is taken to address another question and regarding that question, namely (46), the proposition associated with the epithet is indeed not-at-issue and projects. However, with regard to the overt question, the situation is different – thus the impression that the conventional implicature of (45) is at-issue.

- (46) Have you ever met Bailey?

By analogy, then, we could say that the (41) case involves an example in which the apparently not-at-issue inference that Ashton smoked is in fact at-issue with regard to another question that the speaker B actually addresses. Consider, for example, that the actual discourse context contains a larger question, namely (47a). Crucially, when B answers, B does not only provide the answer to (47a) but also to the super-question, (47) by also addressing (47b), thus, making the non-projecting inference at-issue.

- (47) Who danced and who smoked?

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- a. Who danced?
- b. Who smoked?

This raises the final point in this discussion. We have suggested above that there are conventional means to signal the QUD addressed by an utterance. We will briefly consider some of them, mainly focus-marking (see Roberts 1996, Beaver & Clark 2008, Zimmermann & Onea 2011), i.e. the requirement that the QUD has to be focus-congruent to the answer. And thus, we can never assume a QUD which is not focus-congruent to the utterance given. We briefly explain this idea. If an overt question is present and a direct answer to this question is given, the focus alternative value of the answer needs to be a superset of the set of question alternatives. In other words, the answer needs to be focus-congruent to the question. This idea has been spelled out in different ways in various theoretical frameworks. Example (48) follows Rooth (1992) (see also Onea & Zimmermann 2019 for a soft introductory discussion of question answer congruence). In (48), we show how the so-called focus-semantic value $\llbracket \cdot \rrbracket^F$ of some assertion is a superset of the question-alternatives provided by the congruent question.

- (48) a. $\llbracket \text{Who danced?} \rrbracket = \{\lambda w. \text{danced}(x, w) \mid x \in C\}$
 b. $\llbracket [\text{Jaden}]_F \text{ danced} \rrbracket = \lambda w. \text{danced}(J, w)$
 c. $\llbracket [\text{Jaden}]_F \text{ danced} \rrbracket^F = \{\lambda w. \text{danced}(x, w) \mid x \in D_e\}$

Crucially, if we now return to (41), we realise that in fact this sentence can be pronounced with a variety of intonational patterns. The most natural one, it seems to us, is the one in (49), where the brackets mark prosodic and not merely syntactic units.

- (49) a. Who danced?
 b. It is possible that $\llbracket [\text{Jaden}]_F \text{ danced} \rrbracket$ and $\llbracket [\text{Ashton}]_{CT} [\text{smoked}]_F \rrbracket$

Here, the embedded disjunction is taken to involve two separate intonational phrases, each with its own information structural partition. In particular, the first phrase has a clear focus on Jaden. As we have already seen, this makes this part of the embedded clause focus-congruent to the immediately preceding question. Crucially, however, the second part has a contrastive topic intonation that is usually assumed, following Büring (2003), Rojas-Esponda (2015), Kamali & Krifka (2020), to signal the presence of a higher question within a discourse strategy. In this case the relevant super-question is the one in (50). This question now comes with two different strategies that could be used to answer it: the one in (50) and the one in (51). What makes the example somewhat odd is, indeed,

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that here the two strategies are mixed. The first part seems to follow the strategy in (51), whereas the second part seems to follow the strategy in (50). But notice that in the specific case in which there are only two individuals and two actions in the domain, the mixing of the strategies seems to be innocent.

- (50) Who did what?
 - a. What did Ashton do?
 - b. What did Jaden do?
- (51) Who did what?
 - a. Who danced?
 - b. Who smoked?

Importantly, now, *Ashton smoked* is at-issue with regard to both (50a) and (51b) and is thus not predicted to project. It conventionally signals that it addresses both questions.

Before concluding this section, we need to mention two additional issues. Firstly, we consider an example that shows the strength of the proposal, and secondly we consider a peculiarity of the theory.

Consider (52), from Beaver (2010). This example involves a factive verb that is usually taken to presuppose its clausal complement. However, the example nicely shows that the actual focus-background structure of the sentence determines whether projection obtains.

- (52) a. If the TA discovers that [your work is plagiarised]F, I will be [forced to notify the Dean]F.
- b. If the TA [discovers]F that your work is plagiarised, I will be [forced to notify the Dean]F.

The logic of the example is as follows: in (52a) the focus on the embedded clause signals that the question addressed, the QUD, is (53a). Now, this is surely not the only question that is being addressed, and sticking to classical theories of discourse structure with questions (be it in the version of Roberts 1996, Onea 2016 and most other theories), it will not be easy to pinpoint how exactly conditionals relate and should be integrated into question-answering strategies. However, whatever the correct answer to this problem may be, it should be clear that the question (53a) is expected to be a highly relevant discourse question if this intonation contour is chosen. Indeed, in this case, the embedded clause is both at-issue and does not project. As opposed to this, if the focus is on the main verb in the conditional clause, the question addressed appears to be a question like (53b). Here, the embedded clause is not-at-issue and will indeed project.

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- (53) a. What will the TA discover?
b. Will the TA discover that your work is plagiarised?

For classical theories of presupposition projection, it is not obvious that predicting this contrast could ever be possible without incorporating some notion of at-issueness. We will have a brief discussion in section (4) about this problem.

We now turn to the peculiarity of the theory that we indicated above without becoming more specific about: it may seem that the theory is circular or at least non-falsifiable. The worry goes like this: you are saying that that which projects is not at-issue. But how can I create a counter-example to your theory? I tried to create a case where something that appeared to be at-issue projected. Then you said, roughly speaking, that it projected because it was not *meant* to be at-issue. Then I gave an example in which something that was in fact not-at-issue did not project, and then you said: well that is the wrong question with regard to which I said that it is not-at-issue. It was in fact at-issue. In fact, you even said, because it is marked grammatically for not being at-issue it will project even if it directly answers the QUD! It is important to articulate such worries and address them head on.

There are several ways to address this problem: One perspective is a purely empirical one: is there a correlation between at-issueness and projection? If there is, the way in which the theory needs to be spelled out in detail can be further discussed, but at least the worry of circularity (which would amount to zero predictions) is out of the way. At least if we take the diagnostics used in Tonhauser et al. (2018) at face value, it seems that the answer to this question is clearly positive.

The second perspective is a more theoretical one: any theory that has empirical predictions will depend on some parameters. The more parameters a theory depends on (which can be fitted) the weaker the theory. For this theory, the crucial parameter seems to be an overall theory of question-answer congruence and of QUDs in discourse. For example, when we say that an NRRC or an epithet cannot be at-issue and will always need to project, we could mean that the compositional semantics of these items is simply not allowing focus to project in such a way as to make these congruent answers to a question. Hence, even if they appear in a sentence evidently uttered as an answer to a question with respect to which the respective utterance would be at-issue, we will still need to assume another QUD simply because otherwise the focus-congruence rule would be violated. Thus, in fact, it is – in many cases – impossible to construct a counter-example. But this is not because the theory is not falsifiable but because the theory is using the question-answer congruence rule which indepen-

dently rules out certain question-answer configurations. More importantly with respect to the purposes of this paper, while certain constructions seem to be inherently incompatible with focus, such as epithets or non-clause-final-NRRCs, others, such as clausal arguments of embedding verbs, are and thus do in fact provide an ideal testing ground for the theory. As suggested by the example in (53), here, the predictions are borne out.

Summing up the results of this section, we conclude that the at-issueness hypothesis is made explicit mainly by considering the focus-background structure of utterances, which determines the range of possible QUDs the speaker could be addressing. Once the QUD is established we can decide whether an entailment is at-issue or not. Once this is established, it is predicted that the not-at-issue inferences will project.

3.2 What is the explanandum?

As discussed above, the at-issueness theory of projection is mainly a generalisation and not yet an explanation. In fact, we can entirely accept the generalisation and yet not even agree on what is the explanandum for a theory of projection. Does a theory of projection need to explain the cases in which projection takes place or should a theory of projection rather explain those cases in which projection does not happen? It may come as a surprise that this is not a rhetorical question.

One possibility is that a projective content projects because it has a specific conventional projective property, i.e. terms such as *presupposition* or *conventional implicature*, or simply *potential to project*, are compositional semantic notions. When we say that this is a conventional property, we mean that it is determined by the linguistic system itself. By default, then, that which has the potential to project, projects. In this case, if there is anything to explain it is this: why is it that in some cases some of the instances of projective content fail to project? The answer then appears to be: projective content fails to project whenever it is at-issue.

The radically opposite view is this: there is no conventional difference between projective and non-projective content. These simply constitute inferences of a sentence and whether or not they end up projecting or not depends on something else – the primal candidate being their pragmatic status with regard to the question under discussion. Let us call this the pragmatic view. In this case, the question to answer is: why do some entailments project? The theory of Simons et al. (2010) hints at this, but this point never really gets substantiated in the literature: entailment-cancelling operators somehow fail to see content that is

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not at-issue. One possibility would be that entailment-cancelling operators do in fact compositionally interact with focus and it is then indeed at-issueness that literally drives the computation of projection. A recent proponent of a radically pragmatic theory of presuppositions, at least, is Simons (2001, 2007). One could take Simons et al. (2017) to also advocate such a theory, inasmuch as they discuss cases in which inferences associated with non-factive verbs such as *to believe* seem to project in certain pragmatic contexts. But we simply do not see how such a view could apply to conventional implicatures, which – for all extents and purposes – appear to be conventionally marked as projective. Moreover, the systematic differences of different types of projective content, as classified in Tonhauser et al. (2013) and experimentally shown in Tonhauser et al. (2018) also seem to be hard to explain on such an approach. If such a radically pragmatic perspective is to be a sustainable line of attack, in our view, it will have to be limited to presuppositions of so called soft triggers.

Notice, however, that both the pragmatic and the semantic view outlined above essentially take a causal approach. But an alternative is also conceivable. There might be a general pragmatic correlation between not-at-issueness and projection, but the correlation itself may not be causal in nature. Instead, there may be different explanations for the ways in which at-issueness and projection interact for different phenomena. Let us call this the hybrid view.

In what follows, we attempt to briefly explain how the hybrid view could be substantiated.⁶ Suppose the speaker has two thoughts that she wishes to express: the propositions in (54). Suppose, further, that she knows what question she wishes to address, i.e. one of the three options in (55):

- (54) a. George is an idiot beyond imagination.
- b. George is a notorious criminal.
- (55) a. Why is George a successful politician?
- b. Why did George not pass his exam?
- c. How is George?

By virtue of knowing which question is being addressed, the speaker knows which of the two propositions is at-issue, hence, the speaker can decide to express the other one as an appositive, thereby marking it as a conventional implicature, or to express them as a plain conjunction. The hybrid view now essentially boils down to suggesting that the projection itself may be a conventional property

⁶We gratefully acknowledge that parts of this idea goes back to a discussion with David Beaver, p.c. 2017.

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of the conventional implicature, the appositive, but one can only use such a structure if the corresponding proposition is not at-issue. This gives the options in (56).

- (56) Q: Why is George a successful politician?
A: George, an idiot beyond imagination, is a notorious criminal.
- (57) Q: Why did George not pass his exam?
A: George, a notorious criminal, is an idiot beyond imagination.
- (58) Q: How is George?
A: George is a notorious criminal and an idiot beyond imagination.

Hence, projection, at least in such cases, is not necessarily derived from not-at-issueness, but from the very structural properties of the appositives. Encoding something as an appositive of the kind in (56) is, however, a speaker-decision that is correlated to the speaker-decision to address a question such that the very same proposition ends up as not at-issue. A possible – intuitive – bridge is the notion of importance: what is at issue-should be more important and what is more important should be the main clause rather than the appositive. The perspective suggested above for appositives is very closely related to Ott & Onea (2015) and Beaver (2020), who think of appositives and conventional implicatures in general as independent speech acts of different types. We will not go into further detail of this idea, but we will briefly return to it in Section ???. What is important is rather that in this case, the explanandum is neither the projection (as facilitated by not-at-issueness), nor the lack of projection (blocked by at-issueness), but rather the speaker-level necessity to encode not-at-issue material as potentially projective, i.e. the correlation between not-at-issueness and projection.

The case discussed above involves two distinct, logically independent propositions and, thus, suggests that their encoding as potentially projective (in correlation with at-issueness) is clearly a decision problem of the speaker. But suppose what the speaker wants to express is not two independent thoughts, but something that is (at least somewhat) lexically constrained, i.e. one thought depends on another. An example would be, indeed, the thought that the TA discovers that the hearer cheated on the exam, as in example (52) above. In this case, the thought that the TA discovers *p*, cannot be divorced from *p* itself. While in the case of appositives, there was a structural asymmetry between the main clause and the appositive that may be seen as a marker of the potential to project, in the case of (factive) presuppositions, there seems to be a lexical/semantic asymmetry that is responsible for marking the potential to projection. This view seems to be a version of the theory proposed by Abrusán (2011, 2016).

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The core idea in Abrusán (2011) is that the *explanation* for why factive verbs presuppose is that the embedded proposition is not *about* the main event introduced by the factive verb. Her idea, inspired by research on vision, is that such entailments are not in the focus of attention and thus become presupposed. Abrusán (2011) is not interested in projection but in triggering of presuppositions. This paper is not the right venue to evaluate and discuss the triggering mechanism proposed by Abrusán in detail. Our point here is a simpler one: it appears possible to rephrase Abrusán (2011) as a variant of Simons et al. (2010) without losing any essential component of her theory. Abrusán (2011) claims that a presupposition of a factive verb is triggered *because* the complement of a factive verb is not about the main event of the matrix verb. But this suggests that the main event is by default more important than other events, which are not about it. Importance, however, can be spelled out in many ways. One prominent option is this: what is important should be at-issue. Then, the default question addressed by a sentence containing a factive verb is not about the truth of the embedded proposition. Say, if you want to know whether it is raining, you usually do not speak about people's attitudes toward this proposition. And if you speak about people's attitudes you are not usually primarily interested in whether the content of their attitudes is actually correct.

Crucially, then, on the hybrid view, there seems to be a similarity between conventional implicatures and presuppositions. In both cases, an asymmetry (albeit a different one) is responsible for their structural or semantic marking of potentially projective content and is also exploited by the speaker in marking what issue is being addressed, by making what is – intuitively – more important the answer to the QUD.

But there is a big difference as well. While appositives are independent of each other and the speaker may find it easy to encode either of the independent propositions as the appositive or the main clause, in the case of (factive) presuppositions, the lexically coded dependency between the two propositions cannot be changed by the speaker (at least not easily). Hence, if the proposition which is marked for the potential to project is intended to be at issue, the speaker may use a more economical device to achieve this goal: focus. The semantic structure marks one proposition as dependent on another but focus gives the hearer a handle on at-issueness, which has a direct impact on projection. Indeed, Abrusán (2011) observes that the triggering mechanism of factive presuppositions needs to be focus-sensitive, i.e. presuppositions for so called soft triggers, in our case factive verbs, are only ever generated if the sentence complement is not focused. This is because focus would act as an independent device that shifts the attention to another entailment. In Abrusán (2016) the idea is extended to projection and

cancellation phenomena, where she argues that while the cancellation of presuppositions for soft triggers is driven by focus, blocking their very triggering, in the case of some other presupposition triggers such as some discourse particles, presuppositions are always triggered and they invariably project – as also noted by Tonhauser et al. (2013).

Hence, it would seem that on a hybrid view focus could in some way be responsible for the non-projection of (factive) presuppositions, and in the case of conventional implicatures the structure itself determines what projects (but the choice of the structure is determined by at-issueness). It turns out, however, that, despite initial evidence, the situation is more complicated and focus itself cannot be the whole answer to the problem of presupposition projection even under the at-issueness hypothesis.

The overall idea that focus signals the question addressed by an utterance in a compositional way goes back to the seminal paper of Roberts (1996). However, Roberts (2012) forcefully argues that it is not only the focus-congruent question that plays a role in the interpretation of utterances but also communicative goals and general domain questions.

Importantly, this claim is not limited to contrastive topics, which are claimed to associate with entire discourse strategies, and to some discourse particles, such as German *überhaupt* as analysed in Rojas-Esponda (2015). Instead, Roberts (2012) explicitly wants to incorporate the effects of more remote domain questions that cannot interfere with the semantic composition, at least not directly. She argues:

The more general intentional structure of discourse outlined above also suggests a generalisation of the notion of Relevance to other goals and intentions, as well as to the QUD.

Roberts (2012) gives the following general example, where the patient's question is relevant to the goal proposed by the nurse's suggestion and not a QUD.

- (59) Nurse: Don't eat on the morning before your test.
 Patient: May I drink water?

With this in mind the question naturally arises whether at-issueness effects are limited to cases in which the QUD is focus-congruent to the relevant assertion. If remote, non-congruent QUDs still play a role, this has important consequences for the exact nature of the explanation of the role of at-issueness in projection. For if projection was in some way governed by focus at the level of compositional semantics, non-focus-congruent questions should not be responsible for projection-effects. Indeed, Simons et al. (2017) suggest that such effects may exist. They discuss the example in (60):

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(60) Polly: Why is it taking Phil so long to get back here?

Amy: [with sudden realisation] He doesn't know the car's parked in the garage!

The intended intonation in this case is with a pitch accent on the embedded clause. This allows for a range of various foci, given the fact that focus tends to project from deeply embedded constituents. Crucially, however, the verb *know* cannot be the sole focus in the example and it is by all means very likely that the entire embedded clause or parts of it will be interpreted as the focus. Thus the congruent question will be a question like the ones in (61):

(61) a. What does Phil know?

b. What does Phil not know?

c. etc.

Importantly, if this is the congruent question, i.e. the compositionally relevant QUD, the proposition that the car is in the garage can be seen as at-issue (though it depends somewhat on the exact analysis of the question). More importantly, Simons et al. (2017) argue that the reason why projection occurs in the example (60) is that without projection the answer would not be relevant to the overt question, the domain question. Thus, in determining whether something projects or not, it not only matters what the congruent question is but also what the domain question is. In fact they argue that the same line of reasoning could also lead to the prediction that not only *know* but also *believe* can embed projective content.

The fact that a domain question that is not focus-congruent can also be relevant to projection creates a serious problem for hybrid accounts. If a hybrid account is viable, it cannot be compositional since the domain question stands in no compositional relation with the relevant expression.

4 Pushing the limits of semantic theories

While the original hypothesis of Simons et al. (2010) would seem to suggest that all types of projective content work the same way, the classification in Tonhauser et al. (2013) suggests that in fact a systematic distinction between different types of projective content is reasonable. Moreover, we have suggested above that it is hard to imagine an entirely pragmatic explanation that handles conventional implicatures and presuppositions in the same way. Finally, we have also seen that a compositional interaction between focus and projection seems unlikely given

the observation that a) in the case of conventional implicatures focus seems to be irrelevant and b) QUDs that are not focus-congruent can also impact projection. Thus, in this section, we wish to consider how the at-issueness-projection hypothesis could relate to existing theories of projection both in the domain of presupposition and conventional implicature. Of course, given the huge range of literature on both topics, we will need to limit ourselves to a small selection of ideas which we deem particularly interesting for future research.

In the domain of presupposition projection, we wish to make two points. Firstly, we wish to suggest that the at-issueness theory could in fact provide a missing element in an old projection theory developed in Gazdar (1979). Secondly, we will discuss how this idea could significantly improve on the binding theory of presuppositions developed by van der Sandt (1992). In a final subsection, we briefly connect the theory of appositives and non-restrictive relative clauses from a QUD perspective with extant literature.

4.1 QUD and presupposition cancellation

One of the first truly predictive formal theories of presupposition projection was developed in Gazdar (1979). Gazdar famously proposes a fairly complicated system that manages all embedded inferences generated by an embedded clause. He distinguishes potential implicatures (im-plicatures) and potential presuppositions (pre-suppositions) from actual implicatures and actual presuppositions utterances may have. Pre-suppositions are triggered locally in some potentially deeply embedded environment. A filtering mechanism is used to decide which of the locally generated inferences will eventually survive and become true presuppositions.⁷ One remarkable rule of the filtering mechanism that was deemed empirically necessary but which does not appear very intuitive is important: if the sentence has a potential presupposition that contradicts an implicature, the presupposition ‘loses’ and is filtered out, while the implicature remains.

A well-known example is the filtering with conditionals, as in (62). Ever since at least Karttunen (1977), Karttunen & Peters (1979), it is received knowledge that if the antecedent of a conditional entails the presupposition of the consequent, this presupposition will not project. Otherwise, presuppositions of the consequent of the conditional project. Thus, both presuppositions in (63) are expected to project, whereas in (62), none do.

- (62) a. If Justice has kids, Justice’s kids will be linguists.

⁷For the details of the theory we have to refer to (Gazdar 1979) and general textbooks on pragmatics, e.g. (Levinson 1983, 2000, Huang 2014).

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- b. If Mary loves Jada, Michael surely found out that she does.
- (63) a. If Justice decides to learn even just one more language, Justice's kids will be linguists.
- b. If Mary keeps on talking, Michael surely finds out that she loves Jada.

According to Gazdar (1979), a conditional implicates that it is unknown to the speaker whether the antecedent is true. In (62) there is, hence, an implicature that the speaker does not know whether Justice has kids and Mary loves Jada respectively. Hence, for (62), the presupposition of the consequence will be cancelled, because it contradicts the ignorance implicature. As opposed to this, the ignorance inference associated with (63) does not contradict the presupposition of the consequent and thus the pre-supposition of the consequent can become a true presupposition.

The theory proposed in Gazdar (1979) has its obvious historical merits but also some well-known issues we will not discuss here. However, even in introductions to pragmatics, students usually find it mysterious that implicatures would win against presuppositions – are they not supposed to be weaker inferences? It seems to us that the at-issueness hypothesis could help clarify that intuition in some way.

From a communicative perspective, presupposition projection does not amount to presupposed propositions becoming new information but rather to a requirement on the common ground. If *p* becomes a presupposition, it will be assumed that it is common ground that *p* is true. But if an inference is at-issue, it cannot be common ground and vice versa.⁸ An ignorance implicature of the sort exemplified above would seem to obtain precisely in cases in which the inference the speaker is ignorant about is at-issue and, thereby, naturally blocks it from becoming a presupposition. Indeed, it has been shown in erotetic logics (Wiśniewski 1995) that the question about the truth of the antecedent of a conditional is often an erotetic inference, and thus a natural follow-up question to a conditional statement. Put this way, the theory of at-issueness could in fact be viewed as a powerful update of an old idea by Gazdar, by integrating it with erotetic theories of discourse.

⁸While this does not immediately follow from the theory of questions and answers sketched above, the theory can easily be accommodated to guarantee this. One way to do this, see e.g. Onea (2016), is to intersect all question alternatives with the context set (Stalnaker 1978). Thus, each question alternative entails the context set and thus information that is in the context set cannot be relevant to the question, i.e. cannot be at-issue.

4.2 QUD and binding of presuppositions

Another classical theory of presuppositions that exhibits a remarkable, and as far as we know hitherto unknown, connection to the at-issueness theory of presupposition projection is the binding theory of presuppositions. Van der Sandt (1992) suggests that there is a deep connection between anaphora and presupposition. The core observation of van der Sandt (1992) is that, in the very same way dynamic semantics models the anaphoric binding of a pronoun, the satisfaction of presuppositions can be modelled. In particular, they distinguish two main types of presupposition satisfaction: binding and accommodation. A presupposition is bound if in discourse an explicitly represented antecedent can be found, and a presupposition is accommodated if binding is not possible. For example, in (64a) the first sentence acts as an anaphoric antecedent and thus binds the presupposition of the second sentence. As opposed to this, in the second case, there is no binder available, and thus the presupposition needs to be accommodated. Importantly, whenever binding is an option, we no longer need to accommodate.

- (64) a. Justice has kids. If his kids are happy, he is happy.
 b. Justice is a simple guy. If his kids are happy, he is happy.

Thus, accommodation is governed by a global geometry constraint intrinsic to DRT (Kamp 1981); a current formulation is given in Kamp et al. (2011). In particular, DRT has a hierarchic system of discourse representation, such that a presupposition starts out as a designated entailment at the lowest possible starting location and floats upwards, i.e. tries to project in search of an antecedent to bind. If binding is successful the process stops. This also explains the filtering function of, e.g. conditional antecedents, as within the DRT-architecture used by van der Sandt (1992) and Geurts & van der Sandt (1999), conditional antecedents are anaphorically accessible from the consequence – which famously explains donkey anaphora. If the attempt to find an appropriate binder is unsuccessful, the presupposition floats to the global level and then tries to be accommodated from there, in the highest slot in which a contradiction does not emerge. If such contradictions eventually would lead to contradictions in all projection sites, the final option is always local accommodation which oftentimes means no projection whatsoever, as the presupposition is limited to a local effect in this case. We illustrate the idea in example (65). In (65a) no contradiction arises, and thus we can globally accommodate, thus project, the presupposition that Justice has a car. In (65b), global accommodation is blocked by the information that the speaker does not know whether Justice has a car, and thus only local accommodation

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remains an option, hence yielding: If Justice has a car and Mary discovers that he has one ...

- (65) a. If Mary discovers that Justice has a car, he will have to drive her home.
- b. I don't know whether Justice has a car. But if Mary discovers that he has one, he will have to drive her home.

Now, the binding theory of presuppositions has been extended to a theory of focus as a presupposition-inducing device in itself; see Geurts & van der Sandt (2004). However, the theory has absolutely no obvious built-in means to deal with the variability captured by the at-issueness theory of presupposition projection. In fact, DRT itself is not particularly well suited for question-semantics and question-answer patterns never have been a main topic of investigation in DRT to start with.

However, at least a very rudimentary theory of questions can be given. Take Hamblin-question-alternatives and represent them as individual propositions marked in some way such as to avoid taking them as inferences. If we now assume that question-alternatives are available discourse referents and thus possible binders for presuppositions, we end up with a surprising prediction: whenever a presupposition is at-issue in the sense that it could be anaphorically bound to a question-alternative, it will be bound to a question-alternative and not accommodated globally. Since question-alternatives are mere possibilities and do not license global entailments, it follows that at-issue inferences do not project. Not-at-issue inferences, on the contrary, need to project. We summarise this idea in terms of a simplified but respectable unified theory of at-issueness and presupposition binding.

- (66) A unified theory of at-issueness and presupposition binding
 - a. All presuppositions need to be bound or accommodated.
 - b. Binding has priority over accommodation.
 - c. Question-alternatives are possible binders.

This theory now comes with a remarkable prediction: suppose that question-alternatives disappear from a discourse representation structure (DRS) once the question is resolved. Suppose, further, that all open questions in the QUD-stack in the sense of Roberts (1996) are represented in the global DRS. Then, it is immediately predicted that not only the immediate focus-congruent question but also the entire question strategy, and thus the domain question, are responsible

for determining the at-issue status and thus the projection of potential presuppositions.

Let us return to the example (65a). Here, the presupposition that Justice has a car needs to be bound somewhere or become accommodated. But where it can be bound depends on what questions we encounter. Suppose the QUD contains the open question (67a). This does not simply mean that it is unknown whether Justice has a car (which would not block accommodation) but that this question has been recognised as relevant for and thus as part of the discourse strategy at hand. In this case, the presupposition is bound to the positive alternative of the question and since the alternative to which it is bound is not entailed, the presupposition is not entailed either. But suppose the question is (67b) and (67a) is not part of the QUD stack. Now, there is no antecedent for the presupposition to be bound to. Hence, accommodation obtains.

- (67) a. Does Justice have a car?
 b. Under which conditions does Justice have to drive Mary home?

Of course, we do not suggest this as a fully worked theory, but rather as a fairly obvious and hitherto unexplored path for future research. Such a theory could finally have both the merits of the at-issue-projection hypothesis and a true-explanative power as presupposition projection would be tied to an independently motivated and needed mechanism of anaphoric binding.

4.3 QUD and non-restrictive relative clauses

Finally, in this very brief section, we turn to the way in which existing theories of appositives can relate to the at-issueness-based projection hypothesis for appositives and non-restrictive relative clauses. Here, the main question to be asked is this: are for example non-restrictive relative clauses actually syntactically embedded at all?

Both theoretical options have been proposed in the literature and while there may be some debate about what the nature of syntactic dependence and independence means, the view that non-restrictive relative clauses and appositives are generated deep in the syntactic structure of the main clause is a well-established view advocated by Potts (2005) and subsequent literature, e.g., AnderBois et al. (2015). However, a larger portion of the literature going back to McCawley (1988), Gobbo (2003), Ott & Onea (2015), Ott (2016), Onea & Ott (2022) and many others (see Schlenker 2020 for a recent overview) suggests that these items are in fact, at least in many cases, generated high in the syntactic structure or as independent speech acts only linearly interposed into the structure of the host clause.

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With this in mind, under such theories predicting projection turns out to be a simple exercise, for they are not embedded in the first place under any operator that could in fact cancel them as entailments. As already argued, however, there is still truth to the observation that these items are in fact not-at-issue. It just so happens that projection, in this case, is not a puzzle. The reason why a speaker may have chosen to use a construction that needs to be globally entailed can still be not-at-issueness. But this is only a tendency. Thus, not surprisingly, Onea & Ott (2022) argue that in fact, at least for nominal appositives, there is a systematic type of examples which is usually at-issue and in fact does not project.

5 Conclusion and outlook

With respect to clausal embedding, the most interesting property of presuppositions and conventional implicatures is their projective behaviour. Even if their triggers are embedded deep in the sentence structure, they can emerge as independent truth-conditional or expressive content that is not affected by the embedding linguistic material. This raises the issue of whether projective phenomena can be understood in a uniform way. Questions under discussion and the associated notion of at-issueness are core ingredients of newer approaches that try to achieve such a uniform understanding. This chapter introduced the main ideas and discussed the conceptual and empirical problems that arise when one tries to spell out in detail the meaning of at-issueness and the mechanics of projection. Most theories assume that sentences are subordinated to a focus-congruent question that is again subordinated to higher discourse structuring questions (see, for example, Klein & von Stutterheim 1987, van Kuppevelt 1995, Roberts 1996; see also Benz & Jasinskaja 2017). Hence, progress about the questions whether at-issueness can be defined in terms of superordinated questions and whether, in turn, projective behaviour is conditioned by at-issueness can be expected from annotated corpora. Some work has been done in this direction (in particular Kuthy et al. 2018, Riester 2019, Westera et al. 2020, Hesse et al. 2022); however, there is still work to be done before the theoretical questions can be answered.

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Chapter 9

Embedding in discourse

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The chapter addresses the question of what constitutes embedding in discourse and argues that discourse-structural subordination is a case in point. The concept of discourse-structural subordination is discussed both from the viewpoint of different theories applying the notion of “rhetorical” or “discourse relations”, as well as in goal-driven approaches to discourse structure. The chapter also provides an overview of how discourse-structural embedding is realized in language with an emphasis on the relationship between discourse-structural and syntactic subordination. Finally, the notion of discourse subordination is revisited from the viewpoint of the semantic and pragmatic processes that may lead to the establishment of a subordinating relation between discourse segments.

1 Introduction

This chapter addresses the question of whether the notion of embedding is relevant in discourse structure and if so, what constitutes embedding in discourse. We will argue that *discourse-structural subordination* is a case in point, discussing this concept both from the viewpoint of different theories applying the notion of rhetorical or discourse relations, as well as in goal-driven approaches to discourse structure. Furthermore, we will provide an overview of how discourse-structural embedding is realized in language with an emphasis on the relationship between discourse-structural and syntactic subordination. Finally, we will



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revisit the notion of discourse subordination from the viewpoint of the semantic and pragmatic processes that may be taken to lead to the establishment of a subordinating relation between discourse segments.

The chapter is structured as follows. In Section 2 we first introduce the notions of rhetorical/discourse relations and the distinction between coordinating and subordinating discourse relations before providing an overview of theories of discourse structure that contribute an implicit or explicit notion of discourse-structural subordination. Section 3 discusses parallels and mismatches between embedding in syntax and discourse, while Section 4 is devoted to semantic and pragmatic mechanisms that establish subordination in discourse. Section 5 provides a summary and points at further related issues.

2 Discourse-structural subordination

While there is no widely accepted definition of discourse-structural subordination, one of the most prominent concepts related to subordination in discourse is the notion of subordinating discourse relations. In what follows, we first introduce the distinction between subordinating and coordinating discourse relations, tracking down its deployment throughout some of the most prominent theories of discourse that utilize the notion of rhetorical or discourse relations in an attempt to pin down the exact notion of discourse subordination that these theories implicitly or explicitly provide. We then explore the view on discourse subordination that emerges from a second set of theories of discourse structure that regard discourse goals or questions under discussion as the central organizing principle in discourse. The section ends with a summary and a discussion of more recent work providing comparison or integration of the two sets of approaches to discourse structure.

2.1 Subordinating vs. coordinating discourse relations

In one of the first influential approaches to the study of discourse, Rhetorical Structure Theory (RST; Mann & Thompson 1988), *discourse* (or *rhetorical, coherence*) *relations*¹ are defined on the basis of the following definition of discourse coherence:

¹The use of these terms varies across theoretical approaches. While RST uses the notion of “rhetorical relations”, Hobbs (1985) speaks of “coherence relations”, and later work like Asher & Lascarides (2003) and Asher & Vieu (2005) uses the term “discourse relations”. Throughout this chapter, we will use “discourse relations” as a general term for the purpose of discussing general issues, and “rhetorical” and “coherence relations” when we deal with the respective theory.

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... For every part of a coherent text, there is some function, some plausible reason for its presence, evident to readers, and furthermore, there is no sense that some parts are somehow missing. RST focuses on the first part—an evident role for every part.

(Mann, RST website, <http://www.sfu.ca/rst/>)

A rhetorical relation is thus a pragmatic function that an utterance or a larger part of a text fulfills with respect to another. For instance, *Elaboration* holds between two discourse units that describe the same state of affairs at different levels of abstraction (e.g. Hobbs 1979, 1985, Kehler 2002), see (1). *Parallel* holds between discourse units in virtue of the similarity or uniformity of their content along some relevant dimension (Hobbs 1985, Kehler 2002, Asher & Lascarides 2003), as in (2), where the similarity condition is satisfied by the predication of the same action, *setting*, of the two data structures (“stack A” and “link variable P”). Contrast connects discourse units whose content is “opposite” or “contradictory” in some respect, and is typically marked by the connective *but*, see (3).²

- (1) Elaboration:
 Go down First Street.
 Just follow First Street three blocks to A Street. (Hobbs 1985: 18)
- (2) Parallel:
 Set stack A empty and
 set link variable P to T. (Hobbs 1985: 15)
- (3) Contrast: John is tall, but Bill is short. (Lakoff 1971: 133)

The theory of Hobbs (1985: 25) was one of the first in which what he called *coherence relations* was divided into two categories of relations, coordinating and subordinating. However, Hobbs didn’t relate these notions to discourse-structural embedding. Rather, such a distinction is suggested as an approach to the problem of identifying the content of larger discourse segments composed of single clauses: when two segments are linked by some relation, they compose to a single unit which is then linked to a new discourse unit by means of a discourse relation. Hobbs contends that in order to be able to apply his definitions of coherence relations to discourse units larger than a single clause, one needs to be able

²For a more comprehensive list and informal definitions of coherence relations, see, for instance, Zeevat (2011), Jasinskaja & Karagjosova (2020), as well as the catalogue of relations on the RST-web site (<https://www.sfu.ca/rst/01intro/definitions.html>) and the glossary of SDRT-relations in Asher & Lascarides (2003: 459–471).

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to say “what is asserted by those segments” in terms of a joint, overall content, a “kind of summary” or “discourse topic” (ibid.: 25). This requirement follows from the way Hobbs defines coherence relations in terms of inferences drawn by the listener on the basis of the content of the segments and the knowledge of speaker and hearer. Without computing a joint assertion for a composed discourse unit, there is no way in which a unit can be attached to it by means of a coherence relation. The difference between coordinating and subordinating relations is then described as follows. “To recognize a coordinating relation, one must generally discover some common proposition inferable from each segment”. When one of the two segments is subordinated to the other, however, “the assertion of the composed segment is the assertion of the dominant segment” (ibid.). According to Hobbs, in a Contrast relation S_0 , *but* S_1 , where S_0 and S_1 are discourse segments, one (usually the second) segment is informationally more dominant with respect to the overall discourse topic of S_0 and S_1 . Thus, in spite of the structural parallelism displayed in (3), applying Hobbs’ informational dominance criterion would lead to characterizing the relation between S_0 and S_1 as subordinating. An Explanation relation S_0 , *because* S_1 receives a similar treatment, based on the argument that the first segment determines the content (discourse topic) of the entire complex segment (ibid.). In contrast, an Elaboration relation is coordinating according to Hobbs, since the two segments contribute a common proposition as the assertion of the composed segment (ibid.). For instance, the common proposition for (1) would be that the addressee should go three blocks down First Street until she reaches A Street. Similarly, Parallel involves the inference of a common proposition which counts as the assertion of the composed segment. Thus, from each of the clauses in (2) one can infer that a data structure is being set to a value (ibid: 15). From this point of view, subordination is defined in terms of informational dominance, and relations like Contrast, Explanation and Violated expectation count as subordinating, whereas Elaboration and Parallel as coordinating.

At the same time, Hobbs (1985) provides a classification of coherence relations based on four aspects of the discourse situation – (i) relations reflecting real world coherence between events (such as Cause), (ii) relations connecting discourse segments to conversational goals (Evaluation), (iii) relations connecting segments to the listener’s prior knowledge (Background, Explanation), and (iv) inferential relations between discourse segments (including Parallel, Elaboration, Contrast and Violated expectation). This classification cuts across the above coordination/subordination-distinction. Thus, Hobbs’ fourth class which he calls “expansion relations”, involves relations that “expand the discourse in place”, “rather than carrying it forward and filling the background” the way relations

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like Cause, Explanation and Background do. Thus, whereas some expansion relations like Parallel and Elaboration are coordinating, others, such as Contrast and Violated expectation, are categorized as subordinating, just like non-expansion relations like Background and Explanation. Hobbs' distinction between expansion relations and relations that carry the discourse forward comes closer to a structural definition of/view on discourse subordination, and, together with his subordination/coordination distinction, has significantly influenced subsequent work on the notion of subordination in discourse that we discuss below.

Let us however first turn to the rather functionally motivated division between two types of relations that is put forward in Rhetorical Structure Theory (RST; Mann & Thompson 1983, Mann & Thompson 1987, Mann & Thompson 1988). In general, RST is characterized by Mann & Thompson (1988: 243) as a "theory of text organization" that "identifies hierarchical structure in text" in describing "the relations between text parts in functional terms", with the aim to provide a "general way to describe the relations among clauses in a text" (ibid., 244). More specifically, RST views discourse as structured along *rhetorical relations* between non-overlapping text spans, where a text span is a contiguous linear interval of text, that is, a clause or a portion of text with an RST structure. Only one single relation may hold between two text spans. Relations are defined in terms of the writer's intentions and the effect that text spans are attempted to achieve with respect to the reader's attitudes (knowledge, beliefs, desires) on the subject matter described by the text spans, which in many respects is comparable to Hobbs' inferential view on discourse coherence. In contrast to Hobbs, however, segments in RST are linked to a larger segment without the requirement of computing a common semantic core of the segments constituting the larger segment. Furthermore, within a particular RST relation, a text span serves as either a Nucleus or a Satellite. This distinction is based on the observation that the relations are mostly asymmetric. For instance, in an *Evidence* relation, if it holds that A is evidence for B, it does not hold that B is evidence for A (ibid., 266). Nuclearity is taken to be a central principle of the hierarchically organized discourse structure that is represented as an RST-tree, as well as a central notion for a theory of text structure (ibid., 268). It is defined in functional terms: the Nuclei are comprehensible without the Satellite, are not replaceable and more essential to the writer's purpose, whereas Satellites are incomprehensible without the Nucleus and replaceable by other material. In addition, whereas a Satellite supports the purpose of the Nucleus, it is not strictly necessary for the coherence of the text. Thus if "we delete all units that function as nuclei anywhere in the text, the result should be incoherent and the central message difficult or impossible to comprehend" (Mann & Thompson 1987: 32). On the other hand, deleting units that only function as

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Satellites in a text should not have an effect on its coherence, and the result is a synopsis of the original text.³

Based on the distinction between Nucleus and Satellite two principally different types of relations are assumed. While most relations are of the type Nucleus-Satellite, a few relations are Multinuclear, with no Satellite. This is illustrated in Figure 1, where the diagram to the left represents the Nucleus-Satellite relation Circumstance and the diagram to the right represents the Contrast relation. Graphically, the Nucleus is represented by a vertical (or oblique) line, and the arrow visualizes the relation. In a Circumstance relation, the Satellite unit “sets a framework, for instance, a temporal or spacial framework, within which to interpret the nucleus” (ibid.: 48), as in (4), where the Satellite in 2 provides the temporal framework within which the Nucleus in 1 should be interpreted. On the other hand, a multinuclear Contrast relation connects two Nuclei each of which has an independent standing in discourse, see (3).

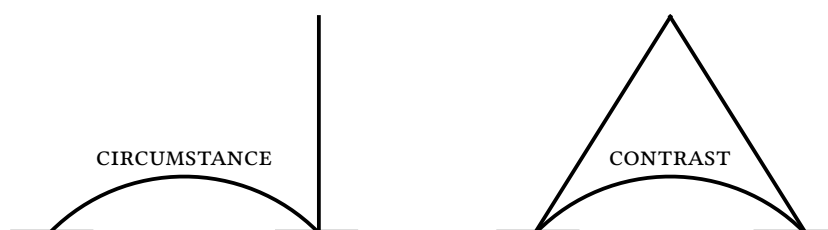


Figure 1: Nucleus-Satellite (left) and Multinuclear (right) RST-structure

- (4) a. 1. Probably the most extreme case of Visitors Fever I have ever witnessed was a few summers ago ...
b. 2. ...when I visited relatives in the Midwest. (Mann & Thompson 1987: 48)

The notion of nuclearity in RST is thus one way of looking at discourse subordination or embedding in discourse: a Satellite is a subordinated text span that is embedded into a Nucleus-span. As a matter of fact, Mann and Thompson draw

³Marcu (1997, 2000) uses this functional distinction between Nucleus and Satellite units for the task of automatic text summarization. As an anonymous reviewer noted, however, one should obviously only delete atomic Satellite units in order to get a synopsis of a text. Deleting a larger Satellite unit in an RST-tree will also involve the deletion of all nuclear units that are part of the complex Satellite.

an explicit link between nuclearity and syntactic embedding, stating that nuclearity is grammaticalized in hypotaxis (Mann & Thompson 1988: 270), as examples like (4) suggest, where the main clause represents the Nucleus and the subordinate one the Satellite. However, as Matthiessen & Thompson (1988: 290) argue, nuclearity and hypotaxis are clearly distinct notions, as there are many Nucleus-Satellite relations which do not involve hypotaxis, instead being realized by mere juxtaposition of clauses, see for instance (1).

Comparing the notion of nuclearity with Hobbs' subordinating coherence relations (and his class of expanding coherence relations), the two seem to be orthogonal. Since a Nucleus is comprehensible without a Satellite, the Nucleus must represent the informationally dominant text unit. Nevertheless, while Hobbs classifies Contrast as a subordinating relation in which one of the spans is informationally subordinate and thus not contributing to the overall content of the discourse units connected by it, Contrast is a multinuclear relation in RST, where both spans are essential to the writer's purpose and to the overall "message" of the complex discourse unit. Similarly, whereas RST conceives of Elaboration as a Nuclear-Satellite relation where the elaborating Satellite does not contribute to a summary of the overall discourse content, Hobbs views it as a coordinating relation since both segments would contribute their joint content to the overall content of the complex unit. Other relations like Violated expectation and Background, which Hobbs considers subordinating, are of the Nucleus-Satellite type in RST. Finally, whereas Hobbs' expansion relations Parallel⁴ and Contrast are multinuclear in RST, others, like Explanation and Violated expectation, are Nucleus-Satellite relations in RST.

A distinction between two kinds of relations between discourse units, coordinating and subordinating, is also employed within the Linguistic Discourse Model (LDM) (Polanyi 1988). At the same time, LDM offers a concept of subordination in discourse that is broader than the distinction between Nucleus and Satellite in RST (cf. also Blühdorn 2008), or Hobbs' notion of subordinating relations, as LDM views discourse structure as emerging from both structural and semantic relationships between units of discourse. Thus, a central premise on which LDM is based, is the view on discourse structure as resulting from the process of "recursive sequencing and embedding of discourse units of various types" (Polanyi 1988: 603). The hierarchical structure that results from this process has the form of a Discourse Parse Tree built up "on a left to right clause by clause basis" (ibid.). The basic unit of discourse formation in LDM is the *discourse constituent unit* (henceforth, DCU), which in turn is composed by clauses

⁴The Parallel relation corresponds roughly to the original RST concept of a Joint, or, in later RST developments, to the relations List and Conjunction, see Stede et al. (2017).

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(where an individual clause is considered an elemental DCU). Four main types of DCUs are distinguished, partly drawing upon Hobbs' dichotomy between relations that locally expand the discourse and those that carry it forward, at the same time relating the former to the notion of structural subordination. Thus, Sequences are formed by conjoining arbitrarily many constituent elements of the same type, where the elements are considered to be coordinated with each other, with sub-types like List and Narrative.⁵ Expansions, on the other hand, are described as DCUs composed of a unit and an immediately following unit that in some way expands the content of the former unit. Here, Polanyi draws on Hobbs' class of expansion relations. However, she treats the expanding units uniformly as units that are subordinated to the unit they expand in terms of structural dominance in the Discourse Parse Tree. Polanyi (1988: 609) explicitly adopts Hobbs' definition of his (expansion but coordinating) *Elaboration* relation to formalize this type of unit. Thus, subordination in LDM is on the one hand defined in semantic terms as subordinated DCUs that expand the propositional content of the immediately preceding clause in discourse. On the other hand, it is also defined in terms of structural attachment: When processing discourse and building the Discourse Parse Tree on a left-to-right clause-by-clause basis, phenomena like elaborations, digressions and interruptions that interrupt the completion of an ongoing discourse unit are treated as subordinated or embedded. This sets them apart from units that "continue the development of an ongoing unit", as in the case of stories or proposals for a course of action (*ibid.*, 611). This distinction is implemented in terms of differently labelled nodes indicating that a coordination or subordination relation obtains between an incoming unit and an accessible node in the tree. In Figure 2, the left hand structure represents a "coordination node" C with two clauses *a* and *b* that are coordinated with each other. In the "subordination node" S on the right hand, however, the *a* clause is superordinate to or embeds the *b* clause.

The structural attachment of new information in discourse is governed by what later became known as the Right Frontier Constraint (Asher & Lascarides 2003), postulating that only the nodes at the right frontier of the discourse tree are accessible for attachment of new discourse units, see Figure 3 which shows the accessible ("open") and inaccessible ("closed") nodes in the Discourse Parse Tree. The "right frontier" metaphor is based on the way discourse structure is represented in LDM as a graph constructed from left to right as discourse is

⁵Note that the List relation corresponds to Hobbs' coordinating Parallel relation which however belongs to his class of expansion relations. These LDM units are moreover described in terms similar to corresponding coordinating SDRT relations such as *List* and *Narration*, see the paragraph on SDRT below.

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Figure 2: Coordinated and subordinated structures in a Discourse Parse Tree (Polanyi 1988: 612)

processed incrementally clause by clause (Polanyi 1988).⁶ On the other hand, the

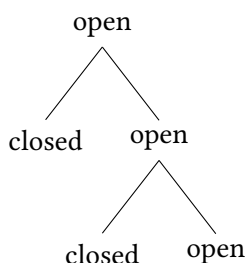


Figure 3: Open and closed nodes in a Discourse Parse Tree (Polanyi 1988: 613)

decision of whether to attach new units in terms of subordination or coordination is made on the basis of computing the semantic content of the units (ibid.: 617), a process which is not further specified but which involves inferences from world knowledge and linguistic knowledge.⁷

Subordination in terms of structural attachment is most prominently dealt with in the framework of Segmented Discourse Representation Theory (SDRT; Asher 1993, Asher & Lascarides 2003, Lascarides & Asher 2007), a formal theory of the representation and interpretation of discourse that accounts for the hierarchical organization of discourse, relying on the concept of discourse relations between text spans. Just like in RST, each text segment in SDRT is connected to

⁶Cf. Polanyi (1988: 613): “A new constituent is attached to the Discourse Parse Tree as the rightmost constituent at a structurally accessible existing level in the Tree.”

⁷This process is described in terms similar to the process involved in recognizing coherence relations described in Hobbs (1985) and the processing of utterance content in Asher & Lascarides (2003), as well as the derivation of a particular relation (dominance or satisfaction-precedence) holding between two DSPs in Grosz & Sidner’s model, see the paragraph below as well as Section 2.2.1.

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another via discourse relations with no overlap between text segments. Unlike RST, however, SDRT does not address speakers' intentions. Instead, discourse relations are defined in terms of their semantic effects on the interpretation of discourse. Thus, the relation *Explanation* between a segment π_2 and its predecessor π_1 is defined in terms of its indicating that the event described by π_2 causes the one described by π_1 . Discourse relations are derived from the utterance content (e.g. lexical semantics, connectives) and (non-linguistic) domain knowledge by means of commonsense reasoning defined in terms of a so-called glue logic.⁸ More than one relation can obtain between two segments, and one text segment can relate to more than one other text segment. Similar to the structure of discourse envisaged by Hobbs, SDRT allows complex discourse units that can function as a single complex node in a relation to another node, where in order to be able to infer a particular discourse relation, the common content of the two units within the complex node needs to be computed. (How this is done is, however, an issue that, to our knowledge, has not been explicitly addressed in the SDRT literature, while Hobbs (1985: 26) envisages "rules for assigning assertions to larger segments of discourse" like the ones mentioned for examples (1)-(3) above).⁹ In SDRT, discourse relations are classified into two categories, coordinating or subordinating, based on criteria like the availability of structural attachment points and anaphoric accessibility. This distinction is based on the observation that the hierarchical structure imposed by discourse relations constrains the interpretation of anaphora by restricting the parts of discourse to which new information is attached (Asher & Lascarides 2003: 146, see also Section 3.1.2), thus offering a similar structural notion of discourse subordination as in Polanyi 1988. Thus, Asher (1993) and Asher & Lascarides (2003) assume that discourse relations like *Elaboration*, *Explanation* and *Source* induce subordination in discourse segmentation and are therefore regarded as subordinating, whereas *Parallel*, *Contrast*, *Narration* and *Result* induce coordination and count as coordinating.

Subordination is at the same time defined in semantic terms: "[α subordinates β] iff the main eventuality described in β is a subsort of the main eventuality

⁸Glue logic may be briefly characterized as a logic for constructing the logical forms of discourse, supplemented by a separate logic for representing and interpreting the logical forms called the logic of information content, see Lascarides & Asher (2007).

⁹Note that in SDRT, a notion of discourse topic plays a role for some relations such as *Narration* and *Background*: They require a common topic summarizing the content of the two related constituents, cf. Asher & Lascarides (2003: 164) where *Narration* is argued to hold between two discourse units only if there is some information that summarizes the two units such that it is a topic for the two, and where topic is defined as a subordinating relation, rendering at the same time the two units as subordinated to their common topic.

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described in α or the proposition associated with β defeasibly implies that associated with α .” (Asher 1993). Semantically, subordinated discourse units are assumed to provide a more detailed description of some aspect of the superordinated unit, thus introducing a “new level of detail in discourse” (Asher & Lascarides 2003: 8). In this regard, SDRT draws on Hobbs’s (1985: 14) class of expansion relations.¹⁰

In their attempt to sharpen the distinction between coordinating and subordinating discourse relations, Asher & Vieu (2005) argue that a discourse relation is not coordinating or subordinating by virtue of its semantic content but rather in terms of how it is “presented in the discourse”, that is, the two notions are a feature of the structure of the discourse representation (*ibid.*, 594, 606). Thus, although Result is coordinating by default, this default can be overridden in specific contexts, whereby it may come out as a subordinating relation. Based on such observations, they argue for the need of a systematic way of distinguishing between coordinating and subordinating discourse relations, suggesting the following general test (next to 3 more specific ones) that can help categorizing relations in terms of this distinction (where $\langle \alpha, \beta, \gamma \rangle$ are discourse units and $R_1(\alpha, \beta)$ is a discourse relation already established between α and β):

- (5) If you can attach some γ to α , then R_1 is subordinating. If you can attach only to β , R_1 is coordinating. Asher & Vieu (2005: 600)

Based on this test, the difference between a coordinating relation like *Parallel* and a subordinating relation like *Source* can be captured in terms of different attachment possibilities of a new discourse unit to an existing unit. Thus in (6), the new unit (6c) cannot be attached to (6a), which indicates that the *Parallel* relation between segments (6a) and (6b) is coordinating, whereas in (7), (7c) can be attached to (7a), which makes the relation *Source* between (7a) and (7b) subordinating.

- (6) *Parallel*, a coordinating relation: cf. Figure 4
 a. John broke the vase.
 b. Bill broke the mirror.
 c. This is unfortunate.
- (7) *Source*, a subordinating relation: cf. Figure 5
 a. John broke the vase.

¹⁰Note, however, that Hobbs counts to this class both relations like *Parallel* and *Contrast*, which in SDRT are considered coordinating, and *Elaboration*, which counts as the prototypical example for a subordinating relation in SDRT.

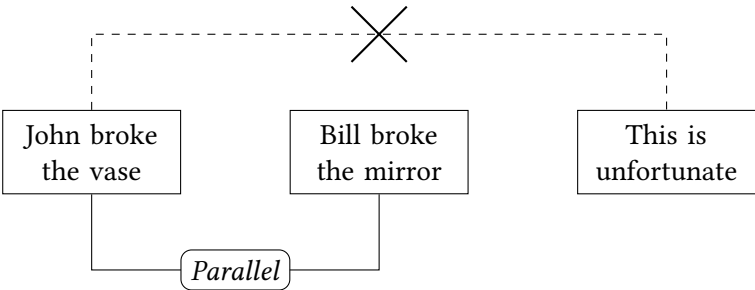


Figure 4: *Parallel*, a coordinating relation

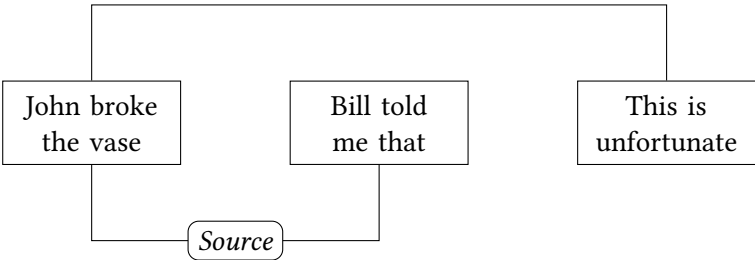


Figure 5: *Source*, a subordinating relation

- b. Bill told me that.
- c. This is unfortunate.

The test in (5) is based on the “Right Frontier Constraint” first formulated in Polanyi (1988) and refined in SDRT. As argued in Asher & Vieu (2005: 593), this constraint in its original form does not make enough elements available for attachment, attributing this drawback to the lack of distinction between different types of discourse relations, i.e. relations that are defined as either subordinating or coordinating. As a matter of fact, we saw that Polanyi makes a similar distinction between coordination and subordination, structurally implemented in terms of coordination and subordination nodes in the discourse tree (labelled C and S, respectively, cf. discussion above), indicating that a coordination or subordination relation obtains between an incoming unit and an accessible node in the discourse tree. However, the way the discourse tree is constructed does not allow for the right attachment points, as Asher & Vieu (2005) argue. Thus, in (7), (7c) could not be attached to (7a) in Polanyi’s discourse tree, since (7a) would represent a closed node, see the left hand structure in Figure 6. Instead, the correct hierarchical representation of discourse structure would distinguish graphically between coordinating and subordinating relations, as in the right hand structure

in Figure 6, Asher and Vieu argue. In such structures, coordinating relations are depicted by horizontal and subordinating relations by vertical arrows between nodes. Consequently, discourse structure is represented in form of directed two-dimensional graphs rather than trees. In the right hand structure in Figure 6, the subordinating relation between (7a) and (7b) is represented graphically in terms of a vertical arrow, leaving the *a*-node still accessible for the attachment of unit *c*.¹¹ Hence, the right frontier in Asher & Vieu (2005) consists of the last processed node and all the nodes to which it is connected by a subordinating relation, which would be nodes (7a) and (7b) in the right hand structure in Figure 6. As already mentioned, the right frontier constraint not only governs the way new material is attached to existing discourse but also captures the influence of discourse subordination on anaphoric reference in terms of the “look-left-one-step-only-or-look-up rule” (Asher & Vieu 2005), predicting that subordination works against recency in anaphora resolution. Thus, the propositional anaphor *this* in (6) cannot be resolved to the proposition expressed by the segment in (6a) since the corresponding discourse segment is not at the right frontier and thus not available for attachment of the new material, in contrast to (7) where the segment corresponding to (7b) is subordinated to the one in (7a) and (7a) is thus accessible for the resolution of the propositional anaphor.

2.2 Discourse-structural subordination in goal and question based approaches to discourse

A different view on discourse structure and discourse subordination is provided in goal- and question-based approaches of discourse. Below, we will briefly review the most influential approaches to discourse structure that (implicitly or explicitly) provide a notion of discourse subordination based on the notions of goals and questions in discourse.

¹¹Note that the adequacy of the Right Frontier Constraint and the tree-like representation of discourse has been challenged in work by Wolf & Gibson (2004) who argue that discourse involves a lot of crossed dependencies and nodes with multiple parents. This view, however, is questioned in Egg & Redeker (2010), who show that much of the cases discussed by Wolf & Gibson are due to particular theoretical choices rather than to the actual structure of discourse. At the same time, in comparing RST and SDRT, Danlos (2008) reaches the conclusion that none of these frameworks displays a “strong generative capacity”: None of them allow the representation of all the discourse structures corresponding to felicitous discourses, while also excluding those corresponding to infelicitous ones. Behrens & Fabricius-Hansen (2010) argue at the same time that a SDRT-relation like Background seems both subordinating and coordinating when different criteria are applied: subordinating according to the RFC but coordinating with respect to the topic criterion, cf. also fn. 9.



Figure 6: Two structures for example (7): Polanyi's (1988) discourse tree on the left and Asher and Vieu's (2005) discourse graph on the right.

2.2.1 Grosz & Sidner (1986)

In the discourse model proposed in Grosz & Sidner (1986), discourse structure is based on intentions (as opposed to the discourse relations in RST and SDRT). Discourse consists of three distinct, but interacting components, dealing with different aspects of the utterances made in discourse:

1. *Linguistic structure*: consists of the actual utterances comprising a discourse, aggregated into discourse segments (DS), and an embedding relationship that can hold between segments. This relationship is only a “surface reflection” of relationships on the deeper level of intentional structure.
2. *Intentional structure*: consists of discourse segment purposes (DSP) that specify how the individual segments contribute to the overall purpose of the discourse, defined as the intention underlying the engagement in a particular discourse with a particular content. Examples include the intention that some agent intend to perform some physical activity, or the intention that some agent come to believe some facts.
3. *Attentional state*: comprised of a set of focus spaces containing information about objects, relations, properties and intentions that are salient at each point of the discourse and thus crucial for the processing of subsequent utterances. A focus space is associated with each discourse segment. Changes in attentional state are modeled by a set of transition rules, implemented in terms of push and pop operations defined on a focus space stack.

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In Grosz & Sidner (1986), embedding in discourse relates to the intentional structure. The model envisages only two relations between discourse segments, dominance and satisfaction-precedence, which are structural relations holding between Discourse Segment Purposes (DSP). Satisfaction-precedence corresponds to linear precedence in parsing at sentence level (ibid. 204). A DSP1 satisfaction-precedes a DSP2 whenever DSP1 must be satisfied before DSP2. On the other hand, when an action satisfying one intention DSP1 is intended as part of satisfying some other intention DSP2, DSP1 contributes to DSP2, and DSP2 dominates DSP1. The dominance relation thus induces a partial ordering on DSPs, called the *dominance hierarchy*. This hierarchy provides a record of the discourse purposes of all segments and their dominance (and relevant satisfaction-precedence) relations (ibid: 180).¹² Both dominance and satisfaction-precedence relations depend on domain knowledge, but may be reflected by the linguistic structure (e.g. in terms of cue phrases like *first ...then, moreover* etc.).

Figure 7 shows how the three components interact for a (slightly modified) fragment of a discourse from Grosz & Sidner (1986: 183) in which an author discusses the influence of movies on young people, see (8):

- (8)
- | | | |
|---|---|---|
| { | { | DS1: 1. Parents ought not to permit their children to attend a movie picture show often or without being quite certain of the show they permit them to see. |
| | { | DS2: 2. No one can deny, of course, that great educational and ethical gains may be made through the movies [...] |
| | { | DS3: 3. But the important fact to be determined is the total result of continuous and indiscriminate attendance on shows of this kind. |
| | | 4. Can it be other than harmful? |

The linguistic structure is graphically represented in terms of left square brackets and indentation of the embedded units. The individual discourse segments

¹²Grosz & Sidner (1986: 193) argue that embedding occurs only in the attentional structure. However, the stack relationships arise from the ways in which the DSs relate, and information about these relationships is represented in the dominance hierarchy (ibid., 180). See also Moser & Moore (1996: 411), who observe that the relation of dominance between DSPs determines “the embeddedness relation of the discourse segments that realize them”, since segments are produced in order to satisfy certain intentions.

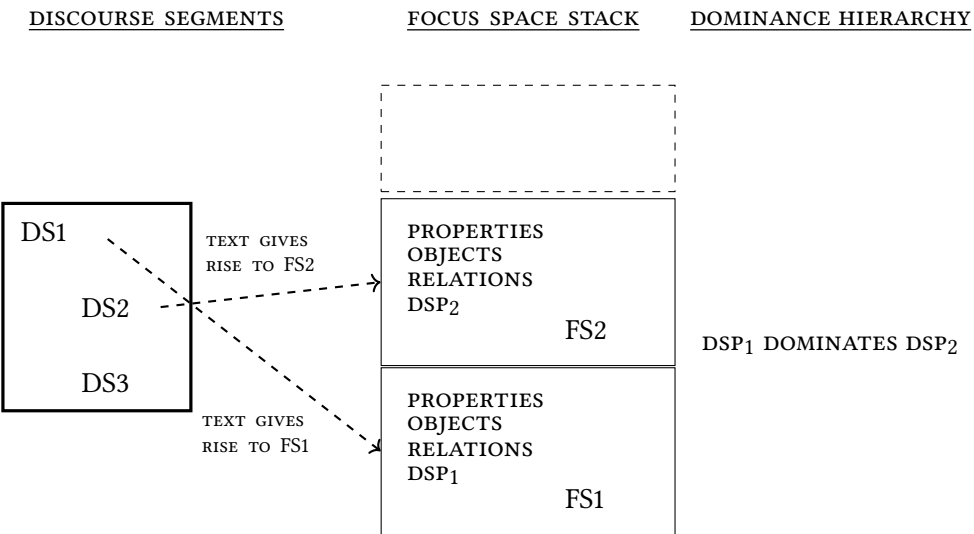


Figure 7: A snapshot of a stage in discourse processing (cf. Grosz & Sidner 1986: 181)

DS1 and DS2 give rise to their respective focus spaces, FS1 and FS2, which in turn are parts of a stack structure containing the focus spaces of the relevant segments. The stack of focus spaces thus represents the attentional state at the given stage of the discourse. Each focus space contains the salient entities (objects, their properties and relations between them) of the respective segment, including its discourse segment purpose DSP. When DS2 is being processed, its focus space FS2 is stacked on top of FS1 because the associated DSP1 dominates DSP2, i.e. satisfying DSP2 (intend the reader to believe that the writer is aware of the gains of watching movies) contributes to satisfying DSP1 (intend the reader to believe that it is not harmless for children to watch movies often or without parental control). When DSP2 is satisfied, FS2 is popped off the stack. Then DS3 is being processed, and its focus space FS3 is pushed onto the stack, since also this new segment contributes to DSP1 (where DSP3=intend the reader to believe that continuous and indiscriminate attendance of movies is harmful). After satisfying DSP3, FS3 is popped off the stack, and the focus space of the superordinating DS1 is processed. The dominance relation between DSP1 and DSP2 and between DSP1 and DSP3 represents the intentional structure of the current discourse. When the discourse is completed, the focus stack is empty, and the intentional structure (the dominance hierarchy) will be fully constructed (Grosz & Sidner 1986: 180).

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The hierarchical relationship between focus spaces influences furthermore the interpretation of linguistic expressions occurring in the discourse segments in terms of e.g. reference resolution, where an entity belonging to the linguistic structure of a segment DS1 (and represented in its FS1) is less salient and thus less available for anaphoric resolution of a pronoun in a consequent segment DS2, due to the fact that FS1 is not processed until FS2 is processed and popped off the stack. This effect of segmentation on the use of referring expressions becomes evident when we consider that the pronoun *it* in sentence 4 of the discourse in (8) is used within the same DS to refer to the NP *continuous and indiscriminate attendance on shows of this kind*, while a generic NP like *shows of this kind* is used in DS3 – instead of a pronoun – to refer to the NP *the movies* in the *preceding* DS2. Notably, SDRT would predict the same anaphoric attachment site for the pronoun *it* in sentence 4, where, however, sentences 3 and 4 would be segmented into separate discourse units: While a relation of Contrast holds between 2 and 3, 4 can only be attached to 3, which represents the right frontier of (this portion) of the discourse. On the other hand, the right frontier constraint in SDRT would not exclude the use of a pronoun like *them* in 3 to refer to the NP *the movies* instead of the generic NP *shows of this kind*, as 3 represents an available attachment site for 4.

The three components of discourse structure provide the discourse participants with the necessary information needed to figure out how a new utterance in discourse is to be integrated in terms of both content and purpose (and also serves as a basis for certain expectations on what new utterances will follow and how to accommodate them). Intentions thus play a primary role in explaining discourse structure, defining discourse coherence.

Comparing their model to theories of discourse coherence such as Hobbs (1985) and Mann & Thompson (1988), Grosz & Sidner (1986: 202) argue that although intentions underlie most of the coherence relations these theories assume, they are unable to account for the vast diversity of intentions that underlie discourse and the relationships between them. Nevertheless, Moser & Moore (1996) propose a synthesis of RST and Grosz and Sidner's model, arguing that both theories make similar claims about the role of speaker's intentions in determining the structure of discourse, even though intentional structure is only implicitly contained in the functional distinction between Nucleus and Satellite in RST. In particular, Moser & Moore (1996: 413) argue that nuclearity in RST and the dominance relation between intentions in Grosz & Sidner (1986) correspond to each other, as "the asymmetry between nucleus and satellite originates with the speaker's intentions": "the nucleus expresses a belief or action that the hearer is intended to adopt", whereas the satellite provides information that is intended

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to facilitate adopting this intention. Thus in either model, text is structured depending on how spans/segments are intended to support other spans/segments in achieving their purpose (*ibid.*).

2.2.2 Questions under Discussion

The concept of Questions under Discussion (QUD) has been introduced in various frameworks with somewhat varying aims and underlying assumptions. The central idea on which these approaches are based is that sentences in discourse address an explicit or implicit question, either by representing an answer to it or by giving rise to a sub-question that can help answering the superordinated question, cf. also Benz & Jasinskaja (2017: 177).

The first to suggest that texts are structured around an explicit or implicit “question which the text in its entirety is produced to answer” were von Steutterheim & Klein (1989: 41). They call the implicit question that an utterance answers the *quaestio* and argue that a *quaestio* may be derived from the general context in which the utterance is made or from a higher *quaestio* which the entire text is intended to answer. Thus, a question like “What happened to you last week?” may be subdivided into several sub-questions corresponding to the sub-events happening within the time span of the entire event (Q1: *What happened to you at t1? ...at t2?*), thus representing the “backbone” of the narrative. Notably, such questions partition the text into a “main structure” – “those utterances in the text which directly contribute to answering the question”, and a “side structure” – those parts of the text that provide additional (but not necessarily unimportant) material (*ibid.*: 40). In this approach, a notion of discourse subordination is involved in both the distinction between questions and subquestions and the distinction between main structure and side structure, the latter being reminiscent of the distinction between Nucleus and Satellite in RST, where a Satellite represents information that is not essential for the coherence of the discourse and may be omitted or replaced by other material.

In van Kuppevelt (1995), the distinction between different structural levels of discourse is further elaborated and related to the notions of foreground and background in discourse. The foreground is made up of the events forming the storyline in narratives, where the order of presentation matches the temporal order, while the background contains material consisting of attached elaborations that lack such temporal properties. According to van Kuppevelt, topicality represents the basic organizing principle of discourse (*ibid.*: 812). Topic (and subtopic) is defined as every contextually induced explicit or implicit (sub)question that is answered in discourse. Such a question represents “a set of possible extensional

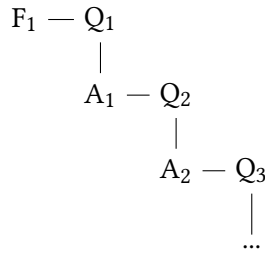
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Figure 8: A question–answer structure in van Kuppevelt (1995: 818);
F=Feeder, Q=Question, A=Answer

values” consisting of entities evoked in the discourse, one of which is selected by the answer to the question, the comment. As a result, discourse is seen as structured around a topic-comment hierarchy and discourse structure as the result of the contextual induction of explicit and/or implicit (sub)topic-forming questions (ibid.: 813). More specifically, subquestions are contextually induced in a recursive fashion: “when a topic-constituting question has been answered unsatisfactorily, it gives rise to a further subquestion that, if also answered unsatisfactorily, gives rise to a further subquestion, and so on recursively, until the original, topic-constituting question has been answered satisfactorily.” (ibid.: 816), cf. Figure 8 where F is a “feeder”, i.e. an utterance that triggers such a recursive process of subquestioning.¹³ This is the structure representing for instance the question-answer discourse structure in (9) (adapted from van Kuppevelt 1995: 815), where Q_1 is triggered by the opening sentence F_1 which serves as a feeder, and the subsequent questions are triggered by indeterminacies contained in the respective answers A_1 and A_2 respectively:¹⁴

- (9) F_1 A: Last Tuesday our company got a new president.
 Q_1 B: Who is it?
 A_1 A: It is the former manager of a successful software house.
 Q_2 B: Which software house?

¹³Even a non-linguistic “event” can have this function. More specifically, a linguistic feeder F_n is defined as “a topicless unit of discourse or one the topic of which is no longer prominent at the moment of questioning” which provides a set of indeterminacies that is contextually unrestricted by preceding questions (ibid.: 814).

¹⁴Note that an answer to a question may also serve as a feeder. It may, for instance, introduce an indeterminacy that does not contribute to answering the original topic constituting question, but leads to the introduction of a new topic, resulting in a topic shift or topic digression (ibid.: 821).

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A₂ A: The one where your friend works.

Q₃ B: Which friend?

...

Based on this, van Kuppevelt distinguishes three structural levels in discourse, corresponding to 3 different types of topic-forming questions: (i) a main structure, representing an answer to the “leading topic-forming question” (which also defines the global discourse topic, e.g. A₁ in (9)), (ii) a substructure (“subordinated” structure) resulting from subquestions and contributing directly or indirectly (in terms of e.g. justifications) to an answer to the main topic-forming question (e.g. A₂ in (9), and (iii) a side structure, which corresponds to topic digression (ibid.: 822) and which arises when for instance in (9), A₂ would give rise to the alternative Q₃ “By the way, did you hear that Peter got married yesterday?”.¹⁵ In addition, a functional distinction is made between two types of subquestions in the substructure: those that are contextually induced as the result of a preceding unsatisfactory answer because the answer is (a) quantitatively unsatisfactory (in terms of completeness, as in “Which members of the gang have been arrested? Unfortunately not the leader”, ibid.: 817) or (b) qualitatively unsatisfactory (not specific enough or not accepted by the addressee and calling for support justification, motivation, evidence, e.g. A₁ and A₂ in (9) which are not specific enough to answer the question Q₁). The “quantitative elaborations” constitute a goal-satisfying part of the answer to the main question, whereas the “qualitative elaborations” form the goal-subservient parts of it, that is, the parts that can be deleted while preserving the goal of the question and the coherence of the text that forms an answer to it. Discourse subordination in van Kuppevelt’s model is thus explicitly related to the process of subquestioning, where subquestions are not autonomous but “subservient to the program associated with a preceding topic-constituting question”. On the other hand, the distinction between goal-satisfying and goal-subservient substructures is reminiscent of the Nucleus-Satellite distinction in RST and the notion of subordination provided by it.

Roberts’ (1996) formal model of discourse and information structure is probably the most influential QUD-based model of discourse.¹⁶ It relies on the assumption (originally suggested in Stalnaker 1978) that the primary goal of every discourse is in general to share information about “the way things are”. Roberts suggests that this goal can be formulated in terms of a question like *What is the*

¹⁵Note that the question of the topic digression is not a subquestion of the topic-constituting question but achieves a topic shift (ibid.: 822).

¹⁶Equally influential in research on dialogue semantics and computational modeling of dialogue is the QUD-based model developed in Ginzburg (1996).

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way things are? This question is answered by breaking it down into a number of subquestions that are logically related and addressed one by one, along a strategy of inquiry that the speaker develops towards answering the “big question”. The strategy represents a hierarchy of questions under discussion (QUDs) that are subsequently answered and that reflect the hierarchical structure of discourse. The set of QUDs at a given point in a discourse is modelled by the QUD-stack, a push-down store (similar to Grosz & Sidner’s 1986 stack structure containing the discourse segments’ focus spaces) representing the ordered set of all as yet unanswered questions. The question on top of the stack represents the immediate QUD. Higher questions on the stack are subquestions of the lower. When a question is pursued by asking a subquestion, the subquestion is added to the stack, such that the stack partly reflects a strategy of questions (since a strategy to answer a question involves conducting the set of subinquiries). When a question is answered, it is popped off the stack. A superquestion is answered when enough subquestions are answered. Consider the example in (10), which represents a discourse that realizes a strategy to answer question 1 by means of the set of subquestions $\{a, a_i, a_{ii}, b, b_i, b_{ii}\}$.

- (10) Who ate what? (Roberts 1996: 16)
- a. What did Hilary eat?
 - i. Did Hilary eat bagels?
Ans(a_i) = yes
 - ii. Did Hilary eat tofu?
Ans(a_{ii}) = yes
 - b. What did Robin eat?
 - i. Did Robin eat bagels?
Ans(b_i) = yes
 - ii. Did Robin eat tofu?
Ans(b_{ii}) = yes

Roberts’ QUD-theory can thus be seen as a specific implementation of Grosz & Sidner (1986) where conversational goals and strategies to achieve these are seen as the central organizing principle of discourse and where questions correspond to goals and the subquestion relation to Grosz & Sidner’s (1986) dominance relation. In contrast to Grosz and Sidner, however, Roberts provides a formal definition of subordination, capturing the subquestion relation in terms of entailment: a question q_1 entails another q_2 iff every proposition that answers q_1 answers q_2 as well (Roberts 1996: 7). In contrast to van Kuppevelt, where the

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questioning process is triggered by indeterminacies (contextually determined or linguistically provided by feeders or by unsatisfactorily answered subquestions of the topic-constituting question), the strategy of inquiry that the speaker develops is determined on the basis of the information structure of the discourse in which the utterances occur. Following Carlson's (1983) insight that questions that are implicit can be retrieved/inferred from context, Roberts thus claims that intonational focus presupposes that an assertion is a congruent answer to the QUD at the time of its utterance (*ibid.*: 33), relating the information structure of individual utterances to their role in discourse. While the questioning process in van Kuppevelt functions in a top-down fashion, being triggered by indeterminacies related to the current utterance in discourse that give rise to the question the following utterance provides an answer to, a QUD is reconstructed in a bottom-up fashion from the information structure of the utterance that provides an answer to this question.

2.2.3 Jasinskaja & Karagjosova (2020)

In their attempt to capture the notion of subordination in discourse in more precise terms, Jasinskaja & Karagjosova (2020) suggest a goal-based definition of discourse subordination in which discourse structure is defined in terms of relationships between communicative goals. In line with Grosz & Sidner (1986), Roberts (1996) and work by Farkas & Bruce (2010), Jasinskaja & Karagjosova (2020) assume that typical communicative goals are to make the addressee answer a question, believe some proposition (or accept it to a degree sufficient for the purposes of the exchange, i.e. make that proposition part of the common ground Stalnaker 2002), or make the addressee perform (or commit to performing) some action. Accordingly, they define discourse subordination in terms of super- and subordinated discourse goals.¹⁷

- (11) **Subordination in discourse:** In a sequence of discourse units $\langle U_1, U_2 \rangle$, U_2 is subordinate to U_1 whenever the communicative goal of U_1 cannot be reached before the communicative goal of U_2 is reached. (Jasinskaja & Karagjosova 2020)

¹⁷Note that RST employs a similar notion of functional goals in terms of particular effects of rhetorical relations on the addressee's knowledge, beliefs and desires, as part of the definitions of the individual rhetorical relations. Thus, the effect of the Elaboration relation is specified as the the addressee (i) recognizing the situation referred to in the Satellite as providing additional detail for the Nucleus and (ii) identifying the element for which detail is provided (Mann & Thompson 1988: 273). However, these effects are not given an account in RST in discourse structural terms.

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This definition states that the speaker first attempts to reach the goal associated with U_1 , which, however, cannot be reached before that of U_2 . Having started working towards the goal of U_1 , the speaker needs to interrupt this process, deal with U_2 instead, and, in or after doing so, reach the primary goal of U_1 . The idea of interruption is argued to be essential to discourse subordination, since it explains why U_1 has to be kept in memory while U_2 is processed in order to be able to eventually go back to U_1 after processing U_2 .¹⁸ This reflects the way Grosz and Sidner's focus stack model applies to the notion of subordination and explains why subordination works against recency as in Asher & Vieu's (2005) version of the Right Frontier Constraint (see also Section 2.1 and the discussion of examples (6) and (7)). The influence of discourse-structural subordination on anaphoric reference captured by the Right Frontier Constraint is viewed in Jasinskaja & Karagjosova (2020) as a structural reformulation of the idea formulated by Grosz & Sidner (1986) that the processing of discourse requires keeping track of a stack of focus spaces (cf. Section 2.2.1). In a certain class of cases which correspond to subordinating discourse relations, the focus space of the subordinate unit is pushed on the stack while the focus space of the subordinating unit also stays there until the subordinate unit is processed. Thus in (7), repeated below as (12), the subordinate unit (12b) would be pushed on the stack after the subordinating unit (12a) and processed before (12a), since (12b) plays a role in reaching the communicative goal of (12a) (e.g., convincing the addressee that John broke the vase) by providing the source of the information given in (12a). Once the processing of the subordinate unit is complete, its focus space is popped off the stack so the focus space of the subordinating unit becomes topmost again and available for various operations such as anaphoric reference. This accounts for the resolution of the propositional anaphor *this* in (12c) to (12a) (instead of (12b) which is popped off the stack and no longer available as an antecedent for pronoun resolution).

- (12) a. John broke the vase.
 b. Bill told me that.
 c. This is unfortunate.

In the case of coordination like (6), the focus spaces are processed on a “first come, first served” basis, that is, previously processed units are not kept in mem-

¹⁸Note that this notion of interruption is not to be equated with the notion of interruptions in discourse employed in Grosz & Sidner (1986) and defined as “pieces of discourse that break the flow of preceding discourse”. Interrupting discourse units in Grosz and Sidner's sense correspond to van Kuppevelt's digressions which belong to the side structure of discourse. Cf. also Section 4.3.

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ory, so there is no way to go back and access the entities associated with those units.

2.3 Subordination in theories of discourse structure: summary and further developments

Summing up, subordination in discourse can be viewed in terms of (i) functional asymmetry between text spans (Nucleus vs. Satellite in RST, goal-satisfying vs. goal-subservient parts of discourse in van Kuppevelt, informational dominance in Hobbs), (ii) structural attachment (LDM, SDRT), (iii) a semantic relation between a subordinated unit and its superordinated unit (defeasible implication, SDRT) or between questions addressed by utterances in discourse (entailment, Roberts), and (iv) in terms of a structural relation of dominance between intentions or satisfaction of communicative goals (Grosz and Sidner, Jasinskaja and Karagjosova) or between topic-constituting and subtopic-constituting questions (van Kuppevelt).

There have been continuous efforts to bring together the different approaches to discourse structure. Thus, the Relational Discourse Analysis developed in Moore & Pollack (1992) and Moser & Moore (1996) is an attempt to merge RST with Grosz and Sidner's theory into a theory of discourse structure in which the intentional structure of discourse is linked to its rhetorical structure in terms of defining a set of intentional relations (such as convince, enable, concede) next to informational relations (such as cause–effect). In more recent research, special attention has been given to the combination of approaches underlying theories of discourse relations and question-based theories of discourse. Thus, Jasinskaja & Zeevat (2008) and Jasinskaja (2010) have argued that additive and contrastive markers such as *and* and *but*, which mark discourse relations such as *Narration* and *Contrast*, also convey information about the QUDs addressed by the clauses they connect. Karagjosova (2011) explored the interplay between discourse relations and QUDs in capturing the meaning of the German connective *nämlich* ('namely'). An interesting link between discourse relations and her QUD-model of information structure is already suggested by Roberts (1996: 62–63), who notes that particular types of RST-relations can be characterized as types of strategies and that discourse relations in general can be characterized in terms of questions and answers, suggesting that one could use a *Why*-question and its answer to characterize *Explanation*. At the same time, Roberts points at some differences between the two concepts, noting that the goal of discourse is only partly to offer more information by addressing a hierarchy of questions, the other part being to "achieve consensus about the value of the information contributed", a

task that is accomplished by discourse relations intended to convince the hearer to accept the information offered and add it to the common ground by showing, for instance, how this information explains other known facts (*ibid.*). This distinction between two kinds of goals of discourse is reminiscent of the two different functions that Mann & Thompson (1988) ascribe to Nucleus and Satellite units in discourse, as well as of van Kuppevelt's distinction between main structure and (qualitative) goal-subservient structure as two different hierarchical levels of discourse. Comparing QUD-models like van Kuppevelt's with discourse relation based models like SDRT, Karagjosova (2013) argues that the two approaches deal with two distinct dimensions of discourse structure, showing that a discourse consisting of two units related by Elaboration may give rise to two different question-answer structures depending on the way in which the topic-constituting question is defined. Benz & Jasinskaja (2017) observe that a straightforward link between rhetorical structure and QUD-theories of discourse structure is contained in Mann and Thompson's definition of discourse relations cited in Section 2.1: While discourse relations capture one aspect of discourse coherence – “an evident role of every part”, QUDs capture the second – “no sense that a part is missing”. This is illustrated in (13) where the intuition that the discourse is incomplete as it stands cannot be accounted for in terms of discourse relations, since the discourse is incoherent in spite of the fact that the two sentences are related by *Elaboration*. The incompleteness of the discourse can however be captured by a notion of a purpose or goal of a discourse, conceptualized as a QUD like “Which two things did the speaker do?” triggered by (13a), and requiring that the discourse is completed only when the question is fully answered in terms of providing information on “two things”, see (14) (*ibid.*: 183).¹⁹ Benz & Jasinskaja (2017) take this as a strong argument for the contribution of the notion of QUDs to a general theory of discourse coherence that goes beyond discourse relations.

- (13) a. I did two things on my seventy-fifth birthday.
b. I visited my wife's grave.

- (14) Then I joined the army.

¹⁹Remember that according to van Kuppevelt, assertions in discourse not only answer questions but also raise questions themselves, an idea implemented in terms of the so-called feeders. Onea (2016) develops this idea further, arguing that certain linguistic expressions, such as indefinite expressions and nominal appositives, systematically license “potential questions” in discourse which play an additional role in explicating discourse structure next to Robert's QUDs.

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At the same time, it has been argued in Hunter & Abrusán (2017) that Roberts' definition of subordination in terms of question entailment is too rigid to account for the rhetorical structure of discourse, this being one of the reasons why the two structures cannot be mapped directly to one another without loss of information. An attempt to loosen this constraint is presented in Riester (2019) and Riester et al. (2018), where QUDs are integrated in a theory of discourse and information structure represented by means of discourse trees, combining features from Roberts' QUD theory, SDRT and van Kuppevelt's model of discourse. In Riester's QUD trees, subordinating discourse relations are represented by questions – *Why*-questions for *Explanation*, *What-about*-questions for *Elaboration*, whereas coordinating discourse relations such as *Narration* are captured as a series of parallel questions about different times, constituting thus a temporal progression in the spirit of Klein & von Stutterheim (1992). The entailment relation between QUDs is loosened by allowing subordinated units of discourse to answer questions that are not entailed subquestions but are triggered by unsatisfactory answers to subquestions in a top-down fashion, adopting van Kuppevelt's notion of feeders as answers introducing topic-forming questions that do not belong to the main structure (cf. Figure 8. Further research on this topic is needed to fully explore the possibilities of developing an integrated theory of discourse structure that adequately captures the various notions of subordination in discourse.

3 Embedding in syntax and discourse

In this section, we proceed to discuss the relation between syntactic embedding (or lack thereof) and discourse-structural subordination and coordination. In doing so, we will largely assume the notion of discourse-structural subordination which underlies theories employing the concept of discourse relations, such as RST and SDRT, and which is presupposed in most current research on the topic. The two frameworks diverge in the criteria underlying the distinction between coordination and subordination in SDRT and between mono- and multinuclear relations in RST. Still, there is a rough correspondence in the way central subordinating relations like *Elaboration* and *Background* on the one hand and coordinating relations such as *Narration/Sequence* and *Contrast* are categorized with respect to the hierarchical structure of discourse (for a consensus list of discourse relations, cf. e.g. Jasinskaja & Karagjosova 2020).

Subordinating relations such as *Explanations* may take on a number of different syntactic structures with a varying degree of clausal integration, see (15). In all these cases, the underlined material provides an explanation of the occurrence

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of the mental state of annoyance, establishing a discourse-structural subordinating relation between two segments:²⁰

- (15) a. *Adverbial clause:*
Peter annoyed Mary because he sang loudly.
- b. *Non-restrictive relative clause:*
Peter, who sang loudly, annoyed Mary.
- c. *Gerund:*
Singing loudly, Peter annoyed Mary.
- d. *Independent sentence (without explicit marker):*
Peter annoyed Mary. He sang loudly.
- e. *Sentential complement:*
It annoyed Mary that Peter sang loudly.

There is a certain tendency for subordinate clauses to go along with discourse-structural subordination. Thus, Matthiessen & Thompson (1988: 308) argued that subordinate clausal syntax (hypotaxis) is a grammaticalization of the relationship of discourse subordination. Their counts based on a small corpus of 18 short texts revealed that a subordinate clause would realize a subordinating discourse relation in 45 out of 48 cases, whereas only 3 (= 6%) subordinate clauses realized a coordinating discourse relation.²¹ In Loock's (2007) study of 450 English non-restrictive relative clauses (NRRC) collected from texts of four different genres, only 20 (= 4.4%) were continuative (i.e., discourse-structurally coordinated).²²

The opposite clearly does not hold. First, subordinating discourse relations may span across sentence boundaries, as shown by (15d), partly because there are other, more conventional means of encoding subordinating relations that do not involve syntactic subordination (such as connectives). What is more, some cases of syntactic subordination raise the interesting question of what should count as a discourse segment, see the sentential complement in (15e). In the following, we will first discuss parallels between syntactic and discourse-structural embedding

²⁰It should be noted that the status of sentential complements as in (15e) as (independent) discourse segments is debatable, but for the parallelism in interpretation, we include them in our discussion. See the paragraph "Complementation" in Section 3.1 for further discussion.

²¹Matthiessen & Thompson (1988) base their analysis on RST (Mann & Thompson 1988) and operate with the notions of 'nucleus-satellite relation' and 'hypotaxis', which are not identical but closely correspond to the notions of discourse-structural subordination and subordinate clause, respectively, cf. also Section 3.1 for further details on the distinction between syntactic embedding and hypotaxis.

²²See also Jasinskaja & Poschmann (2018) who argue that there is a statistical tendency for an alignment between hierarchical structures in syntax and discourse.

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(Section 3.1) before we turn to what may be viewed to be mismatches between the two domains (Section 3.2).

3.1 Parallels between syntactic and discourse structure

In this section, we first investigate the correspondences between syntactically non-embedding structures and discourse-structural coordination on the one hand, and syntactically embedding structures (in terms of adjunction and complementation) and subordinating discourse structures on the other. We then proceed to discuss some parallels between discourse prominence and syntactic subordination.

3.1.1 Syntactic relations between clauses and their discourse structural counterparts

3.1.1.1 Coordination

There are many cases in which discourse-structural coordination corresponds to clausal coordination (see Chapter 5 Coordination). In general, syntactic coordination, or conjunction, holds between syntactic units that have the same status or play the same role in a given syntactic context (cf. e.g. Fabricius-Hansen & Ramm 2008: 5 and the literature cited there). It is usually taken to be a symmetric relation in the sense that if unit A is coordinated with unit B, then B is also coordinated with A. However, various cases of asymmetric coordination are reported in the literature, such as *and*-constructions performing “subordinate conjoining” (Lang 1984) or “No Backward Anaphoricity” (ibid.), that is, the fact that the first conjunct is an accessible anaphoric antecedent for anaphoric elements in the second conjunct, but not the other way around.²³ Clausal coordination is signalled by coordinating conjunctions, prototypically *and*, *but* and *or*, thus representing a syndetic variety of paratactic clause combining (Fabricius-Hansen & Ramm 2008: 6).²⁴ The same coordinating conjunctions are also typical markers of coordinating relations such as *Parallel* (16), *Narration* (17), *Contrast* (18) or *Alternation* (19).

(16) Mary plays football and John plays tennis.

²³Some researchers even assume a subordination-coordination continuum (cf. e.g., Kortmann 1997: 56).

²⁴Malchukov (2004) and Mauri (2008) offer an account of the particular semantics (functions, uses) of various coordination markers within the semantic map approach: adversative, contrastive, additive, disjunctive, etc. conjunctions, converbs (non-finite verb forms expressing adverbial subordination), etc.

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- (17) Max fell and John helped him up.
 (18) John is tall, but Bill is short. (Lakoff 1971: 133)
 (19) Either Jane likes football or she has a crush on David Beckham.
 (Asher & Lascarides 2003: 169)

On the other hand, coordinating discourse relations need not be syntactically realized in terms of syndetic clause combination but may take the form of asyndetic clause combination or mere juxtaposition of independent clauses in discourse, as in the *Narration*-example (20) below. Moreover, coordinating discourse relations may be (additionally) marked by adverbs and particles. Thus, typical markers of the *Parallel* relation are the particles *also* and *too* that may be used in syndetic or asyndetic parataxis/clause combining or mere juxtaposition, as in the *Parallel*-example (21).

- (20) Max fell. John helped him up. (Asher & Lascarides 2003: 6)
 (21) Mary drives a Toyota $\left\{ \begin{array}{c} \cdot \\ , \\ \text{and} \end{array} \right\}$ John also drives a Toyota.

3.1.1.2 Adjunction

Discourse-structural subordination corresponds roughly to clause combining in terms of adjoined, primarily adverbial clauses, a relation also known as adverbial subordination (see Chapter 3 Adverbial clauses).

Adverbial clauses are viewed in Sæbø (2011) as subordinate clauses that modify their superordinate clauses at various syntactic phrasal levels (e.g., verb phrase, tense phrase, mood phrase) and in various dimensions (such as times and worlds). They are mostly marked by a subordinator. As a matter of fact, there is a controversy as to the extent to which adverbial clauses are syntactically subordinated or embedded into a main (matrix) clause. Thus, Matthiessen & Thompson (1988) argue for a distinction between clausal (syntactic) embedding where one clause is a constituent part of another, and clause combination based on some circumstantial, that is, temporal, conditional or other relation and syntactically realized as either parataxis or hypotaxis. They argue further that adverbial subordination, as well as non-restrictive relative clauses, represent a case of clause combination rather than clausal embedding, thus arguing against grouping embedding and hypotaxis together as ‘subordination’.²⁵ Based on this distinction between embed-

²⁵One argument for this involves the fact that adverbial clauses may be syntactically embedded within a noun phrase, as in [_{NP} the happy days [before the Magistrate had been invited]] (Matthiessen & Thompson 1988: 279).

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ded and hypotactic adverbial clauses, Matthiessen and Thompson view hypotaxis as a grammaticalization of the hierarchical organization of discourse, in terms of a special case of the mono-nuclear relation in RST. Others, like Fabricius-Hansen & Ramm (2008: 15), equate hypotaxis with subordination in the general sense of “subordinate clauses that are not embedded” (and parataxis with explicitly marked coordination and mere juxtaposition of mutually independent clauses). See, however, the distinction between core and peripheral adverbial clauses suggested in Haegeman (2002, 2004, 2006) and discussed in Frey (2011) for German. See also the detailed discussion of correspondences between various degrees of clausal integration and the Sweetser (1990) sub-types of explanatory (and concessive) relations in Frey (2020).

In adverbial hypotaxis, the circumstantial relation to the clause that an adverbial clause is adjoined to is marked by a corresponding adverbial subordinator. An extensive typology of adverbial subordinators such as *because*, *if* and *while* in the languages of Europe is presented in Kortmann 1997. The same items that serve as subordinators at the syntactic level also function as markers of subordinating relations such as *Explanation* (*because*), *Flashback* (*after*) or *Background* (*while*), see also Zeevat (2011) who provides a list of grammaticalized English cues for RST relations. Among them are markers of Nucleus-Satellite relations such as *because*, *since* for *Evidence*, *Justify*, *Motivation* and *Non-volitional Cause*, *but* and *though* for *Concession*, *if* for *Condition* and *namely* for *Elaboration*.

Next to adverbial subordination, there is a further case of syntactic adjunction that is relevant to the notion of subordination in syntax and discourse. In their study of the semantics of non-restrictive relative clauses (NRRCs; see also Chapter 4 Relative clauses), Jasinskaja & Poschmann (2018) and Jasinskaja & Poschmann (2020) observe that NRRCs tend to realize subordinating discourse relations since they are syntactically subordinate, and are therefore interpreted globally (i.e., as independent sentences, outside the scope of semantic operators in the embedding sentence). Thus in (22), adapted from Jasinskaja & Poschmann (2020), the NRRC constitutes an *Explanation* and the content of the NRRC is entailed by the discourse, showing that it is outside the scope of the conditional operator. However, they contend, an NRRC may also be interpreted locally if the discourse relation it establishes to its host is coordinating (e.g., *Narration*, *Parallel*, *Contrast*). See (23), where the NRRC establishes a *Narration* relation to its host, and the content of the NRRC is not entailed by the discourse since it is within the scope of the conditional operator in the main clause.²⁶

²⁶Jasinskaja & Poschmann (2020) draw on earlier work by Schlenker (2013) who first observed that the two readings are related to two different discourse relations, however without elaborating on the nature of the difference.

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- (22) If Peter praised John,
 who designed the new prototype, ↗ John designed the new prototype
 he would be doing the right thing.
- (23) If Peter praised John,
 who then was delighted, ↘ John was delighted
 he would be doing the right thing.

Finally, participial and converb clauses also represent a relevant case of syntactic adjunction which at the level of discourse translates into discourse structural subordination in terms of correspondence to subordinating discourse relations (cf. e.g. Behrens et al. 2012). As noted by Sæbø (2011) (citing Kortmann 1995, König 1995), although present (*gerund*) or past participial clauses in English allow in general for a wide range of interpretations, there is a clear tendency to interpret these clauses as expressing certain discourse relations, based on the lexical content and the context of utterance. Thus in (24), the converb clause *wriggling and writhing* is taken to elaborate on the matrix clause, as the adjunct events represent subevents of the main event (Behrens 1998, cf. also König 1995, according to whom the two converbs and the main verb describe different aspects of the same event).

- (24) The trout struggled, wriggling and writhing. (Sæbø 2011: 1436)

In particular, preposed *-ing* adjuncts may under given circumstances convey largely the same relations as conditional *if*-clauses (25), causal *because*-clauses (26) or temporal *when/while*-clauses (27) (Sæbø 2011: 1437):²⁷

- (25) Driving slowly through Thorpe, you will see signs for Dovedale on the way.
- (26) Having confessed to having sex with the girl, the man was sentenced to one year on an abandoned island.
- (27) Reaching the coast, they pick up the scent of their home river.

²⁷ See also Behrens et al. (2012) on further structures competing with non-finite co-eventive adjuncts.

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3.1.1.3 Complementation

It is less clear that there is a discourse-structural counterpart to complementation (see Chapter 2 Argument clauses). Traditionally, complement clauses are considered to be part of the discourse unit of the corresponding main clause. Thus, Mann & Thompson (1988: 248) state that discourse units should have “independent functional integrity” (see also the notion of “conceptual dependency” in Schilperoord & Verhagen 1998). However, there are some interesting exceptions to this (see for instance discussion in Schilperoord & Verhagen 1998 and Hoek 2018). In particular, Carlson & Marcu (2001) observe that what they characterize as *attribution verbs* (such as *say*, *tell*, *state*) may establish a discourse relation they call *Attribution* between the matrix and subordinate clause, if the latter is finite, cf. (28), where the clause containing the attribution verb *say* represents the Satellite in an Attribution relation to the second clause representing the Nucleus.

- (28) [Mercedes officials said] [they expect flat sales next year] (Carlson & Marcu 2001: 8)

Also in SDRT, a subordinating Attribution relation is assumed to hold between a communicative act and the agent of the act (Reese et al. 2007; cf. also Hunter & Asher 2016 and Bimpikou et al. 2021 on the discourse structure of reports in discourse). In Carlson & Marcu (2001: 9), the class of attribution verbs consist of speech verbs and various cognitive predicates like *think*, *predict*, and *fear*. While we are not aware of any principled discussion of this feature of certain verb classes in the literature, we think that for some of these cognitive predicates, a more specific (subordinating) discourse relation than *Attribution* is established.²⁸ In particular, we believe it is reasonable to assume that – as clause-embedding predicates – psychological predicates such as *annoy* or *fear* involve a subordinating *Explanation* relation between the matrix clause and an embedded *that* clause.²⁹

²⁸See, however, work by Danlos (n.d.) on what she calls “discourse verbs” such as *cause* and *precede* which take eventualities/facts as arguments and express particular discourse relations as part of their semantics, thus entering into competition with the corresponding discourse connectives *as a result* and *next*.

²⁹As pointed out by a reviewer, treating the relation that sentential complements in these cases establish to the main clause as a discourse relation requires redefining what counts as a discourse segment, suggesting an alternative treatment of the connection inferred between a matrix event and its complement in terms of (a kind of) implicature. Clarifying the exact status of the relation between matrix clauses and embedded clauses goes beyond the scope of the present Chapter, however.

The subject of much research on the syntax-semantics interface, psychological predicates select for an Experiencer and a Stimulus argument (where the Stimulus argument can alternate between a DP and a CP argument; for empirical evidence in English and German, see for instance Salkoff 2002, Engelberg 2018). The mental state of the Experiencer is causally dependent on the Stimulus (Bott & Solstad 2021, Kutscher 2009, Solstad & Bott 2022). Thus, the annoyance of the Experiencer (*us*) in (29) is caused by Jim’s late arrival.

(29) That Jim arrived late annoyed us. (Salkoff 2002: 105)

Evidence for the explanatory nature of stimulus specifications has been provided by Bott & Solstad (2014, 2021), Kehler et al. (2008), Solstad & Bott (2022). Thus, when participants are asked to provide continuations for simple prompts like *Jim annoyed Jane*, they show a strong tendency ($\approx 60\%$) to provide explanations (e.g., *He was late*). These explanations are of the same type as those provided after *because* prompts (*Jim annoyed Jane because ...*), see Bott & Solstad (2014), Solstad & Bott (2022).

While not concerned with the syntactic status of subordinated clauses or the relation between syntactic and discourse-structural subordination in particular, Asher & Lascarides (2003: 267 ff.) argued that Stimulus-Experiencer verbs like *annoy* evoke the interpretation of an *Explanation* relation in (30) (adapted from Asher & Lascarides 2003: 252, 269):

(30) Fred annoyed Mary. He ate soup last night.

We will return to the role of psychological predicates in establishing discourse coherence in Section 4, where we discuss semantic and pragmatic processes involved in discourse-structural embedding.

3.1.2 Discourse prominence

There is a similar effect of subordinating discourse relations and subordinate clauses on discourse prominence (von Heusinger & Schumacher 2019) in terms of accessibility for anaphora. As we saw in Section 2.1, anaphoric reference follows largely the same principles as discourse attachment: In searching for an antecedent for an anaphoric expression, we are allowed to “look left one step only or look up” (Asher & Vieu 2005). For example, the 1st and the 2nd sentence in (31) are connected by a coordinating relation *Parallel*. Therefore, only the 2nd sentence is on the right frontier and the pronoun *he* in the following sentence is preferably resolved to a referent mentioned in the 2nd sentence, i.e. Bill. In contrast, in (32) the second sentence provides evidence for the 1st by reference to

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an authoritative source. *Evidence* is a subtype of *Explanation*, i.e. a subordinating relation. Both sentences in (32) are therefore on the right frontier, so the following sentence could continue giving the evidence, in which case the pronoun *he* could refer either to Bill or to John; alternatively, it could continue telling the story about John, in which case *he* would refer to John.

(31) *Parallel*:

John broke the vase. Bill broke the mirror. He_{Bill} ...

(32) *Explanation (Evidence)*:

John broke the vase. Bill told me that. He_{John/Bill} ...

There is evidence that syntactic subordination has a similar effect on anaphoric resolution like subordination in discourse. Experimental studies like Frazier & Clifton (2005) and Cooreman & Sanford (1996) show that a pronoun is less likely to refer to an antecedent in a subordinate clause than to one in a main clause, regardless of the recency status of the subordinate clause. Thus in (33a), the pronoun *she* is more likely to be resolved to the subject of the main clause rather than to the one of the subordinate clause, whereas in (33c), *she* can only be resolved to the subject of the main clause (cf. Jasinskaja & Poschmann 2018: 448).

- | | | | |
|------|----|--|------------------|
| (33) | a. | Mary left before Sue arrived. She _{Mary>Sue} ... | main-subordinate |
| | b. | Mary left. Then Sue arrived. She _{Sue} ... | main-main |
| | c. | After Mary left, Sue arrived. She _{Sue} ... | subordinate-main |

3.2 Mismatches between syntactic and discourse structure

Although the previous section pointed at a number of parallels between syntactic embedding and discourse-structural subordination, it is evident that discourse-structural subordination cannot be identified with syntactic subordination, that is, that discourse-structurally subordinate discourse material need not be realized in a subordinate clause, and a subordinate clause does not always encode a relation of discourse subordination. For detailed discussion of the non-equivalence of these notions see in particular Blühdorn (2008).

Example (7) above is an instance of discourse-structural subordination without syntactic subordination: The subordinating discourse relation *Source* holds between two independent sentences. On the other hand, syntactically coordinate structures can also realize subordinating discourse relations. For instance, the second conjunct of *and* in (34) gives an *Explanation* of the first, and the second disjunct of *or* in (35) is a *Reformulation* (a special case of *Elaboration*) of the first disjunct.

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- (34) *Explanation:*
Max fell, *and* it was Bill who pushed him. (Asher & Vieu 2005)

- (35) *Reformulation:*
The hearer must identify the speech act performed by an utterance, or (in other words), she must identify its illocutionary force.³⁰

Conversely, subordinate clauses are syntactically embedded, but can realize coordinating discourse relations. One prominent example is described in Holler (2008) and involves continuative relative clauses, a subclass of non-restrictive relative clauses in German. As argued by Holler, continuative relative clauses like the ones in (36) and (37) are attached to the main clause by means of coordinating, in the SDRT-sense, discourse relations, here Narration and Contrast, signalled by the discourse markers *dann* and *aber* respectively, while ordinary non-restrictive relative clauses like (38) are connected to the main clause by means of subordinating relations (examples from Holler 2008):³¹

- (36) *Narration:*
Oskar traf einen Bauern, den er *dann* nach dem Weg fragte.
'Oskar met a farmer, whom he then asked for the way.'

- (37) *Contrast:*
Oskar machte einen Versuch, der *aber* restlos scheiterte.
'Oskar made an attempt, which however completely failed.'

- (38) *Elaboration:*
Oskar machte diesen Versuch, der nicht scheitern konnte.
'Oskar made an attempt, which could not fail.'

As a matter of fact, the relation signalled by *aber* in (37) is better characterized as Violated expectation or Concession rather than in terms of the general Contrast relation provided by the SDRT inventory of discourse relations. In SDRT, Contrast is by default coordinating, unlike Mann & Thompson (1986), where both a multi-nuclear Contrast relation and two Nucleus-Satellite relations Antithesis and Concession are assumed. Similarly, Hobbs (1985) views both his Contrast and Violated expectation relation as subordinating, while the latter belongs to

³⁰This is a modified version of Blakemore (1993: 109)'s original Reformulation example "The hearer must identify the speech act performed by an utterance. In other words, she must identify its illocutionary force."

³¹Cf. also Jasinskaja & Poschmann (2018) and Jasinskaja & Poschmann (2020) and the discussion of NRRCs in Section 3.1.1 and Chapter 4 Relative clauses for an overview.

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the class of expansion relations. Moreover, applying the test in (5) yields that the relation holding between the main clause and the relative clause in (37) is subordinating: A discourse continuation like *Das nächste Mal wollte er sich besser vorbereiten* ('Next time he wanted to be better prepared.') would naturally attach to the main clause which contains the anaphoric antecedent for the pronoun. Nevertheless, although the interpretation of the data may in some cases be theory-dependent, unambiguous examples like (36) do support the observation of a mismatch between syntactic and discourse structure.³²

Among adverbial clauses, *whereas*-clauses (39) are syntactically subordinate, but express *Parallel* or *Contrast*, which are coordinating discourse relations:

- (39) *Parallel*:
John is tall, *whereas* Bill is small.

On the other hand, clauses syntactically coordinated by means of conjunctions like *but* may involve subordinating (i.e. Nucleus-Satellite) RST-relations like *Antithesis* and *Concession*. Similarly, the German coordinating conjunction *denn*, as well as its English counterpart *for*, marks the subordinating relation *Explanation*.

Finally, a further case of mismatch between syntactic subordination and subordination in discourse is discussed in Riester et al. (2018: 34) who treat (following Simons et al. 2011) embedding matrix verbs of sentential arguments as non-at-issue material (of type evidentials) which does not answer the current QUD, thus belonging to van Kuppevelt's subordinated layer of discourse (cf. Figure 8. See (40) where it is the sentential complement that answers the QUD and thus belongs to the main discourse structure.

- (40) Q: {What is the current status of the bill?}
Jetzt heißt es, das Kanzleramt hat diese Verordnung gestoppt. (Riester 2019)
'Now they are saying that the Chancellery has stopped the bill.'

4 Semantic and pragmatic processes leading to the establishment of subordination in discourse

In this section, we will discuss how semantic and pragmatic aspects of the discourse beyond domain or world knowledge (as discussed in Asher & Lascarides

³²Note also that examples like this do not pose a problem for SDRT since, as mentioned in Section 2, the test in (5) is motivated by the observation that a relation that is coordinating by default may in some contexts come out a subordinating.

2003) may lead to the establishment of discourse-structural subordination. In doing so, we will revisit some of the issues discussed in Sections 2.2.3 and 3.1.1 from the viewpoint of what motivates the establishment of discourse-structural subordination, both in terms of subordinating discourse relations and in terms of questions and goals. We will first investigate the relationship between discourse expectations triggered by the semantics of certain verbs and the establishment of discourse subordination in terms of subordinating discourse relations. Then we briefly introduce a case where derivational morphology can be taken to establish subordination in discourse in terms of introducing subquestions. Finally, we turn to the role of the pragmatic process of grounding on the establishment of subordination in discourse in the sense of Jasinskaja & Karagjosova (2020).

4.1 Verb semantics and discourse expectations

In the discussion of the sentential complements of psychological verbs (e.g. *annoy*) in Section 3.1.1, it was argued that such clause-embedding predicates establish an *Explanation* relation between the matrix clause and an embedded (or independent) clause:

- (41) a. That John ate soup in the evening annoyed Mary.
 b. John annoyed Mary. He ate soup in the evening.

As already mentioned in Section 3.1.1, Asher & Lascarides (2003) argued that Stimulus-Experiencer verbs like *annoy* evoke the interpretation of an *Explanation* relation in (41b). Moreover, they argue that these verbs are lexically determined to do so. They contended that although the second sentence in (41b) does not constitute a ‘plausible’ explanation for Mary’s annoyance based on domain knowledge, it is still preferably understood to be an explanation exactly due to the semantics of *annoy*. More precisely, Asher & Lascarides attributed this preference to the semantic nature of the Stimulus/subject argument of *annoy*, which, as stated above, introduces the cause of Mary’s attitude of annoyance. Asher & Lascarides analysed the subject as being semantically propositional in nature, in the sense that it carries a presupposition that the subject was involved in some eventuality which resulted in the annoyance in the Experiencer argument. The second clause in (41b) is thus interpreted as an *Explanation* because it may be taken to justify the presupposition that is introduced by *annoy* and left unspecified otherwise. The fact that the Stimulus argument may be realized as a *that* clause is consistent with this analysis (see also Bott & Solstad 2014, 2021).

Asher & Lascarides limited their observations to psychological predicates of the Stimulus-Experiencer type. However, Bott & Solstad (2014, 2021), Solstad

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& Bott (2022) argued that Experiencer-Stimulus predicates such as *fear* can be given a similar analysis (see also Cheung & Larson 2015) and provided experimental evidence that leaving what Asher & Lascarides characterize as a presupposition unverified (in Bott & Solstad’s 2014, 2021 analysis an underspecified propositional argument) evokes the expectation for an explanation. Thus, as mentioned above, when asked to provide a written continuation for prompts such as *John annoyed Mary* or *John feared Mary*, participants provided around 60% explanations (Bott & Solstad 2014, Kehler et al. 2008, Solstad & Bott 2022), which was around three times as much as the second-most frequent relation, *Result* (Kehler et al. 2008, Solstad & Bott 2022).

Interestingly, psychological verbs constitute a subclass of *Implicit Causality verbs* (henceforth, IC; Garvey & Caramazza 1974, Brown & Fish 1983, Ferstl et al. 2011), which have been linked to *discourse expectations* in psycholinguistic research (Kehler et al. 2008, Bott & Solstad 2021). As mentioned above, these verbs display interesting coherence patterns in discourse continuation studies (in addition to coreference patterns, that will not concern us here; see, for instance Ferstl et al. 2011, Garnham et al. 2021): Prompted to provide a continuation for, for instance, *John annoyed Mary*, participants produce overwhelmingly more *Explanations* than other continuations (Kehler et al. 2008, Bott & Solstad 2014, Kehler & Rohde 2019) as opposed to “non-IC verbs” such as ordinary agent-patient verbs. Beyond psychological verbs, similar properties have been observed for another class of IC verbs, so-called Agent-Evocator verbs, which include, among others, a subset of Fillmore’s (1969) judgment verbs:

(42) Mary punished Peter. He had stolen the money.

In (42), *punish* presupposes that Peter was involved in some eventuality which Mary considered to be punish-worthy (Fillmore 1969). This eventuality is a cause of Mary’s intention to punish Peter (see the analysis of reasons in Solstad 2010). Thus, all IC verbs evoking significantly more explanations may be taken to involve an explanatory relation between underspecified or presuppositional content and the eventuality denoted by the predicate itself. Put differently, the involved predicates come with an explanatory *slot* (Bott & Solstad 2021, Solstad & Bott 2022), which is assumed to be filled by material in discourse. Bott & Solstad (2021) argue that it is precisely the discourse expectations associated with the underspecified, or presupposed, semantic content that leads to the establishment of *Explanation* relations. As a means to avoid accommodation (Altmann & Steedman 1988, Zeevat 2000), subsequent or preceding discourse will be taken to verify the presupposition, leading to causal inferences even in far-fetched contexts as in (41).

Since both clause-embedding psychological predicates and the so-called ‘agent-evocator’ predicates display these preference patterns, the establishment of *Explanation* relations cannot be attributed to a correlation with syntactic subordination. Admittedly, it is an open question whether only subordinating discourse relations can be subject to discourse expectations or whether for instance *Narration* and *Occasion* relations may be expected for specific end-point-oriented predicates, such as transfer-of-possession predicates (Stevenson et al. 1994, Kehler et al. 2008).

4.2 Derivational morphology

Next to verb semantics, there are also cases of word derivational means leading to the establishment of subordination in discourse. One such case is described in Salfner (2018), who argues that the German suffix *-mäßig* (‘pertaining to’/‘in terms of’/‘concerning’) is used to mark delimiters (cf. Krifka 2008), which signal that only a discourse subquestion will be addressed by the speaker. For instance, the last utterance in (43) which contains the adverb *hardwaremäßig* (‘with regard to the hardware’) does not fully answer any of the possible questions in (43a–43c). Rather, it makes salient a set of alternative subquestions to a superordinated QUD and at the same time signals that the rest of the utterance is only going to address the subquestion specified by the *-mäßig*-adverb.

- (43) a. Wie ist der Laptop? (Salfner 2018: 232)
 ‘How is the laptop?’
 b. Ist der Laptop gut?
 ‘Is the laptop good?’
 c. Hat es sich gelohnt, einen so teuren Laptop anzuschaffen?
 ‘Was it worth buying such an expensive laptop?’
 d. Hardwaremäßig ist der Laptop super.
 ‘As for the hardware, the laptop is great.’

Moreover, depending on the propositional content of the utterance, a *-mäßig*-adverb may signal that the host utterance represents an answer to the current QUD that is quantitatively and, in some cases, also qualitatively unsatisfying.

4.3 Grounding in communication

We saw in Section 2.2.3 that one way of capturing what it means for a discourse unit to be subordinate to another discourse unit is by stating that the communicative goal of the superordinate unit cannot be reached before the goal of the

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subordinated unit is achieved. In defining what it means to achieve a communicative goal, Jasinskaja & Karagjosova (2020) follow Clark & Schaefer (1989), Clark (1996), Traum (1994) and other works that draw upon a notion of *grounding*, i.e. the process by which an addressee signals to the speaker that the communicative goal of the latter has been reached, for instance, by immediately performing the requested action or answering the question asked by the speaker. In particular, they adopt Clark's (1996) idea that grounding proceeds in four stages that correspond to four levels of action in communication, and consider the ultimate goal of an utterance to be reaching grounding at level 4:

1. vocalisation by the speaker, which should result in attention by the hearer;
2. presentation of the acoustic signal, which should result in the hearer's identification of the phonological string;
3. the speaker's meaning, which should result in the hearer's understanding;
4. the speaker's proposal, which should result in the hearer's uptake of the proposal.

Based on this notion of grounding in discourse, Jasinskaja & Karagjosova (2020) argue that the primary function of discourse subordination in terms of subordinating discourse relations is to handle encountered or anticipated grounding problems at different levels. Grounding failure at levels 1 and 2 would typically make the speaker simply repeat U_1 . Plain repetition is (in Hobbs' view) the trivial case of *Elaboration*. More interesting cases pertain to level 3 (problems of understanding) and level 4 (disagreement). Two of the more frequent reasons for failures to understand are problems of reference resolution (44) and lexical access (45). *Reformulation*, which according to Jasinskaja & Karagjosova (2020) is a subordinating relation, is a way to repair for this kind of problem.

(44) It's there.

[Where is 'there'? / It's WHERE?]

Behind the refrigerator.

(45) This piece begins with an anacrusis,

[What is 'anacrusis'?]

an unaccented note which is not part of the first full bar.

As for handling disagreement, Jasinskaja & Karagjosova (2020) argue that subordinating relations like *Evidence* or *Motivation* address actual or potential disagreements: A piece of information can be made easier to believe by pointing to

observable *Evidence* (46) or by reference to an authoritative source, while *Motivation* is needed if the hearer might refuse to fulfil a request or answer a question.

(46) John must have been here recently.

[Why do you think so?]

There are his footprints.

In all these cases, the discourse flow is “interrupted” because there is a problem that needs to be solved before the speaker can continue. It is in this sense that Jasinskaja & Karagjosova (2020) understand discourse subordination as motivated by the pragmatic process of grounding and involving interruption of a communicative plan to handle a communicative problem. This notion of interruption corresponds to Polanyi’s view on subordinated discourse units as interrupting the completion of an ongoing unit of discourse, rather than to the “true” interruptions in Grosz & Sidner (1986) and topic digressions discussed in van Kuppevelt (1995). On the other hand, the pragmatic notion of subordination proposed in (11) also captures van Kuppevelt’s notion of “satisfaction” involved in the question-triggering mechanism described in Section 2.2.2 where a topic-constituting question that has been answered unsatisfactorily gives rise to further subquestions until the original question has been answered. Reformulated in terms of (11), this subquestioning procedure is repeated until there are no potential grounding problems left that need to be dealt with in order to fully answer the main question.

5 Summary and outlook

In this chapter, we argued that subordination in discourse is a notion comparable to the syntactic phenomenon of clausal embedding. Although there is no uniform notion of embedding in discourse, we identified and discussed various concepts that have been proposed so far to capture different aspects of the hierarchical structure of discourse. As we attempted to show, the notions of subordination offered in the theories we discussed are both comparable and compatible with one another, and we believe that the attempt to integrate them into a unified approach to discourse structure will constitute a considerable and hopefully fruitful effort in future research in the field. Especially when it comes to the task of annotating discourse, several promising research projects which attempt to analyse discourse in terms of both QUD-structure, topical organization and (RST or SDRT) discourse relations are currently on their way, cf. Riester et al. (2021), Scheffler (2021), Hesse et al. (2020). Based on the various notions of subordination

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in discourse, we furthermore identified and discussed various correspondences and mismatches between syntactic embedding and discourse subordination.

While syntactic relations may be one way of realizing subordination in discourse, we showed that various semantic and pragmatic phenomena may also lead to the establishment of discourse subordination, such as verb semantics, derivational morphology and grounding in communication. There are further non-syntactic ways of expressing discourse-structural subordination or coordination that we did not discuss in this chapter. One such phenomenon that has implications for the notion of discourse subordination is modal subordination (Roberts 1989). At the same time, discourse structure in oral and written discourses is reflected by prosodic parameters such as pause duration and pitch range (cf. Mayer et al. 2006, Jasinskaja et al. 2004) and typographic means, such as bullet-lists (which can be described as coordinating structures that are altogether embedded) or headings and subheadings (which are treated in RST by means of the *Preparation*-relation that relates a heading as a Satellite unit to the entire text as a Nucleus, cf. RST website <http://www.sfu.ca/rst/>). Future research on discourse subordination will also have to take these aspects of discourse structure into consideration.

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Chapter 10

Processing embedded clauses

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This chapter highlights key issues in sentence processing that are informed by, and inform, the study of clause embedding. Topics include center-embedding, head-final embedding structures, gap filling, islands, and relative clause processing and modifier attachment ambiguities and preferences. Much of the study of how humans parse sentences has involved embedded structures of one kind or another. Psycholinguists have exploited the complexity created by embedding to probe mechanisms of incremental parsing, including the role of memory, prediction, and language experience (usage). The overview is presented through a 'linguistic' lens through which developments in theories of human sentence processing are viewed. The chapter highlights areas where nuanced and formally-informed grammatical properties of embedding cross-linguistically have impacted our understanding of sentence processing.

1 Introduction

To understand how a system works, it is a good strategy to push it to its limits. As with other domains of inquiry, this is true in the psycholinguistic study of sentence processing. Complex sentences tax a human's parsing abilities, their comprehension, and their working memory capacity in ways that can reveal the underlying mechanisms of sentence processing. Embedded clauses, our topic, are perhaps the richest source of complexity in natural language. Like other types of embedded phrases, embedded clauses allow for sentences of arbitrary length and complexity. In fact, embedded clauses do this better than any other kind of recursive syntactic option available because they admit the widest range of elements within them. These expansions interrupt relationships which help hold



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a sentence together, like subject-verb agreement and anaphoric and movement dependencies. For this reason, embedded clauses are so routinely used as a tool to investigate sentence processing that they could be classified as the *drosophila* of the science of sentence processing.

Since embedded clauses have played such a key role in the development of sentence processing, this survey can touch on just a few issues that stand out, both for their impact on the processing literature and their interest to the linguist studying clause embedding. This chapter is not a general introduction to the field of sentence processing. For that I can recommend, for example, the chapters in van Gompel (2013). This chapter, moreover, does not evaluate theories of sentence processing, because the significance of research on embedded clauses is too varied to serve this purpose. Rather, the purpose of this chapter is (i) to highlight some of the key debates in sentence processing that hinge intimately on the syntax and semantics of clause embedding, and (ii) to promote interaction between sentence processing research and the wealth of knowledge from deep analyses of the grammatical structures of embedding. We will see a number of cases where such interaction has been insightful.

It is useful to start with a few key principles of sentence processing, in particular the incremental nature of the task and the information that parsing decisions might be made upon. This will constitute the first demonstration of the way psycholinguists have exploited clause embedding to probe the nature of the sentence processor. Section 3 delves into the topic of doubly centre-embedded sentences. Satisfactorily capturing the difficulty of such sentences is a kind of litmus test for every sentence processing theory that has come along. So the approaches to this one phenomenon provide a good overview of some prominent theories of sentence processing. Section 4 discusses the role of clause embedding in the processing of filler-gap dependencies (e.g. *wh*-movement), in particular those filler-gap dependencies that cross clause boundaries and islands. In Section 5, the role of embedding in syntactic attachment ambiguities is discussed, both in terms of whether material is attached to matrix or embedded clauses and where embedded clauses themselves preferentially attach. Throughout, we will see that the rich syntactic and discourse-pragmatic properties of embedded clauses play a key role in how they are used by psycholinguists to investigate sentence processing.

2 Some basics of sentence processing with reference to embedding

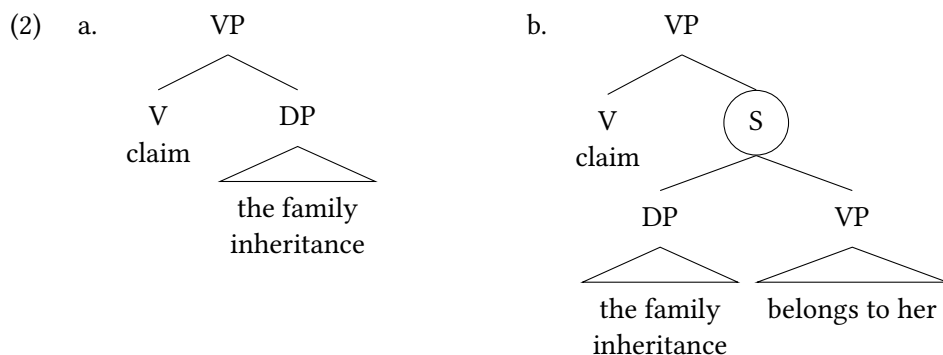
The investigation of human sentence processing (what I will simply call ‘processing’) seeks to uncover the cognitive mechanisms that give structure and meaning to incoming linguistic material, at the phrase and sentence level.¹ One of the central challenges for the processor is that material comes incrementally. This limitation can often lead to temporary ambiguities, so that the material available at one point in an utterance is compatible with more than one continuation. These are *Garden Path Sentences* (Bever 1970): sentences with a temporary ambiguity that can lead the reader to pursue a syntactic parse that is ultimately incorrect once the sentence unfolds. The classic example, not irrelevant to embedding phenomena, is one in which an embedded modifier (a reduced relative) is mistakenly taken for a matrix clause: *The horse raced past the barn fell* (See Section 5).

The abundance of research into Garden Path Sentences demonstrates that the processor does not entirely “wait” to assign an analysis at points of ambiguity (e.g. *raced*), but either commits to or privileges one of the possible analyses. What guides these commitments, and the extent to which these commitments are made or deeply encoded, and whether competitor analyses are entertained, constitute the major topics in this area for the last 40 years. Relevant to the present chapter are temporary ambiguities of the sort shown in (1) (Frazier 1978): the verb *claim* can take either a DP complement or an embedded clause complement. These are named, after terminology of earlier times, NP/S ambiguities.

- (1) Frazier & Rayner (1982) *Minimal Attachment sentences*
 - a. The lawyers think his second wife will claim *the family inheritance*. (DP complement)
 - b. The lawyers think his second wife will claim [*the family inheritance belongs to her*]. (Embedded clause)

¹I do not mean to suggest a clear division between processing and “grammar”, a set of abstract rules separate from the implementation in comprehension and performance. See Lewis & Phillips (2015) for recent discussion and opinions. I should also note that this chapter draws from ‘on-line’ studies of real-time sentence comprehension (e.g. reading time measures) as well as acceptability judgment studies. I make no attempt to compare these methodologies, since in the studies discussed below, the research goals are varied. Moreover, the kinds of models I refer to will only be explored at a basic level, highlighting linguistic aspects primarily over foundational psycholinguistic matters, like the role of memory. For recent computational approaches to sentence processing, see for instance Vasishth & Engelmann (2021).

In a landmark study, which recorded readers' eye-movements as they read (eye-tracking while reading), Frazier & Rayner (1982) found longer reading times for the clausal complement sentences overall and, of particular interest, longer at the point of disambiguation. This suggests that readers were "garden-pathed" in (1b), pursuing an initial analysis where the DP encountered after the verb is taken to be an object, not the subject of the embedded clause. In one influential theory, the Garden Path Model (Frazier 1978, 1987a), this outcome is predicted from general structure building principles. One of these principles is *Minimal Attachment*, which compels the processor to prefer structures with fewer nodes. The DP complement parse (which is correct for (1a)) involves fewer nodes than the embedded clause parse (1b), the latter requiring *at least* a sentential node (S) as is shown in (2b).²



At the point of processing the DP *the family inheritance* in (1), Minimal Attachment favours the structure in (2a) over (2b). When the verb *belongs* is encountered, reanalysis is required, leading to elevated reading times.³

Classic Garden Path theory prioritizes syntactic parsing principles, and in its strictest formulation, only allows other information to influence parsing in a later stage (so-called *two-stage* or *serial processing*). Other approaches allow a variety of linguistic and non-linguistic information to guide parsing, even in the earliest

²This is the most conservative of phrase structures; most theories recognize a head for the S-node, which is T (Tense) or Infl (inflection). Furthermore, these sentences may contain a null complementizer, hence may in fact be embedded CPs not TPs. We return to the role of the complementizer below.

³A key feature of the garden path is the lack of an overt complementizer, which leads to other interesting predictions and manipulations (Holmes et al. 1987, Rayner & Frazier 1987, Ferreira & Henderson 1990, Trueswell et al. 1993).

moments, including lexical and pragmatic information (Ford et al. 1982, MacDonald et al. 1994, Altmann & Steedman 1988). These are broadly categorized as *constraint based* approaches, since they let a diversity of constraints be assessed at the same time. One important piece of information that the processor could use is syntactic subcategorization, such as whether a verb selects for a DP/NP or an embedded S. A number of studies from the 1990s argued that when the subcategorization preferences of verbs were taken into account, garden paths were ameliorated or disappeared. Trueswell et al. (1993) established, through an independent norming study, a collection of verbs that are biased toward embedded clausal complements ('S-biased' like *insist*, *reveal*) and those that are biased toward nominal complements ('NP-biased' like *observe*). Using a number of methodologies, including reading while eye-tracking, they found that garden path effects were present at the disambiguating predicate in the case of NP-biased verbs, but not S-biased verbs. This suggests that subcategorization information is used early (see also Mitchell & Holmes 1985, Osterhout et al. 1994, Garnsey et al. 1997, Jennings et al. 1997).

The experimental record, however, is not so clear cut. Pickering et al. (2000) examined the NP/S ambiguity but unlike previous experiments (cf. Garnsey et al. 1997) they compared manipulations just between S-biased verbs, such as *realize* in (3).⁴

(3) Pickering et al. (2000) Experiment 1 stimuli

- a. The young athlete realized her potential one day might make her a world-class sprinter.
- b. The young athlete realized her exercises one day might make her a world-class sprinter.

The crucial manipulation is the DP following *realize*. In (3a) there is a plausible NP object complement, but not in (3b) (*#She realized her exercises*). On a sentential complement analysis, both NPs are plausible. The logic of the experiment is this: if subcategorization information is available early, then readers will pursue a sentential complement analysis and the plausibility difference will not matter since these are S-biased verbs. This is not what Pickering et al. (2000) found however: there were increased reading times for the implausible (3b) at the post-verbal NP, suggesting readers pursued an complement clause analysis. Furthermore, downstream reading times were reversed, such that (3a) showed greater reading times

⁴Their stimuli also controlled for prior context and the likelihood of the matrix subject (cf. Garnsey et al. 1997).

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than (3b) at the disambiguating modal *might*. This ‘cross-over’ pattern suggests a difficult reanalysis in (3a) from a plausible DP object analysis to a sentential complement analysis (see also Pickering & Traxler 1998). Pickering et al. (2000) propose a processing model that does incorporate frequency information but is not wholly guided by it. They suggest that pursuing an NP complement analysis is more readily falsifiable, since verbs and their objects are subject to pragmatic, syntactic, and semantic co-occurrence restrictions (e.g. selection) whereas verbs and the subjects within their complement clauses are not.

The question of what information is used in incremental sentence processing remains a big topic in the field, and garden path effects like the NP/S ambiguity continue to play a role in their development and formulation (see van Gompel 2013 for a recent overview). In the next section, we turn to another – perhaps *the* – embedded clause phenomenon central to the development of psycholinguistics.

3 Centre-embedding

A touchstone topic in psycholinguistics are sentences with doubly centre-embedded (“nested”) relative clauses (Chomsky 1957, Yngve 1960, Chomsky & Miller 1963a,b, Miller & Isard 1964). The example in (25a) has a relative clause *who the nurse supervised* modifying a noun *intern* which is itself part of a relative clause *who the intern...bothered* modifying the matrix subject *the administrator*. Structures with nesting of two-levels such as this (compared to one-level of centre embedding in (4b)) become impossible for most people to process, even if the propositional content is coherent – note for instance that (4c) has the same words and the same meaning but does not involve double centre-embedding.

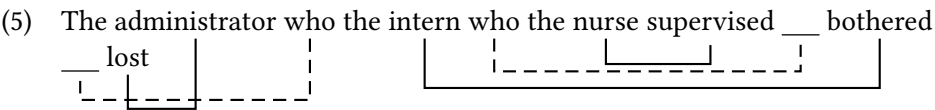
- (4)
- a. The administrator who the intern who the nurse supervised bothered lost the medical reports.
 - b. The administrator who the intern bothered lost the medical reports. sprinter.
 - c. The intern who the nurse supervised had bothered the administrator who lost the medical reports.

The breakdown experienced in (4a) is a type of *processing overload*. It is distinct from other difficulties in sentence processing, such as classic garden paths like *The horse raced past the barn fell*, in that readers and hearers simply cannot overcome it.⁵ A skilled reader or a trained linguist, aware of the intended structure,

⁵There are certainly centre-embeddings that also garden path the reader. Blumenthal (1966), for

typically cannot reach a successful parse, although we will see how several types of manipulations can ameliorate such sentences, up to the point of acceptability and comprehensibility. In the early days of generative grammar, doubly centre-embedded sentences were important for Chomsky’s argument for the distinction between *competence* and *performance*. From a competence perspective, a doubly centre-embedded sentence is the result of a legitimate, iterative application of the rule of relative clause formation. The problem, it was surmised, must then be with the performance systems.

Most approaches attribute the processing overload to the fact that such sentences involve suspended syntactic dependencies which tax our working memory beyond its capacity.⁶ The term *syntactic dependency* is a theory-neutral way of describing relations that typically hold between heads and dependent phrases. One syntactic dependency is that between a predicate and its semantic and/or syntactic arguments (subject, object, etc.). Another syntactic dependency, an unbounded one, is that between a *wh*-phrase (such as a relative pronoun) and its gap. In doubly centre-embedded sentences, subjects are separated from their predicates by complex material and relative pronouns (or the associated heads of the relative clauses) are separated from their gaps by complex material. The syntactic dependencies of a doubly centre-embedded sentence are diagrammed in (5); the predicate-argument dependencies are indicated by solid lines and the dependencies between relative pronouns and their gaps (indicated with an underscore) are shown by a dashed line.



Chomsky & Miller (1963a,b), Yngve (1960) and Kimball (1973) explained the difficulty of centre-embedding in terms of incomplete phrase-structure rules that must be maintained during the processing of the sentence. In the formalism of the time, subject-predicate relations were generated by a phrase structure rule of a sort where S expands into daughters NP and VP ($S \rightarrow NP VP$). At the point just

instance, pointed out that without relative pronouns or complementizers, centre-embeddings give rise to a conjunction parse garden path (e.g. *The administrator, the intern, and the nurse*). But as Fodor & Garrett (1967) points out, this option is precluded with overt relative pronouns. Most of the studies reported retain the relativizers.

⁶Some accounts of centre-embedding difficulty might fall outside this classification, including the idea of *perspective shift* in MacWhinney 1977, which essentially penalizes shifting the subject from one clause to the next, as happens with object relatives modifying a subject.

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before reaching the verb *supervised*, there are three opened S-rules that have not been completed with a VP. Keeping this many unresolved dependencies open is just too much for the processor to handle.

Hawkins (1990, 1994) emphasized how the size of constituents – measured in terms of word count – delays what he calls the “recognition of phrases”, the CONSTITUENCY RECOGNITION DOMAIN. The preferred order or nominal and clausal argument in (6) illustrates this (cf. Stowell 1981). Recognizing the domain of the VP in (6a) – getting evidence for each of its immediate constituents V, PP and CP – only takes three words: *to*, *everyone*, *that*. In contrast, recognizing the VP in (6b) requires parsing all the words in the CP (four) plus the head of the PP (*to*).

- (6) a. They [_{VP} said [_{PP} to everyone] [that it was raining]]
 b. They [_{VP} said [that it was raining] [_{PP} to everyone]]

Complexity increases with the amount of words before a constituent is “recognized.” In centre-embedded sentences, sentential constituents cannot be recognized until their main predicate is encountered. The intervention of complex NPs (those with relative clauses) massively delays this and hence the complexity. Hawkins develops a far-reaching program that relates processing considerations such as these to the grammar of the world’s languages. The appeal of this program is that it recognizes that performance constraints can be “conventionalized” or “grammaticalized” to generate the distribution of word order patterns in natural languages.

Returning more concretely to centre-embedding, these early characterizations just outlined turn out not to be sufficient. There are four key observations about centre-embedding that a processing theory needs to capture. I take each in turn.

3.1 Locus of processing difficulty

If the difficulty of centre-embedded structures arises from unresolved S-rules, we would expect that the processing overload would emerge as the opened rules “pile up” – that is, before the verbs are encountered. This is not, however, what the experimental record shows. Rather, processing breakdown occurs as the reader exits the most deeply embedded clause, at the point of the medial verb (e.g. *bothered* in (5)). This has been confirmed by various methodologies (Grodner & Gibson 2005, Christiansen & MacDonald 2009, Gibson & Thomas 1999, Vasishth et al. 2010). In fact, the difficulty at the medial verb in a doubly centre-embedded sentence in English is so great that readers will perceive an ungrammatical sentence that simply omits this verb (7b) as more acceptable than its grammatical

counterpart in (7a). This is known as the Missing VP effect (Frazier 1985, Gibson & Thomas 1999):

- (7) a. The apartment that the maid who the service had sent over was cleaning every week was well decorated.
- b. *The apartment that the maid who the service had sent over was well decorated.sprinter.

The Missing VP effect is one piece of evidence that the processing overload occurs at the verbs that complete the subject-predicate dependencies. A theory of the processing complexity of English double centre-embedding needs to capture this.⁷

3.2 The role of relative clauses

A second important fact about double centre-embedding that speaks against a simple account involving unresolved subject-predicate dependencies is that a doubly centre-embedded sentence involving noun complement clauses (CC), like the complements *fact* and *rumour* in (8), are relatively easier to parse (Gibson & Thomas 1997):

- (8) The fact that the rumour that the PM resigned is false should not surprise you.

If the processing difficulty were simply due to the fact there are three open expectations for predicates, then we would expect doubly centre-embedded complement clause sentences like (8) to show the same processing difficulty as doubly centre-embedded relative clause sentences. This is not the case, which means that relative clauses are crucial to the processing overload. Of course, a relative pronoun introduces its own requirements, for a gap (see Section 4 for a fuller discussion of gap filling across clauses). We could add that as an open, unresolved expectation. This would mean that doubly centre-embedded relative clauses like (5) in fact introduce five unresolved dependencies before the most deeply embedded verb is encountered. This would capture the fact that (5) is harder to process than (8).

⁷The Missing VP effect has been found in French (Gimenes et al. 2009) and German (Häussler & Bader 2015).

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3.3 The distance of the dependency

Still, even this is not a sufficient characterization of the difficulty of double centre-embedding. There is a third important observation about centre-embedding due to Cowper (1976) that is a major desiderata for processing theories. Both sentences in (9) involve centre-embedding to a depth of two, but while (9a) causes processing overload, (9b) does not (Gibson & Thomas 1997). What makes them different is that in (9a), the inner-most clause is a relative clause (RC) while the outer-most clause is a complement clause (CC) to the noun *fact*. In (9b) it is the reverse: the inner-most clause is a CC embedded in a RC.

- (9) a. The fact [that the employee [who the manager hired ____] stole office supplies] worried the executive.
 b. # The executive [who the fact [that the employee stole office supplies] worried ____] hired the manager.

Both sentences involve the same number of unresolved expectations just before encountering the most deeply embedded verb (*hired* in (9a), *stole* in (9b)), namely four (three subject seeking a predicate and one relative pronoun seeking a gap). And yet (9b) causes processing difficulties while (9a) does not.

The important difference is the distance across which the relative pronoun-gap dependency spans. In (9a) it is a local *wh*-dependency whereas in (9b) it is long-distance. The role this plays in centre-embedding, then, must be captured by processing theories.

3.4 Nature of intervening DPs

A final and equally influential observation is that doubly centre-embedded relatives in which the inner-most relative has an indexical pronoun (*I*, *we*, *you*) as a subject are easier to process (Bever 1970, Bever 1974, Kac 1981, Gibson 1991, Klender 1998, Warren & Gibson 2002):

- (10) a. The reporter [everyone [*I* met] trusts] said the president won't resign yet.
 b. The student [who the professor [who *I* collaborated with] had advised] copied the article.

For instance, Warren & Gibson (2002) found that a doubly centre-embedded sentence where the innermost subject was an indexical pronoun was rated as more acceptable than those in which the same position contained a proper name, definite description or third person pronoun.

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Perhaps the first explicit theory to capture each of the above observations about centre-embedding is Gibson's *Dependency Locality Theory* (DLT) (Gibson 1998, 2000).⁸ DLT is a general processing theory with broad coverage beyond just centre-embedding. DLT has two components: a storage component and an integration component. Integration involves the retrieval of the head of a dependency: an argument for a predicate or a relative pronoun (or head noun) for a gap position in a relative clause. Retrieving an element from memory involves reactivating it. The level of reactivation needed will depend on the extent to which the element's activation level has decayed, and this can be affected by the amount and complexity of intervening material. Gibson capitalizes on the fact that the kind of intervening material that seems to most tax the processor are discourse referents – entities like individuals and events that are introduced in a discourse structure (Kamp 1981, Heim 1982). It has been independently established that processing a DP depends on the nature of the referent – in particular how “accessible” (Gundel et al. 1993) it is (Haviland & Clark 1974, Halliday & Hasan 1980, Garrod & Sanford 1982, Garrod 1994). In DLT, processing new discourse referents incurs a cost.⁹ These include referential DPs, proper nouns, third person pronouns, as well as tensed verbs (which introduce reference to times Partee 1973). Importantly, indexical pronouns are not discourse referents so they do not incur a processing cost. Integration costs, defined in (11), are dependent on the number of discourse referents that intervene between the site of integration and the target of retrieval.

(11) *DLT structural integration cost* (Gibson 2000: 105: (9))

The structural integration cost associated with connecting the syntactic structure for a newly input head h_2 to a projection of a head h_1 that is part of the current structure for the input is dependent on the complexity of the computations that took place between h_1 and h_2 . For simplicity it is assumed that 1 EU [Energy Unit] is consumed for each new discourse referent in the intervening region.

Object relative clauses provide a good demonstration of DLT's structural integration costs. It has been overwhelmingly established that object-gap relatives as in (12a) incur greater processing difficulty than subject gap relatives (12b)

⁸Gibson (1998) proposed two variants of the theory, one of which was the Syntactic Prediction Locality Theory (SPLT). I concentrate on DLT as it is reported in Gibson (2000).

⁹Gibson uses the term “new discourse referent.” Typically we use that term for expressions like indefinites that introduce a discourse referent. Definite descriptions often have familiar referents, although the role of accommodation in psycholinguistics studies is relevant here (Schwarz 2007). I think “new discourse referent” should simply be understood here as discourse referent.

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(Holmes 1973, Hakes et al. 1976, Wanner & Maratsos 1978, Ford 1983, Waters et al. 1987, King & Just 1991, among many others).

- (12) a. The intern who the nurse supervised ____
 b. The nursed who ____ supervised the intern.

At the gap site, the object relative pronoun *who* in (12a) is retrieved and integrated as the object of *supervised*, but there are two discourse references within the span of this dependency (the DP *the nurse* and the verb *supervised*). The structural integration cost at the gap is therefore 2 EUs. At the gap site in a subject relative (12b), there are no integration costs since there are no discourse referents between the head and tail of this dependency.

In doubly centre-embedded object relative clauses, repeated below from (5), the second gap is the site of maximal integration cost because there are five discourse referents intervening: *the intern*, *the nurse*, the second relative pronoun *who*, and the verbs *supervised* and *bothered*.

- (13) The administrator who **the intern who the nurse supervised** ____
 bothered ____ lost |
 |_ _ _ _ _|

Gibson's account captures each of the four observations discussed above. The fact that costs are highest at the gap site, which is adjacent to the embedded verbs, predicts the location of the processing difficulty. Cowper's (1976) contrast in (9) between CCs embedded in RCs versus RCs embedded in CCs is predicted, since the number of discourse referents within the span of the relative pronoun-gap dependency is greater in the former than the latter. Finally, the ameliorating role of indexicals (10) follows from the definition of discourse referents: indexical are, in a certain sense, "cost-free" since the speaker and hearer are always easily accessible. The presence of an indexical pronoun instead of a third person pronoun, a definite description or a proper noun, will reduce the integration cost.

The second component of DLT, the storage component, bears more similarity to earlier theories of centre-embedding difficulty. There is a cost for storing a prediction needed to resolve a syntactic dependency and this can accrue. For instance, each subject requires a predicate, and this sets up an expectation which is added to working memory. Relative pronouns also set up an expectation for a gap. In (5), just before the word *supervised*, there are 5 expectations – which in DLT incur a "5-unit" storage cost. This should be compared to the non-nested relatives in (4b), where the storage costs never surpass 3 units – before reaching *bothered* there are two DP arguments requiring a predicate and one gap expectation. The contrast in (9) is also reflected in different storage costs: since a CC

does not involve a relative pronoun–gap dependency, it will incur one less unit of storage costs than an RC.

Support for storage costs is less robust, but has been argued for by Chen et al. (2005) and Gibson et al. (2005). Chen et al. (2005) looked at the reading time costs associated with expectations initiated by a DP for its predicate. What makes this study particularly informative is that sentential complements were used as the intervening material, which avoids the added complexity of a relative which would itself have a gap. In (14a), the expectations of each subject DP for a predicate are fulfilled immediately. In (14b), however, the DP subject *the implication* sets up an expectation for a predicate but this is only reached after processing its sentential complement. In (14b) there are two DP subjects whose expectation for a predicate must be held while processing the innermost sentential complement (which is identical to that in (14b))

- (14) a. The employee realized that the boss implied that *the company planned a layoff* and so he sought alternative employment.
 b. The employee realized that **the implication** that *the company planned a layoff* was not just a rumour.
 c. **The realization** that **the implication** that *the company planned a layoff* was not just a rumour caused a panic.

As predicated by storage costs, reading times at the critical region *the company planned a layoff* were greater in (14b) than (14a) and greatest in (14c), where two subjects' expectations had to be held. In the section below on cross-linguistic investigations of embedded clauses, we return to the role of storage costs vs. the role of integration (or retrieval costs).

Since Gibson's seminal work on centre-embedding, the retrieval component of integration has led to the greatest interest among psycholinguistics. Over the last two decades or so, increased attention has been paid to research on memory in other domains of cognitive psychology. One prominent model holds that retrieval of items in memory involves cue-based recall from a content-addressable memory (Lewis & Vasishth 2005, Lewis et al. 2006). This is best demonstrated with pronominal anaphora: a pronoun, such as *she*, bears the features *singular* and *feminine* and these serve as a cue to recall antecedents in memory. So items in memory that bear either of these features will be activated, although the one that matches best (i.e. *the girl* vs. *the boy*) will attain the highest activation level and so be more readily retrieved. One thing that makes this model different from DLT is that a content-addressable memory does not involve a search related to a

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syntactic parse tree. Items in memory are not “syntactically” structured. Furthermore, elements in working memory compete for retrieval, with retrieval being easier the better and more recent the match, and therefore more highly activated, the target. So when a gap site cues for the retrieval of a filler, other DPs constitute potential targets. This will make object gap relatives harder than subject gap relatives because there is simply a more recently activated competitor for the head noun (i.e. the embedded subject) in object gap relatives. In addition, competitors that are more similar to the target can interfere (Gordon et al. 2001, Traxler et al. 2002, Gordon et al. 2004), thus causing processing difficulty. This *similarity-based interference* straightforwardly captures why indexicals ameliorate double centre-embedding: they are so distinct from the kinds of DPs that serve as heads of relative clauses (or their associated relative pronouns) that they do not compete in cue-based recall. Gordon et al. (2001, 2004) examine how DPs of different types (not just indexical pronouns) do and do not interfere in the retrieval of the head of the dependency. Lewis & Nakayama (2002) explore the effects of similarity factors, such as morphological case, in Japanese centre-embeddings. Models of sentence processing that implement these observations within broader research on memory can be found in Dyke & Lewis (2003), Lewis & Vasishth (2005), Dyke & McElree (2006), Vasishth & Lewis (2006) among others. As these theories are general and applicable to a range of dependencies (not just those that arise in centre-embedding) I refer to the reader to these works. Nonetheless, it can safely be said that the work by Gibson and others on centre-embedding greatly motivated work that embeds sentences processing in the larger cognitive science of memory.

Yet another approach to the difficulty of centre-embedding, among other processing difficulties, has been proposed within *Surprisal* theory (Hale 2001, Levy 2008). In this framework, the frequency of a structure matters to the expectations of the processor. When incoming material requires pursuing a less-frequent structure, processing difficulties arise. Object-gap relatives following a matrix subject are rare; encountering two such structures consecutively is even rarer (Gildea & Temperley 2007, Karlsson 2007). Space prevents doing justice to these approaches; for an overview of Surprisal Theory, see Levy (2013).

3.5 Embedding cross-linguistically and cross-constructionally

The preponderance of experimental work on centre-embedding in English is limited to clausal dependents of nominals: relative clauses and sentential com-

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plements.¹⁰ In robustly head-final languages, centre-embedding is routine. In Japanese, for instance, which has SOV word order, even one level of clause-embedding constitutes centre-embedding. In (15a) the embedded clause intervenes between the matrix subject and its verb. The example in (15b) involves two levels of centre-embedding:

- (15) a. *Bebiisittaa-ga [ani-ga imooto-o ijimeta to]*
 babysitter-NOM [older.brother-NOM younger.sister-ACC bullied that
 itta
 said
 ‘The babysitter said that my older brother bullied my younger sister.’
- b. *#Obasan-ga [bebiisittaa-ga [ani-ga imooto-o*
 aunt-NOM [babysitter-NOM [older.brother-NOM younger.sister-ACC
 ijimeta to] itta to] omotteiru
 bullied that] said that] thinks
 ‘My aunt thinks that the babysitter said that my older brother bullied
 my younger sister.’

Babyonyshev & Gibson (1999: 424) report that the doubly nested structure is perceived as more complex (and append the # sign) than the singly nested structure (see also Uehara & Bradley 1996). Consistent with DLT, Babyonyshev & Gibson (1999) found an effect of transitivity, such that centre-embedded sentences whose inner-most clause was transitive were rated as harder to understand than minimally different sentences where the inner-most clause was intransitive. This follows on Gibson (1998)’s DLT, since it reduces the number of dependencies being processed and the number of discourse referents introduced within the over-arching dependencies. Centre-embedding have also been examined from the perspective of cue-based retrieval models, both in Japanese (Lewis & Nakayama 2002) and Hindi (Vasishth 2003).

¹⁰These are not the only sources for centre-embedding in English, Adjunct clauses can be self-embedded:

- (i) Because if when the baby is crying, the mother gets upset, the father will help, the grandmother can rest easily.

Processing overload is similarly felt in these cases. The dependencies here are at the clausal level, rather than of the predicate-argument type in the classic centre-embedded sentences discussed above. Moreover, they vary in whether there is a filler-gap dependency: *if*-clauses do not contain a gap whereas temporal *when*-clauses do (Geis 1970). To my knowledge, these embeddings have not received the same amount of attention in the recent psycholinguistics literature.

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In recent years, a number of studies have revealed that sometimes centre-embedding can facilitate processing. These findings have been given the moniker *anti-locality effects*. Konieczny (2000) found in a self-paced reading experiment of German, a verb-final language, that the matrix verb *hingelget* was read faster when it followed the embedded relative clause as in (16a) than when it preceded the (extraposed) relative clause in (16b):

- (16) a. Er hat das Buch, [das Lisa gestern gekauft hatte], hingelegt.
He has the book, that Lisa yesterday bought had laid.down.
'He has laid down the book that Lisa had bought yesterday.'
- b. Er hat das Buch hingelegt, [das Lisa gestern gekauft hatte].
He has the book, laid.down that Lisa yesterday bought had .
'He has laid down the book that Lisa had bought yesterday.'

Konieczny (2000: 644) suggests that a relative clause “preceding the verb might just provide enough time” for arguments that are being processed to help build predictions about the verb to come (that it is coming and what its argument structure might be), and thus facilitates reading time when that verb is encountered (See also Konieczny & Döring 2003). Vasishth & Lewis (2006) found similar effects in Hindi doubly centre-embedded constructions. In a self-paced reading experiment they compared doubly centre-embedded control constructions like (17) to similar constructions in which material was added between the object and verb of the inner-most clause, indicated by “intervening material” in (17). The intervening material was of several types, including a subject-gap relative clause with a DP object.

- (17) Sita-ne Hari-ko Ravi-ko kitaab-ko *intervening material* khariid-neko
Sita-ERG Hari-DAT Ravi-DAT book-ACC buy-INF
bol-neko kahaa.
tell-INF told
'Sita told Hari to tell Ravi to buy the book.'

Gibson (2000)'s DLT predicts increased processing difficulty at the inner-most verb *buy* when intervening material is present, particularly intervening material like the relative clause that contains discourse referents. Contrary to this prediction, Vasishth & Lewis (2006) found faster reading times on this verb in the presence of intervening material. They develop a model in which intervening material heightens the expectation for a verb, thus facilitating its integration. Ensuing research has established the existence of both locality and anti-locality

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effects in the same language and study (Jaeger et al. 2008, Demberg & Keller 2008, Vasishth & Drenhaus 2011, Levy & Keller 2013).

Retrieval, recall, is not the only possible processing difficulty that centre-embedding poses. Gibson's DLT also included a storage cost: encountering a DP subject predicts a verb, and as such predictions stack up (as they do with centre-embedding) processing difficulty might increase. This is a different prediction from retrieval approaches – that is, the integration component of DLT or more recent cue-based retrieval models – which predict difficulty at the retrieval site (the verbs or their close-by gaps). These predictions were tested by Nakatani & Gibson (2010), which is notable as the first on-line processing study of Japanese centre-embedding. For a doubly centre-embedded sentences like (18) (with a schematization just above it), a retrieval-based account predicts equal costs on the DPs but increasing costs on the verbs, with V1 being the most difficult, since it has to retrieve its subject across two other candidate NPs (NP2, NP3).

- (18) [NP1 [NP2 [NP3 V3 COMP] V2 COMP] V1]
 [syoki-ga [daigisi-ga [syusyoo-ga utatanesita
 [secretary-NOM [congressman-NOM [prime minister-NOM dozed
 to] koogisita to] hookokusita]
 COMP] protested COMP] reported]
 'The secretary reported that the congressman had protested that the
 prime minister had dozed.'

In contrast, storage predicts processing difficulty as NPs accrue: for (18) the most costly region would be that corresponding to NP3 because here there are the stored expectations of three verbs. We would expect difficulty to decrease as each dependency is completed. So unlike the retrieval approach, the verbs should be increasingly easy to process moving rightward. The results of a self-paced reading experiment revealed that participants slowed down as the NPs stacked up and sped up as the verbs were encountered, supporting a storage-based view.¹¹

The overall picture, then, shows both effects of storage (the maintenance of syntactic predictions) and effects of retrieval, but in different languages. While Hindi, German and Japanese centre-embedding do not pose processing difficulty for retrieving the heads of the dependencies, there remains the evidence of retrieval costs in English centre-embedding (as noted above from Grodner

¹¹Nakatani & Gibson (2010) also discuss the predictions of *Surprisal Theory*, which is compatible with this result but other details of their study were more equivocal on separating the predictions of a DLT-style storage theory with *Surprisal theory*.

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et al. 2005; see Santi et al. 2019 for a recent replication of the effect for single embeddings). One possibility, suggested by Nakatani & Gibson (2010: 108) is that the type of dependency matters: retrieval difficulty arises in relative clause constructions precisely because they are filler-gap configurations. The German, Hindi and Japanese dependencies, where no retrieval effects are found, involve predicate-argument dependencies. The idea then is that storage costs correlate with predicate-argument dependencies and retrieval costs correlate with gap-filler dependencies. This suggestion, however, is somewhat at odds with the full experimental record: both Grodner et al. (2005) and Santi et al. (2019) find elevated reading times at the matrix predicate in English sentences with relative clause embeddings, and this is presumably where the subject-predicate dependency is resolved. At present, I do not know of a resolution to the various findings of locality and anti-locality.

The value of cross-linguistic experimentation should be apparent by now. What must not be overlooked is the value of examining a range of embedded constructions, even in one language. As noted in footnote 9, relative and complement clauses are by far the most studied types of embedding, especially in terms of centre-embedding. But interesting insights – both for time-course processing theories and structural analyses – arise from studying other types of embedding. A case in point are comparative clauses. Grant (2013) examined deviant sentences like (19a), in which a comparative clause modifies a matrix subject. The deviance is surprising since the version where the comparative clause is extraposed (19b) is perfectly acceptable.

- (19) a. *?/# More boys than girls ordered steak ordered salad
b. More boys ordered salad than girls ordered steak

Osborne (2009) develops a grammatical theory to rule out (19). Grant shows that rather than complicate the grammar, the unacceptability of (19a) can be profitably understood as a type of garden path. The comparative *than* is preferentially parsed as taking a DP sister rather than a clausal element, and so upon encountering another VP the parser aborts. When the element after *than* disambiguates, as with the proper noun in (20a) in comparison to the bare plural in (20b), processing is easier.

- (20) *Grant (2013) Experiment 1 stimuli (subset)*
a. More nurses than Percival thanked cared a great deal about their jobs.
b. More nurses than patients thanked cared a great deal about their jobs.

Structural theorists can profit from results such as these, as they both clarify and, in this case, simplify what needs to be hard-coded in the grammar. We will see an analogous situation – island phenomena – where the reduction of a perceived unacceptability to processing difficulty is more controversial.

While we are on comparatives, it is relevant to highlight another fact that Grant (2013) discovered. Unlike in the canonical relative clauses as we saw above, subject gaps in comparative clauses are more difficult to process than object gaps. In eye-tracking studies, Grant compared subject vs. object gaps in comparatives in sentence internal (or “base”) position (21a), (21b) with subject vs. object gaps in extraposed comparative clauses (21c), (21d):

(21) *Grant (2013) Experiment 2 stimuli*

- a. Object Extraction, Base Position
More friends than we talked to were at the party, according to the pictures on Facebook.
- b. Object Extraction, Extraposed
More friends were at the party than we talked to, according to the pictures on Facebook.
- c. Subject Extraction, Base Position
More friends than talked to us were at the party, according to the pictures on Facebook.
- d. Subject Extraction, Extraposed
More friends were at the party than talked to us, according to the pictures on Facebook

There were faster reading times for object gap comparatives, and an overall penalty against subject gap comparatives in base position. This is quite striking since extraposed clauses tend to be more difficult (Levy et al. 2012). Grant shows that these processing differences between canonical subject-gap relatives and subject-gap comparatives relate to underlying structural differences in the way the gaps are represented: essentially, the gap in a subject comparative is *within* a subject, which following Pancheva & Tomaszewicz (2011) constitutes a type of island violation (see Section 4 on islands). The lesson here, which will be repeated a few times below, is that deep understanding of the compositional details of the syntactic and semantic structures can offer insight into processing observations.

3.6 The role of prosody in embedding

Before leaving the topic of centre-embedding, one final ameliorating factor needs to be discussed. Fodor (2013) has argued that prosody plays a role in the diffi-

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culty – and amelioration – of centre-embedding. Fodor starts by pointing out the tension between the syntax of a doubly centre-embedded sentence – being a deeply-nested structure – and the requirements of prosodic phrasing – which is typically thought to require a flatter structure. In a sense, there is no optimal prosody for a doubly centre-embedded structure. This can pose processing difficulty because, as extensively argued Janet Fodor and colleagues (see Frazier & Gibson 2015 for an overview), silent reading involves implicit prosody. If there is no natural prosody that can be applied to such sentences, this could register as processing difficulty in reading studies.

Fodor (2013) and Fodor et al. (2018) lay out the prosodic options for double centre-embedding. Given the length of such sentences, a prosodic break is required. The break forms two phonological phrases (see, e.g. Selkirk 1996 on distinctions among phonological and intonational phrases). Phonological phrasing is sensitive to major syntactic boundaries (e.g. between subject and predicate), as well as phonological considerations. A relevant phonological consideration is that sister prosodic phrases prefer to be of equal weight. So while a natural place for a prosodic break in English is between subject and predicate, this delivers unbalanced phonological phrases in the case of double centre-embedding as in (22a). (The double pipe “||” indicates a phonological phrase break.) To create balanced phrases, the predicate needs to be longer and the RCs need to be shortened, the latter being achieved in (22b) with a pronoun (see above for a different interpretation of the pronoun manipulation).

- (22) a. The girl the man the cat scratched kicked || died.
 b. The girl the man I love met || died of cholera in 1960.
 (Fodor et al. 2018: (8)–(9))

Fodor et al. (2018) suggest that cases like (22a) are going to be hard to come by. So a more generally applicable solution requires dividing a double centre-embedding into three phonological phrases. This will alleviate some of the pressures on the length and weight, but here too there are pitfalls. Syntactic considerations pressure a second break between the matrix subject and the RCs (23a), but in order to have phrases of equal length the subject needs to be relatively heavy and the RCs relatively short – although here even a non-pronominal inner-most subject appears possible (23b).

- (23) a. The girl || that the young man I love met in Barcelona || died.
 b. The elegant woman || that the man Jill loves met || moved to Barcelona.
 (Fodor et al. 2018: (10),(12))

If the RCs are any longer, however, a fourth phrase is required. But this appears not to be helpful: (24) is on the verge of a list interpretation (see footnote 5 on Blumenthal 1966).

- (24) The elegant woman || that the man Jill loves || met on a cruise ship || moved to Barcelona.

The three-phrase prosody, then, seems to be the best compromise. Prosody should then be included as yet one more factor that interacts with the processing difficulty of centre-embedding. This avenue of research is relatively new, and experimental evidence is accumulating (I refer the reader to the studies cited in Fodor et al. 2018). But these prosodic manipulations have not, to my knowledge, been extensively tested with respect to the manipulations and measures discussed in the previous sections.

4 Extraction from embedded clauses

We have looked at syntactic dependencies that span embedded clauses. Syntactic dependencies can also enter into embedded clauses, as in long-distance *wh*-movement.

- (25) a. Who did they report [_{CP} that the police arrested ____].
 b. This is the criminal which they reported [_{CP} that the police arrested ____].

Frazier & Clifton (1989) reports the first experimental studies that examine long-distance extraction, finding that *wh*-dependencies that cross clause boundaries are more difficult to process than short, mono-clausal extraction. This work builds on an important finding in the processing of filler-gap dependencies known as the *Filled Gap Effect* (FGE) (Crain & Fodor 1985, Stowe 1986). A FGE arises in (26a): the filler *who* expects to find a gap after the verb *bring*. When this gap is ‘filled’ by material, like *us*, processing difficulty is incurred in comparison to the same lexical NP in a non filler-gap sentence like (26b).

- (26) a. My brother wanted to know *who* Ruth will bring us home to ____ at Christmas.
 b. My brother wanted to know if Ruth will bring us home to Mom at Christmas.

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Frazier (1987b) and Frazier & Clifton (1989) attribute FGEs to an *Active Filler Strategy* (28): when the processor encounters a filler (like a *wh*-phrase) it prioritizes finding a gap site, and so expects a gap as soon as possible, such as the direct object position in (26a).¹² When that position is filled, the processor's strategy is thwarted and processing difficulty registers.

(27) *Active Filler Hypothesis* (Frazier & Clifton 1989: 95)

When a filler has been identified, rank the option of assigning it to a gap above all other options.

Frazier & Clifton (1989) tested whether sentences with extraction from an embedded clause show FGEs (at *the guests* in (28b)) just as it is expected to in the short extraction in (28a). (The sentences in (28c), (28d) are non filler-gap controls, where no FGE is expected.)

(28) Frazier & Clifton (1989) Experiment 1 stimuli

- a. Who did the housekeeper from Germany urge the guests to consider ____?
- b. Who did the housekeeper say [_{CP} she urged the guests to consider ____]?
- c. The housekeeper from Germany urged the guests to consider the new chef.
- d. The housekeeper said [_{CP} she urged the guests to consider the new chef].

In self-paced reading measures, both the short and bi-clausal extractions elicited a FGE. Nonetheless, in this and other studies they report, longer reading times were found for long-distance extraction as well as reduced acceptability at an end-of-sentence grammaticality detection task. In another study (their Experiment 4), Frazier & Clifton (1989) directly compared short (29a) vs. long extraction (29b) with short vs. long non-extraction (yes/no) questions.

(29) Frazier & Clifton (1989) Experiment 4 stimuli (subset)

- a. What did your beautifully dressed niece mutter ____ to Willy in the house?
- b. What did you think your niece muttered ____ to Willy in the house?
- c. Did your beautifully dressed niece mutter something to Willy in the house?

¹²See Stowe 1986 for discussion of the matrix subject in (26a).

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- d. Did you think your niece muttered something to Willy in the house? chef].

They found an interaction in acceptability judgments, such that there was no difference between long and short non-extraction questions, but there was a difference between the long and short extractions, the former being judged less acceptable. Frazier & Clifton (1989) take these result to suggest both that fillers are active across clause boundaries, and that maintaining a filler across a clause boundary is costly. They claim that there is nothing about the complexity of the bi-clausal structure itself that could be driving the differences they observed, given the lack of an effect with the non-extraction sentences which were also bi-clausal questions (although see below for possible differences between content questions and yes/no questions). They offer an interpretation in terms of successive cyclic movement (Chomsky 1977, 1981, 1986), whereby long-distance movement proceeds in shorter steps, moving to the edge of each (finite) CP.

- (30) Who did you say [_{CP} ____ that Fred thinks [_{CP} ____ that Mary hired ____]]
- 

Since cross-clausal extraction requires creating an additional filler-gap dependency, this creates a substantial processing load. Gibson & Warren (2004) present evidence that successively cyclic movement can actually aide in processing, as it breaks up the span of material across which the filler has to be held in memory.

In more recent work, Wagers & Phillips (2014) also examined extraction from embedded clauses. They too used FGEs as a probe for active-filling, but using a refined version of this method due to Lee (2004). Instead of comparing filler-gap sentences to non filler-gap sentences as above, fillers of two different types are compered: NPs fillers (31a) vs. PP fillers (31b):

- (31) a. The stones *which* the pilgrim toppled **the cairn** for ____.
b. The stones *for which* the pilgrim toppled **the cairn** ____.

The unmarked word order in English requires NPs to precede PPs. So while the processor will predict a gap for the NP filler in (31a) just following the verb (and hence we expects a FGE at *the cairn*), the PP filler in (31b) makes no such prediction hence no FGE. This design has the advantage of comparing sentences with the same interpretation that both have filler-gap dependencies.

Wagers & Phillips (2014) import this design to test three types of dependency lengths: short mono-clausal (32), long mono-clausal where the subject has a modifying PP (32b), and bi-clausal (32c) where the gap is inside an embedded declar-

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ative clause. This three-level factor was crossed with whether the filler was an NP or PP (as illustrated in (31)).

- (32) Wagers & Phillips (2014), Experiment 1 stimuli
- a. ...*which/for which* the technician carefully prepared **the clean tubes** (for) ____ while wearing a mask...
 - b. ...*which/for which* the technician [_{PP} at the medical research facility] carefully prepared **the clean tubes** (for) ____ while wearing a mask...
 - c. ...*which/for which* the young biologist said [_{CP} that the technician carefully prepared **the clean tubes** (for) while wearing] a mask....

They indeed found elevated reading times (using a self-paced reading methodology) at the ‘filled’ direct object position for NP fillers, and they found this across all conditions. This is consistent with Frazier & Clifton (1989)’s findings, an encouraging result since the filled-gap design is more reliable here. They also report overall that the CP conditions were read more slowly, but this does not interact with filler type.

In addition to manipulating the category of the filler, Wagers & Phillips (2014) conducted a second study in which the plausibility of the filler with respect to the verb was tested (Traxler & Pickering 1996), e.g. *The acorns which the squirrels quickly crammed their small puffy cheeks with* (plausible) vs. *The cats which the squirrels quickly crammed their small puffy cheeks with*. If the processor is engaged in actively resolving the filler gap dependency, the semantic anomaly of the implausible condition should be detected at the verb itself. This is indeed what Wagers & Phillips (2014) found for short, mono-clausal extraction but not for the bi-clausal extractions. In that case, and to some extent in the PP case, the processing difficulty registered at a later region where the gap position was unambiguously present (object of a stranded preposition). Wagers & Phillips (2014) interpret these results within a broader question of whether filler-gap dependencies are processed with the Active Filler Strategy (where the filler is maintained in memory until it finds a satisfying gap) or a “retrieval”-based approach, whereby at a gap site, fillers that have been shunted out of working memory are retrieved (and this is susceptible to decay and interference in the ways suggested in Section 3). Wagers and Phillips argue that their results show that a hybrid account is warranted, in which dependency length plays a role. Gross, category-based predictions about a gap are, as is consistent with the Active Filler Strategy, maintained in working memory over both short and long dependencies, including cross-clausal dependencies. This is what the Filled Gap Effects showed. Finer grained information about the filler like certain semantic details, however, are

not actively maintain and therefore have to be retrieved. This retrieval strategy is sensitive to dependency distance, and so we do not see the same early effects in long-distance movement at the gap site. What remains to be sorted out is whether it is simply distance that regulates these memory resources or whether clausal embedding itself plays a role above and beyond distance.

4.1 Islands

One of the most well studied aspects of extraction are island phenomena (Ross 1967), where movement dependencies are blocked. Many islands are, of course, embedded clauses: *if*- and *whether*-clauses (33a), *wh*-islands (33b), complex NP islands (33c), sentential subject islands (33d) and factive islands (33e):

- (33) a. *? Who did you wonder if/whether they hired ____?
 b. *? Who did you wonder who to hire ____?
 c. * Who did you reject the claim that the should hire ____?
 d. * Who did [that you hired ____] surprise most?
 e. * How did you hate [that they played the song ____]

There is notoriously a great deal of gradation and variation in judgments about island violating sentences, something that has inspired many to seek processing explanations for the unacceptability of extraction from islands (Kluender & Kutas 1993, Kluender 1998, Hofmeister & Sag 2010 among others). Kluender (1998) and Hofmeister & Sag (2010), argue that islands are the result of a type of processing overload, much like centre-embedding. The idea is that in island-violating sentences, several factors conspire, each hard in themselves, to make such sentences unacceptable. One of those factors, they claim, is the general difficulty of long-distance extraction.¹³ The nature of the clause boundary also appears to incur a cost, independently of movement. Kluender & Kutas (1993) show that even in *yes-no* questions, in which no *wh*-element crosses a clause boundary, different types of embedded clauses are processed differently. Using Event-related potentials (ERPs, see Kutas & Federmeier 2007 for a handbook overview), they found neurophysiological responses which they interpreted as indicating a processing load to be highest with *wh*-complements, followed by *if*-complements, and lowest with *that*-complements.

- (34) a. Has she forgotten [that he dragged her to a movie on Christmas Eve]?

¹³Other factors they cite include the referential complexity of the *wh*-filler, and the kinds of referents within the span of the filler-gap dependency, as with Dependency Locality Theory.

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- b. Has she forgotten [if he dragged her to a movie on Christmas Eve]?
- c. Has she forgotten [who he dragged ____ to a movie on Christmas Eve]?

In a judgment study they also found a cline in acceptability, whereby declarative complements were judged more acceptable than embedded *if*-clauses, which in turn were more acceptable than embedded *wh*-questions. Kluender (1998) interprets these differences in relation to the “referential” load of the complements; Hofmeister & Sag (2010) point out that question complements are semantically more complex since, on most semantic theories, they invoke sets of alternative propositions (e.g. Hamblin 1973) whereas declarative complements simply denote simple propositions. The general idea is that the processing load of embedded questions, combined with the processing load of long-distance extraction, have a “super-additive” effect – a processing load that is more than the sum of its parts. This is enough to overload the processor. In particular, both of these features tax working memory beyond its resources, leading to the effect of reduced acceptability. Hofmeister & Sag (2010) are not precise about how the semantic complexity of embedded questions as compared to embedded declaratives taxes working memory, although one could imagine that the difficulty involves maintaining in working memory the kinds of alternatives required by questions (two such alternatives in embedded polar questions, possibly many such alternatives in *wh*-questions).

The processing-based approach to islands has been tested by Sprouse et al. (2012a,b). An important component of this work is their attempt to verify the proposed connection between acceptability and working memory. They charted participants’ working memory capacity, and asked whether there was a correlation between a participant’s working memory capacity and the strength of the super-additive effects of islands from ratings. They found no such correlation (although see the reply in Hofmeister et al. 2012, challenging both the statistical analyses in Sprouse et al. 2012a and their choice of working memory tests). What interests us in the context of embedding is the proposed cost of island structures independently of extraction. Sprouse et al. (2012a) systematically constructed factorial designs that help separate the difficulty of extraction from the difficulty of structures that involve islands. Below is an example of a 2×2 factorial design from their acceptability studies, in which all sentences are *wh*-questions. One factor was island status, whether the embedded clause is an island structure (here a *whether*-clause) as in (35c), (35d) or a non-island declarative clause (35a), (35b). The second factor was the nature of the filler-gap dependency: within a clause (35a), (35b) or cross-clausal. They tested four types of islands: *whether*-islands

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and Complex NP islands (shown (35) and (36)) as well as subject islands, adjunct islands, and complex NP islands (36).

(35) *whether islands* stimuli (Sprouse et al. 2012a)

- a. Who ___ thinks that John bought a car NON-ISLAND, MATRIX GAP
- b. What do you think that John bought ___ NON-ISLAND, EMBEDDED GAP
- c. Who ___ wonders whether John bought a car ISLAND, MATRIX GAP
- d. What do you wonder whether John bought ___ ISLAND, EMBEDDED GAP

(36) *Complex NP islands* stimuli (Sprouse et al. 2012a)

- a. Who ___ claimed that John bought a car NON-ISLAND, MATRIX GAP
- b. What do you claim that John bought ___ NON-ISLAND, EMBEDDED GAP
- c. Who ___ made the claim that John bought a car ISLAND, MATRIX GAP
- d. What do you make the claim that John bought ___
ISLAND, EMBEDDED GAP

Like the studies reported from Kluender & Kutas (1993), the contrast between (36a) and (36c) serves as test of whether islands in general – even if they do not involve extraction from the island – bear an independent processing load. In an acceptability judgement task, Sprouse et al. (2012a) found an independent cost for *whether*-islands as well as subject and adjunct islands, but not complex NP islands. That is, there was no significant difference in acceptability between (36a) and (36c). Importantly, the island violating condition in (36d) was still judged just as unacceptable as the other island violation conditions in the experiment. This finding calls into question the idea that the processing load of island structures, independent of extraction, contributes to a super-additive effect that leads to the unacceptability of all islands. There has been a recent expansion of experimental work using the factorial definition of islands to different languages, examining variation within and across languages and within and across island types (Sprouse et al. 2016, Kush et al. 2018, Pañeda et al. 2020, Lu et al. 2020).

There are also accounts of islands that reduce unacceptability to discourse-pragmatic properties (Erteschik-Shir 1973). Following Goldberg (2006), the studies in Ambridge & Goldberg (2008), Goldberg (2013), Abeillé et al. (2020) pursue the idea that backgrounded or discourse-given constituents block extraction of question phrases they contain because givenness is at odds with the pragmatic function of questions as seeking new information. Many islands constructions correspond to material that is often backgrounded in the discourse, e.g. subjects,

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factive complements. This approach needs to address whether non-question extraction (e.g. relatives, degree, topicalization) shows similar effects, and if so why, given that the discourse properties of questions is absent with such extractions. At the moment of writing this chapter, I believe that these questions are under active investigation.

Another “reductionist” approach to island phenomena is that proposed in Liu et al. (2022). They argue that frequency can explain the perceived unacceptability of island effects in factive and manner-of-speaking verbs. These are claimed in the formal literature to resist extraction, at least of certain elements (**How tall did John know/whisper that the mountain was?*; cf. *How tall did John think that the mountain was?*). Liu et al. (2022)’s claim is that acceptability of extraction in these sentences depends on the language user’s exposure to clausal complements generally, not any grammatical or even pragmatic feature of these constructions. Their verb frame frequency account posits that the more likely a verb is to appear with a clausal complement, the higher the acceptability of extraction from the complement. Liu et al. (2022), however, examined only wh-object NP extraction from factive and manner-of-speaking complements. Since at least Cinque (1990), if not earlier, it has been known that these types of ‘weak’ islands are *selective*, allowing some kinds of wh-phrase to extract rather readily but not so others, including adjuncts and degree phrases (Szabolcsi & Lohndal 2017). Indeed, Potter & Carlson (2019) and Williams et al. (2022) report that when these kinds of extractees are taken into consideration, an island effect emerges. Williams et al. (2022) further show that while frequency does affect acceptability, alongside this an effect of factivity remains. Further work, however, needs to pin down the etiology of factive islands. The consensus in the formal linguistics literature is that some form of semantic or pragmatic (Abrusán 2011, Schwarz et al. 2019) deviance is at play, and this is often tied to the nature of questions. As with other accounts noted above, further experimental work could be helpful in teasing about the role of question phrase extraction versus extraction in general. Another area of interest is possible differences among embedding verbs. Very recently, Degen & Tonhauser (2022) contend that there is no crisp category of factive verb; rather, the extent to which a factive presupposition projects varies greatly by verb and is gradient. In my opinion, experimental work on clause embedding going forward will benefit from much more attention to subtle discourse-pragmatic effects, teasing these apart from purely syntactic phenomena. For some recent experimental and corpus work on issues involving the discourse-pragmatics of embedding (including factivity and embedded verb second), see Djärv (2019).

5 Attachment

5.1 Attachment to embedded clauses

Some of the earliest psycholinguist work in sentence processing was devoted to showing that the clause is privileged as a major unit of processing (Jarvella 1971, Jarvella & Herman 1972, Caplan 1972, Frazier 1978, Marslen-Wilson et al. 1978, Bever & Townsend 1979). In these early studies, much of the attention was paid to what is retained after clausal and sentential boundaries. Generally the gross message of a clause is retained but not its linguistic form (Sachs 1967). For instance, Jarvella (1971) had participants hear passages like those in (37).

- (37) a. With this possibility, Taylor left the capital. After *he had returned to Manhattan*, he explained the offer to his wife.
 b. Taylor did not reach a decision until after *he had returned to Manhattan*. He explained the offer to his wife.

The clause of interest is in italics above: in (37a) it is part of the second sentence; in (37b) it is part of the first sentence. Participants were better at repeating verbatim this particular clauses in cases like (37a), suggesting that the linguistic form of a clause is shunted from memory after it is closed.¹⁴

A relevant, but “little known” (Frazier & Clifton 2005) study by Walker et al. (1968) cited in Fodor et al. (1974), highlights the privileged status of the matrix clause. Walker et al. (1968) used a two-word probe experiment in which participants responded to whether the probe words were part of a previously seen sentence, like (38).

- (38) The scout the Indians saw killed a buffalo.

If the probe words were from different clauses (e.g. *Indians* and *buffalo*) participants responded more slowly than if they were drawn from the same clause.

¹⁴This era saw a number of interesting studies concerning main and subordinate clauses. One popular question was the linear order of main (matrix) vs. subordinate clause. Clark & Clark (1968, 1977), Fodor et al. (1974) present evidence that the main-subordinate order is preferred, and memory is better when clauses are present in that order (but see Townsend & Bever 1978, Bever & Townsend 1979 on interesting reversals in terms of subordinator type). Clark (1973) found that when children begin to use embedding structures, they use the main-subordinate order. Jou & Harris (1990) found, in a sentence recall method, a bias toward main-subordinate structures. Another set of studies suggests that the temporal (or causal) order of the events described factors into the linguistic order (Opacic 1973), including developmental studies by Johnson (1975).

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Interestingly, they were faster to respond to two words from the main clause (*scout, killed*) than two words drawn from the embedded clause. Frazier & Clifton (2005) suggest that this reflects that the matrix clause typically carries the main assertion of the sentence – “most accessible or salient part of the sentence.”

The Walker et. al. results are about the “message” of a clause and therefore its informational relevance. In terms of syntactic processing, by contrast, recent and embedded clauses typically define the domain of accessible material. For example, languages have a strong locality preference in attachment ambiguities involving two VP attachment sites, as in (39) Carreiras & Jr. (1993):

- (39) Juan dijo que Bill se murió (#morirá) ayer.
 Juan said that Bill died (#will die) yesterday.

There is strong preference, in both Spanish and English, to attach the temporal adjunct *yesterday* to the embedded clause, which is also the more recent clause. This holds even when the tense of the embedded clause – which is processed in advance of the adjunct – is semantically incompatible with *yesterday*. In classic Garden Path Theory (Frazier 1978, 1987a), this parsing preference falls under the principle of *Late Closure*, which requires the processor to attach incoming material to the phrase currently being processed, which in this case would be the embedded clause. (Dependency Locality Theory also prioritizes local attachment, Gibson 2000.)

There is a tension, then, in the processing of complex embedded structures. The syntax likes local, low attachment whereas discourse pressures privilege matrix clause material. Inspired by a range of findings, including those in Walker et al. (1968), Frazier & Clifton (2005) propose the *Main Assertion Hypothesis* (see also Frazier 1990’s notion of relativized relevance).

- (40) *Main Assertion Hypothesis* (Frazier & Clifton 2005)

Other things being equal, comprehenders prefer to relate material in a new sentence to the main assertion of the preceding sentence.

In one interesting study, Traxler & Frazier (2008) test the Main Assertion Hypothesis by comparing attachment preferences with main vs. embedded clauses. In their stimuli, shown in (41), the phrase of relevance for attachment is the PP *to the store*: it can attach to the lower verb *checked* or the higher verb *delivered*. A lower attachment is consistent with syntactic attachment preferences (such as Late Closure) while a higher attachment is consistent with the Main Assertion Hypothesis. The prediction is that the Main Assertion Hypothesis will exert

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greater pressure if the higher verb belongs to a matrix clause (as in (41c), (41d)) and so higher attachments will be preferred here. If the higher of the two verbs itself belongs to an embedded clause as in (41a), (41b), that clause does not serve as the main assertion, so the Main Assertion Hypothesis will not prefer high PP attachment. In addition to this clause-type factor, the grammatical attachment site of the PP was manipulated via verb selection properties: (41a), (41c) require higher verb attachment (since *checked to the store* is ungrammatical), whereas (41b), (41d) require low attachment.

- (41) a. Before the baker **delivered** the bread that he **checked [to the store]** on the corner, the customer stopped by for some donuts.
 b. Before the baker **checked** the bread that he **delivered [to the store]** on the corner, the customer stopped by for some donuts.
 c. The baker **delivered** the bread that he **checked [to the store]** early this morning, but it was too late.
 d. The baker **checked** the bread that he **delivered [to the store]** early this morning, but it was too late.

In contrast to predictions of the Main Assertion Hypothesis, if syntactic attachment alone matters, readers should have difficulty with sentences in (41a), (41c) since low attachment leads to an ill-formed sentence. In an eye-tracking while reading experiment, Traxler & Frazier (2008) found difficulty for the embedded clause condition (41a) but crucially not for the matrix clause condition (41c). That is, when the PP needed to be attached high, this was easier for readers if the clause it attached to was a matrix clause. A matrix clause is more likely to serve as a main assertion, and so the Main Assertion Hypothesis acts a counter-vailing pressure against low attachment.

Frazier & Clifton (2005) report evidence for the Main Assertion Hypothesis from verb phrase ellipsis. In both sentences in (42), the elided VP could correspond to the whole matrix clause (Tina laughed after she made a joke) or the embedded clause (Tina made a joke).

- (42) a. Mary laughed after she made a joke about the supervisor. Then Tina did too.
 b. After Mary laughed she made a joke about the supervisor. Then Tina did too.

They found that participants chose to resolve the ellipsis with the matrix clause antecedent over 70% of the time in both (42a), (42b).

The moral of the story is that both the syntax and discourse-pragmatic properties of embedded clauses influence processing, often in divergent and competing ways. These insights tie into a wealth of research on how discourse properties influence sentence processing, perhaps the earliest and most prominent of which is the contention by Crain & Steedman (1985) that the famous garden path in (43) is a result of pragmatic considerations, not context-blind syntactic parsing preferences.

(43) The horse raced past the barn fell. (Bever 1970)

The target restrictive relative (RR) parse requires the listener to not only imagine a (unique) horse that was raced past the barn, but also a contrast set of horses which do not have the property ascribed by the relative.¹⁵ This is a complex discourse model. The more-readily pursued main verb (MV) analysis only requires the reader to allow the existence of a horse, not a contrast set. In this sense, the MV parse requires a more parsimonious or simpler discourse model. Crain & Steedman (1985) propose a Principle of Parsimony: the reading which carries fewer unsupported presuppositions¹⁶ will be favored over one that carries more. Grodner et al. (2005) found that even without ambiguous input, readers find restriction difficult to process in a context that does not motivate restriction. They compared (unambiguous) restrictive and (unambiguous) non-restrictive relatives and found that in null contexts, the restrictive relatives were read more slowly. (When supporting contexts were provided, the effect reversed direction.) They argue that the results support the Principle of Parsimony, in that the restrictive relative requires a more elaborate discourse model (evoking a contrast set) which the non-restrictive does not.¹⁷ Additionally, a number of studies have shown that individuals prefer unmodified NPs in a number of ambiguous constructions and that the Principle of Parsimony is responsible (Ni et al. 1996, Sedivy 2002, Spivey-Knowlton & Sedivy 1995). Findings such as these are used to support constraint-based models of processing, in which a variety of linguistic and extra-linguistic bits of information can guide processing. A particularly stunning demonstration

¹⁵Modifiers do not always implicate contrast – see below – but the claim here is that readers are more likely to pursue a contrastive interpretation for the relative clause (Sedivy 2003).

¹⁶Crain & Steedman (1985) used the word *presupposition* to describe the contrast set component, but Portner (1989) and Sedivy (2003) show that the contrast set evoked by a restrictive relative is a conversational/Gricean implicature.

¹⁷The principle of parsimony relies, in turn, on a more general pressure following from the Gricean Maxim of Quantity: Speakers should say as much as needed to be informative, without saying more than is necessary (Grice 1975). That is, since the simple discourse model is preferred, any additional modification in the description is redundant.

of this are the studies in Schwarz (2007), which show that presupposition triggers can guide readers to otherwise syntactically dispreferred structures.

5.2 Attachment of embedded clauses

In the previous section, we examined ambiguities that arise when material can attach to either a matrix or embedded clause. This section looks at ambiguities about where embedded clauses themselves attach. This serves as a case study revealing the importance of careful syntactic and semantic analyses to the study of embedded clause processing.

As we saw above, the processor generally likes to attach material low in the syntactic tree, which constitute more local or recent material. And this appears to hold up cross-linguistically, e.g. the low attachment of the adjunct in (39) in Spanish and English. Cuetos & Mitchell (1988), however, discovered an apparent exception to this. They found that English and Spanish contrast in the preferred attachment site of the relative clause in sentences like those in (44).

- (44) a. Someone shot the maid₁ of the actress₂ [who was standing on the balcony].
 b. Alguien disparó contra la criada₁ de la actriz₂ [que estaba en el balcón].
 Someone shot at the maid of the actress that was on the balcony

The issue is the attachment of the relative clause *who was standing on the balcony*: it can grammatically attach to the higher noun phrase (NP₁, which is headed by *maid*) or to the lower NP₂ (*actress*).

English speakers prefer low attachment, while Spanish speakers prefer high attachment (Cuetos & Mitchell 1988). A number of languages show similar preferences for high attachment: French (Mitchell et al. 1990, Frenck-Mestre & Pynte 2000, Zagar et al. 1997, Colonna & Pynte 2001), Dutch (Brysbaert & Mitchell 1996, Mitchell & Brysbaert 1998, Mitchell et al. 2000, Desmet et al. 2002), Greek (Papadopoulou & Clahsen 2003), and Serbo-Croatian (Lovrić 2003). Other languages behave like English, including Basque (Gutierrez-Ziardegi et al. 2004), Romanian (Ehrlich et al. 1999), and Chinese (Shen 2006). These findings call into question the universality of parsing principles: why would some languages show preferences for high attachment and some low? Moreover the presence of high attachment preference languages is puzzling from the perspective of locality: the attachment that requires minimal effort is NP₂. For decades, this cross-linguistic difference posed a major puzzle for processing.

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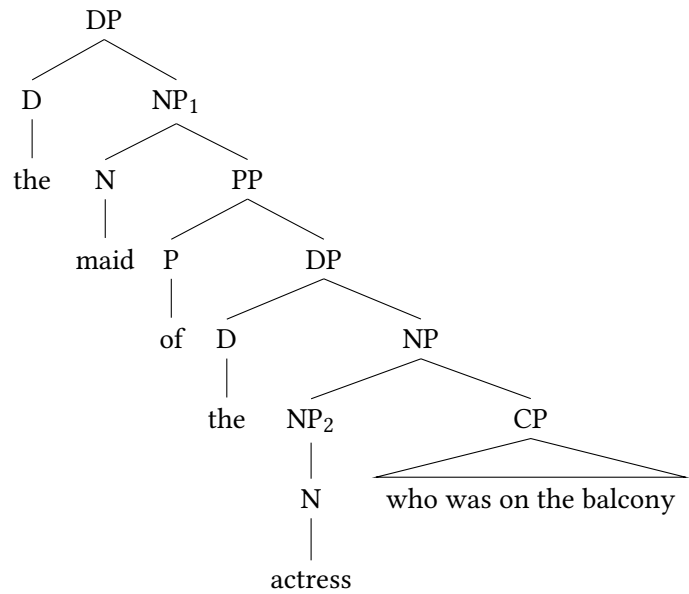


Figure 1: Low attachment

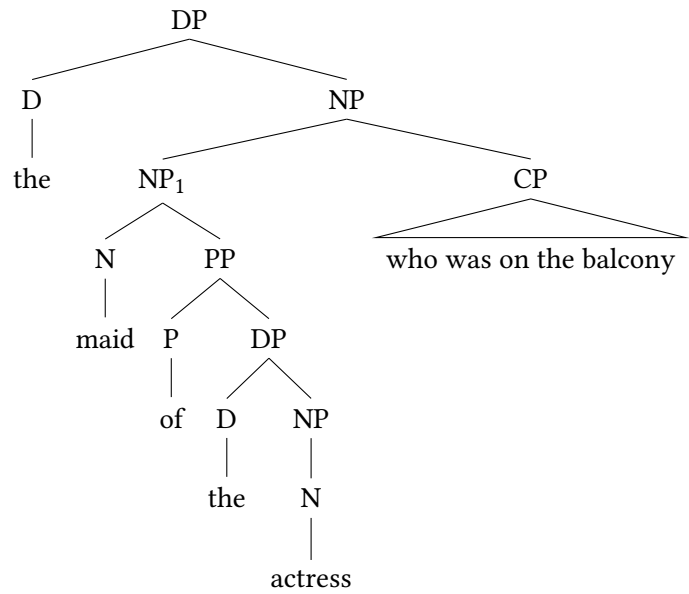


Figure 2: High attachment

10 Processing embedded clauses

In recent work, Grillo and colleagues (Grillo & Costa 2014) have revealed that the stimuli used to test RC attachment suffer from a confound in certain languages. The problem is that languages like Spanish have a distinct construction called a pseudo-relative (PR) that can often be string-identical to a RC. Many of the sentences used to test RC attachment are actually ambiguous between a true RC structure and a PR. The PR structure however is only compatible, for reasons discussed below, with high attachment. If the sentence is parsed as a PR and not an RC – which Grillo shows is very often the case – then the high attachment behaviour does not have any bearing on attachment preferences.

Examples of pseudo-relatives from Spanish, Italian, French and Greek are given below. PRs are superficially similar to relatives, since they involve noun phrases followed by tensed clause with a gap. The easiest way to distinguish PRs from RCs is to use a proper noun in the place of the head noun, as below. This forces either a PR parse or an appositive relative. The appositive parse, however, has a distinctive prosody, which the sentences in (45) do not need to have.

- (45) a. He visto a **Juan que** corría. (Spanish)
 I.have seen to Juan that ran.IMPF
 ‘I saw Juan running.’
 b. Ho visto **Gianni che** correva. (Italian)
 I.have seen Gianni that ran.IMPF
 ‘I saw Gianni running.’
 c. J’ai vue **Jean qui** courait. (French)
 I.have seen Jean that ran.IMPF
 ‘I saw Jean running.’
 d. I Maria evlepe **ton Jani pu** etrexe. (Greek)
 the Mary watch.PAST.IMP the John.ACC that run.PAST.IMPF
 ‘Mary was watching John running.’

(Grillo & Costa 2014)

Unlike (appositive) relatives PRs are constructions in which the “head” and the finite clause are in a subject-predicate relation, and the PR as a whole describes an event, not the individual denoted by the “head.” This can be appreciate by the Italian cleft sentences in (46): the *wh*-element in (46a) is the inanimate *cio* which refers to the event of Mario crying, not Mario. If an animate *wh*-word (*chi*) is used then the post-copular phrase cannot be a pseudo-relative and must be an appositive.

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- (46) a. *Ciò₁ che ho visto è [Mario che piangeva]₁.*
 That which I.have seen is Mario that cry.IMPF
 ‘What I saw was Mario crying’
 b. **Chi₂ ho visto è [Mario₂ che piangeva].*
 Who I.have seen is Mario that cry.IMPF
 ‘Who I saw was Mario crying’ (after Radford 1977: 160(98))¹⁸

The event-denoting nature of PRs restricts the types of predicates that can select it, which are typically event-selecting perception verbs. There are a number of other distinctions between RCs and PRs, including the fact that PRs allow only subject gaps and that the tense of the PR and the matrix clause are highly restricted (they must have simultaneous interpretations).

In the above examples, the PR is shown with a proper noun head, but a common noun is possible as well. And, indeed that common noun can be a complex NP of the sort employed in the RC attachment sentences, as in (47), a French example.

- (47) a. *J’ai vu le fils de l’homme qui courait.*
 I.have seen the son of the’man that ran.
 ‘I saw the son of the man running.’

Recall that as in Spanish, sentences such as these were argued to show a high attachment preference. But (47) is in fact three-ways ambiguous. It could involve an RC attached to either of the NPs. Or it could involve a PR in which the “head” *the son of the man* serves as a PR subject whose predicate is the CP *qui courait* (Kayne 1975, Moulton & Grillo 2015, Grillo & Moulton 2016).

If the PR parse is pursued for Figure 3, it must be the *son* who is running since the complex DP is the subject of the PR. It is not possible for a PR to be a complement of *fils* ‘son’ (via the preposition *de* ‘of’) since ‘son’ does not select for event-denoting phrases.¹⁹ In this sense, PRs force a “high attachment” of the clause.

Grillo & Costa (2014) showed that removing the possibility of a PR parse in Italian reduced the high attachment preference. They compared sentences where

¹⁸(OK if appositive rel.)

¹⁹If the noun itself denotes an event, like the object language *evento* ‘event’, a PR can be introduced by a preposition.

- (i) *l’evento di Maria che ballava.*
 the’event of Maria that danced
 ‘the event of Maria dancing’

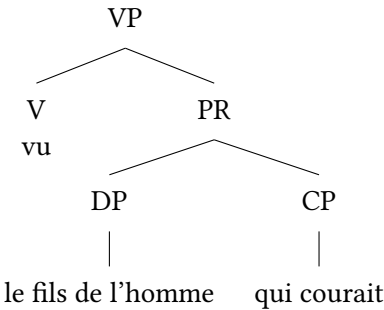


Figure 3: Pseudo-relative structure

the matrix verb was perceptual like *see* (which licenses either an RC or a PR) to those in which the matrix verb was a stative verb like *live with* (which does not license a PR, only an RC). For the perceptual verbs, the rate of high attachment was 78.6%. For the non-PR-taking stative verbs, the rate of high attachment went down to 24.2%. Similar results were found for other PR-languages (Grillo & Spathas 2014, Fernandes 2012, Tomaz et al. 2014, Aguilar & Grillo 2016). Pozniak et al. (2019) reach similar conclusions for French when PRs and RCs are disambiguated using tense properties. These results offer a lesson: informed linguistics is integral to understanding processing results and creating the appropriate stimuli. Attachment differences among languages lead to decades of research devoted to a finding that was, in part if not entirely, due to a simple linguistic confound between RCs and PRs.

Of course, if the presence of a PR analysis is what is responsible for the high attachment preferences found in languages like Spanish and Italian, then that means that readers are pursuing, in the face of ambiguity, a PR analysis over an RC analysis (see Pozniak et al. 2019 for recent discussion and ideas). Here there is an interesting connection to recent work exploring the difference between relative clause and clausal complements of nouns. We already saw, in Cowper (1976)’s examples (see (9)), that clausal complements of nouns, even if centre-embedded, are easier to process than relative clauses (and their relative nesting is crucially important). Staub et al. (2018) have recently presented evidence that a complement clause (CC) parse (48a) is generally preferred over a relative clause (RC) structure (48b):

- (48) Staub et al. (2018) Experiment 1 stimuli
- a. The information [_{CC} that the health department provided a cure] re-
assured the tour operators. CC

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- b. The information [$_{RC}$ that the health department provided] reassured
the tour operators. RC

Their processing results suggest that the processor “avoids” an RC parse. They relate this to Vincenzi’s (1991) *Minimal Chain Principle* which militates against postulating filler-gap dependencies quite generally. This would favour CC analysis (which does not involve a filler gap dependency) over the RC analysis. What is particularly interesting about Staub et al. (2018) is that the CC preference did not correlate with the general bias of the selecting noun (cf. Trueswell et al. 1993 discussed above). Staub et al. do note, however, that the role of context might modulate the differences, suggesting that a context that supports a contrast set for the head noun (e.g. the existence of multiple pieces of information) might better motivate a RC, potentially reducing the CC preference or reversing the preference altogether. This question, I think, is particularly warranted in light of the earlier discussion in Section 5.1 of Grodner et al. (2005)’s finding that non-restrictive relatives show advantages over restrictive relatives in impoverished contexts. Complement clauses in complex noun phrases like (48b) bear a semantic relation to the head noun that, while still poorly understood, is often characterized as similar to an appositive one (see Stowell 1981, Moulton 2009, 2015, Krapova & Cinque 2016, De Cuba 2017). Additionally, it has been argued that definite complex noun phrases with complement clauses (as opposed to those with relative clauses) involve a weaker notion of definiteness (Hankamer & Mikkelsen 2021, Srinivas & Legendre 2024). These subtleties could influence the processing of CCs and RCs independently of the presence of a gap. Comparing CCs with appositive, non-restrictive relatives might help put the two embedded clauses on more equal footing for their contextual requirements. This, of course, would require a language where the appositive, non-restrictive relativizer/complementizer is identical to the CC introducing complementizer.

More recently, Konrad et al. 2021 find further evidence in Italian and French that the parser avoids postulating gaps where possible. These studies are rather notable for their focus on syntactic ambiguities involving embedded clauses that have not been widely explored in the processing literature, including free relatives and indirect questions.

6 Conclusion

This overview has not touched on a number of important topics in clauses embedding. One prominent lacuna are infinitival and other “reduced” embedded clauses.

10 *Processing embedded clauses*

Infinitivals have played a large role in the processing of control structures (Frazier et al. 1983, Boland et al. 1990, Demestre et al. 1999, Green 2018). Another area is the processing of centre-embedded infinitivals, verb clusters, and clause union phenomena (Vasishth 2003, Bayer et al. 2005), something that touches on famous differences between crossed and nested dependencies that arise in these constructions across languages (Bach et al. 1986). *That*-trace effects have also played a prominent role in the literature; for a recent overview see Cowart & McDaniel (2021). This overview has focused exclusively on comprehension rather than production; on this, the reader is referred to the interesting studies concerning multi- and embedded clause production in Ferreira & Swets (2017); for a recent overview of production see Momma (2021).

Embedded clauses are probably one of the psycholinguist's best tools. They are a testing ground for theories of syntactic parsing and memory maintenance and retrieval. In fact, it is rare to find psycholinguistic work on sentence processing that is squarely about embedding itself, rather than using embedded clauses as a tool to probe the architecture of the processor or the memory systems deployed for language comprehension. It is indeed the *drosophila* of sentence processing. This is as it should be: we do not choose the object of scientific study because it's our favourite specimen or construction, but because it offers a window into the things we want to know. But a word of caution: the linguistic landscape of embedding is complex and nuanced and there are many false friends. The conundrum between relative clauses and pseudo-relatives is one good lesson, so are the nuanced theories of centre-embedding that arise from taking into account Cowper's (1976) classic contrast between complement and relative clause nesting and centre-embedding in head-final languages. As noted throughout this review, especially in connection with islands effects (Section 4.1) and the processing of relative versus complement clauses (Section 5.1), some of the interesting questions going forward involve the discourse-pragmatic status of embedded clauses. At the same time, cross-linguistic coverage of work on processing embedding remains limited. Of course, investigating the processing of different types of embedding systems should be motivated by theoretical questions, be they psycholinguistic or formal linguistic.

If this chapter can offer anything, it is the advice to look at a broader range of embedded clause structures in sentence processing. This strategy has a good track record, when it is paired with precise questions about sentence processing. It profits both theories of processing and theories of syntactic and semantic structure. So maybe you can choose your favourite embedded structure to look at, but find a processing friend to explore it with.

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Handbook of Clausal Embedding

The fact that we can embed one clause within another, as in this very sentence, is a curious property of human language. On the one hand, clauses express complete thoughts and can be used to perform separate speech acts; on the other, they can be constituents of such thoughts and contribute to their formation. In this way, they are the prime example of recursivity, which is discussed as one of the distinctive features of the human language faculty.

This *Handbook of Clausal Embedding* aims to introduce and familiarize readers with current research on clausal subordination. It attempts to cover all aspects of embedding configurations – their formal syntactic and morphological markings, their effects on prosody, their semantics, their pragmatics as well as their relation to discourse phenomena. It also discusses clausal embedding in language acquisition and its importance for our understanding of the mind of others, exploring, moreover, its diachronic development and realization across a broad spectrum of linguistic systems. In this, the topic of clausal embedding is relevant for many subdisciplines of linguistics, and it is important for all linguistic frameworks. We as editors, and our authors, have tried to concentrate on the central aspects of their respective subtopics, and make them accessible to students and researchers in linguistics and adjacent fields, like, among others, psycholinguistics and cognitive science. We thought it prudent to have a rather comprehensive notion of clausal embedding that encompasses what might be considered borderline cases. We understand clausal embedding as comprising configurations in which one or more clauses, or expressions that have clause-like denotations like infinitive constructions or deverbal nominalizations, are hosted within another clause or expression with a clause-like denotation. This covers argument clauses, adjunct clauses, and relative clauses. We included coordinated clauses and parenthetical clauses, as they also may constitute parts of larger clauses. Furthermore, we discussed the presence of embedding within discourse, which further contributes to our understanding of the intricate and overarching nature of clausal organization within language.